

The Thomas algorithm, when applied in this fashion, generally avoids buildup of computer truncation errors because usually none of the steps involves subtraction of nearly equal quantities. Furthermore, computed values of $x_{i,j}$ are almost always positive. The algorithm is highly efficient, requires a minimum of computer storage as noted previously, and is superior to alternative matrix-inversion routines. A modified Thomas algorithm for difficult cases is given by Boston and Sullivan [9]. Such cases can occur for columns having large numbers of equilibrium stages and with components whose absorption factors [see (5-38)] are less than unity in one section of stages and greater than unity in another section.

Bubble-Point (BP) Method for Distillation

Frequently, distillation involves species that cover a relatively narrow range of vapor-liquid equilibrium ratios (K -values). A particularly effective solution procedure for this case was suggested by Friday and Smith [7] and developed in detail by Wang and Henke [1]. It is referred to as the bubble-point (BP) method because a new set of stage temperatures is computed during each iteration from bubble-point equations of the type discussed in Section 4.4. In the method, all equations are partitioned and solved sequentially except for the modified M equations, which are solved separately for each component by the tridiagonal matrix technique.

The algorithm for the Wang-Henke BP method is shown in Figure 10.5. A FORTRAN computer program for the method is available [10]. Problem specifications consist of conditions and stage location of all feeds, pressure at each stage, total flow rates of all side streams (note that liquid distillate flow rate, if any, is designated as U_1), heat transfer rates to or from all stages except stage 1 (condenser) and stage N (reboiler), total number of stages, external bubble-point reflux flow rate, and vapor distillate flow rate. A sample problem specification is shown in Figure 10.6.

To initiate the calculations, values for the tear variables are assumed. For most problems, it is sufficient to establish an initial set of V_j values based on the assumption of constant molar interstage flows using the specified reflux, distillate, feed, and side stream flow rates. A generally adequate initial set of T_j values can be provided by computing or assuming both the bubble-point temperature of an estimated bottoms product and the dew-point temperature of an assumed vapor distillate product; or computing or assuming bubble-point temperature if distillate is liquid or a temperature in-between the dew-point and bubble-point temperatures if distillate is mixed (both vapor and liquid); and then determining the other stage temperatures by assuming a linear variation of temperature with stage location.

To solve (10-12) for x_i by the Thomas method, $K_{i,j}$ values are required. When they are composition dependent, initial

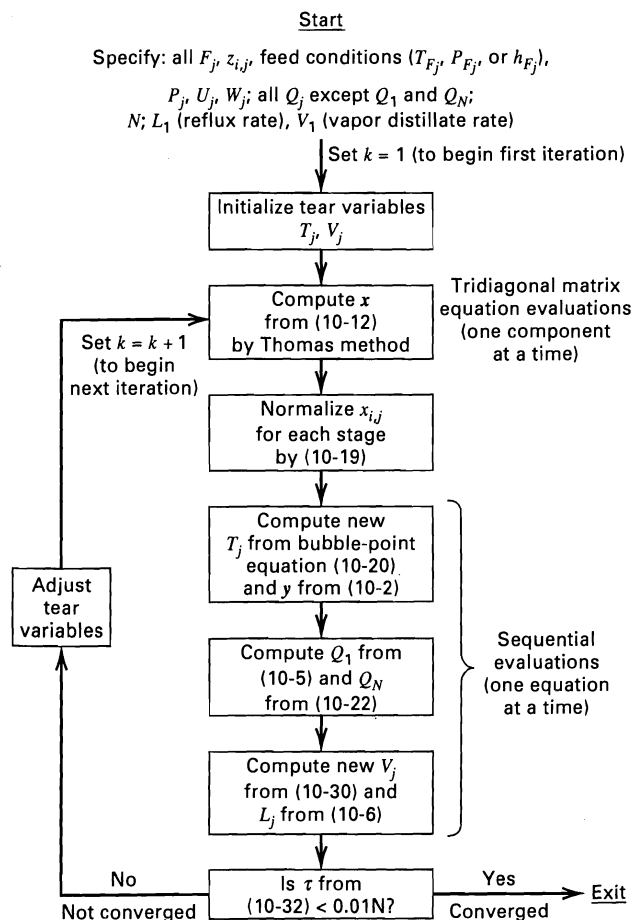


Figure 10.5 Algorithm for Wang-Henke BP method for distillation.

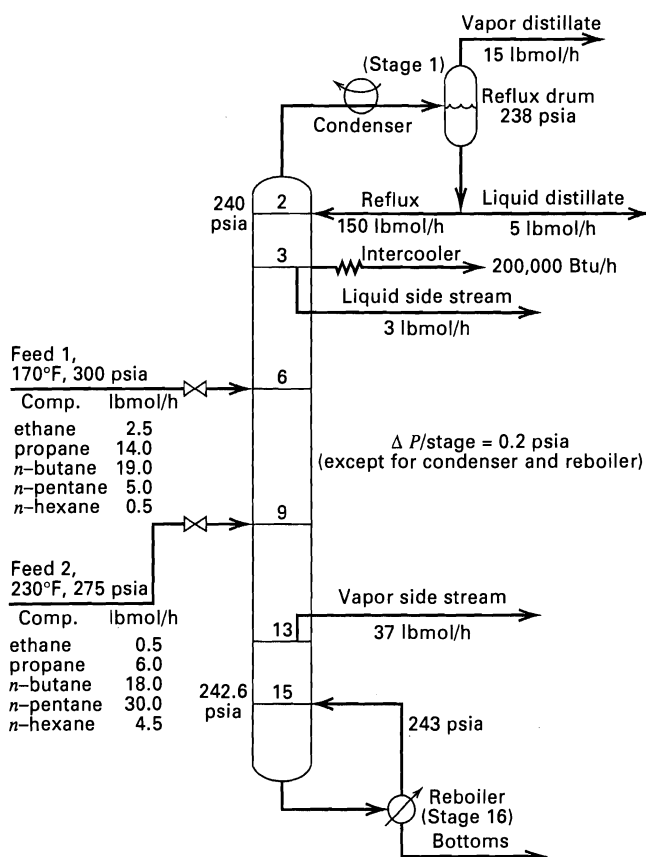


Figure 10.6 Sample specification for application of Wang-Henke BP method to distillation.

assumptions for all $x_{i,j}$ and $y_{i,j}$ values are also needed unless ideal K -values are employed for the first iteration. For each iteration, the computed set of $x_{i,j}$ values for each stage will, in general, not satisfy the summation constraint given by (10-4). Although not mentioned by Wang and Henke, it is advisable to normalize the set of computed $x_{i,j}$ values by the relation

$$(x_{i,j})_{\text{normalized}} = \frac{x_{i,j}}{\sum_{i=1}^C x_{i,j}} \quad (10-19)$$

These normalized values are used for all subsequent calculations involving $x_{i,j}$ during the iteration.

A new set of temperatures T_j is computed stage by stage by computing bubble-point temperatures from the normalized $x_{i,j}$ values. Friday and Smith [7] showed that bubble-point calculations for stage temperatures are particularly effective for mixtures having a narrow range of K -values because temperatures are not then sensitive to composition. For example, in the limiting case where all components have identical K -values, the temperature corresponds to the conditions of $K_{i,j} = 1$ and is not dependent on $x_{i,j}$ values. At the other extreme, however, bubble-point calculations to establish stage temperatures can be very sensitive to composition. For example, consider a binary mixture containing one component with a high K -value that changes little with temperature. The second component has a low K -value that changes very rapidly with temperature. Such a mixture is methane and n -butane at 400 psia. The effect on the bubble-point temperature of small quantities of methane dissolved in liquid n -butane is very large, as indicated by the following results.

Liquid Mole Fraction of Methane	Bubble-Point Temperature, °F
0.000	275
0.018	250
0.054	200
0.093	150

Thus, the BP method is best when components have a relatively narrow range of K -values.

The necessary bubble-point equation is obtained in the manner described in Chapter 4 by combining (10-2) and (10-3) to eliminate $y_{i,j}$, giving

$$\sum_{i=1}^C K_{i,j} x_{i,j} - 1.0 = 0 \quad (10-20)$$

which is nonlinear in T_j and must be solved iteratively. Wang and Henke prefer to use Muller's iterative method [11] because it is reliable and does not require the calculation of derivatives. Muller's method requires three initial assumptions of T_j . For each assumption, the value of S_j is computed from

$$S_j = \sum_{i=1}^C K_{i,j} x_{i,j} - 1.0 = 0 \quad (10-21)$$

The three sets of (T_j, S_j) are fitted to a quadratic equation for S_j in terms of T_j . The quadratic equation is then employed to predict T_j for $S_j = 0$, as required by (10-20). The validity of

this value of T_j is checked by using it to compute S_j in (10-21). The quadratic fit and S_j check are repeated with the three best sets of (T_j, S_j) until some convergence tolerance is achieved, say $|T_j^{(n)} - T_j^{(n-1)}|/T_j^{(n)} \leq 0.0001$, with T in absolute degrees, where n is the iteration number for the temperature loop in the bubble-point calculation, or one can use $S_j \leq 0.0001$ C, which is preferred.

Values of $y_{i,j}$ are determined along with the calculation of stage temperatures using the E equations, (10-2). With a consistent set of values for $x_{i,j}$, T_j , and $y_{i,j}$, molar enthalpies are computed for each liquid and vapor stream leaving a stage. Since F_1 , V_1 , U_1 , W_1 , and L_1 are specified, V_2 is readily obtained from (10-6), and the condenser duty, a (+) quantity, is obtained from (10-5). Reboiler duty, a (-) quantity, is determined by summing (10-5) for all stages to give

$$Q_N = \sum_{j=1}^N (F_j h_{F_j} - U_j h_{L_j} - W_j h_{V_j}) - \sum_{j=1}^{N-1} Q_j - V_1 h_{V_1} - L_N h_{L_N} \quad (10-22)$$

A new set of V_j tear variables is computed by applying the following modified energy balance, which is obtained by combining (10-5) and (10-6) twice to eliminate L_{j-1} and L_j . After rearrangement,

$$\alpha_j V_j + \beta_j V_{j+1} = \gamma_j \quad (10-23)$$

where

$$\alpha_j = h_{L_{j-1}} - h_{V_j} \quad (10-24)$$

$$\beta_j = h_{V_{j+1}} - h_{L_j} \quad (10-25)$$

$$\gamma_j = \left[\sum_{m=1}^{j-1} (F_m - W_m - U_m) - V_1 \right] (h_{L_j} - h_{L_{j-1}}) + F_j (h_{L_j} - h_{F_j}) + W_j (h_{V_j} - h_{L_j}) + Q_j \quad (10-26)$$

and enthalpies are evaluated at the stage temperatures last computed rather than at those used to initiate the iteration. Written in didiagonal matrix form (10-23) applied over stages 2 to $N-1$ is:

$$\begin{bmatrix} \beta_2 & 0 & 0 & 0 & \dots & 0 \\ \alpha_3 & \beta_3 & 0 & 0 & \dots & 0 \\ 0 & \alpha_4 & \beta_4 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & 0 & \alpha_{N-3} & \beta_{N-3} & 0 \\ 0 & \dots & 0 & 0 & \alpha_{N-2} & \beta_{N-2} \\ 0 & \dots & 0 & 0 & 0 & \alpha_{N-1} & \beta_{N-1} \end{bmatrix} \begin{bmatrix} V_3 \\ V_4 \\ V_5 \\ \dots \\ V_{N-2} \\ V_{N-1} \\ V_N \end{bmatrix} = \begin{bmatrix} \gamma_2 - \alpha_2 V_2 \\ \gamma_3 \\ \gamma_4 \\ \dots \\ \gamma_{N-3} \\ \gamma_{N-2} \\ \gamma_{N-1} \end{bmatrix} \quad (10-27)$$

Matrix equation (10-27) is readily solved one equation at a time by starting at the top where V_2 is known and working down recursively. Thus,

$$V_3 = \frac{\gamma_2 - \alpha_2 V_2}{\beta_2} \quad (10-28)$$

$$V_4 = \frac{\gamma_3 - \alpha_3 V_3}{\beta_3} \quad (10-29)$$

or, in general

$$V_j = \frac{\gamma_{j-1} - \alpha_{j-1} V_{j-1}}{\beta_{j-1}} \quad (10-30)$$

and so on. Corresponding liquid flow rates are obtained from (10-6).

The solution procedure is considered to be converged when sets of $T_j^{(k)}$ and $V_j^{(k)}$ values are within some prescribed tolerance of corresponding sets of $T_j^{(k-1)}$ and $V_j^{(k-1)}$ values, where k is the iteration index. One possible convergence criterion is

$$\sum_{j=1}^N \left[\frac{T_j^{(k)} - T_j^{(k-1)}}{T_j^{(k)}} \right]^2 + \sum_{j=1}^N \left[\frac{V_j^{(k)} - V_j^{(k-1)}}{V_j^{(k)}} \right]^2 \leq \epsilon \quad (10-31)$$

where T is the absolute temperature and ϵ is some prescribed tolerance. However, Wang and Henke suggest that the following simpler criterion, which is based on successive sets of T_j values only, is adequate.

$$\tau = \sum_{j=1}^N [T_j^{(k)} - T_j^{(k-1)}]^2 \leq 0.01 N \quad (10-32)$$

Successive substitution is often employed for iterating the tear variables; that is, values of T_j and V_j generated from (10-20) and (10-30), respectively, during an iteration are used directly to initiate the next iteration. However, experience indicates that it is desirable frequently to adjust the values of the generated tear variables prior to beginning the next iteration. For example, upper and lower bounds should be placed on stage temperatures, and any negative values of interstage flow rates should be changed to near-zero positive values. Also, to prevent oscillation of the iterations, damping can be employed to limit changes in the values of V_j and absolute T_j from one iteration to the next to, say, 10%.

EXAMPLE 10.1

For the distillation column shown in Figure 10.7, do one iteration of the BP method up to and including the calculation of a new set of T_j values from (10-20). Use composition-independent K -values.

SOLUTION

By overall total material balance

$$\text{Liquid distillate} = U_1 = F_3 - L_5 = 100 - 50 = 50 \text{ lbmol/h}$$

Then

$$L_1 = (L_1/U_1)U_1 = (2)(50) = 100 \text{ lbmol/h}$$

By total material balance around the total condenser

$$V_2 = L_1 + U_1 = 100 + 50 = 150 \text{ lbmol/h}$$

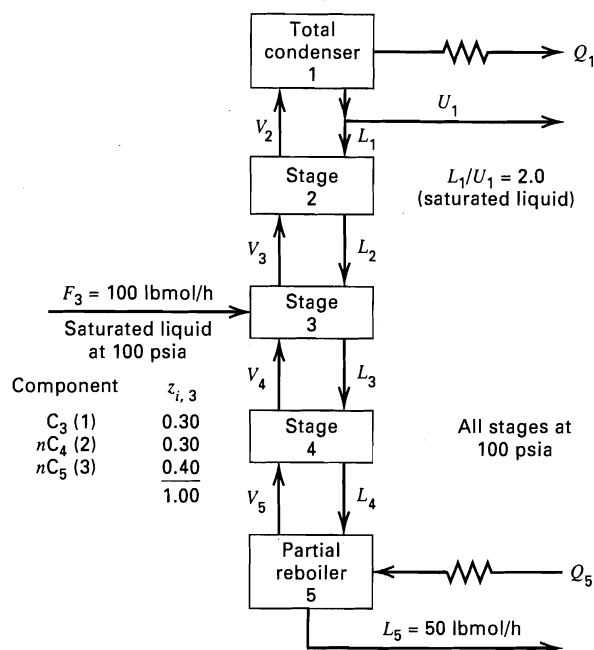


Figure 10.7 Specifications for distillation column of Example 10.1.

Initial guesses of tear variables are

Stage j	V_j , lbmol/h	T_j , °F
1	(Fixed at 0 by specifications)	65
2	(Fixed at 150 by specifications)	90
3	150	115
4	150	140
5	150	165

At 100 psia, the estimated K -values at the assumed stage temperatures are

Stage	$K_{i,j}$				
	1	2	3	4	5
$C_3(1)$	1.23	1.63	2.17	2.70	3.33
$nC_4(2)$	0.33	0.50	0.71	0.95	1.25
$nC_5(3)$	0.103	0.166	0.255	0.36	0.49

The matrix equation (10-12) for the first component C_3 is developed as follows. From (10-8) with $V_1 = 0$, $W = 0$,

$$A_j = V_j + \sum_{m=1}^{j-1} (F_m - U_m)$$

Thus, $A_5 = V_5 + F_3 - U_1 = 150 + 100 - 50 = 200 \text{ lbmol/h}$.

Similarly, $A_4 = 200$, $A_3 = 100$, and $A_2 = 100$ in the same units.

From (10-9) with $V_1 = 0$, $W = 0$,

$$B_j = - \left[V_{j+1} + \sum_{m=1}^j (F_m - U_m) + U_j + V_j K_{1,j} \right]$$

Thus, $B_5 = - [F_3 - U_1 + V_5 K_{1,5}] = - [100 - 50 + (150)(3.33)] = -549.5 \text{ lbmol/h}$.

Similarly, $B_4 = -605$, $B_3 = -525.5$, $B_2 = -344.5$, and $B_1 = -150$ in the same units.

From (10-10), $C_j = V_{j+1} K_{1,j+1}$. Thus, $C_1 = V_2 K_{1,2} = 150(1.63) = 244.5 \text{ lbmol/h}$.

Similarly, $C_2 = 325.5$, $C_3 = 405$, and $C_4 = 499.5$ in the same units.

From (10-11), $D_j = -F_j z_{1,j}$. Thus, $D_3 = -100(0.30) = -30$ lbmol/h.

Similarly, $D_1 = D_2 = D_4 = D_5 = 0$.

Substitution of the above values in (10-7) gives

$$\begin{bmatrix} -150 & 244.5 & 0 & 0 & 0 \\ 100 & -344.5 & 325.5 & 0 & 0 \\ 0 & 100 & -525.5 & 405 & 0 \\ 0 & 0 & 200 & -605 & 499.5 \\ 0 & 0 & 0 & 200 & -549.5 \end{bmatrix} \begin{bmatrix} x_{1,1} \\ x_{1,2} \\ x_{1,3} \\ x_{1,4} \\ x_{1,5} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -30 \\ 0 \\ 0 \end{bmatrix}$$

Using (10-14) and (10-15), we apply the forward step of the Thomas algorithm as follows.

$$p_1 = \frac{C_1}{B_1} = 244.5/(-150) = -1.630$$

$$q_1 = \frac{D_1}{B_1} = 0/(-150) = 0$$

$$p_2 = \frac{C_2}{B_2 - A_2 p_1} = \frac{325.5}{-344.5 - 100(-1.630)} = -1.793$$

By similar calculations, the matrix equation after the forward-elimination procedure is

$$\begin{bmatrix} 1 & -1.630 & 0 & 0 & 0 \\ 0 & 1 & -1.793 & 0 & 0 \\ 0 & 0 & 1 & -1.170 & 0 \\ 0 & 0 & 0 & 1 & -1.346 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{1,1} \\ x_{1,2} \\ x_{1,3} \\ x_{1,4} \\ x_{1,5} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0.0867 \\ 0.0467 \\ 0.0333 \end{bmatrix}$$

Applying the backward steps of (10-17) and (10-18) gives

$$x_{1,5} = q_5 = 0.0333$$

$$x_{1,4} = q_4 - p_4 x_{1,5} = 0.0467 - (-1.346)(0.0333) = 0.0915$$

Similarly,

$$x_{1,3} = 0.1938, x_{1,2} = 0.3475, x_{1,1} = 0.5664$$

The matrix equations for nC_4 and nC_5 are solved in a similar manner to give

Stage	$x_{i,j}$				
	1	2	3	4	5
C_3	0.5664	0.3475	0.1938	0.0915	0.0333
nC_4	0.1910	0.3820	0.4483	0.4857	0.4090
nC_5	0.0191	0.1149	0.3253	0.4820	0.7806
$\sum_i x_{i,j}$	0.7765	0.8444	0.9674	1.0592	1.2229

After these compositions are normalized, bubble-point temperatures at 100 psia are computed iteratively from (10-20) and compared to the initially assumed values,

Stage	$T^{(2)}, ^\circ\text{F}$	$T^{(1)}, ^\circ\text{F}$
1	66	65
2	94	90
3	131	115
4	154	140
5	184	165

The rate of convergence of the BP method is unpredictable, and, as shown in Example 10.2, it can depend drastically on the assumed initial set of T_j values. In addition, cases with high reflux ratios can be more difficult to converge than cases with low reflux ratios. Orbach and Crowe [12] describe a generalized extrapolation method for accelerating convergence based on periodic adjustment of the tear variables when their values form geometric progressions during at least four successive iterations.

EXAMPLE 10.2

Calculate stage temperatures, interstage vapor and liquid flow rates and compositions, reboiler duty, and condenser duty by the BP method for the distillation column specifications given in Example 5.4.

SOLUTION

The computer program of Johansen and Seader [10] based on the Wang-Henke procedure was used. In this program, no adjustments to the tear variables are made prior to the start of each iteration, and the convergence criterion is (10-32). The K -values and enthalpies are computed from correlations for hydrocarbons. The only initial assumptions required are distillate and bottoms temperatures shown previously for four cases.

The significant effect of initially assumed distillate and bottoms temperatures on the number of iterations required to satisfy (10-32) is indicated by the following results.

Case	Assumed Temperatures, $^\circ\text{F}$		Number of Iterations for Convergence
	Distillate	Bottoms	
1	11.5	164.9	29
2	0.0	200.0	5
3	20.0	180.0	12
4	50.0	150.0	19

The terminal temperatures of Case 1 were within a few degrees of the exact values and were much closer estimates than those of the other three cases. Nevertheless, Case 1 required the largest number of iterations. Figure 10.8 is a plot of τ from (10-32) as a function of the number of iterations for each of the four cases. Case 2 converged rapidly to the criterion of $\tau < 0.13$. Cases 1, 3, and 4 converged rapidly for the first three or four iterations, but then moved only slowly toward the criterion. This was particularly true of Case 1, for which application of a convergence-acceleration method would be particularly desirable. In none of the four cases did oscillations of values of the tear variables occur; rather the values approached the converged results in a monotonic fashion.

The overall results of the converged calculations, as taken from Case 2, are shown in Figure 10.9. Product-component flow rates were not quite in material balance with the feed. Therefore, adjusted values that do satisfy overall material-balance equations were determined by averaging the calculated values and are included in Figure 10.9. A smaller value of τ would have improved the overall material balance. Figures 10.10 to 10.13 are plots of converged values for stage temperatures, interstage flow rates, and mole-fraction compositions from the results of Case 2. Results from the other

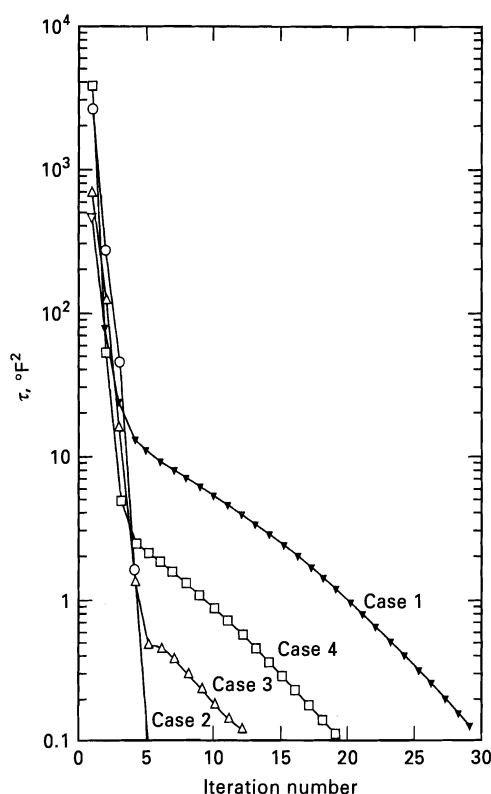


Figure 10.8 Convergence patterns for Example 10.2.

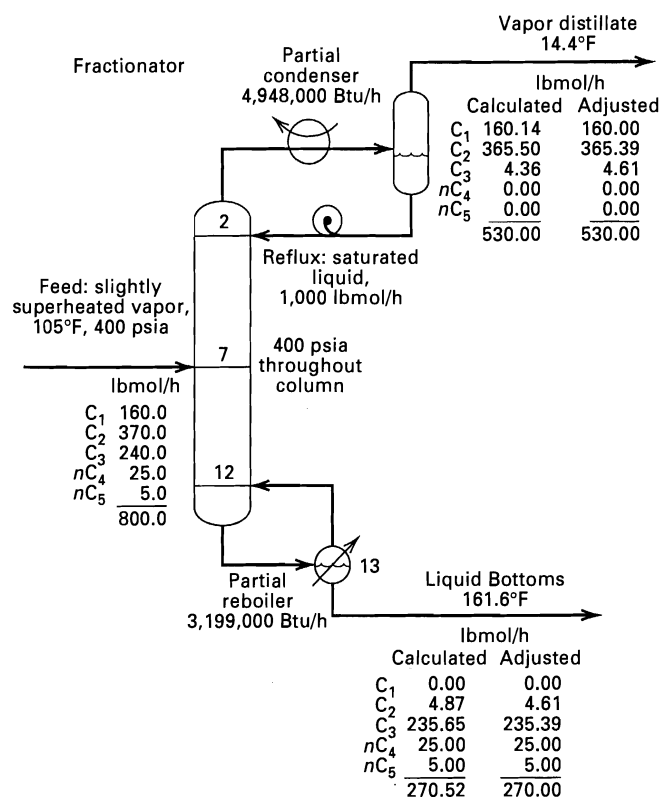


Figure 10.9 Specifications and overall results for Example 10.2.

three cases were almost identical to those of Case 2. Included in Figure 10.10 is the initially assumed linear temperature profile. Except for the bottom stages, it does not deviate significantly from the converged profile. A jog in the profile is seen at the feed stage. This is a common occurrence.

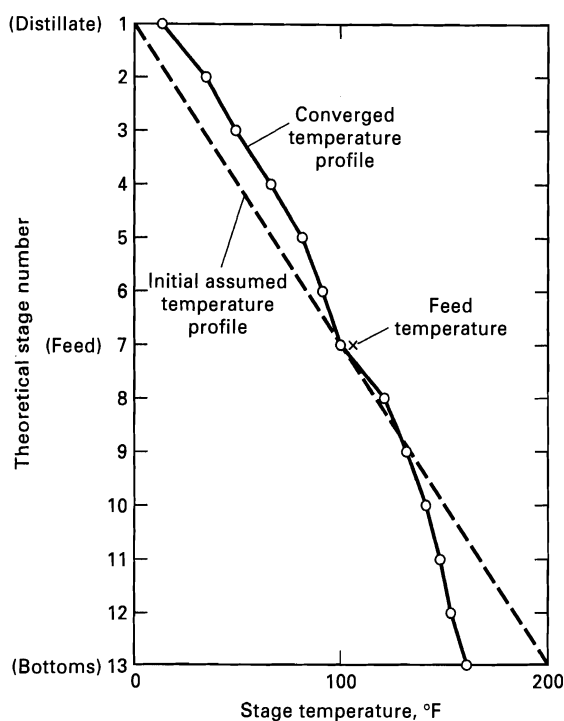


Figure 10.10 Converged temperature profile for Example 10.2.

In Figure 10.11, it is seen that the assumption of constant interstage molar flow rates does not hold in the rectifying section. Both liquid and vapor flow rates decrease in moving down from the top stage toward the feed stage. Because the feed is vapor near the dew point, the liquid rate changes only slightly across the feed stage. Correspondingly, the vapor rate decreases across the feed stage by an amount equal to the feed rate. For this problem, the interstage molar flow rates are almost constant in the stripping section. However, the assumed vapor flow rate in this section based on adjusting the rectifying-section rate across the feed zone is approximately 33% higher than the average converged vapor rate. A much better initial estimate of the vapor rate in the stripping section can be made by first computing the reboiler duty from the condenser duty based on the specified reflux rate and then determining the corresponding vapor rate leaving the partial reboiler.

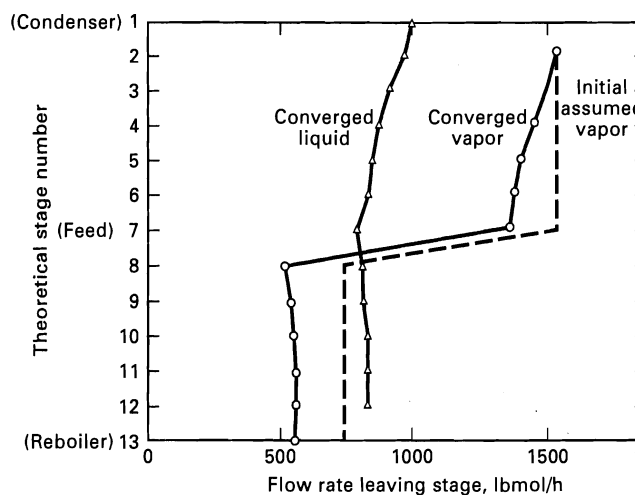


Figure 10.11 Converged interstage flow rate profiles for Example 10.2.

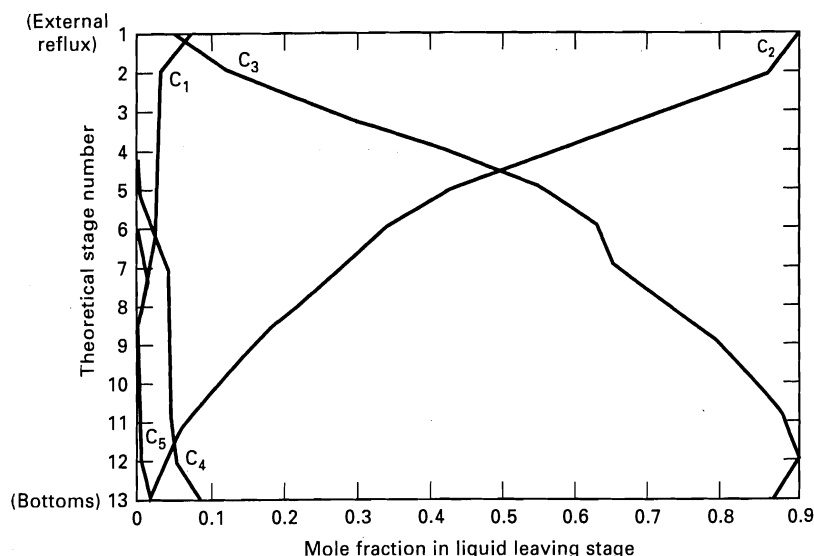


Figure 10.12 Converged liquid composition profiles for Example 10.2.

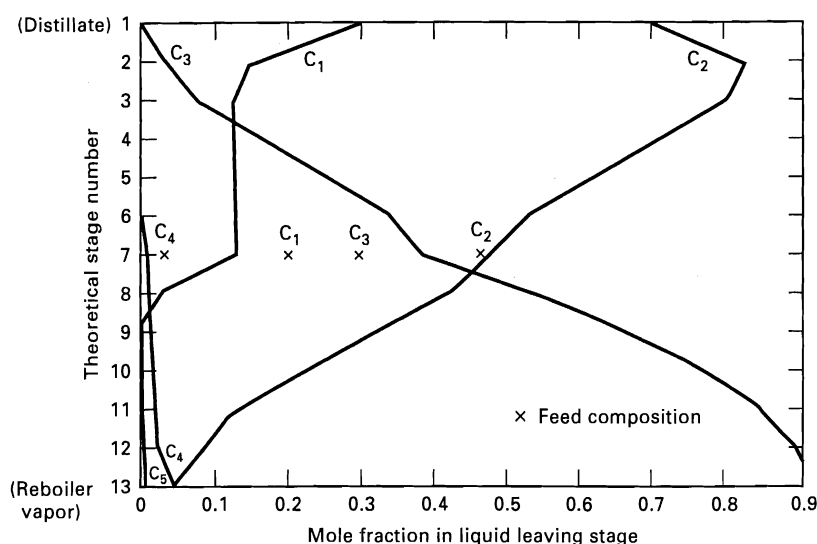


Figure 10.13 Converged vapor composition profiles for Example 10.2.

For this problem, the separation is between C_2 and C_3 . Thus, these two components can be designated as the light key (LK) and heavy key (HK), respectively. Thus C_1 is a lighter-than-light key (LLK), and nC_4 and nC_5 are heavier than the heavy key (HHK). Each of these four designations exhibits a different type of composition-profile curve as shown in Figures 10.12 and 10.13. Except at the feed zone and at each end of the column, both liquid and vapor mole fractions of the light key (C_2) decrease smoothly and continuously from the top of the column to the bottom. The inverse occurs for the heavy key (C_3). Mole fractions of methane (LLK) are almost constant over the rectifying section except near the top. Below the feed zone, methane rapidly disappears from both vapor and liquid streams. The inverse is true for the two HHK components. In Figure 10.13, it is seen that the feed composition is somewhat different from the composition of either the vapor entering the feed stage from the stage below or the vapor leaving the feed stage.

For problems where a specification is made of the distillate flow rate and the number of theoretical stages, it is difficult to specify the feed-stage location that will give the highest degree of separation. However, once the results of a rigorous calculation are available, a modified McCabe–Thiele plot

based on the key components [13] can be constructed to determine whether the feed stage is optimally located or whether it should be moved. For this plot, mole fractions of the light-key component are computed on a nonkey-free basis. The resulting diagram for Example 10.2 is shown in Figure 10.14. It is seen that the trend toward a pinched-in region is more noticeable in the rectifying section just above stage 7 than in the stripping section just below stage 7. This suggests that a better separation between the key components might be made by shifting the feed entry to stage 6. The effect of feed-stage location on the percent loss of ethane to the bottoms product is shown in Figure 10.15. As predicted from Figure 10.14, the optimal feed stage is stage 6.

Sum-Rates Method for Absorption and Stripping

The chemical components present in most absorbers and strippers cover a relatively wide range of volatility. Hence, the BP method of solving the MESH equations will fail because calculation of stage temperature by bubble-point

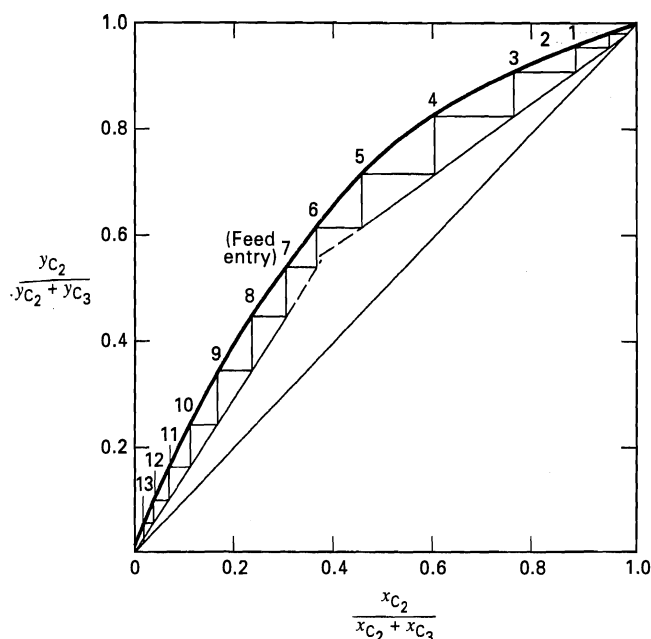


Figure 10.14 Modified McCabe-Thiele diagram for Example 10.2.

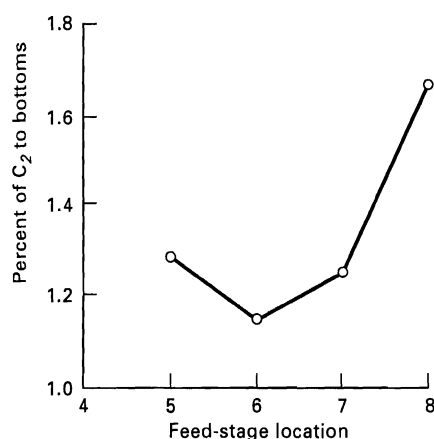


Figure 10.15 Effect of feed-stage location on separation for Example 10.2.

determination (10-20) is too sensitive to liquid-phase composition, and the stage energy balance (10-5) is much more sensitive to stage temperatures than to interstage flow rates. In this case, Friday and Smith [7] showed that an alternative procedure devised by Sujata [14] could be successfully applied. This procedure, termed the *sum-rates (SR) method*, was further developed in conjunction with the tridiagonal-matrix formulation for the modified *M* equations by Burningham and Otto [15].

Figure 10.16 shows the algorithm for the Burningham-Otto SR method. A FORTRAN computer program for the method is available [16]. Problem specifications consist of conditions and stage locations for all feeds, pressure at each stage, total flow rates of any side streams, heat-transfer rates to or from any stages, and total number of stages.

An initial set of tear variables T_j and V_j is assumed to initiate the calculations. For most problems it is sufficient to

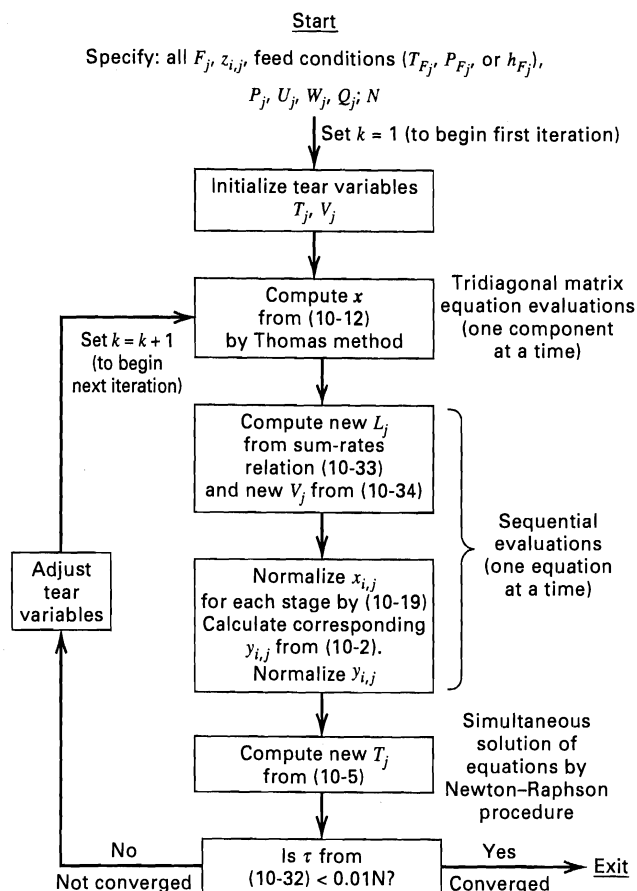


Figure 10.16 Algorithm for Burningham-Otto SR method for absorption/stripping.

assume a set of V_j values based on the assumption of constant-molar interstage flows, working up from the bottom of the absorber using specified vapor feeds and any vapor side stream flows. Generally, an adequate initial set of T_j values can be derived from assumed top-stage and bottom-stage values and a linear variation with stages in-between.

Values of $x_{i,j}$ are obtained by solving (10-12) by the Thomas algorithm. However, the values obtained are not normalized at this step but are utilized directly to produce new values of L_j by applying (10-4) in the form referred to as the *sum-rates equation*:

$$L_j^{(k+1)} = L_j^{(k)} \sum_{i=1}^C x_{i,j} \quad (10-33)$$

where values of $L_j^{(k)}$ are obtained from values of $V_j^{(k)}$ by (10-6). Corresponding values of $V_j^{(k+1)}$ are obtained from a total material balance, which is derived by summing (10-1) over the C components, combining the result with (10-3) and (10-4), and summing that result over stages j through N to give

$$V_j = L_{j-1} - L_N + \sum_{m=j}^N (F_m - W_m - U_m) \quad (10-34)$$

Normalized values of $x_{i,j}$ are next calculated from (10-19). Corresponding values of $y_{i,j}$ are computed from (10-2).

A new set of values for stage temperatures T_j is obtained by solving the simultaneous set of energy-balance relations for the N stages given by (10-5). The temperatures are embedded in the specific enthalpies corresponding to the unspecified vapor and liquid flow rates. In general, these enthalpies are nonlinear in temperature. Therefore, an iterative solution procedure is required, such as the Newton-Raphson method [10].

In the Newton-Raphson method, the simultaneous, nonlinear equations in terms of variables x_i are written in zero form:

$$f_i\{x_1, x_2, \dots, x_n\} = 0, \quad i = 1, 2, \dots, n \quad (10-35)$$

Initial guesses, marked by asterisks, are provided for the n variables and each function is expanded about these guesses in a Taylor's series that is terminated after the first derivatives to give

$$0 = f_i\{x_1, x_2, \dots, x_n\} \quad (10-36)$$

$$\approx f_i\{x_1^*, x_2^*, \dots, x_n^*\} + \left. \frac{\partial f_i}{\partial x_1} \right|_* \Delta x_1 + \left. \frac{\partial f_i}{\partial x_2} \right|_* \Delta x_2 + \dots + \left. \frac{\partial f_i}{\partial x_n} \right|_* \Delta x_n \quad (10-37)$$

where $\Delta x_j = x_j - x_j^*$.

Equations (10-36) are linear and can be solved directly for the corrections Δx_i . If the corrections are all found to be zero, the guesses are correct and equations (10-35) have been solved; if not, the corrections are added to the guesses to provide a new set of guesses that are applied to (10-36). The procedure is repeated, in a series of r iterations, until all the corrections, and thus the functions, become zero to within some tolerance. In recursion form (10-36) and (10-37) are

$$\sum_{j=1}^n \left[\left(\frac{\partial f_i}{\partial x_j} \right)^{(r)} \Delta x_j^{(r)} \right] = -f_i^{(r)}, \quad i = 1, 2, \dots, n \quad (10-38)$$

$$x_j^{(r+1)} = x_j^{(r)} + \Delta x_j^{(r)}, \quad j = 1, 2, \dots, n \quad (10-39)$$

EXAMPLE 10.3

Solve the simultaneous, nonlinear equations

$$x_1 \ln x_2 + x_2 \exp(x_1) = \exp(1)$$

$$x_2 \ln x_1 + 2x_1 \exp(x_2) = 2 \exp(1)$$

for x_1 and x_2 to within ± 0.001 , by the Newton-Raphson method.

SOLUTION

In the form of (10-35), the two equations are

$$f_1\{x_1, x_2\} = x_1 \ln x_2 + x_2 \exp(x_1) - \exp(1) = 0$$

$$f_2\{x_1, x_2\} = x_2 \ln x_1 + 2x_1 \exp(x_2) - 2 \exp(1) = 0$$

From (10-38), the linearized recursive form of these equations is

$$\left(\frac{\partial f_1}{\partial x_1} \right)^{(r)} \Delta x_1^{(r)} + \left(\frac{\partial f_1}{\partial x_2} \right)^{(r)} \Delta x_2^{(r)} = -f_1^{(r)}$$

$$\left(\frac{\partial f_2}{\partial x_1} \right)^{(r)} \Delta x_1^{(r)} + \left(\frac{\partial f_2}{\partial x_2} \right)^{(r)} \Delta x_2^{(r)} = -f_2^{(r)}$$

The solution of these two equations is readily obtained by the method of determinants to give

$$\Delta x_1^{(r)} = \frac{\begin{vmatrix} f_2^{(r)} & \left(\frac{\partial f_1}{\partial x_2} \right)^{(r)} \\ f_1^{(r)} & \left(\frac{\partial f_2}{\partial x_2} \right)^{(r)} \end{vmatrix}}{D}$$

and

$$\Delta x_2^{(r)} = \frac{\begin{vmatrix} f_1^{(r)} & \left(\frac{\partial f_2}{\partial x_1} \right)^{(r)} \\ f_2^{(r)} & \left(\frac{\partial f_1}{\partial x_1} \right)^{(r)} \end{vmatrix}}{D}$$

where

$$D = \begin{vmatrix} \left(\frac{\partial f_1}{\partial x_1} \right)^{(r)} & \left(\frac{\partial f_2}{\partial x_2} \right)^{(r)} \\ \left(\frac{\partial f_2}{\partial x_1} \right)^{(r)} & \left(\frac{\partial f_1}{\partial x_2} \right)^{(r)} \end{vmatrix}$$

and the derivatives as obtained from the equations are

$$\left(\frac{\partial f_1}{\partial x_1} \right)^{(r)} = \ln(x_2^{(r)}) + x_2^{(r)} \exp(x_1^{(r)}),$$

$$\left(\frac{\partial f_2}{\partial x_1} \right)^{(r)} = \frac{x_2^{(r)}}{x_1^{(r)}} + 2 \exp(x_2^{(r)})$$

$$\left(\frac{\partial f_1}{\partial x_2} \right)^{(r)} = \frac{x_1^{(r)}}{x_2^{(r)}} + \exp(x_1^{(r)}),$$

$$\left(\frac{\partial f_2}{\partial x_2} \right)^{(r)} = \ln(x_1^{(r)}) + 2x_1^{(r)} \exp(x_2^{(r)})$$

As initial guesses, take $x_1^{(1)} = 2$, $x_2^{(1)} = 2$. Applying the Newton-Raphson procedure, one obtains the following results where at the sixth iteration, values of $x_1 = 1.0000$ and $x_2 = 1.0000$ correspond closely to the required values of zero for f_1 and f_2 .

r	$x_1^{(r)}$	$x_2^{(r)}$	$f_1^{(r)}$	$f_2^{(r)}$	$(\partial f_1 / \partial x_1)^{(r)}$	$(\partial f_1 / \partial x_2)^{(r)}$	$(\partial f_2 / \partial x_1)^{(r)}$	$(\partial f_2 / \partial x_2)^{(r)}$	$\Delta x_1^{(r)}$	$\Delta x_2^{(r)}$
1	2.0000	2.0000	13.4461	25.5060	15.4731	8.3891	15.7781	30.2494	-0.5743	-0.5436
2	1.4247	1.4564	3.8772	7.3133	6.4354	5.1395	9.6024	12.5880	-0.3544	-0.3106
3	1.0713	1.1457	0.7720	1.3802	3.4806	3.8541	7.3591	6.8067	-0.0138	-0.1878
4	1.0575	0.9579	-0.0059	0.1290	2.7149	3.9830	6.1183	5.5679	-0.0591	0.0417
5	0.9984	0.9996	-0.0057	-0.0122	2.7126	3.7127	6.4358	5.4244	0.00159	0.000368
6	1.0000	1.0000	5.51×10^{-6}	2.86×10^{-6}	2.7183	3.7183	6.4366	5.4366	12.1×10^{-6}	-3.0×10^{-6}
7	1.0000	1.0000	0.0	-2×10^{-9}	2.7183	3.7183	6.4366	5.4366	—	—

As applied to the solution of a new set of T_j values from the energy equation (10-5), the recursion equation for the Newton-Raphson method is

$$\left(\frac{\partial H_j}{\partial T_{j-1}}\right)^{(r)} \Delta T_{j-1}^{(r)} + \left(\frac{\partial H_j}{\partial T_j}\right)^{(r)} \Delta T_j^{(r)} + \left(\frac{\partial H_j}{\partial T_{j+1}}\right)^{(r)} \Delta T_{j+1}^{(r)} = -H_j^{(r)} \quad (10-40)$$

where

$$\Delta T_j^{(r)} = T_j^{(r+1)} - T_j^{(r)} \quad (10-41)$$

$$\frac{\partial H_j}{\partial T_{j-1}} = L_{j-1} \frac{\partial h_{L_{j-1}}}{\partial T_{j-1}} \quad (10-42)$$

$$\frac{\partial H_j}{\partial T_j} = -(L_j + U_j) \frac{\partial h_{L_j}}{\partial T_j} - (V_j + W_j) \frac{\partial h_{V_j}}{\partial T_j} \quad (10-43)$$

$$\frac{\partial H_j}{\partial T_{j+1}} = V_{j+1} \frac{\partial h_{V_{j+1}}}{\partial T_{j+1}} \quad (10-44)$$

The partial derivatives depend upon the enthalpy correlations that are utilized. For example, if composition-independent polynomial equations in temperature are used, then

$$h_{V_j} = \sum_{i=1}^C y_{i,j} (A_i + B_i T + C_i T^2) \quad (10-45)$$

$$h_{L_j} = \sum_{i=1}^C x_{i,j} (a_i + b_i T + c_i T^2) \quad (10-46)$$

and the partial derivatives are

$$\frac{\partial h_{V_j}}{\partial T_j} = \sum_{i=1}^C y_{i,j} (B_i + 2C_i T) \quad (10-47)$$

$$\frac{\partial h_{L_j}}{\partial T_j} = \sum_{i=1}^C x_{i,j} (b_i + 2c_i T) \quad (10-48)$$

The N relations given by (10-40) form a tridiagonal-matrix equation that is linear in $\Delta T_j^{(r)}$. The form of the matrix equation is identical to (10-12) where, for example, $A_2 = (\partial H_2 / \partial T_1)^{(r)}$, $B_2 = (\partial H_2 / \partial T_2)^{(r)}$, $C_2 = (\partial H_2 / \partial T_3)^{(r)}$, $x_{i,2} \leftarrow \Delta T_2^{(r)}$, and $D_2 = -H_2^{(r)}$. The matrix of partial derivatives is called the *Jacobian correction matrix*. The Thomas algorithm can be employed to solve for the set of corrections $\Delta T_j^{(r)}$. New guesses of T_j are then determined from

$$T_j^{(r+1)} = T_j^{(r)} + t \Delta T_j^{(r)} \quad (10-49)$$

where t is a scalar attenuation factor that is useful when initial guesses are not reasonably close to the true values. Generally, as in (10-39), t is taken as 1, but an optimal value can be determined at each iteration to minimize the sum of the squares of the functions,

$$\sum_{j=1}^N [H_j^{(r+1)}]^2$$

When all the corrections $\Delta T_j^{(r)}$ have approached zero, the resulting values of T_j are used with criteria such as (10-31) or (10-32) to determine whether convergence has been achieved. If not, before beginning a new k iteration, one can adjust values of V_j and T_j as indicated in Figure 10.16 and

previously discussed for the BP method. Rapid convergence is generally observed for the sum-rates method.

EXAMPLE 10.4

Calculate stage temperatures and interstage vapor and liquid flow rates and compositions by the rigorous SR method for the absorber column specifications given in Figure 5.11.

SOLUTION

The digital computer program of Shinohara et al. [16], based on the Burningham-Otto solution procedure, was used. Initial assumptions for the top-stage and bottom-stage temperatures were 90°F (32.2°C) (entering liquid temperature) and 105°F (40.6°C) (entering gas temperature), respectively. The corresponding number of iterations to satisfy the convergence criterion of (10-32) was seven. Values of τ were as follows.

Iteration Number	τ , (°F) ²
1	9,948
2	2,556
3	46.0
4	8.65
5	0.856
6	0.124
7	0.0217

The overall results of the converged calculations are shown in Figure 10.17. Adjusted values of product-component flow rates that satisfy overall material-balance equations are included. Figures 10.18 to 10.20 are plots of converged values for stage temperatures, interstage total flow rates, and interstage component vapor flow rates, respectively. Figure 10.18 shows that the initial assumed linear-temperature profile is grossly in error. Because of the substantial degree of absorption and accompanying high heat released by

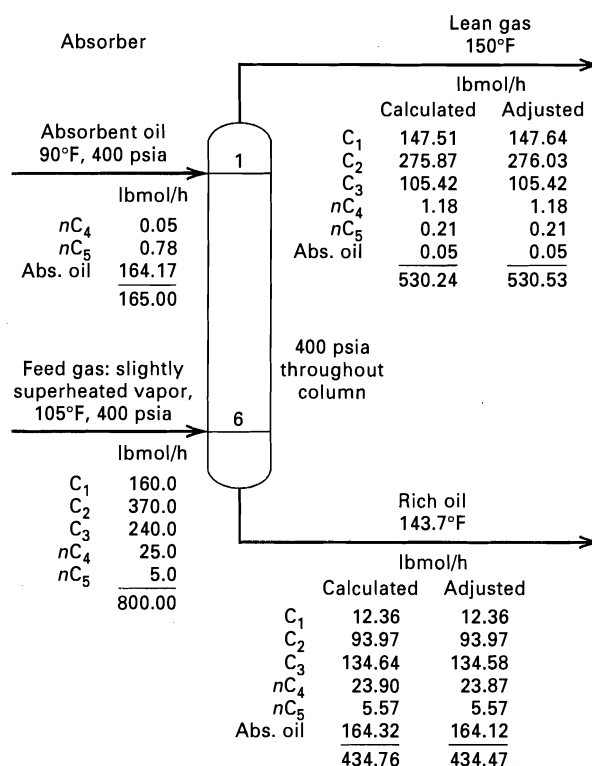


Figure 10.17 Specifications and overall results for Example 10.4.

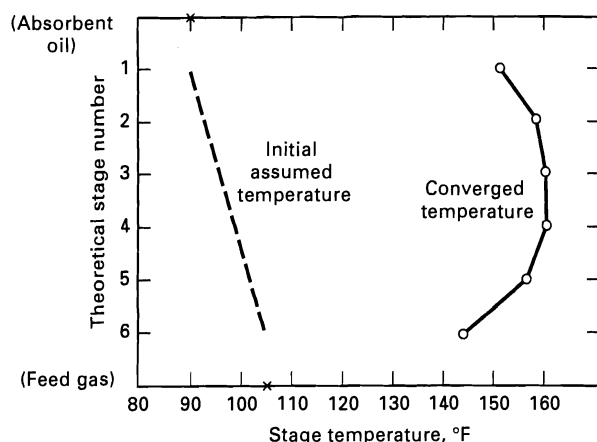


Figure 10.18 Converged temperature profile for Example 10.4.

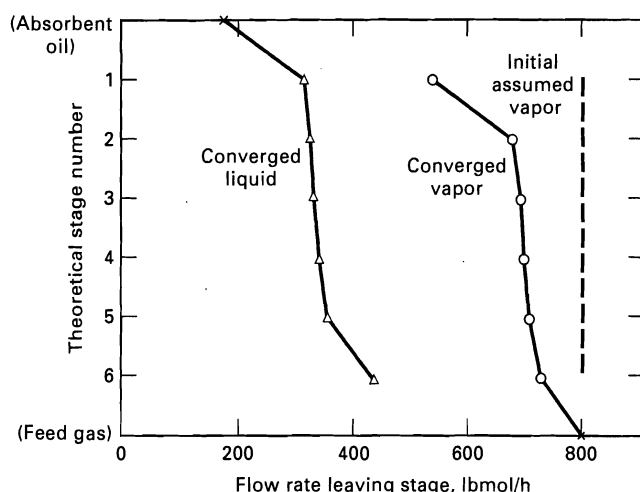


Figure 10.19 Converged interstage flow rate profiles for Example 10.4.

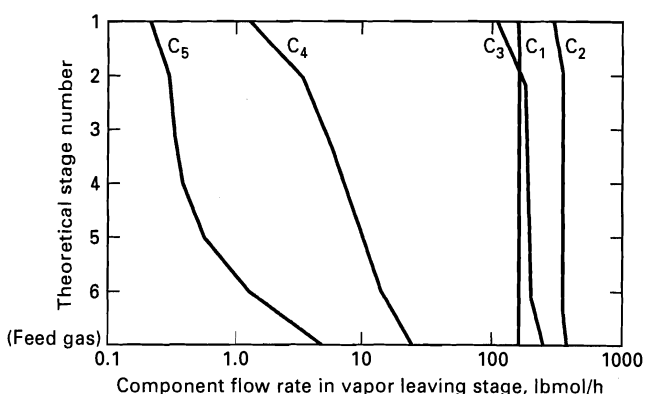


Figure 10.20 Converged component vapor flow rate profiles for Example 10.4.

absorption, stage temperatures are considerably greater than the two entering stream temperatures. The heat is absorbed by both the vapor and liquid streams. The peak stage temperature is essentially at the midpoint of the column. Figure 10.19 shows that the bulk of the overall absorption occurs at the two terminal stages. In Figure 10.20, it is seen that absorption of C_1 and C_2 occurs almost exclusively at the top and bottom stages. Absorption of C_3 occurs throughout the column, but mainly at the two terminal stages. Absorption of C_4 and C_5 also occurs throughout the column, but for C_5 mainly at the bottom where vapor first contacts absorption oil.

Isothermal Sum-Rates Method for Liquid-Liquid Extraction

Multistage liquid-liquid extraction equipment is operated frequently in an adiabatic manner. When entering streams are at the same temperature and heat of mixing is negligible, the operation is also isothermal. For this condition, or when stage temperatures are specified, as indicated by Friday and Smith [7] and shown in detail by Tsuboka and Katayama [17], a simplified isothermal version of the sum-rates method (ISR) can be applied. It is based on the same equilibrium-stage model presented in Section 10.1. However, with all stage temperatures specified, values of Q_j can be computed from stage energy balances, which can be partitioned from the other equations and solved in a separate step following the calculations discussed here. In the ISR method, particular attention is paid to the possibility that phase compositions may strongly influence K_{ij} values.

Figure 10.21 shows the algorithm for the Tsuboka-Katayama ISR method. Liquid-phase and vapor-phase symbols correspond to raffinate and extract, respectively. Problem specifications consist of flow rates, compositions, and stage locations for all feeds; stage temperatures (frequently all equal); total flow rates of any side streams; and total number of stages. Stage pressures need not be specified but are understood to be greater than corresponding stage bubble-point pressures to prevent vaporization.

With stage temperatures specified, the only tear variables are V_j (extract flow rates) values. An initial set is obtained by assuming a perfect separation among the components of the feed and neglecting mass transfer of the solvent to the raffinate phase. This gives approximate values for the flow rates of the exiting raffinate and extract phases. Intermediate values of V_j are obtained by linear interpolation over the N stages. Modifications to this procedure are necessary for side streams or intermediate feeds. As shown in Figure 10.21, the tear variables are reset in an outer iterative loop.

The effect of phase compositions is often considerable on K_D -values (distribution coefficients) for liquid-liquid extraction. Therefore, it is best also to provide initial estimates of $x_{i,j}$ and $y_{i,j}$ from which initial values of $K_{i,j}$ are computed. Initial values of $x_{i,j}$ are obtained by linear interpolation, with stage, of the compositions of the known entering and assumed exit streams. Corresponding values of $y_{i,j}$ are computed by material balance from (10-1). Values of $\gamma_{iL,j}$ and $\gamma_{iV,j}$ are determined from an appropriate correlation—for example, the van Laar, NRTL, UNIQUAC, or UNIFAC equations discussed in Chapter 2. Corresponding K_D -values are obtained from the following equation, which is equivalent to (2-30).

$$K_{i,j} = \frac{\gamma_{iL,j}}{\gamma_{iV,j}} \quad (10-50)$$

A new set of $x_{i,j}$ values is obtained by solving (10-12) by the Thomas algorithm of Section 10.3. These values are compared to the assumed values by computing

$$\tau_1 = \sum_{j=1}^N \sum_{i=1}^C \left| x_{i,j}^{(r-1)} - x_{i,j}^{(r)} \right| \quad (10-51)$$

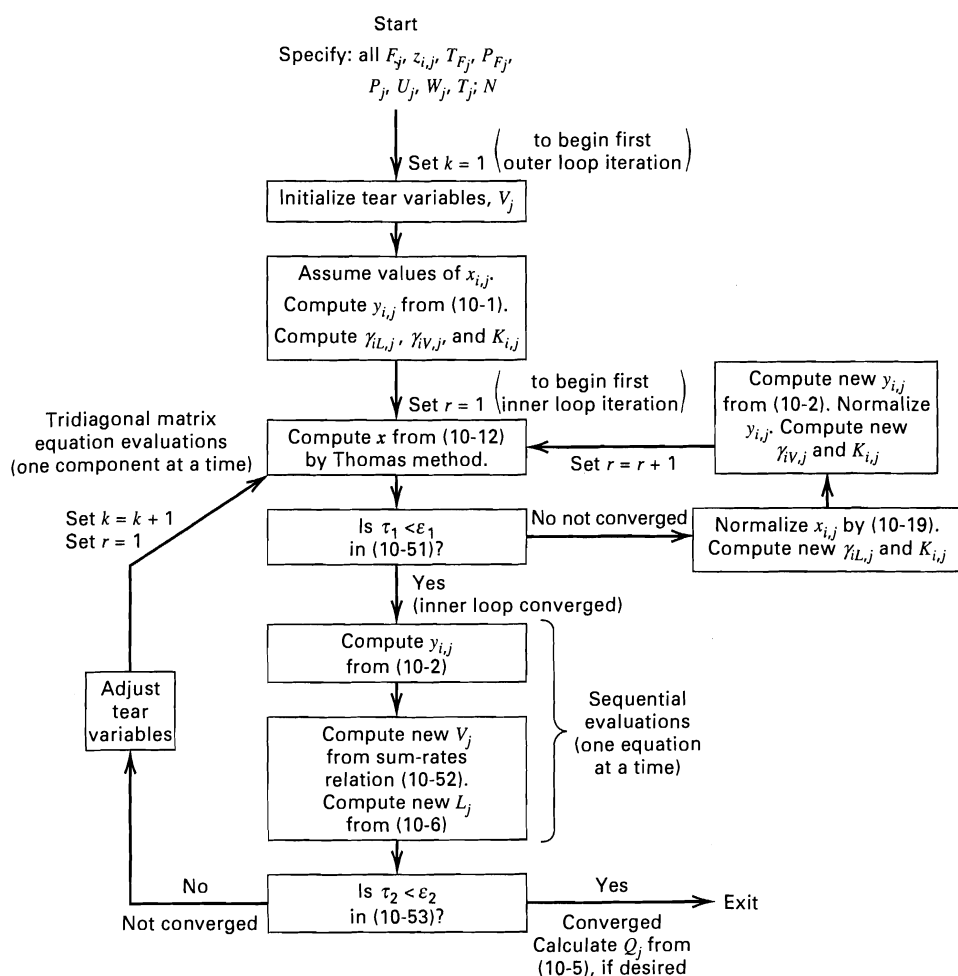


Figure 10.21 Algorithm for Tsuboka-Katayama ISR method for liquid-liquid extraction.

where r is an inner loop index. If $\tau_1 > \epsilon_1$, where, for example, the convergence criterion ϵ_1 might be taken as 0.01 NC, the inner loop is used to improve values of $K_{i,j}$ by using normalized values of $x_{i,j}$ and $y_{i,j}$ to compute new values of $\gamma_{iL,j}$ and $\gamma_{iV,j}$.

When the inner loop is converged, values of $x_{i,j}$ are used to calculate new values of $y_{i,j}$ from (10-2). A new set of tear variables V_j is then computed from the sum-rates relation

$$V_j^{(k+1)} = V_j^{(k)} \sum_{i=1}^C y_{i,j} \quad (10-52)$$

where k is an outer-loop index. Corresponding values of $L_j^{(k+1)}$ are obtained from (10-6).

The outer loop is converged when

$$\tau_2 = \sum_{j=1}^N \left(\frac{V_j^{(k)} - V_j^{(k-1)}}{V_j^{(k)}} \right)^2 \leq \epsilon_2 \quad (10-53)$$

where, for example, the convergence criterion ϵ_2 may be taken as 0.01 N .

Before beginning a new k iteration, we can adjust values of V_j as previously discussed for the BP method. Convergence of the ISR method is generally rapid but is subject to the extent to which $K_{i,j}$ depends upon composition.

EXAMPLE 10.5

The separation of benzene (B) from n -heptane (H) by ordinary distillation is difficult. At atmospheric pressure, the boiling points differ by 18.3°C. However, because of liquid-phase nonideality, the relative volatility decreases to a value less than 1.15 at high benzene concentrations [18]. An alternative method of separation is liquid-liquid extraction with a mixture of dimethylformamide (DMF) and water [19]. The solvent is much more selective for benzene than for n -heptane at 20°C. For two different solvent compositions, calculate interstage flow rates and compositions by the rigorous ISR method for the countercurrent, liquid-liquid extraction cascade, which contains five equilibrium stages and is shown schematically in Figure 10.22.

SOLUTION

Experimental phase-equilibrium data for the quaternary system [19] were fitted to the NRTL equation by Cohen and Renon [20]. The resulting binary-pair constants in (2-92) and (2-93) are

Binary Pair, ij	τ_{ij}	τ_{ji}	α_{ji}
DMF, H	2.036	1.910	0.25
Water, H	7.038	4.806	0.15
B, H	1.196	-0.355	0.30
Water, DMF	2.506	-2.128	0.253
B, DMF	-0.240	0.676	0.425
B, Water	3.639	5.750	0.203

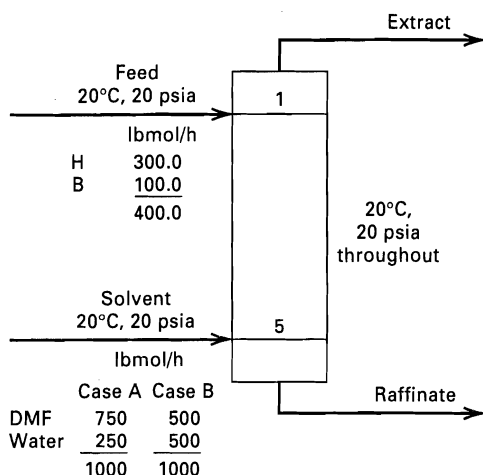


Figure 10.22 Specifications for Example 10.5.

For Case A, estimates of V_j (the extract phase), $x_{i,j}$, and $y_{i,j}$ are as follows, based on a perfect separation and linear interpolation by stage.

Stage j	V_j	$y_{i,j}$				$x_{i,j}$			
		H	B	DMF	Water	H	B	DMF	Water
1	1100	0.0	0.0909	0.6818	0.2273	0.7895	0.2105	0.0	0.0
2	1080	0.0	0.0741	0.6944	0.2315	0.8333	0.1667	0.0	0.0
3	1060	0.0	0.0566	0.7076	0.2359	0.8824	0.1176	0.0	0.0
4	1040	0.0	0.0385	0.7211	0.2404	0.9375	0.0625	0.0	0.0
5	1020	0.0	0.0196	0.7353	0.2451	1.0000	0.0	0.0	0.0

The converged solution is obtained by the ISR method with the following corresponding stage flow rates and compositions:

Stage j	V_j	$y_{i,j}$				$x_{i,j}$			
		H	B	DMF	Water	H	B	DMF	Water
1	1113.1	0.0263	0.0866	0.6626	0.2245	0.7586	0.1628	0.0777	0.0009
2	1104.7	0.0238	0.0545	0.6952	0.2265	0.8326	0.1035	0.0633	0.0006
3	1065.6	0.0213	0.0309	0.7131	0.2347	0.8858	0.0606	0.0532	0.0004
4	1042.1	0.0198	0.0157	0.7246	0.2399	0.9211	0.0315	0.0471	0.0003
5	1028.2	0.0190	0.0062	0.7316	0.2432	0.9438	0.0125	0.0434	0.0003

Computed products for the two cases are:

	Extract, lbmol/h		Raffinate, lbmol/h	
	Case A	Case B	Case A	Case B
H	29.3	5.6	270.7	294.4
B	96.4	43.0	3.6	57.0
DMF	737.5	485.8	12.5	14.2
Water	249.9	499.7	0.1	5.0
	1113.1	1034.1	286.9	365.9

On a percentage extraction basis, the results are:

	Case A	Case B
Percent of benzene feed extracted	96.4	43.0
Percent of <i>n</i> -heptane feed extracted	9.8	1.87
Percent of solvent transferred to raffinate	1.26	1.45

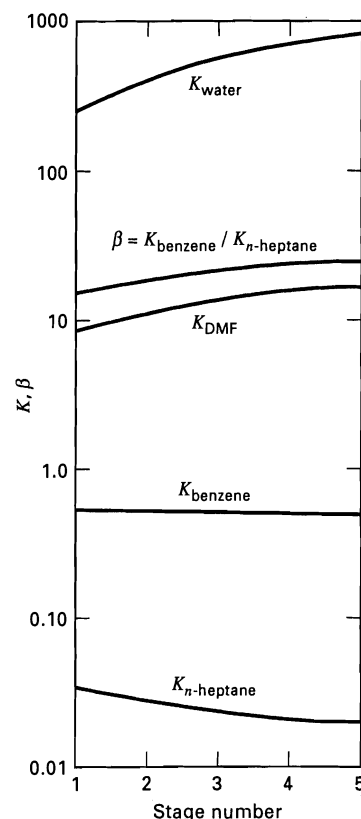


Figure 10.23 Variation of distribution coefficient and relative selectivity for Example 10.5, Case A.

Thus, the solvent with 75% DMF extracts a much larger percentage of the benzene, but the solvent with 50% DMF is more selective between benzene and *n*-heptane.

For Case A, the variations with stage of K -values and the relative selectivity are shown in Figure 10.23, where the relative selectivity is $\beta_{B,H} = K_B/K_H$. The distribution coefficient for *n*-heptane varies by a factor of almost 1.75 from stage 5 to stage 1, while the coefficient for benzene is almost constant. The relative selectivity varies by a factor of almost 2.

10.4 NEWTON-RAPHSON METHOD

The BP and SR methods for vapor-liquid contacting converge only with difficulty or not at all for separations involving very nonideal liquid mixtures or for cases where the separator is like an absorber or stripper in one section and a fractionator in another section (e.g., a reboiled absorber). Furthermore, BP and SR methods are generally restricted to the very limited specifications stated previously. More general procedures capable of solving all types of multicomponent, multistage, separation problems are based on the simultaneous solution of all the MESH equations, or combinations thereof, by simultaneous-correction (SC) techniques, often using the Newton-Raphson method.

In order to develop an SC procedure that uses the Newton-Raphson method, one must select and order the unknown variables and the corresponding functions (MESH equations) that contain them. As discussed by Goldstein and Stanfield [21], grouping of the functions by type is computationally most efficient for problems involving a large number of components,

but few stages. Alternatively, it is most efficient to group the functions according to stage location for problems involving many stages, but relatively few components. The latter grouping, presented here, is described by Naphtali [22] and was implemented by Naphtali and Sandholm [23].

The SC procedure of Naphtali and Sandholm is developed in detail because it utilizes many of the mathematical techniques presented in Section 10.3 on tearing methods. A computer program for their method is given by Fredenslund, Gmehling, and Rasmussen [24]. However, that program does not have the flexibility of specifications in Newton-Raphson implementations found in commercial simulators for computer-aided process design.

The equilibrium-stage model of Figures 10.1 and 10.3 is again employed. However, rather than solving the $N(2C + 3)$ MESH equations simultaneously, we combine (10-3) and (10-4) with the other MESH equations to eliminate $2N$ variables and thus reduce the problem to the simultaneous solution of $N(2C + 1)$ equations. This is done by first multiplying (10-3) and (10-4) by V_j and L_j , respectively, to give

$$V_j = \sum_{i=1}^C v_{i,j} \quad (10-54)$$

$$L_j = \sum_{i=1}^C l_{i,j} \quad (10-55)$$

where we have used the mole-fraction definitions

$$y_{i,j} = \frac{v_{i,j}}{V_j} \quad (10-56)$$

$$x_{i,j} = \frac{l_{i,j}}{L_j} \quad (10-57)$$

Equations (10-54) to (10-57) are now substituted into (10-1), (10-2), and (10-5) to eliminate V_j , L_j , $y_{i,j}$, and $x_{i,j}$ and introduce component flow rates $v_{i,j}$ and $l_{i,j}$. As a result, the following $N(2C + 1)$ equations are obtained, where $s_j = U_j/L_j$ and $S_j = W_j/V_j$ are dimensionless side stream flow rates:

Material Balance

$$M_{i,j} = l_{i,j}(1 + s_j) + v_{i,j}(1 + S_j) - l_{i,j-1} - v_{i,j+1} - f_{i,j} = 0 \quad (10-58)$$

Phase Equilibria

$$E_{i,j} = K_{i,j} l_{i,j} \frac{\sum_{\kappa=1}^C v_{\kappa,j}}{\sum_{\kappa=1}^C l_{\kappa,j}} - v_{i,j} = 0 \quad (10-59)$$

Energy Balance

$$\begin{aligned} H_j &= h_{L_j}(1 + s_j) \sum_{i=1}^C l_{i,j} + h_{V_j}(1 + S_j) \sum_{i=1}^C v_{i,j} \\ &\quad - h_{L_{j-1}} \sum_{i=1}^C l_{i,j-1} - h_{V_{j+1}} \sum_{i=1}^C v_{i,j+1} \\ &\quad - h_{F_j} \sum_{i=1}^C f_{i,j} - Q_j = 0 \end{aligned} \quad (10-60)$$

where $f_{i,j} = F_j z_{i,j}$.

If N and all $f_{i,j}$, T_{F_j} , P_{F_j} , P_j , s_j , S_j , and Q_j are specified, the M , E , and H functions are nonlinear in the $N(2C + 1)$ unknown (output) variables $v_{i,j}$, $l_{i,j}$, and T_j for $i = 1$ to C and $j = 1$ to N . Although other sets of specified and unknown variables are possible, we consider these sets first.

Equations (10-58), (10-59), and (10-60) are solved simultaneously by the Newton-Raphson iterative method in which successive sets of the output variables are completed until the values of the M , E , and H functions are driven to within some tolerance of zero. During the iterations, nonzero values of the functions are called *discrepancies* or *errors*. Let the functions and output variables be grouped by stage in order from top to bottom. As will be shown, this is done to produce a block-tridiagonal structure for the Jacobian matrix of partial derivatives so that a matrix form of the Thomas algorithm can be applied. Let

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_j, \dots, \mathbf{X}_N]^T \quad (10-61)$$

and

$$\mathbf{F} = [\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_j, \dots, \mathbf{F}_N]^T \quad (10-62)$$

where $\mathbf{X}_j$ is the vector of output variables for stage j arranged in the order

$$\mathbf{X}_j = [v_{1,j}, v_{2,j}, \dots, v_{i,j}, \dots, v_{C,j}, T_j, l_{1,j}, l_{2,j}, \dots, l_{i,j}, \dots, l_{C,j}]^T \quad (10-63)$$

and $\mathbf{F}_j$ is the vector of functions for stage j arranged in the order

$$\mathbf{F}_j = [H_j, M_{1,j}, M_{2,j}, \dots, M_{i,j}, \dots, M_{C,j}, E_{1,j}, E_{2,j}, \dots, E_{i,j}, \dots, E_{C,j}]^T \quad (10-64)$$

The Newton-Raphson iteration is performed by solving for the corrections $\Delta \mathbf{X}$ to the output variables from (10-38), which in matrix form becomes

$$\Delta \mathbf{X}^{(k)} = - \left[\left(\frac{\partial \mathbf{F}}{\partial \mathbf{X}} \right)^{-1} \right]^{(k)} \mathbf{F}^{(k)} \quad (10-65)$$

These corrections are used to compute the next approximation to the set of output variables from

$$\mathbf{X}^{(k+1)} = \mathbf{X}^{(k)} + \Delta \mathbf{X}^{(k)} \quad (10-66)$$

The quantity $(\partial \mathbf{F} / \partial \mathbf{X})$ is the following Jacobian or $(N \times N)$ matrix of blocks of partial derivatives of all the functions with respect to all the output variables.

$$\frac{\partial \mathbf{F}}{\partial \mathbf{X}} = \begin{bmatrix} \overline{\mathbf{B}}_1 & \overline{\mathbf{C}}_1 & 0 & 0 & \dots & 0 \\ \overline{\mathbf{A}}_2 & \overline{\mathbf{B}}_2 & \overline{\mathbf{C}}_2 & 0 & \dots & 0 \\ 0 & \overline{\mathbf{A}}_3 & \overline{\mathbf{B}}_3 & \overline{\mathbf{C}}_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \overline{\mathbf{A}}_{N-1} & \overline{\mathbf{B}}_{N-1} & \overline{\mathbf{C}}_{N-1} \\ 0 & \dots & 0 & 0 & \overline{\mathbf{A}}_N & \overline{\mathbf{B}}_N \end{bmatrix} \quad (10-67)$$

This Jacobian is of a block-tridiagonal form, like (10-12), because functions for stage j are only dependent on output variables for stages $j - 1$, j , and $j + 1$. Each $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, or $\bar{\mathbf{C}}$ block in (15-67) represents a $(2C + 1)$ by $(2C + 1)$ submatrix of partial derivatives, where the arrangements of output variables and functions are given by (10-63) and (10-64), respectively. Blocks $\bar{\mathbf{A}}_j$, $\bar{\mathbf{B}}_j$, and $\bar{\mathbf{C}}_j$ correspond to submatrices of partial derivatives of the functions on stage j with respect to the output variables on stages $j - 1$, j , and $j + 1$, respectively. Thus, using (10-58), (10-59), and (10-60), and denoting only the nonzero partial derivatives by +, or by row or diagonal strings of + $\cdots$ +, or by the following square or rectangular blocks enclosed by connected strings,

$$\begin{array}{c} + \cdots + \\ \vdots \\ + \cdots + \end{array}$$

we find that the blocks have the following form, where + is replaced by a numerical value (-1 or 1) in the event that the partial derivative has only that value.

$$\bar{\mathbf{A}}_j = \frac{\partial \bar{\mathbf{F}}_j}{\partial \mathbf{X}_{j-1}} = \begin{array}{c} \text{Functions} \\ \left[\begin{array}{c} H_j \\ M_{1,j} \\ \vdots \\ M_{C,j} \\ E_{1,j} \\ \vdots \\ E_{C,j} \end{array} \right] \end{array} \begin{array}{c} \text{Output variables} \\ \left[\begin{array}{cccc} v_{1,j-1} \cdots v_{C,j-1} & T_{j-1} & l_{1,j-1} \cdots l_{C,j-1} \\ & + & + \cdots + \\ & & -1 \\ & & \vdots \\ & & -1 \end{array} \right] \end{array} \quad (10-68)$$

$$\bar{\mathbf{B}}_j = \frac{\partial \bar{\mathbf{F}}_j}{\partial \mathbf{X}_j} = \begin{array}{c} \text{Functions} \\ \left[\begin{array}{c} H_j \\ M_{1,j} \\ \vdots \\ M_{C,j} \\ E_{1,j} \\ \vdots \\ E_{C,j} \end{array} \right] \end{array} \begin{array}{c} \text{Output variables} \\ \left[\begin{array}{cccc} v_{1,j} \cdots v_{C,j} & T_j & l_{1,j} \cdots l_{C,j} \\ + \cdots + & & + \\ + & & + \\ \vdots & & \vdots \\ + & & + \\ + \cdots + & & + \end{array} \right] \end{array} \quad (10-69)$$

$$\bar{\mathbf{C}}_j = \frac{\partial \bar{\mathbf{F}}_j}{\partial \mathbf{X}_{j+1}} = \begin{array}{c} \text{Functions} \\ \left[\begin{array}{c} H_j \\ M_{1,j} \\ \vdots \\ M_{C,j} \\ E_{1,j} \\ \vdots \\ E_{C,j} \end{array} \right] \end{array} \begin{array}{c} \text{Output variables} \\ \left[\begin{array}{cccc} v_{1,j+1} \cdots v_{C,j+1} & T_{j+1} & l_{1,j+1} \cdots l_{C,j+1} \\ & + \cdots + & \\ & -1 & \\ & \vdots & \\ & -1 \end{array} \right] \end{array} \quad (10-70)$$

Thus, (10-65) consists of a set of $N(2C + 1)$ simultaneous, linear equations in the $N(2C + 1)$ corrections $\Delta \mathbf{X}$. For example, the $2C + 2$ equation in the set is obtained by expanding function H_2 (10-60) into a Taylor's series like (10-36) around the $N(2C + 1)$ output variables. The result is as follows after the usual truncation of terms involving derivatives of order greater than one:

$$\begin{aligned} 0(\Delta v_{1,1} + \cdots + \Delta v_{C,1}) - \frac{\partial h_{L_1}}{\partial T_1} \sum_{i=1}^C l_{i,1}(\Delta T_1) \\ - \left(\frac{\partial h_{L_1}}{\partial l_{1,1}} \sum_{i=1}^C l_{i,1} + h_{L_1} \right) \Delta l_{1,1} - \cdots \\ - \left(\frac{\partial h_{L_1}}{\partial l_{C,1}} \sum_{i=1}^C l_{i,1} + h_{L_1} \right) \Delta l_{C,1} \\ + \left[\left(\frac{\partial h_{v_2}}{\partial v_{1,2}} \right) (1 + S_2) \sum_{i=1}^C v_{i,2} + h_{v_2}(1 + S_2) \right] \Delta v_{1,2} + \cdots \\ + \left[\left(\frac{\partial h_{v_2}}{\partial v_{C,2}} \right) (1 + S_2) \sum_{i=1}^C v_{i,2} + h_{v_2}(1 + S_2) \right] \Delta v_{C,2} \\ + \left[\left(\frac{\partial h_{L_2}}{\partial T_2} \right) (1 + s_2) \sum_{i=1}^C l_{i,2} + \left(\frac{\partial h_{v_2}}{\partial T_2} \right) (1 + S_2) \sum_{i=1}^C v_{i,2} \right] \Delta T_2 \\ + \left[\left(\frac{\partial h_{L_2}}{\partial l_{1,2}} \right) (1 + s_2) \sum_{i=1}^C l_{i,2} + h_{L_2}(1 + s_2) \right] \Delta l_{1,2} + \cdots \\ + \left[\left(\frac{\partial h_{L_2}}{\partial l_{C,2}} \right) (1 + s_2) \sum_{i=1}^C l_{i,2} + h_{L_2}(1 + s_2) \right] \Delta l_{C,2} \\ - \left(\frac{\partial h_{v_3}}{\partial v_{1,3}} \sum_{i=1}^C v_{i,3} + h_{v_3} \right) \Delta v_{1,3} - \cdots \\ - \left(\frac{\partial h_{v_3}}{\partial v_{C,3}} \sum_{i=1}^C v_{i,3} + h_{v_3} \right) \Delta v_{C,3} \\ - \frac{\partial h_{v_3}}{\partial T_3} \sum_{i=1}^C v_{i,3} \Delta T_3 + 0(\Delta l_{1,3} + \cdots + \Delta l_{C,N}) = -H_2 \end{aligned} \quad (10-71)$$

Although lengthy, equations such as (10-71) are handled readily in computer programs.

As a further example, the entry in the Jacobian matrix for row $(2C + 2)$ and column $(C + 3)$ is obtained from (10-71) as

$$\frac{\partial H_2}{\partial l_{2,1}} = -\frac{\partial h_{L1}}{\partial l_{2,1}} \sum_{i=1}^C l_{i,1} + h_{L1} \quad (10-72)$$

All partial derivatives are stated by Naphtali and Sandholm [23].

Partial derivatives of enthalpies and K -values depend upon the particular correlation utilized for these properties and are sometimes simplified by including only the dominant terms. For example, suppose that the Chao-Seader correlation is to be used for K -values. In general,

$$K_{i,j} = K_{i,j} \left\{ P_j, T_j, \frac{l_{i,j}}{\sum_{\kappa=1}^C l_{\kappa,j}}, \frac{v_{i,j}}{\sum_{\kappa=1}^C v_{\kappa,j}} \right\}$$

In terms of the output variables, the partial derivatives $\partial K_{i,j}/\partial T_j$; $\partial K_{i,j}/\partial l_{i,j}$; and $\partial K_{i,j}/\partial v_{i,j}$ all exist and can be expressed analytically or evaluated numerically if desired. However, for some problems, the terms that include the first and second of these three groups of derivatives may be the dominant terms so that the third group may be taken as zero.

EXAMPLE 10.6

Derive an expression for $(\partial h_V/\partial T)$ from the Redlich-Kwong equation of state.

SOLUTION

From (2-53),

$$h_V = \sum_{i=1}^C (y_i h_{iV}^\circ) + RT \left[Z_V - 1 - \frac{3A}{2B} \ln \left(1 + \frac{B}{Z_V} \right) \right]$$

where h_{iV}° , Z_V , A , and B all depend on T , as determined from (2-36) and (2-46) to (2-50). Thus,

$$\begin{aligned} \frac{\partial h_V}{\partial T} = & \sum_{i=1}^C \left[y_i \left(\frac{\partial h_{iV}^\circ}{\partial T} \right) \right] + R \left[Z_V - 1 - \frac{3A}{2B} \ln \left(1 + \frac{B}{Z_V} \right) \right] \\ & + RT \left\{ \left(\frac{\partial Z_V}{\partial T} \right) - \frac{3}{2} \left(\frac{\partial(A/B)}{\partial T} \right) \ln \left(1 + \frac{B}{Z_V} \right) \right. \\ & \left. - \frac{3A}{2B} \left[\frac{1}{Z_V} \left(\frac{\partial B}{\partial T} \right) - \frac{B}{Z_V^2} \left(\frac{\partial Z_V}{\partial T} \right) \right] \right\} \end{aligned}$$

From (2-36) and (2-35),

$$\left(\frac{\partial h_{iV}^\circ}{\partial T} \right) = \sum_{k=0}^4 (a_k)_i T^k = (C_{P_V}^\circ)_i$$

From (2-48) and Table 2.5,

$$B = \frac{bP}{RT} \quad \text{and} \quad b = \frac{0.08664RT_c}{P_c}$$

Thus,
$$\frac{\partial B}{\partial T} = -\frac{B}{T}$$

From (2-47) to (2-50) and Table 2.5,

$$\frac{A}{B} = \frac{a}{bRT} \quad \text{and} \quad a = \frac{0.42748R^2T_c^{2.5}}{P_cT^{0.5}}$$

Thus,
$$\frac{\partial(A/B)}{\partial T} = -1.5 \frac{A}{BT}$$

From (2-46),
$$Z_V^3 - Z_V^2 + (A - B - B^2)Z_V - AB = 0$$

By implicit differentiation,

$$\begin{aligned} 3Z_V^2 \frac{\partial Z_V}{\partial T} - 2Z_V \frac{\partial Z_V}{\partial T} + (A - B - B^2) \frac{\partial Z_V}{\partial T} \\ + \frac{Z_V}{T} (-2.5A + B + 2B^2) + \frac{3.5AB}{T} = 0 \end{aligned}$$

which reduces to

$$\frac{\partial Z_V}{\partial T} = \frac{(Z_V/T)(2.5A - B - 2B^2) - 3.5AB/T}{3Z_V^2 - 2Z_V + (A - B - B^2)}$$

Because the Thomas algorithm can be applied to the block-tridiagonal structure of (10-67), submatrices of partial derivatives are computed only as needed. The solution of (10-65) follows the scheme in Section 10.3, given by (10-13) to (10-18) and represented in Figure 10-4, where matrices and vectors $\bar{A}_j$, $\bar{B}_j$, $\bar{C}_j$, $-\bar{F}_j$, and ΔX_j correspond to variables A_j , B_j , C_j , D_j , and x_j , respectively. However, the simple multiplication and division operations in Section 10.3 are changed to matrix multiplication and inversion, respectively. The steps are as follows:

Starting at stage 1, $\bar{C}_1 \leftarrow (\bar{B}_1)^{-1} \bar{C}_1$, $F_1 \leftarrow (\bar{B}_1)^{-1} F_1$, and $\bar{B}_1 \leftarrow I$ (the identity submatrix). Only $\bar{C}_1$ and F_1 are saved. For stages j from 2 to $(N - 1)$, $\bar{C}_j \leftarrow (\bar{B}_j - \bar{A}_j \bar{C}_{j-1})^{-1} \bar{C}_j$, $F_j \leftarrow (\bar{B}_j - \bar{A}_j \bar{C}_{j-1})^{-1} (F_j - \bar{A}_j F_{j-1})$. Then $\bar{A}_j \leftarrow 0$, and $\bar{B}_j \leftarrow I$. Save $\bar{C}_j$ and F_j for each stage. For the last stage, $F_N \leftarrow (B_N - A_N C_{N-1})^{-1} (F_N - A_N F_{N-1})$, $\bar{A}_N \leftarrow 0$, $\bar{B}_N \leftarrow I$, and therefore $\Delta X_N = -F_N$. This completes the forward steps. Remaining values of ΔX are obtained by successive, backward substitution from $\Delta X_j = -F_j \leftarrow -(F_j - \bar{C}_j F_{j+1})$.

For the last stage, $F_N \leftarrow (B_N - A_N C_{N-1})^{-1} (F_N - A_N F_{N-1})$, $\bar{A}_N \leftarrow 0$, $\bar{B}_N \leftarrow I$, and therefore $\Delta X_N = -F_N$. This completes the forward steps. Remaining values of ΔX are obtained by successive, backward substitution from $\Delta X_j = -F_j \leftarrow -(F_j - \bar{C}_j F_{j+1})$. This procedure is illustrated by the following example.

EXAMPLE 10.7

Solve the following matrix equation, which has a block-tridiagonal structure, by the matrix form of the Thomas algorithm.

$$\begin{bmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 1 & 2 & 0 & 0 & 0 & 0 \\ \hline 0 & 1 & 3 & 1 & 2 & 1 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 0 \\ 0 & 0 & 2 & 2 & 1 & 1 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 2 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \\ \hline \Delta x_4 \\ \Delta x_5 \\ \Delta x_6 \\ \hline \Delta x_7 \\ \Delta x_8 \\ \Delta x_9 \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \\ 8 \\ \hline 12 \\ 8 \\ 8 \\ \hline 7 \\ 5 \\ 6 \end{bmatrix}$$

SOLUTION

The matrix equation is in the form

$$\begin{bmatrix} \bar{\mathbf{B}}_1 & \bar{\mathbf{C}}_1 & 0 \\ \bar{\mathbf{A}}_2 & \bar{\mathbf{B}}_2 & \bar{\mathbf{C}}_2 \\ 0 & \bar{\mathbf{A}}_3 & \bar{\mathbf{B}}_3 \end{bmatrix} \cdot \begin{bmatrix} \Delta \mathbf{X}_1 \\ \Delta \mathbf{X}_2 \\ \Delta \mathbf{X}_3 \end{bmatrix} = - \begin{bmatrix} \mathbf{F}_1 \\ \mathbf{F}_2 \\ \mathbf{F}_3 \end{bmatrix}$$

Following the procedure just given, starting at the first block row,

$$\bar{\mathbf{B}}_1 = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 2 & 2 \end{bmatrix}, \quad \bar{\mathbf{C}}_1 = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & 0 \end{bmatrix}, \quad \mathbf{F}_1 = \begin{bmatrix} -9 \\ -7 \\ -8 \end{bmatrix}$$

By standard matrix inversion

$$(\bar{\mathbf{B}}_1)^{-1} = \begin{bmatrix} 0 & 2/3 & -1/3 \\ 1 & -1/3 & -1/3 \\ -1 & 0 & 1 \end{bmatrix}$$

By standard matrix multiplication

$$(\bar{\mathbf{B}}_1)^{-1}(\bar{\mathbf{C}}_1) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix}$$

which replaces $\bar{\mathbf{C}}_1$, and

$$(\bar{\mathbf{B}}_1)^{-1}(\mathbf{F}_1) = \begin{bmatrix} -2 \\ -4 \\ 1 \end{bmatrix}$$

which replaces $\mathbf{F}_1$. Also

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ replaces } \bar{\mathbf{B}}_1$$

For the second block row

$$\bar{\mathbf{A}}_2 = \begin{bmatrix} 0 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \quad \bar{\mathbf{B}}_2 = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 0 \\ 2 & 1 & 1 \end{bmatrix},$$

$$\bar{\mathbf{C}}_2 = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix}, \quad \bar{\mathbf{F}}_2 = \begin{bmatrix} -12 \\ -8 \\ -8 \end{bmatrix}$$

By matrix multiplication and subtraction

$$(\bar{\mathbf{B}}_2 - \bar{\mathbf{A}}_2 \bar{\mathbf{C}}_1) = \begin{bmatrix} 3 & 1 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix}$$

which upon inversion becomes

$$(\bar{\mathbf{B}}_2 - \bar{\mathbf{A}}_2 \bar{\mathbf{C}}_1)^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 3/5 & -6/5 \\ 1 & -1/5 & -3/5 \end{bmatrix}$$

By multiplication

$$(\bar{\mathbf{B}}_2 - \bar{\mathbf{A}}_2 \bar{\mathbf{C}}_1)^{-1} \bar{\mathbf{C}}_2 = \begin{bmatrix} 0 & -1 & -1 \\ 2/5 & 2 & 1 \\ 1/5 & 1 & 1 \end{bmatrix}$$

which replaces $\bar{\mathbf{C}}_2$. In a similar manner, the remaining steps for this and the third block row are carried out to give

$$\begin{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & -1 \\ 2/5 & 2 & 1 \\ 1/5 & 1 & 1 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{bmatrix} \cdot \begin{bmatrix} \Delta X_1 \\ \Delta X_2 \\ \Delta X_3 \\ \Delta X_4 \\ \Delta X_5 \\ \Delta X_6 \\ \Delta X_7 \\ \Delta X_8 \\ \Delta X_9 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ +1 \\ +1 \\ -22/5 \\ -16/5 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

Thus, $\Delta X_7 = \Delta X_8 = \Delta X_9 = 1$.

The remaining backward steps begin with the second block row where

$$\bar{\mathbf{C}}_2 = \begin{bmatrix} 0 & -1 & -1 \\ 2/5 & 2 & 1 \\ 1/5 & 1 & 1 \end{bmatrix}, \quad \bar{\mathbf{F}}_2 = \begin{bmatrix} 1 \\ -22/5 \\ -16/5 \end{bmatrix}$$

$$(\mathbf{F}_2 - \bar{\mathbf{C}}_2 \bar{\mathbf{F}}_3) = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$$

Thus, $\Delta X_4 = \Delta X_5 = \Delta X_6 = 1$. Similarly, for the first block row, the result is

$$\Delta X_1 = \Delta X_2 = \Delta X_3 = 1$$

Usually, it is desirable to specify certain top- and bottom-stage variables other than the condenser duty and/or reboiler duty. (In fact, the condenser and reboiler duties are usually so interdependent that specification of both values is not recommended.) Specifying other variables is readily accomplished by removing heat balance functions H_1 and/or H_N from the simultaneous equation set and replacing them with discrepancy functions depending upon the desired specification(s). Functions for alternative specifications for a column with a partial condenser are listed in Table 10.1.

If desired, (10-54) can be modified to permit real rather than theoretical stages to be computed. Values of the Murphree vapor-phase plate efficiency must then be specified. These values are related to phase compositions by

Table 10.1 Alternative Functions for H_1 and H_N

Specification	Replacement for H_1	Replacement for H_N
Reflux or reboil (boilup) ratio, (L/D) or (V/B)	$\sum l_{i,1} - (L/D) \sum v_{i,1} = 0$	$\sum v_{i,N} - (V/B) \sum l_{i,N} = 0$
Stage temperature, T_D or T_B	$T_1 - T_D = 0$	$T_N - T_B = 0$
Product flow rate, D or B	$\sum v_{i,1} - D = 0$	$\sum l_{i,N} - B = 0$
Component flow rate in product, d_i or b_i	$v_{i,1} - d_i = 0$	$l_{i,N} - b_i = 0$
Component mole fraction in product, y_{iD} or x_{iB}	$v_{i,1} - (\sum v_{i,1}) y_{iD} = 0$	$l_{i,N} - (\sum l_{i,N}) x_{iB} = 0$

the definition

$$\eta_j = \frac{y_{i,j} - y_{i,j+1}}{K_{i,j} x_{i,j} - y_{i,j+1}} \quad (10-73)$$

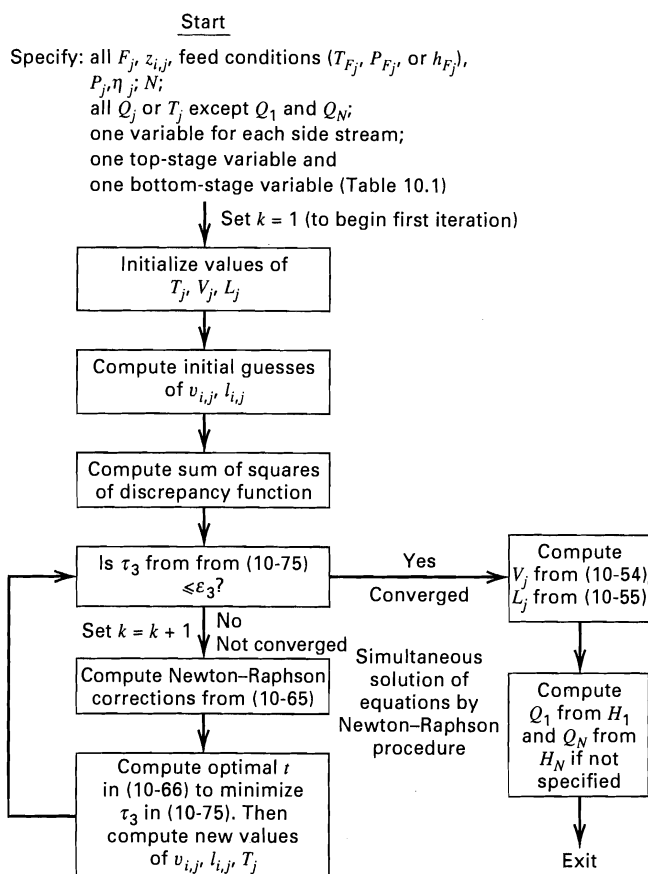
In terms of component flow rates, (10-73) becomes the following discrepancy function, which replaces (10-59).

$$E_{i,j} = \frac{\eta_j K_{i,j} l_{i,j} \sum_{\kappa=1}^C v_{\kappa,j}}{\sum_{\kappa=1}^C l_{\kappa,j}} - v_{i,j} + \frac{(1 - \eta_j) v_{i,j+1} \sum_{\kappa=1}^C v_{\kappa,j+1}}{\sum_{\kappa=1}^C v_{\kappa,j+1}} = 0 \quad (10-74)$$

If a total condenser with subcooling is desired, it is necessary to specify the degrees of subcooling, if any, and to replace (10-59) or (10-74) with functions that express identity of reflux and distillate compositions as discussed by Naphtali and Sandholm [23].

The algorithm for the Naphtali-Sandholm implementation of the Newton-Raphson method is shown in Figure 10.24. Problem specifications are quite flexible. Pressure, compositions, flow rates, and stage locations are necessary specifications for all feeds. The thermal condition of each feed can be given in terms of enthalpy, temperature, or molar fraction vaporized. If a feed is found to consist of two phases, the phases can be sent to the same stage or the vapor can be directed to the stage above the designated feed stage. Stage pressures and stage efficiencies can be designated by specifying top- and bottom-stage values. Remaining values are obtained by linear interpolation. By default, intermediate stages are assumed to be adiabatic unless Q_j or T_j values are specified. Vapor and/or liquid side streams can be designated in terms of total flow rate or flow rate of a specified component, or by the ratio of the side stream flow rate to the flow rate remaining and passing to the next stage. The top- and bottom-stage specifications are selected from Q_1 or Q_N , and/or more generally from the other specifications listed in Table 10.1.

In order to achieve convergence, the Newton-Raphson procedure requires that reasonable guesses be provided for the values of all output variables. Rather than provide all these guesses a priori, we can generate them if T , V , and L are guessed for the bottom and top stages and, perhaps, for one or more intermediate stages. Remaining guessed values of T_j , V_j , and L_j are readily obtained by linear interpolation of the given T_j values and computed (V_j/L_j) values. Initial values for $v_{i,j}$ and $l_{i,j}$ are then obtained by either of two

**Figure 10.24** Algorithm for the Newton-Raphson method of Naphtali-Sandholm for all vapor-liquid separators.

techniques. If K -values are composition independent or can be approximated as such, one technique is to compute $x_{i,j}$ values and corresponding $y_{i,j}$ values from (10-12) and (10-2) as in the first iteration of the BP or SR method. A much cruder estimate is obtained by flashing the combined feeds at some average column pressure and a V/L ratio that approximates the ratio of overheads to bottoms products. The resulting mole-fraction compositions of the equilibrium vapor and liquid phases are assumed to hold for each stage. The second technique works surprisingly well, but the first technique is preferred for difficult cases. For either technique, the initial component flow rates are computed by using the $x_{i,j}$ and $y_{i,j}$ values to solve (10-56) and (10-57) for $l_{i,j}$ and $v_{i,j}$, respectively.

Based on initial guesses for all output variables, the sum of the squares of the discrepancy functions is computed and compared to the convergence criterion

$$\tau_3 = \sum_{j=1}^N \left\{ (H_j)^2 + \sum_{i=1}^C [(M_{i,j})^2 + (E_{i,j})^2] \right\} \leq \epsilon_3 \quad (10-75)$$

In order that the values of all discrepancies be of the same order of magnitude, it is necessary to divide energy-balance functions H_j by a scale factor approximating the latent heat of vaporization (e.g., 1,000 Btu/lbmol). If the convergence criterion is computed from

$$\epsilon_3 = N(2C + 1) \left(\sum_{j=1}^N F_j^2 \right) 10^{-10} \quad (10-76)$$

resulting converged values of the output variables will generally be accurate, on the average, to four or more significant figures. When employing (10-76), most problems are converged in 10 iterations or less.

Generally, the convergence criterion is far from satisfied during the first iteration when guessed values are assumed for the output variables. For each subsequent iteration, the Newton-Raphson corrections are computed from (10-65). These corrections can be added directly to the present values of the output variables to obtain a new set of values for the output variables. Alternatively, (10-66) can be employed where t is a nonnegative, scalar step factor. At each iteration, a single value of t is applied to all output variables. By permitting t to vary from, say, slightly greater than zero up to 2, it can serve to dampen or accelerate convergence, as appropriate. For each iteration, an optimal value of t is sought to minimize the sum of the squares given by (10-75). Generally, optimal values of t proceed from an initial value for the second iteration at between 0 and 1 to a value nearly equal to or slightly greater than 1 when the convergence criterion is almost satisfied. An efficient optimization procedure for finding t at each iteration is the Fibonacci search [25]. If no optimal value of t can be found within the designated range, t can be set to 1, or some smaller value, and the sum of squares can be allowed to increase. Generally, after several iterations, the sum of squares will decrease for every iteration.

If the application of (10-66) results in a negative component flow rate, Naphtali and Sandholm recommend the following mapping equation, which reduces the value of the unknown variable to a near-zero, but nonnegative, quantity.

$$X^{(k+1)} = X^{(k)} \exp \left[\frac{t \Delta X^{(k)}}{X^{(k)}} \right] \quad (10-77)$$

In addition, it is advisable to limit temperature corrections at each iteration.

The Naphtali-Sandholm SC method is readily extended to staged separators involving two liquid phases (e.g., extraction) and three coexisting phases (e.g., three-phase distillation), as shown by Block and Hegner [26], and to interlinked separators as shown by Hofeling and Seader [27].

EXAMPLE 10.8

A reboiled absorber is to be designed to separate the hydrocarbon vapor feed of Examples 10.2 and 10.4. Absorbent oil of the same composition as that of Example 10.4 will enter the top stage. Complete specifications are given in Figure 10.25. The 770 lbmol/h (349 kmol/h) of bottoms product corresponds to the amount of C_3 and heavier in the two feeds. Thus, the column is to be designed as a deethanizer. Calculate stage temperatures, interstage vapor and liquid flow rates and compositions, and reboiler duty by the Newton-Raphson method. Assume all stage efficiencies are 100%. Compare the degree of separation of the feed to that achieved by ordinary distillation in Example 10.2.

SOLUTION

A digital computer program for the method of Naphtali and Sandholm was used. The K -values and enthalpies were assumed independent of composition and were computed by linear interpolation

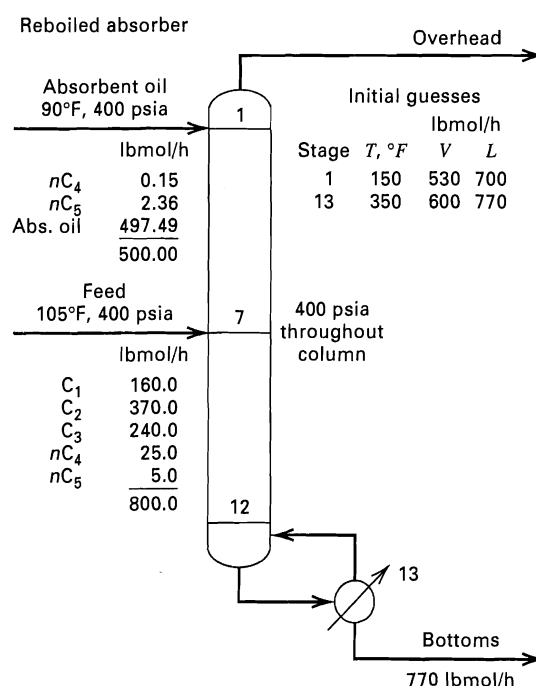


Figure 10.25 Specifications for Example 10.8.

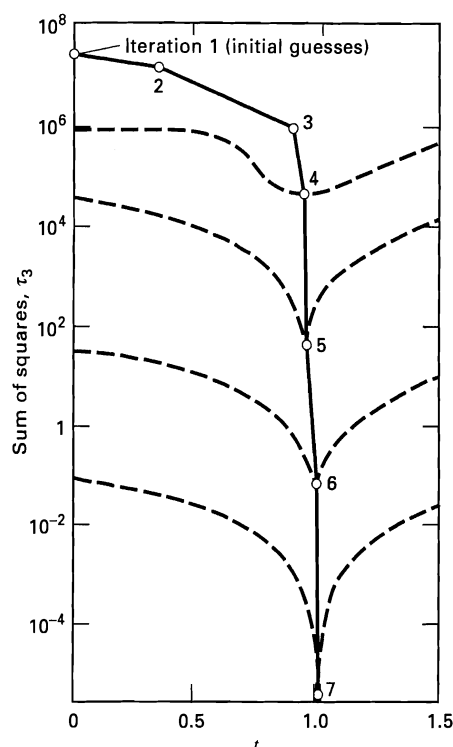


Figure 10.26 Convergence pattern for Example 10.8.

between tabular values given at 100°F increments from 0 to 400°F (−17.8 to 204.4°C).

From (10-76), the convergence criterion is

$$\epsilon_3 = 13[2(6) + 1](500 + 800)^2 10^{-10} = 2.856 \times 10^{-2}$$

Figure 10.26 shows the reduction in the sum of the squares of the 169 discrepancy functions from iteration to iteration. Seven iterations were required to satisfy the convergence criterion. The initial iteration was based on values of the unknown variables computed from interpolation of the initial guesses shown in Figure 10.25 together with a flash of the combined feeds at 400 psia (2.76 MPa) and a V/L ratio of 0.688 (530/770). Thus, for the first iteration, the following mole-fraction compositions were computed and were assumed to apply to every stage.

Species	y	x
C ₁	0.2603	0.0286
C ₂	0.4858	0.1462
C ₃	0.2358	0.1494
nC ₄	0.0153	0.0221
nC ₅	0.0025	0.0078
Abs. Oil	0.0003	0.6459
	1.0000	1.0000

The corresponding sum of squares of the discrepancy functions, τ_3 , of 2.865×10^7 was very large. Subsequent iterations employed the Newton-Raphson method. For iteration 2, the optimal value of t was found to be 0.34. However, this caused only a moderate reduction in the sum of squares. The optimal value of t increased to 0.904 for iteration 3, and the sum of squares was reduced by an order of magnitude. For the fourth and subsequent iterations, the effect of t on the sum of squares is included in Figure 10.26. Following iteration 4, the sum of squares was reduced by at least two orders of magnitude for each iteration. Also, the optimal value of t was rather

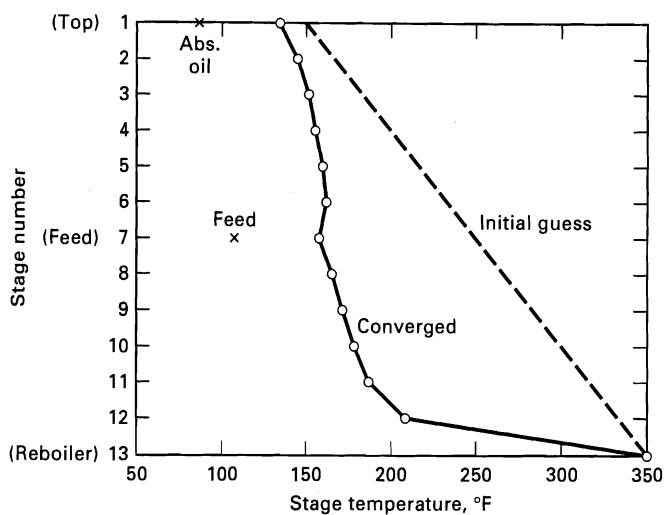


Figure 10.27 Converged temperature profile for Example 10.8.

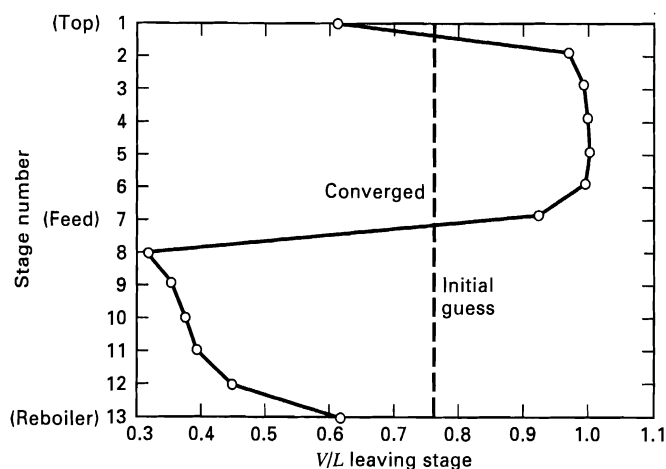


Figure 10.28 Converged vapor-liquid ratio profile for Example 10.8.

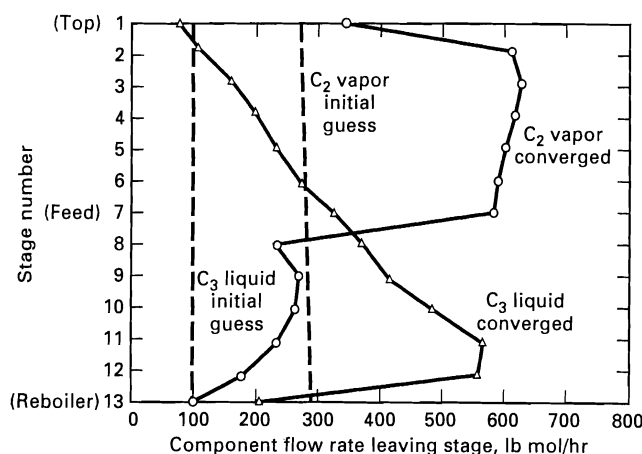
sharply defined and corresponded closely to a value of 1. An improvement of τ_3 was obtained for every iteration.

In Figures 10.27 and 10.28, converged temperature and V/L profiles are compared to the initially guessed profiles. In Figure 10.27, the converged temperatures are far from linear with respect to stage number. Above the feed stage, the temperature profile increases from the top down in a gradual and declining manner. The relatively cold feed causes a small temperature drop from stage 6 to stage 7. Temperature also increases from stage 7 to stage 13. A particularly dramatic increase occurs in moving from the bottom stage in the column to the reboiler, where heat is added. In Figure 10.28, the V/L profile is also far from linear with respect to stage number. Dramatic changes in this ratio occur at the top, middle, and bottom of the column.

Component flow-rate profiles for the two key components (ethane vapor and propane liquid) are shown in Figure 10.29. The initial guessed values are in very poor agreement with the converged values. The propane-liquid profile is quite regular except at the bottom, where a large decrease occurs because of vaporization in the reboiler. The ethane-vapor profile has large changes at the top, where entering oil absorbs appreciable ethane, and at the feed stage, where substantial ethane vapor is introduced.

Table 10.2 Product Compositions and Reboiler Duty for Example 10.8

	Composition-Independent Tabular Properties	Chao-Seader Correlation	Soave-Redlich-Kwong Equation
Overhead component flow rates, lbmol/h			
C ₁	159.99	159.98	159.99
C ₂	337.96	333.52	341.57
C ₃	31.79	36.08	28.12
nC ₄	0.04	0.06	0.04
nC ₅	0.17	0.21	0.18
Abs. oil	0.05	0.15	0.10
	530.00	530.00	530.00
Bottoms component flow rates, lbmol/h			
C ₁	0.01	0.02	0.01
C ₂	32.04	36.4	28.43
C ₃	208.21	203.92	211.88
nC ₄	25.11	25.09	25.11
nC ₅	7.19	7.15	7.18
Abs. oil	497.44	497.34	497.39
	770.00	770.00	770.00
Reboiler duty, Btu/h	11,350,000	10,980,000	15,640,000
Bottoms temperature, °F	346.4	338.5	380.8

**Figure 10.29** Converged flow rates for key components in Example 10.8.

Converged values for the reboiler duty and overhead and bottoms compositions are given in Table 10.2. Also included are converged results for two additional solutions that used the Chao-Seader and Soave-Redlich-Kwong equations for K -values and enthalpies in place of interpolation of composition-independent tabular properties. With the Soave-Redlich-Kwong equation, a somewhat sharper separation between the two key components is predicted. In addition, the Soave-Redlich-Kwong equation predicts a substantially higher bottoms temperature and a much larger reboiler duty. As discussed in Chapter 4, the effect of physical properties on equilibrium-stage calculations can be significant.

It is interesting to compare the separation achieved with the reboiled absorber of this example to the separation achieved by ordinary distillation of the same feed in Example 10.2 as shown in Figure 10.9. The latter separation technique results in a much sharper

separation and a much lower bottoms temperature and reboiler duty for the same number of stages. However, refrigeration is necessary for the overhead condenser, and the reflux flow rate is twice the absorbent oil flow rate. If the absorbent-oil flow rate for the reboiled absorber is made equal to the reflux flow rate, calculations give a separation almost as sharp as for ordinary distillation. However, the bottoms temperature and reboiler duty are increased to almost 600°F (315.6°C) and 60,000,000 Btu/h (63.3 GJ/h), respectively.

10.5 INSIDE-OUT METHOD

In the bubble-point (BP) and sum-rates (SR) methods described in Section 10.3 and the Newton-Raphson method described in Section 10.4, a large percentage of the computational effort is expended in calculating K -values, vapor-phase enthalpies, and liquid-phase enthalpies, particularly when rigorous thermodynamic-property models (e.g., Soave-Redlich-Kwong, Peng-Robinson, Wilson, NRTL, UNIQUAC) are utilized. As seen in Figures 10.30a and 10.30b, these property calculations are made at each iteration. Furthermore, at each iteration, derivatives are required of: (1) all three thermodynamic properties with respect to temperature and compositions of both phases, for the Newton-Raphson method; (2) K -values with respect to temperature for the BP method, unless Mullers method is used to compute bubble points; and (3) vapor and liquid enthalpies with respect to temperature for the SR method.

In 1974, Boston and Sullivan [28] presented an algorithm designed to significantly reduce the time spent in computing thermodynamic properties when designing steady-state,

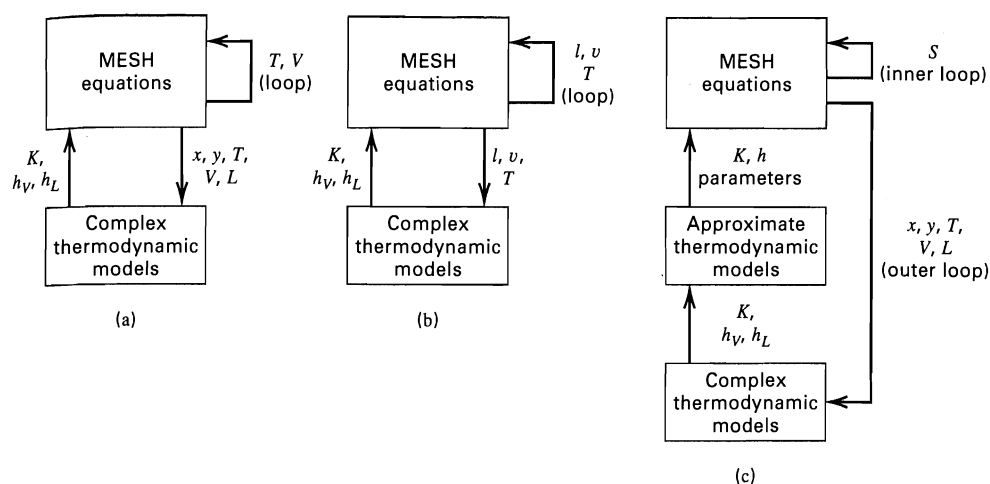


Figure 10.30 Incorporation of thermodynamic property correlations into interactive loops. (a) BP and SR methods. (b) Newton-Raphson method. (c) Inside-out method.

multicomponent separation operations. As shown in Figure 10.30c, two sets of thermodynamic-property models are employed: (1) a simple, approximate empirical set used frequently to converge inner loop calculations, and (2) the rigorous and complex set used less often in the outer loop. The MESH equations are always solved in the inner loop with the approximate set. The parameters in the empirical equations for the approximate set are updated in the outer loop by the rigorous equations, but only at infrequent intervals. A distinguishing feature of the Boston-Sullivan method is these inner and outer loops; hence the name *inside-out* for this class of methods. Another name, less frequently used, is *two-tier* methods.

Another difference that distinguishes the inside-out method, as shown in Figure 10.30, is the choice of iteration variables. For the Newton-Raphson method, the iteration variables are $l_{i,j}$, v_{ij} , T_j . For the BP and SR methods, the choice is $x_{i,j}$, $y_{i,j}$, T_j , L_j , and V_j . For the inside-out method, the iteration variables for the outer loop are the parameters in the approximate equations for the thermodynamic properties. The iteration variables for the inner loop are related to the stripping factors, $S_{i,j} = K_{i,j} V_j / L_j$.

In the original presentation of the inside-out method in 1974, the development and application of the method was restricted to hydrocarbon distillation (moderately nonideal systems) for the Case II variable specification in Table 5.4, but with multiple feeds, side streams, and intermediate heat exchangers. For these applications, the inside-side method was shown to be rapid and robust. Since 1974, the method has been extended and improved in a number of published articles [29, 30, 31, 32, 33, 34] and proprietary implementations in simulation computer programs. These extensions permit the inside-out method to be applied to almost any type of steady-state, multicomponent, multistage vapor-liquid separation operation. In the extensive implementation of the inside-out method by ASPEN technology in ASPEN PLUS, in computer programs called RADFRAC and MULTIFRAC, these applications include:

1. Absorption, stripping, reboiled absorption, reboiled stripping, extractive distillation, and azeotropic distillation

2. Three-phase (vapor-liquid-liquid) systems
3. Reactive systems
4. Highly nonideal systems requiring activity-coefficient models
5. Interlinked systems of separation units, including pumparounds, bypasses, and external heat exchangers
6. Narrow-boiling, wide-boiling, and dumbbell (mostly heavy and light components with little in between) feeds
7. Presence of free water
8. Wide variety of specifications other than Case II of Table 5.2 for the reflux ratio and product rates (e.g. product purities)
9. Use of Murphree-stage efficiencies

The inside-out method takes advantage of the following characteristics of the iterative calculations:

1. Component relative volatilities vary much less than component K -values.
2. Enthalpy of vaporization varies less than phase enthalpies.
3. Component stripping factors combine effects of temperature and liquid and vapor flows at each stage.

The inner loop of the inside-out method uses relative volatility, energy, and stripping factors to improve stability and reduce computing time. A widely used implementation of the inside-out method is that of Russell [31], which is described here together with further refinements suggested and tested by Jelinek [33].

MESH Equations

As with the BP, SR, and Newton-Raphson methods, the equilibrium-stage model of Figures 10.1 and 10.3 is again employed. The form of the equations is similar to the Newton-Raphson method in that component flow rates are utilized. However, in addition, the following inner-loop

variables are defined:

$$\alpha_{i,j} = K_{i,j}/K_{b,j} \quad (10-78)$$

$$S_{b,j} = K_{b,j}V_j/L_j \quad (10-79)$$

$$R_{Lj} = 1 + U_j/L_j \quad (10-80)$$

$$R_{Vj} = 1 + W_j/V_j \quad (10-81)$$

where K_b is the K -value for a base or hypothetical reference component, $S_{b,j}$ is the stripping factor for the base component, R_{Lj} is a liquid-phase withdrawal factor, and R_{Vj} is a vapor-phase withdrawal factor. For stages without side streams, R_{Lj} and R_{Vj} reduce to 1. With the defined variables of (10-78) to (10-81), (10-54) to (10-57) still apply, but the MESH equations, (10-58) to (10-60), become as follows, where (10-83) results from the use of (10-80) to (10-82) to eliminate the variables in V and the side stream ratios s and S :

Phase Equilibria:

$$v_{i,j} = \alpha_{i,j} S_{b,j} l_{i,j}, \quad i = 1 \text{ to } C, \quad j = 1 \text{ to } N \quad (10-82)$$

Component Material Balance:

$$l_{i,j-1} - (R_{Lj} + \alpha_{i,j} S_{b,j} R_{Vj}) l_{i,j} + (\alpha_{i,j+1} S_{b,j+1}) l_{i,j+1} = -f_{i,j}, \quad i = 1 \text{ to } C, \quad j = 1 \text{ to } N \quad (10-83)$$

Energy Balance:

$$H_j = h_{Lj} R_{Lj} L_j + h_{Vj} R_{Vj} V_j - h_{Lj-1} L_{j-1} - h_{Vj+1} V_{j+1} - h_{Fj} F_j - Q_j = 0, \quad j = 1 \text{ to } N \quad (10-84)$$

where $S_{i,j} = \alpha_{i,j} S_{b,j}$.

In addition, discrepancy functions of the type shown in Table 10.1 for the Newton–Raphson method can be added to the MESH equations to permit any reasonable set of product specifications.

Rigorous and Complex Thermodynamic Property Models

The complex thermodynamic models referred to in Figure 10.30 can include any of the types of models discussed in Chapter 2, including those based on P – v – T equations of state (e.g., Soave–Redlich–Kwong and Peng–Robinson) and those based on free-energy models for predicting liquid-phase activity coefficients (e.g., Wilson, NRTL, and UNIQUAC). These models are used to generate parameters in the approximate thermodynamic-property models. In general, the rigorous property models are of the form:

$$K_{i,j} = K_{i,j}\{P_j, T_j, \mathbf{x}_j, \mathbf{y}_j\} \quad (10-85)$$

$$h_{Vj} = h_{Vj}\{P_j, T_j, \mathbf{y}_j\} \quad (10-86)$$

$$h_{Lj} = h_{Lj}\{P_j, T_j, \mathbf{x}_j\} \quad (10-87)$$

Approximate Thermodynamic Property Models

K-Values

The approximate models used in the inside-out method are designed to facilitate the calculation of stage temperatures

and stripping factors. The approximate K -value model of Russell [31] and Jelinek [33], which differs only slightly from the model of Boston and Sullivan [28] and originates from a proposal in the classic textbook by Robinson and Gilliland [35], is (10-78) combined with

$$K_{b,j} = \exp(A_j - B_j/T_j) \quad (10-88)$$

Either a component in the feed or a hypothetical reference component can be selected as the base, b , component, with the latter preferred. For that case, the base component is determined from a vapor-composition weighting using the following relations:

$$K_{b,j} = \exp\left(\sum_i w_{i,j} \ln K_{i,j}\right) \quad (10-89)$$

where $w_{i,j}$ are weighting functions given by

$$w_{i,j} = \frac{y_{i,j} [\partial \ln K_{i,j} / \partial (1/T)]}{\sum_i y_{i,j} [\partial \ln K_{i,j} / \partial (1/T)]} \quad (10-90)$$

A unique K_b model and values of $\alpha_{i,j}$ in (10-78) are derived for each stage j from values of $K_{i,j}$ determined from the rigorous model. At the top stage, the base component will be close to one of the light components, while at the bottom stage, the base component will be close to a heavy component. The derivatives in (10-90) are obtained numerically or analytically from the rigorous model. To determine the values of A_j and B_j in (10-88), two temperatures must be selected for each stage. For example, the estimated or current temperatures of the two adjacent stages, $j-1$ and $j+1$, might be selected. Calling these two temperatures T_1 and T_2 and using (10-88) at each stage, b :

$$B = \frac{\ln(K_{bT_1}/K_{bT_2})}{\left(\frac{1}{T_2} - \frac{1}{T_1}\right)} \quad (10-91)$$

and

$$A = \ln K_{bT_1} + B/T_1 \quad (10-92)$$

If highly nonideal-liquid solutions are involved, it is advisable to separate the rigorous K -value into two parts, as in (2-27). Thus,

$$K_i = \gamma_{iL}(\phi_{iL}/\bar{\phi}_{iV}) \quad (10-93)$$

Then, $(\phi_{iL}/\bar{\phi}_{iV})$ is used to determine K_b and, as proposed by Boston [30], values of γ_{iL} at each stage are fitted at a reference temperature T^* , to the liquid-phase mole fraction by the linear function

$$\gamma_{iL}^* = a_i + b_i x_i \quad (10-94)$$

to obtain the approximate estimates, γ_{iL}^* . Equation (10-83) is then modified by replacing $\alpha_{i,j}$ with $\alpha_{i,j} \gamma_{iL}^*$, where

$$\alpha_{i,j} = \frac{(\phi_{iL}/\bar{\phi}_{iV})_j}{K_{b,j}} \quad (10-95)$$

rather than the $\alpha_{i,j}$ given by (10-78).

Enthalpies

Boston and Sullivan [28] and Russell [31] employ the same approximate enthalpy models. Jelinek [33] does not use approximate enthalpy models, because the additional complexity involved in the use of two enthalpy models may not always be justified to the extent that the use of both approximate and rigorous K -value models is justified.

The basis for the enthalpy calculations is the same as for the rigorous equations discussed in Chapter 2. Thus, for either phase, from Table 2.6,

$$h = h_V^\circ + (h - h_V^\circ) = h_V^\circ + \Delta H \quad (10-96)$$

where h_V° is the ideal-gas mixture enthalpy, as given by the polynomial equations, (2-35) and (2-36), based on the vapor-phase composition for h_V and the liquid-phase composition for h_L . The ΔH term is the enthalpy departure, $\Delta H_V = (h_V - h_V^\circ)$ for the vapor phase, which accounts for the effect of pressure, and $\Delta H_L = (h_L - h_V^\circ)$ for the liquid phase, which accounts for the enthalpy of vaporization and the effect of pressure on both liquid and vapor phases, as indicated in (2-57). Of particular importance is the enthalpy of vaporization, which dominates the ΔH_L term. The time-consuming parts of the enthalpy calculations are the two enthalpy-departure terms, which are complex when an equation of state is used. Therefore, in the approximate enthalpy equations, the rigorous enthalpy departures are replaced by the simple linear functions,

$$\Delta H_{Vj} = c_j - d_j(T_j - T^*) \quad (10-97)$$

and

$$\Delta H_{Lj} = e_j - f_j(T_j - T^*) \quad (10-98)$$

where the departures are modeled in terms of enthalpy per unit mass instead of per unit mole, and T^* is a reference temperature. The parameters c , d , e , and f are evaluated from the rigorous models at each iteration of the outer loop.

Inside-Out Algorithm

The inside-out algorithm of Russell [31] involves an initialization procedure, inner-loop iterations, and outer-loop iterations.

Initialization Procedure

Before inner- or outer-loop calculations can begin, it is necessary to provide reasonably good estimates of all stage values of $x_{i,j}$, $y_{i,j}$, T_j , V_j , and L_j . Boston and Sullivan [28] suggest the following procedure:

1. Specify the number of theoretical stages, conditions of all feeds, feed-stage locations, and column pressure profile.
2. Specify stage locations for each product withdrawal (including side streams) and for each heat exchanger.
3. Provide an additional specification for each product and each intermediate heat exchanger.

4. If not specified, estimate each product withdrawal rate, and estimate each value of V_j . Estimate values of L_j from the total material-balance equation, (10-6).
5. Estimate an initial temperature profile, T_j , by combining all feed streams (composite feed) and determining the bubble- and dew-point temperatures at the average column pressure. The dew-point temperature is taken as the top-stage temperature, T_1 , whereas the bubble-point temperature is taken as the bottom-stage temperature, T_N . Intermediate-stage temperatures are estimated by linear interpolation. Reference temperatures T^* for use with (10-94), (10-97), and (10-98) are set equal to T_j .
6. Flash the composite feed isothermally at the average column pressure and average column temperature. The resulting vapor and liquid compositions, y_i and x_i , are the estimated compositions for each stage.
7. Using the initial estimates from Steps 1 through 7, use the selected complex thermodynamic-property correlation to determine values of the stagewise outside-loop K and h parameters A_j , B_j , $a_{i,j}$, $b_{i,j}$, c_j , d_j , e_j , f_j , $K_{b,j}$, and $\alpha_{i,j}$ of the approximate models.
8. Compute initial values of $S_{b,j}$, R_{Lj} , and R_{Vj} from (10-79), (10-80), and (10-81).

Inner-Loop Calculation Sequence

An iterative sequence of inner-loop calculations begins with a set of values for the outside-loop parameters listed in Step 7, obtained initially from the initialization procedure and later from outer-loop calculations, using results from the inner loop, as shown in Figure 10.30c.

9. Compute component liquid flow rates, $l_{i,j}$, from the set of N equations (10-83) for each of the C components by the tridiagonal-matrix algorithm.
10. Compute component vapor flow rates, v_{ij} , from (10-82).
11. Compute a revised set of total flow rates, V_j and L_j , from the component flow rates by (10-54) and (10-55), respectively.
12. To calculate a revised set of stage temperatures, T_j , as follows, compute a set of x_i values for each stage from (10-57), then a revised set of $K_{b,j}$ values from a combination of the bubble-point equation, (4-12), $\sum_i K_i x_i = 1$, with (10-78), which gives

$$K_{b,j} = 1 / \sum_{i=1}^C (\alpha_{i,j} x_{i,j}) \quad (10-99)$$

From this new set of $K_{b,j}$ values, compute a new set of stage temperatures from the following rearrangement of (10-88):

$$T_j = \frac{B_j}{A_j - \ln K_{b,j}} \quad (10-100)$$

At this point in the inner-loop iterative sequence, we have a revised set of values for $v_{i,j}$, $l_{i,j}$, and T_j , which satisfy the component material-balance and phase-equilibria equations for the estimated thermodynamic properties. However, these values do not satisfy the energy balance and specification equations unless the estimated base-component stripping factors and product-withdrawal rates are correct.

13. Select inner-loop iteration variables as

$$\ln S_{b,j} = \ln(K_{b,j} V_j / L_j) \quad (10-101)$$

together with any other iteration variables. For a simple distillation column of the type shown in Figure 10.9, no other inner-loop iteration variables would be needed if the condenser and reboiler duties were specified. If the reflux ratio (L/D) and bottoms flow rate (B) are specified in place of the two duties, which is the more common situation, one adds, in place of the two (10-84) equations for H_1 and H_N , the following two specification equations from Table 10.1 in the form of discrepancy functions, D_1 and D_2 :

$$D_1 = L_1 - (L/D)V_1 = 0 \quad (10-102)$$

$$D_2 = L_N - B = 0 \quad (10-103)$$

For each side stream, a side-stream-withdrawal factor is added as an inner-loop iteration variable, e.g., $\ln(U_j/L_j)$ and $\ln(W_j/V_j)$, together with a specification equation on purity or some other variable.

14. Compute enthalpies of all streams from (10-96) to (10-98).
15. Compute normalized discrepancies of H_j , D_1 , D_2 , etc., from the energy balances (10-84) and (10-102), (10-103), etc., except compute Q_1 from H_1 and Q_N from H_N where appropriate. A typical normalization is discussed in Section 10.4 for the Newton-Raphson method.
16. Compute the Jacobian of partial derivatives of H_j , D_1 , D_2 , etc., with respect to the iteration variables of (10-101), etc. This is done by successive perturbation of each iteration variable and recalculation of the discrepancies through Steps 9 to 15, numerically or by differentiation.
17. Compute corrections to the inner-loop iteration variables by a Newton-Raphson iteration of the type discussed for the SR method in Section 10.3 and the Newton-Raphson method in Section 10.4.
18. Compute new values of the iteration variables from the sum of the previous values and the corrections with (10-66), using damping if necessary to reduce the sum of the squares of the normalized discrepancies.
19. Check whether the sum of the squares is sufficiently small. If so, proceed to the outer-loop calculation procedure given next. If not, repeat Steps 15 to 18

using the latest values of the iteration variables. For any subsequent cycles through Steps 15 to 18, Russell [31] uses Broyden [36] updates to avoid reestimation of the Jacobian partial derivatives, whereas Jelinek [33] recommends the standard Newton-Raphson method of recalculating the partial derivatives for each inner-loop iteration.

20. Upon convergence of Steps 15 to 19, Steps 9 through 12 will have produced an improved set of primitive variables $x_{i,j}$, $v_{i,j}$, $l_{i,j}$, T_j , V_j , and L_j . From (10-56), corresponding values of $y_{i,j}$ can be computed. The values of these variables are not correct until the approximate thermodynamic properties are in agreement with the properties from the rigorous models. The primitive variables are input to the outer-loop calculations to bring the approximate and complex models into successively better agreement.

Outer-Loop Calculation Sequence

21. Using the values of the primitive variables from Step 20, compute relative volatilities and stream enthalpies from the complex thermodynamic models. If they are in close agreement with the previous values used to initiate a set of inner-loop iterations, both the outer-loop and inner-loop iterations are converged and the problem is solved. If not, proceed to Step 22.
22. Determine values of the stagewise outside-loop K and h parameters of the approximate models from the complex models as in initialization Step 7.
23. Compute values of $S_{b,j}$, $R_{L,j}$, and $R_{V,j}$, as in initialization Step 8.
24. Repeat the inner-loop calculation sequence of Steps 9 through 20.

Convergence of the inside-out method is not guaranteed. However, for most problems, the method is robust and rapid. Convergence can encounter difficulty because of poor initial estimates, resulting in negative or zero flow rates at certain locations in the column. To counteract this tendency, all component stripping factors are scaled with a scalar multiplier, S_b , sometimes called the base stripping factor, to give

$$S_{i,j} = S_b \alpha_{i,j} S_{b,j} \quad (10-104)$$

The value of S_b is initially chosen to force the results of the initialization procedure to give a reasonable distribution of component flows throughout the column. Russell recommends that S_b be chosen only once, whereas Boston and Sullivan compute a new value S_b for each new set of $S_{b,j}$ values.

For highly nonideal-liquid mixtures, use of the inside-out method may become quite difficult. When that occurs, the Newton-Raphson method may be preferred. If the Newton-Raphson method also fails to converge, relaxation or continuation methods, described by Kister [37], are usually

successful, but computing time may be an order of magnitude longer than that for similar problems converged successfully with the inside-out method.

EXAMPLE 10.9

For the conditions of the distillation column shown in Figure 10.7, obtain a converged solution by the inside-out method, using the SRK equation-of-state for thermodynamic properties.

SOLUTION

A computer solution was obtained with the equipment module TOWR (an inside-out method) of the CHEMCAD process simulation program of Chemstations, Inc. The only initial assumptions are a condenser outlet temperature of 65°F and a bottoms-product temperature of 165°F. The bubble-point temperature of the feed is computed as 123.5°F. In the initialization procedure, the constants A and B in (10-88), with T in °R, are determined from the SRK equation, with the following results:

Stage	T , °F	A	B	K_b
1	65	6.870	3708	0.8219
2	95	6.962	4031	0.7374
3	118	7.080	4356	0.6341
4	142	7.039	4466	0.6785
5	165	6.998	4576	0.7205

Values of the enthalpy coefficients c , d , e , and f in (10-97) and (10-98) are not tabulated here but are also computed for each stage, based on the initial temperature distribution.

In the inner-loop calculation sequence, component flow rates are computed from (10-83) by the tridiagonal-matrix method. The resulting bottoms-product flow rate deviates somewhat from the specified value of 50 lbmol/h. However, by modifying the component stripping factors with a base stripping factor, S_b , in (10-104) of 1.1863, the error in the bottoms flow rate is reduced to 0.73%.

The initial inside-loop error from the solution of the normalized energy-balance equations, (10-84), is found to be only 0.04624. This is reduced to 0.000401 after two iterations through the inner loop.

At this point in the inside-out method, the revised column profiles of temperature and phase compositions are used in the outer loop with the complex SRK thermodynamic models to compute updates of the approximate K and h constants. Only one inner-loop iteration is required to obtain satisfactory convergence of the energy equations. The K and h constants are again updated in the outer loop. After one inner-loop iteration, the approximate K and h

values are found to be sufficiently close to the SRK values that overall convergence is achieved. Thus, a total of only three outer-loop iterations and four inner-loop iterations are required.

To illustrate the efficiency of the inside-out method to converge this example, the results from each of the three outer-loop iterations are summarized in the following table:

Outer-Loop Iteration	Stage Temperatures, °F				
	T_1	T_2	T_3	T_4	T_5
Initial guess	65	—	—	—	165
1	82.36	118.14	146.79	172.66	193.20
2	83.58	119.50	147.98	172.57	192.53
3	83.67	119.54	147.95	172.43	192.43

Outer-Loop Iteration	Total Liquid Flows, lbmol/h				
	L_1	L_2	L_3	L_4	L_5
Specification	100	—	—	—	—
1	100.00	89.68	187.22	189.39	50.00
2	100.03	89.83	188.84	190.59	49.99
3	100.0	89.87	188.96	190.56	50.00

Outer-Loop Iteration	Component Flows in Bottoms Product, lbmol/h			
	C_3	nC_4	nC_5	L_5
1	0.687	12.045	37.268	50.000
2	0.947	12.341	36.697	49.985
3	0.955	12.363	36.683	50.001

From this table it is seen that the stage temperatures and total liquid flows are already close to the converged solution after only one outer-loop iteration. However, the composition of the bottoms product, specifically with respect to the lightest component, C_3 , is not close to the converged solution until after two iterations. The inside-out method does not always converge so dramatically but is usually quite efficient, as shown in the following table.

Problem	Total Number of Inner Loops	Number of Outer-Loop Iterations
Exercise 10.11	7	6
Exercise 10.25	6	3
Exercise 10.37	17	9
Exercise 10.41	16	5

Computing times for each of these four exercises was less than 1 second on a PC with a Pentium 4 processor at 2.4 GHz.

SUMMARY

1. Rigorous methods are readily available for computer-solution of equilibrium-based models for multicomponent, multistage absorption, stripping, distillation, and liquid-liquid extraction.
2. The equilibrium-based model for a countercurrent-flow cascade provides for multiple feeds, vapor side streams, liquid side streams, and intermediate heat exchangers. Thus, the model can handle almost any type of column configuration.

3. The model equations include component material balances, total material balances, phase equilibria relations, and energy balances.
4. Some or all of the model equations can usually be grouped so as to obtain tridiagonal-matrix equations, for which an efficient solution algorithm is available.

5. Widely used methods for iteratively solving all of the model equations are the bubble-point (BP) method, the sum-rates (SR) method, the Newton–Raphson method, and the inside-out method.
6. The BP method is generally restricted to distillation problems involving narrow-boiling feed mixtures.
7. The SR method is generally restricted to absorption and stripping problems involving wide-boiling feed mixtures or in the ISR form to extraction problems.
8. The Newton–Raphson and inside-out methods are designed to solve any type of column configuration for any type of feed

mixture. Because of its computational efficiency, the inside-out method is often the method of choice; however, it may fail to converge when highly nonideal-liquid mixtures are involved, in which case the slower Newton–Raphson method should be tried. Both methods permit considerable flexibility in specifications.

9. When both the Newton–Raphson and inside-out methods fail, resort can be made to much slower relaxation and continuation methods.

REFERENCES

1. WANG, J.C., and G.E. HENKE, *Hydrocarbon Processing* **45**(8), 155–163 (1966).
2. MYERS, A.L., and W.D. SEIDER, *Introduction to Chemical Engineering and Computer Calculations*, Prentice-Hall, Englewood Cliffs, NJ, 484–507 (1976).
3. LEWIS, W.K., and G.L. MATHESON, *Ind. Eng. Chem.* **24**, 496–498 (1932).
4. THIELE, E.W., and R.L. GEDDES, *Ind. Eng. Chem.* **25**, 290 (1933).
5. HOLLAND, C.D., *Multicomponent Distillation*. Prentice-Hall, Englewood Cliffs, NJ (1963).
6. AMUNDSON, N.R., and A.J. PONTINEN, *Ind. Eng. Chem.* **50**, 730–736 (1958).
7. FRIDAY, J.R., and B.D. SMITH, *AIChE J.* **10**, 698–707 (1964).
8. BOSTON, J.F., and S.L. SULLIVAN, JR., *Can. J. Chem. Eng.* **52**, 52–63 (1974).
9. BOSTON, J.F., and S.L. SULLIVAN, JR., *Can. J. Chem. Eng.* **50**, 663–669 (1972).
10. JOHANSON, P.J., and J.D. SEADER, *Stagewise Computations—Computer Programs for Chemical Engineering Education* (ed. by J. Christensen), Aztec Publishing, Austin, TX, pp. 349–389, A-16 (1972).
11. LAPIDUS, L., *Digital Computation for Chemical Engineers*, McGraw-Hill, New York, pp. 308–309 (1962).
12. ORBACH, O., and C.M. CROWE, *Can. J. Chem. Eng.* **49**, 509–513 (1971).
13. SCHEIBEL, E.G., *Ind. Eng. Chem.* **38**, 397–399 (1946).
14. SUJATA, A.D., *Hydrocarbon Processing* **40**(12), 137–140 (1961).
15. BURNINGHAM, D.W., and F.D. OTTO, *Hydrocarbon Processing* **46**(10), 163–170 (1967).
16. SHINOHARA, T., P.J. JOHANSEN, and J.D. SEADER, *Stagewise Computations—Computer Programs for Chemical Engineering Education*, J. Christensen, Ed., Aztec Publishing, Austin, TX, pp. 390–428, A-17 (1972).
17. TSUBOKA, T., and T. KATAYAMA, *J. Chem. Eng. Japan* **9**, 40–45 (1976).
18. HÁLA, E., I. WICHTERLE, J. POLAK, and T. BOUBLIK, *Vapor–Liquid Equilibrium Data at Normal Pressures*, Pergamon, Oxford, p. 308 (1968).
19. STEIB, V.H., *J. Prakt. Chem.* **4**, Reihe, Bd. 28, 252–280 (1965).
20. COHEN, G., and H. RENON, *Can. J. Chem. Eng.* **48**, 291–296 (1970).
21. GOLDSTEIN, R.P., and R.B. STANFIELD, *Ind. Eng. Chem., Process Des. Develop.* **9**, 78–84 (1970).
22. NAPHTALI, L.M., “The distillation column as a large system,” paper presented at the AIChE 56th National Meeting, San Francisco, May 16–19, 1965.
23. NAPHTALI, L.M., and D.P. SANDHOLM, *AIChE J.* **17**, 148–153 (1971).
24. FREDENSLUND, A., J. GMEHLING, and P. RASMUSSEN, *Vapor–Liquid Equilibria Using UNIFAC, A Group Contribution Method*. Elsevier, Amsterdam (1977).
25. BEVERIDGE, G.S.G., and R.S. SCHECHTER, *Optimization: Theory and Practice*, McGraw-Hill, New York, pp. 180–189 (1970).
26. BLOCK, U., and B. HEGNER, *AIChE J.* **22**, 582–589 (1976).
27. HOFELING, B., and J.D. SEADER, *AIChE J.* **24**, 1131–1134 (1978).
28. BOSTON, J.F., and S.L. SULLIVAN, JR., *Can. J. Chem. Eng.* **52**, 52–63 (1974).
29. BOSTON, J.F., and H.I. BRITT, *Comput. Chem. Engng.* **2**, 109–122 (1978).
30. BOSTON, J.F., *ACS Symp. Ser. No. 124*, 135–151 (1980).
31. RUSSELL, R.A., *Chem. Eng.* **90**(20), 53–59 (1983).
32. TREVINO-LOZANO, R.A., T.P. KISALA, and J.F. BOSTON, *Comput. Chem. Engng.* **8**, 105–115 (1984).
33. JELINEK, J., *Comput. Chem. Engng.* **12**, 195–198 (1988).
34. VENKATARAMAN, S., W.K. CHAN, and J.F. BOSTON, *Chem. Eng. Prog.* **86**(8), 45–54 (1990).
35. ROBINSON, C.S., and E.R. GILLILAND, *Elements of Fractional Distillation*, 4th edition, pp. 232–236. McGraw-Hill, New York (1950).
36. BROYDEN, C.G., *Math Comp.* **19**, 577–593 (1965).
37. KISTER, H. Z., *Distillation Design*, McGraw-Hill, Inc., New York (1992).

EXERCISES

The exercises for this chapter are most conveniently divided into two groups: (1) those that can be solved manually, and (2) those that are best solved with computer implementation of the methods discussed in this chapter. The first group is referenced to section numbers of this chapter. The second group of problems follows the first group and is referenced to the type of separator. Computer implementations for use with the second group are found in the

following widely available programs and simulators:

ASPEN PLUS of Aspen Technology

CHEMCAD of Chemstations

HYSYS of Aspen Technology

PRO/II of SimSci-Esscor

Section 10.1

10.1 Show mathematically that (10-6) is not independent of (10-1), (10-3), and (10-4).

10.2 Revise the MESH equations to account for entrainment, occlusion, and chemical reaction.

Section 10.2

10.3 Revise the MESH equations (10-1) to (10-6) to allow for pumparounds of the type shown in Figure 10.2 and discussed by Bannon and Marple [*Chem. Eng. Prog.* **74**(7), 41–45 (1978)] and Huber [*Hydrocarbon Processing* **56**(8), 121–125 (1977)]. Combine the equations to obtain modified M equations similar to (10.7). Can these equations still be partitioned in a series of C tridiagonal-matrix equations?

10.4 Use the Thomas algorithm to solve the following matrix equation for x_1, x_2 , and x_3 .

$$\begin{bmatrix} -160 & 200 & 0 \\ 50 & -350 & 180 \\ 0 & 150 & -230 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -50 \\ 0 \end{bmatrix}$$

10.5 Use the Thomas algorithm to solve the following tridiagonal matrix equation for the $\mathbf{x}$ vector.

$$\begin{bmatrix} -6 & 3 & 0 & 0 & 0 \\ 3 & -4.5 & 3 & 0 & 0 \\ 0 & 1.5 & -7.5 & 3 & 0 \\ 0 & 0 & 4.5 & -7.5 & 3 \\ 0 & 0 & 0 & 4.5 & -4.5 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 100 \\ 0 \\ 0 \end{bmatrix}$$

Section 10.3

10.6 On page 162 of their article, Wang and Henke [1] claim that their method of solving the tridiagonal matrix for the liquid-phase mole fractions does not involve subtraction of nearly equal quantities. Prove or disprove their statement.

10.7 Derive an equation similar to (10-7), but with $v_{i,j} = y_{i,j} V_j$ as the variables instead of the liquid-phase mole fractions. Can the resulting equations still be partitioned into a series of C tridiagonal-matrix equations?

10.8 In a computer program for the Wang–Henke bubble-point method, 10,100 storage locations are wastefully set aside for the four indexed coefficients of the tridiagonal-matrix solution of the component material balances for a 100-stage distillation column.

$$A_j x_{i,j-1} + B_j x_{i,j} + C_j x_{i,j+1} - D_j = 0$$

Determine the minimum number of storage locations required if the calculations are conducted in the most efficient manner.

10.9 Solve by the Newton–Raphson method the simultaneous, nonlinear equations

$$\begin{aligned} x_1^2 + x_2^2 &= 17 \\ (8x_1)^{1/3} + x_2^{1/2} &= 4 \end{aligned}$$

for x_1 and x_2 to within ± 0.001 . As initial guesses, assume

- $x_1 = 2, x_2 = 5$.
- $x_1 = 4, x_2 = 5$.
- $x_1 = 1, x_2 = 1$.
- $x_1 = 8, x_2 = 1$.

10.10 Solve by the Newton–Raphson method the simultaneous, nonlinear equations

$$\begin{aligned} \sin(\pi x_1 x_2) - \frac{x_2}{2} - x_1 &= 0 \\ \exp(2x_1) \left[1 - \frac{1}{4\pi} \right] + \exp(1) \left[\frac{1}{4\pi} - 1 - 2x_1 + x_2 \right] &= 0 \end{aligned}$$

for x_1 and x_2 to within ± 0.001 . As initial guesses, assume

- $x_1 = 0.4, x_2 = 0.9$.
- $x_1 = 0.6, x_2 = 0.9$.
- $x_1 = 1.0, x_2 = 1.0$.

10.11 One thousand kilogram-moles per hour of a saturated liquid mixture of 60 mol% methanol, 20 mol% ethanol, and 20 mol% n -propanol is fed to the middle stage of a distillation column having three equilibrium stages, a total condenser, a partial reboiler, and an operating pressure of 1 atm. The distillate rate is 600 kmol/h, and the external reflux rate is 2,000 kmol/h of saturated liquid. Assuming that ideal solutions are formed such that K -values can be obtained from vapor pressures and assuming constant molar overflow such that the vapor rate leaving the reboiler and each stage is 2,600 kmol/h, calculate one iteration of the BP method up to and including a new set of T_j values. To initiate the iteration, assume a linear-temperature profile based on a distillate temperature equal to the normal boiling point of methanol and a bottoms temperature equal to the arithmetic average of the normal boiling points of the other two alcohols.

Section 10.4

10.12 Solve the following nine simultaneous linear equations, which have a block tridiagonal matrix structure, by the Thomas algorithm.

$$\begin{aligned} x_2 + 2x_3 + 2x_4 + x_6 &= 7 \\ x_1 + x_3 + x_4 + 3x_5 &= 6 \\ x_1 + x_2 + x_3 + x_5 + x_6 &= 6 \\ x_4 + 2x_5 + x_6 + 2x_7 + 2x_8 + x_9 &= 11 \\ x_4 + x_5 + 2x_6 + 3x_7 + x_9 &= 8 \\ x_5 + x_6 + x_7 + 2x_8 + x_9 &= 8 \\ x_1 + 2x_2 + x_3 + x_4 + x_5 + 2x_6 + 3x_7 + x_8 &= 13 \\ x_2 + 2x_3 + 2x_4 + x_5 + x_6 + x_7 + x_8 + 3x_9 &= 14 \\ x_3 + x_4 + 2x_5 + x_6 + 2x_7 + x_8 + x_9 &= 10 \end{aligned}$$

10.13 Naphtali and Sandholm group the $N(2C + 1)$ equations by stage. Instead, group the equations by type (i.e., enthalpy balances, component balances, and equilibrium relations). Using a three-component, three-stage example, show whether the resulting matrix structure is still block tridiagonal.

10.14 Derivatives of properties are needed in the Naphtali–Sandholm SC method. For the Chao–Seader correlation, determine analytical derivatives for

$$\frac{\partial K_{i,j}}{\partial T_j}, \quad \frac{\partial K_{i,j}}{\partial v_{i,k}}, \quad \frac{\partial K_{i,j}}{\partial l_{i,k}}$$

10.15 A rigorous partial NR method for multicomponent, multi-stage vapor–liquid separations can be devised that is midway between the complexity of the BP/SR methods on the one hand and the NR methods on the other hand. The first major step in the

procedure is to solve the modified M equations for the liquid-phase mole fractions by the usual tridiagonal-matrix algorithm. Then, in the second major step, new sets of stage temperatures and total vapor flow rates leaving a stage are computed simultaneously by a Newton–Raphson method. These two major steps are repeated until a sum-of-squares criterion is satisfied. For this partial NR method:

- Write the two indexed equations you would use to simultaneously solve for a new set of T_j and V_j .
- Write the truncated Taylor series expansions for the two indexed equations in the T_j and V_j unknowns, and derive complete expressions for all partial derivatives, except that derivatives of physical properties with respect to temperature can be left as such. These derivatives are subject to the choice of physical property correlations.
- Order the resulting linear equations and the new variables ΔT_j and ΔV_j into a Jacobian matrix that will permit a rapid and efficient solution.

10.16 Revise equations (10-58) to (10-60) to allow two interlinked columns of the type shown in Figure 10.31 to be solved simultaneously by the NR method. Does the matrix equation that results from the Newton–Raphson procedure still have a block tridiagonal structure?

10.17 In Equation (10-63), why is the variable order selected as v, T, l ? What would be the consequence of changing the order to l, v, T ? In Equation (10-64), why is the function order selected as H, M, E ? What would be the consequence of changing the order to E, M, H ?

Section 10.5

10.18 Suggest in detail a method for determining the scalar multiplier, S_b , in (10-104).

10.19 Suggest in detail an error function, similar to (10-75), that could be used to determine convergence of the inner-loop calculations for the inside-out method.

Distillation Problems

10.20 Calculate product compositions, stage temperatures, interstage vapor and liquid flow rates and compositions, reboiler duty, and condenser duty for the following distillation-column specifications.

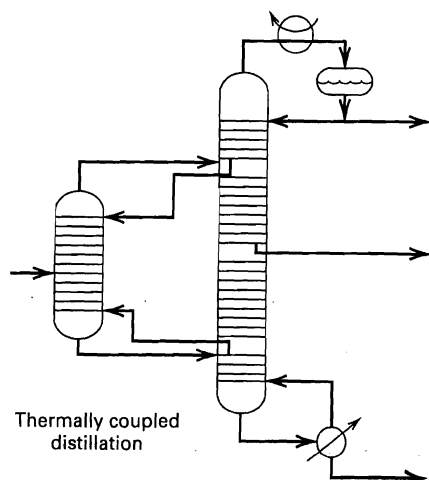


Figure 10.31 Data for Exercise 10.15.

Feed (bubble-point liquid at 250 psia and 213.9°F):

Component	Lbmol/h
Ethane	3.0
Propane	20.0
<i>n</i> -Butane	37.0
<i>n</i> -Pentane	35.0
<i>n</i> -Hexane	5.0

Column pressure = 250 psia

Partial condenser and partial reboiler

Distillate rate = 23.0 lbmol/h

Reflux rate = 150.0 lbmol/hr

Number of equilibrium stages (exclusive of condenser and reboiler) = 15

Feed is sent to middle stage

For this system at 250 psia, K -values and enthalpies may be computed by the Soave–Redlich–Kwong equations.

10.21 Determine the optimal feed stage location for Exercise 10.20.

10.22 Revise Exercise 10.20 so as to withdraw a vapor side stream at a rate of 37.0 lbmol/h from the fourth stage from the bottom.

10.23 Revise Exercise 10.20 so as to provide an intercondenser on the fourth stage from the top with a duty of 200,000 Btu/h and an interreboiler on the fourth stage from the bottom with a duty of 300,000 Btu/h.

10.24 Using the Peng–Robinson equations for thermodynamic properties, calculate the product compositions, stage temperatures, interstage vapor and liquid flow rates and compositions, reboiler duty, and condenser duty for the following multiple-feed distillation column, which has 30 equilibrium stages exclusive of a partial condenser and a partial reboiler and operates at 250 psia.

Feeds (both bubble-point liquids at 250 psia):

Component	Pound-moles per Hour	
	Feed 1 to Stage 15 from the Bottom	Feed 2 to Stage 6 from the Bottom
Ethane	1.5	0.5
Propane	24.0	10.0
<i>n</i> -Butane	16.5	22.0
<i>n</i> -Pentane	7.5	14.5
<i>n</i> -Hexane	0.5	3.0

Distillate rate = 36.0 lbmol/hr.

Reflux rate = 150.0 lbmol/hr.

Determine whether the feed locations are optimal.

10.25 Use the Chao–Seader or Grayson–Streed correlation for thermodynamic properties to calculate product compositions, stage temperatures, interstage flow rates and compositions, reboiler duty, and condenser duty for the distillation specifications in Figure 10.32.

Compare your results with those given in the *Chemical Engineers' Handbook*, Sixth Edition, pp. 13-42 to 13-45. Why do the two solutions differ?

10.26 Solve Exercise 10.11 using the UNIFAC method for K -values and obtain the converged solution.

10.27 Calculate with the Peng–Robinson equations for thermodynamic properties, the product compositions, stage temperatures,

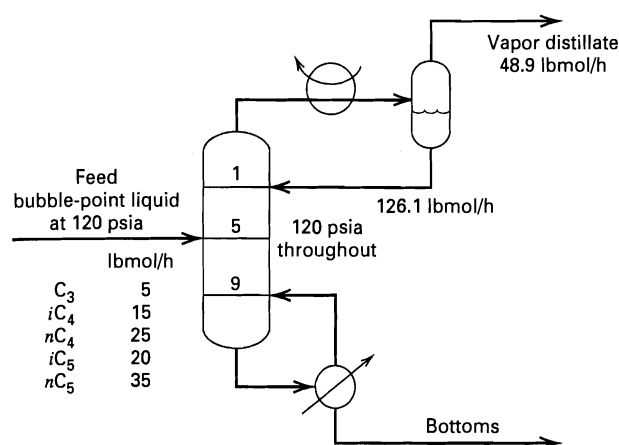


Figure 10.32 Data for Exercise 10.25.

interstage flow rates and compositions, reboiler duty, and condenser duty for the distillation specifications in Figure 10.33, which represent an attempt to obtain four nearly pure products from a single distillation operation. Reflux is a saturated liquid. Why is such a high reflux ratio required?

10.28 Repeat Exercise 10.25, but substitute the following specifications for the specifications of vapor distillate rate and reflux rate:

Recovery of nC_4 in distillate = 98%

Recovery of iC_5 in bottoms = 98%

If the calculations fail to converge, the number of stages may be less than the minimum value. If so, increase the number of stages, revise the feed location, and repeat until convergence is achieved.

10.29 A saturated liquid feed at 125 psia contains 200 lbmol/h of 5 mol% iC_4 , 20 mol% nC_4 , 35 mol% iC_5 , and 40 mol% nC_5 . This feed is to be distilled at 125 psia with a column equipped with a total condenser and partial reboiler. The distillate is to contain 95% of the nC_4 in the feed, and the bottoms is to contain 95% of the iC_5 in the feed. Use the SRK equation for thermodynamic properties to determine a suitable design. Twice the minimum number of equilibrium stages, as estimated by the Fenske equation in Chapter 9, should provide a reasonable number of equilibrium stages.

10.30 A depropanizer distillation column is designed to operate at an average total pressure of 315 psia for separating a feed into distillate and bottoms with the flow rates shown next:

	lbmol/h		
	Feed	Distillate	Bottoms
Methane (C_1)	26	26	
Ethane (C_2)	9	9	
Propane (C_3)	25	24.6	0.4
n -Butane (C_4)	17	0.3	16.7
n -Pentane (C_5)	11		11
n -Hexane (C_6)	12		12
Totals	100	59.9	40.1

The thermal condition of the feed is such that it is 66 mol% vapor at tower pressure. Steam at 315 psia and cooling water at 65°F are available for the reboiler and condenser. The total pressure drop across the column may be taken to be 2 psi as a first approximation.

(a) Should a total condenser be used for this column?

(b) What are the feed temperature, K -values, and relative volatilities (with reference to C_3) at the feed temperature and pressure?

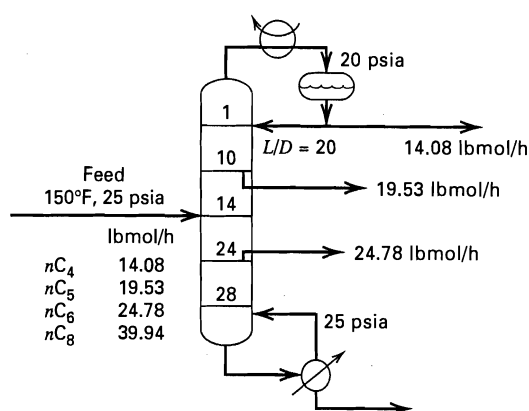


Figure 10.33 Data for Exercise 10.27.

(c) If the reflux ratio is 1.3 times the minimum reflux, what is the actual reflux ratio? How many theoretical plates are needed in the rectifying and stripping sections?

(d) Compute the separation of species. How will the separation differ, if a reflux ratio of 1.5, 15 theoretical plates, and feed at the 9th plate are chosen.

(e) For part (c), compute the temperature and concentrations on each stage. What is the effect of feed plate location? How will the results differ if a reflux ratio of 1.5 and 15 theoretical plates are used?

10.31 Toluene is to be separated from biphenyl by ordinary distillation. The specifications for the separation are as follows:

	lbmol/h		
	Feed	Distillate	Bottoms
Benzene	3.4		
Toluene	84.6		2.1
Biphenyl	5.1	1.0	

Temperature = 264°F; Pressure = 37.1 psia for the feed
Reflux ratio = 1.3 times minimum reflux with total condenser
Top pressure = 36 psia; bottom pressure = 38.2 psia

(a) Determine the actual reflux ratio and the number of theoretical trays in the rectifying and stripping sections.

(b) For a D/F ratio of $(3.4 + 82.5 + 1.0)/93.1$, compute the separation of species. Compare the results to the preceding specifications.

(c) If the separation of species computed in part (b) is not sufficiently close to the specified split, adjust the reflux ratio to achieve the specified toluene flow in the bottoms.

10.32 The following stream at 100°F and 480 psia is to be separated by two ordinary distillation columns into the indicated products.

	lbmol/h			
Species	Feed	Product 1	Product 2	Product 3
H_2	1.5	1.5		
CH_4	19.3	19.2	0.1	
C_6H_6 (benzene)	262.8	1.3	258.1	3.4
C_7H_8 (toluene)	84.7		0.1	84.6
$C_{12}H_{10}$ (biphenyl)	5.1			5.1

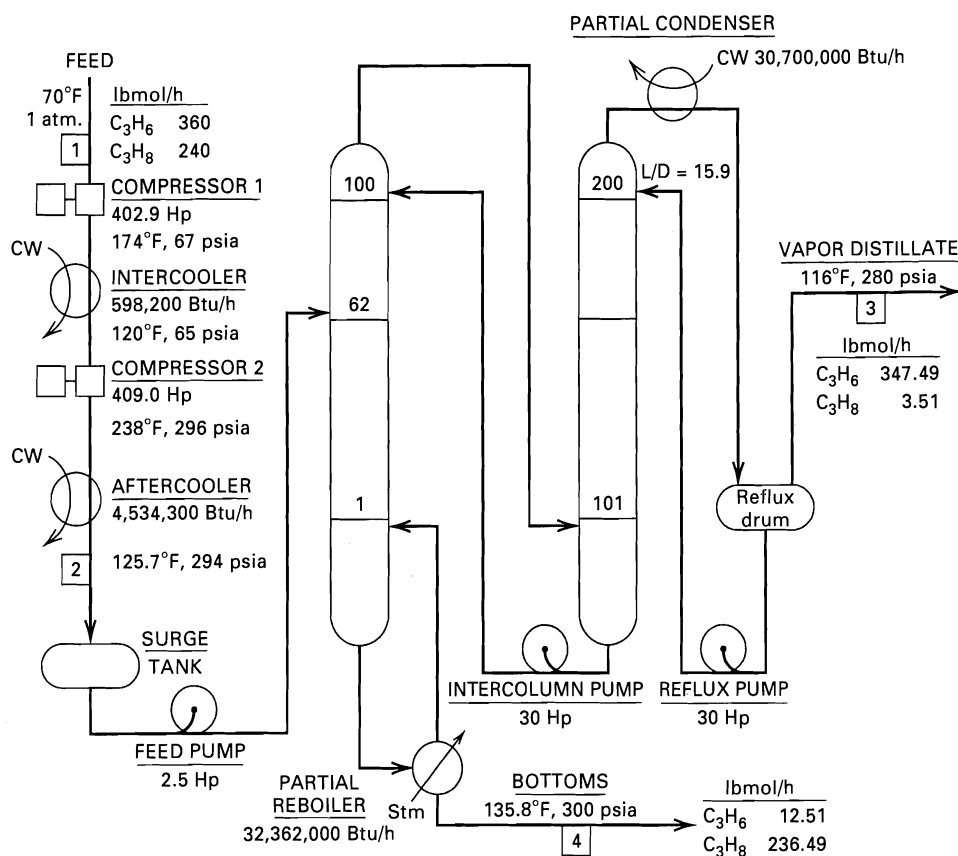


Figure 10.34 Data for Exercise 10.33.

Two different distillation sequences are to be examined. In the first sequence, CH_4 is removed in the first column. In the second sequence, toluene is removed in the first column. Compute the two sequences in the following manner: Estimate the actual reflux ratio and theoretical-tray requirements for both sequences. Specify a reflux ratio equal to 1.3 times the minimum. Adjust isobaric column pressures so as to obtain distillate temperatures of about 130°F ; however, no column pressure should be less than 20 psia. Specify total condensers, except that a partial condenser should be used when methane is taken overhead.

10.33 A process for the separation of a propylene–propane mixture to produce 99 mol% propylene and 95 mol% propane is shown in Figure 10.34. Because of the high product purities and the low relative volatility, 200 stages may be required. Assuming a tray efficiency of 100% and tray spacing of 24 inches, this will necessitate the two columns shown in series, because a single tower would be too tall. Assume a vapor distillate pressure of 280 psia, a pressure drop of 0.1 psi per tray, and a 2-psi drop through the condenser. The stage numbers and reflux ratio shown are only approximate. Determine the necessary reflux ratio for the stage numbers shown. Pay close attention to the determination of the proper feedstage location so as to avoid pinch or near-pinch conditions wherein several adjacent trays may not be accomplishing anything.

10.34 So-called stabilizers are distillation columns that are often used in the petroleum industry to perform relatively easy separations between light components and considerably heavier components when one or two single-stage flashes are inadequate. An example of a stabilizer is shown in Figure 10.35 for the separation of H_2 , methane, and ethane from benzene, toluene, and xylenes. Such

columns can be difficult to calculate because a purity specification for the vapor distillate cannot be readily determined. Instead, it is more likely that the designer will be told to provide a column with 20 to 30 trays and a water-cooled partial condenser to provide 100°F reflux at a rate that will provide sufficient boilup at the bottom of the column to meet the purity specification there. It is desired to more accurately design the stabilizer column. The number of theoretical stages shown are just a first approximation and may

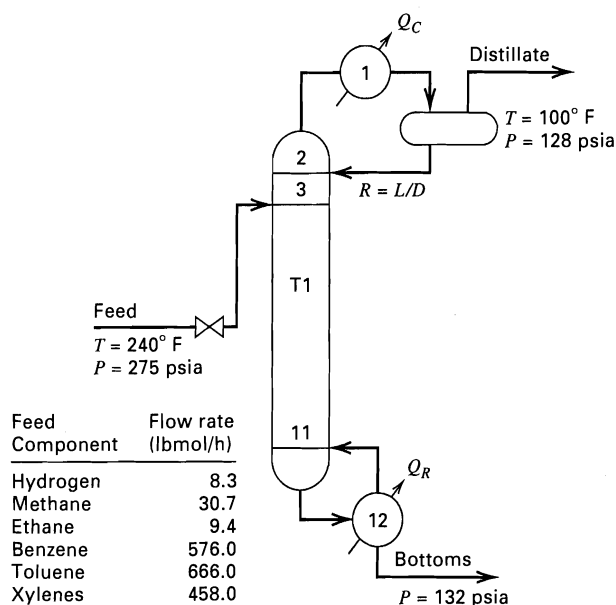


Figure 10.35 Data for Exercise 10.34.

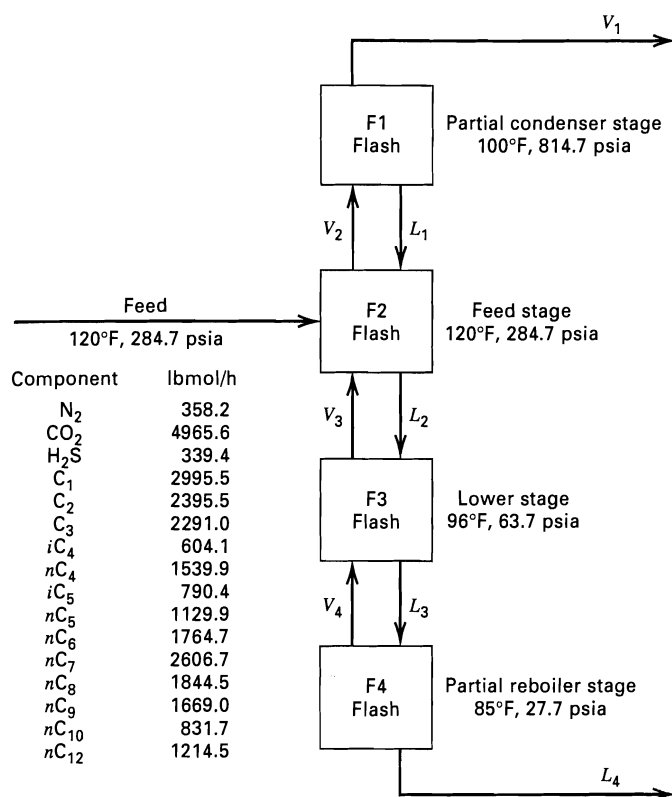


Figure 10.36 Data for Exercise 10.35.

be varied. Strive to achieve a bottoms product with no more than 0.05 mol% methane plus ethane and a vapor distillate temperature of about 100°F.

These specifications may be achieved by varying the distillate rate and the reflux ratio. Reasonable initial estimates for these two quantities are 49.4 lbmol/h and 2. Assume a tray efficiency of 70%.

10.35 A multiple recycle-loop problem formulated by Cavett¹ and shown in Figure 10.36 has been used extensively to test tearing, sequencing, and convergence procedures. The flowsheet is the equivalent of a four-theoretical-stage, near-isothermal distillation (rather than the conventional near-isobaric type), for which a patent by Gunther² exists. The flowsheet does not include necessary mixers, compressors, pumps, valves, and heat exchangers to make it a practical system. For the specifications shown on the drawing, determine the component flow rates for all streams in the process.

Absorber and Stripper Problems

10.36 An absorber is to be designed for a pressure of 75 psia to handle 2,000 lbmol/h of gas at 60°F having the following composition.

Component	Mole Fraction
Methane	0.830
Ethane	0.084
Propane	0.048
n-Butane	0.026
n-Pentane	0.012

¹R. H. Cavett, *Proc. Am. Petrol. Inst.* **43**, 57 (1963).

²A. Gunther, U.S. Patent 3,575,077 (April 13, 1971).

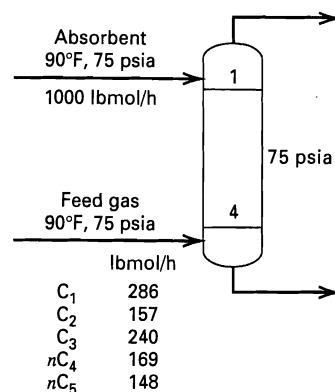


Figure 10.37 Data for Exercise 10.37.

The absorbent is an oil, which can be treated as a pure component having a molecular weight of 161. Calculate product rates and compositions, stage temperatures, and interstage vapor and liquid flow rate and compositions for the following conditions.

	Number of Equilibrium Stages	Entering Absorbent Flow Rate lbmol/h	Entering Absorbent Temperature, °F
(a)	6	500	90
(b)	12	500	90
(c)	6	1,000	90
(d)	6	500	60

10.37 Calculate product rates and compositions, stage temperatures, and interstage vapor and liquid flow rates and compositions for an absorber having four equilibrium stages with the specifications in Figure 10.37. Assume the oil is nC₁₀.

10.38 In Example 10.4, temperatures of the gas and oil, as they pass through the absorber, increase substantially. This limits the extent of absorption. Repeat the calculations with a heat exchanger that removes 500,000 Btu/h from:

- Stage 2.
- Stage 3.
- Stage 4.
- Stage 5.

How effective is the intercooler? Which stage is the preferred location for the intercooler? Should the duty of the intercooler be increased or decreased assuming that the minimum-stage temperature is 100°F using cooling water? Assume the absorber oil is nC₁₂.

10.39 Calculate product rates and compositions, stage temperatures, and interstage vapor and liquid flow rates and compositions for the absorber shown in Figure 10.38.

10.40 Determine product compositions, stage temperatures, interstage flow rates and compositions, and reboiler duty for the reboiled absorber shown in Figure 10.39. Repeat the calculations without the interreboiler and compare both sets of results. Is the interreboiler worthwhile? Should an intercooler in the top section of the column be considered?

10.41 Calculate the product compositions, stage temperatures, interstage flow rates and compositions, and reboiler duty for the reboiled stripper shown in Figure 10.40.

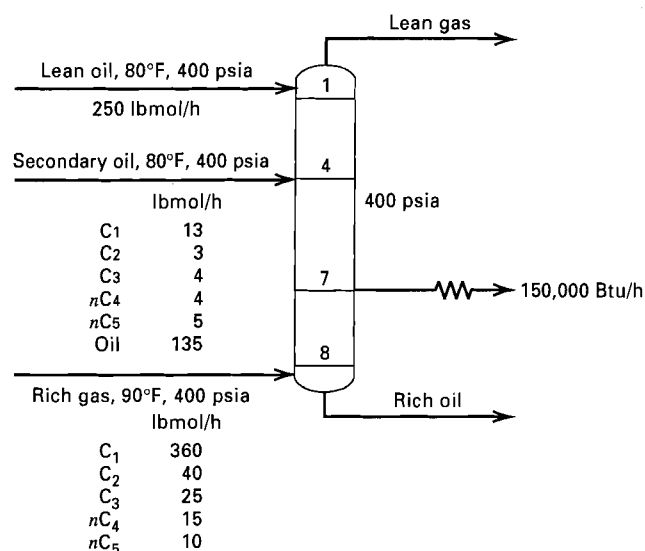


Figure 10.38 Data for Exercise 10.39.

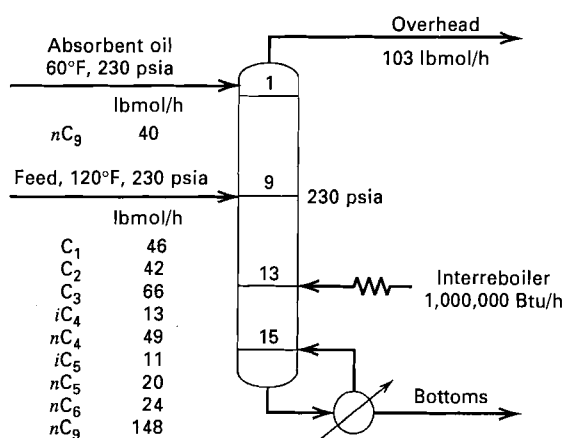


Figure 10.39 Data for Exercise 10.40.

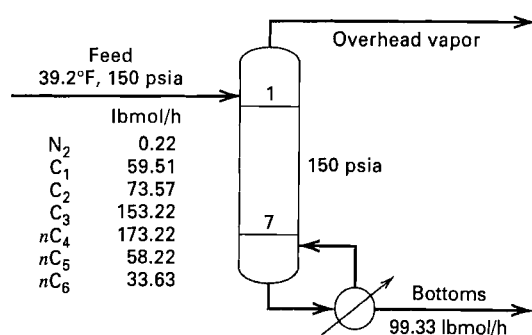


Figure 10.40 Data for Exercise 10.41.

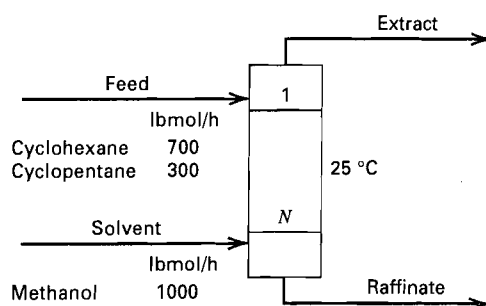


Figure 10.41 Data for Exercise 10.42.

Liquid-Liquid Extraction Problems

10.42 A mixture of cyclohexane and cyclopentane is to be separated by liquid-liquid extraction at 25°C with methanol. Phase equilibria for this system may be predicted by the NRTL or UNIQUAC equations. Calculate product rates and compositions and interstage flow rates and compositions for the conditions in Figure 10.41 with:

- $N = 1$ equilibrium stage.
- $N = 2$ equilibrium stages.
- $N = 5$ equilibrium stages.
- $N = 10$ equilibrium stages.

10.43 The liquid-liquid extractor in Figure 8.1 operates at 100°F and a nominal pressure of 15 psia. For the feed and solvent flows shown, determine the number of equilibrium stages to extract 99.5% of the acetic acid, using the NRTL equation for activity coefficients. The NRTL constants may be taken as follows:

- 1 = ethyl acetate
2 = water
3 = acetic acid

<i>I</i>	<i>J</i>	<i>B_{ij}</i>	<i>B_{ji}</i>	α_{ij}
1	2	166.36	1190.1	0.2
1	3	643.30	-702.57	0.2
2	3	-302.63	-1.683	0.2

Compare the computed compositions of the raffinate and extract products to those of Figure 8.1.

Enhanced Distillation and Supercritical Extraction

When two or more components differ in boiling point by less than approximately 50°C and form a nonideal-liquid solution, the relative volatility may be below 1.10. Then, separation by ordinary distillation may be uneconomical and if an azeotrope forms even impossible. In that event, the following separation techniques, referred to as *enhanced distillation* by Stichlmair, Fair, and Bravo [1], should be explored:

1. *Extractive Distillation:* A method that uses a large amount of a relatively high-boiling solvent to alter the liquid-phase activity coefficients of the mixture, so that the relative volatility of the key components becomes more favorable. The solvent enters the column above the feed entry and a few trays below the top, and exits from the bottom of the column without causing an azeotrope to be formed. If the feed to the column is an azeotrope, the solvent breaks it. Also, the solvent may reverse volatilities.
2. *Salt Distillation:* A variation of extractive distillation in which the relative volatility of the key components is altered by dissolving a soluble, ionic salt in the top reflux. Because the salt is nonvolatile, it stays in the liquid phase as it passes down the column.
3. *Pressure-Swing Distillation:* A method for separating a pressure-sensitive azeotrope that utilizes two columns operated in sequence at two different pressures.
4. *Homogeneous Azeotropic Distillation:* A method of separating a mixture by adding an entrainer that forms a homogeneous minimum- or maximum-boiling azeotrope with one or more feed components. The entrainer is added near the top of the column, to the feed, or near the bottom of the column, depending upon whether the azeotrope is removed from the top or bottom.
5. *Heterogeneous Azeotropic Distillation:* A more useful azeotropic-distillation method in which a minimum-boiling heterogeneous azeotrope is formed by the entrainer. The azeotrope splits into two liquid phases in the overhead condensing system. One liquid phase is sent back to the column as reflux, while the other liquid phase is sent to another separation step or is a product.
6. *Reactive Distillation:* A method that adds a separating agent to react selectively and reversibly with one or more of the constituents of the feed. The reaction product is subsequently distilled from the nonreacting components. The reaction is then reversed to recover the separating agent and the other reacting components. Reactive distillation also refers to the case where a chemical reaction and multistage distillation are conducted simultaneously in the same apparatus to produce other chemicals. This combined operation, sometimes referred to as *catalytic distillation* if a catalyst is used, is especially suited to chemical reactions limited by equilibrium constraints, since one (or more) of the products of the reaction is (are) continuously separated from the reactants.

For ordinary distillation of multicomponent mixtures, the determination of feasible distillation sequences, the design of the columns in the sequence by rigorous methods described in Chapters 10 and 12, and the optimization of the column operating conditions are tedious, but are relatively straightforward. In contrast, determining and optimizing feasible, enhanced-distillation sequences is a considerably more difficult task. In particular, rigorous calculations of enhanced distillation frequently fail because of liquid-solution nonidealities and/or the difficulty of specifying feasible separations. To significantly reduce the chances of failure, especially for ternary systems, graphical techniques, described by Partin [2] and developed largely by Doherty and co-workers, and by Stichlmair and co-workers, as referenced later, provide valuable guidance for the development of feasible, enhanced-distillation sequences prior to making rigorous calculations. This chapter presents an introduction to the principles of these graphical methods and applies them to enhanced distillation. Doherty and Malone [94], Stichlmair and Fair [95], and Siirola and Barnicki [96] give more detailed treatments of enhanced distillation.

Also included is a discussion of supercritical extraction, which differs considerably from conventional liquid-liquid extraction because of strong nonideal effects, and also requires considerable care in the development of an optimal

system. The principles and techniques in this chapter are largely restricted to ternary systems; enhanced distillation and supercritical extraction are most commonly applied to

such systems because the expense of these operations often requires that a multicomponent mixture first be reduced, by distillation or other means, to a binary or ternary system.

11.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Explain how enhanced-distillation methods work and how they differ from ordinary distillation.
- Explain how supercritical-fluid extraction differs from liquid-liquid extraction.
- Describe what residue-curve maps and distillation-curve maps represent on triangular diagrams for a ternary system.
- Explain how residue-curve maps and distillation-curve maps are constructed and under what conditions they are identical.
- Explain how residue-curve maps limit feasible product-composition regions in ordinary distillation and enhanced, ternary distillation.
- List requirements for an effective solvent in extractive distillation.
- Calculate, with a simulation program, a separation by extractive distillation.
- Explain how salt distillation differs from extractive distillation.
- Explain how pressure-swing distillation is used to separate a binary azeotropic mixture.
- Explain why it is difficult to find an entrainer that will permit use of homogeneous azeotropic distillation.
- Calculate, with a simulation program and a residue-curve map, a separation by homogeneous azeotropic distillation.
- List characteristics of an entrainer for heterogeneous azeotropic distillation.
- Calculate, with a simulation program, but using a residue-curve map and a bimodal curve, a separation by heterogeneous azeotropic distillation.
- List conditions necessary for carrying out reactive distillation.
- Calculate, with a simulation program, a separation by reactive distillation.
- Explain why enormous changes in properties can occur in the critical region.
- Calculate, with a simulation program, a separation by supercritical-fluid extraction.

11.1 USE OF TRIANGULAR GRAPHS

When a binary mixture at a given pressure is separated by continuous distillation in equilibrium stages, all possible equilibrium compositions are uniquely located on a vapor-liquid (y - x) equilibrium curve. Figure 11.1 shows typical isobaric vapor-liquid equilibrium curves in terms of the mole

fractions of the lowest-boiling component, A. In Figure 11.1a, possible compositions of the distillate and bottoms cover the entire range from pure B to pure A for a *zeotropic* (nonazeotropic) system. Temperatures, although not shown on the isobaric equilibrium curve, range from the boiling point of A to the boiling point of B. As the liquid and vapor

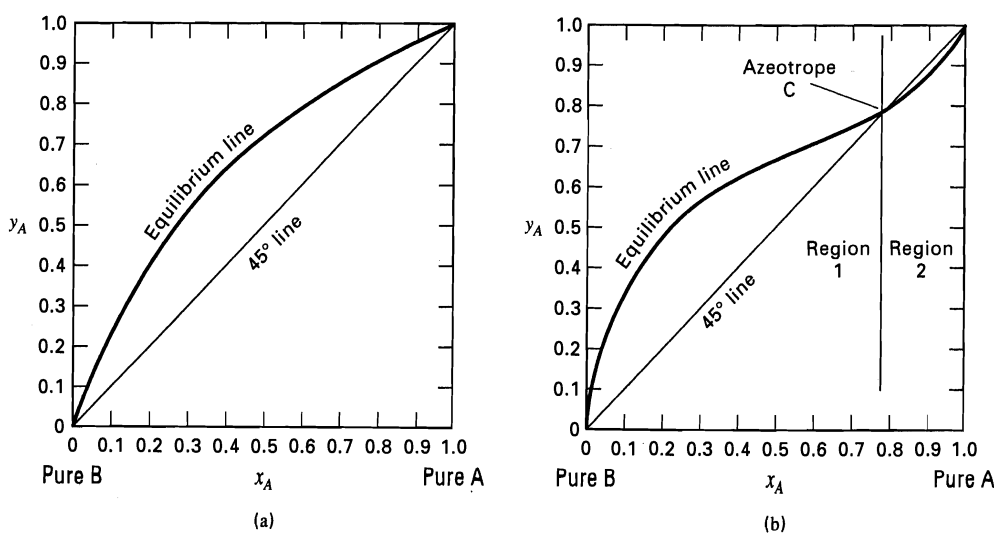


Figure 11.1 Vapor-liquid equilibria for binary systems. (a) Zeotropic system. (b) Azeotropic system.

compositions change from pure B to pure A, the temperature decreases.

In Figure 11.1b, a minimum-boiling azeotrope is formed at C and divides the plot into two regions. For distillation in Region 1, distillate and bottoms compositions can only vary from pure B to azeotrope C, and in Region 2, only from pure A to azeotrope C. For Region 1, as the composition changes from pure B to azeotrope C, the temperature decreases, as shown for example in Figure 4.6, where B is isopropyl alcohol, A is isopropyl ether, and the minimum-boiling azeotrope occurs at 78 mol% isopropyl ether and 66°C at 1 atm. In Region 2, the temperature also decreases as the composition changes from pure A to azeotrope C. Thus, a single distillation column operating at 1 atm cannot separate the mixture into two nearly pure products. Depending upon whether the feed composition lies in Region 1 or 2, the column, at best, can only produce a distillate of azeotrope C and a bottoms of either pure B or pure A. However, all possible equilibrium compositions still lie on the equilibrium curve. These results are consistent with the Gibbs phase rule, as discussed in Chapter 4. From (4-1), for two components and two phases, that rule gives two degrees of freedom. Thus, if the pressure and temperature are fixed, the equilibrium vapor and liquid compositions are fixed. However, as shown in Figure 11.2, for the case of an azeotrope-forming binary mixture, two feasible solutions exist within a certain temperature range. The particular solution observed depends on the overall composition of the two phases.

In the distillation of a ternary system, possible equilibrium compositions do not lie uniquely on a single, isobaric equilibrium curve because the Gibbs phase rule gives an additional degree of freedom. The other compositions are determined only if the temperature, pressure, and composition of one component in one phase are fixed.

As discussed in Chapters 4 and 8, the composition of a ternary mixture can be represented on a triangular diagram,

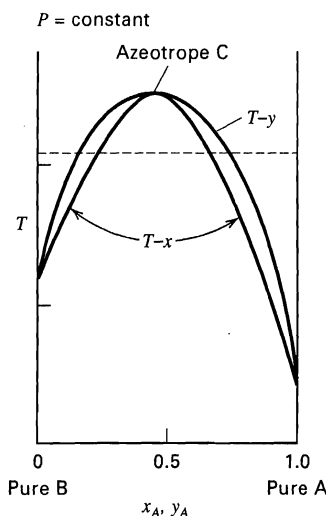


Figure 11.2 Multiple equilibrium solutions for an azeotropic system.

either equilateral or right, where the three apexes of the triangle represent the pure components. Although Stichlmair [3] shows that vapor-liquid phase equilibria at a fixed pressure can be plotted by letting the triangular grid represent the liquid phase and superimposing lines of constant equilibrium-vapor composition for two of the three components, this representation is seldom used. It is more useful, when developing a feasible-separation process, to plot only equilibrium-liquid-phase compositions on the triangular diagram. Typical plots of this type at 1 atm, for three different ternary systems, are shown in Figure 11.3, where compositions are in mole fractions. Each curve in each diagram is the locus of possible equilibrium liquid-phase compositions that occur during distillation of a mixture, starting from any point on the curve. The boiling points of the three components and their binary and/or ternary azeotropes are included on the diagrams. The zeotropic alcohol system of Figure 11.3a does not form any azeotropes. If a mixture of these three alcohols is distilled, there is only one distillation region, similar to the binary system of Figure 11.1a. Accordingly, the distillate product can be nearly pure methanol (A) or the bottoms product can be nearly pure 1-propanol (C). However, nearly pure ethanol (B), the intermediate-boiling component, cannot be produced either as a distillate or bottoms. To separate this ternary mixture into the three components, a sequence of two ordinary-distillation columns is used, as shown in Figure 11.4, where the feed, distillate, and bottoms product compositions must lie on a straight (total material-balance) line within the triangular diagram. Thus, in the so-called *direct sequence* of Figure 11.4a, the feed, F, is first separated into distillate A and a bottoms of B and C; then B is separated from C in the second column. In the *indirect sequence* of Figure 11.4b, a distillate of A and B and a bottoms of C is produced in the first column, followed by the separation of A from B in the second column.

When a ternary mixture forms an azeotrope, the possible products from a single ordinary distillation column depend on the feed composition, as for a binary mixture. However, unlike the case of the binary mixture, where two distillation regions, shown in Figure 11.1b, are simple and well defined, the determination of possible distillation regions for azeotrope-forming ternary mixtures is complex. Consider first the example of Figure 11.3b, for a mixture of acetone (A), methanol (B), and ethanol (C), which are in the order of increasing boiling point. The only azeotrope formed at 1 atm is a minimum-boiling binary azeotrope, at 55.7°C, of the two lower-boiling components, acetone and methanol. The azeotrope contains 78.4 mol% acetone. For this type of system, as will be shown later, no distillation boundaries for the ternary mixture exist, even though an azeotrope is present. Thus, a feed composition located within the triangular diagram can be separated into two binary products, consistent with the straight (total material-balance) line. That is, ternary distillate or bottoms products can be avoided if the column split is properly selected. For example, the following five feed compositions can all produce, at a high reflux ratio and for a

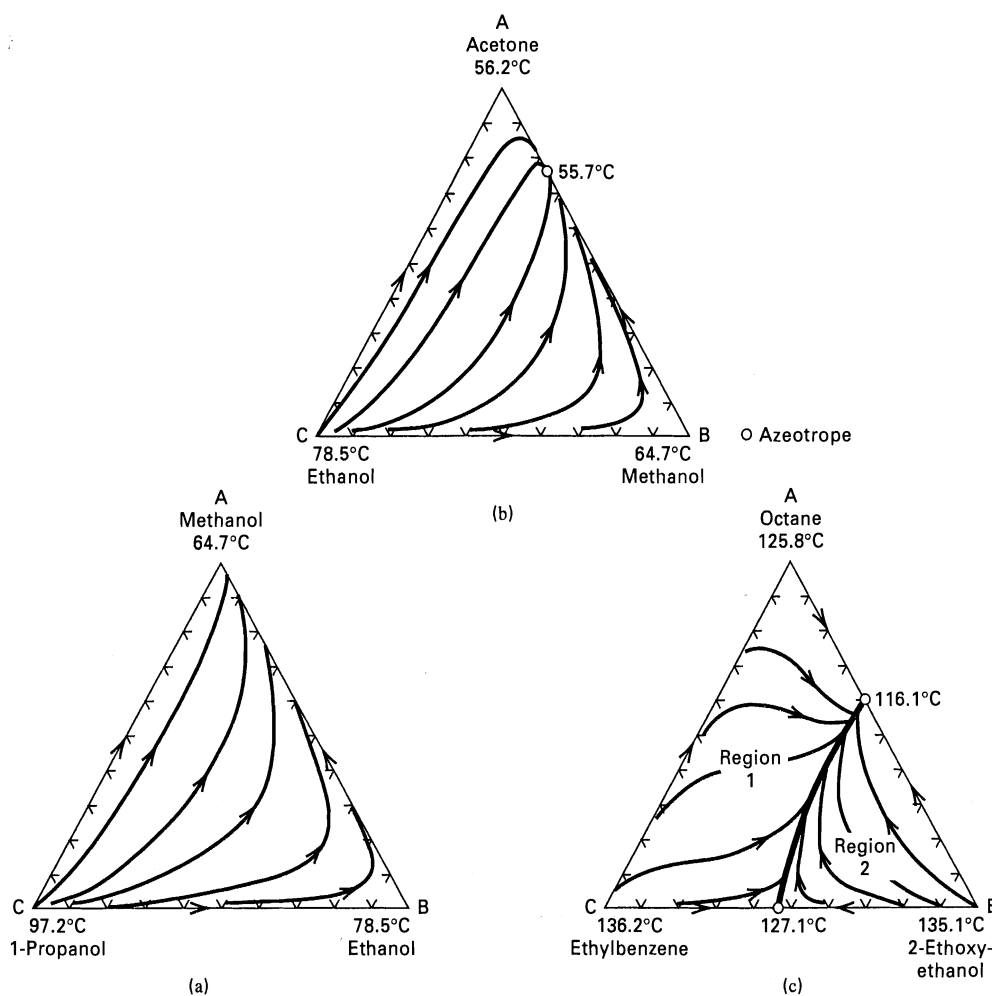


Figure 11.3 Distillation curves for liquid-phase compositions of ternary systems at 1 atm. (a) Mixture not forming an azeotrope. (b) Mixture forming one minimum-boiling azeotrope. (c) Mixture forming two minimum-boiling azeotropes.

large number of stages, a distillate of the minimum-boiling azeotrope of acetone and methanol, and a bottoms product containing methanol and ethanol. That is, little or no ethanol will be in the distillate and little or no acetone in the bottoms.

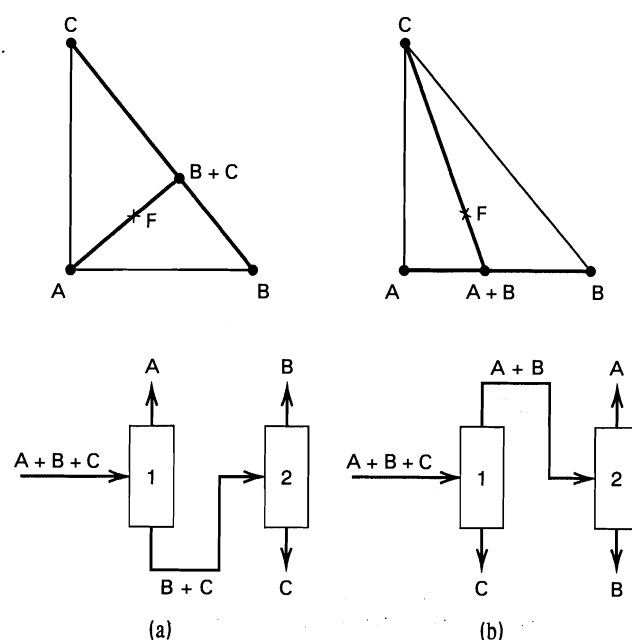


Figure 11.4 Distillation sequences for ternary zeotropic mixtures. (a) Direct sequence. (b) Indirect sequence.

Case	Feed:		Distillate:		Bottoms:	
	x_{acetone}	x_{methanol}	x_{acetone}	x_{methanol}	x_{acetone}	x_{methanol}
1	0.1667	0.1667	0.7842	0.2158	0.0000	0.1534
2	0.1250	0.3750	0.7837	0.2163	0.0000	0.4051
3	0.2500	0.2500	0.7837	0.2163	0.0000	0.2658
4	0.3750	0.1250	0.7837	0.2163	0.0000	0.0412
5	0.3333	0.3333	0.7837	0.2163	0.0000	0.4200

Alternatively, the column split can be selected to obtain a bottoms of nearly pure ethanol and a distillate of acetone and methanol. For either split, the straight, total material-balance line passing through the feed point can extend to the sides of the triangle.

Next, consider the more complex case of the ternary mixture of *n*-octane (A), 2-ethoxyethanol (B), and ethylbenzene (C), shown in Figure 11.3c. A and B form a minimum-boiling binary azeotrope at 116.1°C, and B and C do the same at 127.1°C. A triangular diagram for this type of system is separated by a *distillation boundary* (shown as a bold curved line) into regions 1 and 2. A material-balance line connecting the feed to the distillate and bottoms cannot cross this distillation boundary, thus restricting the possible products of ordinary distillation of the ternary feed mixture. For example, a mixture with a feed composition inside Region 2 cannot produce a bottoms of ethylbenzene, the highest-boiling component in the mixture. It can be distilled to produce a distillate of the A–B azeotrope and a bottoms of a

mixture of B and C, or a bottoms of B and a distillate of all three components. If the feed composition lies in Region 1 of Figure 11.3c, ordinary distillation can produce the A–B azeotrope and a bottoms of a mixture of A and C, or a bottoms of C and a distillate of a mixture of A and B. Thus, each region produces unique products.

To further illustrate the restriction in product compositions caused by a distillation boundary, consider ordinary distillation with a feed mixture of 15 mol% A, 70 mol% B, and 15 mol% C. If the mixture is that in Figure 11.3a or b, a bottoms product of nearly pure C, the highest-boiling component, is obtained with a feed split corresponding to a distillate-to-bottoms ratio of 85/15. If, however, the mixture is that in Figure 11.3c, the same feed split ratio results in a bottoms of nearly pure B, the second-highest-boiling component.

Thus, products of a ternary mixture cannot be predicted merely from the boiling points of the components and azeotropes and a specified distillate-to-bottoms molar ratio, when distillation boundaries are present. These boundaries, as well as the mappings of distillation curves in the ternary plots of Figure 11.3, can be determined by either of the methods described in the next two sections.

Residue-Curve Maps

Consider the simple Rayleigh batch or differential distillation (no trays, packing, or reflux) shown schematically in Figure 13.1. For any component of a ternary mixture, a material balance for its vaporization from the liquid in the still, assuming that the liquid is perfectly mixed and at its bubble-point temperature, is given by (13-1), which can be written as

$$\frac{dx_i}{dt} = (y_i - x_i) \frac{dW}{Wdt} \quad (11-1)$$

where

x_i = mole fraction of component i in W moles of perfectly mixed liquid residue in the still

y_i = mole fraction of component i in the vapor (instantaneous distillate) in equilibrium with x_i

Because W changes (decreases) with time, t , it is possible to combine W and t into a single variable. Following the treatment of Doherty and Perkins [4], let this variable be ξ , such that

$$\frac{dx_i}{d\xi} = x_i - y_i \quad (11-2)$$

Combining (11-1) and (11-2) to eliminate $dx_i/(x_i - y_i)$:

$$\frac{d\xi}{dt} = -\frac{1}{W} \frac{dW}{dt} \quad (11-3)$$

Let the initial condition be $x = 0$ and $W = W_0$ at $t = 0$. Then the solution to (11-3) for ξ at time t is

$$\xi\{t\} = \ln[W_0/W\{t\}] \quad (11-4)$$

Because $W\{t\}$ decreases monotonically with time, $\xi\{t\}$ must increase monotonically with time and is considered a dimensionless, warped time. Thus, for the ternary mixture, the simple distillation process can be modeled by the following set of differential-algebraic equations (DAEs), assuming

that a second liquid phase does not form:

$$\frac{dx_i}{d\xi} = x_i - y_i, \quad i = 1, 2 \quad (11-5)$$

$$\sum_{i=1}^3 x_i = 1 \quad (11-6)$$

$$y_i = K_i x_i, \quad i = 1, 2, 3 \quad (11-7)$$

and the bubble-point-temperature equation:

$$\sum_{i=1}^3 K_i x_i = 1 \quad (11-8)$$

where, in the general case, $K_i = K_i\{T, P, \mathbf{x}, \mathbf{y}\}$.

Thus, the system consists of seven equations in nine variables: $P, T, x_1, x_2, x_3, y_1, y_2, y_3$, and ξ . If the pressure is fixed, the next seven variables can be computed from (11-5) to (11-8) as a function of ξ , from a specified initial condition. The calculations can proceed in either the forward or backward direction of ξ . The results, when plotted on a triangular graph, are called a *residue curve* because the plot follows, with time, the liquid-residue composition in the still. A collection of residue curves, for a given ternary system at a fixed pressure, is a *residue-curve map*. A simple, but inefficient, procedure for calculating a residue curve is illustrated in the following example. Better, but more elaborate, procedures are given by Doherty and Perkins [4] and Bossen, Jørgensen, and Gani [5]. The last procedure is also applicable when two separate liquid phases form, as is a procedure by Pham and Doherty [6].

EXAMPLE 11.1

Calculate and plot a portion of a residue curve for the ternary system, n -propanol (1), isopropanol (2), and benzene (3) at 1 atm, starting from a bubble-point liquid with a composition of 20 mol% each of 1 and 2, and 60 mol% of component 3. For K -values, use the modified Raoult's law (see Table 2.3) with regular-solution theory [see (2-64)] for estimating the liquid-phase activity coefficient as a function of composition and temperature. The normal boiling points of the three components in $^{\circ}\text{C}$ are 97.3, 82.3, and 80.1, respectively. Minimum-boiling azeotropes are formed at 77.1 $^{\circ}\text{C}$ for components 1,3 and at 71.7 $^{\circ}\text{C}$ for 2,3.

SOLUTION

A bubble-point calculation, using (11-7) and (11-8), gives starting values of y of 0.1437, 0.2154, and 0.6409, respectively, and a value of 79.07 $^{\circ}\text{C}$ for the starting temperature, from the ChemSep program of Taylor and Kooijman [7].

For a specified increment in the dimensionless time, ξ , the differential equations (11-5) can be solved for x_1 and x_2 using Euler's method with a spreadsheet. Then x_3 is obtained from (11-6). The corresponding values of y and T are then obtained from (11-7) and (11-8). This procedure is repeated for the next increment in ξ . Thus, from (11-5), for component 1:

$$\begin{aligned} x_1^{(1)} &= x_1^{(0)} + (x_1^{(0)} - y_1^{(0)})\Delta\xi \\ &= 0.2000 + (0.2000 - 0.1437)0.1 = 0.2056 \end{aligned}$$

where superscripts (0) indicate starting values and a superscript of (1) indicates the value after the first increment in ξ . The value of

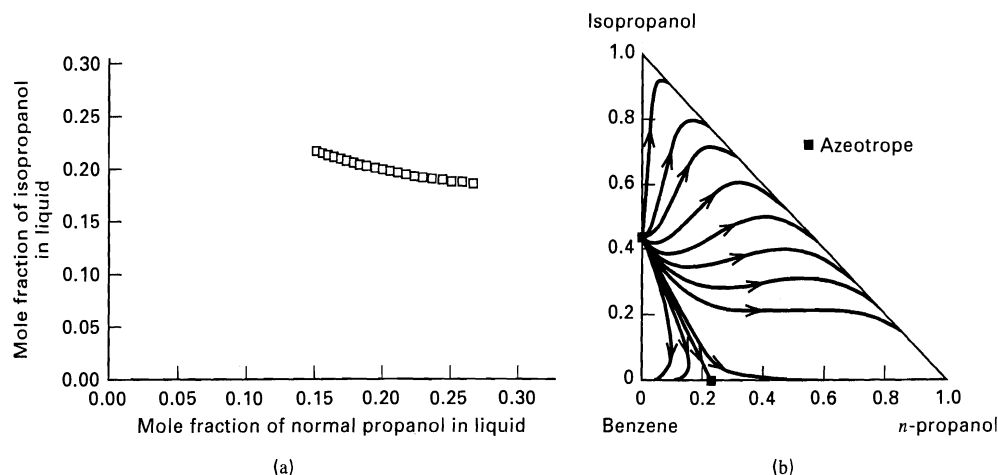


Figure 11.5 Residue curves for the normal propanol–isopropanol–benzene system at 1 atm for Example 11.1. (a) Calculated partial residue curve. (b) Residue-curve map.

0.1 for $\Delta\xi$ gives reasonable accuracy, since the change in x_1 is seen to be only 2.7%. Similarly:

$$x_2^{(1)} = 0.2000 + (0.2000 - 0.2154)0.1 = 0.1985$$

From (11-6):

$$x_3^{(1)} = 1 - x_1^{(1)} - x_2^{(1)} = 1 - 0.2056 - 0.1985 = 0.5959$$

From a bubble-point calculation using (11-7) and (11-8), from ChemSep,

$$y^{(1)} = [0.1474, 0.2134, 0.6392]^T \quad \text{and} \quad T^{(1)} = 79.14^\circ\text{C}$$

The calculations are continued in the forward direction of ξ to $\xi = 1.0$. The calculations are also carried out in the backward direction back to $\xi = -1.0$. The results are in the table below, and the partial residue curve is plotted in Figure 11.5a. For comparison, the complete residue-curve map for this system, from Doherty [8], is given on a right-triangle diagram in Figure 11.5b.

ξ	x_1	x_2	y_1	y_2	$T, ^\circ\text{C}$
-1.0	0.1515	0.2173	0.1112	0.2367	78.67
-0.9	0.1557	0.2154	0.1141	0.2344	78.71
-0.8	0.1600	0.2135	0.1171	0.2322	78.75
-0.7	0.1644	0.2117	0.1201	0.2300	78.79
-0.6	0.1690	0.2099	0.1232	0.2278	78.83
-0.5	0.1737	0.2081	0.1264	0.2256	78.87
-0.4	0.1786	0.2064	0.1297	0.2235	78.91
-0.3	0.1837	0.2047	0.1331	0.2214	78.95
-0.2	0.1889	0.2031	0.1365	0.2194	79.00
-0.1	0.1944	0.2015	0.1401	0.2173	79.05
0.0	0.2000	0.2000	0.1437	0.2154	79.07
0.1	0.2056	0.1985	0.1474	0.2134	79.14
0.2	0.2115	0.1970	0.1512	0.2115	79.19
0.3	0.2175	0.1955	0.1550	0.2095	79.24
0.4	0.2237	0.1941	0.1589	0.2076	79.30
0.5	0.2302	0.1928	0.1629	0.2058	79.24
0.6	0.2369	0.1915	0.1671	0.2041	79.41
0.7	0.2439	0.1902	0.1714	0.2023	79.48
0.8	0.2512	0.1890	0.1758	0.2006	79.54
0.9	0.2587	0.1878	0.1804	0.1989	79.61
1.0	0.2665	0.1867	0.1850	0.1973	79.68

The residue-curve map in Figure 11.5b shows residue curves with arrows. The curves include the three border sides of the triangular diagram. The arrow on each curve points from a lower-

boiling component or azeotrope to a higher-boiling component or azeotrope. In Figure 11.5b, all residue curves of the ternary mixture originate from the isopropanol–benzene azeotrope (lowest boiling point of 71.7°C). One of the curves terminates at the other azeotrope (n -propanol–benzene, which has a higher boiling point of 77.1°C). This is a special residue curve, called a *simple distillation boundary* because it divides the ternary region into two separate distillation regions. All residue curves lying above and to the right of the distillation boundary terminate at the n -propanol apex, which has the highest boiling point (97.3°C) for that region. All residue curves lying below and to the left of the distillation boundary are deflected to the benzene apex, whose boiling point of 80.1°C is the highest for this second region.

On the triangular diagram, all pure-component vertices and azeotropic points, whether binary azeotropes on the borders of the triangle, as in Figure 11.5b, or a ternary azeotrope within the triangle, are singular or fixed points of the residue curves because at these points, $dx/d\xi = 0$. In the vicinity of these points, the behavior of a residue curve depends upon the two eigenvalues of (11-5). At each pure-component vertex, the two eigenvalues are identical. At each azeotropic point, the two eigenvalues are different. Three cases, illustrated by each of three pattern groups in Figure 11.6, are possible:

Case 1: Both eigenvalues are negative. This is the point reached as ξ tends to infinity. It is the point at which all residue curves in a given region terminate. Thus, it is the component or azeotrope with the highest boiling point in the region. This point is a *stable node* because it is like the low point of a valley, in which a rolling ball finds a stable position. In Figure 11.6b, the stable node is pure n -propanol.

Case 2: Both eigenvalues are positive. This is the point from which all residue curves in a given region originate. Thus, it is the component or azeotrope with the lowest boiling point in the region. This point is an *unstable node* because it is like the top of a peaked mountain from which a ball rolls toward a stable position. In Figure 11.6b, the unstable node is the isopropanol–benzene azeotrope.

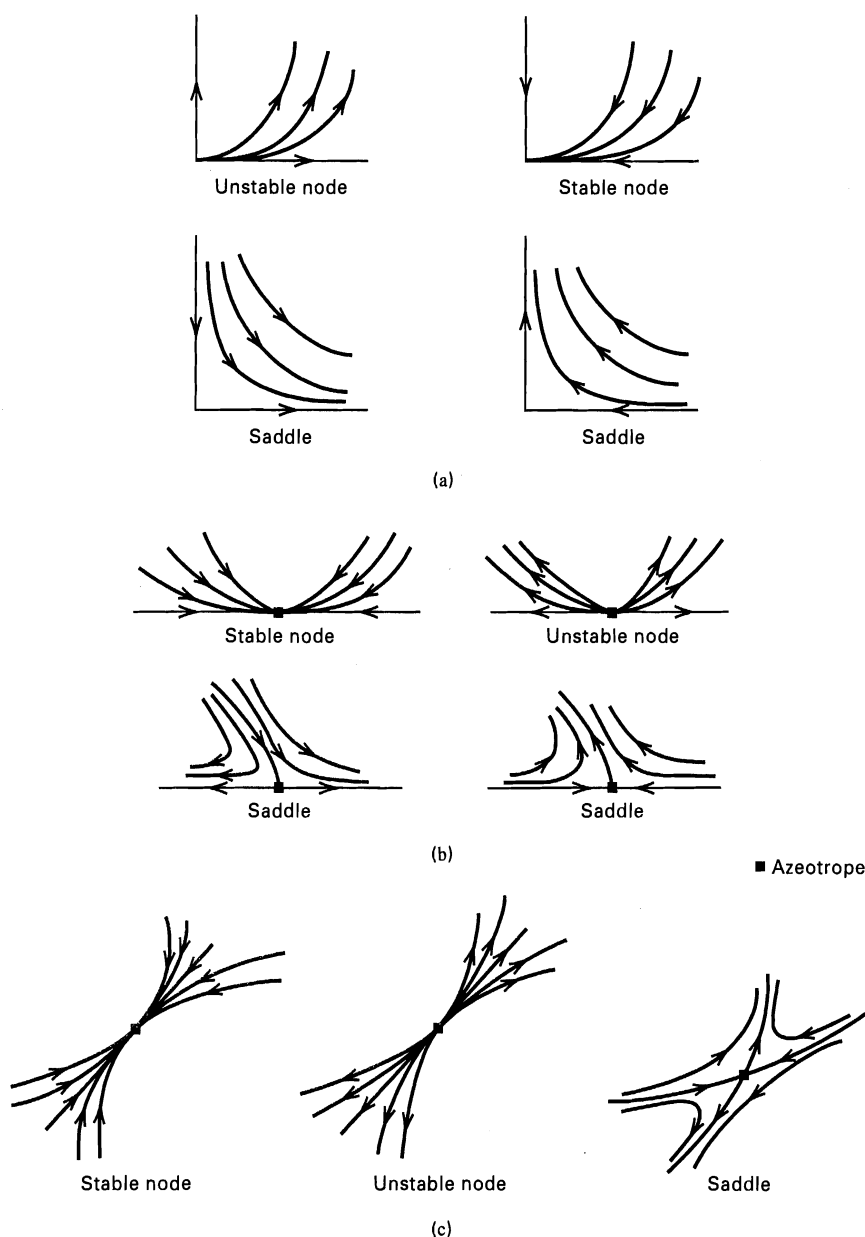


Figure 11.6 Residue-curve patterns (a) near pure-component vertices; (b) near binary azeotropes; (c) near ternary azeotropes.

[From M.F. Doherty and G.A. Caldarola, *IEC Fundam.*, 24, 477 (1985) with permission.]

Case 3: One eigenvalue is positive and one is negative.

Residue curves within the triangle move toward and then away from such points, which are *saddles*. For a given distillation region, all pure components and azeotropes intermediate in boiling point between the stable node and the unstable node are saddles. In Figure 11.5b, the upper distillation region has one saddle at the isopropanol vertex and another saddle at the normal propanol–benzene azeotrope.

From Example 11.1, it is clear that calculation of a residue-curve map requires a considerable effort. However, computer-aided simulation programs such as ASPEN PLUS [9] and CHEMCAD compute residue maps. Alternatively, as developed by Doherty and Perkins [10] and Doherty [8], the classification of singular points as stable nodes, unstable nodes, and saddles provides a rapid method for approximating a residue-curve map, including approximate distillation boundaries, from just the pure-component boiling points and

azeotrope boiling points and compositions. Boiling points of pure components are readily found in handbooks and component data banks of computer-aided simulation programs. Extensive listings of binary azeotropes are found in Horsley [11] and Gmehling et al. [12]. The former lists more than 1,000 binary azeotropes. The latter includes experimental data for more than 20,000 systems involving approximately 2,000 compounds, as well as material on selecting enhanced distillation systems. The listings of ternary azeotropes are undoubtedly quite incomplete. However, in lieu of experimental data, a homotopy-continuation method for estimating all homogeneous azeotropes of a multicomponent mixture from a thermodynamic model (e.g., Wilson, NRTL, UNIQUAC, UNIFAC) has been developed by Fidkowski, Malone, and Doherty [13]. Eckert and Kubicek [97] present an extension for computing heterogeneous azeotropes.

Based on experimental evidence, for ternary mixtures, with very few exceptions, there are at most three binary

azeotropes and one ternary azeotrope. Accordingly, the following set of restrictions apply to a ternary system:

$$N_1 + S_1 = 3 \quad (11-9)$$

$$N_2 + S_2 = B \leq 3 \quad (11-10)$$

$$N_3 + S_3 = 1 \text{ or } 0 \quad (11-11)$$

where N is the number of stable and unstable nodes, S is the number of saddles, B is the number of binary azeotropes, and the subscript is the number of components at the node (stable or unstable) or saddle. Thus, S_2 is the number of binary azeotrope saddles. Doherty and Perkins [10] developed the following topological relationship among N and S :

$$2N_3 - 2S_3 + 2N_2 - B + N_1 = 2 \quad (11-12)$$

For the system of Figure 11.5b, which has no ternary azeotrope, we see that $N_1 = 2$, $N_2 = 1$, $N_3 = 0$, $S_1 = 1$, $S_2 = 1$, $S_3 = 0$, and $B = 2$. Applying (11-12) gives $0 - 0 + 2 - 2 + 2 = 2$. Equation (11-9) gives $2 + 1 = 3$, (11-10) gives $1 + 1 = 2$, and (11-11) gives $0 + 0 = 0$. Thus, all four relations are satisfied.

The topological relationships are especially useful for rapidly sketching, on a ternary diagram, an approximate residue-curve map, including distillation boundaries, as described in detail by Foucher, Doherty, and Malone [14]. Their procedure involves the following nine steps (0–8), which are partly illustrated by a hypothetical example taken from their article and shown in Figure 11.7. The procedure is summarized in Figure 11.8. Approximate maps are usually developed from data at 1 atm. In the description of the steps, the term *species* refers to pure components and azeotropes.

Step 0: Label the ternary diagram with the pure-component, normal-boiling-point temperatures. It is preferable to designate the top vertex of the triangle as the low boiler (L), the bottom-right vertex as the high boiler (H), and the bottom-left vertex as the intermediate boiler (I). Plot composition points for the binary and ternary azeotropes and add labels for their normal boiling points. This determines the value of B . See Figure 11.7, Step 0, where two minimum-boiling and one maximum-boiling binary azeotropes and one ternary azeotrope are designated by filled square markers. Thus, $B = 3$.

Step 1: Draw arrows on the edges of the triangle, in the direction of increasing temperature, for each pair of adjacent species. See Figure 11.7, Step 1, where there are six species on the edges of the triangle and six arrows have been added.

Step 2: Determine the type of singular point for each pure-component vertex, by using Figure 11.6 with the arrows drawn in Step 1. This determines the values for N_1 and S_1 . If a ternary azeotrope exists, go to Step 3; if not, go to Step 5. In Figure 11.7, Step 2, L is a saddle because one arrow points toward L and one points away from L; H is a stable node because both arrows point toward H, and I is a saddle. Therefore, $N_1 = 1$ and $S_1 = 2$.

Step 3 (for a ternary azeotrope): Determine the type of singular point for the ternary azeotrope, if one exists. The point is a node if (a) $N_1 + B < 4$, and/or (b) excluding the pure-component saddles, the ternary azeotrope has the highest, second-highest, lowest, or second-lowest boiling point of all species. Otherwise, the point is a saddle. This determines the values for N_3 and S_3 . If the point is a node, go to Step 5; if a saddle, go to Step 4. In Figure 11.7, Step 3, $N_1 + B = 1 + 3 = 4$.

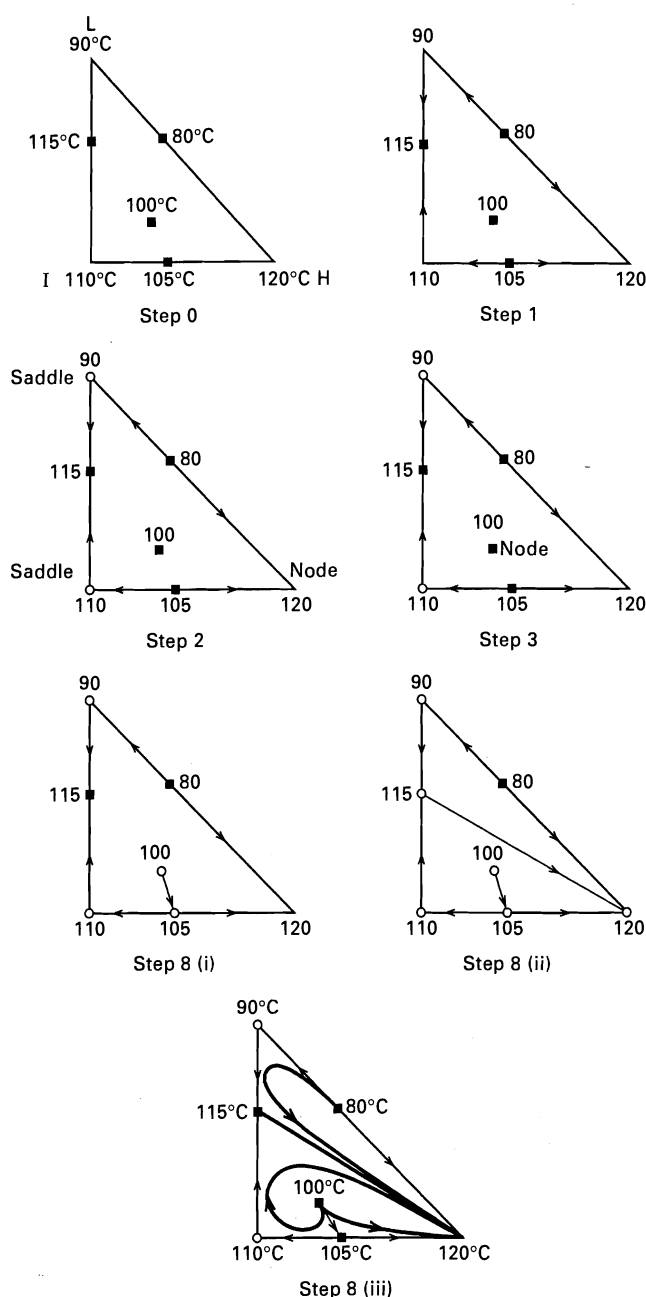


Figure 11.7 Step-by-step development of an approximate residue-curve map for a hypothetical system with two minimum-boiling binary azeotropes, one maximum-boiling binary azeotrope and one ternary azeotrope.

[From E.R. Foucher, M.F. Doherty, and M.F. Malone, *IEC Res.* 30, 764 (1991) with permission.]

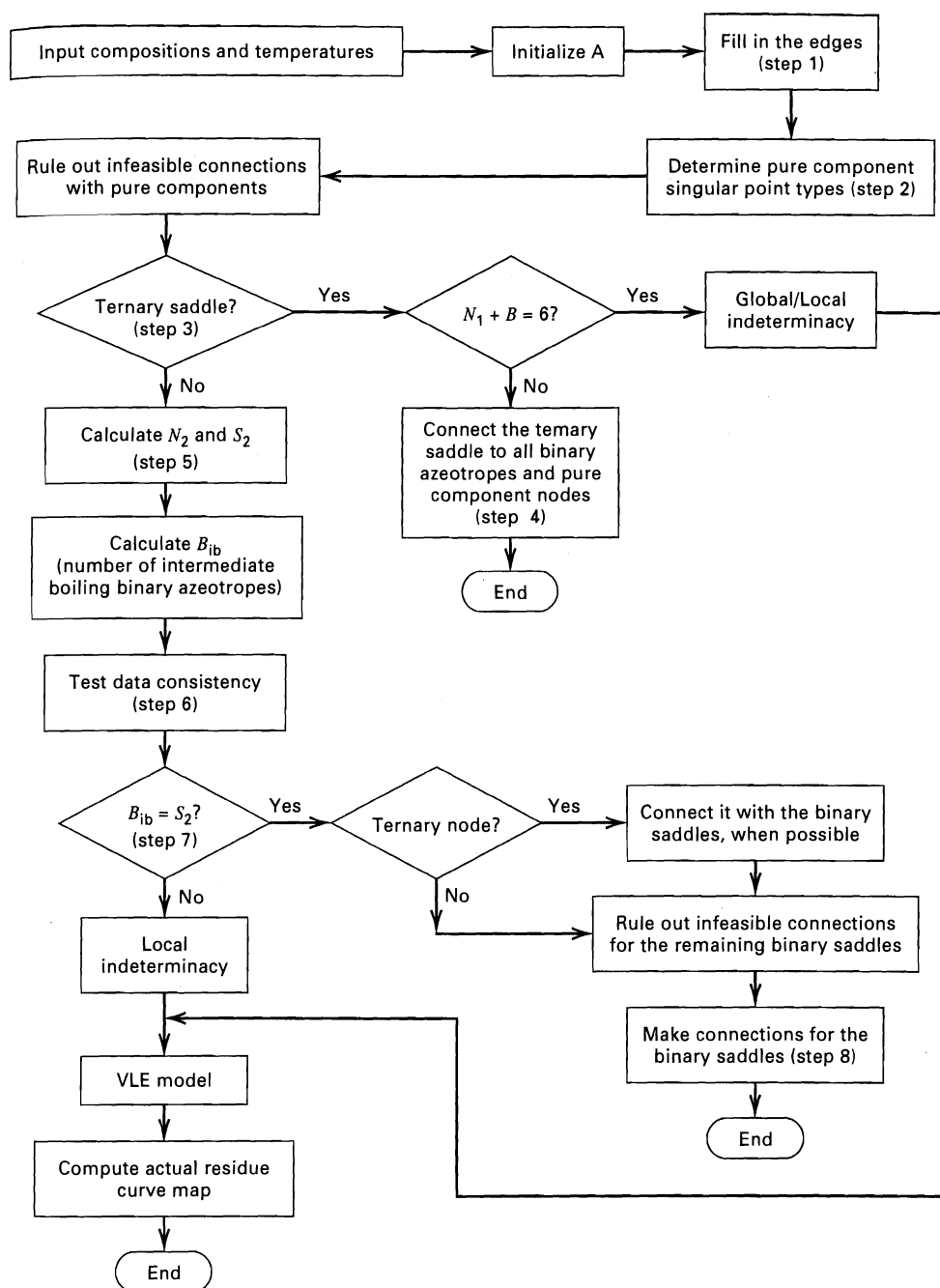


Figure 11.8 Flowchart of algorithm for sketching an approximate residue-curve map. [From E.R. Foucher, M.F. Doherty, and M.F. Malone, *IEC Res.* **30**, 763 (1991) with permission.]

However, excluding L and I because they are saddles, the ternary azeotrope has the second-lowest boiling point. Therefore, the point is a node, and $N_3 = 1$ and $S_3 = 0$. The type of node, stable or unstable, is still to be determined.

Step 4 (for a ternary saddle): Connect the ternary saddle, by straight lines, to all binary azeotropes and to all pure-component nodes (but not to pure-component saddles) and draw arrows on the lines to indicate the direction of increasing temperature. Determine the type of singular point for each binary azeotrope, by using Figure 11.6 with the arrows drawn in this step. This determines the values for N_2 and S_2 . These values should be consistent with (11-10) and (11-12). This completes

the development of the approximate residue-curve map, with no further steps needed. However, if $N_1 + B = 6$, then special checks must be made, as given in detail by Foucher, Doherty, and Malone [14]. This step does not apply to the example in Figure 11.7, because the ternary azeotrope is not a saddle.

Step 5 (for a ternary node or no ternary azeotrope): Determine the number of binary nodes, N_2 , and binary saddles, S_2 , from (11-10) and (11-12), where (11-12) can be solved for N_2 to give

$$N_2 = (2 - 2N_3 + 2S_3 + B - N_1)/2 \quad (11-13)$$

For the example of Figure 11.7, $N_2 = (2 - 2 + 0 + 3 - 1)/2 = 1$. From (11-10), $S_2 = 3 - 1 = 2$.

- Step 6:** Count the binary azeotropes that are intermediate boilers (i.e. that are not the highest- or the lowest-boiling species), (and call that number B_{ib}). Make the following two data consistency checks: (a) The number of binary azeotropes, B , less B_{ib} must equal N_2 , and (b) S_2 must be $\leq B_{ib}$. For the system in Figure 11.7, both checks are satisfied because $B_{ib} = 2$, $B - B_{ib} = 1$, $N_2 = 1$, and $S_2 = 2$. If these two consistency checks are not satisfied, one or more of the species' boiling points may be in error.
- Step 7:** If $S_2 \neq B_{ib}$, this procedure cannot determine a unique residue-curve-map structure, which therefore must be computed from (11-5) to (11-8). If $S_2 = B_{ib}$, there is a unique structure, which is completed in Step 8. For the example in Figure 11.7, $S_2 = B_{ib} = 2$; therefore, there is a unique map.
- Step 8:** In this final step for a ternary node or no ternary azeotrope, the distillation boundaries (connections), if any, are determined and entered on the triangular diagram as straight lines, and, if desired, one or more representative residue curves are sketched as curved lines within each distillation region. This step applies to cases of $S_3 = 0$, $N_3 = 0$ or 1, and $S_2 = B_{ib}$. In all cases, the number of distillation boundaries equals the number of binary saddles, S_2 . Each binary saddle must be connected to a node (pure component, binary, or ternary). A ternary node must be connected to at least one binary saddle. Thus, a pure-component node cannot be connected to a ternary node, and an unstable node cannot be connected to a stable node. The connections are made by determining a connection for each binary saddle such that (a) a minimum-boiling binary saddle connects to an unstable node that boils at a lower temperature and (b) a maximum-boiling binary saddle connects to a stable node that boils at a higher temperature. It is best to first consider connections with the ternary node and then examine the possible connections for the remaining binary saddles. In the example of Figure 11.7, $S_2 = 2$, with these saddles denoted as L-I, a maximum-boiling azeotrope at 115°C, and as I-H, a minimum-boiling azeotrope at 105°C. Therefore, we make two connections to establish two distillation boundaries. The ternary node at 100°C cannot connect to L-I because 100°C is not greater than 115°C. The ternary node can, however, connect, as shown in Step 8 (i), to I-H because 100°C is lower than 105°C. This marks the ternary node as unstable. The connection for L-I can only be to H, as shown in Step 8 (ii) because it is a node (stable), and 120°C is greater than 115°C. This completes the connections. Finally, as shown in Step 8 (iii) of Figure 11.7, three typical, but approximate, residue curves are added to the diagram. These curves originate from unstable nodes and terminate at stable nodes.

Residue-curve maps are used to determine feasible distillation sequences for nonideal ternary systems. Matsuyama and Nishimura [15] showed that the topological constraints just discussed limit the number of possible maps to about 113. However, Siirola and Barnicki [96] show 12 additional maps; all 125 maps are called distillation region diagrams (DRD). Doherty and Calderola [16] provide sketches of 87 maps that contain at least one minimum-boiling binary azeotrope. These maps cover most of the cases found in industrial applications, since minimum-boiling azeotropes are much more common than maximum-boiling azeotropes.

Distillation-Curve Maps

A residue curve represents the liquid-residue composition with time as the result of a simple, one-stage batch distillation. The curve is pointed in the direction of increasing time, from a lower-boiling state to a higher-boiling state for the liquid residue. An alternative representation for distillation on a ternary diagram is a *distillation curve* for continuous, rather than batch, distillation. The curve is most readily determined for total reflux (infinite reflux ratio) at a constant pressure, usually 1 atm. The calculations are made down or up the column starting from any composition. Suppose we choose to make the calculations by moving up the column, starting from a stage designated as Stage 1, and numbering the stages upward. At any location between equilibrium stages j and $j + 1$, it will be recalled, from the McCabe-Thiele method for binary mixtures in Chapter 7 or the Fenske equation for multicomponent systems from Chapter 9, that passing vapor and liquid streams have the same composition. Thus,

$$x_{i,j+1} = y_{i,j} \quad (11-14)$$

Also, liquid and vapor streams leaving the same stage are in equilibrium. Thus,

$$y_{i,j} = K_{i,j} x_{i,j} \quad (11-15)$$

To calculate a distillation curve for a fixed pressure, an initial liquid-phase composition, $x_{i,1}$, is assumed. This liquid is at its bubble-point temperature, which is determined from (11-8), which also gives the equilibrium-vapor composition, $y_{i,1}$ in agreement with (11-15). The composition, $x_{i,2}$, of the passing liquid stream is equal to $y_{i,1}$ by (11-14). The process is then repeated to obtain $x_{i,3}$, then $x_{i,4}$, and so forth. The sequence of liquid-phase compositions, which corresponds to the operating line for the total-reflux condition, is plotted on the triangular diagram. The procedure is essentially that of Fenske for the determination of the minimum number of equilibrium stages for operation at total reflux to achieve a specified split of two key components, as discussed in Chapter 9. The distillation curve is analogous to the 45° line on a McCabe-Thiele diagram for a binary mixture. The calculation of a portion of a distillation curve is illustrated next.

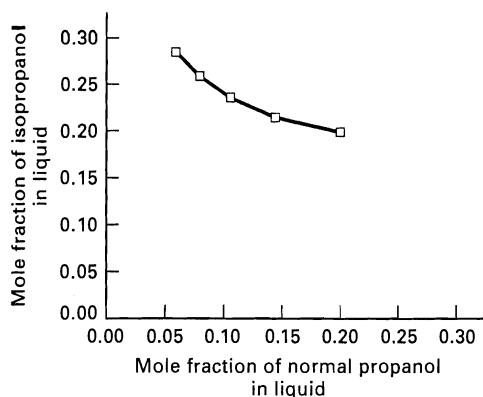


Figure 11.9 Calculated distillation curve for the normal propanol-isopropanol-benzene system at 1 atm for Example 11.2.

EXAMPLE 11.2

Calculate and plot a portion of a distillation curve for the same starting conditions as Example 11.1.

SOLUTION

The starting values, $x^{(1)}$, are 0.2000, 0.2000, and 0.6000 for components 1, 2, and 3, respectively. From Example 11.1, the bubble-point calculation gives a temperature of 79.07°C and the following values for $y^{(1)}$: 0.1437, 0.2154, and 0.6409. From (11-14), values of $x^{(2)}$ are 0.1437, 0.2154, and 0.6409. A bubble-point calculation for this composition gives $T^{(2)} = 78.62^\circ\text{C}$ and $y^{(2)} = 0.1063, 0.2360$, and 0.6577. Subsequent calculations are summarized in the following table:

Equilibrium

Stage	x_1	x_2	y_1	y_2	$T, ^\circ\text{C}$
1	0.2000	0.2000	0.1437	0.2154	79.07
2	0.1437	0.2154	0.1063	0.2360	78.62
3	0.1063	0.2360	0.0794	0.2597	78.29
4	0.0794	0.2597	0.0592	0.2846	78.02
5	0.0592	0.2846	0.0437	0.3091	77.80

The resulting distillation curve is plotted in Figure 11.9, where points represent equilibrium stages and are connected by straight lines.

Distillation curves can be computed more rapidly than residue curves, and closely approximate them for reasons noted by Fidkowski, Doherty, and Malone [17]. If (11-5), which must be solved numerically as in Example 11.1, is written in a forward-finite-difference form, we obtain

$$(x_{i,j+1} - x_{i,j})/\Delta\xi = x_{i,j} - y_{i,j} \quad (11-16)$$

In Example 11.1, $\Delta\xi$ was set to +0.1 for calculations that give increasing values of T and to -0.1 to give decreasing values of T . If we choose the latter direction to be consistent with the direction used in Example 11.2, but set $\Delta\xi$ equal to -1.0, (11-16) becomes identical to (11-14). Thus, residue curves (which are true, continuous curves) are equal to distillation curves (which are discrete points, through which a smooth curve is drawn), when the residue curves are approximated by a crude forward-finite-difference formulation, using $\Delta\xi = -1.0$.

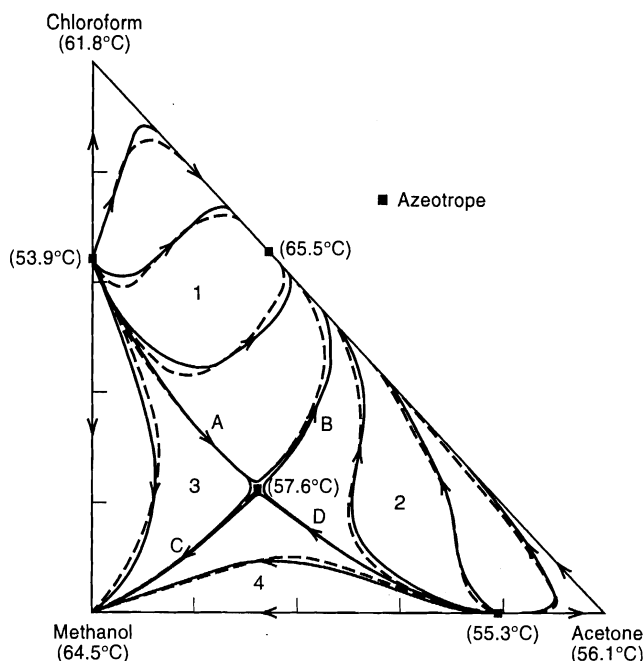


Figure 11.10 Comparison of residue curves to distillation curves.

[From Z.T. Fidkowski, M.F. Malone, and M.F. Doherty, *AIChE J.*, **39**, 1303 (1993) with permission.]

A collection of distillation curves, including lines for distillation boundaries, is a *distillation-curve map*, an example of which from Fidkowski et al. [17] is shown in Figure 11.10 for the acetone-chloroform-methanol system at 1 atm. The Wilson equation was used to compute liquid-phase activity coefficients for the system. The dashed lines are the distillation curves; they approximate the residue curves, which are solid lines. This system has two minimum-boiling binary azeotropes, one maximum-boiling binary azeotrope, and a ternary saddle azeotrope. The map shows four distillation boundaries, designated by A, B, C, and D, consistent with Step 4 earlier. These computed boundaries, which define four distillation regions (1 to 4), are all curved lines rather than the approximate straight lines in the sketches of Figure 11.7.

Distillation-curve maps have been used extensively by Stichlmair and associates [1, 3, 18] for the development of feasible-distillation sequences. In their maps, arrows on the distillation curves are directed toward the lower-boiling species, rather than the higher-boiling species as in residue-curve maps.

Product-Composition Regions at Total Reflux (Bow-Tie Regions)

As mentioned above, the possible distillation regions for azeotrope-forming ternary mixtures are not obvious. Fortunately, residue-curve maps and distillation-curve maps can be used to make preliminary estimates of regions of *feasible-product compositions* for distillation of nonideal ternary mixtures. The product regions are determined by superimposing a column material-balance line on either curve-map diagram. Consider first the simpler zeotropic ternary system in Figure 11.11a, which shows a typical isobaric residue-curve

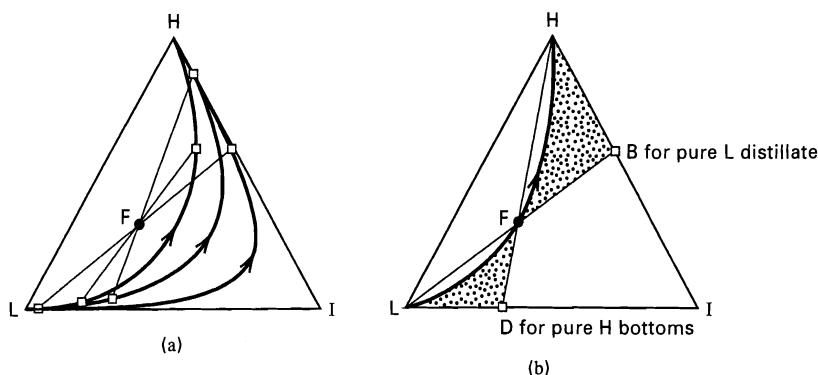


Figure 11.11 Product-composition regions for a zeotropic system. (a) Material-balance lines and distillation curves. (b) Product-composition regions shown shaded.

[From S. Widagdo and W.D. Seider, *AIChE J.*, **42**, 96–130 (1996) with permission.]

map with three residue curves. Assume that this map is identical to a corresponding distillation-curve map for total-reflux conditions and to a map for a finite, but very high reflux ratio. Suppose a ternary feed, denoted by F in Figure 11.11a, is to be continuously distilled in a column, operating isobarically at a high reflux ratio, to produce a distillate, D , and a bottoms, B . As shown in Chapters 4 and 8, if a straight line is drawn that connects distillate and bottoms compositions, the line must pass through the feed composition at some intermediate point to satisfy overall and component material-balance equations. This line is a *material-balance line*, three of which are included on Figure 11.11a. For a given material-balance line, a set of D and B composition points, designated by open squares, must lie on the same distillation curve. This causes the material-balance line to intersect the distillation curve at these two points and be a chord to the distillation curve.

The limiting distillate-composition point for this zeotropic system is pure low-boiling component, L . From the material-balance line passing through F , as shown in Figure 11.11b, the corresponding bottoms composition with the least amount of component L is point B . At the other extreme, the limiting bottoms-composition point is pure high-boiling component, H . A material-balance line from this point, through feed point F , ends at D . These two

lines and the distillation curve define the feasible product-composition regions, shown shaded. Note that because, for a given feed, both the distillate and bottoms compositions must lie on the same distillation curve, the shaded feasible regions lie on the convex side of the distillation curve that passes through the feed point. Because of its appearance, the feasible-product-composition region is referred to as a *bow-tie-region*.

For an azeotropic system, where distillation boundaries are present, a feasible-product-composition region can be found for each distillation region. Two examples are shown in Figure 11.12. The first, in Figure 11.12a, has two distillation regions caused by two minimum-boiling binary azeotropes. A curved distillation boundary connects the two minimum-boiling azeotropes. In the lower, right-hand distillation region (1), the lowest-boiling species is the n -octane–2-ethoxy-ethanol minimum-boiling azeotrope, while the highest-boiling species is 2-ethoxy-ethanol. Accordingly, for feed F_1 , straight lines are drawn from the points for each of these two species, through the point F_1 , and to a boundary (either a distillation boundary or a side of the triangle). Shaded, feasible-product-composition regions are then drawn on the outer side of the distillation curve that passes through the feed point. The result is that distillate compositions are confined to shaded region D_1 and bottoms compositions are

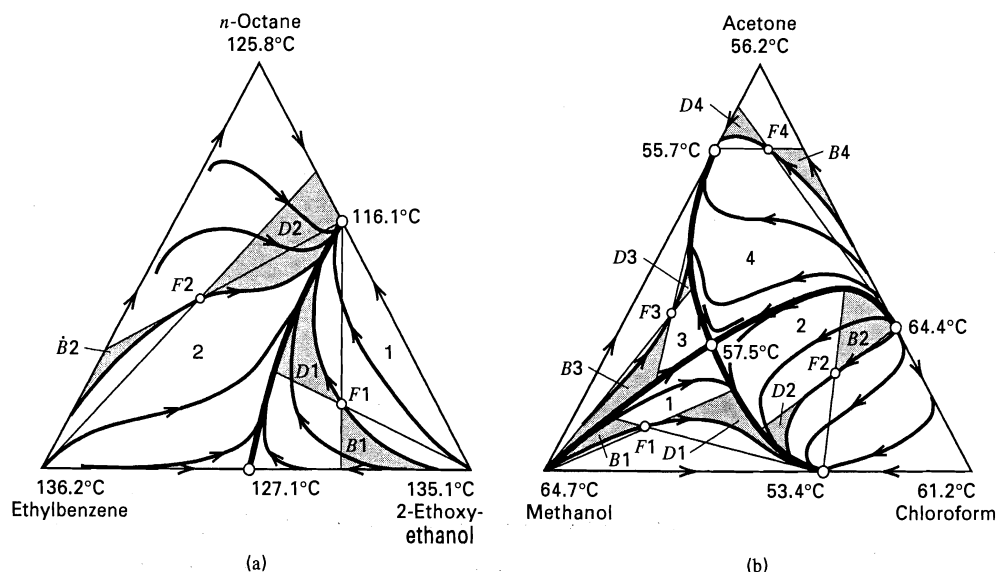


Figure 11.12 Product-composition regions for given feed compositions. (a) Ternary mixture with two minimum-boiling binary azeotropes at 1 atm. (b) Ternary mixture with three binary and one ternary azeotrope at 1 atm.

confined to shaded region B_1 . For a given D_1 , B_1 must lie on a straight line that passes through D_1 and F_1 . At total reflux, D_1 and B_1 must also lie on the same distillation curve. A more complex distillation-curve map, with four distillation regions, is shown in Figure 11.12b for the acetone-methanol-chloroform system with two minimum-boiling binary azeotropes, one maximum-boiling binary azeotrope, and one ternary azeotrope. One shaded bow-tie region, determined in the same way as for region 1 in Figure 11.12a, is present for each distillation region. For this system, feasible-product-composition regions are highly restricted.

A complicated situation is observed in distillation region 1 on the left side of Figure 11.12a. In that region, the lowest-boiling species is the binary azeotrope of octane and 2-ethoxy-ethanol, while the highest-boiling species is the ethylbenzene. The complicating factor in distillation region 1 is that feed F_2 lies on or close to an inflection point of an S-shaped distillation curve. In this case, as discussed by Wahnschafft et al. [20], feasible-product-composition regions may lie on either side of the distillation curve that passes through the feed point. The feasible regions shown are similar to those determined by Stichlmair et al. [1], while other feasible regions are shown for this system by Wahnschafft et al. [20]. As they point out, for a situation such as this, mass-balance lines of the type drawn in Figure 11.12b do not limit the feasible regions. Hoffmaster and Hauan [98] provide a method for determining extended-product-feasibility regions in the presence of S-shaped distillation curves.

In Figures 11.11b, 11.12a, and 11.12b, each bow-tie region is confined to its distillation region, as defined by the distillation boundaries. In all cases, the feed, distillate, and bottoms points on the material-balance line lie within a distillation region, with the feed point between the distillate and bottoms points. The material-balance lines do not cross the distillation-boundary lines. Is this always so? The answer is no! Under conditions where the distillation-boundary line is highly curved, it can be crossed by material-balance lines to obtain feasible-product compositions. That is, a feed point can be on one side and the distillate and bottoms points on the other side of the distillation-boundary line. Consider the example in Figure 11.13, taken from Widagdo and Seider [19]. The distillation-boundary line, which is highly curved, extends from a minimum-boiling azeotrope K of H-I to the pure component L. This line divides the triangular diagram into two distillation regions, 1 and 2. Feed F_1 can be separated into products D_1 and B_1 , which lie on distillation curve (a). In this case, the material-balance line and the distillation curve are both on the convex side of the distillation-boundary line. However, because the feed point F_1 lies close to the highly curved boundary line, F_1 can also be separated into D_2 and B_2 (or B_3), which lie on a distillation curve in region 2 on the concave side of the boundary. Thus, the material-balance line crosses the boundary from the convex to the concave side. Feed F_2 can be separated into D_4 and B_4 , but not into D and B . In the latter case, the material-balance

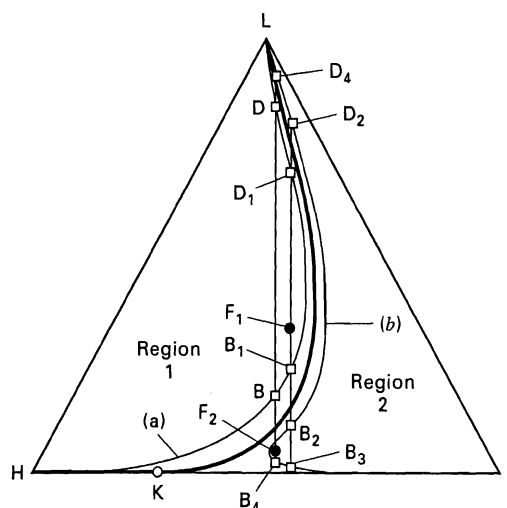


Figure 11.13 Feasible and infeasible crossings of distillation boundaries for an azeotropic system.

[From S. Widagdo and W.D. Seider, *AIChE J.*, **42**, 96–130 (1996) with permission.]

line cannot cross the boundary from the concave to the convex side, because the point F_2 does not lie between D and B on the material-balance line. The determination of the feasible-product-composition regions for Figure 11.13 is left for an exercise at the end of this chapter. A detailed treatment of product composition regions is given by Wahnschafft et al. [20].

11.2 EXTRACTIVE DISTILLATION

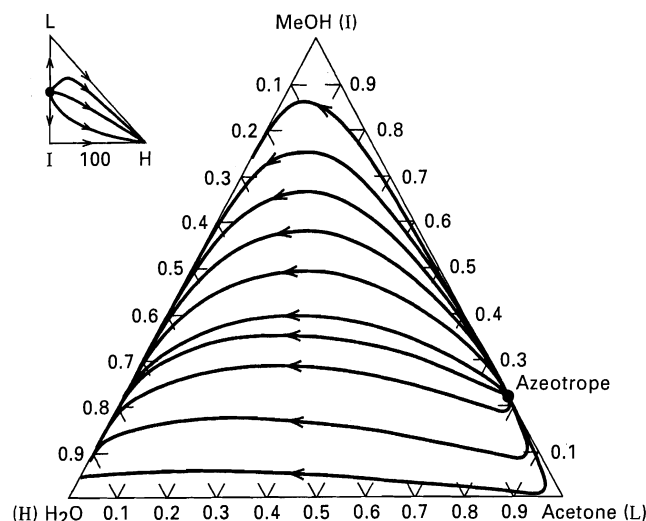
Extractive distillation is used to separate azeotropes and other mixtures that have key components with a relative volatility below about 1.1 over an appreciable range of concentration. If the feed is a minimum-boiling azeotrope, a solvent, with a lower volatility than the key components of the feed mixture, is added to a tray above the feed stage and a few trays below the top of the column so that (1) the solvent is present in the downflowing liquid phase to the bottom of the column, and (2) little solvent is stripped and lost to the overhead vapor. If the feed is a maximum-boiling azeotrope, the solvent enters the column with the feed. The components in the feed must have different affinities for the solvent so that the solvent causes an increase in the relative volatility of the key components, to the extent that separation becomes feasible and economical. The solvent should not form an azeotrope with any components in the feed. Generally, a molar ratio of solvent-to-feed on the order of 1 is required to achieve this goal. The bottoms from the extractive distillation column is processed further to recover the solvent for recycle and complete the feed separation. The name, extractive distillation, was introduced by Dunn et al. [21] in connection with the commercial separation of toluene from a paraffin-hydrocarbon mixture, using phenol as solvent.

Table 11.1 Some Industrial Applications of Extractive Distillation

Key Components in Feed Mixture	Solvent
Acetone–methanol	Aniline, ethylene glycol, water
Benzene–cyclohexane	Aniline
Butadienes–butanes	Acetone
Butadiene–butene-1	Furfural
Butanes–butenes	Acetone
Butenes–isoprene	Dimethylformamide
Cumene–phenol	Phosphates
Cyclohexane–heptanes	Aniline, phenol
Cyclohexanone–phenol	Adipic acid diester
Ethanol–water	Glycerine, ethylene glycol
Hydrochloric acid–water	Sulfuric acid
Isobutane–butene-1	Furfural
Isoprene–pentanes	Acetonitrile, furfural
Isoprene–pentenenes	Acetone
Methanol–methylene bromide	Ethylene bromide
Nitric acid–water	Sulfuric acid
<i>n</i> -Butane–butene-2s	Furfural
Propane–propylene	Acrylonitrile
Pyridine–water	Bisphenol
Tetrahydrofuran–water	Dimethylformamide, propylene glycol
Toluene–heptanes	Aniline, phenol

Table 11.1 lists a number of industrial applications of extractive distillation. Consider the case of the acetone–methanol system. At 1 atm, acetone (nbp = 56.2°C) and methanol (nbp = 64.7°C) form a minimum-boiling azeotrope of 80 mol% acetone at a temperature of 55.7°C. The UNIFAC program was used to predict the vapor–liquid equilibria for this system at 1 atm. The azeotrope was estimated to occur at 55.2°C with 77.1 mol% acetone. At infinite dilution with respect to methanol, the relative volatility of acetone (A) with respect to methanol (M), $\alpha_{A,M}$, is predicted to be 0.74, with a liquid-phase activity coefficient for methanol of 1.88. At infinite dilution with respect to acetone, $\alpha_{A,M}$ is 2.48; by coincidence, the liquid-phase activity coefficient for acetone is 1.88 also. Water is a possible solvent for the system because at 1 atm: (1) it does not form a binary or ternary azeotrope with acetone and/or methanol, and (2) it boils (100°C) at a higher temperature. The resulting residue-curve map with arrows directed from the azeotrope to pure water, computed by ASPEN PLUS using UNIFAC, is shown in Figure 11.14, where it is seen that no distillation boundaries exist. As discussed by Doherty and Caldarola [16], this is an ideal situation for the selection of an extractive distillation process. Their schematic residue-curve map for this type system (designated 100) is included as an insert in Figure 11.14.

Ternary mixtures of acetone, methanol, and water at 1 atm give the following separation factors, estimated from the UNIFAC equation, when appreciable solvent is present.

**Figure 11.14** Residue-curve map for acetone–methanol–water system at 1 atm.

Mol% Water	Relative Volatility, $\alpha_{A,M}$			Liquid-Phase Activity Coefficient at Infinite Dilution	
	Methanol-rich	Acetone-rich	Equimolar	Acetone	Methanol
40	2.48	2.57	2.03	2.12	0.70
50	2.56	2.86	2.29	2.41	0.72

Thus, the presence of appreciable water increases the liquid-phase activity coefficient of acetone and decreases that of methanol, with the result that, over the entire concentration range of acetone and methanol, the relative volatility of acetone to methanol is at least 2.0. This makes it possible, with extractive distillation, to obtain a distillate of acetone and a bottoms of methanol and water. Furthermore, the relative volatilities of acetone to water and methanol to water average about 4.5 and 2.0, respectively. Thus, it is relatively easy to prevent an appreciable amount of water from reaching the distillate, and, in subsequent operations, to separate methanol from water by ordinary distillation.

EXAMPLE 11.3

Forty moles per second of a bubble-point mixture of 75 mol% acetone and 25 mol% methanol at 1 atm is separated by an extractive-distillation process, using water as the solvent, to produce an acetone product of not less than 95 mol% acetone, a methanol product of not less than 98 mol% methanol, and a water stream for recycle of at least 99.9 mol% purity. Prepare a preliminary process design, using the traditional sequence consisting of ordinary distillation followed by extractive distillation, and then ordinary distillation to recover the solvent, as shown for another system in Figure 11.15.

SOLUTION

In the first column, the feed mixture of acetone and methanol would be partially separated by ordinary distillation, where the distillate composition approaches that of the binary azeotrope. The bottoms

Table 11.2 Material and Energy Balances for Extractive Distillation Process of Example 11.3

Material Balances						
Species	Flow rate, mol/s: Column 2 Feed	Column 2 Solvent	Column 2 Distillate	Column 2 Bottoms	Column 3 Distillate	Column 3 Bottoms
Acetone	30	0	29.86	0.14	0.14	0.0
Methanol	10	0	0.016	9.984	9.926	0.058
Water	0	60	1.35	58.65	0.06	58.59
Total	40	60	31.226	68.774	10.126	58.648

Energy Balances		
	Column 1	Column 2
Condenser duty, MW	4.71	1.07
Reboiler duty, MW	4.90	1.12

would be nearly pure acetone or nearly pure methanol depending upon whether the feed contains more or less than 80 mol% acetone, respectively. However, in this example, the feed composition is already close to the azeotrope composition; therefore, the first column is omitted.

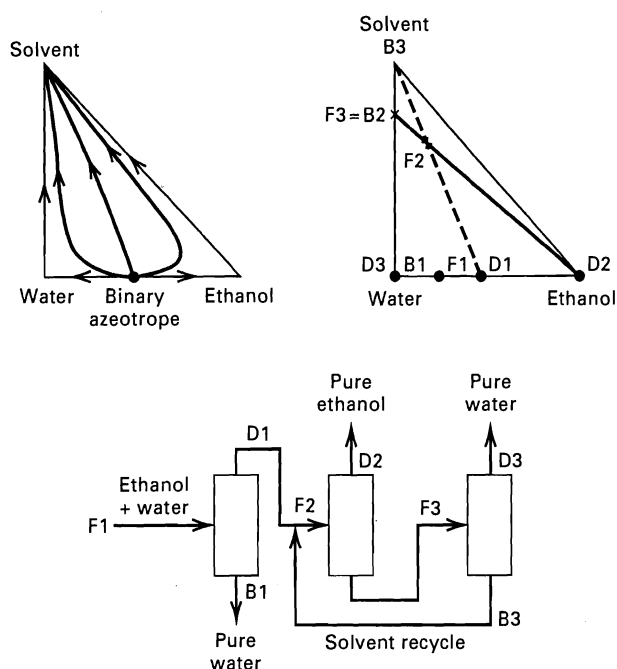
Accordingly, the acetone-methanol feed is sent to the second column, an extractive-distillation column equipped with a total condenser and a partial reboiler to produce a distillate of at least 95 mol% acetone. The acetone recovery is better than approximately 99% to achieve methanol purity in the third column. The bottoms from the extractive-distillation column is pumped to that column, where methanol and water are separated by ordinary distillation to achieve the specified purities.

The ChemSep and ChemCAD programs were used to make the calculations, with the UNIFAC method for activity coefficients.

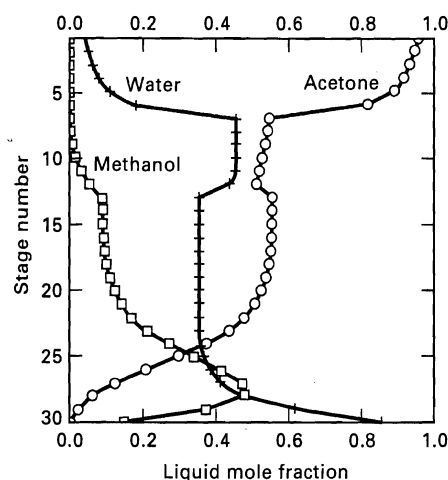
Equilibrium-stage, feed-stage, solvent entry-stage, solvent flow-rate, and reflux-ratio requirements were varied until a satisfactory design was achieved. The resulting material and energy balances are summarized in Table 11.2.

For the extractive-distillation column, a solvent flow rate of 60 mol/s of water is suitable. Using 28 theoretical trays, a 50°C solvent entry at Tray 6 from the top, a feed entry at Tray 12 from the top, and a reflux ratio of 4, a distillate composition of 95.6 mol% acetone is achieved. The impurity is mainly water. The acetone recovery is 99.5%. A 6-ft-diameter column with 60 sieve trays on 2-ft tray spacing is adequate for the operation. A liquid-phase composition profile is shown in Figure 11.16. The mole fraction of water (the solvent) in the liquid phase is appreciable, at least 0.35 for all of the stages below the solvent-entry stage. A distillation-curve map for the actual extractive distillation operation is given in Figure 11.17, where both the vapor and liquid curves are plotted. The arrows are directed from the bottom of the column to the top.

For the ordinary-distillation column, operation with 16 theoretical stages, a bubble-point feed-stage location of Stage 11 and a reflux ratio of 2 is adequate to achieve a methanol distillate of 98.1 mol% purity and a water bottoms, suitable for recycle, of 99.9 mol% purity. A McCabe-Thiele diagram in Figure 11.18

**Figure 11.15** Distillation sequence for extractive distillation.

[From M.F. Doherty and G.A. Calderola, *IEC Fundam.*, 24, 479 (1985) with permission.]

**Figure 11.16** Liquid composition profile for extractive-distillation column of Example 11.3.

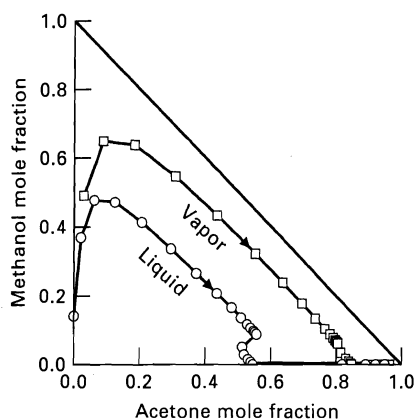


Figure 11.17 Distillation-curve map for Example 11.3. Data points are for theoretical stages.

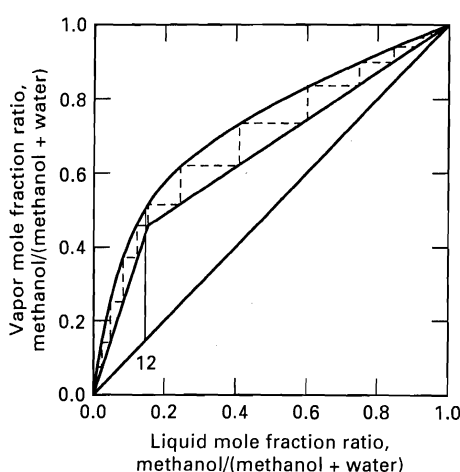


Figure 11.18 McCabe-Thiele diagram for methanol-water distillation in Example 11.3.

shows the locations of the theoretical stages. The feed stage is optimally located. Water makeup is less than 1.5 mol/s. A 2.5-ft-diameter column packed with 48 feet of 50-mm-diameter metal Pall rings is suitable for the separation.

One unfortunate aspect in the design of the extractive-distillation column in Example 11.3 is caused by the relatively low boiling point of water. With a solvent entry point of Tray 6 from the top, 1.35 mol/s (2.25% of the water solvent) is stripped from the liquid into the distillate. The use of two other higher-boiling solvents listed in Table 11.1, aniline (nbp = 184°C) or ethylene glycol (nbp = 198°C), results in far less stripping of solvent. Other possible solvents for the separation of acetone from methanol by extractive distillation include methylethylketone (MEK) and ethanol. MEK behaves in a fashion opposite to that of water: MEK causes the volatility of methanol to be greater than that of acetone. Thus, methanol becomes the distillate in the extractive-distillation column, leaving acetone to be separated from MEK in the subsequent column.

In selecting a solvent for extractive distillation, a number of factors are considered, including availability, cost,

corrosivity, vapor pressure, thermal stability, heat of vaporization, reactivity, toxicity, infinite-dilution activity coefficients in the solvent of the components to be separated, and ease of recovery for recycle. In addition, the solvent should not form azeotropes. Initial screening is based on the measurement or prediction of infinite-dilution activity coefficients. Berg [22] discusses, in detail, the selection of separation agents for both extractive and azeotropic distillation. He points out that all successful solvents for extractive distillation are highly hydrogen-bonded liquids, such as (1) water, amino alcohols, amides, and phenols that form three-dimensional networks of strong hydrogen bonds, and (2) alcohols, acids, phenols, and amines that are composed of molecules containing both active hydrogen atoms and donor atoms (oxygen, nitrogen, and fluorine). In general, it is very difficult or impossible to find a suitable solvent to economically separate components having the same functional groups.

Extractive distillation is also used to separate binary mixtures that form a maximum-boiling azeotrope, as shown in the following example.

EXAMPLE 11.4

Acetone (nbp = 56.16°C) and chloroform (nbp = 61.10°C) form a maximum-boiling homogeneous azeotrope at 1 atm and 64.43°C that contains 37.8 mol% acetone. Thus, they cannot be separated by ordinary distillation at 1 atm. Instead, it is proposed to separate them using extractive distillation in a two-column sequence, shown in Figure 11.19, with benzene (nbp = 80.24°C) as the solvent. Benzene does not form azeotropes with either of the feed components.

In the first column, the feed, blended with recycled solvent, is distilled to produce a distillate of 99 mol% acetone. The bottoms is sent to the second column, where 99 mol% chloroform leaves as distillate and the bottoms, which is rich in benzene, is recycled to the inlet of the first column with a small flow of makeup benzene. If the fresh feed is 21.8858 mol/s of 54.83 mol% acetone, with the balance chloroform, design a feasible two-column system using a ratio of 3.1667 moles of benzene per mole of acetone + chloroform in the combined feed to the first column. Both columns operate at a nominal pressure of 1 atm with total condensers, saturated liquid reflux, and partial reboilers. Use the UNIFAC method for

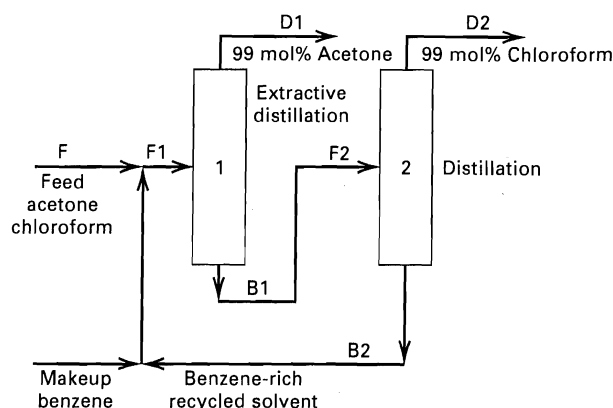


Figure 11.19 Process for the separation of acetone and chloroform in Example 11.4.

Table 11.3 Material and Energy Balances for Homogeneous Azeotropic Distillation of Example 11.4

Material Balances with Flows in mol/s						
Species	F	F_1	D_1	$B_1 = F_2$	D_2	B_2
Acetone	12.0000	12.0000	11.9948	0.0052	0.0052	0.0000
Chloroform	9.8858	12.0000	0.1046	11.8954	9.7812	2.1142
Benzene	0.0000	76.0000	0.0207	75.9793	0.0934	75.8859

Energy Balances		
Heat duty, kcal/h	Column 1	Column 2
Condenser	950,000	891,600
Reboiler	958,400	1,102,000

estimating activity coefficients. The combined feed to the first column is brought to the bubble point before entering the feed stage.

SOLUTION

The residue-curve map for the ternary system acetone–chloroform–benzene at 1 atm is shown in Figure 11.20. The only azeotrope is that formed by acetone and chloroform. A curved distillation boundary extending from that azeotrope to the pure benzene apex divides the diagram into two distillation regions. The first column, which produces nearly pure acetone, operates in Region 1, whereas the second column operates in upper Region 2.

This ternary system was studied in detail by Fidkowski, Doherty, and Malone [17]. A design, based on their studies and using the ChemCAD process simulator, is summarized in Table 11.3. The first column contains 65 theoretical stages with the combined feed entering Stage 30 from the top. With a reflux ratio of 10, the acetone distillate purity is achieved with an acetone recovery of better than 99.95%. In Column 2, which contains 50 theoretical stages with the feed entering at Stage 30 from the top, a reflux ratio of 11.783 gives the required chloroform purity in the distillate, but with a recovery of only 82.23%. However, this is not serious

because the remaining chloroform leaving in the bottoms is recycled with the benzene to Column 1, with the result being an overall recovery of chloroform of 98.9%. The benzene makeup rate is 0.1141 mol/s. Feed, distillate, and bottoms compositions are designated in Figure 11.20.

11.3 SALT DISTILLATION

The use of water as a solvent in the extractive distillation of acetone and methanol in Example 11.3 has the two disadvantages that a large amount of water is required to adequately alter the relative volatility and, even though the solvent is introduced into the column several trays below the top tray, enough water is stripped by vapor traffic into the distillate to reduce the acetone purity to 95.6 mol%. The water vapor pressure can be lowered, and thus the purity of acetone distillate increased, by use of an aqueous inorganic-salt solution as the solvent. For example, a 1927 patent application by Othmer [23] describes the use of a concentrated, calcium-chloride brine. Not only does calcium chloride, which is highly soluble in water, reduce the volatility of water, but it also has a strong affinity for methanol. Thus, the relative volatility of acetone with respect to methanol is further enhanced. The separation of the brine solution from methanol is easily accommodated in the subsequent distillation step, with the brine solution recycled to the extractive distillation column. The vapor pressure of the dissolved salt is so small that it never enters the vapor phase, provided that entrainment is avoided. An even earlier patent by Van Raymbeke [24] describes the extractive distillation of ethanol from water by using solutions of calcium chloride, zinc chloride, or potassium carbonate in glycerol.

Rather than using a solvent that contains a dissolved salt, the salt can be added as a solid or melt directly into the column by dissolving it in the liquid reflux before it enters the column. This technique was demonstrated experimentally by Cook and Furter [25] in a 4-inch-diameter, 12-tray rectifying column with bubble caps for the separation of ethanol from water using potassium acetate. At salt concentrations

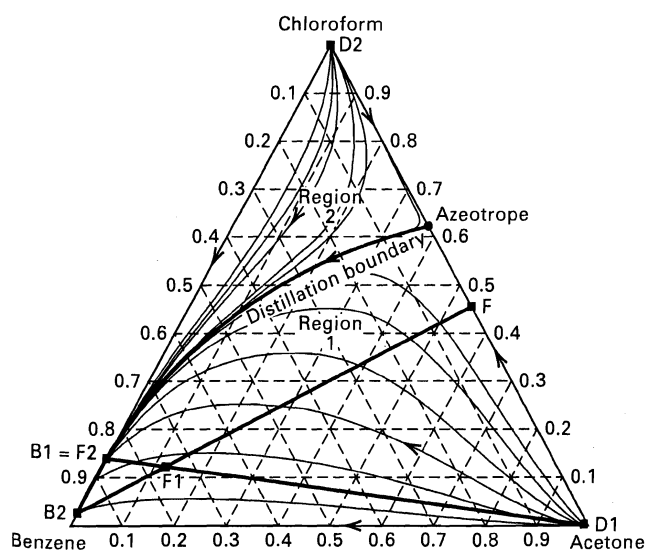


Figure 11.20 Residue-curve map for Example 11.4.

below saturation and between 5 and 10 mol%, an almost pure ethanol distillate was achieved. The salt, which must be soluble in the reflux, is recovered from the aqueous bottoms by evaporation and crystallization.

Salt distillation is accompanied by several potential problems. First and foremost is corrosion, particularly with aqueous chloride-salt solutions, which may require stainless steel or a more corrosion-resistant material. The feeding and dissolving of a salt into the reflux stream poses many problems, as described by Cook and Furter [25]. The solubility of the salt will be low in the reflux because it is rich in the more volatile component, while the salt will be most soluble in the less-volatile component. Consequently, the solid salt must be metered at a constant rate. The salt-feeding mechanism must avoid bridging and must prevent the entry of vapor, which could cause clogging upon condensation. The salt must be rapidly dissolved. The reflux must be maintained near the boiling point to avoid precipitation of already-dissolved salt in lines. In the column, the presence of the dissolved salt may increase the potential for foaming, which may require the addition of antifoaming agents and/or an

increase in column diameter. Some concern has been voiced for the possibility of crystallization of salt within the column. However, the concentration of the less-volatile component (e.g., water) increases down the column. Thus, the solubility of the salt increases down the column, while its concentration remains relatively constant. Therefore, the possibility of clogging and plugging due to solids formation in the column is highly unlikely.

In aqueous alcohol solutions, both *salting out* and *salting in* have been observed by Johnson and Furter [26], as shown in the vapor-liquid equilibrium data in Figure 11.21: in (a), sodium-nitrate salts out methanol, but in (b), mercuric chloride salts in methanol. Even low concentrations of potassium acetate can eliminate the ethanol-water azeotrope, as shown in Figure 11.21c. Mixed potassium- and sodium-acetate salts were used in Germany and Brazil from 1930 to 1965 for the separation of ethanol and water.

Surveys of the use of inorganic salts for extractive distillation, including effects on vapor-liquid equilibria, are given by Johnson and Furter [27], Furter and Cook [28], and Furter [29, 30]. A survey of methods for predicting the effect of

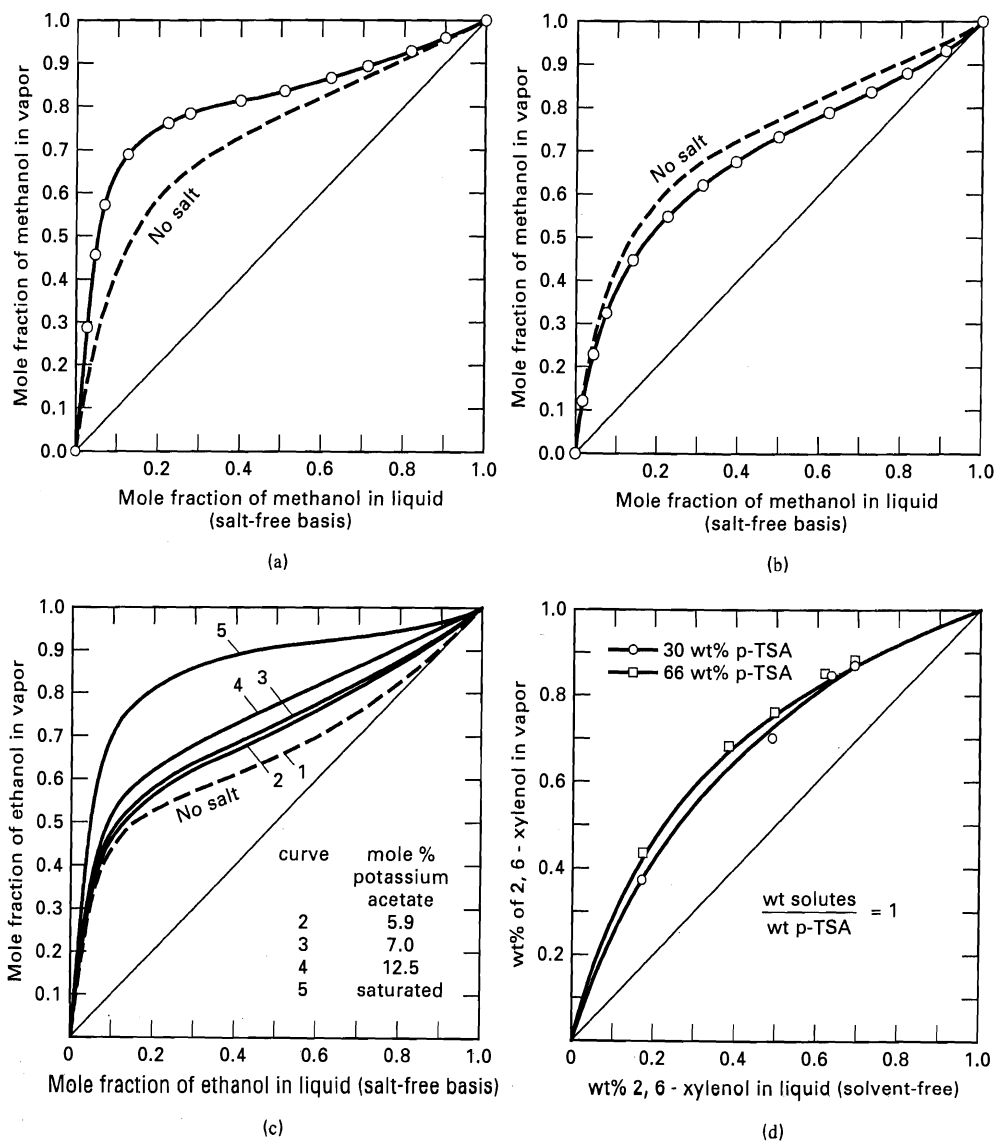


Figure 11.21 Effect of dissolved salts on vapor-liquid equilibria at 1 atm. (a) Salting-out of methanol by saturated aqueous sodium nitrate. (b) Salting-in of methanol by saturated aqueous mercuric chloride. (c) Effect of salt concentration on ethanol-water equilibria. (d) Effect of *p*-toluenesulfonic acid (*p*-TSA) on phase equilibria of 2,6 xyleneol-*p*-cresol. [From A.I. Johnson and W.F. Furter, *Can. J. Chem. Eng.*, **43**, 356-358 (1965) with permission.]

inorganic salts on vapor-liquid equilibria is given by Kumar [31]. Column-simulation results, using the Newton-Raphson method, are presented by Llano-Restrepo and Aguilar-Arias [99] for the ethanol-water-calcium chloride system and by Fu [100], for the ethanol-water-ethanediol-potassium acetate system, who shows that his simulation results compare favorably with measurements on an industrial column.

Salt distillation can also be applied to the separation of organic compounds that have little capacity alone for dissolving inorganic salts by using a special class of organic salts called *hydrotropes*. Typical hydrotropic salts are alkali and alkaline-earth salts of the sulfonates of toluene, xylene, or cymene, and the alkali benzoates, thiocyanates, and salicylates. For example, Mahapatra, Gaikar, and Sharma [32] showed that the addition of aqueous solutions of 30 and 66 wt% *p*-toluenesulfonic acid to mixtures of 2,6-xlenol and *p*-cresol at 1 atm increased the relative volatility from approximately 1 to about 3, as shown in Figure 11.21d. Hydrotropes can also be used to enhance separations by liquid-liquid extraction, as shown by Agarwal and Gaikar [33].

11.4 PRESSURE-SWING DISTILLATION

When a binary azeotrope disappears at some pressure or changes composition by 5 mol% or more over a moderate range of pressure, consideration should be given to using, without a solvent, two distillation columns operating in series at different pressures. This process is referred to as *pressure-swing distillation* or *two-column distillation*. Knapp and Doherty [34] list 36 pressure-sensitive, binary azeotropes, taken mainly from the compilation of Horsley [11]. The effect of pressure on the temperature and composition of two minimum-boiling azeotropes is shown in Figure 11.22. The mole fraction of ethanol in the ethanol-water azeotrope increases from 0.8943 at 760 torr to more than 0.9835 at 90 torr. Although not shown in Figure 11.22b, the azeotrope finally disappears at below about 70 torr. A much more dramatic change in azeotropic composition with pressure is seen in Figure 11.22b for the ethanol-benzene system, which forms a minimum-boiling azeotrope at 44.8 mol%

ethanol and 1 atm. Applications of pressure-swing distillation, which was first noted by Lewis [35] in a 1928 patent, include the separations of the minimum-boiling azeotrope of tetrahydrofuran-water and the maximum-boiling azeotropes of hydrochloric acid-water and formic acid-water.

Consider the case, described by Van Winkle [36], of a minimum-boiling azeotrope for the mixture A-B, with T - y - x curves as shown in Figure 11.23a. As the pressure is decreased from P_2 to P_1 , the azeotropic composition moves toward a smaller percentage of A. An operable pressure-swing sequence is shown in Figure 11.23b. The total feed, F_1 , to Column 1, operating at the lower pressure, P_1 , is the sum of the fresh feed, F , whose composition is richer in A than the azeotrope, and the recycled distillate, D_2 , whose composition is close to that of the azeotrope at pressure, P_2 . The compositions of D_2 and, consequently, F_1 are both richer in A than the azeotrope composition at P_1 . The bottoms, B_1 , leaving Column 1 is almost pure A. The distillate, D_1 , which is slightly richer in A than the azeotrope, but less rich in A than the azeotrope at P_2 , is fed to Column 2, where the bottoms, B_2 , is almost pure B. Robinson and Gilliland [37] provide an example of the separation of ethanol and water, where the fresh-feed composition is less rich in ethanol than the azeotrope. For that case, the products are still removed as bottoms, but nearly pure B is taken from the first column and A from the second.

Pressure-swing distillation can also be applied to the separation of the less-common, maximum-boiling binary azeotropes. The sequence is shown in Figure 11.23c, where both products are withdrawn as distillates, rather than as bottoms. In this case, the composition of the azeotrope becomes richer in A as the pressure is decreased. The fresh feed, which is richer in A than the azeotrope at the higher pressure, is first distilled in Column 1 at the higher pressure, P_1 , to produce a distillate of nearly pure A and a bottoms slightly richer in A than the azeotrope at the higher pressure. The bottoms is fed to Column 2, operating at the lower pressure, P_2 , where the azeotrope composition is richer in A than the feed to that column. Accordingly, the distillate is nearly pure B, while the recycled bottoms from Column 2 is slightly less rich in A than the azeotrope at the lower pressure.

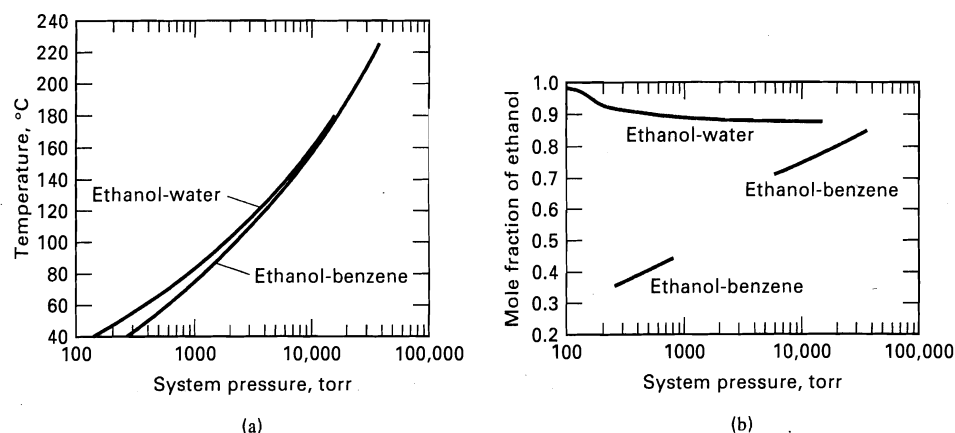


Figure 11.22 Effect of pressure on azeotrope conditions. (a) Temperature of azeotrope. (b) Composition of azeotrope.

[From *Perry's Chemical Engineers' Handbook* 6th ed., R.H. Perry and D.W. Green, Eds. McGraw-Hill (1984) with permission.]

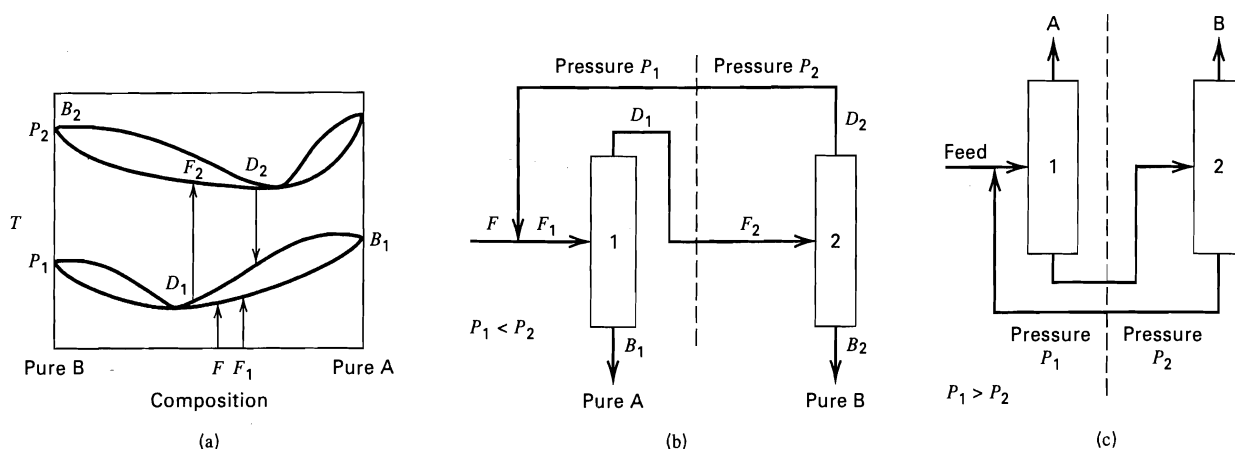


Figure 11.23 Pressure-swing distillation. (a) T - y - x curves at pressures P_1 and P_2 for minimum-boiling azeotrope. (b) Distillation sequence for minimum-boiling azeotrope. (c) Distillation sequence for maximum-boiling azeotrope.

For all pressure-swing-distillation sequences, the recycle ratio is an important factor in the design and depends on the difference in azeotropic composition for the column pressures. The following example illustrates the calculation and importance of the recycle stream.

EXAMPLE 11.5

Ninety mol/s of a mixture of two-thirds by moles ethanol and one-third benzene at the bubble point at 101.3 kPa is to be separated into 99 mol% ethanol and 99 mol% benzene. Ordinary distillation is not feasible because the mixture forms a minimum-boiling azeotrope at 760 torr with a composition of 44.8 mol% ethanol and a temperature of 68°C. If the pressure is reduced to 200 torr, as shown in Figure 11.22b, the azeotrope composition shifts to 36 mol% ethanol at 35°C. This magnitude of shift makes this a candidate for pressure-swing distillation.

Apply the sequence shown in Figure 11.23b. Let the first column operate with a top-tray pressure of 30 kPa (225 torr). Because the feed composition is greater than the azeotrope composition at the pressure of this column, the distillate composition approaches the minimum-boiling azeotrope at the top-tray pressure, and 99 mol% ethanol can be withdrawn as bottoms. The distillate is sent to the second column, which operates with a top-tray pressure of 106 kPa. The feed to this column has an ethanol content greater than that of the azeotrope at the pressure of the second column. Accordingly, the distillate composition approaches the azeotrope at the top-tray pressure, and 99 mol% benzene can be withdrawn as bottoms. The distillate is recycled to the first column. Design a pressure-swing distillation system for this separation.

SOLUTION

For the first column, which operates under vacuum, the reflux-drum and reboiler pressures are set at 26 and 40 kPa, respectively. For the second column, which operates just slightly above ambient pressure, the reflux-drum and reboiler pressures are set at 101.3 and 120 kPa, respectively. The bottoms compositions are specified at the required purities. The distillate composition for the first column is set at 37 mol% ethanol, slightly greater than the azeotrope

composition at 30 kPa. The distillate composition for the second column is set at 44 mol% ethanol, slightly less than the azeotrope composition at 106 kPa. With these composition specifications, material-balance calculations on ethanol and benzene give the following flow rates in moles per second.

Component	F	D_2	F_1	B_1	D_1	B_2
Ethanol	60.0	67.3	127.3	59.7	67.6	0.3
Benzene	30.0	85.6	115.6	0.6	115.0	29.4
Totals:	90.0	152.9	242.9	60.3	182.6	29.7

It is seen that the recycle molar flow rate, D_2 , is about 10% greater than that of the fresh feed, F .

Equilibrium-stage calculations for the two columns were made with the ChemSep program, using total condensers and partial reboilers. For Column 1, a number of runs were made in an attempt to find optimal feed-tray locations for the fresh feed and the recycle, using a reasonable reflux rate that avoided any near-pinch conditions. The selected design uses seven theoretical trays (not counting the partial reboiler), with the recycle stream, at a temperature of 68°C, sent to Tray 3 from the top and the fresh feed to Tray 5 from the top. A reflux ratio of 0.5 is sufficient to achieve specifications. The resulting liquid-phase composition profile is shown in Figure 11.24a, where the desirable lack of composition pinch points is observed. The McCabe-Thiele diagram for Column 1 is given in Figure 11.24b, where the three operating lines are evident and optimal-feed locations are indicated. Because of the azeotrope, the operating lines and equilibrium curve all lie below the 45° line. The condenser duty is 9.88 MW, while the reboiler duty is 8.85 MW. The bottoms temperature is 56°C. This column was sized with the ChemCAD program for sieve trays on 24-inch tray spacing and a 1-inch weir height to minimize pressure drop. The resulting diameter is 3.2 meters (10.5 ft). A tray efficiency of about 47% is predicted, making the required number of trays equal to 15.

For the design of Column 2, a similar procedure was used to establish the optimal feed tray, total trays, and reflux ratio. The selected design turned out to be a refluxed stripper with only three theoretical stages (not counting the partial reboiler). A reflux rate of only 25.5 mol/s achieves the product specifications, with most of the liquid traffic in the stripper coming from the feed. The resulting liquid-phase composition profile is shown in Figure 11.25a, where,

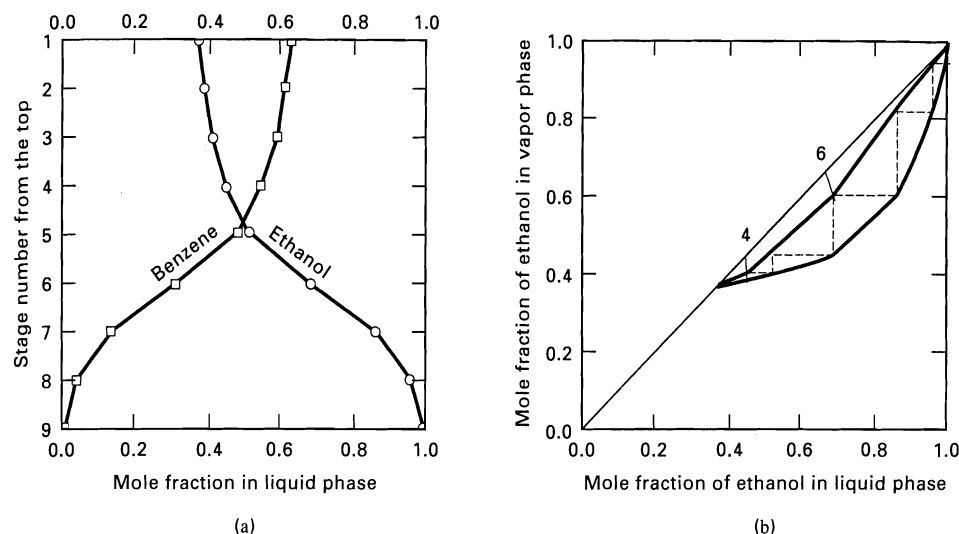


Figure 11.24 Computed results for Column 1 of pressure-swing distillation system in Example 11.5. (a) Liquid composition profiles. (b) McCabe-Thiele diagram.

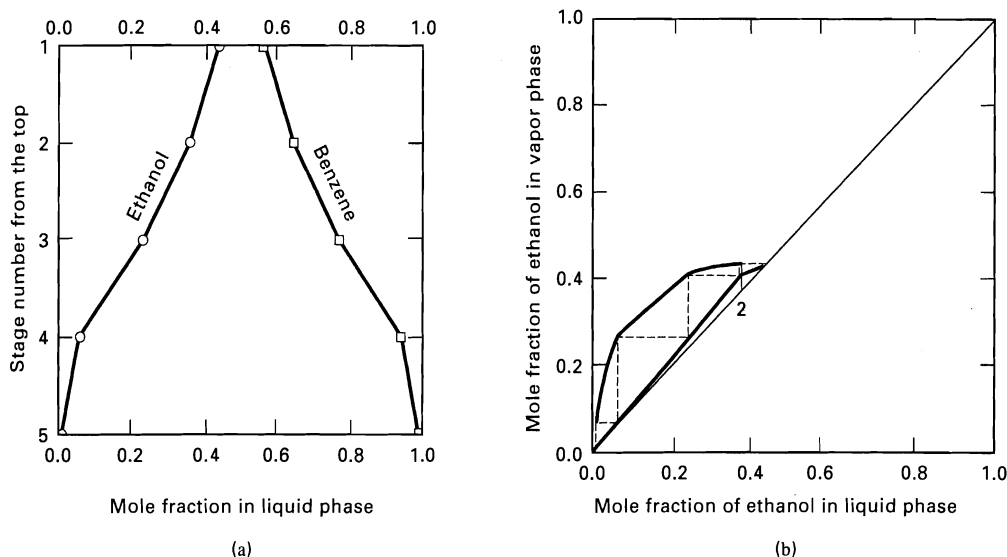


Figure 11.25 Computed results for Column 2 of pressure-swing distillation system in Example 11.5. (a) Liquid composition profiles. (b) McCabe-Thiele diagram.

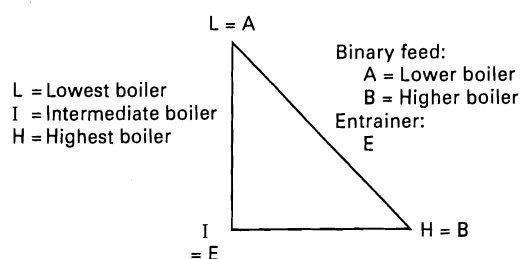
again, no composition pinches are evident. The McCabe-Thiele diagram for Column 2 is given in Figure 11.25b, where an optimal-feed location is indicated. The condenser duty is 6.12 MW, while the reboiler duty is 7.07 MW. The bottoms temperature is 84°C. This column was sized for the same conditions as Column 1, resulting in a column diameter of 2.44 meters (8 ft). A tray efficiency of 50% results in 6 actual trays.

11.5 HOMOGENEOUS AZEOTROPIC DISTILLATION

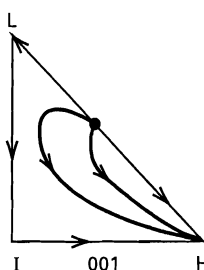
As discussed earlier, an azeotrope can be separated by extractive distillation, using a solvent that is higher boiling (less volatile) than the components in the feed, and that does not form an azeotrope with any of them. Alternatively, the separation can be made by *homogeneous azeotropic distillation*, using an *entrainer* that is not subject to such restrictions. Like extractive distillation, a sequence of two or three distillation columns is used. Alternatively, the sequence is a hybrid system that includes separation operations other than distillation, such as liquid-liquid extraction.

The conditions that a potential entrainer must satisfy for homogeneous azeotropic distillation to be feasible has been a subject of study by a number of investigators, including Doherty and Caldarola [16], Stichlmair, Fair, and Bravo [1], Foucher, Doherty, and Malone [14], Stichlmair and Herguieuja [18], Fidkowski, Malone, and Doherty [13], Wahnschafft and Westerberg [38], and Laroche, Bekiaris, Andersen, and Morari [39]. If it is assumed that a distillation boundary, if any, of a residue-curve map is straight or cannot be crossed, the conditions of Doherty and Caldarola apply. These conditions are based on the rule that for a potential entrainer, E, the two components, A and B, to be separated, or any product azeotrope, must lie in the same distillation region of the residue-curve map. Thus, a distillation boundary cannot be connected to the A-B azeotrope. Furthermore, A or B, but not both, must be a saddle. The maps suitable for a sequence that includes homogeneous azeotropic distillation together with ordinary distillation are classified into the five groups illustrated in Figure 11.26a, b, c, d, and e. The figure for each group includes the applicable residue-curve maps and the sequence of separation columns used to

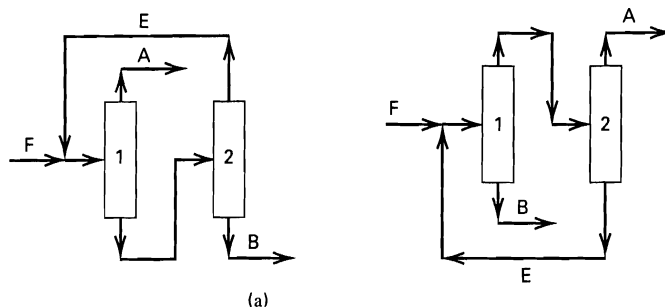
Residue-curve map arrangement



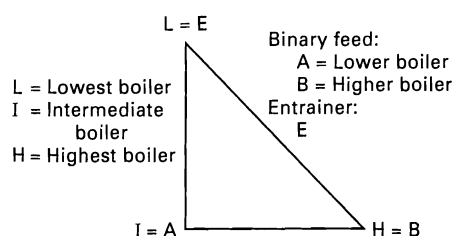
Applicable residue-curve map



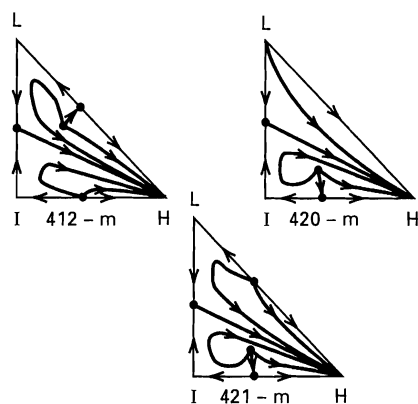
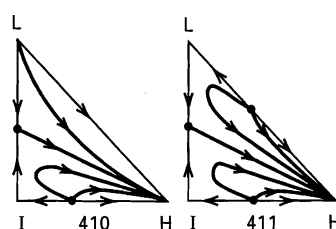
Sequences



Residue-curve map arrangement



Applicable residue-curve maps



Typical sequence

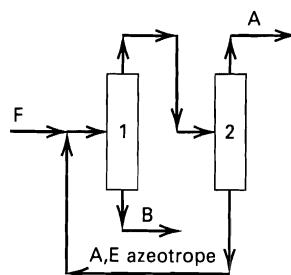


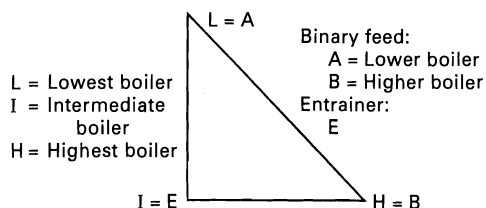
Figure 11.26 Residue-curve maps and distillation sequences for homogeneous azeotropic distillation. (a) Group 1: A and B form a minimum-boiling azeotrope, $I = E$, E forms no azeotropes. (b) Group 2: A and B form a minimum-boiling azeotrope, $L = E$, E forms a maximum-boiling azeotrope with A. (continued)

separate A from B and recycle the entrainer. For all groups, the residue-curve map is drawn in the manner of Doherty and Caldarola [16], with the lowest-boiling component, L, at the top vertex; the intermediate-boiling component, I, at the bottom-left vertex; and the highest-boiling component, H, at the bottom-right vertex. Component A is the lower-boiling component of the binary mixture and B the higher-boiling. For the first three groups, A and B form a minimum-boiling azeotrope; for the other two groups, they form a maximum-boiling azeotrope.

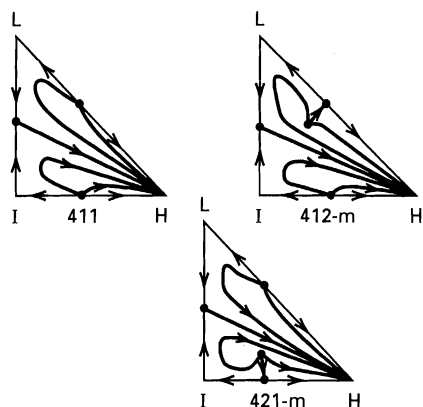
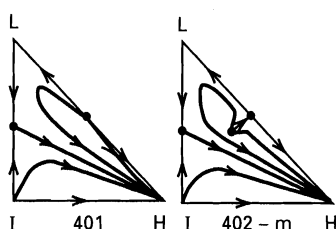
In Group 1, the intermediate boiler, I, is E, which forms no azeotropes with A and/or B. As shown in Figure 11.26a,

this case, like extractive distillation, involves no distillation boundary. Two sequences are shown, both of which assume that the fresh feed, F, of A and B, as fed to Column 1, is close to the azeotropic composition. Thus, this feed may be the distillate from a previous column used to produce the azeotrope from the original mixture of A and B. Either the *direct sequence*, in which Column 2 is fed by the bottoms from Column 1, or the *indirect sequence*, in which Column 2 is fed by the distillate from Column 1, may be used. In the first sequence, the entrainer is recovered as distillate from Column 2 and recycled to Column 1. In the second sequence, the entrainer is recovered as bottoms from Column 2 and

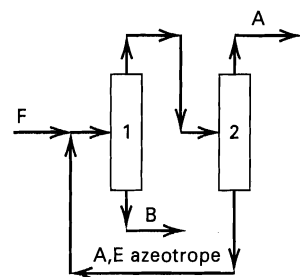
Residue-curve map arrangement



Applicable residue-curve maps

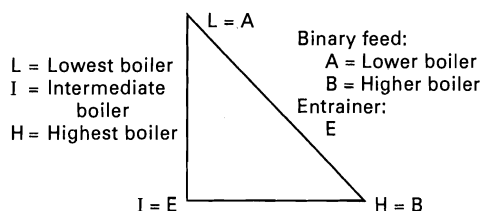


Typical sequence

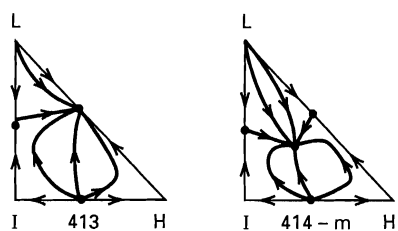
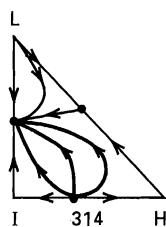


(c)

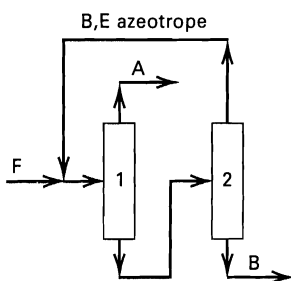
Residue-curve map arrangement



Applicable residue-curve maps



Sequence



(d)

Figure 11.26 (Continued) (c) Group 3: A and B form a minimum-boiling azeotrope, $I = E$, E forms a maximum-boiling azeotrope with A. (d) Group 4: A and B form a maximum-boiling azeotrope, $I = E$, E forms a minimum-boiling azeotrope with B.

recycled to Column 1. Although both sequences show the entrainer being combined with the fresh feed before being fed to Column 1, the fresh feed and the recycled entrainer can be fed to different trays to enhance the separation.

In Group 2, the low boiler, L, is E, which forms a maximum-boiling azeotrope with A. Entrainer E may also form a minimum-boiling azeotrope with B, and/or a minimum-boiling (unstable node) ternary azeotrope. Thus, in Figure 11.26b, any of the five residue-curve maps shown may apply. In all five cases, a distillation boundary exists,

which is directed from the maximum-boiling azeotrope of A-E to pure B, the high boiler. A feasible indirect or direct sequence is restricted to the subtriangle bounded by the vertices of pure components A, B, and the binary azeotrope of A-E. An example of an indirect sequence is included in Figure 11.26b. In this case, the azeotrope of A-E is recycled to Column 1 from the bottoms of Column 2. Alternatively, as shown in Figure 11.26c for Group 3, A and E may be switched to make A the low boiler and E the intermediate boiler, which again forms a maximum-boiling azeotrope

Benzene (nbp = 80.13°C) and cyclohexane (nbp = 80.64°C) form a minimum-boiling homogeneous azeotrope at 1 atm and 77.4°C that contains 54.2 mol% benzene. Thus, they cannot be separated by ordinary distillation at 1 atm. Instead, it is proposed to separate them by using acetone as the entrainer in the separation sequence shown in Figure 11.27. The fresh feed to the azeotropic column consists of 100 kmol/h of 75 mol% benzene and 25 mol% cyclohexane. Determine a feasible acetone-addition rate to the feed so that nearly pure benzene can be obtained as the bottoms product. Acetone (nbp = 56.14°C) forms a minimum-boiling azeotrope with cyclohexane (but not benzene) at 53.4°C and 1 atm at 74.6 mol% acetone. The residue-curve map at 1 atm is given in Figure 11.28.

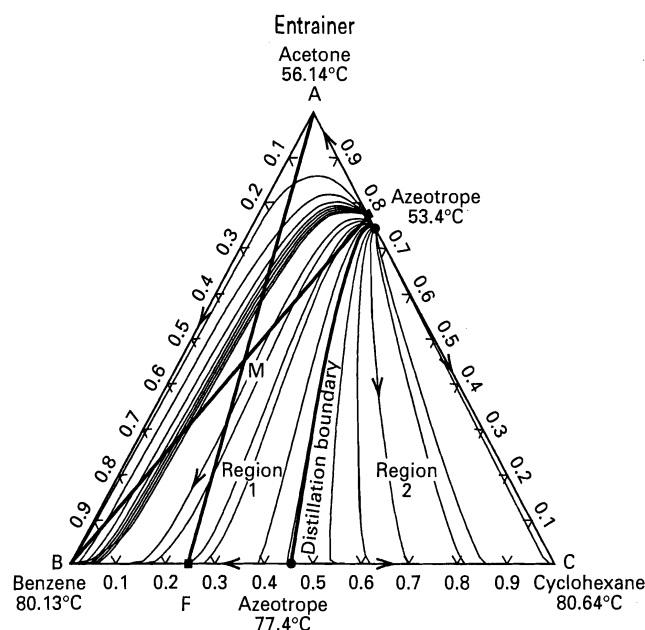


Figure 11.28 Residue-curve map for Example 11.6.

SOLUTION

The residue-curve map shows a slightly curved distillation boundary connecting the two azeotropes and dividing the diagram into distillation regions, 1 and 2. The fresh-feed composition is designated in Figure 11.28 by a filled-in box labeled F. If a straight line is drawn from F to the pure acetone apex, A, the mixture of the fresh feed and the acetone entrainer must lie somewhere in Region 1 on this line. Suppose that the 100 kmol/h of fresh feed is combined with an equal flow rate of entrainer. The mixing point, M, is

Table 11.4 Effect of Acetone-Entrainer Flow Rate on Benzene Purity for the Homogeneous-Azeotropic-Distillation Process of Example 11.6

Case	Acetone Flow Rate, kmol/h	Benzene Purity in Bottoms, %
1	50	88.69
2	75	94.21
3	100	99.781
4	125	99.779

located at the midpoint of the line connecting F and A. If a line is drawn from the benzene apex, B, through M and to the side of the triangle that connects the acetone apex to the cyclohexane apex, it does not cross the distillation boundary separating the two regions, but lies completely in Region 1. Thus, the separation into a nearly pure benzene bottoms and a distillate mixture containing mainly acetone and cyclohexane is possible. This is confirmed by calculations with the ASPEN PLUS process simulator for a column operating at 1 atm with 38 theoretical stages, a total condenser, a partial reboiler, a reflux ratio of 4, a bottoms product flow rate of 75 kmol/h (equivalent to the benzene flow rate in the feed to the column), and a bubble-point combined feed sent to Stage 19 from the top. The resulting product flow rates are listed in Table 11.4 as Case 3, where it is seen that a bottoms of 99.8 mol% benzene is achieved with a benzene recovery of the same value. A higher entrainer flow rate of 125 kmol/h, included in Table 11.4 as Case 4, is also successful in achieving high benzene-bottoms-product purity and recovery. However, if only 75 kmol/h (Case 2) or 50 kmol/h (Case 1) of entrainer is used, a nearly pure benzene bottoms is not achieved because of the distillation boundary restriction.

11.6 HETEROGENEOUS AZEOTROPIC DISTILLATION

The requirement, for a distillation sequence based on homogeneous azeotropic distillation, that A and B must lie in the same distillation region of the residue-curve map with entrainer E, is so restrictive that it is usually difficult, if not impossible, to find a feasible entrainer. The Group 1 map in Figure 11.26a requires that the entrainer not form an azeotrope but yet be the intermediate-boiling component, while the other two components form a minimum-boiling azeotrope. Such systems are rare, because most intermediate-boiling entrainers form an azeotrope with one or both of the other two azeotrope-forming components. The other four groups in Figure 11.26 all require that at least one maximum-boiling azeotrope be formed. However, such azeotropes are far less common than minimum-boiling azeotropes. The result is that industrial applications of distillation sequences based on homogeneous azeotropic distillation are not common.

An alternative technique that does find wide industrial application is heterogeneous azeotropic distillation, which is used to separate close-boiling binary mixtures and minimum-boiling binary azeotropes by employing an entrainer that

forms a binary and/or ternary heterogeneous azeotrope. As discussed in Section 4.3, a heterogeneous azeotrope is one involving more than one liquid phase. The overall composition of the liquid phases is equal to that of the vapor phase. Thus, all three phases have different compositions. The overhead vapor from the column is close to the composition of the heterogeneous azeotrope. When condensed, two liquid phases form in a decanter downstream of the condenser. After separation in the decanter, most or all of the entrainer-rich liquid phase is returned to the column as reflux, while most or all of the other liquid phase is sent to the next column for further separation. Because these two phases usually lie in different distillation regions of the residue-curve map, the restriction that usually dooms distillation sequences based on homogeneous azeotropic distillation is overcome. Thus, in heterogeneous azeotropic distillation, the components to be separated need not lie in the same distillation region.

Heterogeneous azeotropic distillation has been practiced for almost a century, first by batch and then by continuous processing. Two of the most widely used applications are (1) the use of benzene or one of a number of other entrainers to separate the minimum-boiling azeotrope of ethanol and

water, and (2) the use of ethyl acetate or one of a number of other entrainers to separate the close-boiling mixture of acetic acid and water. Other applications, cited by Widagdo and Seider [19], include dehydrations of isopropanol with isopropylether, *sec*-butyl-alcohol with *disec*-butyl-ether, chloroform with mesityl oxide, formic acid with toluene, and acetic acid with toluene. Also, dehydration of tanker-transported feedstocks such as styrene and benzene is a major application.

Consider the separation of the azeotrope of ethanol and water by heterogeneous azeotropic distillation. The two most widely used entrainers are benzene and diethyl ether. A number of other entrainers are feasible, including *n*-pentane, illustrated later in Example 11.7, and cyclohexane. In 1902, Young [40] discussed the use of benzene as an entrainer for the batch dehydration of ethanol, in perhaps the first application of heterogeneous azeotropic distillation. In 1928, Keyes obtained a patent [41] on a continuous process, discussed in a 1929 article [42]. A residue-curve map, computed by Bekiaris, Meski, and Morari [43] for the ethanol (E)–water (W)–benzene (B) system at 1 atm, using the UNIQUAC equation (with parameters from ASPEN PLUS) for liquid-phase activity coefficients, is shown in Figure 11.29. Superimposed on the residue-curve map is a bold-dashed binodal curve for the boundary of the two liquid-phase region. The normal boiling points of E, W, and B are 78.4, 100, and 80.1°C, respectively. The UNIQUAC equation predicts that homogeneous minimum-boiling azeotropes AZ1 and AZ2 are formed by E and W at 78.2°C and 10.0 mol% W, and by E and B at 67.7°C and 44.6 mol% E, respectively. A heterogeneous minimum-boiling azeotrope AZ3 is predicted for W and B at 69.3°C, with a vapor composition of 29.8 mol% W. The overall composition of the two liquid phases is the same as that of the vapor, but each liquid phase is almost pure. The

B-rich liquid phase is predicted to contain 0.55 mol% W, while the W-rich liquid phase contains only 0.061 mol% B. A ternary minimum-boiling heterogeneous azeotropic AZ4 is predicted at 64.1°C, with a vapor composition of 27.5 mol% E, 53.1 mol% B, and 19.4 mol% W. The overall composition of the two liquid phases of the ternary azeotrope is the same as that of the vapor, but a thin, dashed tie line through the AZ4 point shows that the benzene-rich liquid phase contains 18.4 mol% E, 79.0 mol% B, and 2.6 mol% W, while the water-rich liquid phase contains 43.9 mol% E, 6.3 mol% B, and 49.8 mol% W.

In Figure 11.29, the map is divided into three distillation regions by three, thick, solid-line distillation boundaries that each extend from the ternary azeotrope to a binary azeotrope. Each distillation region contains one pure component. Because the ternary azeotrope is the lowest-boiling azeotrope, it is an unstable node. Because all three binary azeotropes boil below the boiling points of the three pure components, the binary azeotropes are saddles and the pure components are stable nodes. Accordingly, all residue curves begin at the ternary azeotrope and terminate at a pure component apex. Liquid–liquid solubility is shown as a thick, dashed, curved line. However, this curve is not like the usual ternary solubility curve, because it is for isobaric, rather than isothermal, conditions. Superimposed on the distillation boundary that separates distillation regions 2 and 3 are thick dashes that represent the vapor composition in equilibrium with two liquid phases. The compositions of the two equilibrium liquid phases for a particular vapor composition are obtained from the two ends of the straight tie line that passes through the vapor composition point and terminates at the liquid solubility curve. The only tie line shown in Figure 11.29 is a thin, dashed line that passes through the ternary azeotrope. Other tie lines, which would represent other temperatures, could be added; however, in most heterogeneous azeotropic distillation operations, an attempt is made to restrict the formation of two liquid phases to just the decanter downstream of the condenser where the composition approaches the ternary azeotrope.

Figure 11.29 clearly shows how a distillation boundary is crossed by the tie line through AZ4 to form two liquid phases in the decanter. This phase split is utilized in the following manner by a typical azeotropic-tower operation for the dehydration of ethanol by benzene. The tower is treated as a column with no condenser, a main feed that enters a few trays below the top of the column, and the reflux of benzene-rich liquid as a second feed. The composition of the combined two feeds lies in distillation region 1. Thus, from the directions of the residue curves, the products of the tower can be a bottoms of nearly pure ethanol and an overhead vapor approaching the composition of the ternary azeotrope. When that vapor is condensed, phase splitting occurs to give a water-rich phase that lies in distillation region 3 and an entrainer-rich phase in distillation region 2. Thus, if the water-rich phase is sent to a reboiled stripper, the residue curves indicate that a nearly pure-water bottoms can be

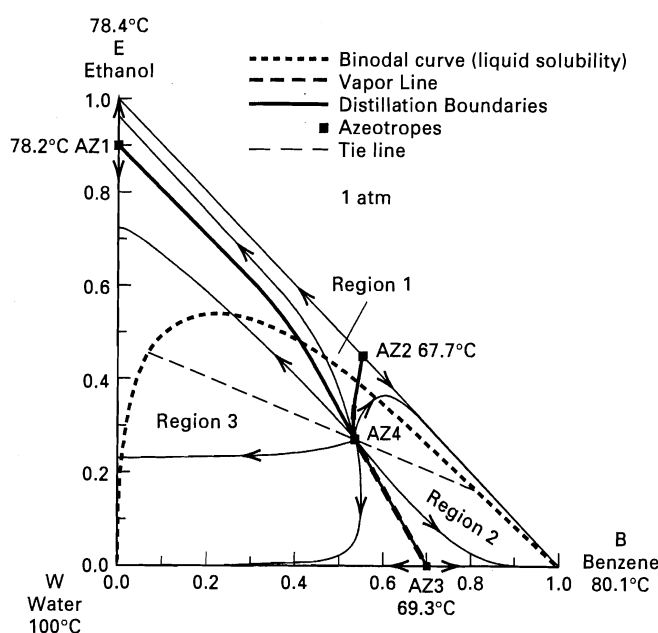


Figure 11.29 Residue-curve map for the ethanol–water–benzene system at 1 atm.

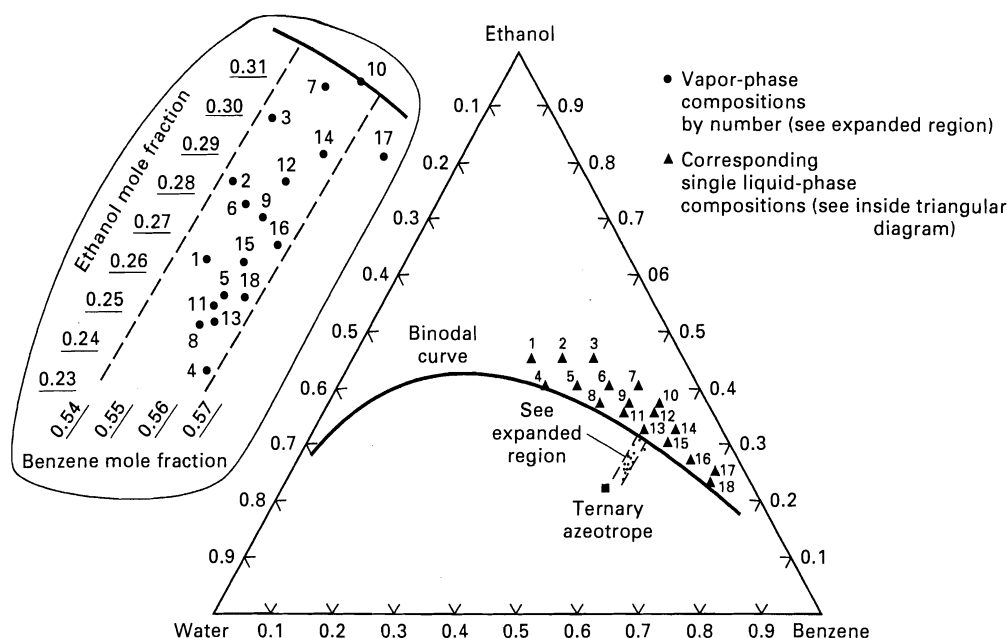


Figure 11.30 Overhead vapor compositions not in equilibrium with two liquid phases.

[From J. Prokopakis and W.D. Seider, *AIChE J.*, **29**, 49–60 (1983) with permission.]

produced, with the overhead vapor, rich in ethanol, recycled to the decanter. When the entrainer-rich phase in distillation region 2 is added to the main feed, which lies in distillation region 1, the overall composition lies in region 1.

It is preferable to restrict the formation of two liquid phases to the decanter. To avoid formation of two liquid phases on the top trays of the azeotropic tower, the composition of the vapor leaving the top tray must be such that the equilibrium liquid lies outside of the two-phase liquid region enclosed by the binodal curve and the base of the triangle in Figure 11.29. As shown in a comprehensive study by Prokopakis and Seider [44], vapor compositions that form two liquid phases when condensed, but are in equilibrium with only one liquid phase on the top tray, are restricted to a very small window, as shown in Figure 11.30. Furthermore, that window can only be achieved by adding to the reflux of entrainer-rich liquid phase a portion of the water-rich liquid phase or a portion of the condensed vapor prior to separation in the decanter.

A variety of column sequences for heterogeneous azeotropic distillation have been proposed. Three of these that utilize only distillation, taken from a study by Ryan and Doherty [45], are shown in Figure 11.31. Most common is the three-column sequence, in which an aqueous feed dilute in ethanol is first preconcentrated in Column 1 to obtain a nearly pure water-bottoms product and a distillate with composition approaching that of the binary azeotrope. The latter is fed to the azeotropic tower, Column 2, where nearly pure ethanol is recovered as the bottoms product and the tower is refluxed by most or all of the entrainer-rich liquid phase from the decanter. The water-rich phase, which contains ethanol and a small amount of entrainer, is sent to the entrainer-recovery column, which is a distillation column with both rectifying and stripping sections, or a stripper. The distillate from the recovery column is recycled to the azeotropic column. Alternatively, the distillate from

Column 3 could be recycled to the decanter. As shown in all three sequences of Figure 11.31, portions of either liquid phase from the decanter can be returned to the azeotropic tower or to the next column in the sequence to control phase splitting on the top trays of the azeotropic tower.

A four-column distillation sequence is shown in Figure 11.31b. The first column is identical to the first column of the three-column sequence of Figure 11.31a. The second column is the azeotropic column, which is fed by the near-azeotrope distillate of ethanol and water from Column 1 and by a recycle distillate of about the same composition from Column 4. The purpose of Column 3 is to remove, as distillate, the entrainer from the water-rich liquid phase leaving the decanter and recycle it back to the decanter. Ideally, the composition of this distillate is identical to that of the vapor distillate from Column 2. The bottoms from Column 3 is separated in Column 4 into a bottoms of nearly pure water, and a distillate that approaches the ethanol–water azeotrope and is therefore recycled to the feed to Column 2. A study by Pham and Doherty [46] found no advantage for the four-column sequence over the three-column sequence.

A novel two-column distillation sequence, due to Lynn and described by Ryan and Doherty [45], is shown in Figure 11.31c. The feed is sent to Column 2, which is a combined preconcentrator and entrainer recovery column. The distillate from this column is the feed to the azeotropic column. The bottoms from Column 1 is nearly pure ethanol, while Column 2 produces a bottoms of nearly pure water. For feeds that are very dilute in ethanol, Ryan and Doherty found that the two-column sequence has a lower investment cost, but a higher operating cost, than the three-column sequence. For feeds that are richer in ethanol, these two sequences are economically comparable.

The ethanol–benzene–water residue curve map of Figure 11.29 is only one of a number of different residue-curve maps that can lead to feasible distillation sequences that

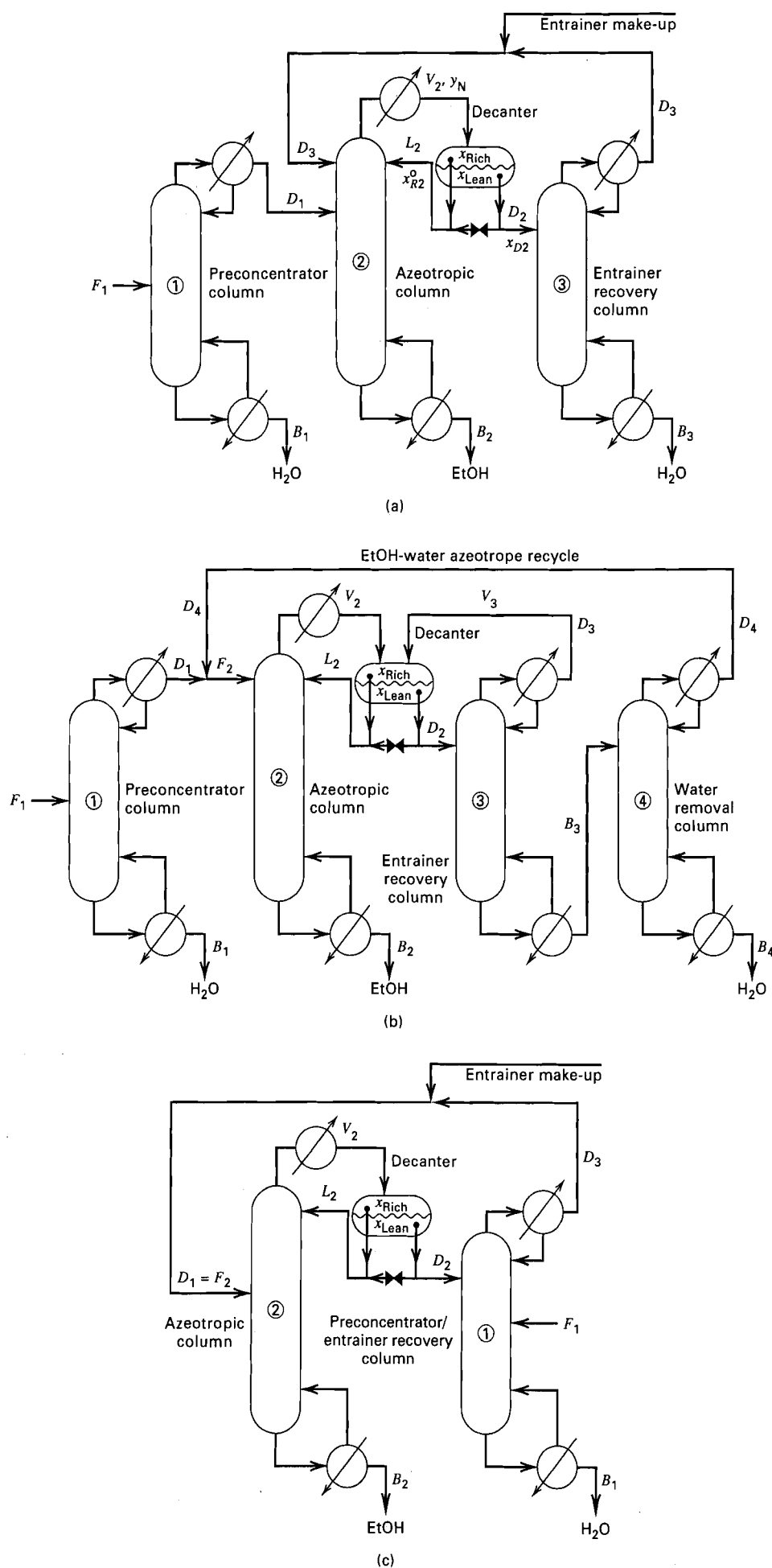


Figure 11.31 Distillation sequences for heterogeneous azeotropic distillation:
 (a) Three-column sequence;
 (b) four-column sequence;
 (c) two-column sequence.

[From P.J. Ryan and M.F. Doherty, *AIChE J.*, **35**, 1592–1601 (1989) with permission.]

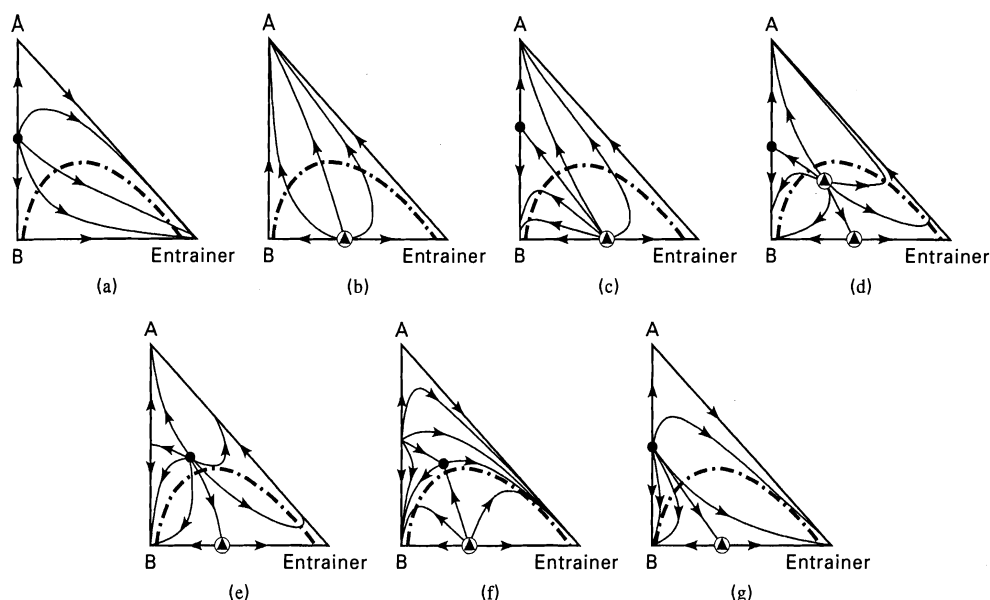


Figure 11.32 Residue-curve maps for heterogeneous azeotropic distillation that lead to feasible distillation sequences.

[H.N. Pham and M.F. Doherty, *Chem. Eng. Sci.*, **45**, 1845–1854 (1990) with permission.]

include heterogeneous azeotropic distillation. Pham and Doherty [46] note that a feasible entrainer is one that causes phase splitting over a portion of the three-component composition region, but does not cause the two components of the feed to be placed in different distillation regions. Figure 11.32 shows seven such maps, where the dash-dot lines are the liquid–liquid solubility (binodal) curves.

The convergence of rigorous calculations for heterogeneous azeotropic distillation columns by the methods described in Chapter 10 can be extremely difficult, especially when the convergence of the entire sequence is attempted. For calculation purposes, it is preferable to uncouple the

columns by using a residue-curve map to establish, by material-balance calculations, the flow rates and compositions of the feeds and products for each column. This procedure is illustrated for a three-column sequence in Figure 11.33, where the dash-dot lines separate the three distillation regions, the short-dash line is the liquid–liquid solubility curve, and the remaining lines are material-balance lines. Each column in the sequence can be computed separately. Even then, the calculations can be so sensitive, because of nonidealities in the liquid phase and possible phase splitting, that it may be necessary to use more robust methods. Among the most successful approaches for the most difficult cases are the boundary-value, tray-by-tray method of Ryan and Doherty [45], the homotopy-continuation method of Kovach and Seider [47], and the collocation method of Swartz and Stewart [48].

Multiplicity of Solutions

Solutions to mathematical models for operations of interest to chemical engineers are not always unique. The existence of multiple, steady-state solutions for the continuous, stirred-tank reactor (CSTR) has been known since at least 1922 and is described in detail in a number of textbooks on chemical reaction engineering. The existence of multiplicity in steady-state separation problems is a relatively new discovery. Gani and Jørgensen [49] define the following three types of multiplicity, all of which can, under certain conditions, occur in distillation simulations:

1. **Output multiplicity**, where all input variables are specified and more than one solution for the set of output variables is found. For example, for a distillation column, the feed condition, number of stages, feed-stage location, distillate flow rate, reflux ratio, type condenser and reboiler, and column-pressure profile might be specified and two or more sets of product compositions and column profiles found.

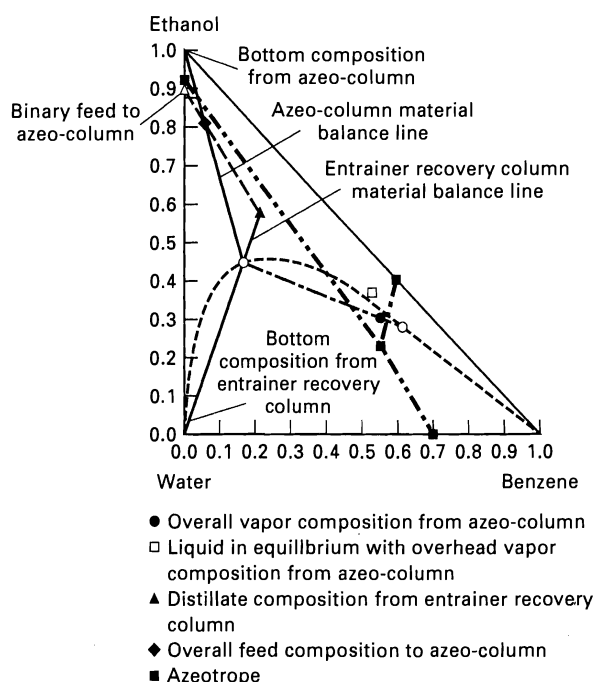


Figure 11.33 Material-balance lines for the three-column sequence of Figure 11.31a.

[From P.J. Ryan and M.F. Doherty, *AIChE J.*, **35**, 1592–1601 (1989) with permission.]

2. *Input multiplicity*, where one or more output variables are specified and multiple solutions are found for the unknown input variables.
3. *Internal-state multiplicity*, where multiple sets of internal conditions or profiles are found for the same values of the input and output variables.

Of particular interest here is output multiplicity for azeotropic distillation, which was first discovered by Shewchuk [50] in 1974. With different starting guesses, he found two steady-state solutions for the dehydration of ethanol by heterogeneous azeotropic distillation with benzene. In a more detailed study for the same system, Magnussen, Michelsen, and Fredenslund [51] found, with difficulty, for a rather narrow range of ethanol flow rate in the top feed to the column, three steady-state solutions, two of which are stable. The unstable solution can not be achieved in an operating plant. A similar multiplicity was found when pentane was used as the entrainer. One of the two stable solutions predicts a far purer ethanol bottoms product than the other stable solution. Thus, from a practical standpoint it is important to obtain all stable solutions when more than one exists. Subsequent studies, some contradictory, show that multiple solutions usually persist only over a narrow range of distillate- or bottoms-flow rate specifications, but may exist over a wide range of reflux rate provided that a sufficient number of stages are present. Composition profiles of five multiple solutions found by Kovach and Seider [47] for a 40-tray ethanol–water–benzene heterogeneous azeotropic distillation are shown in Figure 11.34. The variation in the profiles is extremely large. Again, when multiple solutions exist, it is important to locate them. Unfortunately, the use of current steady-state, computer-aided process design and simulation programs to find multiple solutions is fraught with a number of difficulties because: (1) azeotropic columns are difficult to converge to even one solution, (2) multiple solutions may exist only in a very restricted range, (3) the multiple solutions can only be found

in these programs by changing the initial guesses of the composition profiles, and (4) the choice of activity-coefficient correlation and interaction parameters can be crucial. Accordingly, the best results have been obtained when more advanced techniques such as continuation and bifurcation analysis are employed. These methods are described and applied by Kovach and Seider [47], Widagdo and Seider [19], Bekiaris, Meski, Radu, and Morari [52], and Bekiaris, Meski, and Morari [43]. The last two articles provide reasons why multiple solutions occur in homogeneous and heterogeneous azeotropic distillation.

EXAMPLE 11.7

Design and economic studies by Black and Ditsler [53] and Black, Golding, and Ditsler [54] show that *n*-pentane is a superior entrainer for the dehydration of ethanol. Like benzene, *n*-pentane forms a minimum-boiling heterogeneous ternary azeotrope with ethanol and water. Design a separation system for the dehydration of 16.8176 kmol/h of 80.937 mol% ethanol and 19.063 mol% water as a liquid at 344.3 K and 333 kPa, using *n*-pentane as an entrainer, to produce 99.5 mol% ethanol, and water with less than 100 ppm (by weight) of combined ethanol and *n*-pentane.

SOLUTION

A heterogeneous azeotropic distillation process for this ternary system has been studied extensively by Black [55], who proposed the two-column process flow diagram shown in Figure 11.35. The process consists of an 18-equilibrium-stage heterogeneous azeotropic distillation column (C-1) equipped with a total condenser and a partial reboiler, a decanter (D-1), a 4-equilibrium-stage reboiled stripper (C-2), and a condenser (E-1) to condense the overhead vapor from C-2. Each reboiler adds the equivalent of another equilibrium stage. Column C-1 operates at a bottoms pressure of 344.6 kPa with a column pressure drop of 13.1 kPa. Column C-2 operates at a top pressure of 308.9 kPa, with a column pressure drop of 3.0 kPa. These pressures permit the use of cooling water in the condensers. Purity specifications are placed on the bottoms products. The feed enters C-1 at Stage 3 from the top. The ethanol product is withdrawn from the bottom of C-1. A small *n*-pentane makeup stream, not shown in Figure 11.35, enters Stage 2 from the top. The overhead vapor from C-1 is condensed and sent to D-1, where a pentane-rich liquid phase and a water-rich liquid phase are formed. The pentane-rich phase is returned to C-1 as reflux, while the water-rich phase is sent to C-2, where the water is stripped of residual pentane and ethanol to produce a bottoms of the specified water purity. Twenty percent of the condensed vapor from C-2 is returned to D-1. To ensure that two liquid phases form in the decanter but not on the trays of C-1, the remaining 80% of the condensed vapor from C-2 is combined with the pentane-rich phase from D-1 for use as additional reflux to C-1. The specifications for the problem are included on Figure 11.35.

A very important step in the design of a heterogeneous azeotropic distillation column is the selection of a suitable method for predicting liquid-phase activity coefficients and the determination of the binary interaction parameters. The latter usually involves the regression of both vapor–liquid (VLE) and liquid–liquid (LLE) experimental equilibrium data for all binary pairs. If

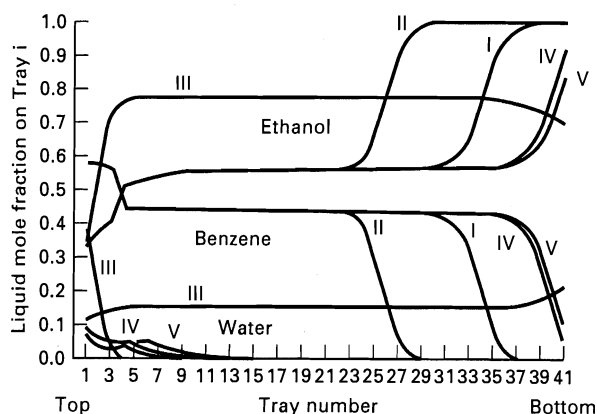


Figure 11.34 Five multiple solutions for a heterogeneous distillation operation.

[From J.W. Kovach III and W.D. Seider, *Computer Chem. Engng.*, **11**, 593 (1987) with permission.]

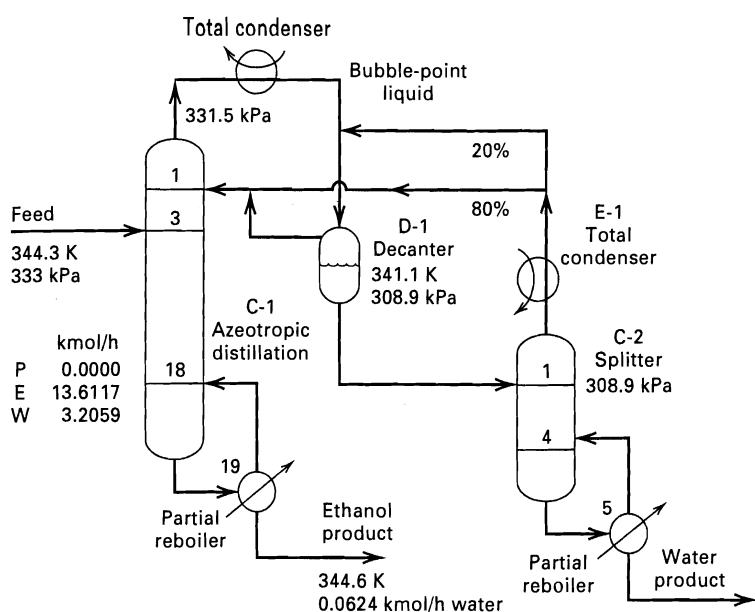


Figure 11.35 Process flow diagram for Example 11.7.
[From *Perry's Chemical Engineers' Handbook*, 6th ed., R.H. Perry and D.W. Green, Eds., McGraw-Hill, New York (1984) with permission.]

available, ternary data can also be included in the regression. Unfortunately, for most activity-coefficient prediction methods, it is difficult to simultaneously fit VLE and LLE data. Accordingly, often different binary-interaction parameters are used for the azeotropic column where VLE is important and for the decanter where LLE is important. This has been found to be particularly desirable for the ethanol–water–benzene system. For this example, however, the use of a single set of binary-interaction parameters with the modification by Black [56] of the Van Laar equation was deemed adequate. The binary-interaction parameters are listed by Black, Golding, and Ditsler [54].

The calculations were made with the Process simulation program of Simulation Sciences, Inc., using their rigorous distillation routine to model the columns and a three-phase-flash routine to model the decanter. Because the entrainer was internal to the system, except for a very small makeup rate, it was necessary to provide a reasonable initial guess for the component flow rates in the combined feed to the decanter. The guessed values in kilomoles per hour were 25.0 for *n*-pentane, 3.0 for ethanol, and 7.5 for water. The converged material balance is given in Table 11.5, where it is seen that the product specifications are met and approximately 22.6 kmol/h of *n*-pentane circulates through the top trays of the azeotropic distillation column. The computed condenser and

reboiler duties for Column C-1 are 1,116.5 and 1,135.0 MJ/h, respectively. The reboiler duty for Column C-2 is 486 MJ/h and the duty for Condenser E-1 is 438 MJ/h.

Because of the large effect of composition on liquid-phase activity coefficients, column profiles for azeotropic columns often show steep fronts. In Figure 11.36a to c, stage temperatures, total vapor and liquid flow rates, and liquid-phase compositions for Column C-1 vary only slightly from the reboiler (Stage 19) up to Stage 13. In this region, the liquid phase is greater than 99 mol% ethanol, whereas the *n*-pentane concentration slowly builds up from a negligible concentration in the bottoms to just less than 0.02 mol% at Stage 13. From Stage 13 to Stage 8, the *n*-pentane mole fraction in the liquid increases very rapidly to 53.8 mol%. In the same region, the temperature decreases sharply from 385.6 K to 348.4 K. Continuing up the column from Stage 8 to Stage 3, where the feed enters, the most significant change is the mole fraction of water in the liquid. Rather drastic changes in all variables take place about Stage 3. The large effects of *n*-pentane concentration on the relative volatility of water to ethanol, and of water concentration on the relative volatility of *n*-pentane to ethanol are shown in Figure 11.36d, where the variation over the column is about 10-fold for each pair.

No phase splitting occurs in either column, but two liquid phases of drastically different composition are formed and separated in the decanter. The light phase, which is almost twice the quantity of the heavy phase, is 95 mol% *n*-pentane, whereas the heavy phase is 90 mol% water. These extremely different compositions are due to the small amount of ethanol in the overhead vapor from C-1. Because of the high concentration of water in the feed to the stripper, C-2, the concentrations of ethanol and *n*-pentane in the liquid phase are quickly reduced to parts-per-million levels. Temperatures, vapor flow rates, and liquid flow rates in the stripper, C-2, are almost constant at 408 K, 15.6 kmol/h, and 12.4 kmol/h, respectively. Because of the large relative volatility of ethanol with respect to water (approximately 9) under the dilute ethanol conditions in C-2, the ethanol mole fraction decreases by almost an order of magnitude for each equilibrium stage. The extremely large relative volatility of *n*-pentane to water (more than 1,000) causes the *n*-pentane to be entirely stripped in just two stages.

Table 11.5 Converged Material Balance for Example 11.7

Stream	Flow rate, kmol/h			
	<i>n</i> -Pentane	Ethanol	Water	Total
C-1 feed	0.0000	13.6117	3.2059	16.8176
C-1 overhead	22.5565	2.1298	10.7269	35.4132
C-1 bottoms	0.0000	13.6117	0.0624	13.6741
C-1 reflux	22.5565	2.1298	7.5834	32.2697
D-1 <i>n</i> C5-rich	22.5500	1.0637	0.1129	23.7266
D-1 water-rich	0.0081	1.3326	12.4816	13.8223
C-2 overhead	0.0081	1.3326	9.3381	10.6788
C-2 bottoms	0.0000	0.0000	3.1435	3.1435

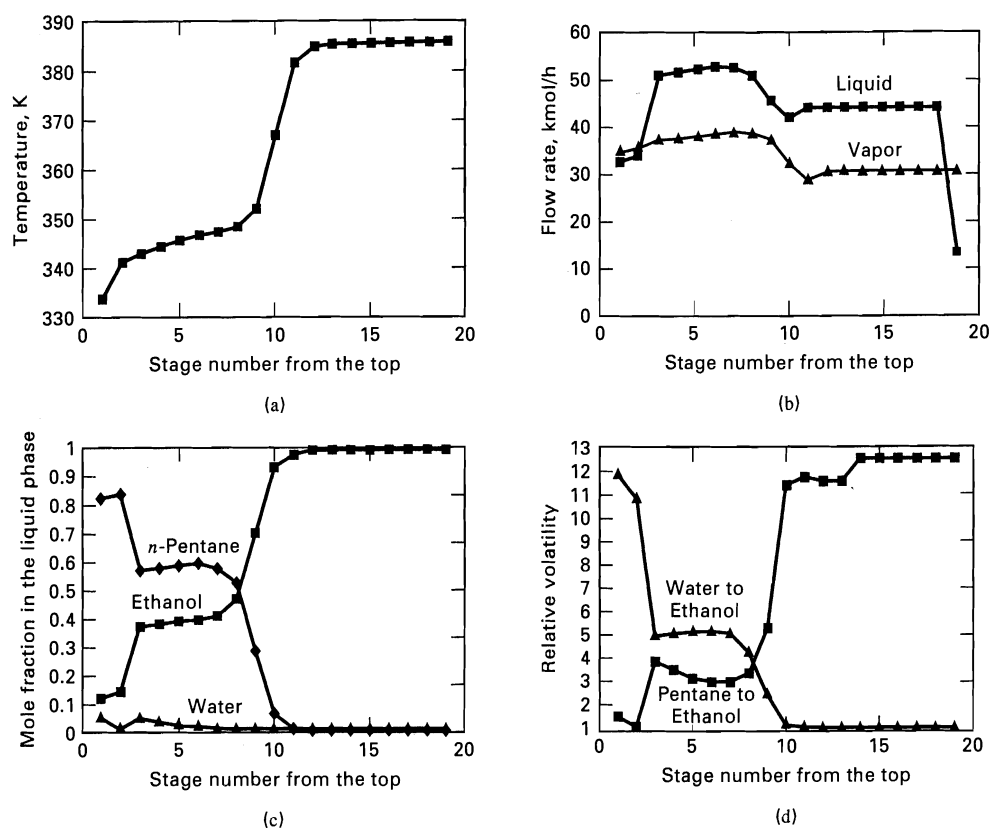


Figure 11.36 Results for azeotropic distillation column of Example 11.7. (a) Temperature profile. (b) Vapor and liquid traffic profiles. (c) Liquid-phase composition profiles. (d) Relative volatility profiles.

11.7 REACTIVE DISTILLATION

Reactive distillation involves simultaneous chemical reaction and distillation. The chemical reaction usually takes place in the liquid phase or at the surface of a solid catalyst in contact with the liquid phase. One general application of reactive distillation, described by Terrill, Sylvestre, and Doherty [57], is the separation of a close-boiling or azeotropic mixture of components A and B, where a reactive entrainer E is introduced into the distillation column. If A is the lower-boiling component, it is preferable that E be higher boiling than B and that it react selectively and reversibly with B to produce reaction product C, which also has a higher boiling point than component A and does not form an azeotrope with A, B, or E. Component A is removed from the distillation column as distillate, and components B and C, together with any excess E, are removed as bottoms. Components B and E are recovered from C in a separate distillation step, where the reaction is reversed to completely react C back to B and E; B is taken off as distillate, and E is taken off as bottoms and recycled to the first column. Terrill, Sylvestre, and Doherty [57] discuss the application of reactive entrainers to the separation of mixtures of *p*-xylene and *m*-xylene, whose normal boiling points differ by only 0.8°C, resulting in a relative volatility of only 1.029. Separation by ordinary distillation is impractical because, for example, to produce 99 mol% pure products from an equimolar feed, more than 500 theoretical stages are required. By reacting the *m*-xylene with a reactive entrainer such as *tert*-butylbenzene accompanied by a solid aluminum chloride catalyst, or

chelated sodium *m*-xylene dissolved in cumene, the stage requirements are drastically reduced.

Closely related to the use of reactive entrainers in distillation is the use of reactive absorbents in absorption, which finds wide application in industry. For example, sour natural gas is sweetened by the removal of hydrogen sulfide and carbon dioxide acid gases by absorption into aqueous alkaline solutions of mono- and di-ethanolamines. Fast and reversible reactions occur to form soluble-salt complexes such as carbonates, bicarbonates, sulfides, and mercaptans. The rich solution leaving the absorber is sent to a reboiled stripper where the reactions are reversed at higher temperatures to regenerate the amine solution as the bottoms and deliver the acid gases as overhead vapor.

A second application of reactive distillation involves taking into account undesirable chemical reactions that may occur during distillation. For example, Robinson and Gilliland [58] present an example involving the separation of cyclopentadiene from C_7 hydrocarbons. During distillation, cyclopentadiene dimerizes. The more volatile cyclopentadiene is taken overhead as distillate, but a small amount dimerizes in the lower section of the column and leaves in the bottoms with the C_7 s. Alternatively, the cyclopentadiene can be dimerized to facilitate its separation by distillation from other constituents of a mixture. Then the dicyclopentadiene is removed as bottoms from the distillation column. However, during distillation, it is also necessary to account for possible depolymerization to produce cyclopentadiene, which would migrate to the distillate.

The most interesting application of reactive distillation, and the only one considered in detail in this section, involves combining chemical reaction(s) and separation by distillation in a single distillation apparatus. This concept appears to have been first pronounced by Backhaus, who, starting in 1921 [59], obtained a series of patents for esterification reactions in a distillation column. This concept of continuous and simultaneous chemical reaction and distillation in a single vessel was verified experimentally by Leyes and Othmer [60] for the esterification of acetic acid with an excess of *n*-butanol in the presence of sulfuric acid catalyst to produce butyl acetate and water. This type of reactive distillation should be considered as an alternative to the use of separate reactor and distillation vessels whenever the following hold:

1. The chemical reaction occurs in the liquid phase, in the presence or absence of a homogeneous catalyst, or at the interface of a liquid and a solid catalyst.
2. Feasible temperature and pressure for the reaction and distillation are the same. That is, reaction rates and distillation rates are of the same order of magnitude.
3. The reaction is equilibrium-limited such that if one or more of the products formed can be removed, the reaction can be driven to completion; thus, a large excess of a reactant is not necessary to achieve a high conversion. This is particularly advantageous when recovery of the excess reagent is difficult because of azeotrope formation. For reactions that are irreversible, it is more economical to take the reactions to completion in a reactor and then separate the products in a separate distillation column. In general, reactive distillation is not attractive for supercritical conditions, for gas-phase reactions, and for reactions that must take place at high temperatures and pressures, and/or that involve solid reactants or products.

Careful consideration must be given to the configuration of the distillation column when employing reactive distillation. Important factors are feed entry and product-removal stages, the possible need for intercoolers and interheaters when the heat of reaction is appreciable, and the method for obtaining required residence time for the liquid phase. In the following ideal cases, it is possible, as shown by Belck [61] and others for several two-, three-, and four-component systems, to obtain the desired products without the need for additional distillation.

Case 1: The reaction $A \leftrightarrow R$ or $A \leftrightarrow 2R$, where R has a higher volatility than A. In this case, only a reboiled rectification section is needed. Pure A is sent to the column reboiler where all or most of the reaction takes place. As R is produced, it is vaporized, passing to the rectification column where it is purified. Overhead vapor from the column is condensed, with part of the condensate returned to the column as reflux. Chemical reaction may also take place in the column. If A and R form a maximum-boiling azeotrope, this configuration

is still applicable if, under steady-state conditions, the mole fraction of R in the reboiler is greater than the azeotropic composition.

Case 2: The reaction $A \leftrightarrow R$ or $2A \leftrightarrow R$, where A has the lower boiling point or higher volatility. In this case, only a stripping section is needed. The feed of pure liquid A is sent to the top of the column, from which it flows down the column, reacting to produce R. The column is provided with a total condenser and a partial reboiler. No product is withdrawn from the top of the column. Product R is withdrawn from the reboiler. This configuration requires close examination because, at a certain location in the column, chemical equilibrium may be achieved, and if the reaction is allowed to proceed below that point, the reverse reaction can occur.

Case 3: The reactions $2A \leftrightarrow R + S$ or $A + B \leftrightarrow R + S$, where A and B are intermediate in volatility to R and S, and R has the highest volatility. In this case, the feed enters an ordinary distillation column somewhere near the middle, with R withdrawn as distillate and S withdrawn as bottoms. If B is less volatile than A, then B may enter the column separately and at a higher level than A.

Commercial applications of reactive distillation include the following

1. The esterification of acetic acid with ethanol to produce ethyl acetate and water
2. The reaction of formaldehyde and methanol to produce methylal and water, using a solid acid catalyst, as described by Masamoto and Matsuzaki [62]
3. The esterification of acetic acid with methanol to produce methyl acetate and water, using sulfuric acid catalyst, as patented by Agreda and Partin [63], and described by Agreda, Partin, and Heise [64]
4. The reaction of isobutene with methanol to produce methyl-*tert*-butyl ether (MTBE), using a solid, strong-acid ion-exchange resin catalyst, as patented by Smith [65–67] and further developed by DeGarmo, Parulekar, and Pinjala [68]

The first widely studied example of reactive distillation is the esterification of acetic acid (A) with ethanol (B) to produce ethyl acetate (R) and water (S). The respective normal boiling points in °C are 118.1, 78.4, 77.1, and 100. Also, minimum-boiling binary homogeneous azeotropes are formed by B–S at 78.2°C with 10.57 mol% B, and by B–R at 71.8°C with 46 mol% B. A minimum-boiling, binary heterogeneous azeotrope is formed by R–S at 70.4°C with 24 mol% S, and a ternary, minimum-boiling azeotrope is formed by B–R–S at 70.3°C with 12.4 mol% B and 60.1 mol% R. Thus, this system is exceedingly complex and nonideal. A number of studies, both experimental and computational, have been published, many of which are cited by Chang and Seader [69], who developed a robust computational procedure for reactive distillation based on a homotopy continuation

method. More recently, other computational procedures, used in computer-aided process design programs, have been reported by Venkataraman, Chan, and Boston [70] and Simandl and Svrcek [71]. Kang, Lee, and Lee [72] obtained binary-interaction parameters for the UNIQUAC equation by fitting experimental data simultaneously for vapor-liquid equilibrium and liquid-phase chemical equilibrium.

In all of the computational procedures, a reaction-rate term must be added to the component material balance for a stage. For example, in the development of Chang and Seader [69], (10-58) for the Newton-Raphson procedure is modified to include a reaction-rate source term for the liquid phase, assuming that at each stage, the liquid phase is completely mixed:

$$M_{i,j} = l_{i,j}(1 + s_j) + v_{i,j}(1 + S_j) - l_{i,j-1} - v_{i,j+1} - f_{i,j} - (V_{LH})_j \sum_{n=1}^{NRX} v_{i,n} r_{j,n}, \quad i = 1, \dots, C \quad (11-17)$$

where

$(V_{LH})_j$ = the volumetric liquid holdup at stage j

$v_{i,n}$ = stoichiometric coefficient for component i and reaction n using the customary convention of positive values for products and negative values for reactants

$r_{j,n}$ = reaction rate for reaction n on stage j , as the increase in moles of a reference reactant per unit time per unit volume of liquid phase

NRX = number of reversible and irreversible chemical reactions.

Typically, each reaction rate is expressed in a power-law form with liquid molar concentrations (where the n subscript is omitted in the following equation):

$$r_j = \sum_{p=1}^2 k_p \prod_{q=1}^{NRC} c_{j,q}^m = \sum_{p=1}^2 A_p \exp\left(-\frac{E_p}{RT_j}\right) \prod_{q=1}^{NRC} c_{j,q}^m \quad (11-18)$$

where

$c_{j,q}$ = concentration of component q on stage j

k_p = reaction rate constant for the p th term, where $p = 1$ indicates the forward reaction and $p = 2$ indicates the reverse reaction; k_1 is positive and k_2 is negative

m = the exponent on the concentration

NRC = number of components in the power-law expression

A_p = preexponential (frequency) factor

E_p = activation energy

With (11-17) and (11-18), a reaction may be treated as irreversible ($k_2 = 0$), reversible (k_2 negative and not equal to zero), or at equilibrium. The last can be achieved by using very large values for the volumetric liquid holdup at each stage in the case of a single, reversible reaction, or by multiplying each of the two frequency factors, A_1 and A_2 , by the same large number, thus greatly increasing the forward and backward reactions, but maintaining the correct value for the chemical-

reaction equilibrium constant. For equilibrium reactions, it is important that the power-law expression for the backward reaction be derived from the power-law expression for the forward reaction and the reaction stoichiometry so as to be consistent with the expression for the chemical-reaction equilibrium constant. The volumetric liquid holdup for a stage, when using a trayed tower, depends on the active bubbling area of the tray, the height of the froth on the tray as influenced by the weir height, and the liquid-volume fraction of the froth. These factors are all considered in the section on pressure-drop calculations in Chapter 6. In general, the liquid backup in the downcomer is not included in the estimate of volumetric liquid holdup. When large holdups are necessary, bubble-cap trays are preferred because they do not allow weeping. When the chemical reaction is in the reboiler, a large liquid holdup can be provided. The following example illustrates the application of the computational procedure to the esterification of acetic acid with ethanol to produce ethyl acetate and water. In this example, the single, reversible chemical reaction is assumed to reach chemical equilibrium at each stage. Thus, no estimate of liquid holdup is needed. In a subsequent example, chemical equilibrium is not achieved and holdup estimates are made, which necessitates an estimate of tower diameter.

EXAMPLE 11.8

A reactive-distillation column containing the equivalent of 13 theoretical stages and equipped with a total condenser and partial reboiler is used to produce ethyl acetate (R) at 1 atm. A saturated liquid feed of 90 lbmol/h of acetic acid (A) enters Stage 2 from the top, while 100 lbmol/h of a saturated liquid of 90 mol% ethanol (B) and 10 mol% water (S) (close to the azeotropic composition) enters Stage 9 from the top. Thus, the acetic acid and ethanol are in stoichiometric ratio for esterification. The other specifications are a reflux ratio of 10 and a distillate rate of 90 lbmol/h in the hope that complete conversion to ethyl acetate (the low boiler) will occur. Kinetic data for the homogeneous reaction are given by Izarraraz, Bentzen, Anthony, and Holland [73], in terms of the rate law:

$$r = k_1 c_A c_B - k_2 c_R c_S$$

with $k_1 = 29,000 \exp(-14,300/RT)$ in L/(mol-min) with T in kelvins, and $k_2 = 7,380 \exp(-14,300/RT)$ in L/(mol-min) with T in kelvins. Because the activation energies for the forward and backward steps are the same, the chemical-equilibrium constant is independent of temperature and equal to $k_1/k_2 = 3.93$. Assume that chemical equilibrium is achieved at each theoretical stage. Thus, very large values of liquid holdup are specified for each stage. Binary-interaction parameters, for all six binary pairs, for predicting liquid-phase activity coefficients from the UNIQUAC equation are as follows, from Kang, Lee, and Lee [72]:

Components in Binary Pair, $i - j$	Binary Parameters	
	$u_{i,j}/R, K$	$u_{j,i}/R, K$
Acetic acid-ethanol	268.54	-225.62
Acetic acid-water	398.51	-255.84
Acetic acid-ethyl acetate	-112.33	219.41
Ethanol-water	-126.91	467.04
Ethanol-ethyl acetate	-173.91	500.68
Water-ethyl acetate	-36.18	638.60

Vapor-phase association of acetic acid is to be accounted for and the possible formation of two liquid phases is to be checked at each stage. Calculate the compositions of the distillate and bottoms products and determine the liquid-phase-composition and reaction-rate profiles.

SOLUTION

The calculations were made with the SCDS model (Newton-Raphson method) of the ChemCAD computer-aided process simulation program, where the total condenser is counted as the first stage. The only initial estimates provided were 163 and 198°F for the temperatures of the distillate and the bottoms, respectively. Convergence of the calculations required 17 iterations. A complete conversion to ethyl acetate was not achieved, as indicated by the following distillate and bottoms:

Component	Product Flow Rates, lbmol/h	
	Distillate	Bottoms
Ethyl acetate	49.52	6.39
Ethanol	31.02	3.07
Water	6.73	59.18
Acetic acid	2.73	31.36
Total	90.00	100.00

All four components appear in both products. The overall conversion to ethyl acetate is only 62.1%, with 88.6% of this going to

the distillate. The distillate is 55 mol% acetate, while the bottoms is 59.2 mol% water. Only small changes in these compositions occur when the feed locations are varied. Two important factors in the failure to achieve a high conversion and nearly pure products are (1) the highly nonideal nature of the quaternary mixture, accompanied by the large number of azeotropes, and (2) the tendency of the reverse reaction to occur on certain stages. The former effect is shown in Figure 11.37a, where the relative volatilities between ethyl acetate and water and between ethanol and water in the top section of the column are no greater than 1.25, making the separations difficult. The liquid-phase mole-fraction distribution is shown in Figure 11.37b, where, in the section between the two feed points, compositions change slowly despite the esterification reaction. In Figure 11.37c, the reaction-rate profile is quite unusual. Above the upper feed stage (now Stage 3), the reverse reaction is dominant. From that feed point down to the second feed entry (now Stage 10), the forward reaction dominates, but mainly at the upper feed stage. The reverse reaction is dominant for Stages 11–13, whereas the forward reaction dominates at Stages 14 and 15 (the reboiler). The largest extents of forward reaction occur at Stages 3 and 15. Even when the number of stages is increased to 60, with the reaction confined to Stages 25 to 35, the distillate contains an appreciable fraction of ethanol and the bottoms contains a substantial fraction of acetic acid. For this example, the development of a reactive-distillation scheme for achieving a high conversion and nearly pure products represents a significant challenge.

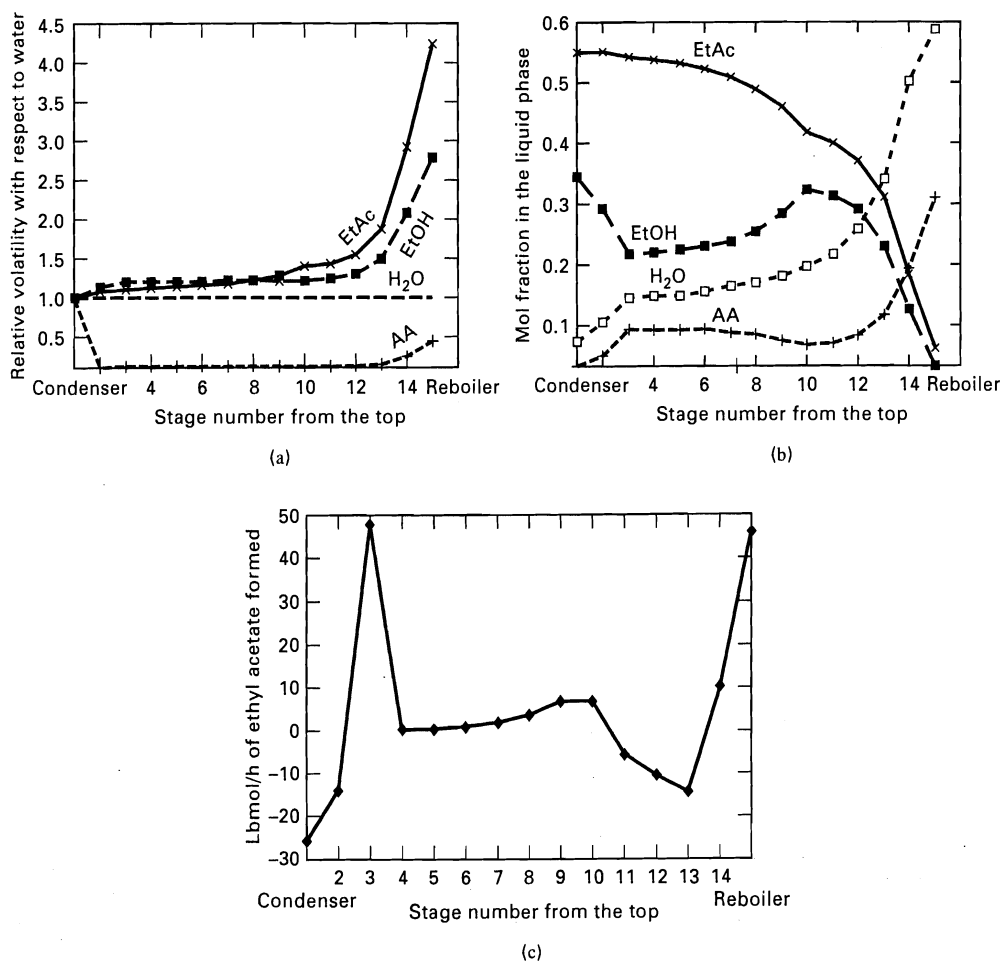


Figure 11.37 Profiles for reactive distillation in Example 11.8. (a) Relative volatility profile. (b) Liquid-phase mole-fraction profiles. (c) Reaction-rate profile.

EXAMPLE 11.9

Using thermodynamic and kinetic data from Rehfinger and Hoffmann [74] for the formation of MTBE from methanol (MeOH) and isobutene (IB), in the presence of *n*-butene (NB), both Jacobs and Krishna [75] and Nijhuis, Kerkhof, and Mak [76] computed, for catalyzed reactive distillation, with the ASPEN PLUS simulator, multiple solutions having drastically different isobutene conversions when the feed stage for methanol was varied. An explanation for these multiple solutions is given by Hauan, Hertzberg, and Lien [77].

Compute a converged solution for the following conditions, taking into account the kinetics of the reaction, but assuming vapor-liquid equilibrium at each stage. The distillation column has a total condenser, a partial reboiler, and 15 equilibrium stages in the column, which operates at 11 bar. Stages are counted down from the top, with the total condenser numbered stage 1 when using a process simulator, even though it is not an equilibrium stage. The mixed butenes feed, consisting of 195.44 mol/s of IB and 353.56 mol/s of NB enters stage 11 as a vapor at 350 K and 11 bar. The methanol at a flow rate of 215.5 mol/s enters stage 10 as a liquid at 320 K and 11 bar. The reflux ratio is 7 and the bottoms flow rate is set at 197 mol/s. The catalyst is provided only for Stages 4 through 11 (8 stages total), with 204.1 kg of catalyst per stage. The catalyst is a strong-acid ion-exchange resin with 4.9 equivalents of acid groups per kilogram of catalyst. Thus, the equivalents per stage are 1,000 or 8,000 for the eight stages. Compute the product compositions and column profiles using the RADFRAC model in ASPEN PLUS.

SOLUTION

The only chemical reaction considered is



with NB inert.

For the forward reaction, the rate law is formulated in terms of mole-fraction concentrations, instead of activities (products of activity coefficient and mole fraction) as in Rehfinger and Hoffmann [74]:

$$r_{\text{forward}} = 3.67 \times 10^{12} \exp(-92,440/RT) x_{\text{IB}}/x_{\text{MeOH}} \quad (1)$$

The corresponding backward rate law is

$$r_{\text{backward}} = 2.67 \times 10^{17} \exp(-134,454/RT) x_{\text{MTBE}}/x_{\text{MeOH}}^2 \quad (2)$$

where r is in moles per second per equivalent of acid groups, $R = 8.314 \text{ J/mol-K}$, T is in kelvins, and x_i is liquid mole fraction.

The Redlich-Kwong equation of state is used to estimate vapor-phase fugacities with the UNIQUAC equation to estimate the liquid-phase activity coefficients. The UNIQUAC binary interaction parameters are as follows, where it is very important to include the inert NB in the system by assuming it has the same parameters as IB and that the two butenes form an ideal solution. The parameters are defined as follows, with all $a_{ij} = 0$.

$$T_{ij} = \exp\left(-\frac{u_{ij} - u_{jj}}{RT}\right) = \exp\left(a_{ij} + \frac{b_{ij}}{T}\right) \quad (3)$$

Components in Binary Pair, ij	Binary Parameters	
	b_{ij} , K	b_{ji} , K
MeOH-IB	35.38	-706.34
MeOH-MTBE	88.04	-468.76
IB-MTBE	-52.2	24.63
MeOH-NB	35.38	-706.34
NB-MTBE	-52.2	24.63

The only initial guesses provided are temperatures of 350 and 420 K, respectively, for Stages 1 and 17; liquid-phase mole fractions of 0.05 for MeOH and 0.95 for MTBE leaving Stage 17; and vapor-phase mole fractions of 0.125 for MeOH and 0.875 for MTBE leaving Stage 17. The ASPEN PLUS input data for release 10.1 are listed in Table 11.6. The converged temperatures for Stages 1 and 17, respectively, are 347 and 420 K. Converged product flow rates are as follows:

Component	Flow Rate, mol/s	
	Distillate	Bottoms
MeOH	28.32	0.31
IB	7.27	1.31
NB	344.92	8.64
MTBE	0.12	186.74
Total	380.63	197.00

Table 11.6 ASPEN PLUS Input Data for Example 11.9

```

TITLE 'mtbe'
IN-UNITS MET VOLUME-FLOW='CUM/HR' ENTHALPY-FLO='MMKCAL/HR' &
HEAT-TRANS-C='KCAL/HR-SQM-K' PRESSURE=BAR TEMPERATURE=C &
VOLUME=CUM DELTA-T=C HEAD=METER MOLE-DENSITY='KMOL/CUM' &
MASS-DENSITY='KG/CUM' MOLE-ENTHALP='KCAL/MOL' &
MASS-ENTHALP='KCAL/KG' HEAT=MMKCAL MOLE-CONC='MOL/L' &
PDROP=BAR
DEF-STREAMS CONVEN ALL
DATABANKS PURECOMP / AQUEOUS / SOLIDS / INORGANIC / &
NOASPENPCD
PROP-SOURCES PURECOMP / AQUEOUS / SOLIDS / INORGANIC
COMPONENTS
  MEOH CH4O MEOH /
  IB C4H8-5 IB /
  NB C4H8-1 NB /
  MTBE C5H12O-D2 MTBE
FLOWSHEET
  BLOCK B1 IN=1 2 OUT=3 4

```

Table 11.6 (Continued)

```

PROPERTIES SYSOP11
PROP-REPLACE SYSOP11 UNIQ-RK
  PROP PHILMX PHILMX11
  PROP HLMX HLMX11
  PROP GLMX GLMX11
  PROP SLMX SLMX11
  PROP MUVMX MUVMX02
  PROP MULMX MULMX02
  PROP KVMX KVMX02
  PROP DV DV01
PROP-DATA UNIQ-1
  IN-UNITS MET VOLUME=FLOW='CUM/HR' ENTHALPY=FLO='MMKCAL/HR' &
    HEAT-TRANS=C='KCAL/HR-SQM-K' PRESSURE=BAR TEMPERATURE=K &
    VOLUME=CUM DELTA-T=C HEAD=METER MOLE-DENSITY='KMOL/CUM' &
    MASS-DENSITY='KG/CUM' MOLE-ENTHALP='KCAL/MOL' &
    MASS-ENTHALP='KCAL/KG' HEAT=MMKCAL MOLE-CONC='MOL/L' &
    PDROP=BAR
  PROP-LIST UNIQ
  BPVAL MEOH IB 0.0 35.38 0.0 0.0 0.0 1000.000
  BPVAL IB MEOH 0.0 -706.34 0.0 0.0 0.0 1000.000
  BPVAL MEOH MTBE 0.0 88.04 0.0 0.0 0.0 1000.000
  BPVAL MTBE MEOH 0.0 -468.76 0.0 0.0 0.0 1000.000
  BPVAL IB MTBE 0.0 -52.2 0.0 0.0 0.0 1000.000
  BPVAL MTBE IB 0.0 24.63 0.0 0.0 0.0 1000.000
  BPVAL MEOH NB 0.0 35.38 0.0 0.0 0.0 1000.000
  BPVAL NB MEOH 0.0 -706.34 0.0 0.0 0.0 1000.000
  BPVAL NB MTBE 0.0 -52.2 0.0 0.0 0.0 1000.000
  BPVAL MTBE NB 0.0 24.63 0.0 0.0 0.0 1000.000
  PROP-SET VLE PHIMX GAMMA PL SUBSTREAM=MIXED PHASE=V L
  PROP-SET VLLE PHMIX GAMMA PL SUBSTREAM=MIXED PHASE=V L1 L2
  STREAM 1
    SUBSTREAM MIXED TEMP=320 <K> PRES=11 NPHASE=1 PHASE=L
    MOLE-FLOW MEOH 215.5 <MOL/SEC> / IB 0. <MOL/SEC> / NB &
      0. <MOL/SEC> / MTBE 0. <MOL/SEC>
  STREAM 2
    SUBSTREAM MIXED TEMP=350 <K> PRES=11 NPHASE=1
    MOLE-FLOW MEOH 0. <MOL/SEC> / IB 195.44 <MOL/SEC> / NB &
      353.56 <MOL/SEC> / MTBE 0. <MOL/SEC>
  BLOCK B1 RADFRAC
    PARAM NSTAGE=17 MAXOL=50 MAXIL=50 ILMETH=NEWTON
    FEEDS 1 10 / 2 11 ON-STAGE
    PRODUCTS 3 1 L / 4 17 L
    P-SPEC 1 11
    COL-SPECS DP-COL=0 MOLE-RDV=0 MOLE-B=197 <MOL/SEC> MOLE-RR=7
    REAC-STAGES 4 11 r-1
    HOLD-UP 4 11 MASS-LHLDP=8000
    T-EST 1 350 <K> / 17 420 <K>
    X-EST 17 MEOH .05 / 17 MTBE .95
    Y-EST 17 MEOH .125 / 17 MTBE .875
    TRAY-REPORT TRAY-OPTION=ALL-TRAYS PROPERTIES=VLE VLLE
  STREAM-REPOR PROPERTIES=VLE VLLE
  REACTIONS R-1 REAC-DIST
    REAC-DATA 1 KINETIC CBASIS=MOLEFRAC
    REAC-DATA 2 KINETIC CBASIS=MOLEFRAC
    RATE-CON 1 PRE-EXP=3.67E12 ACT-ENERGY=92400 <KJ/KMOL>
    RATE-CON 2 PRE-EXP=2.67E17 ACT-ENERGY=134454 <KJ/KMOL>
    STOIC 1 MEOH -1 / IB -1 / NB 0 / MTBE 1
    STOIC 2 MTBE -1 / NB 0 / MEOH 1 / IB 1
    POWLAW-EXP 1 MEOH -1 / IB 1 / NB 0 / MTBE 0
    POWLAW-EXP 2 MTBE 1 / NB 0. / MEOH -2 / IB 0
    ;
    ;
    ;

```

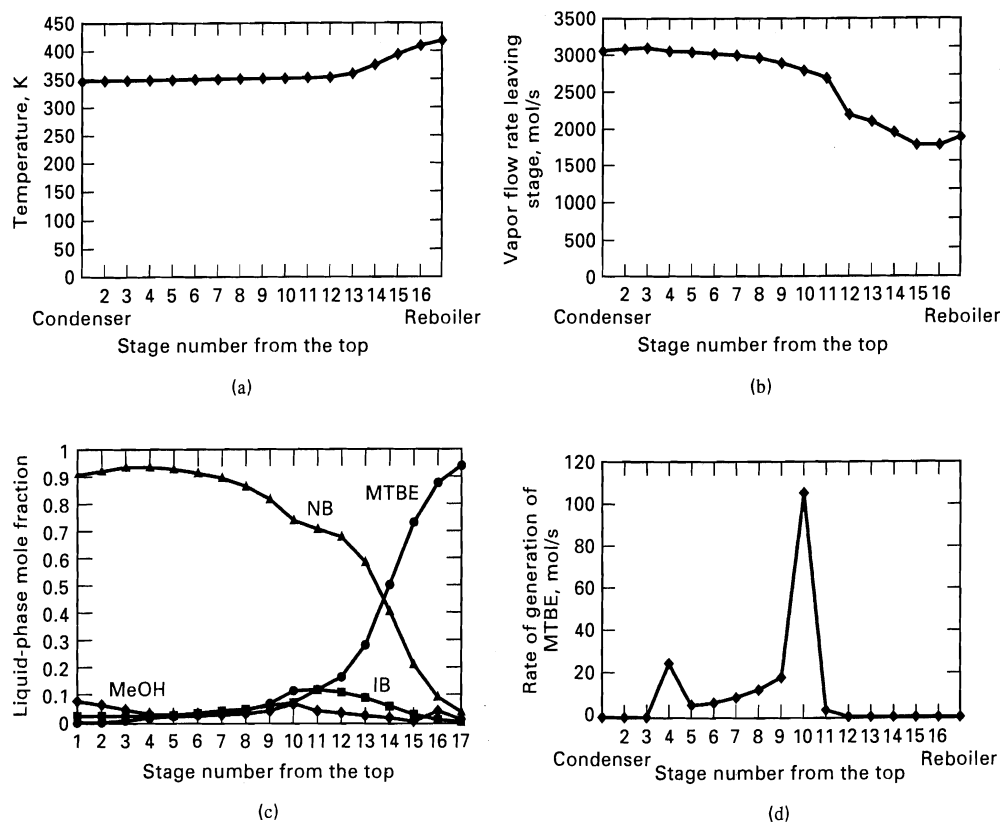


Figure 11.38 Profiles for reactive distillation in Example 11.9. (a) Temperature profile. (b) Vapor traffic profile. (c) Liquid-phase mole-fraction profile. (d) Reaction-rate profile.

The combined feeds to the reactive distillation contained a 10.3% mole excess of MeOH over IB. Therefore, IB was the limiting reactant and the preceding product distribution indicates that 95.6% of the IB, or 186.86 mol/s, reacted to form MTBE. The percent purity of the MTBE in the bottoms is 94.8%. Only 2.4% of the inert NB and 1.1% of the unreacted MeOH are found in the bottoms. The computed condenser and reboiler duties are, respectively, 53.2 and 40.4 MW.

Seven iterations were required to obtain a converged solution. The column profiles are in Figure 11.38. Figure 11.38a shows that most of the reaction occurs in a narrow temperature range of 348.6 to 353 K. The reaction temperature can be varied by adjusting the column pressure. Figure 11.38b shows that the vapor traffic in the column above the two feed entries changes by less than 11%, because of only small changes in temperature. As one moves down the column below the two feed entries, the temperature increases rapidly from 353 to 420 K, causing the vapor traffic to decrease by about 20%. In Figure 11.38c, the liquid-composition profiles show that the liquid is dominated by NB from the top stage down to Stage 13, thus drastically reducing the driving force for the reaction. Below Stage 11, the liquid quickly becomes richer in MTBE as the mole fractions of the other components decrease because of increasing temperature. In the section of the column above the reaction zone, the mole fraction of MTBE quickly decreases as one moves to the top stage. These changes are due mainly to the large differences between the K -values for MTBE and those for the other three components. The relative volatility of MTBE with respect to any of the other components ranges from about 0.24 at the top stage to about 0.35 at the bottom. Nonideality in the liquid phase influences mainly MeOH, whose liquid-phase activity coefficient varies from a high of 10 at Stage 5 to a low of 2.6 at Stage 17. This causes the unreacted MeOH to leave mainly with the NB in the distillate rather than with MTBE in the bottoms. The profile for the rate of reaction is shown in Figure 11.38d, where it is seen that the forward

reaction dominates on every stage of the reaction section. However, 56% of the reaction occurs on Stage 10, which is the MeOH feed stage. The least amount of reaction occurs on Stage 11.

As mentioned earlier, the literature indicates that the percent conversion of IB to MTBE will vary depending upon the stage to which the MeOH is fed. Furthermore, in the range of MeOH feed stages from about 8 to 11, both a low-conversion and a high-conversion solution can be computed. This is shown in Figure 11.39, where the high-conversion solutions are mainly in the 90+ % range, while the low-conversion solutions are all less than 10%. However, if activities are used in the rate expressions, rather than mole fractions, the low-conversion solutions are higher because of the large values for the activity coefficient for MeOH. The results in Figure 11.39 were computed starting with the MeOH feed entering Stage 2. The resulting profiles for this run were used as the initial guesses for the run with MeOH entering Stage 3. Subsequent runs were performed in a similar manner, increasing the

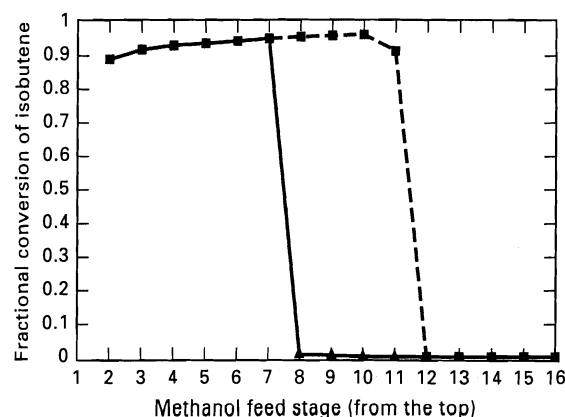


Figure 11.39 Effect of MeOH feed stage location on conversion of IB to MTBE.

MeOH feed stage by 1 each time and initializing with the results of the previous run. High-conversion solutions were obtained for each run until the MeOH feed stage was lowered to Stage 12, at which point the conversion decreased dramatically. Further lowering of the MeOH feed stage to Stage 16 also resulted in a low-conversion solution. However, when the direction of change to the MeOH feed stage was reversed starting from Stage 12, a low-conversion was obtained until the feed stage was decreased to Stage 9, at which point the conversion jumped back to the high-conversion result.

Huss et al. [101] present a detailed study of reactive distillation for the acid-catalyzed reaction of acetic acid and methanol to produce methyl acetate and water, including the effect of the side reaction of methanol dehydration, using simulation models with a comparison to experimental measurements. They consider both finite reaction rates and chemical equilibrium, coupled with phase equilibrium. The results include consideration of reflux limits and multiplicity of solutions (multiple steady states).

11.8 SUPERCRITICAL-FLUID EXTRACTION

Solute extraction from a liquid or solid mixture is usually accomplished with a liquid solvent, as discussed in Chapters 8 and 16, respectively, at conditions of temperature and pressure that lie substantially below the critical temperature and pressure of the solvent. Following the extraction step, the solvent and dissolved solute are subjected to a subsequent separation step, such as distillation, to recover the solvent for recycle and purify the solute.

In 1879, Hannay and Hogarth [78] reported that solid potassium iodide could be dissolved in ethanol, as a dense gas, at supercritical conditions of $T > T_c = 516 \text{ K}$ and $P > P_c = 65 \text{ atm}$. The iodide could then be precipitated from the ethanol by reducing the pressure. This process was later referred to as *supercritical-fluid extraction*, *supercritical-gas extraction*, *supercritical extraction* (SCE), *dense-gas extraction*, or *detraction* (a combination of distillation and extraction). By the 1940s, as chronicled by Williams [79], proposed practical applications of SCE began to appear in the patent and technical literature. Figure 11.40 shows the

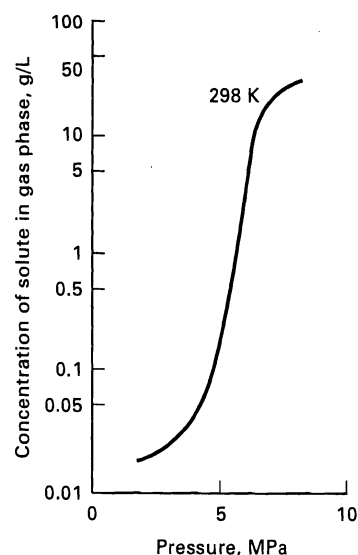


Figure 11.41 Effect of pressure on solubility of pICB in supercritical ethylene.

supercritical fluid region for CO_2 , which has a critical point of 304.2 K and 73.83 bar.

The solvent power of a compressed gas can undergo an enormous change in the vicinity of its critical point. Consider, for example, the solubility of *p*-iodochlorobenzene (pICB) in ethylene, as shown in Figure 11.41, at 298 K for pressures from 2 to 8 MPa. This temperature is 1.05 times the critical temperature of ethylene (283 K) and the pressure range straddles the critical pressure of ethylene (5.1 MPa). At 298 K, pICB is a solid (melting point = 330 K) with a vapor pressure of the order of 0.1 torr. At 2 MPa, if pICB formed an ideal-gas solution with ethylene, the mole fraction of pICB in the gas in equilibrium with pure, solid pICB would be extremely small at about 6.7×10^{-6} or a concentration of 0.00146 g/L. The experimental concentration from Figure 11.41 is 0.015 g/L, which is an order of magnitude higher because of nonideal-gas effects. If the pressure is increased from 2 MPa to almost the critical pressure at 5 MPa (an increase by a factor of 2.5),

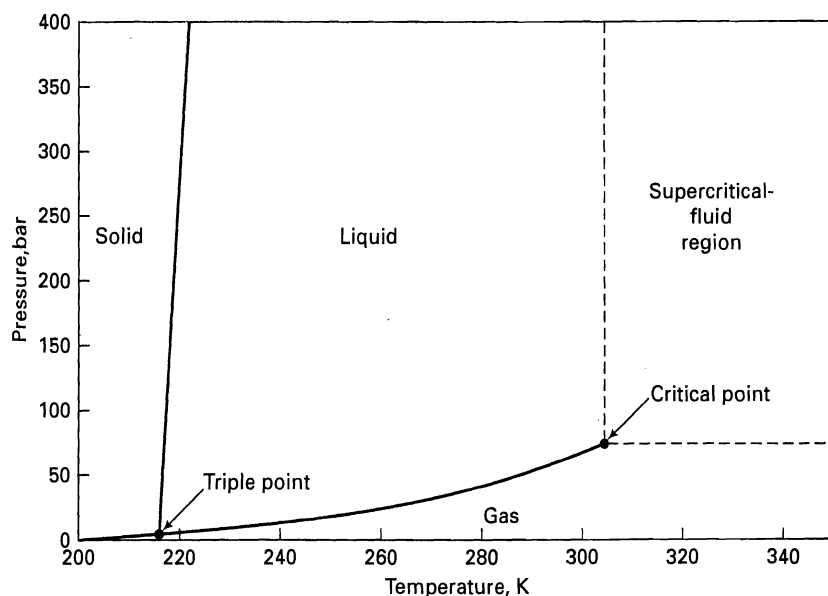


Figure 11.40 Supercritical fluid region for CO_2 .

the equilibrium concentration of pICB is increased about 10-fold to 0.15 g/L. At 8 MPa, the concentration begins to level out at 40 g/L, which is 2,700 times higher than predicted from the vapor pressure for an ideal-gas solution. It is this dramatic increase in solubility of a solute at near-critical conditions of a solvent that makes SCE of interest.

Why such a dramatic increase in solvent power? The explanation lies in the change that occurs to the solvent density while the solubility of the solute increases. A pressure–enthalpy diagram for ethylene is shown in Figure 11.42, which includes the specific volume as a parameter, from which the density can be determined as the reciprocal. The range of variables and parameters straddles the critical point of ethylene. The density of ethylene compared to the solubility of pICB is as follows at 298 K:

Pressure, MPa	Ethylene Density, g/L	Solubility of pICB, g/L
2	25.8	0.015
5	95	0.15
8	267	40

Although there is far from a 1:1 correspondence in the increase of pICB solubility with density for ethylene over this range of pressure, there is a meaningful correlation. As the pressure is increased, closer packing of the solvent molecules allows them to surround and trap solute molecules. This phenomenon is most dramatic and useful at reduced temperatures from about 1.01 to 1.12.

Two other effects in the supercritical region are favorable for SCE. It will be recalled that the molecular diffusivity of a solute in an ambient-pressure gas is about four orders of magnitude higher than for a liquid. For a near-critical fluid, the diffusivity of solute molecules is usually one to two orders of magnitude higher than in a normal liquid solvent, thus resulting in a lower mass-transfer resistance in the solvent phase than might be expected. In addition, the viscosity of the supercritical fluid is about an order of magnitude less than that of a normal liquid solvent.

Many patents have been issued, proposals prepared, and experimental studies conducted on SCE as a possible alternative for distillation, enhanced distillation, and liquid–liquid extraction. However, in general, when these other techniques are feasible, SCE usually cannot compete economically because of high solvent-compression costs to reach near-critical pressure. SCE is most favorable for the extraction of small amounts of large, relatively nonvolatile solutes in solid or liquid mixtures. Such applications are cited by Williams [79] and McHugh and Krukonis [80].

Solvent selection depends on the composition of the feed mixture. If only the chemical(s) to be extracted is (are) soluble in a potential solvent, then high solubility is a key factor. However, if a chemical besides the desired solute is soluble in the potential solvent, then the selectivity of the solvent becomes as important as solubility. A number of light gases and other low-molecular-weight chemicals, including the

following, have received attention as solvents for SCE:

Solvent	Critical Temperature, K	Critical Pressure, MPa	Critical Density, kg/m ³
Methane	192	4.60	162
Ethylene	283	5.03	218
Carbon dioxide	304	7.38	468
Ethane	305	4.88	203
Propylene	365	4.62	233
Propane	370	4.24	217
Ammonia	406	11.3	235
Water	647	22.0	322

Those solvents with a critical temperature below 373 K have been well studied. A particularly desirable solvent, particularly for the extraction of undesirable, valuable, or heat-sensitive chemicals from natural products such as foods, is carbon dioxide, which has a moderate critical pressure, a high critical density, and a critical temperature close to ambient temperature. Carbon dioxide is nonflammable, non-corrosive, nontoxic in low concentrations, readily available, inexpensive, and safe. Also, supercritical carbon dioxide has a relatively low viscosity and high molecular diffusivity. Separation of carbon dioxide from the solute is often possible by simply reducing the extract pressure. According to Williams [79], supercritical carbon dioxide has been used to extract caffeine from coffee, hops oil from beer, piperine from pepper, capsaicin from chilis, oil from nutmeg, and nicotine from tobacco.

Carbon dioxide is not a suitable solvent for all potential applications. McHugh and Krukonis [81] cite the energy crisis of the 1970s that led to substantial research on an energy-efficient separation of ethanol and water. The primary goal, which was to break the ethanol–water azeotrope, was not achieved by SCE with carbon dioxide because, although supercritical carbon dioxide has unlimited capacity to dissolve pure ethanol, water is also dissolved in significant amounts. A liquid–supercritical-fluid phase diagram for the ethanol–water–carbon dioxide ternary system at 308.2 K and 10.08 MPa, based on the experimental data of Takishima, Saiki, Arai, and Saito [82], is shown in Figure 11.43. These conditions correspond to $T_r = 1.014$ and $P_r = 1.366$ for carbon dioxide. For the binary mixture of water and carbon dioxide, two phases exist; a water-rich phase with about 2 mol% carbon dioxide and a carbon dioxide-rich phase with about 1 mol% water. Ethanol and carbon dioxide are completely soluble in each other. Ternary mixtures containing more than 40 mol% ethanol are completely miscible. If a near-azeotropic mixture of ethanol and water, say, 85 mol% ethanol and 15 mol% water, is extracted by carbon dioxide at the conditions of Figure 11.43, a mixing line drawn between this composition and a point for pure carbon dioxide does not appear to cross into the two-phase region. That is, regardless of the amount of solvent used, both water and ethanol are completely soluble in the carbon dioxide and no separation is possible at these temperatures and pressures. Alternatively, consider an ethanol–water broth from a

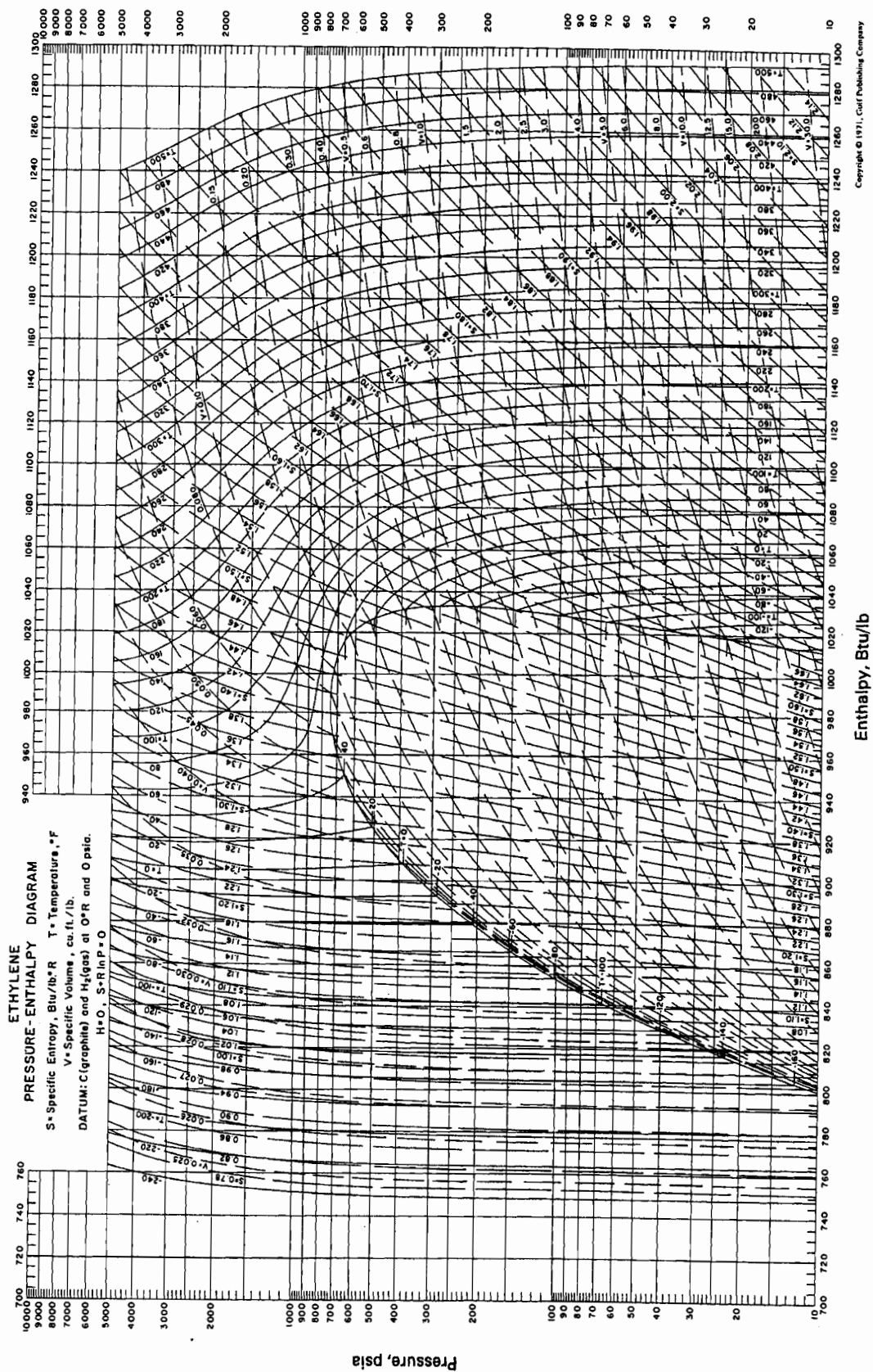


Figure 11.42 Pressure-enthalpy diagram for ethylene.
 [From K.E. Starling, "Fluid Thermodynamic Properties for Light Petroleum Systems," Gulf Publishing, Houston (1973) reprinted with permission.]

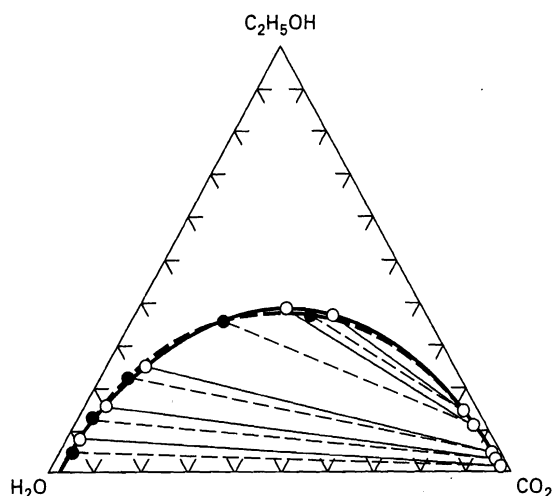


Figure 11.43 Liquid–fluid equilibria for CO_2 – $\text{C}_2\text{H}_5\text{OH}$ – H_2O at 308–313.2 K and 10.1–10.34 MPa.

fermentation reactor with 10 wt% (4.17 mol%) ethanol. If this mixture is extracted with supercritical carbon dioxide, complete dissolution will not occur and a modest degree of separation of ethanol from water can be achieved, as shown in the next example. The separation can be further enhanced by the use of a cosolvent, such as glycerol, that improves the selectivity, as shown by Inomata, Kondo, Ari, and Saito [83].

When CO_2 is used as a solvent, it must be recovered and recycled. Three schemes discussed by McHugh and Krukoni [81] are shown in Figure 11.44. In the first scheme, shown for the separation of ethanol and water, the ethanol–water feed is pumped as a liquid to the pressure of the extraction column, where it is contacted with supercritical carbon dioxide. The raffinate leaving the extractor at the bottom is enriched with respect to water and can be sent to another part of the plant for further processing. The extract stream, which leaves from the top of the extractor and contains most of the carbon dioxide, some ethanol, and a smaller amount of water, is expanded across a valve to a lower pressure. In a flash drum downstream of the valve, ethanol–water condensate is collected and the CO_2 -rich gas is recycled through a gas compressor back to the extractor. However, unless the pressure is greatly reduced across the valve, resulting in large compression costs, little of the ethanol is condensed.

A second CO_2 recovery scheme, due to de Filippi and Vivian [84], is shown in Figure 11.44b. The flash drum is replaced by a high-pressure distillation column, which operates at a pressure just below the pressure of the extraction column to produce a CO_2 -rich distillate and an ethanol-rich bottoms. The distillate is compressed and recycled through the reboiler and back to the extractor. Both the raffinate and the distillate are flashed to recover dissolved CO_2 . This scheme, although more complicated than the first, is more versatile.

A third CO_2 recovery scheme, due to Katz et al. [85] for the decaffeination of coffee, is shown in Figure 11.44c. In the extractor, green, wet coffee beans are mixed with

supercritical CO_2 to extract caffeine. The extract is sent to a second extraction column, where the caffeine is extracted with water. The CO_2 -rich raffinate from this column is recycled through a compressor (not shown) back to the first extraction column, from which the decaffeinated coffee leaves from the bottom and is sent to a roasting tower. The caffeine-rich water leaving the second column is sent to a reverse-osmosis unit, where the water is purified and recycled through a pump (not shown) to the water column. All three separation steps operate at high pressure. The concentrated caffeine–water mixture leaving the osmosis unit is sent to a crystallizer to produce caffeine crystals.

Multiple equilibrium stages in a countercurrent-flow contactor are generally needed to obtain the desired extent of extraction. A major problem in determining the number of stages required is the estimation of liquid–supercritical fluid phase-equilibrium constants. Most commonly, cubic-equation-of-state methods, such as the Soave–Redlich–Kwong (SRK) or Peng–Robinson (PR) equations, are used, but they have two shortcomings. First, their accuracy diminishes in the critical region of the solvent. Second, if the feed contains polar components that form a nonideal-liquid mixture, an appropriate mixing rule, such as that of Wong and Sandler [86], that provides a correct bridge between equation-of-state methods and activity-coefficient methods must be employed.

As discussed in Section 2.5, the SRK and PR equations for pure components both contain two parameters, a and b , that are computed from critical constants. The SRK and PR equations are extended to liquid or vapor mixtures by a *mixing rule* for computing values of a_m and b_m for the mixture from values for the pure components. The simplest mixing rule, due to van der Waals, is:

$$a_m = \sum_{i=1}^C \sum_{j=1}^C x_i x_j a_{ij} \quad (11-19)$$

$$b_m = \sum_{i=1}^C \sum_{j=1}^C x_i x_j b_{ij} \quad (11-20)$$

where x is a mole fraction in the vapor or liquid mixture. Although these two mixing-rule equations are identical in form, the following *combining rules* for a_{ij} and b_{ij} are quite different, with the former being a geometric mean and the latter an arithmetic mean:

$$a_{ij} = (a_i a_j)^{1/2} \quad (11-21)$$

$$b_{ij} = (b_i + b_j)/2 \quad (11-22)$$

As stated by Sandler, Orbey, and Lee [87], (11-19) to (11-22) are usually adequate for nonpolar mixtures of hydrocarbons and light gases when critical temperature and/or size differences between the molecules are not very large.

Molecular-size differences and/or modest degrees of polarity are handled by the following modified combining rules:

$$a_{ij} = (a_i a_j)^{1/2} (1 - k_{ij}) \quad (11-23)$$

$$b_{ij} = [(b_i + b_j)/2] (1 - l_{ij}) \quad (11-24)$$

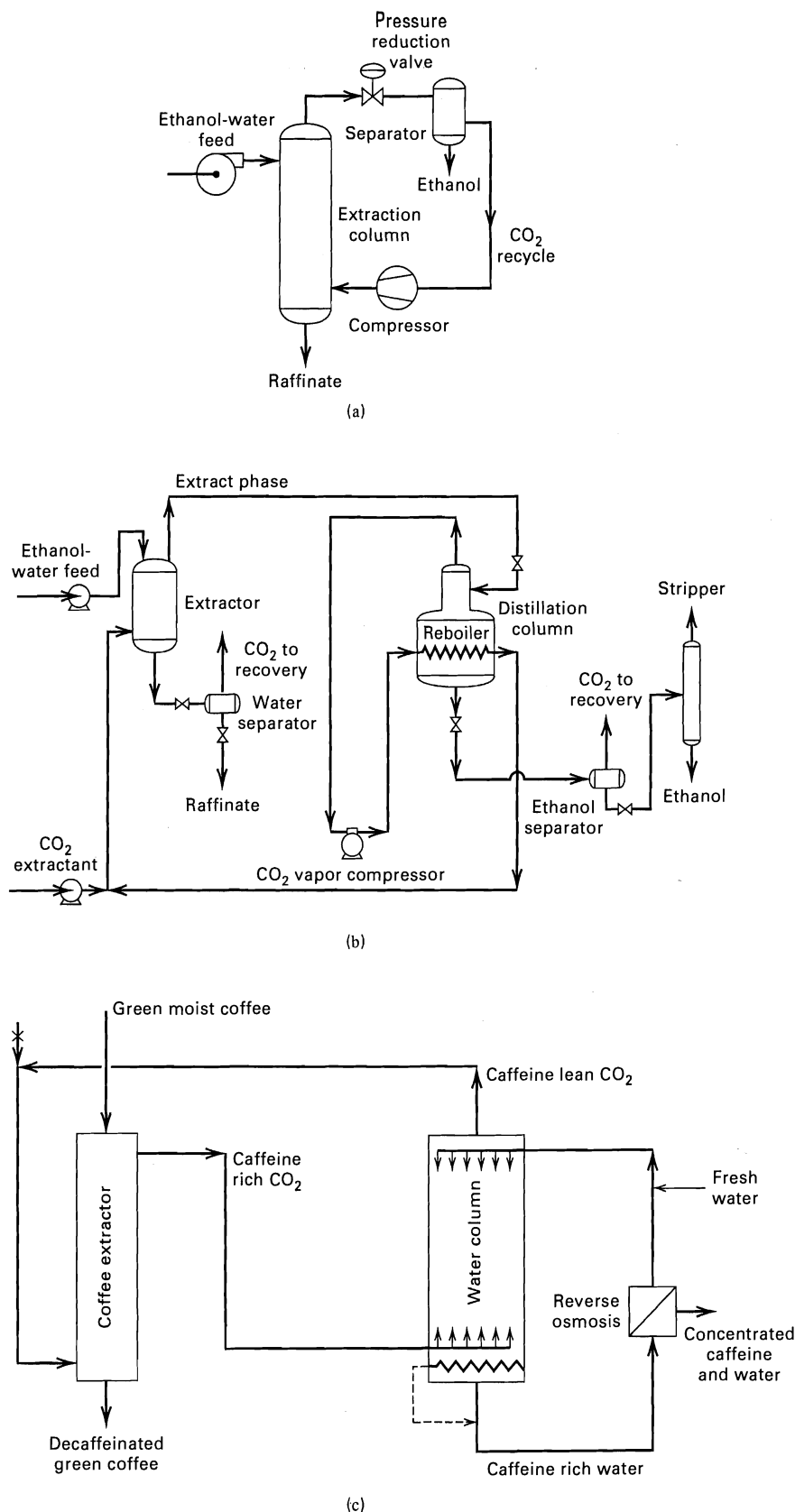


Figure 11.44 Techniques for recovery of CO₂ in supercritical extraction processes. (a) Pressure reduction. (b) High-pressure distillation. (c) High-pressure absorption with water.

where k_{ij} and l_{ij} are binary-interaction parameters back-calculated from experimental vapor-liquid equilibrium and/or density data. Often the latter parameter is set equal to zero. A tabulation of values of k_{ij} , suitable for use with the SRK and PR equations when the mixture contains hydrocarbons

with CO₂, H₂S, N₂, and/or CO, is given by Knapp et al. [88]. In a study by Shibata and Sandler [89], using experimental phase-equilibria and phase-density data for the nonpolar binary system nitrogen-*n*-butane at 410.9 K over a pressure range of about 30 to 70 bar, reasonably good predictions,

except in the critical region, were obtained using (11-19) and (11-20), with (11-23) and (11-24), and values of $k_{ij} = -0.164$ and $l_{ij} = -0.233$ in conjunction with the PR equation. Similar good agreement with experimental data was obtained for the systems nitrogen–cyclohexane, carbon dioxide–*n*-butane, and carbon dioxide–cyclohexane, and the ternary systems nitrogen–carbon dioxide–*n*-butane and nitrogen–carbon dioxide–cyclohexane.

For high pressures and mixtures containing one or more strongly polar components, the preceding rules are inadequate and it would seem desirable, in those cases, to combine the equation-of-state method with the activity-coefficient (free energy) method to handle strong nonidealities in the liquid phase. The following theoretically based mixing rule of Wong and Sandler [86] accomplishes such a bridge between a cubic equation of state and a free-energy or activity-coefficient equation. If, for example, the PR equation of state and the NRTL activity-coefficient equation are used, the Wong and Sandler mixing rule leads to the following expressions for computing a_m and b_m to be used in the PR equation:

$$a_m = RTQD/(1-D) \quad (11-25)$$

$$b_m = Q/(1-D) \quad (11-26)$$

where

$$Q = \sum_{i=1}^C \sum_{j=1}^C x_i x_j \left(b - \frac{a}{RT} \right)_{ij} \quad (11-27)$$

$$D = \sum_{i=1}^C x_i \frac{a_i}{b_i RT} + \frac{G^{ex}(x_i)}{\sigma RT} \quad (11-28)$$

$$\left(b - \frac{a}{RT} \right)_{ij} = \frac{1}{2} \left[\left(b_i - \frac{a_i}{RT} \right) + \left(b_j - \frac{a_j}{RT} \right) \right] (1 - k_{ij}) \quad (11-29)$$

$$\sigma = \frac{1}{\sqrt{2}} [\ln(\sqrt{2} - 1)] \quad (11-30)$$

$$\frac{G^{ex}}{RT} = \sum_{i=1}^C x_i \left(\frac{\sum_{j=1}^C x_j \tau_{ji} g_{ji}}{\sum_{k=1}^C x_k g_{ki}} \right) \quad (11-31)$$

$$g_{ij} = \exp(-\alpha_{ij} \tau_{ij}) \quad (11-32)$$

with $a_{ij} = a_{ji}$.

From Equations (11-25) to (11-32), it is seen that for a binary system, using the NRTL equation, there are four adjustable binary-interaction parameters (BIPs): k_{ij} , α_{ij} , τ_{ij} , and τ_{ji} . These four parameters, for a temperature and pressure range of interest, are best obtained by regression of experimental binary-pair data for VLE, LLE, and/or VLLE. The parameters can then be used to predict phase equilibria for ternary and higher multicomponent mixtures. However, Wong, Orbey, and Sandler [90] show that when values of the latter three parameters are already available, even at just

near-ambient temperature and pressure conditions, from an experimental-data-compilation source such as that of Gmehling and Onken [91], those parameters can be assumed to be independent of temperature and used directly to make reasonably accurate predictions of phase equilibria, even at temperatures to at least 200°C and pressures to 200 bar. Furthermore, regression of experimental data to obtain a value of k_{ij} is not necessary either, because Wong, Orbey, and Sandler show that it can be determined from the other three parameters by choosing its value so that the excess Gibbs free energy computed from the equation of state matches that computed from the activity-coefficient mode. Thus, the application of the Wong–Sandler mixing rule to supercritical extraction conditions is facilitated.

Another phase-equilibrium prediction method applicable to wide ranges of pressure, temperature, molecular size, and polarity is the group-contribution equation of state (GC-EOS) of Skjold-Jørgensen [92]. This method, which combines features of the van der Waals equation of state, the Carnahan–Starling expression for hard spheres, the NRTL activity-coefficient equation, and the group-contribution principle, has been successfully applied to supercritical-extraction conditions. The GC-EOS method is particularly useful when all of the necessary binary data are not available to determine all binary interaction parameters.

When experimental K -values are available, or when the Wong–Sandler mixing rule or the GC-EOS can be applied, equilibrium-stage calculations for supercritical extraction can be made by conventional computer programs, as the following example illustrates.

EXAMPLE 11.10

One mol/s of 10 wt% ethanol in water is extracted by 3 mol/s of carbon dioxide at 305 K and 9.86 MPa in a countercurrent-flow extraction column with the equivalent of five equilibrium stages. Determine the flow rates and compositions of the exiting extract and raffinate.

SOLUTION

This problem, which was taken from Colussi, Fermeglia, Gallo, and Kikic [93], was solved with the Tower Plus model of the ChemCAD process simulator, under conditions of constant temperature and constant pressure, at which composition changes were small enough that K -values could be assumed constant. The following K -values, taken from Colussi et al., who used the GC-EOS method, and which are defined as the mole fraction in the extract divided by the mole fraction in the raffinate, are in reasonably good agreement with experimental data.

Component	K -Value
CO ₂	34.5
Ethanol	0.115
Water	0.00575

Table 11.7 Calculated Flow and Composition Profiles for Example 11.10

Leaving Streams	Stage 1		Stage 2		Stage 3		Stage 4		Stage 5	
	Extract Mole Fraction	Raffinate Mole Fraction	Extract Mole Fraction	Raffinate Mole Fraction	Extract Mole Fraction	Raffinate Mole Fraction	Extract Mole Fraction	Raffinate Mole Fraction	Extract Mole Fraction	Raffinate Mole Fraction
Carbon dioxide	0.98999	0.02870	0.99002	0.02870	0.99012	0.02870	0.99043	0.02870	0.99138	0.02874
Ethanol	0.00466	0.04053	0.00463	0.04023	0.00452	0.03929	0.00419	0.03645	0.00319	0.02775
Water	0.00535	0.93077	0.00535	0.93107	0.00536	0.93201	0.00538	0.93485	0.00543	0.94351
Total flow, gmol/s	3.0013	1.0298	3.0311	1.0294	3.0308	1.0285	3.0298	1.0255	3.0268	0.9987

The percent extraction of ethyl alcohol was computed to be 33.6%, with an extract of 69 wt% pure ethanol (solvent-free basis) and a raffinate containing 93 wt% water (solvent-free basis). The

calculated stage-wise flow rates and component mole fractions are listed in Table 11.7, where stages are numbered from the feed end of the cascade.

SUMMARY

1. Extractive distillation, salt distillation, pressure-swing distillation, homogeneous azeotropic distillation, heterogeneous azeotropic distillation, and reactive distillation are enhanced distillation techniques to be considered when separation by ordinary distillation is uneconomical or impossible. Reactive distillation can also be used to conduct, simultaneously and in the same apparatus, a chemical reaction and a separation by distillation.
2. For ternary systems, a composition plot on a triangular graph is very useful for finding feasible separations, especially when binary and ternary azeotropes form. With such a diagram, distillation paths, called residue curves or distillation curves, are readily tracked. The curves may be restricted to certain regions of the triangular diagram by distillation boundaries. Feasible product compositions at total reflux are readily determined.
3. Extractive distillation, using a low-volatility solvent that enters near the top of the column, is widely used to separate azeotropes and very close-boiling mixtures. Preferably, the solvent should not form an azeotrope with any component in the feed.
4. Certain salts, when added to a solvent, reduce the volatility of the solvent and increase the relative volatility between the two components to be separated. In this process, called salt distillation, the salt is dissolved in the solvent or added as a solid or melt to the reflux.
5. Pressure-swing distillation, utilizing two columns operating at different pressures, can be used to separate an azeotropic mixture when the azeotrope can be made to disappear at some pressure. If not, the technique may still be practical if the azeotropic composition changes by 5 mol% or more over a moderate range of pressure.

6. In homogeneous azeotropic distillation, an entrainer is added to a stage, usually above the feed stage. A minimum- or maximum-boiling azeotrope, formed by the entrainer with one or more feed components, is removed from the top or bottom of the column, respectively. Unfortunately, potential applications of this technique for difficult-to-separate mixtures are not common because of limitations due to distillation boundaries.

7. A more common and useful technique is heterogeneous azeotropic distillation, in which the entrainer forms, with one or more components of the feed, a minimum-boiling heterogeneous azeotrope. When condensed, the overhead vapor splits into organic-rich and water-rich phases. The azeotrope is broken by returning one liquid phase as reflux, with the other sent on as distillate for further processing.

8. A growing application of reactive or catalytic distillation is the combined operation of chemical reaction and distillation in one vessel. To be effective, it must be possible to carry out the reaction and phase separation at the same pressure and range of temperature, with reactants and products favoring different phases so that an equilibrium-limited reaction can go to completion.

9. Liquid-liquid or solid-liquid extraction can be carried out with a supercritical-fluid solvent at temperatures and pressures just above the critical because of favorable values for solvent density and viscosity, solute diffusivity, and solute solubility in the solvent. An attractive supercritical solvent is carbon dioxide, particularly for extraction of certain chemicals from natural products.

REFERENCES

1. STICHLMAIR, J., J.R. FAIR, and J.L. BRAVO, *Chem. Eng. Progress*, **85**(1), 63-69 (1989).
2. PARTIN, L.R., *Chem. Eng. Progress*, **89** (1), 43-48 (1993).
3. STICHLMAIR, J., "Distillation and Rectification," in *Ullmann's Encyclopedia of Industrial Chemistry*, 5th ed., VCH Verlagsgesellschaft Weinheim Vol. B3, pp. 4-1 to 4-94, (1988).

4. DOHERTY, M.F., and J.D. PERKINS, *Chem. Eng. Sci.*, **33**, 281–301 (1978).
5. BOSSEN, B.S., S.B. JØRGENSEN, and R. GANI, *Ind. Eng. Chem. Res.*, **32**, 620–633 (1993).
6. PHAM, H.N. and M.F. DOHERTY, *Chem. Eng. Sci.*, **45**, 1837–1843 (1990).
7. TAYLOR, R., and H.A. KOOIJMAN, *CACHE News*, No. 41, 13–19 (1995).
8. DOHERTY, M.F., *Chem. Eng. Sci.*, **40**, 1885–1889 (1985).
9. ASPEN PLUS, "What's New in Release 9," Aspen Technology, Cambridge, MA (1994).
10. DOHERTY, M.F., and J.D. PERKINS, *Chem. Eng. Sci.*, **34**, 1401–1414 (1979).
11. HORSLEY, L.H., "Azeotropic Data III," in *Advances in Chemistry Series*, American Chemical Society, Washington, D.C., Vol. 116 (1973).
12. GMEHLING, J., J. MENKE, J. KRAFCZYK, and K. FISCHER, *Azeotropic Data*, 2nd edition in 3 volumes, Wiley-VCH, Weinheim, Germany (2004).
13. FIDKOWSKI, Z.T., M.F. MALONE, and M.F. DOHERTY, *Computers Chem. Engng.*, **17**, 1141–1155 (1993).
14. FOUCHER, E.R., M.F. DOHERTY, and M.F. MALONE, *Ind. Eng. Chem. Res.*, **30**, 760–772 (1991) and **30**, 2364 (1991).
15. MATSUYAMA, H., and H.J. NISHIMURA, *J. Chem. Eng. Japan*, **10**, 181 (1977).
16. DOHERTY, M.F., and G.A. CALDAROLA, *Ind. Eng. Chem. Fundam.*, **24**, 474–485 (1985).
17. FIDKOWSKI, Z.T., M.F. DOHERTY, and M.F. MALONE, *AIChE J.*, **39**, 1303–1321 (1993).
18. STICHLMAIR, J.G., and J.-R. HERGUIJUELA, *AIChE J.*, **38**, 1523–1535 (1992).
19. WIDAGDO, S., and W.D. SEIDER, *AIChE J.*, **42**, 96–130 (1996).
20. WAHNSCHAFFT, O.M., J.W. KOEHLER, E. BLASS, and A.W. WESTERBERG, *Ind. Eng. Chem. Res.*, **31**, 2345–2362 (1992).
21. DUNN, C.L., R.W. MILLAR, G.J. PIEROTTI, R.N. SHIRAS, and M. SOUDERS, JR., *Trans. AIChE*, **41**, 631–644 (1945).
22. BERG, L., *Chem. Eng. Progress*, **65** (9), 52–57 (1969).
23. OTHMER, D.F., *AIChE Symp. Series*, **235** (79), 90–117 (1983).
24. VAN RAYMBEKE, U.S. Patent 1,474,216 (1922).
25. COOK, R.A., and W.F. FURTER, *Can. J. Chem. Eng.*, **46**, 119–123 (1968).
26. JOHNSON, A.I., and W.F. FURTER, *Can. J. Chem. Eng.*, **43**, 356–358 (1965).
27. JOHNSON, A.I., and W.F. FURTER, *Can. J. Chem. Eng.*, **38**, 78–87 (1960).
28. FURTER, W.F., and R.A. COOK, *Int. J. Heat Mass Transfer*, **10**, 23–36 (1967).
29. FURTER, W.F., *Can. J. Chem. Eng.*, **55**, 229–239 (1977).
30. FURTER, W.F., *Chem. Eng. Commun.*, **116**, 35 (1992).
31. KUMAR, A., *Sep. Sci. and Tech.*, **28**, 1799–1818 (1993).
32. MAHAPATRA, A., V.G. GAIKAR, and M.M. SHARMA, *Sep. Sci. and Tech.*, **23**, 429–436 (1988).
33. AGARWAL, M., and V.G. GAIKAR, *Sep. Technol.*, **2**, 79–84 (1992).
34. KNAPP, J.P., and M.F. DOHERTY, *Ind. Eng. Chem. Res.*, **31**, 346–357 (1992).
35. LEWIS, W.K., U.S. Patent 1,676,700 (1928).
36. VAN WINKLE, M., *Distillation*, McGraw-Hill, New York (1967).
37. ROBINSON, C.S., and E.R. GILLILAND, *Elements of Fractional Distillation*, 4th ed., McGraw-Hill, New York (1950).
38. WAHNSCHAFFT, O.M., and A.W. WESTERBERG, *Ind. Eng. Chem. Res.*, **32**, 1108 (1993).
39. LAROCHE, L., N. BEKIARIS, H.W. ANDERSEN, and M. MORARI, *AIChE J.*, **38**, 1309 (1992).
40. YOUNG, S., *J. Chem. Soc. Trans.*, **81**, 707–717 (1902).
41. KEYES, D.B., U.S. Patent 1,676,735 (1928).
42. KEYES, D.B., *Ind. Eng. Chem.*, **21**, 998–1001 (1929).
43. BEKIARIS, N., G.A. MESKI, and M. MORARI, *Ind. Eng. Chem. Res.*, **35**, 207–217 (1996).
44. PROKOPAKIS, G.J., and W.D. SEIDER, *AIChE J.*, **29**, 49–60 (1983).
45. RYAN, P.J., and M.F. DOHERTY, *AIChE J.*, **35**, 1592–1601 (1989).
46. PHAM, H.N., and M.F. DOHERTY, *Chem. Eng. Sci.*, **45**, 1845–1854 (1990).
47. KOVACH, III, J.W., and W.D. SEIDER, *Computers and Chem. Engng.*, **11**, 593 (1987).
48. SWARTZ, C.L.E., and W.E. STEWART, *AIChE J.*, **33**, 1977–1985 (1987).
49. GANI, R., and S.B. JØRGENSEN, *Computers Chem. Engng.*, **18**, Suppl., S55 (1994).
50. SHEWCHUK, C.F., "Computation of Multiple Distillation Towers," Ph.D. Thesis, University of Cambridge (1974).
51. MAGNUSSEN, T., M.L. MICHELSEN, and A. FREDENSLUND, *Inst. Chem. Eng. Symp. Series No. 56, Third International Symp. on Distillation*, Rugby, England (1979).
52. BEKIARIS, N., G.A. MESKI, C.M. RADU, and M. MORARI, *Ind. Eng. Chem. Res.*, **32**, 2023–2038 (1993).
53. BLACK, C., and D.E. DITSLER, *Advances in Chemistry Series*, ACS, Washington, D.C., Vol. 115, pp. 1–15 (1972).
54. BLACK, C., R.A. GOLDING, and D.E. DITSLER, *Advances in Chemistry Series*, ACS, Washington, D.C., Vol. 115, pp. 64–92 (1972).
55. BLACK, C., *Chem. Eng. Progress*, **76** (9), 78–85 (1980).
56. BLACK, C., *Ind. Eng. Chem.*, **50**, 403–412 (1958).
57. TERRILL, D.L., L.F. SYLVESTRE, and M.F. DOHERTY, *Ind. Eng. Chem. Proc. Des. Develop.*, **24**, 1062–1071 (1985).
58. ROBINSON, C.S., and E.R. GILLILAND, *Elements of Fractional Distillation*, 4th ed., McGraw-Hill, New York (1950).
59. BACKHAUS, A.A., U.S. Patent 1,400,849 (1921).
60. LEYES, C.E., and D.F. OTHMER, *Trans. AIChE*, **41**, 157–196 (1945).
61. BELCK, L.H., *AIChE J.*, **1**, 467–470 (1955).
62. MASAMOTO, J., and K. MATSUZAKI, *J. Chem. Eng. Japan*, **27**, 1–5 (1994).
63. AGREDA, V.H., and L.R. PARTIN, U.S. Patent 4,435,595 (March 6, 1984).
64. AGREDA, V.H., L.R. PARTIN, and W.H. HEISE, *Chem. Eng. Prog.*, **86** (2), 40–46 (1990).
65. SMITH, L.A., U.S. Patent 4,307,254 (Dec. 22, 1981).
66. SMITH, L.A., U.S. Patent 4,443,559 (April 17, 1984).
67. SMITH, L.A., U.S. Patent 4,978,807 (Dec. 18, 1990).
68. DEGARMO, J.L., V.N. PARULEKAR, and V. PINJALA, *Chem. Eng. Prog.*, **88** (3), 43–50 (1992).
69. CHANG, Y.A., and J.D. SEADER, *Computers Chem. Engng.*, **12**, 1243–1255 (1988).
70. VENKATARAMAN, S., W.K. CHAN, and J.F. BOSTON, *Chem. Eng. Progress*, **86** (8), 45–54 (1990).
71. SIMANDL, J., and W.Y. SVRCEK, *Computers Chem. Engng.*, **15**, 337–348 (1991).
72. KANG, Y.W., Y.Y. LEE, and W.K. LEE, *J. Chem. Eng. Japan*, **25**, 649–655 (1992).
73. IZARRARAZ, A., G.W. BENTZEN, R. G. ANTHONY, and C.D. HOLLAND, *Hydrocarbon Processing*, **59** (6), 195 (1980).

74. REHFINGER, A., and U. HOFFMANN, *Chem. Eng. Sci.*, **45**, 1605–1617 (1990).
75. JACOBS, R., and R. KRISHNA, *Ind. Eng. Chem. Res.*, **32**, 1706–1709 (1993).
76. NIJHUIS, S.A., F.P.J.M. KERKHOFF, and N.S. MAK, *Ind. Eng. Chem. Res.*, **32**, 2767–2774 (1993).
77. HAUAN, S., T. HERTZBERG, and K.M. LIEN, *Ind. Eng. Chem. Res.*, **34**, 987–991 (1995).
78. HANNAY, J.B., and J. HOGARTH, *Proc. Roy. Soc. (London) Sec. A*, **29**, 324 (1879).
79. WILLIAMS, D.F., *Chem. Eng. Sci.*, **36**, 1769–1788 (1981).
80. MCHUGH, M., and V. KRUKONIS, *Supercritical Fluid Extraction—Principles and Practice*, Butterworths, Boston (1986).
81. MCHUGH, M., and V. KRUKONIS, *Supercritical Fluid Extraction—Principles and Practice*, 2nd ed., Butterworth-Heinemann, Boston (1994).
82. TAKISHIMA, S., A. SAIKI, K. ARAI, and S. SAITO, *J. Chem. Eng. Japan*, **19**, 48–56 (1986).
83. INOMATA, H., A. KONDO, K. ARAI, and S. SAITO, *J. Chem. Eng. Japan*, **23**, 199–207 (1990).
84. DE FILIPI, R.P., and J.E. VIVIAN, U.S. Patent 4,349,415 (1982).
85. KATZ, S.N., J.E. SPENCE, M.J. O'BRIAN, R.H. SKIFF, G.J. VOGEL, and R. PRASAD, U.S. Patent 4,911,941 (1990).
86. WONG, D.S.H., and S.I. SANDLER, *AIChE J.*, **38**, 671–680 (1992).
87. SANDLER, S.I., H. ORBEY, and B-I. LEE, in *Models for Thermodynamic and Phase Equilibria Calculations*, S.I. Sandler, Ed., Marcel Dekker, New York, pp. 87–186 (1994).
88. KNAPP, H., R. DORING, L. OELLRICH, U. PLOCKER, and J.M. PRAUSNITZ, *Vapor-Liquid Equilibria for Mixtures of Low Boiling Substances*, Chem. Data Ser., Vol. VI, DECHEMA, pp. 771–793 (1982).
89. SHIBATA, S.K., and S.I. SANDLER, *Ind. Eng. Chem. Res.*, **28**, 1893–1898 (1989).
90. WONG, D.S.H., H. ORBEY, and S.I. SANDLER, *Ind. Eng. Chem. Res.*, **31**, 2033–2039 (1992).
91. GMEHLING, J., and U. ONKEN, *Vapor-Liquid Equilibrium Data Compilation*, DECHEMA Data Series, DECHEMA, Frankfurt (1977).
92. SKJOLD-JØRGENSEN, S., *Ind. Eng. Chem. Res.*, **27**, 110–118 (1988).
93. COLUSSI, I.E., M. FERMEGLIA, V. GALLO, and I. KIKIC, *Computers Chem. Engng.*, **16**, 211–224 (1992).
94. DOHERTY, M.F., and M.F. MALONE, *Conceptual Design of Distillation Systems*, McGraw-Hill, New York (2001).
95. STICHLMAIR, J.G., and J.R. FAIR, *Distillation Principles and Practices*, Wiley-VCH, New York (1998).
96. SIROLA, J.J., and S.D. BARNICKI, in *Perry's Chemical Engineers' Handbook*, Seventh Edition, Perry, R.H., and D.W. Green, eds., McGraw-Hill, New York, pp. 13–54 to 13–85 (1997).
97. ECKERT, E., and M. KUBICEK, *Computers Chem. Eng.*, **21**, 347–350 (1997).
98. HOFFMASTER, W.R., and S. HAUAN, *AIChE J.*, **48**, 2545–2556 (2002).
99. LLANO-RESTREPO, M., and J. AGUILAR-ARIAS, *Computers Chem. Engng.*, **27**, 527–549 (2003).
100. FU, J., *AIChE J.*, **42**, 3364–3372 (1996).

EXERCISES

Section 11.1

11.1 For the ternary system, normal hexane–methanol–methyl acetate at 1 atm find, in suitable references, all the binary and ternary azeotropes, sketch an approximate residue-curve map on a right-triangular diagram, and indicate the distillation boundaries. Determine for each azeotrope and pure component whether it is a stable node, an unstable node, or a saddle.

11.2 For the same ternary system as in Exercise 11.1, use a process-simulation program with the UNIFAC equation to calculate a portion of a residue curve at 1 atm starting from a bubble-point liquid with a composition of 20 mol% normal hexane, 60 mol% methanol, and 20 mol% methyl acetate.

11.3 For the same conditions as Exercise 11.2, use a process-simulation program with the UNIFAC equation to calculate a portion of a distillation curve at 1 atm.

11.4 For the ternary system acetone, benzene, and *n*-heptane at 1 atm find, in suitable references, all the binary and ternary azeotropes, and sketch an approximate distillation-curve map on an equilateral-triangle diagram, and indicate the distillation boundaries. Determine for each azeotrope and pure component whether it is a stable node, an unstable node, or a saddle.

11.5 For the same ternary system as in Exercise 11.4, use a process-simulation program with the UNIFAC equation to calculate a portion of a residue curve at 1 atm starting from a bubble-point liquid with a composition of 20 mol% acetone, 60 mol% benzene, and 20 mol% *n*-heptane.

11.6 For the same conditions as Exercise 11.5, use a process-simulation program with the UNIFAC equation to calculate a portion of a distillation curve at 1 atm.

11.7 Develop the feasible product-composition regions for the system of Figure 11.13, using Feed F_1 .

11.8 Develop the feasible product composition regions for the system of Figure 11.10 if the feed composition is 50 mol% chloroform, 25 mol% methanol, and 25 mol% acetone.

Section 11.2

11.9 Repeat Example 11.3, but with ethanol as the solvent.

11.10 Repeat Example 11.3, but with MEK as the solvent.

11.11 Repeat Example 11.4, but with toluene as the solvent.

11.12 An equimolar mixture of *n*-heptane and toluene at 200°F, 20 psia, and a flow rate of 400 lbmol/h is to be separated by extractive distillation at 20 psia, using phenol at 220°F as the solvent, at a flow rate of 1200 lbmol/h. Design a suitable two-column system, obtaining reasonable product purities, with only a small loss of solvent.

Section 11.4

11.13 Repeat Example 11.5, but with a feed of 100 mol/s of 55 mol% ethanol and 45 mol% benzene.

11.14 Determine the feasibility of separating 100 mol/s of a mixture of 20 mol% ethanol and 80 mol% benzene by pressure-swing distillation. If feasible, design such a system.

11.15 Design a pressure-swing distillation system to produce 99.8 mol% ethanol for 100 mol/s of an aqueous feed containing 30 mol% ethanol.

Section 11.5

11.16 In Example 11.6, a mixture of benzene and cyclohexane is separated in a separation sequence that begins with homogeneous azeotropic distillation using acetone as the entrainer. Can the same separation be achieved using methanol as the entrainer? If not, why not? [Ref.: Ratliff, R.A., and W.B. Strobel, *Petro. Refiner*, **33** (5), 151 (1954)].

11.17 Devise a separation sequence to separate 100 mol/s of an equimolar mixture of toluene and 2,5-dimethylhexane into nearly pure products. Include in the sequence a homogeneous azeotropic distillation column using methanol as the entrainer and determine a feasible design for that column. [Ref.: Benedict, M., and L.C. Rubin, *Trans. AIChE*, **41**, 353–392 (1945)].

11.18 A mixture of 55 wt% methyl acetate and 45 wt% methanol at a flow rate of 16,500 kg/h is to be separated into one product of 99.5 wt% methyl acetate and another product of 99 wt% methanol. It has been suggested that such a separation might be possible by using a sequence of one homogeneous azeotropic distillation column and one ordinary distillation column. Possible entrainers are *n*-hexane, cyclohexane, and toluene. Determine the feasibility of such a sequence. If feasible, prepare a process design. If not feasible, suggest an alternative process and prove its feasibility.

Section 11.6

11.19 Design a three-column distillation sequence to separate 150 mol/s of an azeotropic mixture of ethanol and water at 1 atm into nearly pure ethanol and nearly pure water using heterogeneous azeotropic distillation with benzene as the entrainer.

11.20 Design a three-column distillation sequence to separate 120 mol/s of an azeotropic mixture of isopropanol and water at 1 atm into nearly pure isopropanol and nearly pure water using heterogeneous azeotropic distillation with benzene as the entrainer. [Ref.: Pham, H.N., P.J. Ryan, and M.F. Doherty, *AIChE J.*, **35**, 1585–1591 (1989)].

11.21 Design a two-column distillation sequence to separate 1,000 kmol/h of 20 mol% aqueous acetic acid into nearly pure acetic acid and nearly pure water. The first column should use heterogeneous azeotropic distillation with *n*-propyl acetate as the entrainer.

Section 11.7

11.22 Repeat Example 11.9, with the entire range of methanol feed-stage locations. Compare your results for isobutene conversion with the values shown in Figure 11.39.

11.23 Repeat Exercise 11.22, but with activities, instead of mole fractions, in the reaction rate expressions. How do the results differ? Explain.

11.24 Repeat Exercise 11.22, but with the assumption of chemical equilibrium on stages where catalyst is employed. How do the results differ from Figure 11.39? Explain.

Section 11.8

11.25 Repeat Example 11.10, but with 10 equilibrium stages instead of 5. What is the effect of this change?

11.26 An important application of supercritical extraction is the removal of solutes from particles of porous natural materials. Such applications include the extraction of caffeine from coffee beans and the extraction of ginger oil from ginger root. When CO₂ is used as the solvent, the rate of extraction is found to be independent of the flow rate of CO₂ past the particles, but dependent upon the particle size. Develop a suitable mathematical model for the rate of extraction that is consistent with these observations. What parameter in the model would have to be determined by experiment?

11.27 Cygnarowicz and Seider [*Biotechnol. Prog.*, **6**, 82–91 (1990)] present a process design for the supercritical extraction of β -carotene from water with carbon dioxide using the GC-EOS method of Skjold-Jørgensen to estimate phase equilibria. Repeat the calculations for the conditions of their design using the Peng–Robinson EOS with the Wong–Sandler mixing rules. How do the two designs compare?

11.28 Cygnarowicz and Seider [*Ind. Eng. Chem. Res.*, **28**, 1497–1503 (1989)] present a process design for the supercritical extraction of acetone from water with carbon dioxide using the GC-EOS method of Skjold-Jørgensen to estimate phase equilibria. Repeat the calculations for the conditions of their design using the Peng–Robinson EOS with the Wong–Sandler mixing rules. How do the two designs compare?

Rate-Based Models for Distillation

Chapter 10 contains rigorous, equilibrium-based models for continuous-flow, steady-state, multicomponent, multi-stage distillation, absorption, stripping, and liquid–liquid extraction based on component material balances, energy balances, and thermodynamic correlations and criteria for phase equilibria. These models are extended in Chapter 11 to supercritical extraction and enhanced distillation, including extractive distillation, azeotropic distillation, and reactive distillation. The fundamental equations for the equilibrium-based models were first published by Sorel [1] in 1893. His equations consisted of total and component material balances around top and bottom sections of equilibrium stages (theoretical plates), including a total condenser and a reboiler, and corresponding energy balances that included provision for heat losses, which are an important factor for small laboratory columns, but not for insulated, industrial columns. Sorel used graphs of phase-equilibrium data instead of equations. Because of the complexity of Sorel's model, it was not widely applied until 1921, when it was adapted to graphical-solution techniques for binary systems, first by Ponchon and then by Savarit, who used an enthalpy–concentration diagram. In 1925, a much simpler, but less-rigorous, graphical technique was developed by McCabe and Thiele, who eliminated the energy balances by assuming constant vapor and liquid molar flow rates from equilibrium stage to equilibrium stage except across feed or side-stream withdrawal stages. This is referred to as the constant-molar-overflow assumption. When applicable, the McCabe–Thiele graphical method, developed in detail in Chapter 7, is applied even today for binary distillation, because the method gives valuable insight into changes in phase compositions from stage to stage.

Because some of Sorel's equations are nonlinear, it is not possible to obtain algebraic solutions, unless simplifying assumptions are made. A notable achievement in this respect was made by Smoker [2] in 1938 for the distillation of a binary mixture by assuming not only constant molar overflow, but also constant relative volatility between the two components. Smoker's equation is still useful for superfractionators involving close-boiling binary mixtures, where that assumption is valid. Starting in 1932, two iterative, numerical methods were developed for obtaining a

general solution to Sorel's model for the distillation of multicomponent mixtures. The Thiele–Geddes method [3] requires specification of the number of equilibrium stages, the feed stage, the reflux ratio, and the distillate flow rate, with the resulting distribution of the components between distillate and bottoms being calculated. The Lewis–Matheson method [4] computes the number of stages required and the location of the feed stage for a specified reflux ratio and split between two key components. These two methods were widely used for the simulation and design of single-feed, multicomponent distillation columns prior to the 1960s.

Attempts in the late 1950s and early 1960s to adapt the Thiele–Geddes and Lewis–Matheson methods to computations with a digital computer had limited success. The real breakthrough in computerization of equilibrium-stage calculations occurred when Amundson and co-workers, starting in 1958, applied techniques of matrix algebra. This led to a number of successful computer-aided methods, based on sparse-matrix algebra, for Sorel's equilibrium-based model. The most important of these models are presented in Chapter 10. Today, computer-aided design and simulation programs abound for the rigorous, iterative, numerical solution of Sorel's equilibrium-based model for a wide variety of column configurations and specifications. Although the iterative computations sometimes fail to converge, the methods are widely applied and have become more flexible and robust with each passing year.

The methods presented in Chapters 10 and 11 assume that equilibrium is achieved, at each stage, with respect to both heat and component mass transfer. Except when temperature changes significantly from stage to stage, the assumption of temperature equality for vapor and liquid phases leaving a stage is usually acceptable. However, in most industrial applications, the assumption of equilibrium with respect to exiting-phase compositions is not reasonable. In general, exiting vapor-phase mole fractions are not related to exiting liquid-phase mole fractions by thermodynamic K -values. To overcome this limitation of equilibrium-based models, Lewis [5], in 1922, proposed the use of an overall stage efficiency for converting theoretical stages to actual stages. Unfortunately, experimental data show that this efficiency varies, depending on the application, over a range of about

5 to 120%, where the high values are achieved for distillation in large-diameter, single-liquid-pass trays because of a cross-flow effect, whereas the lower values occur in absorption columns when a high-viscosity, high-molecular-weight absorbent is used.

A preferred procedure for accounting for nonequilibrium with respect to mass transfer, since its introduction by Murphree [6] in 1925, has been to incorporate the Murphree vapor-phase tray efficiency, $(E_{MV})_{i,j}$, directly into Sorel's model as a replacement for the equilibrium equation based on definition of the K -value. Thus, the equation

$$K_{i,j} = y_{i,j}/x_{i,j} \quad (12-1)$$

where i refers to the component and j the stage, is replaced by

$$(E_{MV})_{i,j} = (y_{i,j} - y_{i,j+1})/(y_{i,j}^* - y_{i,j+1}) \quad (12-2)$$

where stages are numbered from the top.

Thus, the efficiency is the ratio of the actual change in vapor-phase mole fraction to the change that would occur if equilibrium were achieved. The equilibrium value, $y_{i,j}^*$, is obtained from (12-1), with substitution into (12-2) giving

$$(E_{MV})_{i,j} = (y_{i,j} - y_{i,j+1})/(K_{i,j}x_{i,j} - y_{i,j+1}) \quad (12-3)$$

Equations (12-2) and (12-3) assume the following:

1. Uniform concentrations of vapor and liquid streams entering into and exiting a tray
2. Complete mixing throughout the liquid flowing across the tray
3. Plug flow of the vapor up through the liquid
4. Negligible resistance to mass transfer in the liquid phase

Application of the Murphree efficiency using empirical correlations has proved to be adequate for binary and close-boiling, ideal, and near-ideal multicomponent vapor-liquid mixtures. However, deficiencies of the Murphree efficiency for general multicomponent vapor-liquid mixtures have long been recognized. Murphree himself stated clearly the deficiencies of his development for multicomponent mixtures and for cases where the efficiency is low. He even stated that the theoretical plate should not be the basis of calculation for ternary mixtures.

When the equilibrium-based model is applied to multicomponent mixtures, a number of problems arise. Values of E_{MV} differ from component to component and vary from stage to stage. But, at each stage, the number of independent values of E_{MV} must be determined so as to force the sum of the mole fractions in the vapor phase to sum to 1. This introduces the possibility that negative values of E_{MV} can result. This is in contrast to binary mixtures for which the values of E_{MV} are always positive and are identical for the two components. When using the Murphree vapor-phase efficiency, the temperatures of the exiting vapor and liquid

phases are assumed to be the same and equal to the bubble-point temperature of the exiting liquid phase. Because the vapor phase is not in equilibrium with the liquid phase, the vapor temperature does not correspond to the dew-point temperature. It is even possible, algebraically, for the vapor temperature to correspond to a value below its dew-point temperature, which is physically impossible.

Values of E_{MV} can be obtained from experimental data or correlations. These values, however, are more likely to be Murphree vapor-point (rather than tray) efficiencies. Point efficiencies only apply to a particular location in the liquid on the tray. To convert these point efficiencies to tray efficiencies, vapor and liquid flow patterns must be assumed after the manner of Lewis [7], as discussed by Seader [8]. However, if the vapor and liquid phases are both completely mixed, the point efficiency equals the tray efficiency.

Walter and Sherwood [9] found that experimentally measured tray efficiencies covered an enormous range: 0.65 to 4.2% for absorption and stripping of carbon dioxide from water and glycerine solutions; 4.7 to 24% for absorption of olefins into oils; and 69 to 92% for absorption of ammonia, humidification of air, and rectification of alcohol.

In 1957, Toor [10] showed that diffusion in a ternary mixture is enormously more complex than in a binary mixture because of coupling among component concentration gradients, especially when components differ widely in size, shape, and polarity. Toor showed that, in addition to diffusion due to the conventional Fickian concentration driving force, the possible consequences of gradient coupling could result in: (1) diffusion against a driving force (reverse diffusion), (2) no diffusion even though a concentration driving force is present (diffusion barrier), and (3) diffusion with zero driving force (osmotic diffusion). Theoretical calculations by Toor and Burchard [11] predicted the possibility of negative values of E_{MV} in multicomponent systems, but values of E_{MV} for binary systems are restricted to the range from 0 to 100%.

In 1977, Krishna et al. [12] extended the theoretical work of Toor and Burchard and showed that when the vapor mole-fraction driving force of a component (call it A) is small compared to the other components in the mixture, the transport rate of A is controlled by the other components, with the result that E_{MV} for A is anywhere in the range from minus infinity to plus infinity. They confirmed this theoretical prediction by conducting experiments with the ethanol/*tert*-butanol/water system and obtained values of E_{MV} for *tert*-butanol ranging from -2,978% to +527%. In addition, values of E_{MV} for ethanol and water sometimes differed significantly.

Two other tray efficiencies are defined in the literature: the vaporization efficiency of Holland, which was first mentioned by McAdams, and the Hausen tray efficiency, which eliminates the assumption in E_{MV} that the exiting

liquid is at its bubble point. The former cannot distinguish the Toor phenomena and can vary widely in a manner that is not ascribable to the particular component. The latter does appear to be superior to E_{MV} , but is considerably more complicated and difficult to use, and it has not found wide application.

Although the equilibrium-based model, modified to incorporate stage efficiency, is adequate for binary mixtures and for the major components in nearly ideal multicomponent mixtures, that model has serious deficiencies for more general cases and the development of a more realistic nonequilibrium, transport- or rate-based model has long been a desirable goal. In 1977, Waggoner and Loud [13] developed a rate-based, mass-transport model limited to nearly ideal, close-boiling, multicomponent systems. However, an energy-transport equation was not included (because thermal equilibrium would be closely approximated for a close-boiling mixture) and the coupling of component mass-transfer rates was ignored.

In 1979, Krishna and Standart [14] showed the possibility of applying rigorous multicomponent mass- and heat-transfer theory to calculations of simultaneous transport. The theory was further developed by Taylor and Krishna [15]. The availability of this theory led to the development in 1985 by Krishnamurthy and Taylor [16] of the first general,

rate-based, computer-aided model for application to trayed and packed columns for distillation and other continuous, countercurrent, vapor-liquid separation operations. This model applies the two-film theory of mass transfer discussed in Chapter 3, with phase equilibria assumed at the interface of the two phases, and provides options for vapor and liquid flow configurations in trayed columns, including plug flow and perfectly mixed flow, on each tray. Although the model does not require tray efficiencies or values of HETP, correlations of mass-transfer and heat-transfer coefficients are needed for the particular type of trays or packing employed. The model was extended in 1994 by Taylor, Kooijman, and Hung [17] to include: (1) effects of entrainment of liquid droplets in the vapor and occlusion of vapor bubbles in the liquid; (2) estimation of the column-pressure profile; (3) interlinking streams; and (4) axial dispersion in packed columns. In addition, unlike the 1985 model, which required the user to specify the column diameter and tray geometry or packing size, the 1994 version includes a design mode that estimates column diameter for a specified fraction of flooding or pressure drop. Rate-based models are implemented in several computer programs, including RATEFRAC [18] of Aspen Technology, ChemSep Release 3.1 [19], and CHEMCAD.

12.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Write equations that model a nonequilibrium vapor-liquid stage, where the assumption of equilibrium is only applied at the interface between two phases.
- Explain component-coupling effects in multicomponent mass transfer.
- Explain the bootstrap problem and how it is handled for distillation.
- Cite available methods for estimating transport coefficients and interfacial areas required for rate-based calculations.
- Explain differences among ideal vapor-liquid flow patterns that are employed for rate-based calculations.
- Apply a simulation program to make a rate-based calculation for a multicomponent, multistage, vapor-liquid separation problem.

12.1 RATE-BASED MODEL

A schematic diagram of a nonequilibrium stage, consisting of a tray, a group of trays, or a segment of a packed section, is shown in Figure 12.1. Entering stage j , at pressure P_j , are liquid F_j^L and/or vapor F_j^V molar flow rates; component i molar flow rates, $f_{i,j}^L$ and $f_{i,j}^V$; and stream molar enthalpies, H_j^{LF} and H_j^{VF} . Also leaving from (+) or entering to (−) the liquid and/or vapor phases in the stage are heat transfer rates Q_j^V and Q_j^L , respectively. Also entering the stage from the stage above is liquid molar flow rate L_{j-1} at temperature T_{j-1}^L and pressure P_{j-1} , with molar enthalpy H_{j-1}^L and component mole fractions $x_{i,j-1}$; and entering the stage from the stage below is vapor molar flow rate V_{j+1} at temperature

T_{j+1}^V and pressure P_{j+1} , with molar enthalpy H_{j+1}^V and component mole fractions $y_{i,j+1}$. Within the stage, mass transfer of components occurs across the phase boundary at molar rates $N_{i,j}$ from the vapor phase to the liquid phase (+) or vice versa (−), and heat transfer occurs across the phase boundary at rates e_j from the vapor phase to the liquid phase (+) or vice versa (−). Leaving the stage is liquid at temperature T_j^L and pressure P_j , with molar enthalpy H_j^L ; and vapor at temperature T_j^V and pressure P_j , with molar enthalpy H_j^V . A fraction, r_j^L , of the liquid exiting the stage may be withdrawn as a liquid side stream at molar flow rate U_j , leaving the molar flow rate L_j to enter the stage below or to exit the column. A fraction r_j^V , of the vapor exiting the stage may

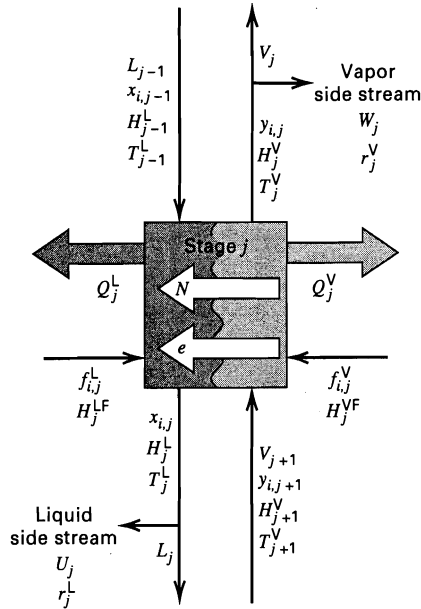


Figure 12.1 Nonequilibrium stage for rate-based method.

be withdrawn as a vapor side stream at molar flow rate W_j , leaving the molar flow rate V_j to enter the stage above or to exit the column. If desired, entrainment, occlusion, interlink flows, a second immiscible liquid phase, and chemical reaction(s) can be added to the model.

Recall that the equilibrium-stage model of Chapter 10 utilizes the $2C + 3$ MESH equations for each stage:

- C mass balances for components
- C phase equilibria relations
- 2 summations of mole fractions
- 1 energy balance

In the rate-based model, the mass and energy balances around each equilibrium stage are each replaced by separate balances for each phase around a stage, which can be a tray, a collection of trays, or a segment of a packed section. In residual form, the equations are as follows, where the residuals are on the left-hand sides and become zero when the computations are converged. When not converged, the residuals are used to determine the proximity to convergence.

Liquid-phase component material balance:

$$M_{i,j}^L \equiv (1 + r_j^L)L_j x_{i,j} - L_{j-1}x_{i,j-1} - f_{i,j}^L - N_{i,j}^L = 0, \quad i = 1, 2, \dots, C \quad (12-4)$$

Vapor-phase component material balance:

$$M_{i,j}^V \equiv (1 + r_j^V)V_j y_{i,j} - V_{j+1}y_{i,j+1} - f_{i,j}^V + N_{i,j}^V = 0, \quad i = 1, 2, \dots, C \quad (12-5)$$

Liquid-phase energy balance:

$$E_j^L \equiv (1 + r_j^L)L_j H_j^L - L_{j-1}H_{j-1}^L - \left(\sum_{i=1}^C f_{i,j}^L \right) H_j^{LF} + Q_j^L - e_j^L = 0 \quad (12-6)$$

Vapor-phase energy balance:

$$E_j^V \equiv (1 + r_j^V)V_j H_j^V - V_{j+1}H_{j+1}^V - \left(\sum_{i=1}^C f_{i,j}^V \right) H_j^{VF} + Q_j^V + e_j^V = 0 \quad (12-7)$$

where at the phase interface, I,

$$E_j^I = e_j^V - e_j^L = 0 \quad (12-8)$$

Equations (12-4) and (12-5) are coupled by the component mass-transfer rates:

$$R_{i,j}^L \equiv N_{i,j} - N_{i,j}^L = 0, \quad i = 1, 2, \dots, C - 1 \quad (12-9)$$

$$R_{i,j}^V \equiv N_{i,j} - N_{i,j}^V = 0, \quad i = 1, 2, \dots, C - 1 \quad (12-10)$$

The equations for the mole-fraction summation for each phase are applied at the vapor-liquid interface:

$$S_j^{LI} \equiv \sum_{i=1}^C x_{i,j}^I - 1 = 0 \quad (12-11)$$

$$S_j^{VI} \equiv \sum_{i=1}^C y_{i,j}^I - 1 = 0 \quad (12-12)$$

A hydraulic equation for stage pressure drop is given by

$$H_j \equiv P_{j+1} - P_j - (\Delta P_j) = 0, \quad j = 1, 2, 3, \dots, N - 1 \quad (12-13)$$

where the stage is assumed to be at mechanical equilibrium such that

$$P_j^L = P_j^V = P_j \quad (12-14)$$

and ΔP_j is the gas-phase pressure drop from stage $j + 1$ to stage j . Equation (12-13) is optional. It is only included when it is desired to compute one or more stage pressures from hydraulics, as discussed in Chapter 6. Phase equilibrium for each component is assumed to exist only at the phase interphase:

$$Q_{i,j}^I \equiv K_{i,j}x_{i,j}^I - y_{i,j}^I = 0, \quad i = 1, 2, \dots, C \quad (12-15)$$

Because only $C - 1$ equations are written for the component mass-transfer rates in (12-9) and (12-10), total phase material balances in terms of total mass-transfer rates, $N_{T,j}$, can be added to the system:

$$M_{T,j}^L \equiv (1 + r_j^L)L_j - L_{j-1} - \sum_{i=1}^C f_{i,j}^L - N_{T,j} = 0 \quad (12-16)$$

$$M_{T,j}^V \equiv (1 + r_j^V)V_j - V_{j+1} - \sum_{i=1}^C f_{i,j}^V + N_{T,j} = 0 \quad (12-17)$$

where

$$N_{T,j} = \sum_{i=1}^C N_{i,j} \quad (12-18)$$

Equations (12-4), (12-5), (12-9), (12-10), (12-16), (12-17), and (12-18) contain terms for component mass-transfer rates, estimated from diffusive and bulk-flow (convective) contributions. The former are based on interfacial area,

average mole-fraction driving forces, and mass-transfer coefficients that account for component coupling effects through binary-pair coefficients. Empirical equations are used for the interfacial area and binary mass-transfer coefficients, based on correlations of experimental data for bubble-cap trays, sieve trays, valve trays, random packings, and structured packings. The average mole-fraction driving forces for diffusion depend upon the assumed vapor and liquid flow patterns. The simplest case is perfectly mixed flow for both the vapor and liquid phases, which simulates small-diameter, trayed columns. The case of countercurrent plug flow for vapor and liquid phases simulates a packed column with no axial dispersion.

Equations (12-6) to (12-8) contain terms for heat-transfer rates. These are estimated from convective and enthalpy-flow contributions, where the former are based on interfacial area, average temperature-driving forces, and convective heat-transfer coefficients, estimated from the Chilton-Colburn analogy for the vapor phase and the penetration theory for the liquid phase.

The K -values in (12-15) are estimated from the same equation-of-state or activity-coefficient models used with equilibrium-based models. Tray or packed-segment pressure drops are estimated from suitable correlations of the type discussed in Chapter 6.

The total number of independent equations, referred to as the MERSHQ equations, for each nonequilibrium stage, is $5C + 5$, as listed in Table 12.1. These equations apply for N stages, that is, $N_E = N(5C + 5)$ equations, in terms of $7NC + 14N + 1$ variables, listed in Table 12.2. The number of degrees of freedom is

$$\begin{aligned} N_D &= N_V - N_E = (7NC + 14N + 1) - (5NC + 5N) \\ &= 2NC + 9N + 1 \end{aligned}$$

Table 12.1 Summary of Independent Equations for Rate-Based Model

Equation	No. of Equations
$M_{i,j}^L$	C
$M_{i,j}^V$	C
$M_{T,j}^L$	1
$M_{T,j}^V$	1
E_j^L	1
E_j^V	1
E_j^I	1
$R_{i,j}^L$	$C - 1$
$R_{i,j}^V$	$C - 1$
S_j^{LI}	1
S_j^{VI}	1
H_j	(optional)
$Q_{i,j}^I$	C
	$5C + 5$

Table 12.2 List of Variables for Rate-Based Model

Variable Type No.	Variable	No. of Variables
1	No. of stages, N	1
2	$f_{i,j}^L$	NC
3	$f_{i,j}^V$	NC
4	T_j^{LF}	N
5	T_j^{VF}	N
6	P_j^{LF}	N
7	P_j^{VF}	N
8	L_j	N
9	$x_{i,j}$	NC
10	r_j^L	N
11	T_j^L	N
12	V_j	N
13	$y_{i,j}$	NC
14	r_j^V	N
15	T_j^V	N
16	P_j	N
17	Q_j^L	N
18	Q_j^V	N
19	$x_{i,j}^I$	NC
20	$y_{i,j}^I$	NC
21	T_j^I	N
22	$N_{i,j}$	NC
		$N_V = 7NC + 14N + 1$

If variable types 1 to 7, 10, 14, and 16 to 18 in Table 12.2 are specified, a total of $2NC + 9N + 1$ variables are assigned values and the degrees of freedom are totally consumed. Then, the remaining $5C + 5$ independent variables in the $5C + 5$ equations are

$$x_{i,j}, y_{i,j}, x_{i,j}^I, y_{i,j}^I, N_{i,j}, T_j^L, T_j^V, T_j^I, L_j, \text{ and } V_j$$

which are the variables to be computed from the equations. Properties $K_{i,j}^I$, H_j^{LF} , H_j^{VF} , H_j^L , and H_j^V are computed from thermodynamic correlations in terms of the remaining independent variables. Transport rates $N_{i,j}^L$, $N_{i,j}^V$, e_j^L , and e_j^V are computed from transport correlations and certain physical properties, in terms of the remaining independent variables. Stage pressures are computed from pressure drops, ΔP_j , stage geometry, fluid-mechanic equations, and certain physical properties, in terms of the remaining independent variables.

For a distillation column, it is preferable to specify $Q_1^L = 0$ and $Q_N^V = 0$. In that case, Q_1^V (heat-transfer rate from the vapor in the condenser) and Q_N^L (heat-transfer rate to the liquid in the reboiler) are not specified, but instead, as in the case of a column with a partial condenser, L_1 (reflux rate) and L_N (bottoms flow rate) are substitute specifications, which are sometimes referred to as standard specifications for ordinary distillation. For an adiabatic absorber or adiabatic stripper, however, all Q_j^L and Q_j^V are set equal to zero, with no substitution of specifications.

12.2 THERMODYNAMIC PROPERTIES AND TRANSPORT-RATE EXPRESSIONS

Rate-based models use the same K -value and enthalpy correlations as equilibrium-based models. However, the K -values apply only at the equilibrium interface between the vapor and liquid phases on trays or in packing. In general, the K -value correlation, whether based on an equation-of-state or an activity-coefficient model, is a function of phase-interface temperature and compositions, and tray pressure. Enthalpies are evaluated only at the conditions of the phases as they exit a tray. For the equilibrium-based model, the vapor is at the dew-point temperature and the liquid is at the bubble-point temperature, where both temperatures are equal and at the stage temperature. For the rate-based model, the liquid is subcooled and the vapor superheated.

The accuracy of enthalpies and, particularly, K -values is crucial to equilibrium-based models. For rate-based models, accurate predictions of heat-transfer rates and, particularly, mass-transfer rates are also required. These rates depend upon transport coefficients, interfacial area, and driving forces. It is important that mass-transfer rates account for component-coupling effects through binary-pair coefficients.

The general forms for component mass-transfer rates across the vapor and liquid films, respectively, on a tray or in a packed segment, are as follows, where both diffusive and convective (bulk-flow) contributions are included:

$$N_{i,j}^V = a_j^I J_{i,j}^V + y_{i,j} N_{T,j} \quad (12-19)$$

and

$$N_{i,j}^L = a_j^I J_{i,j}^L + x_{i,j} N_{T,j} \quad (12-20)$$

where a_j^I is the total interfacial area for the stage and $J_{i,j}^P$ is the molar diffusion flux relative to the molar-average velocity, where P stands for the phase (V or L). For a binary mixture, as discussed in Chapter 3, these fluxes, in terms of mass-transfer coefficients, are given by

$$J_i^V = c_i^V k_i^V (y_i^V - y_i^I)_{\text{avg}} \quad (12-21)$$

and

$$J_i^L = c_i^L k_i^L (x_i^I - x_i^L)_{\text{avg}} \quad (12-22)$$

where c_i^P is the total molar concentration, k_i^P is the mass-transfer coefficient for a binary mixture based on a mole-fraction driving force, and the last terms in (12-21) and (12-22) are the mean mole-fraction driving forces over the stage. The positive direction of mass transfer is assumed to be from the vapor phase to the liquid phase. From the definition of the molar diffusive flux:

$$\sum_{i=1}^C J_i = 0 \quad (12-23)$$

Thus, for the binary system (1, 2), $J_1 = -J_2$.

As discussed in detail by Taylor and Krishna [15], the general multicomponent case for mass transfer is considerably more complex than the binary case because of component-coupling effects. For example, for the ternary system (1, 2, 3),

from Taylor and Krishna [15], the fluxes for the first two components are

$$J_1^V = c_i^V \kappa_{11}^V (y_1^V - y_1^I)_{\text{avg}} + c_i^V \kappa_{12}^V (y_2 - y_2^I)_{\text{avg}} \quad (12-24)$$

$$J_2^V = c_i^V \kappa_{21}^V (y_1^V - y_1^I)_{\text{avg}} + c_i^V \kappa_{22}^V (y_2 - y_2^I)_{\text{avg}} \quad (12-25)$$

The flux for the third component is not independent of the other two, but is obtained from (12-23):

$$J_3^V = -J_1^V - J_2^V \quad (12-26)$$

In these equations, the binary-pair coefficients, κ^P , are complex functions related to inverse-rate functions described below and are called Maxwell–Stefan mass-transfer coefficients in binary mixtures.

For the general multicomponent system (1, 2, ..., C), the independent fluxes for the first $C - 1$ components are given in matrix equation form as

$$\mathbf{J}^V = c_i^V [\kappa^V] (\mathbf{y}^V - \mathbf{y}^I)_{\text{avg}} \quad (12-27)$$

$$\mathbf{J}^L = c_i^L [\kappa^L] (\mathbf{x}^I - \mathbf{x}^L)_{\text{avg}} \quad (12-28)$$

where $\mathbf{J}^P$, $(\mathbf{y}^V - \mathbf{y}^I)_{\text{avg}}$, and $(\mathbf{x}^I - \mathbf{x}^L)_{\text{avg}}$ are column vectors of length $C - 1$ and $[\kappa^P]$ is a $(C - 1) \times (C - 1)$ square matrix. The method for determining the average mole-fraction driving forces depends, as discussed in the next section, upon the flow patterns of the vapor and liquid phases.

The most fundamental theory for multicomponent diffusion is that of Maxwell and Stefan, who, in the period from 1866 to 1871, applied the kinetic theory of ideal gases. Their theory is presented most conveniently in terms of rate coefficients, $\mathbf{B}$, which are defined in reciprocal diffusivity terms [15]. Likewise, it is convenient to determine $[\kappa^P]$ from a reciprocal mass-transfer coefficient function, $\mathbf{R}$, defined by Krishna and Standart [14]. For an ideal-gas solution:

$$[\kappa^V] = [\mathbf{R}^V]^{-1} \quad (12-29)$$

For a nonideal-liquid solution:

$$[\kappa^L] = [\mathbf{R}^L]^{-1} [\Gamma^L] \quad (12-30)$$

where the elements of $\mathbf{R}^P$ in terms of general mole fractions, z_i , are:

$$R_{ii}^P = \frac{z_i}{k_{iC}^P} + \sum_{\substack{k=1 \\ k \neq i}}^C \frac{z_k}{k_{ik}^P} \quad (12-31)$$

$$R_{ij}^P = -z_i \left(\frac{1}{k_{ij}^P} - \frac{1}{k_{iC}^P} \right) \quad (12-32)$$

where here, j refers to the j th component and not the j th stage and the values of k are binary-pair mass-transfer coefficients obtained from correlations of experimental data.

For the four-component vapor-phase system, the combination of (12-27) and (12-29) gives

$$\begin{bmatrix} J_1^V \\ J_2^V \\ J_3^V \end{bmatrix} = c_i^V \begin{bmatrix} R_{11}^V & R_{12}^V & R_{13}^V \\ R_{21}^V & R_{22}^V & R_{23}^V \\ R_{31}^V & R_{32}^V & R_{33}^V \end{bmatrix}^{-1} \begin{bmatrix} (y_1^V - y_1^I)_{\text{avg}} \\ (y_2^V - y_2^I)_{\text{avg}} \\ (y_3^V - y_3^I)_{\text{avg}} \end{bmatrix} \quad (12-33)$$

with

$$J_4^V = -(J_1^V + J_2^V + J_3^V) \quad (12-34)$$

and, for example, from (12-32) and (12-33), respectively:

$$R_{11}^V = \frac{y_1}{k_{14}^V} + \frac{y_2}{k_{12}^V} + \frac{y_3}{k_{13}^V} + \frac{y_4}{k_{14}^V} \quad (12-35)$$

$$R_{12}^V = -y_1 \left(\frac{1}{k_{12}^V} - \frac{1}{k_{14}^V} \right) \quad (12-36)$$

The term $[\Gamma^L]$ in (12-30) is a $(C-1) \times (C-1)$ matrix of thermodynamic factors that corrects for nonideality, which often is a necessary correction for the liquid phase. When an activity-coefficient model is used:

$$\Gamma_{ij}^L = \delta_{ij} + x_i \left(\frac{\partial \ln \gamma_i}{\partial x_j} \right)_{T, P, x_k, k \neq j=1, \dots, C-1} \quad (12-37)$$

For a nonideal vapor, a $[\Gamma^V]$ term can be included in (12-29), but this is rarely necessary. For either phase, if an equation-of-state model is used, (12-37) can be rewritten by substituting $\bar{\phi}_i$, the mixture fugacity coefficient, for γ_i . The term δ_{ij} is the Kronecker delta, which is 1 if $i = j$ and 0 if not. The thermodynamic factor is required because it is generally accepted that the fundamental driving force for diffusion is the gradient of the chemical potential rather than the mole fraction or concentration gradient.

When mass-transfer fluxes are moderate to high, an additional correction term is needed in (12-29) and (12-30) to correct for distortion of the composition profiles. This correction, which can have a serious effect on the results, is discussed in detail by Taylor and Krishna [15]. The calculation of the low mass-transfer fluxes, according to (12-19) to (12-32), is illustrated by the following example.

EXAMPLE 12.1

This example is similar to Example 11.5.1 on page 283 of Taylor and Krishna [15]. The following results were obtained for tray n from a rate-based calculation of a ternary distillation at 14.7 psia, involving acetone (1), methanol (2), and water (3) in a 5.5-ft-diameter column using sieve trays with a 2-inch-high weir. Vapor and liquid phases are assumed to be completely mixed.

Component	y_n	y_{n+1}	y_n^I	K_n^I	x_n
1	0.2971	0.1700	0.3521	2.759	0.1459
2	0.4631	0.4290	0.4677	1.225	0.3865
3	0.2398	0.4010	0.1802	0.3673	0.4676
	1.0000	1.0000	1.0000		1.0000

The computed products of the gas-phase, binary mass-transfer coefficients and interfacial area, using the Chan-Fair correlations of Section 6.6, are as follows in lbmol/(h-unit mole fraction):

$$k_{12} = k_{21} = 1,955; \quad k_{13} = k_{31} = 2,407; \quad k_{23} = k_{32} = 2,797$$

(a) Compute the molar diffusion rates.

(b) Compute the mass-transfer rates.

(c) Calculate the Murphree vapor-tray efficiencies.

SOLUTION

Because rates instead of fluxes are given, the equations developed in this section are used with rates rather than fluxes.

(a) Compute the reciprocal rate functions, $\mathbf{R}$, from (12-31) and (12-32), assuming linear mole-fraction gradients such that z_i can be replaced by $(y_i + y_i^I)/2$.

Thus:

$$z_1 = (0.2971 + 0.3521)/2 = 0.3246$$

$$z_2 = (0.4631 + 0.4677)/2 = 0.4654$$

$$z_3 = (0.2398 + 0.1802)/2 = 0.2100$$

$$R_{11}^V = \frac{z_1}{k_{13}} + \frac{z_2}{k_{12}} + \frac{z_3}{k_{13}} = \frac{0.3246}{2,407} + \frac{0.4654}{1,955} + \frac{0.2100}{2,407} = 0.000460$$

$$R_{22}^V = \frac{z_2}{k_{23}} + \frac{z_1}{k_{21}} + \frac{z_3}{k_{23}} = \frac{0.4654}{2,797} + \frac{0.3246}{1,955} + \frac{0.2100}{2,797} = 0.000408$$

$$R_{12}^V = -z_1 \left(\frac{1}{k_{12}} - \frac{1}{k_{13}} \right) = -0.3246 \left(\frac{1}{1,955} - \frac{1}{2,407} \right) = -0.0000312$$

$$R_{21}^V = -z_2 \left(\frac{1}{k_{21}} - \frac{1}{k_{23}} \right) = -0.4654 \left(\frac{1}{1,955} - \frac{1}{2,797} \right) = -0.0000717$$

$$\text{Thus, in matrix form: } [\mathbf{R}^V] = \begin{bmatrix} 0.000460 & -0.0000312 \\ -0.0000717 & 0.000408 \end{bmatrix}$$

From (12-29), by matrix inversion:

$$[\kappa^V] = [\mathbf{R}^V]^{-1} = \begin{bmatrix} 2,200 & 168.2 \\ 386.6 & 2,480 \end{bmatrix}$$

Because the off-diagonal terms in the preceding 2×2 matrix are much smaller than the diagonal terms, the effect of coupling in this example is small.

From (12-27):

$$\begin{bmatrix} J_1^V \\ J_2^V \end{bmatrix} = \begin{bmatrix} \kappa_{11}^V & \kappa_{12}^V \\ \kappa_{21}^V & \kappa_{22}^V \end{bmatrix} \begin{bmatrix} (y_1 - y_1^I) \\ (y_2 - y_2^I) \end{bmatrix}$$

$$J_1^V = \kappa_{11}^V(y_1 - y_1^I) + \kappa_{12}^V(y_2 - y_2^I) = 2,200(0.2971 - 0.3521) + 168.2(0.4631 - 0.4677) = -121.8 \text{ lbmol/h}$$

$$J_2^V = \kappa_{21}^V(y_1 - y_1^I) + \kappa_{22}^V(y_2 - y_2^I) = 386.6(0.2971 - 0.3521) + 2,480(0.4631 - 0.4677) = -32.7 \text{ lbmol/h}$$

From (12-23):

$$J_3^V = -J_1^V - J_2^V = 121.8 + 32.7 = 154.5 \text{ lbmol/h}$$

(b) From (12-19), but with diffusion and mass-transfer rates instead of fluxes:

$$N_1^V = J_1^V + z_1 N_T^V = -121.8 + 0.3246 N_T^V \quad (1)$$

Similarly:

$$N_2^V = -32.7 + 0.4654 N_T^V \quad (2)$$

$$N_3^V = 154.5 + 0.2100 N_T^V \quad (3)$$

To determine the component mass-transfer rates, it is necessary to know the total mass-transfer rate for the tray, N_T^V . The problem of determining this quantity when the diffusion rates, J , are known is referred to as the *bootstrap problem* (p. 145 in Taylor and Krishna [15]). In chemical reaction with diffusion, N_T is determined by the stoichiometry. In distillation, N_T is determined by an energy balance, which gives the change in molar vapor rate across a tray. For the assumption of constant molar overflow, $N_T = 0$. In this example, that assumption is not valid, and the change is

$$N_T = V_{n+1} - V_n = -54 \text{ lbmol/h}$$

From (1), (2), (3):

$$N_1^V = -121.8 + 0.3246(-54) = -139.4 \text{ lbmol/h}$$

$$N_2^V = -32.7 + 0.4654(-54) = -57.8 \text{ lbmol/h}$$

$$N_3^V = -154.5 + 0.2100(-54) = 143.2 \text{ lbmol/h}$$

(c) Approximate values of the Murphree vapor-tray efficiency are obtained from (12-3), with K -values at phase interface conditions:

$$E_{MV_i} = (y_{i,n} - y_{i,n+1}) / (K_{i,n}^I x_{i,n} - y_{i,n+1}) \quad (4)$$

From (4):

$$E_{MV_1} = \frac{(0.2971 - 0.1700)}{[(2.759)(0.1459) - 0.1700]} = 0.547$$

$$E_{MV_2} = \frac{(0.4631 - 0.4290)}{[(1.225)(0.3865) - 0.4290]} = 0.767$$

$$E_{MV_3} = \frac{(0.2398 - 0.4010)}{[(0.3673)(0.4676) - 0.4010]} = 0.703$$

The general forms for rates of heat transfer across the vapor and liquid films of a stage, respectively, are

$$e_j^V = a_j^I h^V (T^V - T^I) + \sum_{i=1}^C N_{i,j}^V \bar{H}_{i,j}^V \quad (12-38)$$

$$e_j^L = a_j^I h^L (T^I - T^L) + \sum_{i=1}^C N_{i,j}^L \bar{H}_{i,j}^L \quad (12-39)$$

where $\bar{H}_{i,j}^P$ are the partial molar enthalpies of component i for stage j and h^P are convective heat-transfer coefficients. The second terms on the right-hand sides of (12-38) and (12-39) account for the transfer of enthalpy by mass transfer. Temperatures T^V and T^L are the temperatures exiting the stage regardless of the assumed flow patterns for the vapor and liquid.

12.3 METHODS FOR ESTIMATING TRANSPORT COEFFICIENTS AND INTERFACIAL AREA

Equations (12-31) and (12-32) require binary-pair mass-transfer coefficients. In most rate-based model applications, the coefficients are estimated from empirical correlations of experimental data for different contacting devices.

As discussed in Section 6.6, for trayed columns, the most widely used correlations are the AIChE method [20] for bubble-cap trays, the correlations of Chan and Fair [24] for

sieve trays, and the correlations of Scheffe and Weiland [36] for Glitsch V-1 valve trays. Other important correlations are those of Harris [21] and Hughmark [22] for bubble-cap trays; and Zuiderweg [23], Chen and Chuang [25], Taylor and Krishna [15], and Young and Stewart [37, 38] for sieve trays.

Some mass-transfer correlations are presented in terms of the number of transfer units, N_V and N_L , where, by definition:

$$N_V \equiv k^V a h_f / u_s \quad (12-40)$$

$$N_L \equiv k^L a h_f z / (Q_L / W) \quad (12-41)$$

where

a = interfacial area/volume of froth on the tray

h_f = froth height

u_s = superficial vapor velocity based on the bubbling area of the tray

z = length of liquid-flow path across the bubbling area of the tray

Q_L = volumetric liquid flow rate

W = weir length

The interfacial area for a tray, a^I , is related to a by

$$a^I = a h_f A_b \quad (12-42)$$

where

A_b = bubbling area

Thus, k^P and a^I are obtained from correlations in terms of N_V and N_L .

For random (dumped) packings, empirical correlations for mass-transfer coefficients and interfacial-area density (area/packed volume) have been published by Onda, Takeuchi, and Okumoto [26] and Bravo and Fair [27]. For structured packings, the empirical correlations of Bravo, Rocha, and Fair for gauze packings [28] and for a wide variety of structured packings [29] are available. A semi-theoretical correlation by Billet and Schultes [30] based on over 3,500 data points for more than 50 test systems and more than 70 different types of packings, requires five packing parameters and is applicable to both random and structured packings. This correlation is discussed in Section 6.8.

Heat-transfer coefficients for the vapor film are usually estimated from the Chilton-Colburn analogy between heat and mass transfer, described in Chapter 3. Thus,

$$h^V = k^V \rho^V C_P^V (N_{Le})^{2/3} \quad (12-43)$$

where:

$$N_{Le} = \left(\frac{N_{Sc}}{N_{Pr}} \right) \quad (12-44)$$

For the liquid-phase film, a penetration model is preferred, where

$$h^L = k^L \rho^L C_P^L (N_{Le})^{1/2} \quad (12-45)$$

A more detailed heat-transfer model, specifically for sieve trays, is given by Spagnolo et al. [39].

12.4 VAPOR AND LIQUID FLOW PATTERNS

The simplest flow pattern for a stage corresponds to the assumption of a perfectly mixed vapor and a perfectly mixed liquid. Under these conditions, the mass-transfer driving forces in (12-27) and (12-28) are simplified to

$$(y^V - y^I)_{\text{avg}} = (y^V - y^I) \quad (12-46)$$

$$(x^I - x^L)_{\text{avg}} = (x^I - x^L) \quad (12-47)$$

where y^V and x^L are exiting stage mole fractions. These flow patterns are only valid for trayed towers with a short liquid flow path.

A plug-flow pattern for the vapor and/or liquid assumes that the phase moves through the froth without mixing. This pattern requires that the mass-transfer rates be integrated over the froth. An approximation of the integration is provided by Kooijman and Taylor [31], who assume constant mass-transfer coefficients and interface compositions. The resulting expressions for the average mole-fraction driving forces are the same as (12-46) and (12-47) except for a correction factor in terms of N^V or N^L , included on the right-hand side of each equation. Plug-flow patterns are generally more accurate for trayed towers than perfectly mixed flow patterns and are also applicable to packed towers.

The perfectly-mixed-flow and plug-flow patterns are the two patterns presented by Lewis [7] to convert Murphree vapor point efficiencies to Murphree vapor tray efficiencies, as discussed in Section 6.5. They represent the extreme situations. Fair, Null, and Bolles [32] recommend a more realistic partial mixing or dispersion model that utilizes a turbulent Peclet number, whose value can cover a wide range. This model provides a bridge between the two extremes.

For reactive distillation, a rate-based multicell (or mixed pool) model has proved useful. In this model, the liquid on the tray is assumed to flow horizontally across the tray through a series of perfectly mixed cells (perhaps 4 or 5). In the model of Higler, Krishna, and Taylor [40], which is available in v4.3 of the ChemSep program, the vapor phase is also assumed to be perfectly mixed in each cell. If desired, cells for each tray can also be stacked in the vertical direction. Thus, a tray model might consist of a 5×5 cell arrangement for a total of 25 perfectly mixed cells. Higler, Krishna, and Taylor assume that the vapor streams leaving the topmost cells on a tray are collected and mixed before being divided to enter the cells on the next tray. The rate-based multicell model of Pyhalahiti and Jakobsson [41] allows only one set of cells in the horizontal direction, but the vapor streams leaving the cells on a tray may be mixed or not mixed before entering the cells on the next tray and the reversal of liquid flow direction from tray to tray, shown in Figure 6.21 for single-pass trays, is allowed.

12.5 METHOD OF CALCULATION

As indicated in Section 12.1, the number of equations to be solved for the single-cell per tray, rate-based model of Figure 12.1 is $N(5C + 5)$ when the pressure-drop equations

are omitted, as summarized in Table 12.1. The equations contain the variables listed in Table 12.2. Other parameters in the equations are computed from these variables. When the number of equations is subtracted from the number of variables, the number of degrees of freedom is $2NC + 9N + 1$. If the total number of stages and all column feed conditions, including feed-stage locations ($2NC + 4N + 1$ variables) are specified, the number of remaining degrees of freedom, using the variable designations in Table 12.2, is $5N$. A computer program for the rate-based model would generally require the user to specify these $2NC + 4N + 1$ variables. The degree of flexibility provided to the user in the selection of the remaining $5N$ variables depends on the particular rate-based computer algorithm, three of which are widely available: (1) Chem-Sep Release v4.3 from R. Taylor and H. A. Kooijman, (2) RATEFRAC in Release 12 of ASPEN PLUS from Aspen Technology, Cambridge, Massachusetts, and (3) CHEMCAD v5.4. Both algorithms provide a wide variety of correlations for thermodynamic and transport properties. Both programs also provide considerable flexibility in the selection of the remaining $5N$ specifications. The basic $5N$ specifications are

$$r_j^L \text{ or } U_j, r_j^V \text{ or } W_j, P_j, Q_j^L, \text{ and } Q_j^V$$

However, substitutions can be made as discussed next.

ChemSep Program

The ChemSep program applies the transport equations to trays or short heights (called segments) of packing. The condenser and reboiler stages are treated as equilibrium stages. The specification options include:

1. r_j^L and r_j^V : From each stage, either a liquid or a vapor side stream can be specified as (a) a side-stream flow rate or (b) a ratio of the side-stream flow rate to the flow rate of the remaining fluid passing to the next stage, that is,

$$r_j^L = U_j/L_j \quad \text{or} \quad r_j^V = W_j/V_j \text{ in Figure 12.1.}$$
2. P_j : Four options are available:
 - (a) Condenser pressure (if any) and a constant, but different, pressure for all stages in the tower and for the reboiler, if any.
 - (b) Condenser pressure (if any), top tower pressure, and bottom pressure (bottom tower stage or reboiler, if any). Pressures of stages intermediate between top and bottom are obtained by linear interpolation.
 - (c) Condenser pressure (if any), top tower pressure, and specified pressure drop per stage to obtain remaining stage pressures.
 - (d) Condenser pressure (if any) and top tower pressure, with stage pressure drops estimated by ChemSep from hydraulic correlations.

3. Q_j^L and Q_j^V : The heat duty must be specified for all stage heaters and coolers except for the condenser and/or reboiler, if present. In addition, a heat loss for the tower can be specified that is divided equally over all stages. When a condenser (total without subcooling, total with subcooling, or partial) is present, one of the following specifications can replace the heat duty of the condenser: (a) molar reflux ratio, (b) condensate temperature, (c) distillate molar flow rate, (d) reflux molar flow rate, (e) component molar flow rate in distillate, (f) mole fraction of a component in distillate, (g) fractional recovery, from all feeds, of a component in the distillate, (h) molar fraction of all feeds to the distillate, and (i) molar ratio of two components in the distillate.

For distillation, an often-used specification is the molar reflux ratio.

When a reboiler (partial, total with a vapor product, or total with a superheated vapor product) is present, the following list of specification options, similar to those just given for a condenser, can replace the heat duty of the reboiler: (a) molar boilup ratio, (b) reboiler temperature, (c) bottoms molar flow rate, (d) reboiled-vapor (boilup) molar flow rate, (e) component molar flow rate in bottoms, (f) mole fraction of a component in bottoms, (g) fractional recovery, from all feeds, of a component in the bottoms, (h) molar fraction of all feeds to the bottoms, and (i) molar ratio of two components in the bottoms. For distillation, an often-used specification is the molar bottoms flow rate, which must be estimated if it is not specified.

The preceding number of optional specifications is considerable. In addition, ChemSep also provides "flexible" specifications that can substitute for the condenser and/or reboiler duties. These are advanced options supplied in the form of strings that contain values of certain allowable variables and/or combinations of these variables using the five common arithmetic operators (+, -, *, /, and exponentiation). The variables include stage variables (L , V , x , y , and T) and interface variables (x^I , y^I , and T^I) at any stage. Flow rates can be in mole or mass units.

Certain options and advanced options must be used with great care because values can be specified that cannot lead to a converged solution. For example, with a simple distillation column of a fixed number of stages, that number may be less than the minimum number to achieve specified distillate and bottoms purities. As always, it is generally wise to begin a simulation with a standard pair of top and bottom specifications, such as reflux ratio and a bottoms molar flow rate that corresponds to the desired distillate rate. These specifications are almost certain to converge unless interstage liquid or vapor flow rates tend to zero somewhere in the column. A study of the calculated results will provide valuable insight into possible limits in the use of other options.

The equations for the rate-based model, some linear and some nonlinear, are solved by Newton's method in a manner similar to that developed by Naphtali and Sandholm for the

equilibrium-based model described in Chapter 10. Thus, the variables and equations are grouped by stage so that the Jacobian matrix is of the block-tridiagonal form. However, the equations to be solved number $5C + 6$ or $5C + 5$ per stage, depending on whether stage pressures are computed or specified, compared to just $2C + 1$ for the equilibrium-based method.

Calculations of transport coefficients and pressure drops require column diameter and dimensions of column internals. These may be specified (simulation mode) or computed (design mode). In the latter case, default dimensions are selected for the internals, with column diameter computed from a specified value for percent of flooding for a trayed or packed column, or a specified pressure drop per unit height for a packed column.

Computing time per iteration for the design mode is only approximately twice that for the simulation mode, which usually requires less than twice the time for the equilibrium-based model. The number of iterations required for the design mode can be two to three times that for the equilibrium-based model. Overall, the total computing time for the design mode is usually less than an order of magnitude greater than that for the equilibrium-based model. With today's fast workstations and PCs, computing times for the design mode of the rate-based model are usually less than one minute.

Like the Naphtali-Sandholm method, the rate-based model utilizes mainly analytical partial derivations in the Jacobian matrix, and requires initial estimates of all variables. These estimates are generated automatically by the ChemSep program using a method of Powers et al. [33]. In this method, the usual assumptions of constant molar overflow and a linear temperature profile are employed. The initialization of the stage mole fractions is made by performing several iterations of the bubble-point method using ideal K -values for the first iteration and nonideal K -values thereafter. Initial interface mole fractions are set equal to estimated bulk values and initial mass-transfer rates are arbitrarily set to values of $\pm 10^{-3}$ kmol/h with the sign dependent upon the component K -value.

To prevent oscillations and promote convergence of the iterations, corrections to certain variables from iteration to iteration can be limited. Defaults are 10 K for temperature and 50% for flows. When a correction to a mole fraction would result in a value outside of the feasible range of 0 to 1, the default correction is one-half of the step that would take the value to a limit. For very difficult problems, homotopy-continuation methods described by Powers et al. [33] can be applied to promote convergence.

Convergence of Newton's method is determined from values of the residuals of the functions, as in the Naphtali-Sandholm method, or from the corrections to the variables. ChemSep applies both criteria and terminates when either of the following are satisfied:

$$\left[\sum_{j=1}^N \sum_{k=1}^{N_j} f_{k,j}^2 \right]^{1/2} < \epsilon \quad (12-48)$$

$$\sum_{j=1}^N \sum_{k=1}^{N_j} |\Delta X_{k,j}| / X_{k,j} < \epsilon \quad (12-49)$$

where

$f_{k,j}$ = residuals in Table 12.1

N = number of stages

N_j = number of equations for the j th stage

$X_{k,j}$ = unknown variables from Table 12.2

ϵ = a small number with a default value of 10^{-4}

Unlike in the Naphtali–Sandholm method, the residuals are not scaled. Accordingly, the second criterion is usually satisfied first.

From the results of a converged solution, it is highly desirable to back-calculate Murphree vapor-tray efficiencies, component by component and tray by tray, from (12-3) for trayed columns, and HETP values for packed towers.

ChemSep can also perform rate-based calculations for liquid–liquid extraction.

EXAMPLE 12.2

A mixture of *n*-heptane and toluene cannot be separated at 1 atm by ordinary distillation. Accordingly, an enhanced-distillation scheme using methylethyl ketone as a solvent is used. As part of an initial design study, use the rate-based model of ChemSep with the specifications listed in Table 12.3 to calculate a sieve-tray column.

SOLUTION

The information in Table 12.3 was entered via the ChemSep menu and the program was executed. A converged solution was achieved in 8 iterations in 6 seconds on a PC with a Pentium 90 CPU, running the Windows 95 operating system. Initialization of all variables was done by the program.

The predicted separation is as follows:

Component	Distillate, lbmol/h	Bottoms, lbmol/h
<i>n</i> -Heptane	54.87	0.13
Toluene	0.45	44.55
Methylethyl ketone	199.68	0.32

Predicted-column profiles for pressure, liquid-phase temperature, total vapor and liquid flow rates, component vapor and liquid mole fractions, component mass-transfer rates, and Murphree vapor-tray efficiencies are shown in Figure 12.2, where stages are numbered from the top down and stages 2 to 21 are sieve trays. Back-calculated Murphree tray efficiencies are summarized as follows:

Component	Fractional Murphree Efficiencies	
	Range	Median
<i>n</i> -Heptane	0.52 to 1.10	0.73
Toluene	0.70 to 0.79	0.79
Methylethyl ketone	–3.23 to 1.14	0.76

Table 12.3 Specifications for Example 12.2

Total condenser delivering saturated liquid
Partial reboiler
Pressure at condenser outlet = 14.7 psia
Pressure at condenser inlet = 15.0 psia
Reflux ratio = 1.5
Bottoms flow rate = 45 lbmol/h
Total number of trays = 20
Feed 1 to tray 10 from top:
55 lbmol/h of <i>n</i> -heptane
45 lbmol/h of toluene
100 lbmol/h of methylethyl ketone (MEK)
Saturated liquid at 20 psia
Feed 2 to tray 15 from top:
100 lbmol/h of MEK
Saturated liquid at 20 psia
UNIFAC for liquid-phase activity coefficients
Chan–Fair correlation for mass-transfer coefficients
Plug flow for vapor
Mixed flow for liquid
85% of flooding
Tray spacing = 0.5 m (19.7 inches)
Weir height = 2 inches

The median values, based on experience, seem reasonable and give confidence in the rate-based method. The 20 trays are equivalent to approximately 15 equilibrium stages.

For purposes of sizing, the column was divided into three sections: 9 trays above the top feed, 5 trays from the top feed to the bottom feed, and 6 trays below the bottom feed. Computed column diameters are, respectively, 1.75 m (5.74 ft), 1.74 m (5.71 ft), and 1.83 m (6.00 ft). Thus, a 1.83-m (6.00-ft)-diameter column is a reasonable choice. Average predicted pressure drop per tray is 0.06 psi. Computed heat-exchanger duties are as follows:

Condenser: 2.544 MW (8,680,000 Btu/h)

Reboiler: 2.482 MW (8,470,000 Btu/h)

EXAMPLE 12.3

Repeat Example 12.2 for a tower packed with Flexipac 2 structured packing, operating at 75% of flooding. The packing heights are as follows:

Section	Packing Height, ft
Above top feed	13
Between top and bottom feeds	6.5
Below bottom feed	6.5

SOLUTION

Each 6.5 feet of packing was simulated by 50 segments. Because of the large number of segments, mixed flow is assumed for both vapor and liquid. Newton's method could not converge the calculations. Therefore, the homotopy-continuation option was selected.

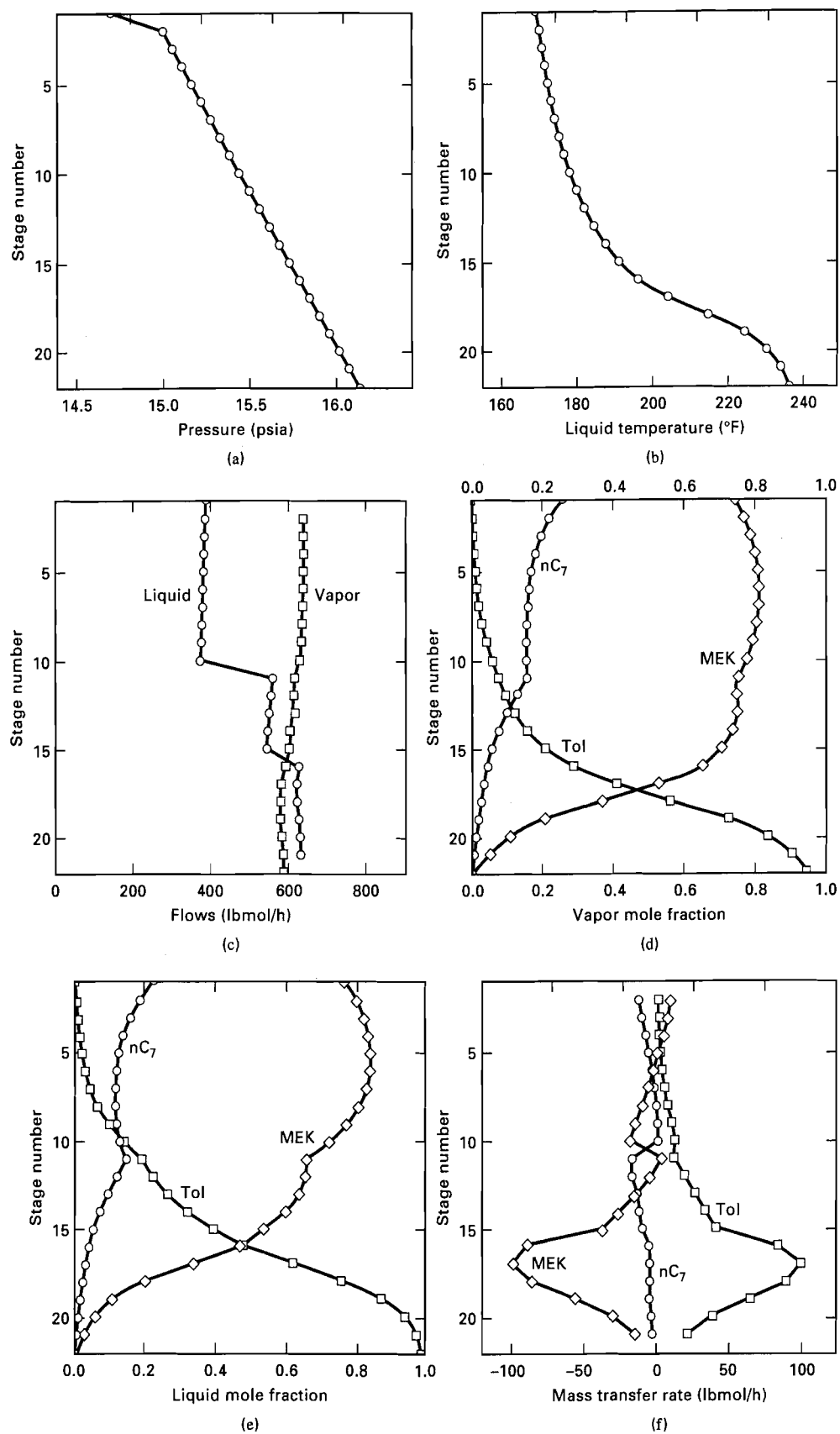


Figure 12.2 Column profiles for Example 12.2: (a) pressure profile; (b) liquid-phase temperature profile; (c) vapor and liquid flow rate profiles; (d) vapor mole-fraction profiles; (e) liquid mole-fraction profiles; (f) mass-transfer rate profiles. (continued)

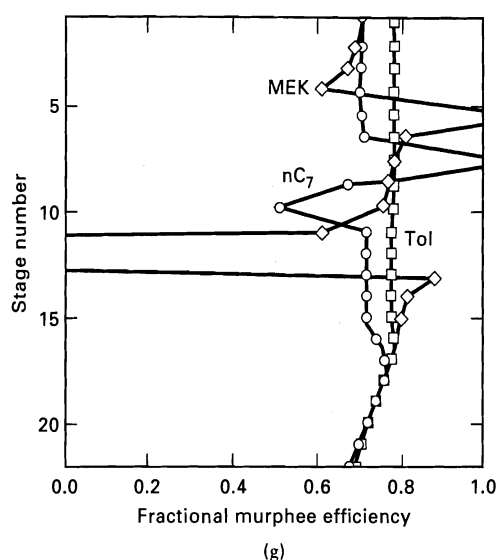


Figure 12.2 (Continued) (g) Murphree vapor-tray efficiencies.

Then convergence was achieved in 73 s after a total of 26 iterations. The predicted separation, which is just slightly better than that in Example 12.2, is as follows:

Component	Distillate, lbmol/h	Bottoms, lbmol/h
<i>n</i> -Heptane	54.88	0.12
Toluene	0.40	44.60
Methylethyl ketone	199.72	0.28

The HETP profile is plotted in Figure 12.3. Median values for *n*-heptane, toluene, and methylethyl ketone, respectively, are approximately 0.55 m (21.7 inches), 0.45 m (17.7 inches), and 0.5 m (19.7 inches). The HETP values for the ketone are seen to vary widely.

Predicted column diameters for the three sections, starting from the top, are 1.65, 1.75, and 1.85 m, which are very close to the predicted sieve-tray diameters.

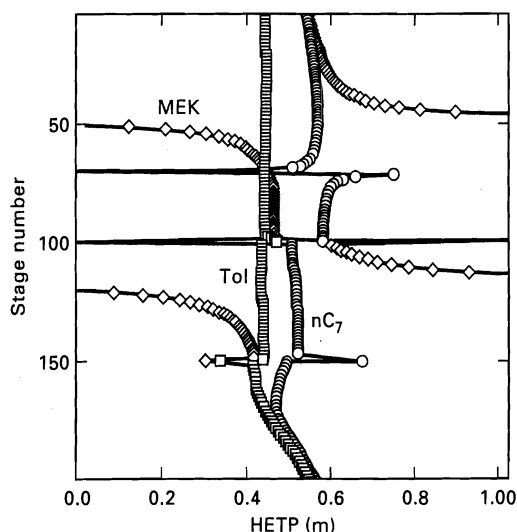


Figure 12.3 Column HETP profiles for Example 12.3.

RATEFRAC Program

The RATEFRAC program of Aspen Technology is designed to model columns used for reactive distillation. The latest version of ChemSep can also model reactive distillation. For RATEFRAC, the reactions can be equilibrium-based or kinetics-based, including reactions among electrolytes. For kinetically controlled reactions, built-in power-law expressions are selected, or the user supplies FORTRAN subroutines for the rate law(s). For equilibrium-based reactions, the user supplies a temperature-dependent equilibrium constant, or RATEFRAC computes reaction-equilibrium constants from free-energy values stored in the data bank. The user specifies the phase in which the reaction takes place. Flow rates of side streams and the column-pressure profile must be provided. The heat duty must be specified for each inter-cooler or interheater. The standard specifications for the rating mode are the reflux ratio and the bottoms flow rate. However, these specifications can be manipulated in the design mode to achieve any of the following substitute specifications:

- Purity of a product or internal stream with respect to one component or a group of components.
- Recovery of a component or group of components in a product stream.
- Flow rate of a component or group of components in a product or internal stream.
- Temperature of a product or internal vapor or liquid stream.
- Heat duty of condenser or reboiler.
- Value of a product or internal stream physical property.
- Ratio or difference of any pair of product or internal stream physical properties, where the two streams can be the same or different.

Mass-transfer correlations are built into RATEFRAC for bubble-cap trays, valve trays, sieve trays, and random packings. Users may provide their own FORTRAN subroutines for transport coefficients and interfacial area. Newton's method is used to converge the calculations.

EXAMPLE 12.4

Use RATEFRAC to predict the column profiles for a 3.5-ft-diameter, 20-bubble-cap tray absorber operating at the conditions listed in Table 12.4.

SOLUTION

No initial estimates of variables were provided. The program was run on a PC with a Pentium 90 CPU running under MS-DOS 6.22. Initial estimates of the values of the variables were provided by RATEFRAC. A total of five iterations were required following an initialization step. Total computing time, including translation,

compiling, linking, and execution, was about 60 s. The following results were obtained for the product streams:

Component	Lean Vapor, lbmol/h	Rich Oil, lbmol/h
Hydrogen	216.6	1.4
Nitrogen	86.1	0.9
Methane	131.4	4.6
Ethane	120.0	19.0
Propane	73.4	44.6
Isobutane	1.1	4.9
<i>n</i> -Butane	0.0	2.0
Isopentane	0.0	43.0
<i>n</i> -Hexane	0.0	14.0
<i>n</i> -Heptane	0.0	4.0
<i>n</i> -Dodecane	0.0	165.0
<i>n</i> -Tridecane	0.0	165.0
	628.6	468.4

The back-calculated fractional Murphree vapor tray efficiencies are as follows:

Component	E_{MV}	
	Range	Median Value
Hydrogen	-1.25 to 1.21	0.43
Nitrogen	-0.51 to 3.25	0.41
Methane	-0.13 to 0.87	0.41
Ethane	-0.03 to 1.02	0.30
Propane	-1.89 to 2.71	0.33
Isobutane	-6.51 to 1.16	0.42
<i>n</i> -Butane	-0.75 to 5.65	0.50
Isopentane	-3.17 to 1.15	0.57
<i>n</i> -Hexane	0.63 to 1.38	0.63
<i>n</i> -Heptane	0.64 to 0.88	0.64
<i>n</i> -Dodecane	-2.92 to 1.01	0.42
<i>n</i> -Tridecane	-5.04 to 0.97	0.42

As chemical engineers become more knowledgeable in the principles of mass and heat transfer, and improved correlations for mass-transfer and heat-transfer coefficients are developed for trays and packings, the use of rate-based models will accelerate. For best results, these models will also benefit from more realistic options for vapor and liquid flow patterns. More comparisons of rate-based models with

Table 12.4 Specifications for Example 12.4

Column top pressure = 182 psia
 Column bottom pressure = 185 psia
 Weir height = 2 inches
 Vapor completely mixed on each tray
 Liquid completely mixed on each tray
 AIChE correlations for binary mass-transfer coefficients and interfacial area
 Chilton-Colburn analogy for heat transfer
 Chao-Seader correlation for K -values

Vapor feed at 123°F and 184 psia:	
Component	lbmol/h
Hydrogen	218
Nitrogen	87
Methane	136
Ethane	139
Propane	118
Isobutane	6
<i>n</i> -Butane	2
Isopentane	43
<i>n</i> -Hexane	14
<i>n</i> -Heptane	4
	767
Liquid absorbent feed at 100°F and 182 psia:	
Component	lbmol/h
<i>n</i> -Dodecane	165
<i>n</i> -Tridecane	165

It is seen that the efficiencies vary widely from component to component and from tray to tray. For absorber simulation and design, a rate-based model is clearly superior to an equilibrium-based model.

industrial operating data are needed to gain confidence in the use of such models. Some recent comparisons are presented by Taylor, Kooijman, and Woodman [34], and Kooijman and Taylor [31]. Comparisons by Ovejero et al. [35], with distillation data obtained in a column packed with spheres and cylinders of known interfacial area, show very good agreement for three binary and two ternary systems.

SUMMARY

1. Rate-based models of multicomponent, multistage, vapor-liquid separation operations became available in the late 1980s. These models are potentially superior to equilibrium-based models for all but near-ideal systems.
2. Rate-based models incorporate rigorous procedures for treating component-coupling effects in multicomponent mass transfer.
3. The number of equations for a rate-based model is greater than for an equilibrium-based model because separate balances are needed for each of the two phases. In addition, rate-based models are influenced by the geometry of the column internals. Correlations are used to predict mass-transfer and heat-transfer rates. Tray or

packing hydraulics are also incorporated into the rate-based model to enable prediction of column-pressure profile. Equilibrium is assumed at the phase interface.

4. Computing time for a rate-based model is not generally more than an order of magnitude greater than that for an equilibrium-based model.

5. Both the ChemSep and RATEFRAC rate-based computer programs offer considerable flexibility in user specifications, so much so that inexperienced users can easily specify impossible conditions. Therefore, it is best to begin simulation studies with standard specifications.

REFERENCES

1. SOREL, ERNEST, *La rectification de l'alcool*, Paris (1893).
2. SMOKER, E.H., *Trans. AIChE*, **34**, 165 (1938).
3. THIELE, E.W., and R.L. GEDDES, *Ind. Eng. Chem.*, **25**, 290 (1933).
4. LEWIS, W.K., and G.L. MATHESON, *Ind. Eng. Chem.*, **24**, 496–498 (1932).
5. LEWIS, W.K., *Ind. Eng. Chem.*, **14**, 492 (1922).
6. MURPHREE, E.V., *Ind. Eng. Chem.*, **17**, 747–750, 960–964 (1925).
7. LEWIS, W.K., *Ind. Eng. Chem.*, **28**, 399 (1936).
8. SEADER, J.D., *Chem. Eng. Prog.*, **85** (10), 41–49 (1989).
9. WALTER, J.F., and T.K. SHERWOOD, *Ind. Eng. Chem.*, **33**, 493–501 (1941).
10. TOOR, H.L., *AIChE J.*, **3**, 198 (1957).
11. TOOR, H.L., and J.K. BURCHARD, *AIChE J.*, **6**, 202 (1960).
12. KRISHNA, R., H.F. MARTINEZ, R. SREEDHAR, and G.L. STANDART, *Trans. I. Chem. E.*, **55**, 178 (1977).
13. WAGGONER, R.C., and G.D. LOUD, *Comput. Chem. Engng.*, **1**, 49 (1977).
14. KRISHNA, R., and G.L. STANDART, *Chem. Eng. Comm.*, **3**, 201 (1979).
15. TAYLOR, R., and R. KRISHNA, *Multicomponent Mass Transfer*, John Wiley and Sons, New York (1993).
16. KRISHNAMURTHY, R., and R. TAYLOR, *AIChE J.*, **31**, 449, 456 (1985).
17. TAYLOR, R., H.A. KOOIJMAN, and J.-S. HUNG, *Comput. Chem. Engng.*, **18**, 205–217 (1994).
18. *ASPEN PLUS Reference Manual—Volume 1*, Aspen Technology, Cambridge, MA (1994).
19. TAYLOR, R., and H.A. KOOIJMAN, *CACHE News*, No. 41, 13–19 (1995).
20. *AIChE, Bubble-Tray Design Manual*, New York (1958).
21. HARRIS, I.J., *British Chem. Engng.*, **10**(6), 377 (1965).
22. HUGHMARK, G.A., *Chem. Eng. Progress*, **61**(7), 97–100 (1965).
23. ZUIDERWEG, F.J., *Chem Eng. Sci.*, **37**, 1441 (1982).
24. CHAN, H., and J.R. FAIR, *Ind. Eng. Chem. Process Des. Dev.*, **23**, 814–827 (1984).
25. CHEN, G.X., and K.T. CHUANG, *Ind. Eng. Chem. Res.*, **32**, 701–708 (1993).
26. ONDA, K., H. TAKEUCHI, and Y.J. OKUMOTO, *J. Chem. Eng. Japan*, **1**, 56–62 (1968).
27. BRAVO, J.L., and J.R. FAIR, *Ind. Eng. Chem. Process Des. Devel.*, **21**, 162–170 (1982).
28. BRAVO, J.L., J.A. ROCHA, and J.R. FAIR, *Hydrocarbon Processing*, **64**(1), 56–60 (1985).
29. BRAVO, J.L., J.A. ROCHA, and J.R. FAIR, *I. Chem. E. Symp. Ser.*, No. 128, A489–A507 (1992).
30. BILLET, R., and M. SCHULTES, *I. Chem. E. Symp. Ser.*, No. 128, B129 (1992).
31. KOOIJMAN, H.A., and R. TAYLOR, *Chem. Eng. J.*, **57**(2), 177–188 (1995).
32. FAIR, J.R., H.R. NULL, and W.L. BOLLES, *Ind. Eng. Chem. Process Des. Dev.*, **22**, 53–58 (1983).
33. POWERS, M.F., D.J. VICKERY, A. AREHOLE, and R. TAYLOR, *Comput. Chem. Engng.*, **12**, 1229–1241 (1988).
34. TAYLOR, R., H.A. KOOIJMAN, and M.R. WOODMAN, *I. Chem. E. Symp. Ser.*, No. 128, A415–A427 (1992).
35. OVEJERO, G., R. VAN GRIEKEN, L. RODRIGUEZ, and J.L. VALVERDE, *Sep. Sci. Tech.*, **29**, 1805–1821 (1994).
36. SCHEFFE, R.D., and R.H. WEILAND, *Ind. Eng. Chem. Res.*, **26**, 228–236 (1987).
37. YOUNG, T.C., and W.E. STEWART, *AIChE J.*, **38**, 592–602 with errata on p. 1302 (1993).
38. YOUNG, T.C., and W.E. STEWART, *AIChE J.*, **41**, 1319–1320 (1995).
39. SPAGNOLO, D.A., E.L. PLAICE, H.J. NEUBURG, and K.T. CHUANG, *Can. J. Chem. Eng.*, **66**, 367–376 (1988).
40. HIGLER, A., R. KRISHNA, and R. TAYLOR, *AIChE J.*, **45**, 2357–2370 (1999).
41. PYHALAHTI, A., and K. JAKOBSSON, *Ind. Eng. Chem. Res.*, **42**, 6188–6195 (2003).

EXERCISES

Section 12.1

12.1 Modify the rate-based model of (12-4) to (12-18) to include entrainment and occlusion.

12.2 Modify the rate-based model of (12-4) to (12-18) to include a chemical reaction in the liquid phase under conditions of:

- (a) Chemical equilibrium
- (b) Kinetic rate law.

12.3 Explain how the number of rate-based modeling equations can be reduced. Would this be worthwhile?

Section 12.2

12.4 The following results were obtained at tray n from a rate-based calculation at 14.7 psia, for a ternary mixture of acetone (1), methanol (2), and water (3) in a sieve-tray column assuming that both phases are perfectly mixed.

Component	y_n	y_{n+1}	y_n^I	K_n^I	x_n
1	0.4913	0.4106	0.5291	1.507	0.3683
2	0.4203	0.4389	0.4070	0.900	0.4487
3	0.0884	0.1505	0.0639	0.3247	0.1830

The products of the computed gas-phase, binary mass-transfer coefficients and interfacial area from the Chan–Fair correlations are as follows in units of lbmol/(h-unit mole fractions).

$$k_{12} = k_{21} = 1,750$$

$$k_{13} = k_{31} = 2,154$$

$$k_{23} = k_{32} = 2,503$$

The computed vapor rates are $V_n = 1,200$ lbmol/h and $V_{n+1} = 1,164$ lbmol/h. Determine:

- (a) The component molar diffusion rates.
- (b) The mass-transfer rates.
- (c) The Murphree vapor-tray efficiencies.

12.5 Write all the expanded equations (12-31) and (12-32) for R^P for a five-component system.

12.6 Repeat the calculations of Example 12.1, but using 1 = methanol, 2 = water, and 3 = acetone. Are the results any different? If not, why not? Prove your conclusion mathematically.

Section 12.3

12.7 Compare and discuss the advantages and disadvantages of the available correlations for estimating binary-pair mass-transfer coefficients for trayed columns.

12.8 Compare and discuss the advantages and disadvantages of the available correlations for estimating binary-pair mass-transfer coefficients for columns with random (dumped) and structured packings.

Section 12.4

12.9 Discuss how the method of Fair, Null, and Bolles [32] might be used to model the flow patterns in a rate-based model. How would the mole-fraction driving forces be computed?

Section 12.5

12.10 A bubble-point mixture of 100 kmol/h of methanol, 50 kmol/h of isopropanol, and 100 kmol/h of water at 1 atm is sent to the 25th tray from the top of a 40-sieve-tray column equipped with a total condenser and partial reboiler, operating at a nominal pressure of 1 atm. If the reflux ratio is 5 and the bottoms flow rate is 150 kmol/h, determine the separation achieved if the UNIFAC method is used to estimate K -values and the Chan–Fair correlations are used for mass transfer. Assume that both phases are perfectly mixed on each tray and that operation is at about 80% of flooding.

12.11 A sieve-tray column, operating at a nominal pressure of 1 atm, is used to separate a mixture of acetone and methanol by extractive distillation using water. The column has 40 trays with a total condenser and partial reboiler. The feed of 50 kmol/h of acetone and 150 kmol/h of methanol at 60°C and 1 atm enters tray 35 from the top, while 50 kmol/h of water at 65°C and 1 atm enters tray 5 from the top. Determine the separation for a reflux ratio of 10 and a bottoms flow rate of 200 kmol/h. Use the UNIFAC method for K -values and the AIChE method for mass transfer. Assume a perfectly mixed liquid and a vapor in plug flow on each tray, with operation at 80% of flooding. Also determine the number of equilibrium stages (to the nearest stage) to achieve the same separation.

12.12 Repeat Exercise 12.10, if a column packed with 2-in. stainless-steel Pall rings is used with 25 ft of rings above the feed and 15 ft below. Be sure to use a sufficient number of segments for the calculations.

12.13 Repeat Exercise 12.10, if a column with structured packing is used with 25 ft above the feed and 15 ft below. Be sure to use a sufficient number of segments.

12.14 Solve Exercise 12.10 for combinations of the following values of percent flooding, weir height, and hole area, respectively:

40, 60 and 80%
1, 2, and 3 inches
6, 10, and 14%

12.15 The upper column of an air-separation system, of the type discussed and shown in Exercise 7.40, contains 48 sieve trays and

operates at a nominal pressure of 131.7 kPa. A feed at 80 K and 131.7 kPa enters the top plate at 1,349 lbmol/h with a composition of 97.868 mol% nitrogen, 0.365 mol% argon, and 1.767 mol% oxygen. A second feed enters tray 12 from the top at 83 K and 131.7 kPa at 1,832 lbmol/h with a composition of 59.7 mol% nitrogen, 1.47 mol% argon, and 38.83 mol% oxygen. The column has no condenser, but has a split reboiler. Vapor distillate leaves the top plate at 2,487 lbmol/h, with remaining products leaving the reboiler as 50 mol% vapor and 50 mol% liquid. Assume that ideal solutions are formed. Determine the effect of percent flooding on the separation and the median Murphree vapor-tray efficiency for oxygen.

12.16 The following bubble-point, organic-liquid mixture at 1.4 atm is distilled by extractive distillation with the following phenol-rich solvent at 1.4 atm and at the same temperature as the main feed:

Component	Feed, kmol/h	Solvent, kmol/h
Methanol	50	0
<i>n</i> -Hexane	20	0
<i>n</i> -Heptane	180	0
Toluene	150	10
Phenol	0	800

The column has 30 sieve trays, with a total condenser and a partial reboiler. The solvent enters the 5th tray and the feed enters tray 15, from the top. The pressure in the condenser is 1.1 atm; the pressure at the top tray is 1.2 atm, and the pressure at the bottom is 1.4 atm. The reflux ratio is 5 and the bottoms rate is 960 kmol/h. Thermodynamic properties can be estimated with the UNIFAC method for the liquid phase and the SRK equation for the vapor phase. The Antoine equation is suitable for vapor pressure. Use the nonequilibrium model of the ChemSep program to estimate the separation. Assume that the vapor and liquid are both well mixed and that the trays operate at 75% of flooding. Specify the Chan–Fair correlation for calculating mass-transfer coefficients. In addition, determine from the tray-by-tray results the average Murphree vapor-tray efficiency for each component (after discarding values that appear to be much different than the majority of values). Try to improve the sharpness of the split by changing the feed and solvent entry tray locations. How can you increase the sharpness of the separation? List as many ideas as you have.

12.17 A bubble-cap tray absorber is designed to absorb 40% of the propane from a rich gas at 4 atm. The specifications for the entering rich gas and absorbent oil are as follows:

	Absorbent Oil	Rich Gas
Flow rate, kmol/s	11.0	11.0
Temperature, °C	32	62
Pressure, atm	4	4
Mole fraction:		
Methane	0	0.286
Ethane	0	0.157
Propane	0	0.240
<i>n</i> -Butane	0.02	0.169
<i>n</i> -Pentane	0.05	0.148
<i>n</i> -Dodecane	0.93	0

(a) Determine the number of equilibrium stages required and the splits of all components.

(b) Determine the actual number of trays required and the splits and Murphree vapor-tray efficiencies of all components.

(c) Compare and discuss the equilibrium-based and rate-based results. What do you conclude?

12.18 A ternary mixture of methanol, ethanol, and water is distilled in a sieve-tray column to obtain a distillate with not more than 0.01 mol% water. The feed to the column is as follows:

Flow rate, kmol/h	142.46
Pressure, atm	1.3
Temperature, K	316
Mole fractions:	
Methanol	0.6536
Ethanol	0.0351
Water	0.3113

For a distillate rate of 93.10 kmol/h, a reflux ratio of 1.2, a condenser outlet pressure of 1.0 atm, and a top-tray pressure of 1.1 atm, determine using the UNIFAC method for activity coefficients:

(a) The number of equilibrium stages required and the corresponding split, if the feed enters at the optimal stage.

(b) The number of actual trays required if the column operates at about 85% of flooding and the feed is introduced to the optimal tray. Compare the split to that in part (a). In addition, compute the component Murphree vapor-tray efficiencies. What do you conclude about the two methods of calculations?

12.19 Repeat Exercise 12.18 for a column packed with 2-in. stainless-steel Pall rings.

12.20 It is required to absorb 96% of the benzene from a gas stream with absorption oil in a sieve-tray column at a nominal pressure of 1 atm. The feed conditions are as follows:

	Vapor	Liquid
Flow rate, kmol/s	0.01487	0.005
Pressure, atm	1.0	1.0
Temperature, K	300	300
Composition, mol fraction:		
Nitrogen	0.7505	0
Oxygen	0.1995	0
Benzene	0.0500	0.005
<i>n</i> -Tridecane (C ₁₃)	0	0.995

Tray geometry is as follows:

Tray spacing, m	0.5
Weir height, m	0.05
Hole diameter, m	0.003
Sheet thickness, m	0.002

Determine column diameter for 80% of flooding, the number of actual trays required, and the Murphree vapor-tray efficiency profile for benzene for the possible combinations of vapor and liquid flow patterns on a tray. Could the equilibrium-based method be used to obtain a reliable solution to this problem?

Chapter 13

Batch Distillation

In batch-separation operations, a feed mixture is charged to the equipment and one or more products are withdrawn. A familiar example is laboratory distillation, shown in Figure 13.1, where a liquid mixture is charged to a still pot, retort, or flask and heated by a flame or electric mantle to boiling. The vapor formed is continuously removed and condensed to produce a distillate.

The composition of both the initial charge and distillate change with time; there is no steady state. The still temperature increases and the relative amount of lower-boiling components remaining in the still pot decreases as distillation proceeds.

Batch operations can be used to advantage under the following circumstances:

1. The capacity of a facility is too small to permit continuous operation at a practical rate.
2. It is necessary, because of seasonal demands, to distill with one unit different feedstocks to produce different products.
3. It is desired to produce several new products with one distillation unit for evaluation by potential buyers.
4. Upstream process operations are batchwise and the composition of feedstocks for distillation vary with time or from batch to batch.
5. The feed contains solids or materials that form solids, tars, or resin that can plug or foul a continuous distillation column.

13.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- List assumptions for and derive the Rayleigh equation for the simplest form of batch distillation (differential distillation).
- Calculate, by graphical and/or algebraic means, batch-still temperature, residue composition, instantaneous distillate composition, and average distillate composition for a binary mixture as a function of time for binary, differential distillation.
- Calculate, by modified McCabe–Thiele methods, residue and distillate compositions for binary, batch rectification under conditions of equilibrium stages, no liquid holdup, and for constant or variable-reflux ratio to achieve constant distillate composition.
- Explain the importance of taking into account liquid holdup.
- Calculate, using shortcut and rigorous equilibrium-stage methods with a simulator, multicomponent, multistage batch rectification that includes a sequence of operating steps to obtain specified products.
- Apply the principles of optimal control to optimize batch distillation.

13.1 DIFFERENTIAL DISTILLATION

The simplest case of batch distillation, as discussed by Lord Rayleigh [1], is *differential distillation*, which involves use of the apparatus shown in Figure 13.1. There is no reflux; at any instant, vapor leaving the still pot with composition y_D is assumed to be in equilibrium with perfectly mixed liquid in the still. For total condensation, $y_D = x_D$. Thus, there is only a single equilibrium stage, the still pot. This apparatus is useful for separating wide-boiling mixtures. The following nomenclature is used for variables that vary with time, t ,

assuming that all compositions refer to a particular species in the multicomponent mixture.

D = instantaneous-distillate rate, mol/h

$y = y_D = x_D$ = instantaneous-distillate composition, mole fraction

W = moles of liquid left in still

$x = x_W$ = composition of liquid left in still, mole fraction

0 = subscript referring to $t = 0$

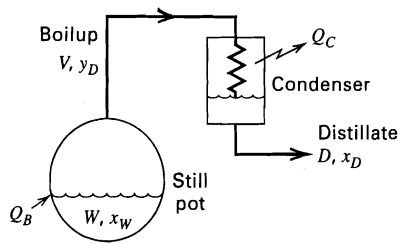


Figure 13.1 Differential (Rayleigh) distillation.

For any component in the mixture:

Instantaneous rate of output = Dy_D

Instantaneous rate of depletion in the still $\left\{ = -\frac{d}{dt}(Wx_W) = -W\frac{dx_W}{dt} - x_W\frac{dW}{dt} \right.$

The distillate rate and, therefore, the rate of depletion of the liquid in the still depend on the rate of heat input to the still. By material balance at any instant:

$$\frac{d}{dt}(Wx_W) = W\frac{dx_W}{dt} + x_W\frac{dW}{dt} = -Dy_D \quad (13-1)$$

Multiplying by dt :

$$Wdx_W + x_WdW = y_D(-Ddt) = y_DdW$$

since by total balance $-Ddt = dW$. Separating variables and integrating from the initial charge condition of W_0 and x_{W_0}

$$\int_{x_{W_0}}^{x_W} \frac{dx_W}{y_D - x_W} = \int_{W_0}^W \frac{dW}{W} = \ln\left(\frac{W}{W_0}\right) \quad (13-2)$$

This is the well-known Rayleigh equation, which was first applied to the separation of wide-boiling mixtures such as $\text{HCl-H}_2\text{O}$, $\text{H}_2\text{SO}_4\text{-H}_2\text{O}$, and $\text{NH}_3\text{-H}_2\text{O}$. Without reflux, y_D and x_W are assumed to be in equilibrium and (13-2) simplifies to

$$\int_{x_0}^x \frac{dx}{y - x} = \ln\left(\frac{W}{W_0}\right) \quad (13-3)$$

where x_{W_0} is replaced by x_0 .

Equation (13-3) is easily integrated only when pressure is constant, temperature change in the still pot is relatively small (close-boiling mixture), and K -values are composition independent. Then $y = Kx$, where K is approximately constant, and (13-3) becomes

$$\ln\left(\frac{W}{W_0}\right) = \frac{1}{K-1} \ln\left(\frac{x}{x_0}\right) \quad (13-4)$$

For a binary mixture, if instead, the relative volatility α is assumed constant, substitution of (4-8) into (13-3), followed by integration and simplification, gives

$$\ln\left(\frac{W_0}{W}\right) = \frac{1}{\alpha-1} \left[\ln\left(\frac{x_0}{x}\right) + \alpha \ln\left(\frac{1-x}{1-x_0}\right) \right] \quad (13-5)$$

If the equilibrium relationship $y = f\{x\}$ is in graphical or tabular form, integration of (13-3) can be performed graphically or numerically. The final liquid remaining in the still pot is often referred to as the *residue*.

The following three examples illustrate some of the applications of the Rayleigh equation to binary mixtures.

EXAMPLE 13.1

A batch still is loaded with 100 kmol of a liquid consisting of a binary mixture of 50 mol% benzene in toluene. As a function of time, make plots of (a) still temperature, (b) instantaneous vapor composition, (c) still-pot composition, and (d) average total-distillate composition. Assume a constant boilup rate of 10 kmol/h and a constant relative volatility of 2.41 at a pressure of 101.3 kPa (1 atm).

SOLUTION

Initially, $W_0 = 100$ kmol, $x_0 = 0.5$. Solving (13.5) for W at values of x from 0.5 in increments of 0.05, and determining corresponding values of time from $t = (W_0 - W)/10$, the following table is generated:

t, h	2.12	3.75	5.04	6.08	6.94	7.66	8.28	8.83	9.35
W, kmol	78.85	62.51	49.59	39.16	30.59	23.38	17.19	11.69	6.52
$x = x_W$	0.45	0.40	0.35	0.30	0.25	0.20	0.15	0.10	0.05

The instantaneous-vapor composition, y , is obtained from (4-8), which is $y = 2.41x/(1 + 1.41x)$, the equilibrium relationship for constant α . The average value of y_D or x_D over the time interval 0 to t is related to x and W at time t by combining overall component and total material balances to give

$$(x_D)_{\text{avg}} = (y_D)_{\text{avg}} = \frac{W_0x_0 - Wx}{W_0 - W} \quad (13-6)$$

Equation (13-6) is much easier to apply than an equation that eventually integrates the distillate composition.

To obtain the temperature in the still, it is necessary to use experimental T - x - y data for benzene-toluene at 101.3 kPa as given in Table 13.1. The temperature and compositions as a function of time are shown in Figure 13.2.

Table 13.1 Vapor-Liquid Equilibrium Data for Benzene (B)-Toluene (T) at 101.3 kPa

x_B	y_B	$T, ^\circ\text{C}$
0.100	0.208	105.3
0.200	0.372	101.5
0.300	0.507	98.0
0.400	0.612	95.1
0.500	0.713	92.3
0.600	0.791	89.7
0.700	0.857	87.3
0.800	0.912	85.0
0.900	0.959	82.7
0.950	0.980	81.4

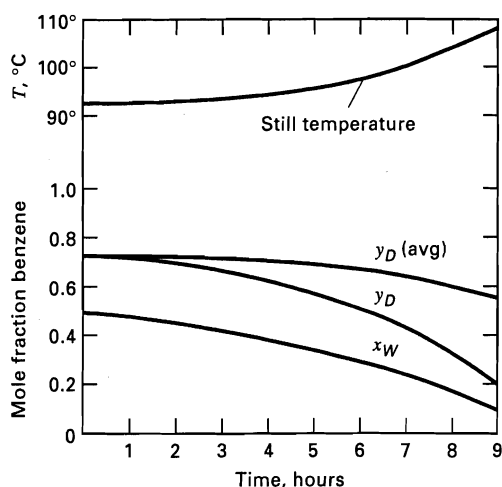


Figure 13.2 Distillation conditions for Example 13.1.

EXAMPLE 13.2

Repeat Example 13.1, except instead of using a constant value of 2.41 for the relative volatility, use the vapor-liquid equilibrium data for benzene-toluene at 101.3 kPa, given in Table 13.1, to solve the problem graphically or numerically with (13-3) rather than (13-5).

SOLUTION

Equation (13-3) can be solved by graphical integration by plotting $1/(y-x)$ versus x with a lower limit of $x_0 = 0.5$. Using the data of Table 13.1 for y as a function of x , points for the plot in terms of benzene are as follows:

x	0.5	0.4	0.3	0.2	0.1
$\frac{1}{y-x}$	4.695	4.717	4.831	5.814	9.259

The area under the plotted curve from $x_0 = 0.5$ to a given value of x is equated to $\ln(W/W_0)$, and W is computed for $W_0 = 100$ kmol. In the region from $x = 0.5$ to 0.3, the value of $1/(y-x)$ changes only slightly. Therefore, a numerical integration by the trapezoidal rule is readily made:

$x = 0.4$:

$$\ln\left(\frac{W}{W_0}\right) = \int_{0.5}^{0.4} \frac{dx}{y-x} \approx \Delta x \left[\frac{1}{y-x} \right]_{\text{avg}}$$

$$= (0.4 - 0.5) \left(\frac{4.695 + 4.717}{2} \right) = -0.4706$$

$$W/W_0 = 0.625, W = -0.625(100) = 62.5 \text{ kmol}$$

$x = 0.3$:

$$\ln\left(\frac{W}{W_0}\right) = \int_{0.5}^{0.3} \frac{dx}{y-x} \approx \Delta x \left[\frac{1}{y-x} \right]_{\text{avg}}$$

$$= (0.3 - 0.5) \left[\frac{4.695 + 4.717 + 4.717 + 4.831}{4} \right] = -0.948$$

$$W/W_0 = 0.388, W = 0.388(100) = 38.8 \text{ kmol}$$

These two values are in good agreement with those in Example 13.1. A graphical integration from $x_0 = 0.4$ to $x = 0.1$ gives $W = 10.7$, which is approximately 10% less than the result in Example 13.1, which uses a constant value of the relative volatility.

The Rayleigh equation (13-1) can be applied to any two components, i and j , of a multicomponent mixture. Thus, if we let M_i be the moles of i in the still pot,

$$\frac{dM_i}{dt} = \frac{d}{dt}(Wx_{w_i}) = -Dy_{D_i}$$

Then

$$dM_i/dM_j = y_{D_i}/y_{D_j} \quad (13-7)$$

For constant $\alpha_{i,j} = y_{D_i}x_{w_j}/y_{D_j}x_{w_i}$, (13-7) becomes

$$dM_i/dM_j = \alpha_{i,j}(x_{w_i}/x_{w_j}) \quad (13-8)$$

Substitution of $M_i = Wx_{w_i}$ for both i and j into (13-8) gives

$$dM_i/M_i = \alpha_{i,j} dM_j/M_j \quad (13-9)$$

Integration from the initial-charge condition gives

$$\ln(M_i/M_{i_0}) = \alpha_{i,j} \ln(M_j/M_{j_0}) \quad (13-10)$$

As shown in the following example, (13-10) is useful for determining the effect of relative volatility on the degree of separation achievable by Rayleigh distillation.

EXAMPLE 13.3

The charge to a simple batch still consists of an equimolar, binary mixture of A and B. For values of $\alpha_{A,B}$ of 2, 5, 10, 100, and 1,000, and 50% vaporization of A, determine the percent vaporization of B and the mole fraction of B in the total distillate.

SOLUTION

For $\alpha_{A,B} = 2$ and $M_A/M_{A_0} = 1 - 0.5 = 0.5$, (13-10) gives

$$M_B/M_{B_0} = (M_A/M_{A_0})^{1/\alpha_{A,B}} = (0.5)^{0.5} = 0.7071$$

Percent vaporization of B = $(1 - 0.7071)(100) = 29.29\%$.

For 200 moles of charge, the amounts of components in the distillate are $D_A = (0.5)(0.5)(200) = 50$ mol and $D_B = (0.2929)(0.5)(200) = 29.29$ mol

$$\text{Mole fraction of B in the total distillate} = \frac{29.29}{50 + 29.29} = 0.3694$$

Similar calculations for other values of $\alpha_{A,B}$ give the following results:

$\alpha_{A,B}$	% Vaporization of B	Mole Fraction of B in Total Distillate
2	29.29	0.3694
5	12.94	0.2057
10	6.70	0.1182
100	0.69	0.0136
1,000	0.07	0.0014

These results show that a sharp separation between A and B for 50% vaporization of A is only achieved if $\alpha_{A,B} \geq 100$. Furthermore, the purity achieved depends on the percent vaporization of A. For $\alpha_{A,B} = 100$, if 90% of A is vaporized, the mole fraction of B in the total distillate increases from 0.0136 to 0.0247. For this reason,

as discussed in detail and illustrated by example in a later section of this chapter, it is common to conduct a binary, batch-distillation separation of light key (LK) and heavy key (HK) in the following manner:

1. Produce a distillate LK cut until the limit of impurity of HK in the total distillate is reached.
2. Continue the batch distillation to produce an intermediate cut of impure LK until the limit of impurity of LK in the liquid in the still is reached.
3. Empty the HK-rich cut from the still.
4. Recycle the intermediate cut to the next still charge.

For desired purities of the LK cut and the HK cut, the fraction of intermediate cut increases as the LK-HK relative volatility decreases.

13.2 BINARY BATCH RECTIFICATION WITH CONSTANT REFLUX AND VARIABLE DISTILLATE COMPOSITION

To achieve a sharp separation and/or reduce the intermediate-cut fraction, a trayed or packed column, located above the still, and a means of sending reflux to the column, is provided as shown for the batch rectifier of Figure 13.3. For a column of a given diameter, the molar vapor-boilup rate is usually fixed at a value safely below the column flooding point. Two modes of operation of batch rectification are cited most frequently because they are the most readily modeled. The first is operation at a constant reflux rate or ratio (same as a constant distillate rate), while the second is operation at a constant distillate composition. With the former, the distillate composition varies with time; with the latter, the reflux ratio or distillate rate varies with time. The first mode is most easily implemented because of the availability of rapidly responding flow sensors. For the second mode, rapidly responding composition sensors may not be readily available. In a third mode of operation, referred to here as the optimal control mode, both reflux ratio (or distillate rate) and distillate composition vary with time so as to maximize the amount of distillate, minimize the operation time, or maximize profit. The first mode of operation is now presented, followed by the second mode. A discussion of optimal control is deferred until the end of this chapter.

If the reflux ratio R or distillate rate D is fixed, instantaneous-distillate and still-bottoms compositions vary

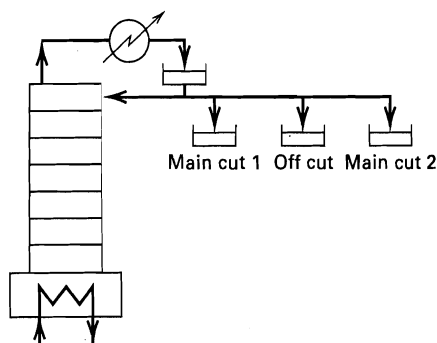


Figure 13.3 Batch rectification.

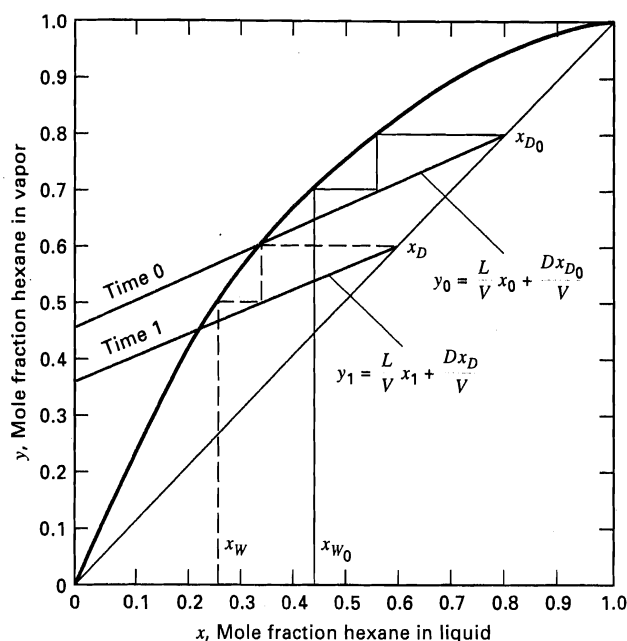


Figure 13.4 Batch binary distillation with fixed L/V and two theoretical stages.

with time. For a total condenser, negligible holdup of vapor and liquid in the condenser and the column, phase equilibrium at each stage, and constant molar overflow, (13-2) still applies with $y_D = x_D$. The analysis of such a batch rectification for a binary system is facilitated by the McCabe-Thiele diagram using the method of Smoker and Rose [2].

Initially, the composition of the light-key component in the liquid in the reboiler of the column in Figure 13.3 is the charge composition, x_{w0} , which is given the value 0.43 in the McCabe-Thiele diagram of Figure 13.4. If there are two theoretical stages, the initial distillate composition x_{D0} at time 0 can be found by constructing an operating line of slope $L/V = R/(R+1)$, such that exactly two stages are stepped off from x_{w0} to the $y = x$ line in Figure 13.4. At an arbitrary later time, say time 1, at still-pot composition $x_w < x_{w0}$, the instantaneous-distillate composition is x_D . A time-dependent series of points for x_D is thus established by trial and error, with L/V and the number of stages held constant.

Equation (13-2) cannot be integrated analytically because the relationship between y_D and x_w depends on the liquid-to-vapor ratio, the number of theoretical stages, and the phase-equilibrium relationship. However, it can be integrated graphically with pairs of values for x_w and $y_D = x_D$ obtained from the McCabe-Thiele diagram for a series of operating lines of the same slope.

The time t required for batch rectification at constant reflux ratio and negligible holdup in the column and condenser can be computed by a total material balance based on a constant boilup rate V , to give the following equation due to Block [3]:

$$t = \frac{W_0 - W_t}{V \left(1 - \frac{L}{V}\right)} = \frac{R+1}{V} (W_0 - W_t) \quad (13-11)$$

With a constant-reflux policy, the instantaneous-distillate purity is above the specification at the beginning of distillation and below specification at the end of the run. By an overall material balance, the average mole fraction of the light-key component in the accumulated distillate at time t is given by

$$x_{D,\text{avg}} = \frac{W_0 x_0 - W_t x_{W_t}}{W_0 - W_t} \quad (13-12)$$

EXAMPLE 13.4

A three-theoretical-stage batch rectifier (first stage is the still pot) is charged with 100 kmol of a 20 mol% n -hexane in n -octane mixture. At a constant reflux ratio of 1 ($L/V = 0.5$), how many moles of charge must be distilled if an average product composition of 70 mol% nC_6 is required? The phase-equilibrium curve at column pressure is given in Figure 13.5. If the boilup rate is 10 kmol/h, calculate the distillation time.

SOLUTION

A series of operating lines and, hence, values of x_W are located by the trial-and-error procedure described earlier, as shown in Figure 13.5 for $x_{W0} = 0.20$ and $x_W = 0.09$. It is then possible to construct the following table:

$y_D = x_D$	0.85	0.60	0.5	0.35	0.3
x_W	0.2	0.09	0.07	0.05	0.035
1	1.54	1.96	2.33	3.33	3.77
$y_D - x_W$					

The graphical integration is shown in Figure 13.6. Assuming a final value of $x_W = 0.1$, for instance, integration of (13-2) gives

$$\ln \frac{100}{W} = \int_{0.1}^{0.2} \frac{dx_W}{y_D - x_W} = 0.162$$

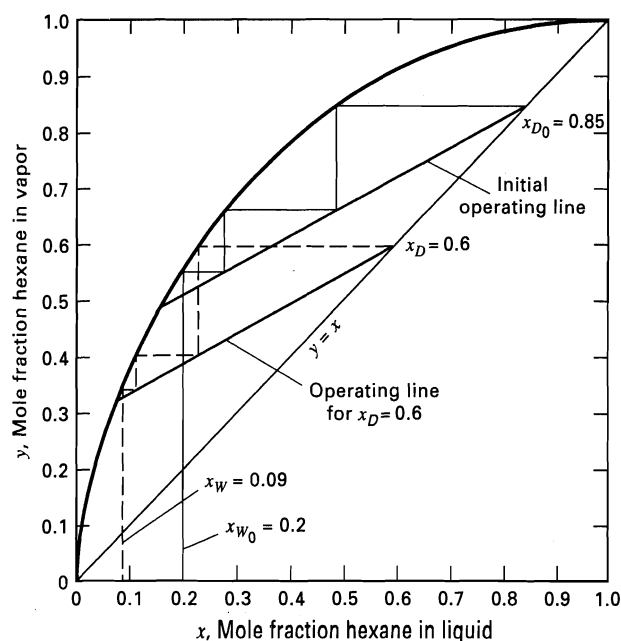


Figure 13.5 Solution to Example 13.4.

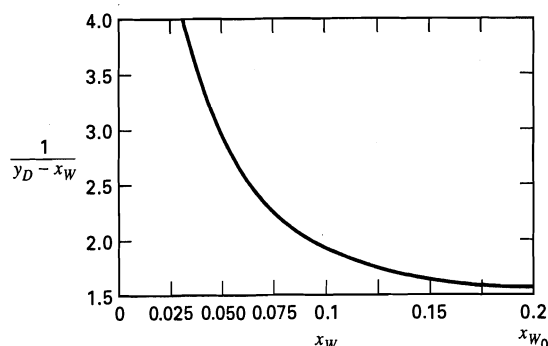


Figure 13.6 Graphical integration for Example 13.4.

Hence,

$$W = 85 \quad \text{and} \quad D = 15.$$

From (13-12):

$$(x_D)_{\text{avg}} = \frac{100(0.20) - 85(0.1)}{(100 - 85)} = 0.77$$

The $(x_D)_{\text{avg}}$ is higher than the desired value of 0.70; hence, another final x_W must be chosen. By trial, the correct answer is found to be $x_W = 0.06$, with $D = 22$, and $W = 78$, corresponding to a value of 0.25 for the integral.

From (13-11), the distillation time is $t = \frac{(1+1)}{10}(100 - 78) = 4.4$ h.

When Rayleigh differential distillation is used, Figure 13.5 shows that a 70 mol% hexane distillate is not achievable because the initial distillate is only 56 mol% hexane.

13.3 BINARY BATCH RECTIFICATION WITH CONSTANT DISTILLATE COMPOSITION AND VARIABLE REFLUX

The constant-reflux-ratio policy described in the previous section is simple and easy to implement. For small batch-rectification systems, it may be the least expensive policy. A more optimal operating policy, as discussed in Section 13.8, is to maintain a constant molar-vapor rate, but continuously vary the reflux ratio to achieve a constant distillate composition that meets the specified purity. This policy requires a more complex control system, including a composition sensor (or suitable substitute) on the distillate, which may be justified only for large batch rectification systems. Other methods of operating batch columns are described by Ellerbe [5].

Calculations for the policy of constant distillate composition can also be made with the McCabe-Thiele diagram, as described by Bogart [4] and illustrated in Example 13.5. The Bogart method assumes negligible liquid holdup and constant molar overflow. An overall material balance for the light-key component, at any time, t , is given by a rearrangement of (13-12) at constant x_D , for W as a function of x_W .

$$W = W_0 \left[\frac{x_D - x_{W0}}{x_D - x_W} \right] \quad (13-13)$$

Differentiating (13-13) with respect to t for varying W and x_W gives

$$\frac{dW}{dt} = W_0 \frac{(x_D - x_{W_0})}{(x_D - x_W)^2} \frac{dx_W}{dt} \quad (13-14)$$

For constant molar overflow, the rate of distillation is given by the rate of loss of charge, or

$$-\frac{dW}{dt} = (V - L) = \frac{dD}{dt} \quad (13-15)$$

where D is now the amount of distillate, not the distillate rate. Substituting (13-15) into (13-14) and integrating:

$$t = \frac{W_0(x_D - x_{W_0})}{V} \int_{x_{W_i}}^{x_0} \frac{dx_W}{(1 - L/V)(x_D - x_W)^2} \quad (13-16)$$

For fixed values of W_0 , x_{W_0} , x_D , V , and the number of equilibrium stages, the McCabe-Thiele diagram is used to determine values of L/V for a series of values of still composition between x_{W_0} and the final value of x_W . These values are then used with (13-16) to determine, by graphical or numerical integration, the time for rectification or the time to reach any intermediate value of still composition. The required number of theoretical stages can be estimated by assuming total-reflux conditions for the final value of x_W . While rectification is proceeding, the instantaneous-distillate rate will vary according to (13-15), which can be expressed in terms of L/V as

$$\frac{dD}{dt} = V(1 - L/V) \quad (13-17)$$

EXAMPLE 13.5

A three-stage batch still (boiler and the equivalent of two equilibrium plates) is loaded with 100 kmol of a liquid containing a mixture of 50 mol% n -hexane in n -octane. A liquid distillate of 0.9 mole fraction hexane is to be maintained by continuously adjusting the reflux ratio, while maintaining a distillate rate of 20 kmol/h. What should the reflux ratio be after 1 h when the accumulated distillate is 20 kmol? Theoretically, when must accumulation of the distillate cut be stopped? Assume negligible holdup on the plates and constant molar overflow.

SOLUTION

When the accumulated distillate = 20 kmol, $W = 80$ kmol, and the still residue composition with respect to the light-key is given by a rearrangement of (13-13):

$$x_W = \frac{W x_D - W_0(x_D - x_{W_0})}{W} = \frac{0.9(80) - 100(0.9 - 0.5)}{80} = 0.4$$

For $y_D = 0.9$, a series of operating lines of varying slope, $L/V = R/(R + 1)$, with three stages stepped off is used to determine the corresponding still residue composition. By trial and error, Line 1 in Figure 13.7 is found for $x_W = 0.4$, corresponding to an $L/V = 0.22$. The reflux ratio = $(L/V)/[1 - (L/V)] = 0.282$.

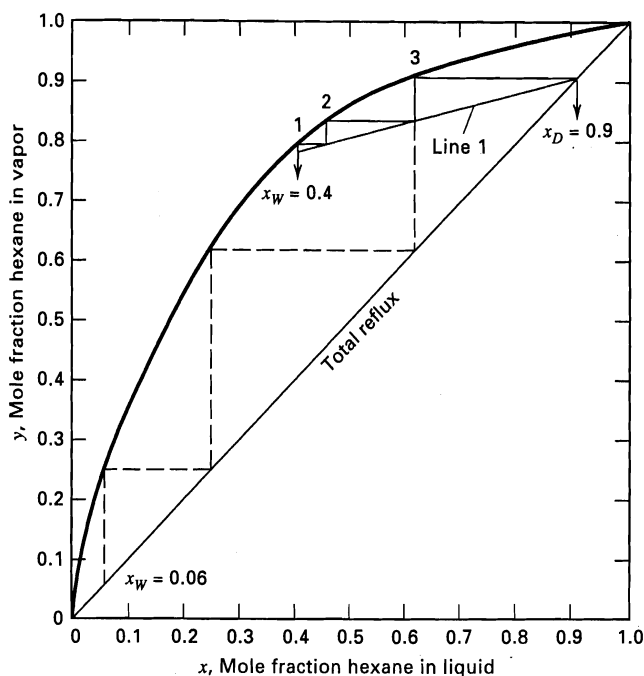


Figure 13.7 Solution to Example 13.5.

At the highest reflux rate possible, $L/V = 1$ (total reflux), and $x_W = 0.06$ according to the dashed-line construction shown in Figure 13.7. The corresponding time by material balance is given by $0.06(100 - 20t) = 50 - 20t(0.9)$. Solving, $t = 2.58$ h.

13.4 BATCH STRIPPING AND COMPLEX BATCH DISTILLATION

A batch stripper consisting of a large accumulator, a trayed or packed stripping column, and a reboiler is shown in Figure 13.8. The initial charge is placed in the accumulator rather than the reboiler. The mixture in the accumulator is fed to the top of the column and the bottoms cut is removed from the reboiler. A batch stripper is useful for removing small quantities of volatile impurities. For binary mixtures, the McCabe-Thiele construction applies, and the graphical methods described in Sections 13.2 and 13.3 can be modified to follow with time the change in composition in the

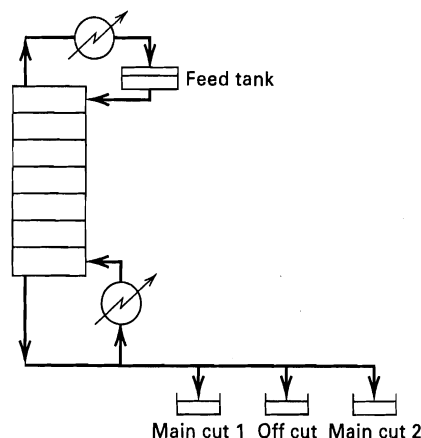


Figure 13.8 Batch stripping.

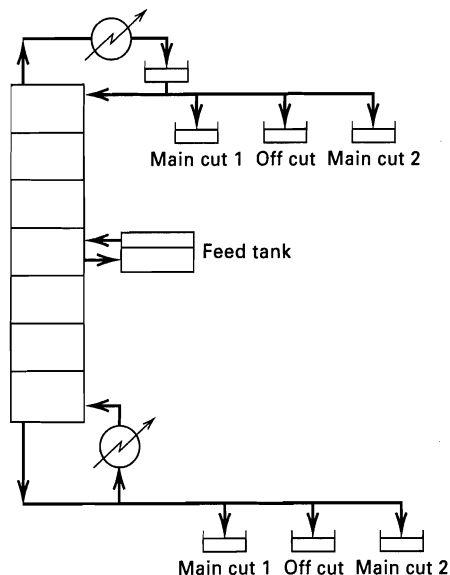


Figure 13.9 Complex batch distillation.

accumulator and the corresponding instantaneous and average composition of the bottoms cut.

A complex batch-distillation unit, of the type described by Hasebe et al. [6], that permits considerable operating flexibility is shown in Figure 13.9. The charge in the feed tank is fed to a suitable column location. Holdups in the reboiler and condenser are kept to a minimum. Products or intermediate cuts are withdrawn from the condenser, the reboiler, or both. In addition, the liquid in the column at the feed location can be recycled to the feed tank if it is desirable to make the composition in the feed tank close to the composition of the liquid at the feed location.

13.5 EFFECT OF LIQUID HOLDUP

Except at high pressure, vapor holdup in a rectifying column is negligible in batch distillation because of the small molar density of the vapor phase. However, the effect of liquid holdup on the trays and in the condensing and reflux system can be significant when the molar ratio of holdup to original charge is more than a few percent. This is especially true when a charge contains low concentrations of one or more of the components to be separated. In general, the effect of holdup in a trayed column is greater than in a packed column because of the lower amount of holdup in the latter. For either type of column, liquid holdup can be estimated by methods described in Chapter 6.

A batch rectifier is usually operated under total-reflux conditions for an initial period of time prior to the withdrawal of distillate product. During this initial time period, liquid holdup in the column increases and approaches a value that is reasonably constant for the remainder of the distillation cycle. Because of the total-reflux concentration profile, the initial concentration of light components in the remaining charge to the still is less than in the original charge. At high liquid holdups, this causes the initial purity and degree of separation to be reduced from estimates based on

methods that ignore liquid holdup. Liquid holdup can reduce the size of product cuts, increase the size of intermediate fractions that are recycled, increase the amount of residue, increase the batch-cycle time, and increase the total energy input. Although approximate methods for predicting the effect of liquid holdup were developed in the past, the complexity of the holdup effect is such that it is now considered best to use the rigorous computer-based, batch-distillation algorithms described in Section 13.7 to study the effect on a case-by-case basis.

13.6 SHORTCUT METHOD FOR MULTICOMPONENT BATCH RECTIFICATION WITH CONSTANT REFLUX

The batch rectification methods presented in Sections 13.2 and 13.3 are limited to binary mixtures, under the assumptions of constant molar overflow, and negligible vapor and liquid holdup. Shortcut methods for handling multicomponent mixtures under the same assumptions have been developed by Diwekar and Madhavan [7] for the two cases of constant distillate composition and constant reflux, and by Sundaram and Evans [8] for constant reflux. Both methods avoid tedious stage-by-stage calculations of vapor and liquid compositions by employing the Fenske–Underwood–Gilliland (FUG) shortcut procedure for continuous distillation, described in Chapter 9, at successive time steps. In essence, they treat batch rectification as a sequence of continuous, steady-state rectifications. As in the FUG method, no estimations of compositions or temperatures are made for intermediate stages.

Sundaram and Evans [8] apply their shortcut method to a column of the type shown in Figure 13.3. An overall mole balance for a constant distillate rate, D , gives

$$-\frac{dW}{dt} = D \quad (13-18)$$

Therefore,

$$-\frac{dW}{dt} = \frac{V}{1+R} \quad (13-19)$$

For any component, i , an instantaneous mole balance around the column gives

$$\frac{d(x_{w_i}W)}{dt} = x_{D_i} \frac{dW}{dt} \quad (13-20)$$

Expanding the LHS of (13-20) and solving for dx_{w_i} :

$$dx_{w_i} = (x_{D_i} - x_{w_i}) \frac{dW}{W} \quad (13-21)$$

In finite-difference form, using Euler's method, (13-19) and (13-21) become, respectively:

$$W^{(k+1)} = W^{(k)} - \left(\frac{V}{1+R} \right) \Delta t \quad (13-22)$$

$$x_{w_i}^{(k+1)} = x_{w_i}^{(k)} + (x_{D_i}^{(k)} - x_{w_i}^{(k)}) \left[\frac{W^{(k+1)} - W^{(k)}}{W^{(k)}} \right] \quad (13-23)$$

where k is the time-increment index. For a given Δt time increment, $W^{(k+1)}$ is computed from (13-22) and then $x_{w_i}^{(k+1)}$ is computed for each component from (13-23). However, (13-23) requires values for $x_{D_i}^{(k)}$.

Calculations are initiated at $k = 0$. The initial charge to the still is $W^{(0)}$. Values of $x_{w_i}^{(0)}$ are equal to the mole fractions of the initial charge. Corresponding values of $x_{D_i}^{(0)}$ depend on the method used to start up the batch rectification still. If total-reflux operation is employed as the start-up method, as mentioned in Section 13.5, the Fenske equation of Chapter 9 can be applied to compute values of $x_{D_i}^{(0)}$ for given values of $x_{w_i}^{(0)}$ if column and condenser holdups are negligible. For a given number of equilibrium stages, N :

$$N = \frac{\log \left[\left(\frac{x_{D_i}}{x_{w_i}} \right) \left(\frac{x_{w_r}}{x_{D_r}} \right) \right]}{\log \alpha_{i,r}} \quad (13-24)$$

Solving,

$$x_{D_i} = x_{w_i} \left(\frac{x_{D_r}}{x_{w_r}} \right) \alpha_{i,r}^N \quad (13-25)$$

where r is an arbitrary reference component of the mixture, such as the least volatile component. Since

$$\sum_{i=1}^C x_{D_i} = 1.0 \quad (13-26)$$

substitution of (13-25) in (13-26) gives

$$x_{D_r} = \frac{x_{w_r}}{\sum_{i=1}^C x_{w_i} \alpha_{i,r}^N} \quad (13-27)$$

The initial distillate composition, $x_{D_r}^{(0)}$, is computed from (13-27). The remaining values of $x_{D_i}^{(0)}$ are computed in turn from (13-25).

Using the initial set of values for $x_{D_i}^{(0)}$, values of $x_{w_i}^{(1)}$ are computed from (13-23) following the calculation of $W^{(1)}$ from (13-22). To compute each subsequent set of $x_{w_i}^{(k+1)}$ for $k > 0$, values of $x_{D_i}^{(k)}$ for $k > 0$ are needed. These are obtained by applying the FUG method. Equation (13-24) applies during batch rectification if N is replaced by $N_{\min} < N$ with $i = \text{LK}$ and $r = \text{HK}$. But $N_{\min}$ is related to N by the Gilliland correlation. A convenient, but approximate, equation for that correlation, due to Eduljee [9], is

$$\frac{N - N_{\min}}{N + 1} = 0.75 \left[1 - \left(\frac{R - R_{\min}}{R + 1} \right)^{0.5668} \right] \quad (13-28)$$

An estimate of the minimum-reflux ratio, $R_{\min}$, is provided by the Class I Underwood equation of Chapter 9, which assumes that all components in the charge distribute between the two products. Thus:

$$R_{\min} = \frac{\left(\frac{x_{D_{\text{LK}}}}{x_{w_{\text{LK}}}} \right) - \alpha_{\text{LK,HK}} \left(\frac{x_{D_{\text{HK}}}}{x_{w_{\text{HK}}}} \right)}{\alpha_{\text{LK,HK}} - 1} \quad (13-29)$$

If one or more components fail to distribute, then Class II Underwood equations should be used. Sundaram and Evans use only (13-29) with LK and HK equal to the lightest component, 1, and the heaviest component, C , in the mixture, respectively.

If (13-25), with $i = 1$, $r = C$, and $N = N_{\min}$, and (13-27) with $r = C$ are substituted into (13-29) with LK = 1 and HK = C , the result is

$$R_{\min} = \frac{\alpha_{1,C}^{N_{\min}} - \alpha_{1,C}}{(\alpha_{1,C} - 1) \sum_{i=1}^C x_{w_i} \alpha_{i,C}^{N_{\min}}} \quad (13-30)$$

For specified values of N and R , (13-28) and (13-30) are solved for $R_{\min}$ and $N_{\min}$ simultaneously by an iterative method. The value of x_{D_C} is then computed from (13-27) with $N = N_{\min}$, followed by the calculation of the other values of x_{D_i} from (13-25). Values of $N_{\min}$ and $R_{\min}$ change with time.

The procedure of Sundaram and Evans involves an inner loop for the calculation of $x_{D_i}^{(k)}$, and an outer loop for $W^{(k+1)}$ and $x_{w_i}^{(k+1)}$. The inner loop requires iterations because of the nonlinear nature of (13-28) and (13-30). Calculations of the outer loop are direct because (13-22) and (13-23) are linear. Application of the method is illustrated in the following example, where relative volatilities are assumed constant.

EXAMPLE 13.6

A charge of 100 kmol of a ternary mixture of A, B, and C with composition $x_{w_A}^{(0)} = 0.33$, $x_{w_B}^{(0)} = 0.33$, and $x_{w_C}^{(0)} = 0.34$ is distilled in a batch rectifier with $N = 3$ (including the reboiler), $R = 10$, and $V = 110$ kmol/h. Estimate the variation of the still, instantaneous distillate, and distillate-accumulator compositions as a function of time for 2 h of operation following an initial start-up period during which a steady-state operation at total reflux is achieved. Use $\alpha_{AC} = 2.0$ and $\alpha_{BC} = 1.5$, and neglect column holdup.

SOLUTION

The method of Sundaram and Evans is applied with $D = V/(1 + R) = 110/(1 + 10) = 10$ kmol/h. Therefore, 10 h would be required to distill the entire charge.

Start-up Period:

From (13-27), with C as the reference r ,

$$x_{D_C}^{(0)} = \frac{0.34}{0.33(2)^3 + 0.33(1.5)^3 + 0.34(1)^3} = 0.0831$$

From (13-25),

$$x_{D_A}^{(0)} = 0.33 \left(\frac{0.0831}{0.34} \right) 2^3 = 0.6449$$

and

$$x_{D_B}^{(0)} = 0.33 \left(\frac{0.0831}{0.34} \right) 1.5^3 = 0.2720$$

Take time increments, Δt , of 0.5 h.

Table 13.2 Results for Example 13.6

Time, h	W, kmol	x_W			$N_{\min}$	$R_{\min}$	x_D			x of Accumulated Distillate		
		A	B	C			A	B	C	A	B	C
0.0	100	0.3300	0.3300	0.3400	—	—	0.6449	0.2720	0.0831	—	—	—
0.5	95	0.3143	0.3329	0.3528	2.6294	1.2829	0.5957	0.2962	0.1081	0.6283	0.2749	0.0968
1.0	90	0.2995	0.3348	0.3657	2.6249	1.3092	0.5803	0.3048	0.1149	0.6045	0.2868	0.1087
1.5	85	0.2839	0.3365	0.3796	2.6199	1.3385	0.5633	0.3142	0.1225	0.5912	0.2932	0.1156
2.0	80	0.2675	0.3378	0.3947	2.6143	1.3709	0.5446	0.3242	0.1312	0.5800	0.2988	0.1212

At $t = 0.5$ h for outer loop:

$$\text{From (13-22), } W^{(1)} = 100 - \left(\frac{110}{1+10} \right) 0.5 = 95 \text{ kmol}$$

From (13-23) with $k = 0$,

$$x_{WA}^{(1)} = 0.33 + (0.6449 - 0.33) \left[\frac{95 - 100}{100} \right] = 0.3143$$

$$x_{WB}^{(1)} = 0.33 + (0.2720 - 0.33) \left[\frac{95 - 100}{100} \right] = 0.3329$$

$$x_{WC}^{(1)} = 0.34 + (0.0831 - 0.34) \left[\frac{95 - 100}{100} \right] = 0.3528$$

At $t = 0.5$ h for inner loop:

From (13-28)

$$\frac{3 - N_{\min}}{3 + 1} = 0.75 \left[1 - \left(\frac{10 - R_{\min}}{10 + 1} \right)^{0.5668} \right]$$

Solving for $R_{\min}$,

$$R_{\min} = 10 - 1.5835 N_{\min}^{1.7643} \quad (1)$$

This equation holds for all values of time t .

From (13-30),

$$R_{\min} = \frac{2^{N_{\min}} - 2}{(2 - 1)[0.3143(2)^{N_{\min}} + 0.3329(1.5)^{N_{\min}} + 0.3528(1)^{N_{\min}}]} \quad (2)$$

Equations (1) and (2) are solved simultaneously for $R_{\min}$ and $N_{\min}$. This can be done by numerical or graphical methods including successive substitution, Newton's method, or with a spreadsheet by plotting each equation as $R_{\min}$ versus $N_{\min}$ and determining the intersection. The result is $R_{\min} = 1.2829$ and $N_{\min} = 2.6294$.

From (13-27), with $N = 2.6294$,

$$x_{DC}^{(1)} = 0.3528 / [0.3143(2)^{2.6294} + 0.3329(1.5)^{2.6294} + 0.3528] = 0.1081$$

From (13-25):

$$x_{DA} = 0.3143 \left(\frac{0.1081}{0.3528} \right) 2^{2.6294} = 0.5959$$

$$x_{DB} = 0.3329 \left(\frac{0.1081}{0.3528} \right) 1.5^{2.6294} = 0.2962$$

Subsequent, similar calculations give the results in Table 13.2.

13.7 STAGE-BY-STAGE METHODS FOR MULTICOMPONENT, BATCH RECTIFICATION

For final design studies or for the simulation of multicomponent, batch rectification, complete stage-by-stage temperature, flow, and composition profiles as a function of time are required. Such calculations are tedious, but can be carried out conveniently with either of two types of computer-based methods. Both methods are based on the same differential-algebraic equations for the distillation model, but differ in the way the equations are solved.

Rigorous Model

Meadows [10] developed the first rigorous, multicomponent, batch-distillation model, based on the assumptions of equilibrium stages, perfect mixing of liquid and vapor phases at each stage, negligible vapor holdup, constant-molar-liquid holdup, M , on a stage and in the condenser system, and adiabatic stages in the column. Distefano [11] extended the model and developed a computer-based method

for solving the set of equations. A more efficient method for solving the equations is presented by Boston et al. [12]. For more rapid calculations, Galindez and Fredenslund [13] developed a quasi-steady-state solution procedure that, at each step, utilizes the simultaneous-correction or inside-out methods for continuous distillation discussed in Chapter 10.

The Distefano model is based on the multicomponent, batch-rectification operation shown in Figure 13.10. The equipment consists of a partial reboiler, a column with N equilibrium stages or equivalent in packing, and a total condenser with a reflux drum. Also included, but not shown in Figure 13.10, are a number of accumulator or receiver drums equal to the desired number of overhead product and intermediate cuts. When product purity specifications cannot be made for successive distillate cuts, then intermediate (waste or slop) cuts are necessary. These cuts are usually recycled. To initiate operation, the feed is charged to the reboiler, to which heat is supplied. Vapor leaving Stage 1 at the top of the column is totally condensed and passes to the reflux drum. At first, a total-reflux condition is established for a steady-state, fixed-overhead vapor-flow rate. Depending upon the amount of liquid holdup in the column and in the condenser system,

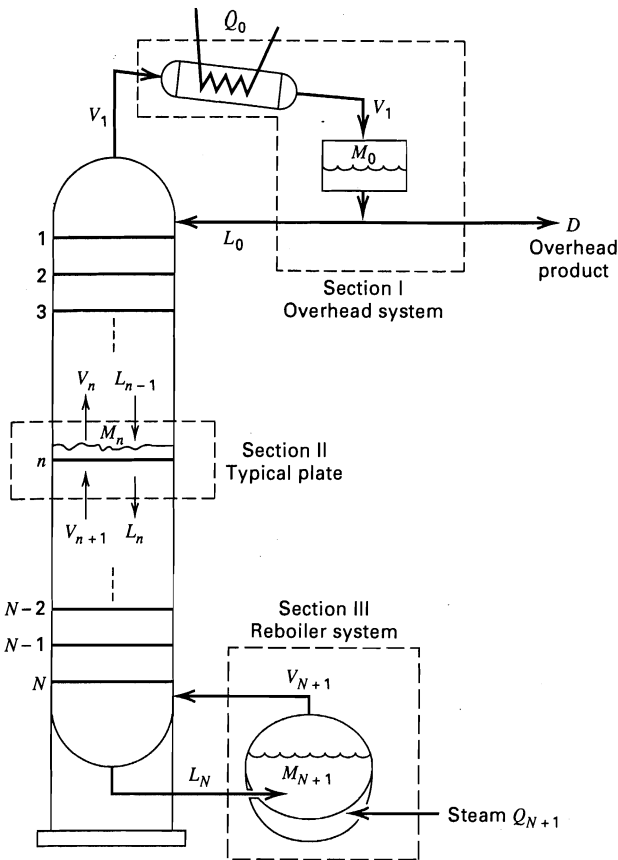


Figure 13.10 Multicomponent, batch-rectification operation.
[From G.P. Distefano, *AIChE J.*, **140**, 190 (1968) with permission.]

the amount and composition of the liquid in the reboiler at total reflux differs to some extent from the original feed.

Starting at time $t = 0$, distillate is removed from the reflux drum and sent to a receiver (accumulator) at a constant molar rate, and a reflux ratio is established. The heat-transfer rate to the reboiler is adjusted so as to maintain the overhead-vapor molar flow rate. Model equations are derived for the overhead condensing system, the column stages, and the reboiler, as illustrated for the overhead condensing system. For section I, in Figure 13.10, component material balances, a total material balance, and an energy balance are given, respectively, by

$$V_1 y_{i,1} - L_0 x_{i,0} - D x_{i,D} = \frac{d(M_0 x_{i,0})}{dt} \quad (13-31)$$

$$V_1 - L_0 - D = \frac{dM_0}{dt} \quad (13-32)$$

$$V_1 h_{V_1} - (L_0 + D) h_{L_0} = Q_0 + \frac{d(M_0 h_{L_0})}{dt} \quad (13-33)$$

where the derivative terms are accumulations due to holdup, which is assumed to be perfectly mixed. Also, for phase equilibrium at Stage 1 of the column:

$$y_{i,1} = K_{i,1} x_{i,1} \quad (13-34)$$

The working equations are obtained by combining (13-31) and (13-34) to obtain a revised component material balance

in terms of liquid-phase compositions, and by combining (13-22) and (13-33) to obtain a revised energy balance that does not include dM_0/dt . Equations for Sections II and III in Figure 13.10 are derived in a similar manner. The resulting working model equations for $t = 0 +$ are as follows, where i refers to the component and j refers to the stage, and M is molar liquid holdup.

1. Component mole balances for the overhead-condensing system, column stages, and reboiler, respectively:

$$\frac{dx_{i,0}}{dt} = - \left[\frac{L_0 + D + \frac{dM_0}{dt}}{M_0} \right] x_{i,0} + \left[\frac{V_1 K_{i,1}}{M_0} \right] x_{i,1}, \quad i = 1 \text{ to } C \quad (13-35)$$

$$\begin{aligned} \frac{dx_{i,j}}{dt} = & \left[\frac{L_{j-1}}{M_j} \right] x_{i,j-1} - \left[\frac{L_j + K_{i,j} V_j + \frac{dM_j}{dt}}{M_j} \right] x_{i,j} \\ & + \left[\frac{K_{i,j+1} V_{j+1}}{M_j} \right] x_{i,j+1}, \quad i = 1 \text{ to } C, j = 1 \text{ to } N \end{aligned} \quad (13-36)$$

$$\begin{aligned} \frac{dx_{i,N+1}}{dt} = & \left(\frac{L_N}{M_{N+1}} \right) x_{i,N} \\ & - \left[\frac{V_{N+1} K_{i,N+1} + \frac{dM_{N+1}}{dt}}{M_{N+1}} \right] x_{i,N+1}, \quad i = 1 \text{ to } C \end{aligned} \quad (13-37)$$

where $L_0 = RD$.

2. Total mole balances for overhead-condensing system and column stages, respectively:

$$V_1 = D(R + 1) + \frac{dM_0}{dt} \quad (13-38)$$

$$L_j = V_{j+1} + L_{j-1} - V_j - \frac{dM_j}{dt}, \quad j = 1 \text{ to } N \quad (13-39)$$

3. Enthalpy balances around overhead-condensing system, adiabatic column stages, and reboiler, respectively:

$$Q_0 = V_1(h_{V_1} - h_{L_0}) - M_0 \frac{dh_{L_0}}{dt} \quad (13-40)$$

$$\begin{aligned} V_{j+1} = & \frac{1}{(h_{V_{j+1}} - h_{L_j})} \\ & \times \left[V_j(h_{V_j} - h_{L_j}) - L_{j-1}(h_{L_{j-1}} - h_{L_j}) + M_j \frac{dh_{L_j}}{dt} \right], \quad j = 1 \text{ to } N \end{aligned} \quad (13-41)$$

$$\begin{aligned} Q_{N+1} = & V_{N+1}(h_{V_{N+1}} - h_{L_{N+1}}) - L_N(h_{L_N} - h_{L_{N+1}}) \\ & + M_{N+1} \left(\frac{dh_{L_{N+1}}}{dt} \right) \end{aligned} \quad (13-42)$$

4. Phase equilibrium on column stages and in the reboiler:

$$y_{i,j} = K_{i,j}x_{i,j}, \quad i = 1 \text{ to } C, \quad j = 1 \text{ to } N + 1 \quad (13-43)$$

5. Mole-fraction sums at column stages and in the reboiler:

$$\sum_{i=1}^C y_{i,j} = \sum_{i=1}^C K_{i,j}x_{i,j} = 1.0, \quad j = 0 \text{ to } N + 1 \quad (13-44)$$

6. Molar holdups in the condenser system and on the column stages based on constant-volume holdups, G_j , respectively:

$$M_0 = G_0\rho_0 \quad (13-45)$$

$$M_j = G_j\rho_j, \quad j = 1 \text{ to } N \quad (13-46)$$

where ρ is liquid molar density.

7. Variation of molar holdup in the reboiler, where M_{N+1}^0 is the initial charge to the reboiler:

$$M_{N+1} = M_{N+1}^0 - \sum_{j=0}^N M_j - \int_0^t D \, dt \quad (13-47)$$

Equations (13-35) through (13-47) constitute an initial-value problem for a system of ordinary differential and algebraic equations (DAEs). The total number of equations is $(2CN + 3C + 4N + 7)$. If variables N , D , $R = L_0/D$, M_{N+1}^0 , and all G_j are specified, and if correlations are available for computing liquid densities, vapor and liquid enthalpies, and K -values, the number of unknown variables, distributed as follows, is equal to the number of equations.

$x_{i,j}$	$CN + 2C$
$y_{i,j}$	$CN + C$
L_j	N
V_j	$N + 1$
T_j	$N + 2$
M_j	$N + 2$
Q_0	1
Q_{N+1}	1
	$2CN + 3C + 4N + 7$

Initial values at $t = 0$ for all these variables are obtained from the steady-state, total-reflux calculation, which depends only on values of N , M_{N+1}^0 , x_{N+1}^0 , G_j , and V_1 .

Equations (13-35) through (13-42) include first derivatives of $x_{i,j}$, M_j , and $h_{L,j}$. Except for M_{N+1} , derivatives of the latter two variables can be approximated with sufficient accuracy by incremental changes over the previous time step. If the reflux ratio is high, as it often is, the derivative of M_{N+1} can also be approximated in the same manner. This leaves only the $C(N + 2)$ ordinary differential equations (ODEs) for the component material balances to be integrated in terms of the $x_{i,j}$ dependent variables.

Rigorous Integration Method

The nonlinear equations (13-35) to (13-37) cannot be integrated analytically. Distefano [11] developed a numerical method of solution based on an investigation of 11 different numerical integration techniques that step in time. Of particular concern were the problems of *truncation* error and *stability*, which make it difficult to integrate the equations rapidly and accurately. Such systems of ODEs or DAEs constitute so-called *stiff systems* as described further below.

Local truncation errors result from using approximations for the functions on the right-hand side of the ODEs at each time step. These errors may be small, but they can propagate through subsequent time steps, resulting in global truncation errors sufficiently large to be unacceptable. As truncation errors become large, the number of significant digits in the computed dependent variables gradually decrease. Truncation errors can be reduced by decreasing the size of the time step.

Stability problems are much more serious. When instability occurs, the computed values of the dependent variables become totally inaccurate, with no significant digits at all. Reducing the time step does not eliminate instability until a time-step criterion, which depends on the numerical method, is satisfied. Even then, a further reduction in the time step is required to prevent oscillations of dependent variables.

Problems of stability and truncation error are conveniently illustrated by comparing results obtained by using the explicit- and implicit-Euler methods, both of which are first-order in accuracy, as discussed by Davis [15] and Riggs [16].

Consider the nonlinear, first-order ODE:

$$\frac{dy}{dt} = f\{t, y\} = ay^2te^{y} \quad (13-48)$$

for $y\{t\}$, where initially $y\{t_0\} = y_0$. The explicit- (forward) Euler method approximates (13-48) with a sequence of discretizations of the form

$$\frac{y_{k+1} - y_k}{\Delta t} = ay_k^2 t_k e^{y_k} \quad (13-49)$$

where Δt is the time step and k the sequence index. The function $f\{t, y\}$ is evaluated at the beginning of the current time step. Solving for y_{k+1} gives the recursion equation:

$$y_{k+1} = y_k + ay_k^2 t_k e^{y_k} \Delta t \quad (13-50)$$

Regardless of the nature of $f\{t, y\}$ in (13-48), the recursion equation can be solved explicitly for y_{k+1} using results from the previous time step. However, as discussed later, this advantage is counterbalanced by a limitation on the magnitude of Δt to avoid instability and oscillations.

The implicit- (backward) Euler method also utilizes a sequence of discretizations of (13-48), but the function, $f\{t, y\}$, is evaluated at the end of the current time step. Thus:

$$\frac{y_{k+1} - y_k}{\Delta t} = ay_{k+1}^2 t_{k+1} e^{y_{k+1}} \quad (13-51)$$

Because the function $f\{t, y\}$ is nonlinear in y , (13-51) cannot be solved explicitly for y_{k+1} . This disadvantage is counterbalanced by unconditional stability with respect to selection of Δt . However, too large a value can result in unacceptable truncation errors.

When the explicit-Euler method is applied to (13-35) to (13-47) for batch rectification, as shown in the following example, the maximum value of Δt can be estimated from the maximum, absolute eigenvalue, $|\lambda|_{\max}$, of the Jacobian matrix of (13-35) to (13-37). To prevent instability, $\Delta t_{\max} \leq 2/|\lambda|_{\max}$. To prevent oscillations, $\Delta t_{\max} \leq 1/|\lambda|_{\max}$. Applications of the explicit- and implicit-Euler methods are compared in the following batch-rectification example.

EXAMPLE 13.7

A charge of 100 kmol of an equimolar mixture of *n*-hexane (A) and *n*-heptane (B) is distilled at 15 psia in a batch rectifier consisting of a total condenser with a constant liquid holdup, M_0 , of 0.10 kmol, a single equilibrium stage with a constant liquid holdup, M_1 , of 0.01 kmol, and a reboiler. Initially the system is brought to the following total-reflux condition, with saturated liquid leaving the total condenser:

Stage	$T, ^\circ\text{F}$	x_A	K_A	K_B	M , kmol
Condenser	162.6	0.85935	—	—	0.1
Plate, 1	168.7	0.70930	1.212	0.4838	0.01
Reboiler, 2	178.6	0.49962	1.420	0.5810	99.89

Distillation begins ($t = 0$) with a reflux rate, L_0 , of 10 kmol/h and a distillate rate, D , of 10 kmol/h. Calculate the mole fractions of *n*-hexane and *n*-heptane at $t = 0.05$ h (3 min), at each of the three rectifier locations, assuming constant molar overflow and constant K -values from above for this small period of elapsed time. Use both the explicit- and implicit-Euler methods to determine the influence of the choice of the time step, Δt .

SOLUTION

Based on the constant molar overflow assumption:

$$V_1 = V_2 = 20 \text{ kmol/h} \quad \text{and} \quad L_0 = L_1 = 10 \text{ kmol/h}$$

Using the K -values and liquid holdups given earlier, (13-35) to (13-37), with all $dM_j/dt = 0$, become as follows:

Condenser

$$\frac{dx_{A,0}}{dt} = -200x_{A,0} + 242.4x_{A,1} \quad (1)$$

$$\frac{dx_{B,0}}{dt} = -200x_{B,0} + 96.76x_{B,1} \quad (2)$$

Plate

$$\frac{dx_{A,1}}{dt} = 1,000x_{A,0} - 3,424x_{A,1} + 2,840x_{A,2} \quad (3)$$

$$\frac{dx_{B,1}}{dt} = 1,000x_{B,0} - 1,967x_{B,1} + 1,162x_{B,2} \quad (4)$$

Reboiler

$$\frac{dx_{A,2}}{dt} = \left(\frac{10}{M_2}\right)x_{A,1} - \left(\frac{28.40}{M_2}\right)x_{A,2} \quad (5)$$

$$\frac{dx_{B,2}}{dt} = \left(\frac{10}{M_2}\right)x_{B,1} - \left(\frac{11.62}{M_2}\right)x_{B,2} \quad (6)$$

where

$$M_2(t = t) = M_2(t = 0) - (V_2 - V_1)t$$

$$\text{or} \quad M_2 = 99.89 - 10t \quad (7)$$

Equations (1) through (6) can be grouped by component into the following two matrix equations:

Component A:

$$\begin{bmatrix} -200 & 242.2 & 0 \\ 1,000 & -3,424 & 2,840 \\ 0 & 10/M_2 & -28.40/M_2 \end{bmatrix} \cdot \begin{bmatrix} x_{A,0} \\ x_{A,1} \\ x_{A,2} \end{bmatrix} = \begin{bmatrix} dx_{A,0}/dt \\ dx_{A,1}/dt \\ dx_{A,2}/dt \end{bmatrix} \quad (8)$$

Component B:

$$\begin{bmatrix} -200 & 96.76 & 0 \\ 1,000 & -1,967 & 1,160 \\ 0 & 10/M_2 & -11.62/M_2 \end{bmatrix} \cdot \begin{bmatrix} x_{B,0} \\ x_{B,1} \\ x_{B,2} \end{bmatrix} = \begin{bmatrix} dx_{B,0}/dt \\ dx_{B,1}/dt \\ dx_{B,2}/dt \end{bmatrix} \quad (9)$$

Although (8) and (9) do not appear to be coupled, they are because at each time step the sums $x_{A,j} + x_{B,j}$ do not equal 1. Accordingly, the mole fractions are normalized at each time step to force them to sum to 1. The initial eigenvalues of the Jacobian matrices (8) and (9) are computed from any of a number of computer programs, such as MathCad, Mathematica, MATLAB, or Maple, to be as follows, using $M_2 = 99.89$ kmol:

	Component A	Component B
λ_0	-126.54	-146.86
λ_1	-3,497.6	-2,020.2
λ_2	-0.15572	-0.03789

It is seen that $|\lambda|_{\max} = 3,497.6$. Thus, for the explicit-Euler method, instability and oscillations can be prevented by choosing:

$$\Delta t \leq 1/3,497.6 = 0.000286$$

If we select $\Delta t = 0.00025$ h (just slightly smaller than the criterion), it takes $0.05/0.00025 = 200$ time steps to reach $t = 0.05$ h (3 min). No such restriction applies to the implicit-Euler method, but too large a Δt may result in an unacceptable truncation error.

Explicit-Euler Method

According to Distefano [11], the maximum step size for integration using an explicit method is nearly always limited by stability considerations, and usually the truncation error is small. Assuming this to be true for this example, the following results were obtained using $\Delta t = 0.00025$ h with a spreadsheet program by converting

(8) and (9) together with (7) for M_2 , to the form of (13-50). Only the results for every 40 time steps are given.

Time, h	Normalized Mole Fractions in Liquid for <i>n</i> -Hexane			Normalized Mole Fractions in Liquid for <i>n</i> -Heptane		
	Distillate	Plate	Still	Distillate	Plate	Still
0.01	0.8183	0.6271	0.4993	0.1817	0.3729	0.5007
0.02	0.8073	0.6219	0.4991	0.1927	0.3781	0.5009
0.03	0.8044	0.6205	0.4988	0.1956	0.3795	0.5012
0.04	0.8036	0.6199	0.4985	0.1964	0.3801	0.5015
0.05	0.8032	0.6195	0.4982	0.1968	0.3805	0.5018

To show the instability effect, a time step of 0.001 h (four times the previous time step) gives the following unstable results during the first five time steps to an elapsed time of 0.005 h. Also included are values at 0.01 h for comparison to the preceding stable results.

Time, h	Normalized Mole Fractions in Liquid for <i>n</i> -Hexane			Normalized Mole Fractions in Liquid for <i>n</i> -Heptane		
	Distillate	Plate	Still	Distillate	Plate	Still
0.000	0.85935	0.7093	0.49962	0.14065	0.2907	0.50038
0.001	0.859361	0.559074	0.499599	0.140639	0.440926	0.500401
0.002	0.841368	0.75753	0.499563	0.158632	0.24247	0.500437
0.003	0.852426	0.00755	0.499552	0.147574	0.99245	0.500448
0.004	0.809963	0.884925	0.499488	0.190037	0.115075	0.500512
0.005	0.874086	1.154283	0.499546	0.125914	-0.15428	0.500454
0.01	1.006504	0.999254	0.493573	-0.0065	0.000746	0.506427

Much worse results are obtained if the time step is increased 10-fold to 0.01 h, as shown in the following table, where at $t = 0.01$ h, a very negative mole fraction has appeared.

Time, h	Normalized Mole Fractions in Liquid for <i>n</i> -Hexane			Normalized Mole Fractions in Liquid for <i>n</i> -Heptane		
	Distillate	Plate	Still	Distillate	Plate	Still
0.00	0.85935	0.7093	0.49962	0.14065	0.2907	0.50038
0.01	0.859456	-0.79651	0.49941	0.140544	1.796512	0.50059
0.02	2.335879	2.144666	0.497691	-1.33588	-1.14467	0.502309
0.03	1.284101	1.450481	0.534454	-0.2841	-0.45048	0.465546
0.04	1.145285	1.212662	8.95373	-0.14529	-0.21266	-7.95373
0.05	1.07721	1.11006	1.191919	-0.07721	-0.11006	-0.19192

Implicit-Euler Method

If (8) and (9) are converted to implicit equations like (13-51), they can be rearranged into a linear, tridiagonal set for each component. For example, the equation for component A on the plate becomes

$$(1,000 \Delta t)x_{A,0}^{(k+1)} - (1 + 3,424 \Delta t)x_{A,1}^{(k+1)} + (2,840 \Delta t)x_{A,2}^{(k+1)} = -x_{A,1}^{(k)}$$

The two tridiagonal equation sets can be solved by the tridiagonal-matrix algorithm or with a spreadsheet program using the iterative, circular-reference technique. For the implicit-Euler method, the selection of the time step, Δt , is not restricted by stability considerations. However, too large a Δt can lead to unacceptable truncation

errors. Normalized, liquid-mole-fraction results at $t = 0.05$ h for just component A are as follows for a number of different choices of Δt , all of which are greater than the 0.00025 h used earlier to obtain stable and oscillation-free results with the explicit-Euler method. Included for comparison is the explicit-Euler result for $\Delta t = 0.00025$ h.

Time = 0.05 h:

Δt , h	Normalized Mole Fractions in Liquid for <i>n</i> -Hexane		
	Distillate	Plate	Still
Explicit-Euler			
0.00025	0.8032	0.6195	0.4982
Implicit-Euler			
0.0005	0.8042	0.6210	0.4982
0.001	0.8042	0.6210	0.4982
0.005	0.8045	0.6211	0.4982
0.01	0.8049	0.6213	0.4982
0.05	0.8116	0.6248	0.4982

The preceding data show acceptable results with the implicit-Euler method using a time step about 200 times the $\Delta t_{\max}$ for the explicit-Euler method.

Another serious computational problem occurs when integrating the equations of batch distillation. Because the liquid holdups on the trays and in the condenser are small, the values of the corresponding liquid mole fractions, $x_{i,j}$, respond quickly to changes. The opposite holds for the reboiler (still) with its large liquid holdup. Hence, the required time step for accuracy is usually small, leading to a very slow response of the overall rectification system. Systems of ODEs having this characteristic constitute so-called stiff systems. For such a system, as discussed by Carnahan and Wilkes [17], an explicit method of solution must utilize a small time step for the entire period even though values of the dependent variables may all be changing slowly for a large portion of the time period. Accordingly, it is preferred to utilize a special implicit-integration technique developed by Gear [14] and others, as contained in the public-domain software package called ODEPACK. Gear-type methods strive for accuracy, stability, and computational efficiency by using multistep, variable order, and variable-step-size implicit techniques.

A commonly used measure of the degree of stiffness is the eigenvalue ratio $|\lambda|_{\max}/|\lambda|_{\min}$, where λ values are the eigenvalues of the Jacobian matrix of the set of ODEs. For the Jacobian matrix of (13-35) through (13-37), the Gerschgorin circle theorem, discussed by Varga [18], can be employed to estimate the eigenvalue ratio. The maximum absolute eigenvalue corresponds to the component with the largest K -value and the tray with the smallest liquid molar holdup. When the Gerschgorin theorem is applied to a row of the Jacobian

matrix based on (13-36):

$$|\lambda|_{\max} \leq \left[\left(\frac{L_{j-1}}{M_j} \right) + \left(\frac{L_j + K_{i,j} V_j}{M_j} \right) + \left(\frac{K_{i,j+1} V_{j+1}}{M_j} \right) \right] \\ \approx 2 \left[\frac{L_j + K_{i,j} V_j}{M_j} \right] \quad (13-52)$$

where i refers to the most-volatile component and j to the stage with the smallest liquid molar holdup. The minimum absolute eigenvalue almost always corresponds to a row of the Jacobian matrix for the reboiler. Thus, from (13-37):

$$|\lambda|_{\min} \leq \left[\left(\frac{L_N}{M_{N+1}} \right) + \left(\frac{V_{N+1} K_{i,N+1}}{M_{N+1}} \right) \right] \\ \approx \left[\frac{L_N + K_{i,N+1} V_{N+1}}{M_{N+1}} \right] \quad (13-53)$$

where i now refers to the least-volatile component and $N + 1$ is the reboiler stage. The largest value of the reboiler holdup is M_{N+1}^0 . The stiffness ratio, SR, is

$$SR = \frac{|\lambda|_{\max}}{|\lambda|_{\min}} \approx 2 \left(\frac{L + K_{\text{lightest}} V}{L + K_{\text{heaviest}} V} \right) \left(\frac{M_{N+1}^0}{M_{\text{tray}}} \right) \quad (13-54)$$

From (13-54), the stiffness ratio depends not only on the difference between tray and initial reboiler molar holdups, but also on the difference between K -values of the lightest and heaviest components in the charge to the still.

Davis [15] states $SR = 20$ is not stiff, $SR = 1,000$ is stiff, and $SR = 1,000,000$ is very stiff. For the conditions of Example 13.7, using (13-54):

$$SR \approx 2 \left[\frac{10 + (1.212)(20)}{10 + (0.581)(20)} \right] \left(\frac{100}{0.01} \right) = 31,700$$

which meets the criterion of a stiff problem. A modification of the computational procedure of Distefano [11], for solving (13-35) through (13-46), is as follows:

Initialization

1. Establish total-reflux conditions, based on vapor and liquid molar flow rates V_j^0 and L_j^0 . V_{N+1}^0 is the desired boilup rate or L_0^0 is based on the desired distillate rate and reflux ratio such that $L_0^0 = D(R + 1)$.
2. At $t = 0$, reduce L_0^0 to begin distillate withdrawal, but maintain the boilup rate established or specified for the total-reflux condition. This involves replacing all L_j^0 with $L_j^0 - D$. Otherwise, the initial values of all variables are those established for total reflux.

Time Step

3. In (13-35) to (13-37), replace liquid-holdup derivatives by total material balance equations:

$$\frac{dM_j}{dt} = V_{j+1} + L_{j-1} - V_j - L_j$$

Solve the resulting equations for the liquid mole fractions using an appropriate implicit-integration

technique and a suitable time step. Normalize the mole fractions if they do not sum to one at each stage.

4. Compute a new set of stage temperatures and corresponding vapor-phase mole fractions from (13-44) and (13-43), respectively.
5. Compute liquid densities and liquid holdups from (13-45) and (13-46), and liquid and vapor enthalpies. Then determine derivatives of enthalpies and liquid holdups with respect to time by forward-finite-difference approximations.
6. Compute a new set of liquid and vapor molar flow rates from (13-38), (13-39), and (13-41).
7. Compute the new reboiler molar holdup from (13-47).
8. Compute condenser and reboiler heat-transfer rates from (13-40) and (13-42).

Iteration to Completion of Operation

9. Repeat Steps 3 through 8 for additional time steps until a specified operation is complete. The specified operation might be for a desired amount of distillate, desired mole fraction of a particular component in the accumulated distillate, etc.

New Operation

10. Dump the accumulated distillate into a receiver, change operating conditions, and repeat Steps 2 through 9. Terminate calculations following the final operation.

The foregoing procedure is limited to narrow-boiling feeds and the simple configuration shown in Figure 13.10. A more flexible and efficient method, specifically designed to cope with stiffness, is that of Boston et al. [12], which uses a modified inside-out algorithm of the type discussed for continuous distillation in Chapter 10, which can handle feeds ranging from narrow-boiling to wide-boiling, even for nonideal-liquid solutions. In addition, the Boston et al. method permits multiple feeds, side streams, tray heat transfer, vapor distillate, and considerable flexibility in operation specifications.

EXAMPLE 13.8

A charge of 100 kmol of 30 mol% acetone, 30 mol% methanol, and 40 mol% water at 60°C and 1 atm is to be distilled in a batch rectifier consisting of a reboiler, a column with five theoretical stages, a total condenser, a reflux drum, and three distillate accumulators. The molar liquid holdup of the condenser-reflux drum is 5 kmol, whereas the molar liquid holdup of each stage is 1 kmol. The pressure is assumed constant at 1 atm throughout the rectifier. The following four events are to occur, each with a reboiler duty of 1 million kcal h:

Event 1: Establishment of total-reflux conditions.

Event 2: Rectification with a reflux ratio of 3 until the purity of the accumulated distillate in the first accumulator drops to 73 mol%.

Event 3: Rectification with a reflux ratio of 3 and a second accumulator for 21 min.

Event 4: Rectification with a reflux ratio of 3 and a third accumulator for 27 min.

Determine accumulator and column conditions at the end of each event. Use the Wilson equation for computing K -values.

SOLUTION

The following results were obtained with the program DESIGN II/Batch of the ChemShare Corporation. The stiffness ratio, SR, may be computed from (13-54) based on total-reflux conditions at the end of Event 1. The conditions are as follows:

Event 1: Total Reflux Conditions:

Stage	$T, ^\circ\text{C}$	$L, \text{kmol/h}$	Mole Fraction in Liquid		
			Acetone	Methanol	Water
Condenser	55.6	138.9	0.770	0.223	0.007
1	55.6	138.6	0.761	0.227	0.012
2	55.7	138.0	0.747	0.235	0.018
3	55.9	137.0	0.722	0.247	0.031
4	56.2	134.8	0.673	0.269	0.058
5	57.3	128.7	0.560	0.306	0.134
Reboiler	62.2	—	0.252	0.307	0.441

The charge remaining in the still is $100 - 5 - 5(1) = 90$ kmol. The most-volatile component is acetone, with a K -value at the bottom stage of 1.203. The least-volatile component is water, with a corresponding K -value of 0.428. The stiffness ratio is

$$\text{SR} \approx 2 \left[\frac{128.7 + (1.203)(134.8)}{128.7 + (0.428)(134.8)} \right] \left(\frac{90}{1} \right) \approx 281$$

Thus, this problem is not very stiff. A time step of 0.06 min is used.

Event 2

The time required to complete Event 2 is computed to be 57.5 min. The accumulated distillate in Tank 1 is 32.0 kmol with a composition as follows: 73.0 mol% acetone, 26.0 mol% methanol, and 1.0 mol% water. The liquid remaining in the reboiler is 58.0 kmol with the following composition: 2.8 mol% acetone, 30.0 mol% methanol, and 67.2 mol% water.

Event 3

The time required to complete this event is specified as 21 min. The accumulated distillate in Tank 2 is 11.3 kmol of the following composition: 47.2 mol% acetone, 51.8 mol% methanol, and 1.0 mol% water. This intermediate cut is recycled for addition to the next charge.

Event 4

At the end of the 27-min specification, the accumulated distillate in Tank 3 is 13.8 kmol of the following composition: 8.3 mol% acetone, 86.2 mol% methanol, and 5.5 mol% water. The composition of the remaining 32.9 kmol in the still is as follows: 0.0 mol% acetone, 0.4 mol% methanol, and 99.6 mol% water.

Rapid-Solution Method

As an alternative to integration of the stiff system of differential equations, the quasisteady-state procedure of Galindez and Fredenslund [13] can be used. With this method, the transient conditions are simulated as a succession of a finite number of continuous steady states of short duration, typically 0.05 h (3 min). Holdup is taken into account, but the stiffness of the problem is of no consequence. Results compare favorably with those from the rigorous integration method.

Consider an intermediate theoretical stage, j , with molar holdup, M_j , in the batch rectifier in Figure 13.11a. A material balance for component i , in terms of component flow rates, rather than mole fractions, is

$$l_{i,j} + v_{i,j} - l_{i,j-1} - v_{i,j+1} + \frac{d(M_j x_{i,j})}{dt} = 0 \quad (13-55)$$

Assume constant molar holdup. Also, assume that during a short time period, $dt = \Delta t = t_{k+1} - t_k$, the component flow rates given by the first four terms in (13-55) remain constant at values corresponding to time t_{k+1} . The component holdup term in (13-55) is

$$\frac{d(M_j x_{i,j})}{dt} = M_j \left[\frac{x_{i,j}\{t_{k+1}\} - x_{i,j}\{t_k\}}{\Delta t} \right] \quad (13-56)$$

But, $x_{i,j} = l_{i,j}/L_j$. Therefore, (13-56) can be rewritten as

$$\frac{d(M_j x_{i,j})}{dt} = \frac{M_j l_{i,j}}{L_j \Delta t} - \frac{M_j l_{i,j}\{t_k\}}{L_j \{t_k\} \Delta t} \quad (13-57)$$

If (13-57) is substituted into (13-55) and terms in the component flow rate $l_{i,j}$ are collected:

$$l_{i,j} \left(1 + \frac{M_j}{L_j \Delta t} \right) + v_{i,j} - l_{i,j-1} - v_{i,j+1} - \frac{M_j l_{i,j}\{t_k\}}{L_j \{t_k\} \Delta t} \quad (13-58)$$

If (13-58) for unsteady-state (batch) distillation is compared to (10-58) for steady-state (continuous) distillation, we see that the term $M_j/(L_j \Delta t)$ in (13-58) corresponds to the

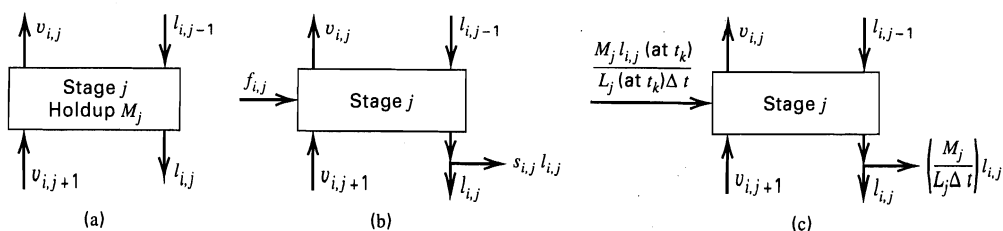


Figure 13.11 Simulation of holdup in a batch rectifier. (a) Stage in a batch rectifier with holdup. (b) Stage in a continuous fractionator. (c) Simulation of batch holdup in a continuous fractionator.

liquid side-stream ratio in (10-58) or that $M_j/\Delta t$ corresponds to a liquid side-stream flow rate. We also see that the term $M_{j,i,j}\{t_k\}/(L_j\{t_k\}\Delta t)$ in (13-58) corresponds to a component feed rate in (10-58). The analogy is shown in parts (b) and (c) of Figure 13.11. Thus, the change in component liquid holdup per unit time, $d(M_jx_{i,j})/dt$ in (13-56), is interpreted for a small, finite-time difference as the difference between a component feed rate into the stage and a component flow rate in a liquid side stream leaving the stage. In a similar manner, the stage enthalpy holdup in the energy balance for the stage is interpreted as the difference over a small, finite-time interval between a heat input to the stage and an enthalpy output in a liquid side stream leaving the stage. The overall result is a system of steady-state equations, identical in form to the equations for the simultaneous-correction and inside-out methods of Chapter 10. Accordingly, as implemented in the Batch Column computer model of Chemstations, Inc., either of those two methods can be used to solve the system of component-material-balance, phase-equilibrium, and energy-balance equations at each time step. The initial guesses used to initiate each time step are the values at the end of the previous time step. Because the variables generally change by only a small amount for each time step, convergence of the Newton-Raphson or inside-out method is generally achieved in a small number of iterations.

EXAMPLE 13.9

A charge of 100 lbmol of a mixture of 25 mol% benzene (B), 50 mol% monochlorobenzene (MCB), and 25 mol% *ortho*-dichlorobenzene (DCB) is distilled in a batch rectifier consisting of a reboiler, 10 equilibrium stages, a reflux drum, and three distillate product accumulators. The condenser-reflux drum holdup is constant at 0.20 ft³, and each stage in the column has a liquid holdup of 0.02 ft³. Pressures are 17.5 psia in the reboiler and 14.7 psia in the reflux drum, with a linear pressure profile in the column from 15.6 psia at the top to 17 psia at the bottom. Following an initialization at total reflux, the batch is distilled in the following three operation steps, each with a vapor boilup rate of 200 lbmol/h and a reflux ratio of 3. Thus, the distillate rate is 50 lbmol/h.

Operation step 1: Terminate when the mole fraction of benzene in the instantaneous distillate drops below 0.100.

Operation step 2: Terminate when the mole fraction of MCB in the distillate drops below 0.40.

Operation step 3: Terminate when the mole fraction of DCB in the reboiler rises above 0.98.

Assume ideal solutions and the ideal-gas law.

SOLUTION

This problem is quite stiff, with a stiffness ratio of approximately 15,000. The quasi-steady-state procedure of Galindez and Fredenslund [13], as implemented in Batch Column, was used with a time increment of 0.005 h for each of the three operation steps. Although 0.05 h is normal for the Galindez and Fredenslund method, the high ratio of distillate rate to charge for this problem necessitated a smaller Δt . Computed results are given in

Table 13.3 Results at the End of Each Operation Step for Example 13.9

	Operation Step		
	1	2	3
Operation time, h	0.605	0.805	0.055
No. of time increments	121	161	11
Accumulated distillate:			
Total lbmol	33.65	41.96	2.73
Mole fractions:			
B	0.731	0.009	0.000
MCB	0.269	0.950	0.257
DCB	0.000	0.041	0.743
Reboiler holdup:			
Total lbmol	66.13	24.19	21.46
Mole fractions:			
B	0.006	0.000	0.000
MCB	0.616	0.044	0.018
DCB	0.378	0.956	0.982
Total heat duties, 10 ⁶ Btu:			
Condenser	1.95	2.65	0.19
Reboiler	2.08	2.63	0.18

Table 13.3, where it is seen that the accumulated distillate cuts from operation steps 1 and 3 are quite impure with respect to benzene and DCB, respectively. The cut from Step 2 is 95 mol% pure MCB. The residual left in the reboiler after Step 3 is quite pure in DCB. A plot of the instantaneous-distillate composition as a function of total-distillate accumulation for all steps is shown in Figure 13.12.

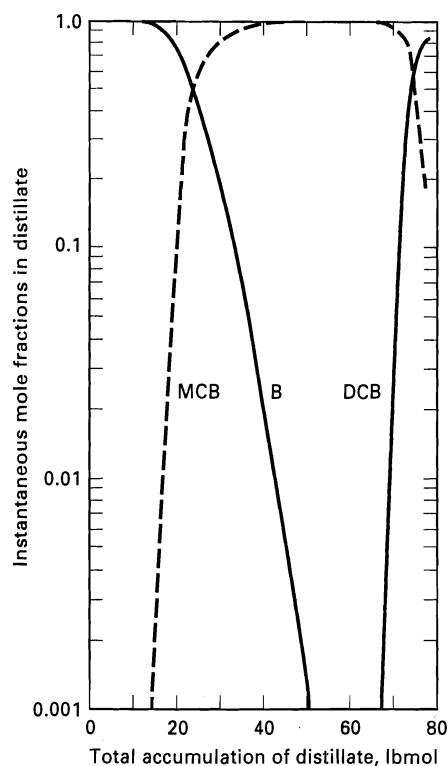


Figure 13.12 Instantaneous-distillate composition profile for Example 13.9.

[Perry's Chemical Engineers' Handbook, 6th ed., R.H. Perry and D.W. Green, Eds., McGraw-Hill, New York (1984) with permission.]

Table 13.4 Results of Alternative Operating Schedule for Example 13.9

Distillate Cut	Amount, lbmol	Composition, Mole Fractions		
		B	MCB	DCB
Benzene-rich	18	0.993	0.007	0.000
Intermediate 1	18	0.374	0.626	0.000
MCB-rich	34	0.006	0.994	0.000
Intermediate 2	8	0.000	0.536	0.464
DCB-rich residual	22	0.000	0.018	0.982
Total	100			

Changes in mole fractions occur very rapidly at certain times during the batch rectification, indicating that relatively pure cuts may be possible. This plot is useful in developing alternative schedules to obtain almost pure cuts. For example, suppose relatively rich distillate cuts of B, MCB, and DCB are desired. From Figure 13.12, an initial benzene-rich cut of, say, 18 lbmol might be taken, followed by an intermediate cut for recycle of, say, 18 lbmol. Then, an MCB-rich cut of 34 lbmol might be taken, followed by another intermediate cut of 8 lbmol, leaving a DCB-rich residual of 22 lbmol. For this series of operation steps, with the same vapor boilup rate of 200 lbmol/h and reflux ratio of 3, the computed results for each distillate accumulation (cut), using Batch Column with a time step of 0.005 h, are given in Table 13.4. As seen, all three product cuts are better than 98 mol% pure. However, $(18 + 8) = 26$ lbmol of intermediate cuts, or about one-fourth of the original charge, would have to be recycled. Further improvements in purities of the cuts or reduction in the amounts of intermediate cuts for recycle can be made by increasing the reflux ratio and/or the number of theoretical stages.

13.8 OPTIMAL CONTROL

As discussed by Luyben [19], design of a batch distillation process can be complex. Two aspects must be considered: (1) the products to be obtained and (2) the control method to be employed. Basic design parameters are the number of trays, the size of the initial charge to the still pot, the boilup ratio, and the reflux ratio as a function of time. Even if the feed is a binary, it may be necessary to take three products: a distillate rich in the most-volatile component, a residue rich in the least-volatile component, and an intermediate (slop or waste) cut containing both components. If the feed is a ternary system, one or two slop cuts may be necessary, in addition to two distillate cuts and the residue.

Slop Cuts

First consider the batch distillation of a binary mixture in the following example.

EXAMPLE 13.10

We wish to batch-distill 100 kmol of an equimolar mixture of *n*-hexane (C6) and *n*-heptane (C7) at 1 atm in a batch-rectification column with a total condenser. It is desired to produce two products, one with 95 mol% C6 and the other with 95 mol% C7. Neglect

Table 13.5 Batch Distillation of a C6-C7 Mixture

	Case 1	Case 2	Case 3	Case 4	Case 5
Reflux ratio	2	3	4	8	9.54
C6 product, kmol	15.1	36.0	42.4	49.2	50.0
C7 product, kmol	34.4	40.7	44.3	49.2	50.0
Slop cut, kmol	50.5	23.3	13.3	1.6	0.0
Mole fraction of C6 in slop cut	0.67	0.59	0.57	0.54	No slop cut
Total operation time, hours	1.97	2.37	2.78	4.57	5.27

holdup and assume a boilup rate of 100 kmol/h. Assume column operation at constant reflux ratio. Thus, the distillate composition will change with time. Determine a reasonable number of equilibrium stages and the effect of reflux ratio on the amount of slop cut.

SOLUTION

To determine the number of equilibrium stages, a McCabe-Thiele diagram, based on *K*-values from the SRK equation of state, is used in the manner of Figures 13.4 and 13.5. For the condition of total reflux ($y = x$, 45° line), the minimum number of stages to achieve a 95 mol% C6 from an initial feed of 50 mol% C6 is 3.1, where one stage is the boiler. For operation at a finite reflux ratio, assume twice the minimum, or five equilibrium plates plus the boiler, for a total of six equilibrium stages.

Now compute the products obtained at different reflux ratios. For each case, the first product is the accumulation of distillate of 95 mol% C6. At this point, if the residue contains less than 95 mol% C7, then, in a second step, a second accumulation of distillate (the slop cut) is made until the residue achieves the desired C7 composition. The reflux ratio is held constant throughout. The results (see Table 13.5) were obtained with the Batch Column model of Chemstations, which is included in the CHEMCAD simulation program. For a perfect separation (by material balance) the C6 and C7 products must each be 50 lbmol at 95 mol% purity. From Table 13.5, this is achieved at a constant reflux ratio of 9.54, with an operating time of 5.27 hours.

For lower reflux ratios, a slop cut whose amount increases as the reflux ratio decreases, is necessary. If the quantity of feed is much larger than the capacity of the still pot, the feed can be distilled in a sequence of charges. Then the slop cut for binary distillation of a batch can be recycled to the next batch. In this manner, each charge consists of fresh feed mixed with recycle slop cut. As discussed by Luyben [19], the composition of the slop cut is often not very different from the feed. This is largely confirmed in Table 13.5. If the number of stages in this example is increased from 6, the reflux ratio for eliminating the slop cut can be reduced. For example, if 10 equilibrium stages are used, the reflux ratio can be reduced from 9.54 to approximately 6.

Slop-cut strategy for batch distillation of a ternary mixture, as discussed by Luyben [19], is considerably more complex, as shown in the following example.

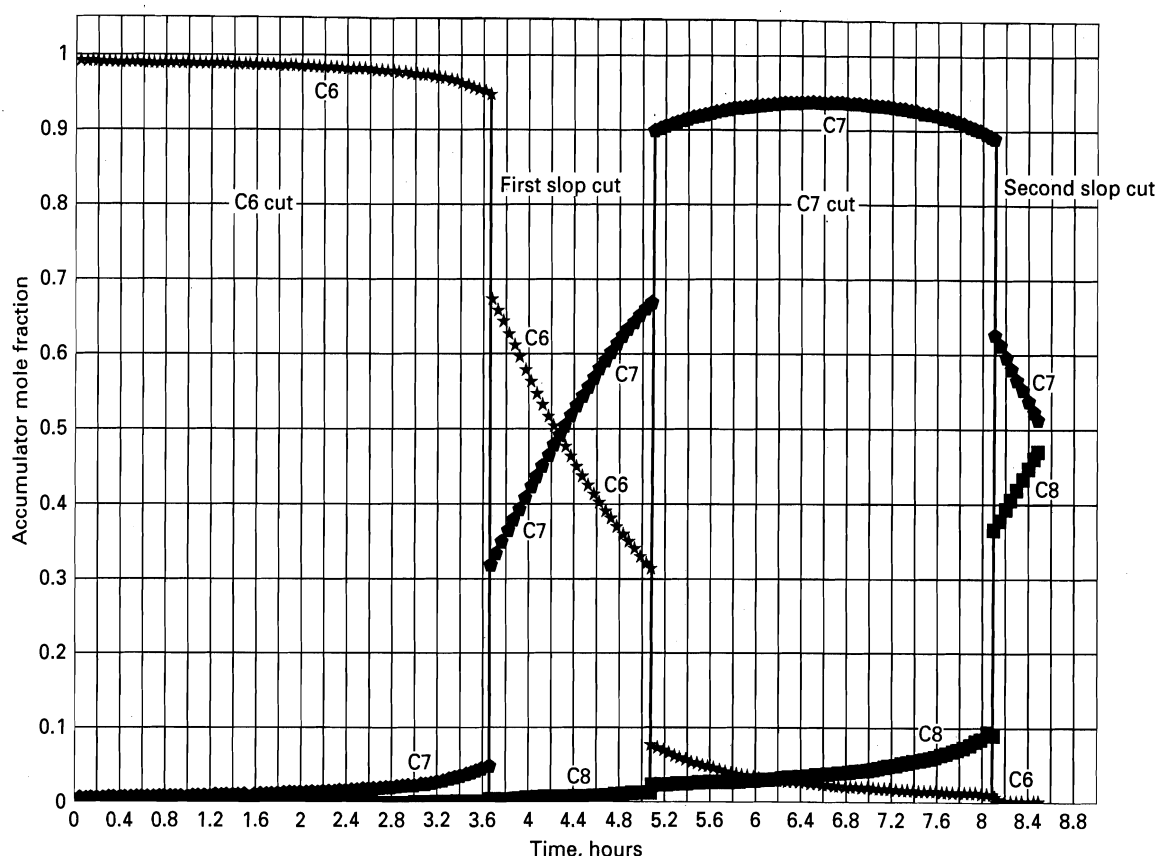


Figure 13.13 Ternary, batch distillation with two slop cuts in Example 13.11.

EXAMPLE 13.11

We wish to batch-distill 150 kmol of an equimolar ternary mixture of C6, C7, and normal octane (C8) at 1 atm in a batch-rectification column with a total condenser. We wish to produce three products: distillates of 95 mol% C6 and 90 mol% C7, and a residue of 95 mol% C8. Neglect holdup and assume a boilup rate of 100 kmol/h. Assume that column operation is controlled at constant reflux ratio. Thus, the distillate composition will change with time. Assume the column will contain five equilibrium stages, which together with the boiler, makes a total of six equilibrium stages. Determine the effect of reflux ratio on the slop cuts.

SOLUTION

The difficulty in this ternary example is to find operating conditions that will permit the purity specification of the intermediate component, C7, to be met. The difficulty lies in the specification for the termination of the second cut, which, unless the reflux ratio is high enough, is a slop cut. For example, suppose the reflux ratio is held constant at 4 and the intention is to terminate the second cut when the mole fraction of C7 in the instantaneous distillate reaches 90 mol% C7. Unfortunately, computer simulations show that only a value of 88 mol% C7 is reached. Therefore, try a higher reflux ratio, e.g., 8. In addition, terminate the third cut (the C7 product) when the mole fraction of C8 in that cut rises to 0.09 in the accumulator; and terminate the second slop cut when the mole fraction of C8 in the residue rises to 0.95, the desired purity. Note that no purity specification has been placed on the C7 product. Instead, it has been assumed that the desired purity of 90 mol% C7 will be achieved with impurities of 9 mol% C8 and 1 mol% C6. Acceptable

Table 13.6 Batch Distillation of a C6-C7-C8 Mixture

	Case 1	Case 2
Reflux ratio	4	8
C6 product, kmol	35.85	46.70
First slop cut:		
Amount, kmol	42.16	16.67
Mole fraction C6	0.373	0.316
Mole fraction C7	0.602	0.672
C7 product:		
Amount, kmol		35.43
Mole fraction C6		0.011
Mole fraction C7	0.877 max	0.898
Second slop cut:		
Amount, kmol		4.38
Mole fraction C6		0.000
Mole fraction C7		0.523
C8 product, kmol		46.82
Total operation time, hr		8.48

results are almost achieved for the reflux ratio of 8, as shown in Table 13.6 and Figure 13.13, where the desired purity of the C7 cut is 89.8 mol%. However, for a reflux ratio of 8, these results may not correspond to the optimal termination specification for the first slop cut. Furthermore, with a small adjustment in the reflux ratio, it may be possible to eliminate the second slop cut. These two considerations are the subject of Exercise 13.29.

Slop (waste, off) cuts and their recycle have been studied by a number of investigators, including Mayur, May, and Jackson [20], Luyben [19], Quintero-Marmol and Luyben [21], Farhat et al. [22], Mujtaba and Macchietto [23], Diehl et al. [24], and Robinson [25].

Optimal Control by Variation of Reflux Ratio

Two methods of controlling a batch distillation, under conditions of constant boilup rate, have been discussed. In Section 13.2, the distillate rate was maintained constant, which, at constant boilup, is equivalent to constant reflux ratio. This resulted in a variable composition for the instantaneous distillate and the accumulated distillate. This method of control is relatively simple and, accordingly, is widely practiced. In Section 13.3, the composition of the instantaneous distillate and, therefore, the accumulated distillate, was maintained constant. This requires a variable reflux ratio and accompanying distillate rate. Although not as simple as the constant reflux ratio method, the constant distillate composition method can be implemented with a rapidly responding composition (or surrogate) sensor.

What is the optimal way to control a batch distillation: (1) by constant reflux ratio, (2) by constant distillate composition, or (3) by some other means? With a simulation program, it is fairly straightforward to compare the first two methods. However, the results depend on the objective for the optimization. Diwekar [26] studied the following three objectives when the accumulated-distillate composition and/or the residual composition is specified:

1. Maximize the amount of accumulated distillate in a given time.

2. Minimize the time to obtain a given amount of accumulated distillate.
3. Maximize the profit.

In the following example, the first two methods of control are compared for the first two objectives.

EXAMPLE 13.12

Repeat Example 13.10 under conditions of constant distillate composition and compare the results to those of Example 13.10 for a constant reflux ratio of 4 with respect to both the amount of distillate and time of operation.

SOLUTION

For Example 13.10, from Table 13.5, for a reflux ratio of 4, the amount of accumulated distillate during the first operation step is 42.4 kmol of 95 mol% C6. The time required for this cut, which is not listed in Table 13.5, is 1.98 hours. Using a simulation program, the operation specifications for a constant composition operation are a boilup rate of 100 kmol/h, as in Example 13.10, with a constant instantaneous-distillate composition of 95 mol% C6. For the maximum distillate objective, the stop time for the first cut is 1.98 hours as in Example 13.10. The amount of distillate obtained is 43.5 kmol, which is 2.6% higher than for operation at constant reflux ratio. The variation of reflux ratio with time for constant composition control is shown in Figure 13.14, where the constant reflux ratio of 4 is also shown. The initial reflux ratio is 1.7; rising gradually at first and rapidly at the end. At 1 hour, the reflux ratio is 4, while at 1.98 hours, it is 15.4. For constant composition control, 42.4 kmol of accumulated distillate are obtained in 1.835 hours, compared to 1.98 hours for reflux-ratio control. Thus, again, constant composition control is more optimal, this time by almost 8%.

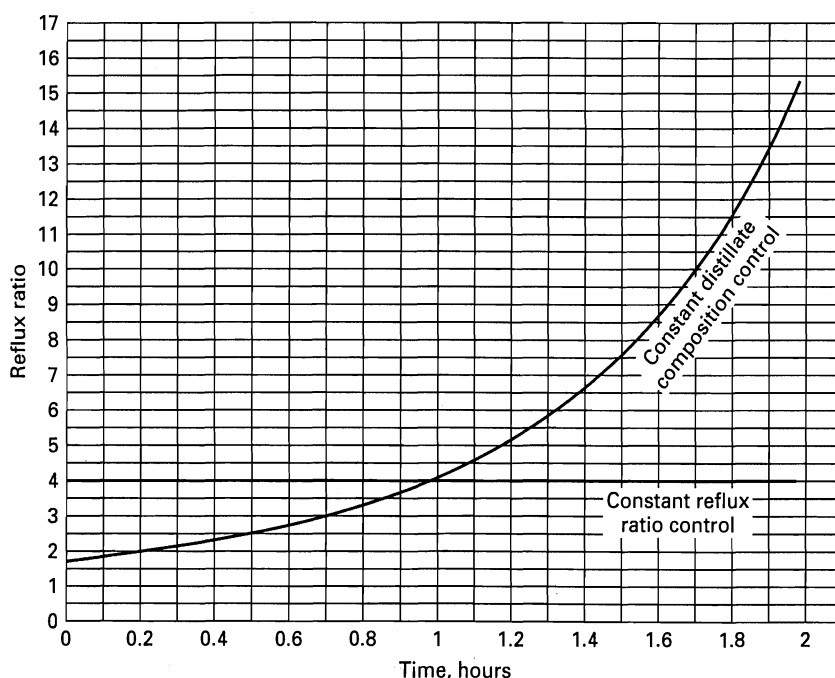


Figure 13.14 Binary, batch distillation under distillate-composition control in Example 13.12.

Studies by Converse and Gross [27], Coward [28, 29], and Robinson [30] for binary systems; by Robinson [30] and Mayur and Jackson [31] for ternary systems; and Diwekar et al. [32] for higher multicomponent systems show that further maximization of the amount of distillate or minimization of operation time, as well as maximization of profit, can be achieved by using an optimal-reflux-ratio policy. Often, this reflux-ratio policy is intermediate between the constant-reflux-ratio and constant-composition controls shown in Figure 13.14 for Example 13.12. Generally, the optimal-reflux-ratio control curve rises less sharply than that for the constant-distillate-composition control, with the result that savings in distillate, time, or money are highest for the more difficult separations. For relatively easy separations, savings for constant-distillate-composition control or optimal-reflux-ratio control may not be justified over the use of the simpler constant-reflux-ratio control.

Determination of optimal-reflux-ratio policy for complex operations requires a much different approach than that used in solving simpler optimization problems, which involve finding the optimal discrete value or set of values that will minimize or maximize some objective with respect to some algebraic function(s). For example, in Chapter 7, a single value of the optimal reflux ratio for a continuous distillation operation is found by plotting, as in Figure 7.22, the total annualized cost versus the reflux ratio, R , and locating the minimum in the curve. The optimal value is found to be approximately 1.24, which corresponds to $R/R_{\min} = 1.1$. In this section, we discuss establishing the optimal reflux ratio as a function of time, $R\{t\}$, for a batch distillation, which is modeled with differential or integral equations rather than algebraic equations. This type of problem requires optimal-control methods that include the calculus of variations, the maximum principle of Pontryagin, dynamic programming of Bellman, and nonlinear programming. Diwekar [33] describes these methods in detail. Their development by mathematicians in Russia and the United States were essential for the success of their respective space programs.

To illustrate one of the approaches to optimal control, consider the classic *Brachistochrone* (Greek for "shortest time") problem of Johann Bernoulli, one of the earliest variational problems, whose investigation by famous mathematicians, including Johann and Jakob Bernoulli, Gottfried Leibnitz, Guillaume de L'Hospital, and Isaac Newton, was the starting point for the development of the *calculus of variations*, a subject considered in detail by Weinstock [34]. A particle, e.g., a bead, is located in the x - y plane at (x_1, y_1) , where the x -axis is horizontal to the right, while the y -axis is vertically downward. The problem is to find the frictionless path, $y = f(x)$, ending at the point (x_2, y_2) , down which the particle will move, subject only to gravity, in the least time. Some possible paths from point 1 to point 2, shown in Figure 13.15, include a straight line, a circular arc, and a broken line consisting of two connected straight lines (one steep followed by one shallow). The shortest distance is the straight line, but Galileo Galilei proposed that the path of shortest

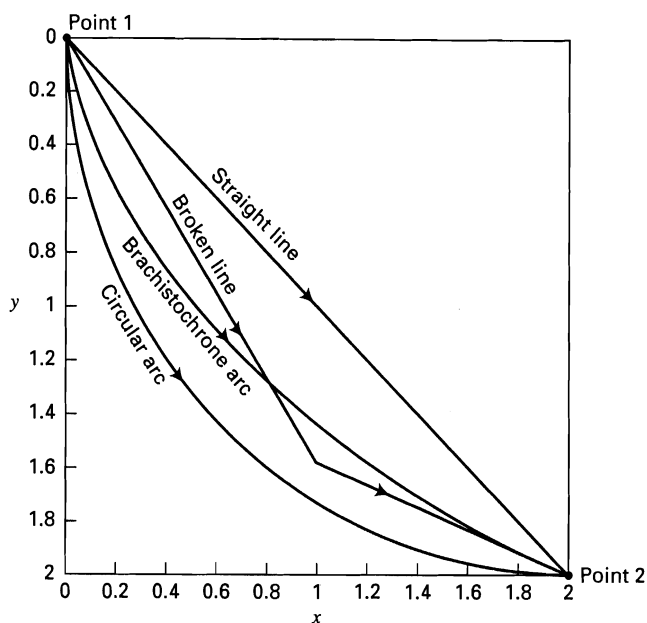


Figure 13.15 Frictionless paths between two points.

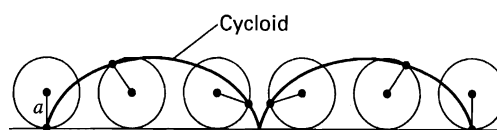


Figure 13.16 Generation of a cycloid from a circle of radius a .

time was the circular arc. However, the other aforementioned mathematicians proved that the solution is the arc of a cycloid, which, as included in Figure 13.15, is the locus of a point on the rim of a circle of radius a rolling along a straight line, as generated in Figure 13.16. The cycloid is given in parametric form as

$$x = a(\theta - \sin \theta) \quad \text{and} \quad y = a(1 - \cos \theta) \quad (13-59)$$

By eliminating θ , the Cartesian equation for the cycloid is

$$x = a \cos^{-1}(1 - y/a) - (2ay - y^2)^{0.5} \quad (13-60)$$

This optimal solution of the Brachistochrone problem is obtained by the calculus of variations as follows.

Let the arc length along the path be s . Then a differential length along the arc is the hypotenuse of a differential triangle, such that

$$ds = [(dy)^2 + (dx)^2]^{0.5} \quad (13-61)$$

or

$$ds = dx \sqrt{\left(\frac{dy}{dx}\right)^2 + 1} \quad (13-62)$$

The time, t_{12} , for the particle to travel from point 1 (P_1) to point 2 (P_2) is given by

$$t_{12} = \int_{P_1}^{P_2} \frac{ds}{v} \quad (13-63)$$

where v is the speed of the particle. By the conservation of energy, as the particle descends, its kinetic energy will increase as its potential energy decreases. Thus, if m is the mass of the particle and g is the acceleration due to gravity, where the velocity increases as the downward distance increases:

$$\frac{1}{2}mv^2 = mgy \quad (13-64)$$

Solving for v ,

$$v = \sqrt{2gy} \quad (13-65)$$

Substituting (13-62) and (13-65) into (13-63) gives

$$t_{12} = \int_{P_1}^{P_2} \sqrt{\frac{\left(\frac{dy}{dx}\right)^2 + 1}{2gy}} dx = \int_{P_1}^{P_2} \sqrt{\frac{1 + (y')^2}{2gy}} dx \quad (13-66)$$

where, $y' = dy/dx$. Thus, the function to be minimized is

$$f = \sqrt{\frac{1 + (y')^2}{2gy}} \quad (13-67)$$

Equation (13.67) is of the following general form of a problem that can be solved by the calculus of variations:

Minimize the integral,

$$I = \int_1^2 f(x, y\{x\}, y'\{x\}) dx \quad (13-68)$$

Weinstock shows that a necessary condition for the solution of (13-68) is the following Euler-Lagrange equation:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad (13-69)$$

There are two special cases of (13-69), resulting in the following simplifications:

1. If f is explicitly independent of y , then,

$$\frac{\partial f}{\partial y'} = C_1 \quad (13-70)$$

2. If f is explicitly independent of x , then

$$y' \left(\frac{\partial f}{\partial y'} \right) - f = C_2 \quad (13-71)$$

The Brachistochrone function of (13-67) is explicitly independent of x , so that (13-71) applies to give

$$(y')^2 [1 + (y')^2]^{-\frac{1}{2}} (2gy)^{-\frac{1}{2}} - [1 + (y')^2]^{\frac{1}{2}} (2gy)^{-\frac{1}{2}} = C_2 \quad (13-72)$$

which simplifies to

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right] y = \frac{1}{2gC_2^2} = 2a \quad (13-73)$$

If point 1 is located at $x = 0, y = 0$, the solution of (13-73) is (13-60), which is the arc of a cycloid.

How much better is the cycloid-arc path compared to the other paths shown in Figure 13.15? If point 2 is taken at $x = 2$ ft and $y = 2$ ft, then with $g = 32.17$ ft/s², the calculated travel times, from the application of (13-66) to move from point 1 to point 2 are as follows, where the cycloid arc is just slightly better than a circular arc:

Path in Figure 13.15	Travel time, seconds
Straight line	0.498
Broken line	0.472
Circular arc of radius = 2 ft	0.460
Cycloid arc with $a = 1.145836$ ft	0.455

Application of the calculus of variations to the determination of the optimal-reflux-control strategy for batch distillation is carried out in a manner similar to the above for the Brachistochrone problem. For example, if it is desired to find the reflux-ratio, R , path that will minimize the time, t , required to obtain an accumulated distillate of given amount and composition for a fixed boilup rate, V , using a column with N equilibrium stages, the integral to be minimized is as follows, where the variables for the distillate are replaced by the variables for the residual, W , remaining in the still:

$$t = \int_{x_{w_0}}^{x_w} \frac{W}{V} \left(\frac{R+1}{y_N - x_w} \right) dx_w \quad (13-74)$$

SUMMARY

1. The simplest case of batch distillation corresponds to the condensation of a vapor rising from a boiling liquid, called differential or Rayleigh distillation. The vapor leaving the liquid surface is assumed to be in equilibrium with the liquid. The compositions of the liquid and vapor continually change as distillation proceeds. The instantaneous vapor and liquid compositions can be computed as a function of time for a given vaporization rate.
2. A batch-rectifier system consists of a reboiler, a column with plates or packing that sits on top of the reboiler, a condenser, a reflux drum, and one or more distillate receivers.

3. For a binary system, a batch rectifier is usually operated at a constant reflux ratio or at a constant distillate composition. For either case, a McCabe-Thiele diagram can be used to follow the progress of the rectification, if the assumptions of constant molar overflow and negligible tray (or packing), condenser, and reflux drum liquid holdups are made.

4. A batch stripper is useful for removing small quantities of impurities from a charge. For complete flexibility, complex batch distillation involving both rectification and stripping can be employed.

5. Liquid holdup on the trays (or packing) and in the condenser and reflux drum can influence the course of batch rectification and the size and composition of the distillate cuts. The complexity of the holdup effect is such that it is best determined by rigorous calculations for specific cases.

6. For multicomponent, batch rectification, with negligible liquid holdup except in the reboiler, the shortcut method of Sundaram and Evans, based on successive applications of the Fenske–Underwood–Gilliland (FUG) method at a sequence of time intervals, can be used to obtain approximate distillate and charge compositions and amounts as a function of time.

7. For accurate and detailed multicomponent, batch-rectification compositions, the model of Distefano as implemented by Boston et al. should be used. It accounts for liquid holdup and permits a sequence of operation steps to produce multiple distillate cuts. The

model consists of algebraic and ordinary differential equations, which, when stiff, are best solved by Gear-type implicit-integration methods. The Distefano model can also be solved by the method of Galindez and Fredenslund, which simulates the unsteady batch process by a succession of steady states of short duration, which are solved by either the Newton–Raphson or the inside-out methods of Chapter 10.

8. Two difficult aspects of batch distillation are (1) determination of the best set of operations for the production of the desired products and (2) the optimal control method to be used. The first, which involves the possibility that slop or waste cuts may be necessary, is solved by computational studies using a simulation program. The second, which requires consideration of the best reflux-rate policy, is solved by applying optimal-control methods.

REFERENCES

1. RAYLEIGH, J.W.S., *Phil. Mag. and J. Sci.*, Series 6, **4** (23), 521–537 (1902).
2. SMOKER, E.H., and A. ROSE, *Trans. AIChE*, **36**, 285–293 (1940).
3. BLOCK, B., *Chem. Eng.*, **68** (3), 87–98 (1961).
4. BOGART, M.J.P., *Trans. AIChE*, **33**, 139–152 (1937).
5. ELLERBE, R.W., *Chem. Eng.*, **80** (12), 110–116 (1973).
6. HASBE, S., B.B. ABDUL AZIZ, I. HASHIMOTO, and T. WATANABE, *Proc. IFAC Workshop, London, Sept. 7–8, 1992*, p. 177.
7. DIWEKAR, U.M., and K.P. MADHAVAN, *Ind. Eng. Chem. Res.*, **30**, 713–721 (1991).
8. SUNDARAM, S., and L.B. EVANS, *Ind. Eng. Chem. Res.*, **32**, 511–518 (1993).
9. EDULJEE, H.E., *Hydrocarbon Processing*, **56** (9), 120–122 (1975).
10. MEADOWS, E.L., *Chem. Eng. Progr. Symp. Ser. No. 46*, **59**, 48–55 (1963).
11. DISTEFANO, G.P., *AIChE J.*, **14**, 190–199 (1968).
12. BOSTON, J.F., H.I. BRITT, S. JIRAPONGPHAN, and V.B. SHAH, in *Foundations of Computer-Aided Chemical Process Design*, Vol. II, AIChE, R.H.S. Mah and W.D. Seider, Eds., pp. 203–237 (1981).
13. GALINDEZ, H., and A. FREDENSLUND, *Comput. Chem. Eng.*, **12**, 281–288 (1988).
14. GEAR, C.W., *Numerical Initial Value Problems in Ordinary Differential Equations*, Prentice-Hall, Englewood Cliffs, NJ (1971).
15. DAVIS, M.E., *Numerical Methods and Modeling for Chemical Engineers*, John Wiley and Sons, New York (1984).
16. RIGGS, J.B., *An Introduction to Numerical Methods for Chemical Engineers*, Texas Tech. Univ. Press, Lubbock, TX (1988).
17. CARNAHAN, B., and J.O. WILKES, “Numerical solution of differential equations—an overview,” in *Foundations of Computer-Aided Chemical Process Design*, R.S.H. Mah and W.D. Seider, Eds., Engineering Foundation, New York, Vol. I, pp. 225–340 (1981).
18. VARGA, R.S., *Matrix Iterative Analysis*, Prentice-Hall, Englewood Cliffs, NJ (1962).
19. LUYBEN, W.L., *Ind. Eng. Chem. Res.*, **27**, 642–647 (1988).
20. MAYUR, D.N., R.A. MAY, and R. JACKSON, *Chem. Eng. Journal*, **1**, 15–21 (1970).
21. QUINTERO-MARMOL, E., and W.L. LUYBEN, *Ind. Eng. Chem. Res.*, **29**, 1915–1921 (1990).
22. FARHAT, S., M. CZERNICKI, L. PIBOULEAU, and S. DOMENECH, *AIChE J.*, **36**, 1349–1360 (1990).
23. MUJTABA, I.M., and S. MACCHIETTO, *Comput. Chem. Eng.*, **16**, S273–S280 (1992).
24. DIEHL, M., A. SCHAFER, H.G. BOCK, J.P. SCHLODER, and D.B. LEINWEBER, *AIChE J.*, **48**, 2869–2874 (2002).
25. ROBINSON, E.R., *Chem. Eng. Journal*, **2**, 135–136 (1971).
26. DIWEKAR, U.M., *Batch Distillation—Simulation, Optimal Design and Control*, Taylor & Francis, Washington, D.C. (1995).
27. CONVERSE, A.O., and G.D. GROSS, *Ind. Eng. Chem. Fundamentals*, **2**, 217–221 (1963).
28. COWARD, I., *Chem. Eng. Science*, **22**, 503–516 (1967).
29. COWARD, I., *Chem. Eng. Science*, **22**, 1881–1884 (1967).
30. ROBINSON, E.R., I. COWARD, *Chem. Eng. Science*, **24**, 1661–1668 (1969).
31. MAYUR, D.N., and R. JACKSON, *Chem. Eng. Journal*, **2**, 150–163 (1971).
32. DIWEKAR, U.M., R.K. MALIK, and K.P. MADHAVAN, *Comput. Chem. Eng.*, **11**, 629–637 (1987).
33. DIWEKAR, U.M., *Introduction to Applied Optimization*, Kluwer Academic Publishers (2003).
34. WEINSTOCK, R., *Calculus of Variations*, McGraw-Hill Book Co., Inc., New York (1952).

EXERCISES

Section 13.1

13.1 (a) A bottle of pure *n*-heptane is accidentally poured into a drum of pure toluene in a commercial laboratory. One of the laboratory assistants, with almost no background in chemistry, suggests that, since heptane boils at a lower temperature than toluene, the following purification procedure can be used:

Pour the mixture (2 mol% *n*-heptane) into a simple still pot. Boil the mixture at 1 atm and condense the vapors until all heptane is boiled away. Obtain the pure toluene from the residue in the still pot.

You, being a chemical engineer, immediately realize that such a purification method will not work. Indicate this by a curve

showing the composition of the material remaining in the pot after various quantities of the liquid have been distilled. What is the composition of the residue after 50 wt% of the original material has been distilled? What is the composition of the cumulative distillate?

(b) When one-half of the heptane has been distilled, what is the composition of the cumulative distillate and of the residue? What weight percent of the original material has been distilled?

Vapor–liquid equilibrium data at 1 atm [*Ind. Eng. Chem.*, **42**, 2912 (1949)] are as follows:

Mole Fraction <i>n</i> -Heptane			
Liquid	Vapor	Liquid	Vapor
0.025	0.048	0.448	0.541
0.062	0.107	0.455	0.540
0.129	0.205	0.497	0.577
0.185	0.275	0.568	0.637
0.235	0.333	0.580	0.647
0.250	0.349	0.692	0.742
0.286	0.396	0.843	0.864
0.354	0.454	0.950	0.948
0.412	0.504	0.975	0.976

13.2 A mixture of 40 mol% isopropanol in water is to be distilled at 1 atm by a simple batch distillation until 70 mol% of the charge has been vaporized (equilibrium data are given in Exercise 7.33). What will be the compositions of the liquid residue remaining in the still pot and of the collected distillate?

13.3 A 30 mol% feed of benzene in toluene is to be distilled in a batch operation. A product having an average composition of 45 mol% benzene is to be produced. Calculate the amount of residue, assuming $\alpha = 2.5$ and $W_0 = 100$.

13.4 A charge of 250 lb of 70 mol% benzene and 30 mol% toluene is subjected to batch, differential distillation at atmospheric pressure. Determine the compositions of the distillate and residue after one-third of the original mass is distilled off. Assume the mixture forms an ideal solution and apply Raoult's and Dalton's laws with vapor-pressure data.

13.5 A mixture containing 60 mol% benzene and 40 mol% toluene is subjected to batch, differential distillation at 1 atm, under three different conditions:

1. Until the distillate contains 70 mol% benzene
2. Until 40 mol% of the feed is evaporated
3. Until 60 mol% of the original benzene leaves in the vapor phase

Using $\alpha = 2.43$, determine for each of the three cases:

- (a) The number of moles in the distillate for 100 mol of feed.
- (b) The compositions of distillate and residue.

13.6 A mixture consisting of 15 mol% phenol in water is to be batch distilled at 260 torr. What fraction of the original batch remains in the still when the total distillate contains 98 mol% water? What is the residue concentration?

Vapor–liquid equilibrium data at 260 torr [*Ind. Eng. Chem.*, **17**, 199 (1925)]:

x , wt% (H_2O):

1.54 4.95 6.87 7.73 19.63 28.44 39.73 82.99 89.95
93.38 95.74

y , wt% (H_2O):

41.10 79.72 82.79 84.45 89.91 91.05 91.15 91.86 92.77
94.19 95.64

13.7 A still is charged with 25 mol of a mixture of benzene and toluene containing 0.35 mole fraction benzene. Feed of the same composition is supplied at a rate of 7 mol/h, and the heat rate is adjusted so that the liquid level in the still remains constant. No liquid leaves the still pot, and $\alpha = 2.5$. How long will it be before the distillate composition falls to 0.45 mole fraction benzene?

13.8 A distillation system consisting of a reboiler and a total condenser (no column) is to be used to separate A and B from a trace of nonvolatile material. The reboiler initially contains 20 lbmol of feed of 30 mol% A. Feed is to be supplied to the reboiler at the rate of 10 lbmol/h, and the heat input is so adjusted that the total moles of liquid in the reboiler remains constant at 20. No residue is withdrawn from the still. Calculate the time required for the composition of the overhead product to fall to 40 mol% A. The relative volatility may be assumed constant at 2.50.

Section 13.2

13.9 Repeat Exercise 13.2 for the case of a batch distillation carried out in a two-stage column with a reflux ratio of $L/V = 0.9$.

13.10 Repeat Exercise 13.3 assuming the operation is carried out in a three-stage column with $L/V = 0.6$.

13.11 One kilomole of an equimolar mixture of benzene and toluene is fed to a batch still containing three equivalent stages (including the boiler). The liquid reflux is at its bubble point, and $L/D = 4$. What is the average composition and amount of product at a time when the instantaneous product composition is 55 mol% benzene? Neglect holdup, and assume $\alpha = 2.5$.

13.12 The fermentation of corn produces a mixture of 3.3 mol% ethyl alcohol in water. If 20 mol% of this mixture is distilled at 1 atm by a simple, batch distillation, calculate and plot the instantaneous-vapor composition as a function of mole percent of batch distilled. If reflux with three theoretical stages (including the reboiler) is used, what is the maximum purity of ethyl alcohol that can be produced by batch distillation?

Equilibrium data are given in Exercise 7.29.

Section 13.3

13.13 An acetone–ethanol mixture of 0.5 mole fraction acetone is to be separated by batch distillation at 101 kPa.

Vapor–liquid equilibrium data at 101 kPa are as follows:

Mole Fraction Acetone										
y	0.16	0.25	0.42	0.51	0.60	0.67	0.72	0.79	0.87	0.93
x	0.05	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90

(a) Assuming an L/D of 1.5 times the minimum, how many stages should this column have if we want the composition of the distillate

to be 0.90 mole fraction acetone at a time when the residue contains 0.1 mole fraction acetone?

(b) Assume the column has eight stages and the reflux rate is varied continuously so that the top product is maintained constant at 0.9 mole fraction acetone. Make a plot of the reflux ratio versus the still-pot composition and the amount of liquid left in the still.

(c) Assume now that the same distillation is carried out at constant reflux ratio (and varying product composition). We wish to have a residue containing 0.1 and an (average) product containing 0.9 mole fraction acetone, respectively. Calculate the total vapor generated. Which method of operation is more energy intensive? Can you suggest operating policies other than constant reflux ratio and constant distillate compositions that might lead to equipment and/or operating cost savings?

13.14 A total of 2,000 gallons of 70 wt% ethanol in water, having a specific gravity of 0.871, is to be separated at 1 atm in a batch rectifier operating at constant distillate composition with a constant molar vapor boilup rate to obtain a distillate product of 85 mol% ethanol and a residual waste water containing 3 wt% ethanol. If the task is to be completed in 24 h, allowing 4 h for charging, start-up, shutdown, and cleaning, determine: (a) the number of theoretical plates required in the column, (b) the reflux ratio when the concentration of ethanol in the pot is 25 mol%, (c) the instantaneous distillate rate in lbmol/h when the concentration of ethanol in the pot is 15 mol%, (d) the lbmol of distillate product, and (e) the lbmol of residual wastewater.

Vapor-liquid equilibrium data are given in Exercise 7.29.

13.15 A charge of 1,000 kmol of a mixture of 20 mol% ethanol in water is to undergo batch rectification at 101.3 kPa at a vapor boilup rate of 100 kmol/h. If the column has the equivalent of six theoretical plates and the distillate composition is to be maintained at 80 mol% ethanol by varying the reflux ratio, determine: (a) the time in hours for the residue to reach an ethanol mole fraction of 0.05, (b) the kmol of distillate obtained when the condition of part (a) is achieved, (c) the minimum- and maximum-reflux ratios during the rectification period, and (d) the variation of the distillate rate in kmol/h during the rectification period. Assume constant molar overflow, neglect liquid holdup, and obtain vapor-liquid equilibrium data from Exercise 7.29.

13.16 A 500 lbmol mixture of 48.8 mol% A and 51.2 mol% B with a relative volatility $\alpha_{A,B}$ of 2.0 is to be separated in a batch rectifier consisting of a total condenser, a column with seven theoretical stages, and a partial reboiler. The reflux ratio is to be varied so as to maintain the distillation composition constant at 95 mol% A. The column can operate satisfactorily with a molar vapor boilup rate of 213.5 lbmol/h. The rectification is to be stopped when the mole fraction of A in the still drops to 0.192. Determine: (a) the time required for rectification, and (b) the total amount of distillate produced.

Section 13.4

13.17 Develop a procedure similar to that of Section 13.2 to calculate a binary batch stripping operation using the equipment arrangement of Figure 13.8.

13.18 A three-theoretical-stage batch stripper (one stage is the reboiler) is charged to the feed tank (see Figure 13.8) with 100 kmol of 10 mol% *n*-hexane in *n*-octane mix. The boilup rate is 30 kmol/h. If a constant boilup ratio (V/L) of 0.5 is used, determine the

instantaneous-bottoms composition and the composition of the accumulated bottoms product at the end of 2 h of operation.

13.19 Develop a procedure similar to that of Section 13.2 to calculate a complex, binary, batch-distillation operation using the equipment arrangement of Figure 13.9.

Section 13.5

13.20 For a batch rectifier with appreciable column holdup:

(a) Why is the composition of the charge to the still higher in the light component than the still composition at the start of rectification, assuming that total-reflux conditions are established before rectification begins?

(b) Why will separation be more difficult than with zero holdup?

13.21 For a batch rectifier with appreciable column holdup, why do tray compositions change less rapidly compared to a rectifier with negligible column holdup, and why is the degree of separation improved?

13.22 Based on the statements in Exercises 13.20 and 13.21, why is it difficult to predict the effect of holdup?

Section 13.6

13.23 Use the shortcut method of Sundaram and Evans to solve Example 13.7, but with zero condenser and stage holdups.

13.24 A charge of 100 kmol of an equimolar mixture of A, B, and C, with $\alpha_{A,B} = 2$ and $\alpha_{A,C} = 4$, is distilled in a batch rectifier containing the equivalent of four theoretical stages, including the reboiler. If holdup can be neglected, use the shortcut method with $R = 5$ and $V = 100$ kmol/h to estimate the variation of the still and instantaneous-distillate compositions as a function of time following a start-up period during which total reflux conditions are established.

13.25 A charge of 200 kmol of a mixture of 40 mol% A, 50 mol% B, and 10 mol% C with $\alpha_{A,C} = 2.0$ and $\alpha_{B,C} = 1.5$ is to be separated in a batch rectifier with a total of three theoretical stages and operating at a reflux ratio of 10, with a molar vapor boilup rate of 100 kmol/h. Holdup is negligible. Use the shortcut method to estimate instantaneous-distillate and bottoms compositions as a function of time for the first hour of operation following start-up to achieve total reflux conditions.

Section 13.7

13.26 A charge of 100 lbmol of 35 mol% *n*-hexane, 35 mol% *n*-heptane, and 30 mol% *n*-octane is to be distilled at 1 atm in a batch rectifier, consisting of a partial reboiler, a column, and a total condenser, at a constant boilup rate of 50 lbmol/h and a constant reflux ratio of 5. Before rectification begins, total reflux conditions are to be established. Then, the following three operation steps are to be carried out to obtain an *n*-hexane-rich cut, an intermediate cut for recycle, an *n*-heptane-rich cut, and an *n*-octane-rich residue:

Step 1: Stop when the accumulated distillate purity drops below 95 mol% *n*-hexane.

Step 2: Empty the *n*-hexane-rich cut produced in Step 1 into a receiver and resume rectification until the instantaneous distillate composition reaches 80 mol% *n*-heptane.

Step 3: Empty the intermediate cut produced in Step 2 into a receiver and resume rectification until the accumulated distillate composition reaches 4 mol% *n*-octane.

For thermodynamic properties, assume ideal solutions and the ideal gas law.

Consider conducting the rectification in two different columns, each with the equivalent of 10 theoretical stages and a condenser-reflux drum liquid holdup of 1.0 lbmol. For each column, determine with a suitable batch-distillation computer program the compositions and amounts in lbmol of each of the four products.

Column 1: A plate column with a total liquid holdup of 8 lbmol.

Column 2: A packed column with a total liquid holdup of 2 lbmol.

Discuss the effect of liquid holdup for the two columns. Are the results what you expected?

13.27 A charge of 100 lbmol of a hydrocarbon mixture containing 10 mol% propane, 30 mol% *n*-butane, 10 mol% *n*-pentane, and the balance *n*-hexane is to be separated in a batch rectifier equipped with a partial reboiler, a total condenser with a liquid holdup of 1.0 ft³, and a column with the equivalent of eight theoretical stages and a total holdup of 0.80 ft³. The pressure in the condenser is 50.0 psia and the column pressure drop is 2.0 psi. The rectification campaign or operating policy, given as follows, is designed to produce cuts of 98 mol% propane, 99.8 mol% *n*-butane, and a residual cut of 99 mol% *n*-hexane, and two intermediate cuts, one of which may be a relatively rich cut of *n*-pentane. All five operating steps are conducted at a molar vapor boilup rate of 40 lbmol/h. Use a suitable batch-distillation computer program to determine the amounts and compositions of all cuts.

Step	Reflux Ratio	Stop Criterion
1	5	98% propane in accumulator
2	20	95% <i>n</i> -butane in instantaneous distillate
3	25	99.8% <i>n</i> -butane in accumulator
4	15	80% <i>n</i> -pentane in instantaneous distillate
5	25	99% <i>n</i> -hexane in the pot

Make suggestions as to how you might alter the operation steps so as to obtain larger amounts of the product cuts and smaller amounts of the intermediate cuts.

13.28 A charge of 100 lbmol of benzene (B), monochlorobenzene (MCB), and *o*-dichlorobenzene (DCB) is being distilled in a batch rectifier that consists of a total condenser, a column with 10 theoretical stages, and a partial reboiler. Following the establishment of total reflux, the first operation step begins for a boilup rate of 200 lbmol/h and a reflux ratio of about 3. At the end of 0.60 h, the following conditions exist for the top three stages in the column:

	Top Stage	Stage 2	Stage 3
Temperature, °F	267.7	271.2	272.5
<i>V</i> , lbmol/h	206.1	209.0	209.5
<i>L</i> , lbmol/h	157.5	158.0	158.1
<i>M</i> , lbmol	0.01092	0.01088	0.01087

Vapor Mole Fractions:

B	0.0994	0.0449	0.0331
MCB	0.9006	0.9551	0.9669
DCB	0.0000	0.0000	0.0000

Liquid Mole Fractions:

B	0.0276	0.0121	0.00884
MCB	0.9724	0.9879	0.99104
DCB	0.0000	0.0000	0.00012

In addition reboiler and condenser holdups at 0.6 h are 66.4 and 0.1113 lbmol, respectively.

For benzene, use the preceding data with (13-36) and (13-39) to estimate the liquid-phase mole fraction of benzene leaving Stage 2 at 0.61 h by using the explicit-Euler method with a Δt of 0.01 h. If the result is unreasonable, explain why with respect to stability and stiffness considerations.

13.29 A mixture of 100 kmol of 30 mol% methanol, 30 mol% ethanol, and 50 mol% *n*-propanol is charged at a pressure of 120 kPa to a batch rectifier, consisting of a partial reboiler, a column containing the equivalent of 10 equilibrium stages, and a total condenser. After establishing a total-reflux condition, the column will begin a sequence of two operating steps, each for a duration of 15 hours at a distillate flow rate of 2 kmol/h and a reflux ratio of 10. Thus, the two accumulated distillates will equal in moles that of the methanol and that of the ethanol in the feed. Neglect the liquid holdup in the condenser and the column. The column pressure drop is 8 kPa, with a pressure drop of 2 kPa through the condenser. Using a process-simulation program with the UNIFAC method for liquid-phase activity coefficients, determine the mole-fraction composition and amount in kmol of each of the three cuts.

13.30 Repeat Exercise 13.29 with the following modifications. Add a third operating step. For all three steps, use the same distillate rate and reflux rate as in Exercise 13.29. Use the following durations for the three steps: 13 hours for Step 1, 4 hours for Step 2, and 13 hours for Step 3. The distillate from Step 2 will be a slop cut. Determine the mole-fraction composition and amount in kmol of each of the four cuts.

13.31 A mixture of 100 kmol of 45 mol% acetone, 30 mol% chloroform, and 25 mol% benzene is charged at pressure of 101.3 kPa to a batch rectifier, consisting of a partial reboiler, a column containing the equivalent of 10 equilibrium stages, and a total condenser. After establishing a total-reflux condition, the column will begin a sequence of two operating steps, each at a distillate flow rate of 2 kmol/h and a reflux ratio of 10. The durations will be 13.3 hours for Step 1 and 24.2 hours for Step 2. Neglect pressure drops and the liquid holdup in the condenser and the column. Using a process-simulation program with the UNIFAC method for liquid-phase activity coefficients, determine the mole-fraction composition and amount in kmol of each of the three cuts.

Section 13.8

13.32 Using a batch-distillation simulation program, make the following modifications to the C6-C7-C8 ternary example in Section 13.8:

- Increase the reflux above 8 to eliminate the second slop cut.
- Change the termination specification on the second step to reduce the amount of the first slop cut, without failing to meet all three product specifications.

Part 3

Separations by Barriers and Solid Agents

In recent years, the number of industrial applications of separations using barriers and solid agents have greatly increased because of progress in producing selective membranes and adsorbents. Chapter 14 presents a discussion of rates of mass transfer through membranes and calculation methods for the more widely used continuous membrane separations for gas and liquid feeds. These include gas permeation, reverse osmosis, dialysis, electrodialysis, pervaporation, ultrafiltration, and microfiltration.

Chapter 15 is concerned with separations of adsorption, ion exchange and chromatography that use solid agents. Discussions of equilibrium and rates of mass transfer in porous adsorbents are followed by calculation methods for batch and continuous equipment for liquid and gaseous feeds, including fixed-bed, pressure-swing, and simulated-moving-bed adsorption.

Membrane Separations

In a membrane-separation process, a feed consisting of a mixture of two or more components is partially separated by means of a semipermeable barrier (the membrane) through which one or more species move faster than another or other species. The most general membrane process is shown in Figure 14.1 where the feed mixture is separated into a *retentate* (that part of the feed that does not pass through the membrane, i.e., is retained) and a *permeate* (that part of the feed that does pass through the membrane). Although the feed, retentate, and permeate are usually liquid or gas, they may also be solid. The barrier is most often a thin, nonporous, polymeric film, but may also be porous polymer, ceramic, or metal materials, or even a liquid or gas. The barrier must not dissolve, disintegrate, or break. The optional sweep, shown in Figure 14.1, is a liquid or gas, used to help remove the permeate. Many of the industrially important membrane-separation operations are listed in Tables 1.2 and 14.1.

In membrane separations: (1) the two products are usually miscible, (2) the separating agent is a semipermeable barrier, and (3) a sharp separation is often difficult to achieve. Thus, membrane separations differ in two or three of these respects from the more common separation operations of absorption, stripping, distillation, and liquid–liquid extraction.

Although membranes as separating agents have been known for more than 100 years [1], large-scale applications have only appeared in the past 50 years. In the 1940s, porous fluorocarbons were used to separate $^{235}\text{UF}_6$ from $^{238}\text{UF}_6$ [2]. In the mid-1960s, reverse osmosis with cellulose acetate was first used to desalinize seawater to produce potable water (drinkable water with less than 500 ppm by weight of dissolved solids) [3]. Commercial ultrafiltration membranes followed in the 1960s. In 1979, Monsanto Chemical Company introduced a hollow-fiber membrane of polysulfone to separate certain gas mixtures—for example, to enrich hydrogen- and carbon dioxide-containing streams [4]. Commercialization of alcohol dehydration by pervaporation began in the late 1980s, as did the large-scale application of emulsion liquid membranes for removal of metals and organics from wastewater.

The replacement of the more-common separation operations with membrane separations has the potential to save large amounts of energy. This replacement requires the production of high-mass-transfer-flux, defect-free, long-life membranes on a large scale and the fabrication of the membrane into compact, economical modules of high surface area per unit volume.

14.0 INSTRUCTIONAL OBJECTIVES

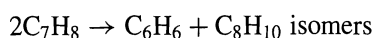
After completing this chapter, you should be able to:

- Explain membrane processes in terms of the membrane, feed, sweep, retentate, and permeate.
- List eight types of industrial membrane-separation processes.
- Discuss industrial, polymeric-membrane materials.
- Differentiate between the membrane mass-transfer parameters of permeability and permeance, and explain their relationship to mass-transfer coefficients.
- Differentiate between asymmetric and thin-layer composite membranes, and between dense and microporous membranes.
- Describe four membrane shapes.
- Describe six membrane modules.
- Explain mechanisms of mass transfer through membranes.
- Derive mass-transfer-rate equations for the solution-diffusion mechanism for liquid and gas mixtures.
- Explain four common idealized flow patterns in membrane modules.
- Explain use and advantages/disadvantages of recycle cascades of membrane modules.
- Explain concentration polarization.

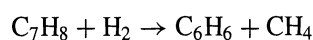
- Calculate mass-transfer rates for dialysis and electrodialysis.
- Explain osmosis and how reverse osmosis can be achieved.
- Calculate mass-transfer rates for reverse osmosis.
- Calculate mass-transfer rates for gas permeation.
- Calculate mass-transfer rates for pervaporation.
- Calculate mass-transfer rates for ultrafiltration and microfiltration.

Industrial Example

A common large-scale application of membranes is to the separation of hydrogen from methane. Following World War II, during which large amounts of toluene were required to produce TNT (trinitrotoluene) explosives, petroleum refiners sought other markets for toluene. One potential market was the use of toluene as a feedstock for the manufacture of benzene, a precursor for nylon, and xylenes, precursors for a number of other chemicals, including polyesters. Toluene can be catalytically disproportionated to benzene and mixed xylenes in an adiabatic reactor with the feed entering at 950°F and a pressure greater than 500 psia. The main reaction is



To suppress the formation of coke, which fouls the catalyst, the reactor feed must contain a substantial fraction of hydrogen at a partial pressure of at least 215 psia. Unfortunately, the hydrogen takes part in a side reaction for the hydrodealkylation of toluene to benzene and methane:



Makeup hydrogen is usually not pure, but contains perhaps 15 mol% methane and 5 mol% ethane. Thus, typically, the reactor effluent contains H_2 , CH_4 , C_2H_6 , C_6H_6 , unreacted C_7H_8 , and C_8H_{10} isomers. As shown in Figure 14.2a, for just the reaction section of the process, this effluent is cooled and partially condensed to 100°F at a pressure of 465 psia. At these conditions, a reasonably good separation is achieved between C_2H_6 and C_6H_6 in the flash drum. Thus, the vapor leaving the flash drum contains most of the H_2 , CH_4 , and C_2H_6 , with most of the aromatic chemicals leaving in the

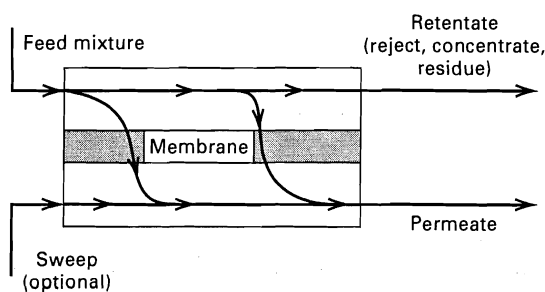


Figure 14.1 General membrane process.

Table 14.1 Industrial Applications of Membrane Separation Processes

1. Reverse osmosis:	Desalinization of brackish water Treatment of wastewater to remove a wide variety of impurities Treatment of surface and ground water Concentration of foodstuffs Removal of alcohol from beer and wine
2. Dialysis:	Separation of nickel sulfate from sulfuric acid Hemodialysis (removal of waste metabolites, excess body water, and restoration of electrolyte balance in human blood)
3. Electrodialysis:	Production of table salt from seawater Concentration of brines from reverse osmosis Treatment of wastewaters from electroplating Demineralization of cheese whey Production of ultrapure water for the semiconductor industry
4. Microfiltration:	Sterilization of drugs Clarification and biological stabilization of beverages Purification of antibiotics Separation of mammalian cells from a liquid
5. Ultrafiltration:	Preconcentration of milk before making cheese Clarification of fruit juice Recovery of vaccines and antibiotics from fermentation broth Color removal from Kraft black liquor in paper-making
6. Pervaporation:	Dehydration of ethanol–water azeotrope Removal of water from organic solvents Removal of organics from water
7. Gas permeation:	Separation of CO_2 or H_2 from methane and other hydrocarbons Adjustment of the H_2/CO ratio in synthesis gas Separation of air into nitrogen- and oxygen-enriched streams Recovery of helium Recovery of methane from biogas
8. Liquid membranes:	Recovery of zinc from wastewater in the viscose fiber industry Recovery of nickel from electroplating solutions

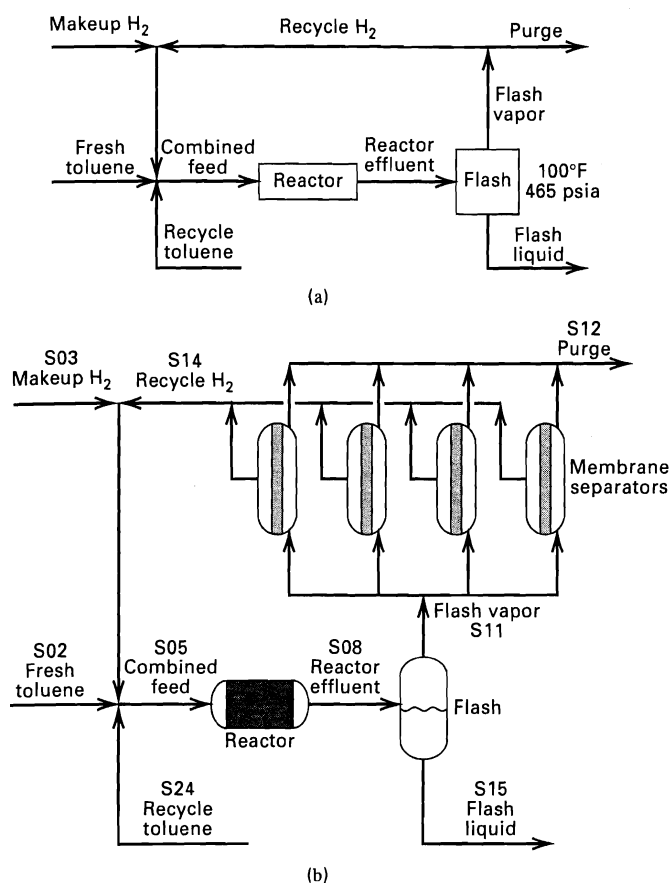


Figure 14.2 Reactor section of process to disproportionate toluene into benzene and xylene isomers. (a) Without a vapor separation step. (b) With a membrane separation step. Note: Heat exchangers, compressors, pump not shown.

liquid. Because of the large amount of hydrogen in the flash-drum vapor, it is important to recycle this stream to the reactor, rather than sending it to a flare or using it as a fuel. However, if all of the vapor were recycled, methane and ethane would build up in the recycle loop, since no other exit is provided. Before the development of acceptable membranes for the separation of H₂ from CH₄ by gas permeation, part of the vapor stream was customarily purged from the process, as shown in Figure 14.2a, to provide an exit for CH₄, and C₂H₆. With the introduction of a suitable

membrane in 1979, it became possible to apply membrane separators, as shown in Figure 14.2b.

Table 14.2 is the steady-state material balance of the reaction section of Figure 14.2b for a plant designed to process 7,750 barrels (42 gal/bbl) per operating day of fresh toluene feed. The gas permeation membrane system separates the flash vapor (stream S11) into an H₂-enriched permeate (S14, the recycled hydrogen), and a methane-enriched retentate (S12, the purge). The flash vapor to the membrane system contains 89.74 mol% H₂ and 9.26 mol% CH₄. No sweep fluid is necessary. The permeate is enriched to 94.46 mol% in H₂. The retentate is enriched in CH₄ to 31.18 mol%. The recovery of H₂ in the permeate is 90%. Thus, only 10% of the H₂ in the vapor leaving the flash drum is lost to the purge. Before entering the membrane separator system, the vapor is heated to a temperature of at least 200°F (the dew-point temperature of the retentate) at a pressure of 450 psia (heater not shown). Because the hydrogen content of the feed is reduced in passing through the membrane separator, the retentate becomes more concentrated in the heavier components. Without the heater, undesirable condensation would occur in the separator. The retentate leaves the separator at about the same temperature and pressure as that of heated flash vapor entering the separator. The permeate leaves at the much-lower pressure of 50 psia and a temperature somewhat lower than 200°F because of gas expansion.

The membrane is an aromatic polyamide polymer consisting of a 0.3-micron-thick, nonporous layer in contact with the feed, and a much-thicker porous support backing to give the membrane strength and ability to withstand the pressure differential of 450 – 50 = 400 psi. This large pressure difference is needed to force the hydrogen through the nonporous membrane, which is in the form of a spiral-wound module made from flat membrane sheets. The average flux of hydrogen through the membrane is 40 scfh (standard ft³/h at 60°F and 1 atm) per ft² of membrane surface area. From the material balance in Table 14.2, the total amount of H₂ transported through the membrane is

$$(1,685.1 \text{ lbmol/h})(379 \text{ scf/lbmol}) = 639,000 \text{ scfh}$$

Table 14.2 Material Balance for Membrane Separation Process in a Toluene Disproportionation Plant; Flow Rates in lbmol/h for Streams in Reactor Section of Figure 14.2b

Component	S02	S03	S24	S14	S05	S08	S15	S11	S12
Hydrogen		269.0		1,685.1	1,954.1	1,890.6	18.3	1,872.3	187.2
Methane		50.5		98.8	149.3	212.8	19.7	193.1	94.3
Ethane		16.8			16.8	16.8	5.4	11.4	11.4
Benzene			13.1		13.1	576.6	571.8	4.8	4.8
Toluene	1,069.4		1,333.0		2,402.4	1,338.9	1,334.7	4.2	4.2
<i>p</i> -Xylene			8.0		8.0	508.0	507.4	0.6	0.6
Total	1,069.4	336.3	1,354.1	1,783.9	4,543.7	4,543.7	2,457.4	2,086.3	302.4

Thus, the required membrane surface area is $639,000/40 = 16,000 \text{ ft}^2$. The membrane is packaged in pressure-vessel modules of $4,000 \text{ ft}^2$ each. Thus, four modules in parallel are used, as shown in Figure 14.2b. A disadvantage of the membrane separator in this application is the need to recompress the recycle hydrogen to the reactor inlet pressure. Unlike distillation, where the energy of separation is usually heat, the energy for gas permeation is the shaft work of gas compression.

Membrane separation is an emerging unit operation. Important progress is still being made in the development of efficient membrane materials and the packaging thereof for the processes listed in Table 14.1. Other novel methods for conducting separation with barriers for a wider variety of mixtures are being researched and developed. Applications covering wider ranges of temperature and types of membrane materials are being found. Already, membrane separation processes have found wide application in such diverse industries as the beverage, chemical, dairy, electronic, environmental, food, medical, paper, petrochemical, petroleum, pharmaceutical, and textile industries. Some of these applications are given in Table 1.2 and included in Table 14.1. Often, compared to other separation equipment, membrane separators are more compact, less capital intensive, and more easily operated, controlled, and maintained. However, membrane separators are usually modular in construction, with many parallel units required for large-scale applications, as contrasted with the more common separation techniques, where larger pieces of equipment are designed as plant size becomes larger.

The key to an efficient and economical membrane separation process is the membrane and the manner in

which it is packaged and modularized. Desirable attributes of a membrane are (1) good permeability, (2) high selectivity, (3) chemical and mechanical compatibility with the processing environment, (4) stability, freedom from fouling, and reasonable useful life, (5) amenability to fabrication and packaging, and (6) ability to withstand large pressure differences across the membrane thickness. Research and development of membrane processes deals mainly with the discovery of suitable membrane materials and their fabrication.

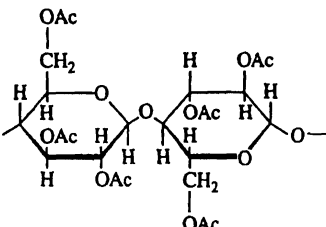
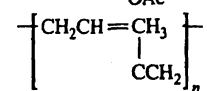
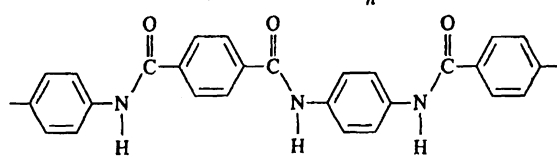
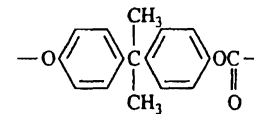
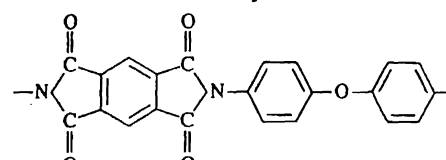
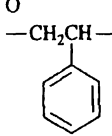
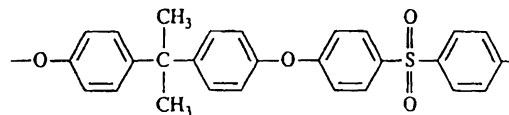
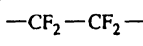
This chapter discusses types of membrane materials, membrane modules, the theory of transport through membrane materials and modules, and the scale-up of membrane separators from experimental performance data. Emphasis is on dialysis, electrodialysis, reverse osmosis, gas permeation, pervaporation, and ultrafiltration, but many of the theoretical principles apply as well to emerging, but not-yet commercialized, membrane processes such as membrane distillation, membrane gas absorption, membrane stripping, membrane solvent extraction, perstraction, and facilitated transport, which are not covered here. The status of industrial membrane separation systems and directions in research to improve existing applications and make possible new applications are considered in detail by Baker et al. [5] in a study supported by the U.S. Department of Energy (DOE) and by a host of contributors in a recent handbook edited by Ho and Sirkar [6], which includes emerging membrane processes. The book, "Membrane Technology and Applications" by Baker [49], is a comprehensive treatment of theory and technology.

14.1 MEMBRANE MATERIALS

Almost all industrial membrane materials are made from natural or synthetic polymers (macromolecules). Natural polymers include wool, rubber, and cellulose. A wide variety of synthetic polymers has been developed and commercialized since 1930. Synthetic polymers are produced by polymerization of a monomer by condensation (step reactions) or addition (chain reactions), or by the copolymerization of two different monomers. The resulting polymer is categorized as having (1) a long linear chain, such as linear polyethylene; (2) a branched chain, such as polybutadiene; (3) a three-dimensional, highly cross-linked structure, such as phenol-formaldehyde; or (4) a moderately cross-linked structure, such as butyl rubber. The linear-chain polymers soften with an increase in temperature, are often soluble in organic solvents, and are referred to as *thermoplastic* polymers. At the other extreme, highly cross-linked polymers do not soften appreciably, are almost insoluble in most organic solvents, and are referred to as *thermosetting* polymers. Of more

interest in the application of polymers to membranes is a classification based on the arrangement or conformation of the polymer molecules. At low temperatures, typically below 100°C , idealized polymers can be classified as *glassy* or *crystalline*. The former refers to a polymer that is brittle and glassy in appearance and lacks any crystalline structure (i.e., *amorphous*), whereas the latter refers to a polymer that is brittle, hard, and stiff, with a crystalline structure. If the temperature of a glassy polymer is increased, a point, called the *glass-transition temperature*, T_g , may be reached where the polymer becomes *rubbery*. If the temperature of a crystalline polymer is increased, a point, called the *melting temperature*, T_m , is reached where the polymer becomes a melt. However, a thermosetting polymer never melts. Many polymers have both amorphous and crystalline regions, that is, a certain degree of crystallinity that varies from 5 to 90%, making it possible for some polymers to have both a T_g and a T_m . Membranes made of glassy polymers can operate below or above T_g ; membranes of crystalline polymers must

Table 14.3 Common Polymers Used in Membranes

Polymer	Type	Representative Repeat Unit	Glass Transition Temp., °C	Melting Temp., °C
Cellulose triacetate	Crystalline			300
Polyisoprene (natural rubber)	Rubbery		-70	
Aromatic polyamide	Crystalline			275
Polycarbonate	Glassy		150	
Polyimide	Glassy		310-365	
Polystyrene	Glassy		74-110	
Polysulfone	Glassy		190	
Polytetrafluoroethylene (Teflon)	Crystalline			327

operate below T_m . Table 14.3 lists *repeat units* and values of T_g and/or T_m for several natural and synthetic polymers, from which membranes have been fabricated. Included are crystalline, glassy, and rubbery polymers. Cellulose triacetate is the reaction product of cellulose and acetic anhydride. Cellulose is the most readily available organic raw material in the world. The repeat unit of cellulose is identical to that shown for cellulose triacetate in Table 14.3, except that the acetyl, Ac (CH_3CO) groups are replaced by H. Typically, the number of repeat units (*degree of polymerization*) in cellulose is 1,000 to 1,500, whereas that in cellulose triacetate is about 300. Partially acetylated products are cellulose acetate and cellulose diacetate, with blends of two or three of the acetates being common. The triacetate is highly crystalline, of uniformly high quality, and hydrophobic.

Polyisoprene (natural rubber) is obtained from at least 200 different plants, with many of the rubber-producing

countries being located in the Far East. Compared to the other polymers in Table 14.3, polyisoprene has a very low glass-transition temperature. Natural rubber has a degree of polymerization of from about 3,000 to 40,000 and is hard and rigid when cold, but soft, easily deformed, and sticky when hot. Depending on the temperature, it slowly crystallizes. To increase the strength, elasticity, and stability of rubber, it is vulcanized with sulfur, a process that introduces cross-links, but still allows unrestricted local motion of the polymer chain.

Aromatic polyamides (also called aramids) are high-melting crystalline polymers that have better long-term thermal stability and higher resistance to solvents than do aliphatic polyamides, such as nylon. Some aromatic polyamides are easily fabricated into fibers, films, and sheets. The polyamide structure shown in Table 14.3 is that of Kevlar, a trade name of DuPont.

Polycarbonates, which are characterized by the presence of the $-\text{OCOO}-$ group in the chain, are mainly amorphous in structure. The polycarbonate shown in Table 14.3 is an aromatic form, but aliphatic forms also exist. Polycarbonates differ from most other amorphous polymers in that they possess ductility and toughness below T_g . Because polycarbonates are thermoplastic, they can be extruded into various shapes, including films and sheets.

Polyimides are characterized by the presence of aromatic rings and heterocyclic rings containing nitrogen and attached oxygen. The structure shown in Table 14.3 is only one of a number available. Polyimides are tough, amorphous polymers with high resistance to heat and excellent wear resistance. They can be fabricated into a wide variety of forms, including fibers, sheets, and films.

Polystyrene is a linear, amorphous, highly pure polymer of about 1,000 units of the structure shown in Table 14.3. Above a relatively low T_g , which depends on molecular weight, polystyrene becomes a viscous liquid that is easily fabricated by extrusion or injection molding. Like many other polymers, polystyrene can be annealed (heated and then cooled slowly) to convert it to a crystalline polymer with a melting point of 240°C . Styrene monomer can be copolymerized with a number of other organic monomers, including acrylonitrile and butadiene to form ABS copolymers.

Polysulfones are relatively new synthetic polymers, first introduced in 1966. The structure in Table 14.3 is just one of many, all of which contain the SO_2 group, which gives the polymers high strength. Polysulfones are easily spun into hollow fibers.

Polytetrafluoroethylene is a straight-chain, highly crystalline polymer with a very high degree of polymerization of the order of 100,000, which gives it considerable strength. It possesses exceptional thermal stability and can be formed into sheets, films, and tubing.

To be effective for separating a mixture of chemical components, a polymer membrane must possess high *permeance* and a high permeance ratio for the two species being separated by the membrane. The permeance for a given species diffusing through a membrane of given thickness is analogous to a mass-transfer coefficient, i.e., the flow rate of that

species per unit cross-sectional area of membrane per unit driving force (concentration, partial pressure, etc.) across the membrane thickness. The molar transmembrane flux of species i is

$$N_i = \left(\frac{P_{M_i}}{l_M} \right) (\text{driving force}) = \bar{P}_{M_i} (\text{driving force}) \quad (14-1)$$

where $\bar{P}_{M_i}$ is the permeance, which is defined as the ratio of P_{M_i} , the *permeability*, to l_M , the membrane thickness.

Polymer membranes can be characterized as dense or microporous. For dense amorphous membranes, pores of microscopic dimensions may be present, but they are generally less than a few Angstroms in diameter, such that most, if not all, diffusing species must dissolve into the polymer and then diffuse through the polymer between the segments of the macromolecular chains. Diffusion can be difficult, but highly selective for glassy polymers. If the polymer is partly crystalline, diffusion will occur almost exclusively through the amorphous regions, with the crystalline regions decreasing the diffusion area and increasing the diffusion path.

A microporous membrane contains interconnected pores that are small (on the order of $0.001\text{--}10\text{ }\mu\text{m}$; $10\text{--}100,000\text{ }\text{\AA}$), but large in comparison to the size of the molecules to be transferred. The pores are formed by a variety of proprietary techniques, some of which are described by Baker et al. [5]. Such techniques are especially valuable for producing symmetric, microporous, crystalline membranes. Permeability for microporous membranes is high, but selectivity is low, for small molecules. However, when molecules both smaller and larger than the pore size are in the feed to the membrane, the molecules may be separated almost perfectly by size.

Thus, for the separation of small molecules, we seem to be presented with a dilemma. We can have high permeability or a high separation factor, but not both. The beginning of the resolution of this dilemma occurred in 1963 with the fabrication by Loeb and Sourirajan [7] of an asymmetric membrane of cellulose acetate by a novel casting procedure. As shown in Figure 14.3a, the resulting membrane consists of a thin dense skin about $0.1\text{--}1.0\text{ }\mu\text{m}$ in. thick, called the *permselective* layer, formed over a much thicker microporous

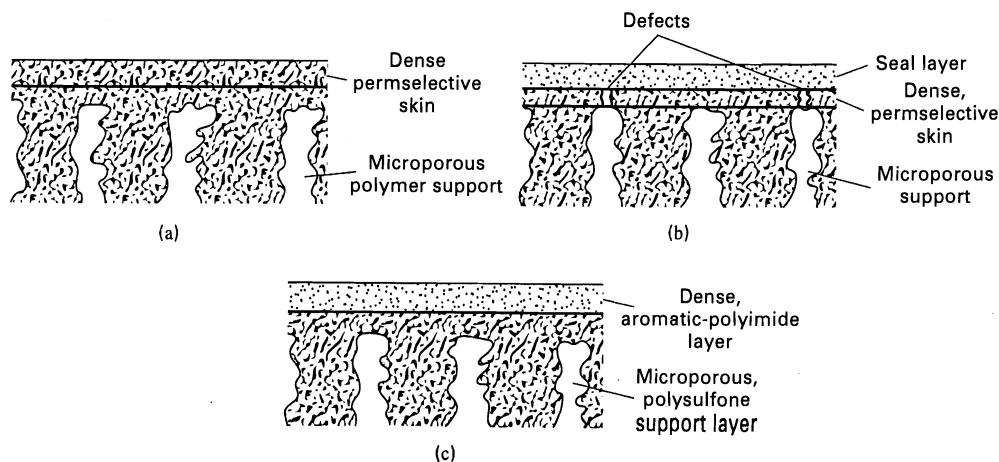


Figure 14.3 Polymer membranes: (a) asymmetric, (b) caulked asymmetric, and (c) typical thin-film composite.

layer that provides support for the skin. The flux rate of a species is controlled by the permeance of the very thin permselective skin. From (14-1), the permeance of species i can be high because of the very small value of l_M even though the permeability, P_{M_i} , is low because of the absence of pores. When large differences of P_{M_i} exist among molecules, both high permeance and high selectivity can be achieved with asymmetric membranes.

A very thin, asymmetric membrane is subject to minute defects or pinholes in the permselective skin, which can render the membrane useless for the separation of a gas mixture. A practical solution to the defect problem for an asymmetric polysulfone membrane was patented by Henis and Tripodi [8] of the Monsanto Company in 1980. They pulled silicone rubber, from a coating on the surface of the skin, into the defects by applying a vacuum. The resulting membrane is sometimes referred to as a *caulked membrane*, as shown in Figure 14.3b.

Another patent by Wrasidlo [9] in 1977, introduced the thin-film composite membrane as an alternative to the asymmetric membrane. In the first application, as shown in Figure 14.3c, a very thin, dense film of polyamide polymer, 250 to 500 Å in thickness, was formed on a thicker microporous polysulfone support. Today, both asymmetric and thin-film composite membranes are fabricated from a variety of polymers by a variety of techniques.

The application of polymer membranes is generally limited to temperatures below about 200°C and to the separation of mixtures that are chemically inert. When operation at high temperatures and/or with chemically active mixtures is necessary, membranes made of inorganic materials can be used. These include mainly microporous ceramics, metals, and carbon; and dense metals, such as palladium, that allow the selective diffusion of very small molecules such as hydrogen and helium.

Some examples of inorganic membranes are (1) asymmetric, microporous α -alumina tubes with 40–100 Å pores at the inside surface and 100,000 Å pores at the outside; (2) microporous glass tubes, the pores of which may or may not be filled with other oxides or the polymerization-pyrolysis product of trichloromethylsilane; (3) silica hollow fibers with extremely fine pores of 3–5 Å; (4) porous ceramic, glass, or polymer materials coated with a thin, dense film of palladium metal that is just a few microns thick; (5) sintered metal; (6) pyrolyzed carbon; and (7) zirconia on sintered carbon. Extremely fine pores (<10 Å) are necessary to separate gas mixtures. Larger pores (>50 Å) may be satisfactory for the separation of large molecules or solid particles from solutions containing small molecules.

EXAMPLE 14.1

A silica-glass membrane of 2 μm thickness and with very fine pores less than 10 Å in diameter has been developed for separating H_2 from CO at a temperature of 500°F. From laboratory data, the membrane permeabilities for hydrogen and carbon monoxide,

respectively, are 200,000 and 700 barrer, where the barrer, a commonly used unit for gas permeation, is defined by:

$$1 \text{ barrer} = 10^{-10} \text{ cm}^3 (\text{STP})\text{-cm}/(\text{cm}^2\text{-s-cmHg})$$

where $\text{cm}^3 (\text{STP})/(\text{cm}^2\text{-s})$ refers to the volumetric transmembrane flux of the diffusing species in terms of standard conditions of 0°C and 1 atm, cm refers to the membrane thickness, and cmHg refers to the transmembrane, partial-pressure driving force for the diffusing species.

The barrer unit is named for R. M. Barrer, who published an early article [10] on the nature of diffusion in a membrane, followed later by a widely referenced monograph on diffusion in and through solids [11].

If the transmembrane, partial-pressure driving forces for H_2 and CO, respectively, are 240 psi and 80 psi, calculate the transmembrane fluxes in $\text{kmol}/(\text{m}^2\text{-s})$. Compare the hydrogen flux to that for hydrogen in the commercial application discussed at the beginning of this chapter.

SOLUTION

At 0°C and 1 atm, 1 kmol of gas occupies $22.42 \times 10^6 \text{ cm}^3$. Also, 2 μm thickness = $2 \times 10^{-4} \text{ cm}$ and 1 cmHg $\Delta P = 0.1934 \text{ psi}$. Therefore, using (14-1):

$$N_{\text{H}_2} = \frac{(200,000)(10^{-10})(240/0.1934)(10^4)}{(22.42 \times 10^6)(2 \times 10^{-4})} = 0.0554 \frac{\text{kmol}}{\text{m}^2\text{-s}}$$

$$N_{\text{CO}} = \frac{(700)(10^{-10})(80/0.1934)(10^4)}{(22.42 \times 10^6)(2 \times 10^{-4})} = 0.000065 \frac{\text{kmol}}{\text{m}^2\text{-s}}$$

In the application discussed at the beginning of this chapter, the flux of H_2 for the polymer membrane is

$$\frac{(1685.1)(1/2.205)}{(16,000)(0.3048)^2(3600)} = 0.000143 \frac{\text{kmol}}{\text{m}^2\text{-s}}$$

Thus, the flux of H_2 through the ultramicroporous-glass membrane is more than 100 times higher than the flux through the dense-polymer membrane. Large differences in molar fluxes through different membranes are common.

The following are useful factors for converting barrers to SI and American Engineering units:

$$1.00 \text{ barrer} = 10^{-10} \frac{\text{cm}^3 (\text{STP}) \cdot \text{cm}}{\text{cm}^2 \cdot \text{s} \cdot \text{cmHg}}$$

Multiply barrers by 3.348×10^{-19} to obtain units of $(\text{kmol} \cdot \text{m})/(\text{m}^2 \cdot \text{s} \cdot \text{Pa})$.

Multiply barrers by 5.584×10^{-12} to obtain units of $(\text{lbmol} \cdot \text{ft})/(\text{ft}^2 \cdot \text{h} \cdot \text{psi})$.

14.2 MEMBRANE MODULES

The asymmetric and thin-film, composite, polymer-membrane materials described in the previous section are available in one or more of the three shapes shown in Figure 14.4a, b, and c. Flat sheets have typical dimensions of 1 m by 1 m by 200 μm thick, with a dense skin or thin, dense layer 500 to 5,000 Å in thickness. Tubular membranes are

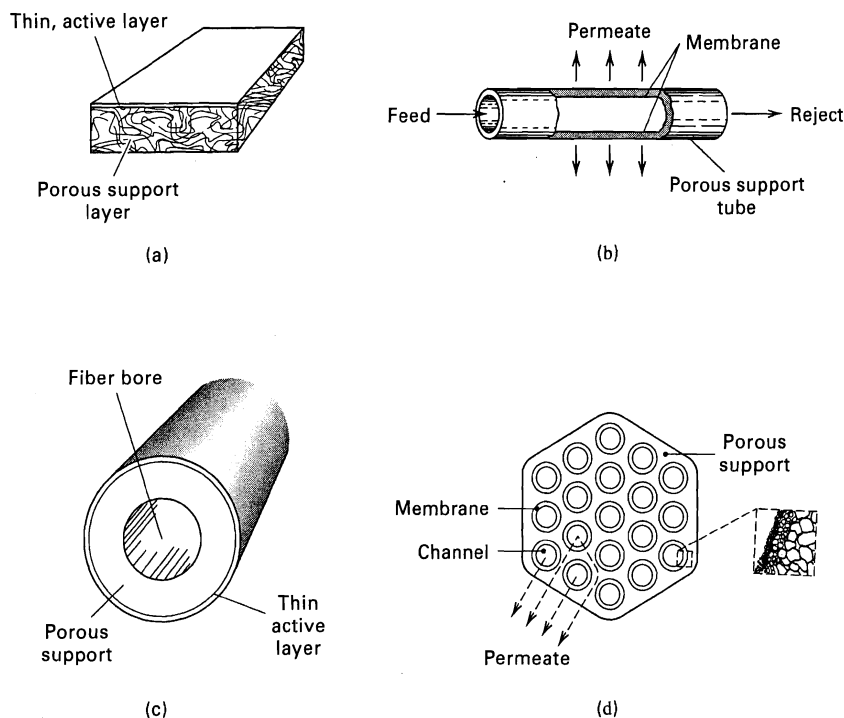


Figure 14.4 Common membrane shapes: (a) flat asymmetric or thin-film composite sheet; (b) tubular; (c) hollow fiber; (d) monolithic.

typically 0.5 to 5.0 cm in diameter and up to 6 m in length. The thin, dense layer is on either the inside, as shown in Figure 14.4b, or the outside surface of the tube. The porous supporting part of the tube is fiberglass, perforated metal, or other suitable porous material. Very small-diameter hollow fibers, first reported by Mahon [12, 13] in the 1960s, are typically 42 μm i.d. by 85 μm o.d. by 1.2 m long with a 0.1- to 1.0- μm -thick dense skin. Hollow fibers, shown in Figure 14.4c, provide a large membrane surface area per unit volume. A honeycomb, monolithic element for inorganic oxide membranes is included in Figure 14.4d. Elements of both hexagonal and circular cross-section are available [14]. The circular flow channels are typically 0.3 to 0.6 cm in diameter, with a 20- to 40-mm-thick membrane layer. The hexagonal element in Figure 14.4d has 19 channels and is 0.85 m long. Both the bulk support and the thin membrane layer are porous, but the pores of the latter can be very small, down to 40 Å. Still in the research stage are membranes based on nanotechnology.

The membrane shapes of Figure 14.4 are incorporated into compact, commercial modules and cartridges, some of which are shown in Figure 14.5. Flat sheets used in plate-and-frame modules are circular, square, or rectangular in cross-section. The sheets are separated by support plates that channel the permeate. In Figure 14.5a, a feed of brackish water flows across the surface of each membrane sheet in the stack. Pure water is the permeate product, whereas the retentate is a concentrated-brine solution.

Flat sheets are also fabricated into spiral-wound modules shown in Figure 14.5b. A laminate, consisting of two membrane sheets separated by spacers for the flow of the feed and permeate, is wound around a central, perforated, collection tube to form a module that is inserted into a pressure vessel.

The feed flows axially in the channels created between the membranes by the porous spacers. Permeate passes through the membrane, traveling inward in a spiral path to the central collection tube. From there, the permeate flows in either axial direction through and out of the central tube. A typical spiral-wound module is 0.1 to 0.3 m in diameter and 3 m long. Six such modules are often placed in series. The four-leaf modification in Figure 14.5c minimizes the pressure drop of the permeate because the permeate travel is less for the same membrane area.

The hollow-fiber module shown in Figure 14.5d, for a gas-permeation application, resembles a shell-and-tube heat exchanger. The pressurized feed enters the shell side at one end. While flowing over the fibers toward the other end, permeate passes through the fiber walls into the central fiber channels. Typically the fibers are sealed at one end and embedded into a tube sheet with epoxy resin at the other end. A commercial module might be 1 m long and 0.1 to 0.25 m in diameter and contain more than one million hollow fibers.

A tubular module is shown in Figure 14.5e. This module also resembles a shell-and-tube heat exchanger, but the feed flows through the tubes. Permeate passes through the wall of the tubes into the shell side of the module. Tubular modules contain up to 30 tubes.

The monolithic module in Figure 14.5f contains from 1 to 37 monolithic elements in a module housing. The feed flows through the circular channels and permeate passes through the membrane and porous support and into the open region between elements.

Table 14.4 is a comparison of the characteristics of four of the modules shown in Figure 14.5. The packing density is the membrane surface area per unit volume of module, for which the hollow-fiber membrane modules are clearly superior.

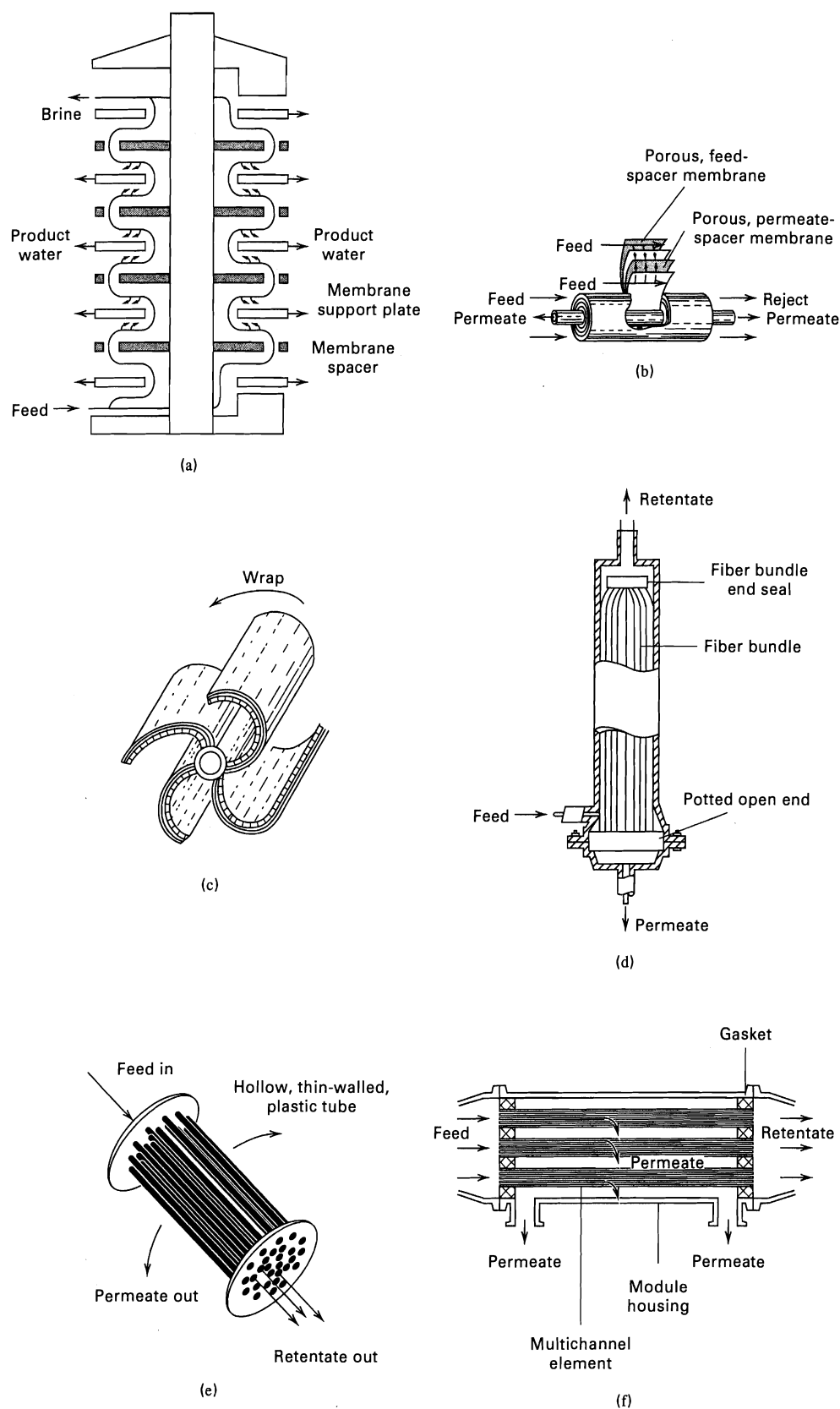


Figure 14.5 Common membrane modules: (a) plate-and-frame, (b) spiral-wound, (c) four-leaf spiral-wound, (d) hollow-fiber, (e) tubular, (f) monolithic.

Although the plate-and-frame module has a high cost and a moderate packing density, it finds use in all membrane applications except gas permeation. It is the only module widely used for pervaporation. The spiral-wound module is very popular for most applications because of its low cost and

reasonable resistance to fouling. Tubular modules are only used for small applications or when a high resistance to fouling and/or ease of cleaning are essential. Hollow-fiber modules, with a very high packing density and low cost, are popular where fouling does not occur and cleaning is not necessary.

Table 14.4 Typical Characteristics of Membrane Modules

	Plate and Frame	Spiral-Wound	Tubular	Hollow-Fiber
Packing density, m ² /m ³	30 to 500	200 to 800	30 to 200	500 to 9,000
Resistance to fouling	Good	Moderate	Very good	Poor
Ease of cleaning	Good	Fair	Excellent	Poor
Relative cost	High	Low	High	Low
Main applications	D, RO, PV, UF, MF	D, RO, GP, UF, MF	RO, UF	D, RO, GP, UF

Note: D, dialysis; RO, reverse osmosis; GP, gas permeation; PV, pervaporation; UF, ultrafiltration; MF, microfiltration.

14.3 TRANSPORT IN MEMBRANES

For a given application, the calculation of membrane surface area is based on laboratory data for the selected membrane. Although permeation can occur by one or more of the mechanisms discussed in this section, these mechanisms are all consistent with (14-1) in either its permeance form or its permeability form, with the latter being applied more widely. However, because both the driving force and the permeability or permeance depend markedly on the mechanism of transport, it is important to understand the nature of transport in membranes, which is the subject of this section. Applications to dialysis, reverse osmosis, gas permeation, pervaporation, ultrafiltration, and microfiltration are presented in subsequent sections.

Membranes can be macroporous, microporous, or dense (nonporous). Only microporous or dense membranes are permselective. However, macroporous membranes are widely used to support thin microporous and dense membranes when significant pressure differences across the membrane are necessary to achieve a reasonable throughput. The theoretical basis for transport through microporous membranes is more highly developed than that for dense membranes, so porous-membrane transport is discussed first, with respect to bulk flow, liquid diffusion, and then gas diffusion. This is followed by nonporous (dense)-membrane transport, including solution diffusion for liquid mixtures and solution diffusion for gas mixtures. External mass-transfer resistances in the fluid films on either side of the membrane are treated where appropriate. It is important to note that, because of the wide range of pore sizes in membranes, the distinction between porous and nonporous membranes is not always obvious. The distinction can be made

based only on the relative permeabilities for diffusion through the pores of the membrane and diffusion through the solid, amorphous regions of the membrane.

Porous Membranes

Mechanisms for the transport of liquid and gas molecules through a porous membrane are depicted in Figure 14.6a, b, and c. If the pore diameter is large compared to the molecular diameter, and a pressure difference exists across the membrane, bulk or convective flow through the pores occurs, as shown in Figure 14.6a. Such a flow is generally undesirable because it is not permselective and, therefore, no separation between components of the feed occurs. If fugacity, activity, chemical potential, concentration, or partial pressure differences exist across the membrane for the various components, but the pressure is the same on both sides of the membrane so as not to cause a bulk flow, permselective diffusion of the components through the pores will take place, effecting a separation as shown in Figure 14.6b. If the pores are of the order of molecular size for at least some of the components in the feed mixture, the diffusion of those components will be restricted (hindered) as shown in Figure 14.6c, resulting in an enhanced separation. Molecules of size larger than the pores will be prevented altogether from diffusing through the pores. This special case is highly desirable and is referred to as *size exclusion* or *sieving*. Another special case exists for gas diffusion where the pore size and/or pressure (typically a vacuum) is such that the mean free path of the molecules is greater than the pore diameter, resulting in so-called *Knudsen diffusion*, which is dependent on molecular weight.

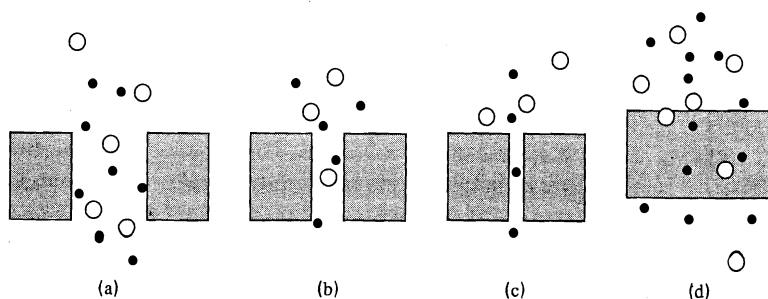


Figure 14.6 Mechanisms of transport in membranes. (Flow is downward.) (a) Bulk flow through pores; (b) diffusion through pores; (c) restricted diffusion through pores; (d) solution-diffusion through dense membranes.

Bulk Flow

Bulk flow is the principle mechanism of transfer through microporous membranes used for ultrafiltration and microfiltration, where separation is achieved mainly by sieving. Consider the bulk flow of a fluid, due to a pressure difference, through an idealized straight, cylindrical pore. If the flow is in the laminar regime ($N_{Re} = Dv\rho/\mu < 2,100$), which is almost always the case for flow in small-diameter pores, the flow velocity, v , is given by the Hagen–Poiseuille law [15] as being directly proportional to the transmembrane pressure drop:

$$v = \frac{D^2}{32\mu L}(P_0 - P_L) \quad (14-2)$$

where D is the pore diameter, large enough to pass all molecules, μ is the viscosity of the fluid, and L is the length of the pore. This law assumes that a parabolic velocity profile exists across the pore radius for the entire length of the pore, that the fluid is Newtonian, and, if a gas, that the mean free path of the molecules is small compared to the pore diameter. If the membrane contains n such pores per unit cross-section of membrane surface area normal to flow, the porosity (void fraction) of the membrane is

$$\epsilon = n\pi D^2/4 \quad (14-3)$$

Then the superficial fluid bulk-flow flux (mass velocity), N , through the membrane is

$$N = v\rho\epsilon = \frac{\epsilon\rho D^2}{32\mu l_M}(P_0 - P_L) = \frac{n\pi\rho D^4}{128\mu l_M}(P_0 - P_L) \quad (14-4)$$

where l_M is the membrane thickness and ρ and μ are fluid properties.

In real porous membranes, pores may not be cylindrical and straight, making it necessary to modify (14-4). One procedure is that due to Carman and Kozeny, as extended by Ergun [16], where the pore diameter in (14-2) is replaced, as a rough approximation, by the hydraulic diameter:

$$\begin{aligned} d_H &= 4 \left(\frac{\text{Volume available for flow}}{\text{Total pore surface area}} \right) \\ &= \frac{4 \left(\frac{\text{Total pore volume}}{\text{Membrane volume}} \right)}{\left(\frac{\text{Total pore surface area}}{\text{Membrane volume}} \right)} = \frac{4\epsilon}{a} \end{aligned} \quad (14-5)$$

where the membrane volume includes the volume of the pores. The specific surface area, a_v , which is the pore surface area per unit volume of just the membrane material (not including the pores), is

$$a_v = a/(1 - \epsilon) \quad (14-6)$$

Pore length is longer than the membrane thickness and can be represented by $l_M\tau$, where τ is a tortuosity factor >1 . Substituting (14-5), (14-6), and the tortuosity factor into

(14-4) gives

$$N = \frac{\rho\epsilon^3(P_0 - P_L)}{2(1 - \epsilon)^2\tau a_v^2\mu l_M} \quad (14-7)$$

In terms of a bulk-flow permeability, (14-7) becomes

$$N = \frac{P_M}{l_M}(P_0 - P_L) \quad (14-8)$$

where

$$P_M = \frac{\rho\epsilon^3}{2(1 - \epsilon)^2\tau a_v^2\mu} \quad (14-9)$$

Typically, τ is approximately 2.5, whereas a_v is inversely proportional to the average pore diameter, giving it values over a wide range.

Equation (14-7) may be compared to the following semi-theoretical Ergun equation [16], which represents the best fit of experimental data for flow of a fluid through a packed bed:

$$\frac{P_0 - P_L}{l_M} = \frac{150\mu v_0(1 - \epsilon)^2}{D_p^2\epsilon^3} + \frac{1.75\rho v_0^2(1 - \epsilon)}{D_p\epsilon^3} \quad (14-10)$$

where D_p is the mean particle diameter, v_0 is the superficial fluid velocity through the membrane, and v_0/ϵ is the actual velocity in the pores. The first term on the right-hand side of (14-10) applies to the laminar flow region and is frequently referred to as Darcy's law. The second term applies to the turbulent region. For a spherical particle, the specific surface area is

$$\begin{aligned} a_v &= \pi D_p^2 / (\pi D_p^3 / 6) \quad \text{or} \\ D_p &= 6/a_v \end{aligned} \quad (14-11)$$

Substitution of (14-11) into (14-10), for just the laminar-flow region, and rearrangement into the bulk-flow flux form gives

$$N = \frac{\rho\epsilon^3(P_0 - P_L)}{(150/36)(1 - \epsilon)^2 a_v^2\mu l_M} \quad (14-12)$$

Comparing (14-12) to (14-7), we see that the term $(150/36)$ in (14-12) corresponds to the term 2τ in (14-7). Thus, τ appears to have a value of 2.08, which seems reasonable. Accordingly, (14-12) can be used as a first approximation to the pressure drop for flow through a porous membrane when the pores are not straight cylinders. For gas flow, the density may be taken as the arithmetic average of the densities at the upstream and downstream faces of the membrane.

EXAMPLE 14.2

It is desired to pass water at 70°F through a supported, polypropylene membrane, with a skin of 0.003 cm thickness and 35% porosity, at the rate of 200 m³/m²-day. The pores can be considered as straight cylinders of uniform diameter equal to 0.2 micron. If the pressure on the downstream side of the membrane is 150 kPa, estimate the required pressure on the upstream side of the membrane. The pressure drop through the support is negligible.

SOLUTION

Equation (14-4) applies, where in SI units:

$$N/p = 200/(24)(3600) = 0.00232 \text{ m}^3/\text{m}^2\text{-s}, \quad \epsilon = 0.35$$

$$D_P = 0.2 \times 10^{-6} \text{ m}, \quad l_M = 0.00003 \text{ m}$$

$$P_L = 150 \text{ kPa} = 150,000 \text{ Pa}, \quad \mu = 0.001 \text{ Pa-s}$$

From (14-4),

$$\begin{aligned} P_0 &= P_L + \frac{32\mu l_M(N/p)}{\epsilon D_P^2} \\ &= 150,000 + \frac{(32)(0.001)(0.00003)(0.00232)}{(0.35)(0.2 \times 10^{-6})^2} \\ &= 309,000 \text{ Pa} \quad \text{or} \quad 309 \text{ kPa} \end{aligned}$$

Liquid Diffusion in Pores

Consider diffusion through the pores of a membrane from a fluid feed to a sweep fluid when identical total pressures but different component concentrations exist on both sides of the membrane. In that case, bulk flow through the membrane due to a pressure difference does not occur and if species diffuse at different rates, a separation can be achieved. If the feed mixture is a liquid of solvent and solutes i , the transmembrane flux for each solute is given by a modified form of Fick's law:

$$N_i = \frac{D_{ei}}{l_M} (c_{i0} - c_{iL}) \quad (14-13)$$

where D_{ei} is the effective diffusivity, and c_i is the concentration of i in the liquid in the pores at the two faces of the membrane. In general the effective diffusivity is given by

$$D_{ei} = \frac{\epsilon D_i}{\tau} K_{ri} \quad (14-14)$$

where D_i is the ordinary molecular diffusion coefficient (diffusivity) of the solute i in the solution, ϵ is the volume fraction of pores in the membrane, τ is the tortuosity, and K_r is a restrictive factor that accounts for the effect of pore diameter, d_p , in causing interfering collisions of the diffusing solutes with the pore wall, when the ratio of molecular diameter, d_m , to pore diameter exceeds about 0.01. The restrictive factor is approximated by Beck and Schultz [17] with:

$$K_r = \left[1 - \frac{d_m}{d_p} \right]^4, \quad (d_m/d_p) \leq 1 \quad (14-15)$$

From (14-15), when $(d_m/d_p) = 0.01$, $K_r = 0.96$, but when $(d_m/d_p) = 0.3$, $K_r = 0.24$. When $d_m > d_p$, $K_r = 0$, and the solute cannot diffuse through the pore. This is the sieving or size-exclusion effect illustrated in Figure 14.6c. In general, as illustrated in the following example, transmembrane fluxes for liquids through microporous membranes are very small because effective diffusivities are very low.

For solute molecules that are not subject to size exclusion, a useful selectivity ratio can be defined as

$$S_{ij} = \frac{D_i K_{ri}}{D_j K_{rj}} \quad (14-16)$$

This ratio is greatly enhanced by the effect of restrictive diffusion when the solutes differ widely in molecular weight and one or more molecular diameters approach the pore diameter. This is shown in the next example.

EXAMPLE 14.3

Beck and Schultz [18] measured effective diffusivities of urea and several different sugars, in aqueous solutions, through microporous membranes of mica, which were especially prepared to give almost straight, elliptical pores of almost uniform size. Based on the following data for a membrane and two solutes, estimate transmembrane fluxes for the two solutes in $\text{g}/\text{cm}^2\text{-s}$ at 25°C . Assume that the aqueous solutions on either side of the membrane are sufficiently dilute that no multicomponent diffusional effects are present.

Membrane:

Material	Microporous mica
Thickness, μm	4.24
Average pore diameter, Angstroms	88.8
Tortuosity, τ	1.1
Porosity, ϵ	0.0233

Solutes (in aqueous solution at 25°C):

Solute	MW	$D_i \times 10^6 \text{ cm}^2/\text{s}$	molecular diameter, $\text{\AA}$	g/cm^3	
				c_{i0}	c_{iL}
1 Urea	60	13.8	5.28	0.0005	0.0001
2 β -Dextrin	1135	3.22	17.96	0.0003	0.00001

SOLUTION

Calculate the restrictive factor and effective diffusivity from (14-15) and (14-14), respectively. For urea (1):

$$K_{r1} = \left[1 - \left(\frac{5.28}{88.8} \right) \right]^4 = 0.783$$

$$D_{e1} = \frac{(0.0233)(13.8 \times 10^{-6})(0.783)}{1.1} = 2.29 \times 10^{-7} \text{ cm}^2/\text{s}$$

For β -dextrin (2):

$$K_{r2} = \left[1 - \left(\frac{17.96}{88.8} \right) \right]^4 = 0.405$$

$$D_{e2} = \frac{(0.0233)(3.22 \times 10^{-6})(0.405)}{1.1} = 2.78 \times 10^{-8} \text{ cm}^2/\text{s}$$

Because of the large differences in molecular size, the two effective diffusivities differ by almost an order of magnitude. From (14-16), the selectivity is

$$S_{1,2} = \frac{(13.8 \times 10^{-6})(0.783)}{(3.22 \times 10^{-6})(0.405)} = 8.3$$

Calculate transmembrane fluxes from (14.13) noting the given concentrations are at the two faces of the membranes. Concentrations in the bulk solutions on either side of the membrane may differ from the concentrations at the faces depending upon the magnitudes of the external mass-transfer resistances in boundary layers or films adjacent to the two faces of the membrane.

For urea:

$$N_1 = \frac{(2.29 \times 10^{-7})(0.0005 - 0.0001)}{4.24 \times 10^{-4}} = 2.16 \times 10^{-7} \text{ g/cm}^2\text{-s}$$

For β -dextrin:

$$N_2 = \frac{(2.768 \times 10^{-8})(0.0003 - 0.00001)}{(4.24 \times 10^{-4})} = 1.90 \times 10^{-8} \text{ g/cm}^2\text{-s}$$

Note that these fluxes are extremely low.

Gas Diffusion

When the mixture on either side of a microporous membrane is a gas, the rate of species diffusion can again be expressed in terms of Fick's law. If pressure and temperature on either side of the membrane are equal and the ideal-gas law holds, (14-13) can be written in terms of a partial-pressure driving force:

$$N_i = \frac{D_{ei} c_M}{Pl_M} (p_{i0} - p_{iL}) = \frac{D_{ei}}{RTl_M} (p_{i0} - p_{iL}) \quad (14-17)$$

where c_M is the total concentration of the gas mixture given as P/RT by the ideal-gas law.

For a gas, diffusion through a pore may occur by ordinary diffusion, as with a liquid, and/or in series with Knudsen diffusion when pore diameter is very small and/or total pressure is low. In the Knudsen-flow regime, collisions occur primarily between gas molecules and the pore wall, rather than between gas molecules. Thus, in the absence of a bulk-flow effect or restrictive diffusion, (14-14) is modified for gas flow:

$$D_{ei} = \frac{\epsilon}{\tau} \left[\frac{1}{(1/D_i) + (1/D_{K_i})} \right] \quad (14-18)$$

where D_{K_i} is the Knudsen diffusivity, which from the kinetic theory of gases applied to a straight, cylindrical pore of diameter d_p is given by

$$D_{K_i} = \frac{d_p \bar{v}_i}{3} \quad (14-19)$$

where $\bar{v}_i$ is the average molecule velocity given by:

$$\bar{v}_i = (8RT/\pi M_i)^{1/2} \quad (14-20)$$

where M is the molecular weight. Combining (14-19) and (14-20):

$$D_{K_i} = 4,850 d_p (T/M_i)^{1/2} \quad (14-21)$$

where D_K is cm^2/s , d_p is cm , and T is K . When Knudsen flow predominates, as it often does for the micropores in membranes, a selectivity based on the permeability ratio for species A and B is given from a combination of (14-1), (14-17), (14-18), and (14-21):

$$\frac{P_{M_A}}{P_{M_B}} = \left(\frac{M_B}{M_A} \right)^{1/2} \quad (14-22)$$

Except for gaseous species of widely differing molecular weight, the permeability ratio from (14-22) is not large, and the separation of gases by microporous membranes at low to

moderate pressures that are equal on both sides of the membrane to minimize bulk flow is almost always impractical, as illustrated in the following example. However, it is important to note that the separation of the two isotopes of UF_6 by the United States government was accomplished by Knudsen diffusion, with a permeability ratio of only 1.0043, on a large scale at Oak Ridge, Tennessee, using thousands of stages and many acres of membrane surface.

EXAMPLE 14.4

A gas mixture of hydrogen (H) and ethane (E) is to be partially separated with a composite membrane having a 1- μm -thick porous skin with an average pore size of 20 Å and a porosity of 30%. The tortuosity can be assumed to be 1.5. The pressure on either side of the membrane is 10 atm and the temperature is 100°C. Estimate the permeabilities of the two components in Barrers.

SOLUTION

From (14-1), (14-17), and (14-18), the permeability can be expressed in $\text{gmol}\cdot\text{cm}/\text{cm}^2\cdot\text{s}\cdot\text{atm}$:

$$P_{M_i} = \frac{\epsilon}{RT\tau} \left[\frac{1}{(1/D_i) + (1/D_{K_i})} \right]$$

where $\epsilon = 0.30$, $R = 82.06 \text{ cm}^3\cdot\text{atm}/\text{mol}\cdot\text{K}$, $T = 373 \text{ K}$, and $\tau = 1.5$.

At 100°C, the ordinary diffusivity is given by $D_H = D_E = D_{H,E} = 0.86/P$ in cm^2/s with total pressure P in atm. Thus, at 10 atm, $D_H = D_E = 0.086 \text{ cm}^2/\text{s}$. Knudsen diffusivities are given by (14-21) with pore diameter, d_p , equal to $20 \times 10^{-8} \text{ cm}$.

$$D_{K_H} = 4,850(20 \times 10^{-8})(373/2.016)^{1/2} = 0.0132 \text{ cm}^2/\text{s}$$

$$D_{K_E} = 4,850(20 \times 10^{-8})(373/30.07)^{1/2} = 0.00342 \text{ cm}^2/\text{s}$$

For both components, diffusion is controlled mainly by Knudsen diffusion.

$$\text{For hydrogen: } \frac{1}{(1/D_H) + (1/D_{K_H})} = 0.0114 \text{ cm}^2/\text{s}.$$

$$\text{For ethane: } \frac{1}{(1/D_E) + (1/D_{K_E})} = 0.00329 \text{ cm}^2/\text{s}.$$

$$P_{M_H} = \frac{0.30(0.0114)}{(82.06)(373)(1.5)} = 7.45 \times 10^{-8} \frac{\text{mol}\cdot\text{cm}}{\text{cm}^2\cdot\text{s}\cdot\text{atm}}$$

$$P_{M_E} = \frac{0.30(0.00329)}{(82.06)(373)(1.5)} = 2.15 \times 10^{-8} \frac{\text{mol}\cdot\text{cm}}{\text{cm}^2\cdot\text{s}\cdot\text{atm}}$$

To convert to Barrer as defined in Example 14.1, note that

$$76 \text{ cmHg} = 1 \text{ atm and } 22,400 \text{ cm}^3(\text{STP}) = 1 \text{ mol}$$

$$P_{M_H} = \frac{7.45 \times 10^{-8}(22,400)}{(10^{-10})(76)} = 220,000 \text{ Barrer}$$

$$P_{M_E} = \frac{2.15 \times 10^{-8}(22,400)}{(10^{-10})(76)} = 63,400 \text{ Barrer}$$

Nonporous Membranes

The transport of components through nonporous (dense) solid membranes is the predominant mechanism of membrane separators for reverse osmosis (liquid), gas permeation (gas), and pervaporation (liquid and vapor). As indicated in

Figure 14.6d, gas or liquid components absorb into the membrane at the upstream face, diffuse through the solid membrane, and desorb at the downstream face.

Liquid diffusivities are several orders of magnitude less than gas diffusivities, and diffusivities of solutes in solids are a few orders of magnitude less than diffusivities in liquids. Thus, differences between diffusivities in gases and solids are enormous. For example, at 1 atm and 25°C, diffusivities in cm²/s for water are as follows:

Water vapor in air	0.25
Water in ethanol liquid	1.2×10^{-5}
Dissolved water in cellulose acetate solid	1×10^{-8}

As might be expected, small molecules fare better than large molecules for diffusivities in solids. For example, from the *Polymer Handbook* [19], diffusivities in cm²/s for several components in low-density polyethylene at 25°C are

Helium	6.8×10^{-6}
Hydrogen	0.474×10^{-6}
Nitrogen	0.320×10^{-6}
Propane	0.0322×10^{-6}

Regardless of whether a nonporous membrane is used to separate a gas or liquid mixture, the *solution-diffusion model* of Lonsdale, Merten, and Riley [20] is most often applied to analyze experimental permeability data and design membrane separators. This model is based on Fick's law for diffusion through solid, nonporous membranes based on the driving force, $c_{i0} - c_{iL}$ shown in Figure 14.7b, where the concentrations are those for the solute dissolved in the

membrane. The concentrations in the membrane are related to the concentrations or partial pressures in the fluid adjacent to the membrane faces by assuming thermodynamic equilibrium for the solute between the fluid and membrane material at the fluid-membrane interfaces. This assumption has been validated experimentally by Motanedian et al. [21] for the case of permeation of light gases through dense cellulose acetate membranes at up to 90 atm.

Solution-Diffusion for Liquid Mixtures

Figures 14.7a and b show typical solute-concentration profiles for liquid mixtures with porous and nonporous membranes, respectively. Included in these diagrams is the drop in concentration across the membrane and, also, possible drops due to resistances in the fluid boundary layers or films on either side of the membrane. For porous membranes, of the type considered in the previous section, the concentration profile is continuous from the bulk-feed liquid to the bulk-permeate liquid because liquid is present continuously from one side to the other. The concentration c_{i0} is the same in the liquid feed just adjacent to the membrane surface and in the liquid just within the entrance of the pore. This is not the case for the nonporous membrane in Figure 14.7b. Solute concentration c'_{i0} is that in the feed liquid just adjacent to the upstream membrane surface, whereas c_{i0} is that in the membrane just adjacent to the upstream membrane surface. In general, c_{i0} is considerably smaller than c'_{i0} , but the two are related by a thermodynamic equilibrium partition coefficient K_i , defined by

$$K_{i0} = c_{i0}/c'_{i0} \quad (14-23)$$

Similarly, at the other face:

$$K_{iL} = c_{iL}/c'_{iL} \quad (14-24)$$

Fick's law applied to the nonporous membrane of Figure 14.7b is:

$$N_i = \frac{D_i}{l_M} (c_{i0} - c_{iL}) \quad (14-25)$$

where D_i is the diffusivity of the solute in the membrane material. If (14-23) and (14-24) are combined with (14-25), and the partition coefficient is assumed to be independent of concentration, such that $K_{i0} = K_{iL} = K_i$, we obtain for the flux

$$N_i = \frac{K_i D_i}{l_M} (c'_{i0} - c'_{iL}) \quad (14-26)$$

If the mass-transfer resistances in the two fluid boundary layers or films are negligible:

$$N_i = \frac{K_i D_i}{l_M} (c_{iF} - c_{iP}) \quad (14-27)$$

In (14-26) and (14-27), $K_i D_i$ is the permeability, P_{Mi} , for the solution-diffusion model, where K_i accounts for the solubility of the solute in the membrane and D_i accounts for diffusion through the membrane. Because D_i is generally

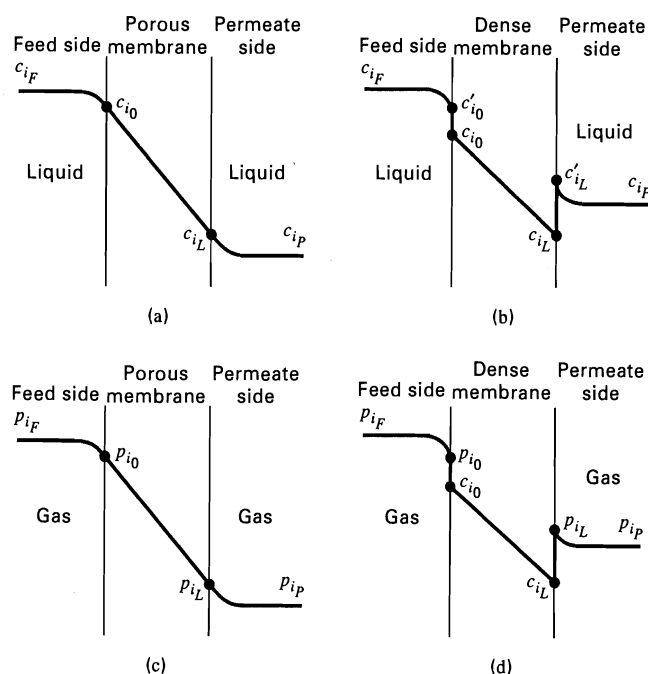


Figure 14.7 Concentration and partial pressure profiles for solute transport through membranes. Liquid mixture with (a) a porous and (b) a nonporous membrane; gas mixture with (c) a porous and (d) a nonporous membrane.

Table 14.5 Factors That Influence Permeability of Solutes in Dense Polymers

Factor	Value Favoring High Permeability
Polymer density	low
Degree of crystallinity	low
Degree of cross-linking	low
Degree of vulcanization	low
Amount of plasticizers	high
Amount of fillers	low
Chemical affinity of solute for polymer (solubility)	high

very small, it is important that the membrane material offers a large value for K_i and/or a small membrane thickness.

Both D_i and K_i , and therefore P_{M_i} , depend on the solute and the membrane. When solutes dissolve in a polymer membrane, it will swell, causing both D_i and K_i to increase. Other polymer membrane factors that can influence D_i , K_i , and P_{M_i} are listed in Table 14.5. However, the largest single factor is the chemical structure of the polymer. Because of the many factors involved, it is important to obtain experimental permeability data on the membrane and feed mixture of interest. The effect of external mass-transfer resistances is considered in a subsection near the end of this section.

Solution-Diffusion for Gas Mixtures

Figures 14.7c and d show typical solute profiles for gas mixtures with porous and nonporous membranes, respectively, including the effect of external-fluid boundary layer or film mass-transfer resistances. For the porous membrane, a continuous partial-pressure profile is shown. For the nonporous membrane, a concentration profile is shown within the membrane where the solute is dissolved in the membrane. Fick's law, given by (14-25), holds for transport through the membrane. Assuming that thermodynamic equilibrium exists at the two fluid-membrane interfaces, the concentrations in Fick's law can be related to the partial pressures adjacent to the membrane faces by Henry's law, which is a linear relation that is most conveniently written for membrane applications as

$$H_{i_0} = c_{i_0}/p_{i_0} \quad (14-28)$$

and

$$H_{i_L} = c_{i_L}/p_{i_L} \quad (14-29)$$

If we assume that H_i is independent of total pressure and that the temperature is the same at both membrane faces:

$$H_{i_0} = H_{i_L} = H_i \quad (14-30)$$

Combining (14-25), (14-28), (14-29), and (14-30), the flux is

$$N_i = \frac{H_i D_i}{l_M} (p_{i_0} - p_{i_L}) \quad (14-31)$$

If the external mass-transfer resistances are neglected, $p_{i_F} = p_{i_0}$ and $p_{i_L} = p_{i_P}$, giving

$$N_i = \frac{H_i D_i}{l_M} (p_{i_F} - p_{i_P}) = \frac{P_{M_i}}{l_M} (p_{i_F} - p_{i_P}) \quad (14-32)$$

where

$$P_{M_i} = H_i D_i \quad (14-33)$$

Thus, the permeability depends on both the solubility of the gas component in the membrane and the diffusivity of that component in the membrane material. An acceptable rate of transport through the membrane can be achieved only by using a very thin membrane and a high pressure on the feed side. The permeability of a gaseous component in a polymer membrane is subject to the factors listed in Table 14.5. Light gases do not interact with the polymer or cause it to swell. Thus, a light-gas-permeant-polymer combination is readily characterized experimentally. Often both solubility and diffusivity are measured. An extensive tabulation is given in the *Polymer Handbook* [19]. Representative data at 25°C are given in Table 14.6. In general, diffusivity decreases and solubility increases with increasing molecular weight of the gas species, making it difficult to achieve a high selectivity. The effect of temperature over a modest range of about 50°C can be represented for both solubility and diffusivity by Arrhenius equations. For example,

$$D = D_0 e^{-E_D/RT} \quad (14-34)$$

In general, the modest effect of temperature on solubility may act in either direction. However, an increase in temperature can cause a substantial increase in diffusivity and, therefore, a corresponding increase in permeability. Typical activation energies of diffusion in polymers, E_D , range from 15 to 60 kJ/mol.

The application of Henry's law for rubbery polymers is well accepted, particularly for low-molecular-weight penetrants, but is less accurate for glassy polymers, for which alternative theories have been proposed. Foremost is the dual-mode model first proposed by Barrer and co-workers [22–24] as the result of a comprehensive study of sorption and diffusion in ethyl cellulose. In this model, sorption of the penetrant occurs by ordinary dissolution in the polymer chains, as described by Henry's law, and by Langmuir sorption into holes or sites between chains of glassy polymers. When the downstream pressure is negligible compared to the upstream pressure, the permeability for Fick's law is given by

$$P_{M_i} = H_i D_i + \frac{D_{L_i} a b}{1 + b P} \quad (14-35)$$

where the second term refers to Langmuir sorption with D_{L_i} = diffusivity of Langmuir sorbed species, P = penetrant pressure, and a, b = Langmuir constants for sorption site capacity and site affinity, respectively.

Koros and Paul [25] found that the dual-mode theory accurately represents data for the sorption of CO₂ in

Table 14.6 Coefficients for Gas Permeation in Polymers

	Gas Species					
	H ₂	O ₂	N ₂	CO	CO ₂	CH ₄
Low-Density Polyethylene:						
$D \times 10^6$	0.474	0.46	0.32	0.332	0.372	0.193
$H \times 10^6$	1.58	0.472	0.228	0.336	2.54	1.13
$P_M \times 10^{13}$	7.4	2.2	0.73	1.1	9.5	2.2
Polyethylmethacrylate:						
$D \times 10^6$	—	0.106	0.0301	—	0.0336	—
$H \times 10^6$	—	0.839	0.565	—	11.3	—
$P_M \times 10^{13}$	—	0.889	0.170	—	3.79	—
Polyvinylchloride:						
$D \times 10^6$	0.5	0.012	0.0038	—	0.0025	0.0013
$H \times 10^6$	0.26	0.29	0.23	—	4.7	1.7
$P_M \times 10^{13}$	1.3	0.034	0.0089	—	0.12	0.021
Butyl Rubber:						
$D \times 10^6$	1.52	0.081	0.045	—	0.0578	—
$H \times 10^6$	0.355	1.20	0.543	—	6.71	—
$P_M \times 10^{13}$	5.43	0.977	0.243	—	3.89	—

Note: Units: D in cm²/s; H in cm³ (STP)/cm³-Pa; P_M in cm³ (STP)-cm/cm²-s-Pa.

polyethylene terephthalate below its glass-transition temperature of about 85°C. Above that temperature, the rubbery polymer obeys just Henry's law. Mechanisms of diffusion for the Langmuir mode have been suggested by Barrer [26].

The ideal, dense-polymer membrane has a high permeance, P_M/l_M , for the penetrant molecules and a high separation factor (selectivity) between the components to be separated. The separation factor is defined similarly to relative volatility in distillation:

$$\alpha_{A,B} = \frac{(y_A/x_A)}{(y_B/x_B)} \quad (14-36)$$

where y_i is the mole fraction in the permeate leaving the membrane, corresponding to the partial pressure p_{i_P} in Figure 14.7d, while x_i is the mole fraction in the retentate on the feed side of the membrane, corresponding to the partial pressure p_{i_F} in Figure 14.7d. Unlike the case of distillation, y_i and x_i are not in equilibrium.

For the separation of a binary gas mixture of species A and B in the absence of external boundary layer or film mass-transfer resistances, the transport fluxes are given by (14-32):

$$N_A = \frac{H_A D_A}{l_M} (p_{A_F} - p_{A_P}) = \frac{H_A D_A}{l_M} (x_A P_F - y_A P_P) \quad (14-37)$$

$$N_B = \frac{H_B D_B}{l_M} (p_{B_F} - p_{B_P}) = \frac{H_B D_B}{l_M} (x_B P_F - y_B P_P) \quad (14-38)$$

When no sweep gas is used, the ratio of N_A to N_B fixes the composition of the permeate so that it is simply the ratio of

y_A to y_B in the permeate gas. Thus,

$$\frac{N_A}{N_B} = \frac{y_A}{y_B} = \frac{H_A D_A (x_A P_F - y_A P_P)}{H_B D_B (x_B P_F - y_B P_P)} \quad (14-39)$$

If the downstream (permeate) pressure, P_P , is negligible compared to the upstream pressure, P_F , such that $y_A P_P \ll x_A P_F$ and $y_B P_P \ll x_B P_F$, (14-39) can be rearranged and combined with (14-36) to give an *ideal separation factor*:

$$\alpha_{A,B}^* = \frac{H_A D_A}{H_B D_B} = \frac{P_{M_A}}{P_{M_B}} \quad (14-40)$$

Thus, a high separation factor can be achieved from a high solubility ratio, a high diffusivity ratio, or both. The separation factor depends on both transport phenomena and thermodynamic equilibria.

When the downstream pressure is not negligible, (14-39) can be rearranged to obtain an expression for $\alpha_{A,B}$ in terms of the pressure ratio, $r = P_P/P_F$, and the mole fraction of A on the feed or retentate side of the membrane. Combining (14-36), (14-40), and the definition of r with (14-39):

$$\alpha_{A,B} = \alpha_{A,B}^* \left[\frac{(x_B/y_B) - r\alpha_{A,B}}{(x_B/y_B) - r} \right] \quad (14-41)$$

Because $y_A + y_B = 1$, we can substitute into (14-41) for x_B , the identity:

$$x_B = x_B y_A + x_B y_B$$

to give

$$\alpha_{A,B} = \alpha_{A,B}^* \left[\frac{x_B \left(\frac{y_A}{y_B} + 1 \right) - r\alpha_{A,B}}{x_B \left(\frac{y_A}{y_B} + 1 \right) - r} \right] \quad (14-42)$$

Table 14.7 Ideal Membrane Separation Factors of Binary Pairs for Two Membrane Materials

	PDMS, Silicon Rubbery Polymer Membrane	PC, Polycarbonate Glassy Polymer Membrane
$P_{M_{He}}$, Barrer	561	14
α_{He,CH_4}^*	0.41	50
$\alpha_{He,C_2H_4}^*$	0.15	33.7
$P_{M_{CO_2}}$, Barrer	4,550	6.5
α_{CO_2,CH_4}^*	3.37	23.2
$\alpha_{CO_2,C_2H_4}^*$	1.19	14.6
$P_{M_{O_2}}$, Barrer	933	1.48
α_{O_2,N_2}^*	2.12	5.12

If we combine (14-36) and (14-42) and replace x_B with $1 - x_A$, we obtain for the separation factor:

$$\alpha_{A,B} = \alpha_{A,B}^* \left[\frac{x_A(\alpha_{A,B} - 1) + 1 - r\alpha_{A,B}}{x_A(\alpha_{A,B} - 1) + 1 - r} \right] \quad (14-43)$$

Equation (14-43) is an implicit equation for $\alpha_{A,B}$, in terms of the pressure ratio, r , and x_A , that is readily solved for $\alpha_{A,B}$ by rearranging the equation into the form of a quadratic equation. In the limit when $r = 0$, (14-43) reduces to (14-40), where $\alpha_{A,B} = \alpha_{A,B}^* = (P_{MA}/P_{MB})$. Many experimental investigators report values of $\alpha_{A,B}^*$. For example, Table 14.7, taken from the *Membrane Handbook* [6], gives data at 35°C for various binary pairs with polydimethyl siloxane (PDMS), a rubbery polymer, and bisphenol-A-polycarbonate (PC), a glassy polymer. For the rubbery polymer, permeabilities are high, but separation factors are low. The opposite is true for the glassy polymer. For a given feed composition, the separation factor places a definite limit on the degree of separation that can be achieved.

EXAMPLE 14.5

Air can be separated into nitrogen-enriched and oxygen-enriched streams by gas permeation with a number of different dense-polymer membranes. In all cases, the membrane is more permeable to oxygen. A total of 20,000 scfm of air is compressed, cooled, and treated to remove moisture and compressor oil prior to being sent to a membrane separator at 150 psia and 78°F. Assume the composition of the air is 79 mol% N_2 and 21 mol% O_2 . A low-density polyethylene membrane in the form of a thin-film composite is being considered with solubilities and diffusivities given in Table 14.6. If the membrane skin is 0.2 μm thick, calculate the material balance and area in ft^2 for the membrane as a function of the cut, θ (fraction of feed permeated). Assume a pressure of 15 psia on the permeate side with perfect mixing on both sides of the membrane, such that compositions on both sides are each uniform and equal to exit compositions. Neglect pressure drop and mass-transfer resistances external to the membrane. Comment on the practicality of the membrane for making a reasonable separation.

SOLUTION

Assume standard conditions are 0°C and 1 atm (359 ft^3/lbmol)

$$n_F = \text{Feed flow rate} = \frac{20,000}{359}(60) = 3,343 \text{ lbmol/h}$$

For the low-density polyethylene membrane, from Table 14.6, and applying (14-33), letting $A = O_2$ and $B = N_2$:

$$P_{MB} = H_B D_B = (0.228 \times 10^{-6})(0.32 \times 10^{-6}) \\ = 0.073 \times 10^{-12} \text{ cm}^3(\text{STP})\text{-cm/cm}^2\text{-s-Pa}$$

or, in American engineering units,

$$P_{MB} = \frac{(0.073 \times 10^{-12})(2.54 \times 12)(3600)(101,300)}{(22,400)(454)(14.7)} \\ = 5.43 \times 10^{-12} \frac{\text{lbmol-ft}}{\text{ft}^2\text{-h-psia}}$$

Similarly, for oxygen: $P_{MA} = 16.2 \times 10^{-12} \frac{\text{lbmol-ft}}{\text{ft}^2\text{-h-psia}}$

Permeance values are based on a 0.2- μm -thick membrane skin ($0.66 \times 10^{-6} \text{ ft}$)

$$\begin{aligned} \text{From (14-1), } \bar{P}_{Mi} &= P_{Mi}/l_M \\ \bar{P}_{MB} &= 5.43 \times 10^{-12}/0.66 \times 10^{-6} \\ &= 8.23 \times 10^{-6} \text{ lbmol/ft}^2\text{-h-psia} \\ \bar{P}_{MA} &= 16.2 \times 10^{-12}/0.66 \times 10^{-6} \\ &= 24.55 \times 10^{-6} \text{ lbmol/ft}^2\text{-h-psia} \end{aligned}$$

Material balance equations

For N_2

$$x_{FB}n_F = y_{PB}n_P + x_{RB}n_R \quad (1)$$

where n = flow rate in lbmol/h and subscripts F , P , and R refer, respectively, to the feed, permeate, and retentate. Let

$$\theta = \text{cut} = n_P/n_F, \quad \text{then } (1 - \theta) = n_R/n_F$$

Note that if all components of the feed have a finite permeability, the cut, θ , can vary from 0 to 1. For a cut of 1, all of the feed becomes permeate and no separation is achieved.

Substituting the definition of θ in (1) gives

$$x_{RB} = \frac{x_{FB} - y_{PB}\theta}{1 - \theta} = \frac{0.79 - y_{PB}\theta}{1 - \theta} \quad (2)$$

Similarly, for O_2 ,

$$x_{RA} = \frac{0.21 - y_{PA}\theta}{1 - \theta} \quad (3)$$

Separation factor

From the definition of the separation factor, (14-36), since both fluid sides are well mixed, such that compositions are those of the retentate and permeate,

$$\alpha_{A,B} = \frac{y_{PA}/x_{RA}}{(1 - y_{PA})/(1 - x_{RA})} \quad (4)$$

Transport equations

The transport of A and B through the membrane of area A_M , with partial pressures at exit conditions because of perfect mixing, can be written as

$$N_B = y_{PB}n_P = A_M \bar{P}_{MB}(x_{RB}P_R - y_{PB}P_P) \quad (5)$$

$$N_A = y_{PA}n_P = A_M \bar{P}_{MA}(x_{RA}P_R - y_{PA}P_P) \quad (6)$$

where A_M is the membrane area normal to flow, n_P , through the membrane. The ratio of (6) to (5) is y_{PA}/y_{PB} , and subsequent

manipulations lead to (14-43), where

$$r = P_P/P_R = 15/150 = 0.1$$

$$\begin{aligned} \text{and } \alpha_{A,B}^* &= \alpha_{O_2,N_2} = \bar{P}_{M_{O_2}}/\bar{P}_{M_{N_2}} \\ &= (24.55 \times 10^{-6})/(8.23 \times 10^{-6}) = 2.98 \end{aligned}$$

From (14-43):

$$\alpha_{A,B} = \alpha = 2.98 \left[\frac{x_{R_A}(\alpha - 1) + 1 - 0.1\alpha}{x_{R_A}(\alpha - 1) + 1 - 0.1} \right] \quad (7)$$

Equations (3), (4), and (7) contain four unknowns: x_{R_A} , y_{P_A} , θ , and $\alpha_{A,B} = \alpha$. The variable θ is bounded between 0 and 1, so values of θ are selected in that range. The other three variables are computed in the following manner. Combine (3), (4), and (7) to eliminate α and x_{R_A} . Solve the resulting nonlinear equation for y_{P_A} . Then solve (3) for x_{R_A} and (4) for α . Solve (6) for the membrane area, A_M . Alternatively, the three equations can be solved simultaneously with a computer program such as Mathcad, Matlab, or Polymath. The following results are obtained:

θ	x_{R_A}	y_{P_A}	$\alpha_{A,B}$	A_M, ft^2
0.01	0.208	0.406	2.602	22,000
0.2	0.174	0.353	2.587	462,000
0.4	0.146	0.306	2.574	961,000
0.6	0.124	0.267	2.563	1,488,000
0.8	0.108	0.236	2.555	2,035,000
0.99	0.095	0.211	2.548	2,567,000

Note that the separation factor remains almost constant, varying by only 2% with a value of about 86% of the ideal value. The maximum oxygen content of the permeate (40.6 mol%) occurs with the smallest amount of permeate ($\theta = 0.01$). The maximum nitrogen content of the retentate (90.5 mol%) occurs with the largest amount of permeate ($\theta = 0.99$). With a retentate equal to 60 mol% of the feed ($\theta = 0.4$), the nitrogen content of the retentate has been increased only from 79 to 85.4 mol%. Furthermore, the membrane area requirements are very large. The low-density polyethylene membrane is not very practical. To achieve a more reasonable separation, say with $\theta = 0.6$ and a retentate of 95 mol% N_2 , it is advisable to use a membrane with an ideal separation factor of 5 in a membrane module that approximates crossflow or countercurrent flow of permeate and retentate with no mixing and a much higher permeance for oxygen. For higher purities, a membrane cascade of two or more stages should be considered. These alternatives are developed in the next two subsections.

Module Flow Patterns

In Example 14.5, perfect mixing was assumed on both sides of the membrane. Three other idealized flow patterns, shown in Figure 14.8, have received considerable attention; all assume no mixing and are comparable to the idealized flow patterns used to design heat exchangers. These patterns are (b) countercurrent flow; (c) cocurrent flow; and (d) crossflow. For a given cut, θ , the flow pattern can significantly affect the degree of separation and the membrane area. For flow patterns (b) to (d), fluid on the feed or retentate side of the membrane flows along and parallel to the upstream surface of the membrane. For countercurrent and cocurrent flow, permeate fluid at a given location on the downstream side of the membrane consists of fluid that has just passed through the membrane at that location plus the permeate fluid flowing to that location. For the crossflow case, there is no flow of permeate fluid along the membrane surface. The permeate fluid that has just passed through the membrane at a given location is the only fluid there. For a given module geometry, it is not always obvious which idealized flow pattern to assume. This is particularly true for the spiral-wound module of Figure 14.5b. If the permeation rate is high, the fluid issuing from the downstream side of the membrane may continue to flow perpendicularly to the membrane surface until it finally mixes with the bulk permeate fluid flowing past the surface. In that case, the idealized crossflow pattern might be appropriate. Hollow-fiber modules may be designed to approximate idealized countercurrent, cocurrent, or crossflow patterns. The hollow-fiber module shown in Figure 14.5d is approximated by a countercurrent flow pattern.

Walawender and Stern [27] present solution methods for all four flow patterns of Figure 14.8 under the assumptions of a binary feed with constant pressure ratio, r , and constant ideal separation factor, $\alpha_{A,B}^*$. Exact analytical solutions are possible for the perfect mixing case (as shown in Example 14.5) and for the crossflow case, but numerical solutions are necessary for the countercurrent and cocurrent flow cases. A reasonably simple, but approximate, analytical solution for the crossflow case, derived by Naylor and Backer [28], is presented here.

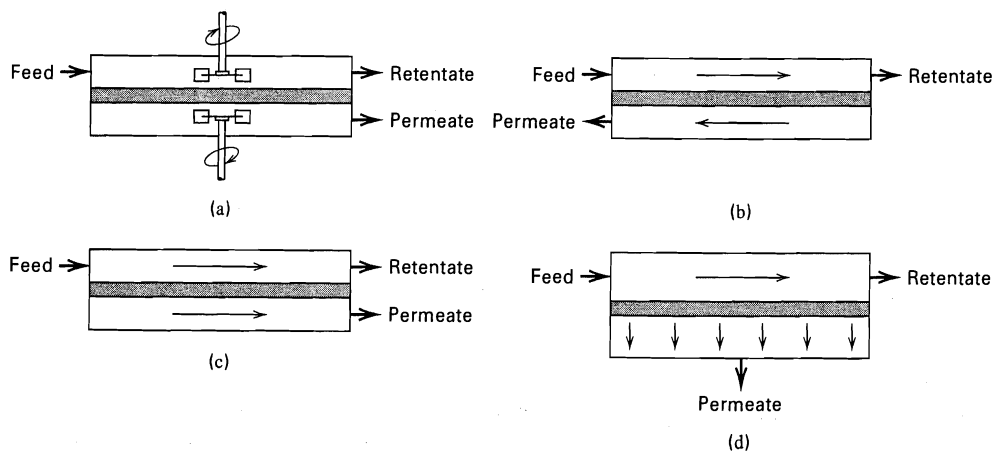


Figure 14.8 Idealized flow patterns in membrane modules: (a) perfect mixing; (b) countercurrent flow; (c) cocurrent flow; (d) crossflow.

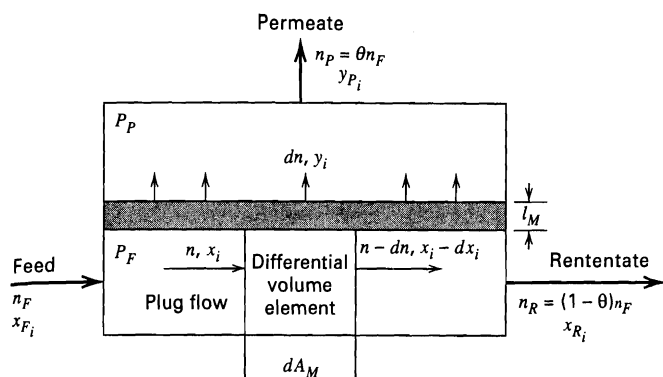


Figure 14.9 Crossflow model for membrane module.

Consider a membrane module with the crossflow pattern shown in Figure 14.9. The feed passes across the upstream membrane surface in plug flow with no longitudinal mixing. The pressure ratio, $r = P_P/P_F$, and the ideal separation factor, $\alpha_{A,B}^*$, are assumed constant. Boundary layer or film mass-transfer resistances external to the membrane are assumed negligible. At the differential element, the local mole fractions in the retentate and permeate, respectively, are x_i and y_i , and the penetrant molar flux is dn/dA_M . Also, the local separation factor is given by (14-43) in terms of the local x_A , r , and $\alpha_{A,B}^*$. An alternative expression for the local permeate composition in terms of y_A , x_A , and r , is obtained by combining (14-36) and (14-41).

$$\frac{y_A}{1 - y_A} = \alpha_{A,B}^* \left[\frac{x_A - r y_A}{(1 - x_A) - r(1 - y_A)} \right] \quad (14-44)$$

A material balance for A around the differential-volume element gives

$$y_A dn = d(n x_A) = x_A dn + n dx_A \quad \text{or} \quad \frac{dn}{n} = \frac{dx_A}{y_A - x_A} \quad (14-45)$$

which is identical to the Rayleigh equation for differential, batch distillation. If (14-36) is combined with (14-45) to eliminate y_A , we obtain

$$\frac{dn}{n} = \left[\frac{1 + (\alpha - 1)x_A}{x_A(\alpha - 1)(1 - x_A)} \right] dx_A \quad (14-46)$$

where $\alpha = \alpha_{A,B}$.

In the solution to Example 14.5, it was noted that $\alpha = \alpha_{A,B}$ is relatively constant over the entire range of cut, θ . Such is generally the case when the pressure ratio, r , is small. If the assumption of constant $\alpha = \alpha_{A,B}$ is made in (14-46) and integration is carried out from the intermediate location of the differential element to the final retentate, that is, from n to n_R and from x_A to x_{R_A} , the result is

$$n = n_R \left[\left(\frac{x_A}{x_{R_A}} \right)^{\left(\frac{1}{\alpha-1} \right)} \left(\frac{1 - x_{R_A}}{1 - x_A} \right)^{\left(\frac{\alpha}{\alpha-1} \right)} \right] \quad (14-47)$$

The mole fraction of A in the final permeate and the total membrane surface area are obtained by integrating the

values obtained from solving (14-44) to (14-46):

$$y_{P_A} = \int_{x_{F_A}}^{x_{R_A}} y_A dn / \theta n_F \quad (14-48)$$

By combining (14-48) with (14-46), (14-47), and the definition of α , the integral in n can be transformed to an integral in x_A , which when integrated gives

$$y_{P_A} = x_{R_A}^{\left(\frac{1}{1-\alpha} \right)} \left(\frac{1 - \theta}{\theta} \right) \times \left[(1 - x_{R_A})^{\left(\frac{\alpha}{\alpha-1} \right)} \left(\frac{x_{F_A}}{1 - x_{F_A}} \right)^{\left(\frac{\alpha}{\alpha-1} \right)} - x_{R_A}^{\left(\frac{\alpha}{\alpha-1} \right)} \right] \quad (14-49)$$

where $\alpha = \alpha_{A,B}$ can be estimated from (14-43) by using $x_A = x_{F_A}$.

The differential rate of mass transfer of A across the membrane is given by

$$y_A dn = \frac{P_{M_A} dA_M}{l_M} [x_A P_F - y_A P_P] \quad (14-50)$$

from which the total membrane surface area can be obtained by integration:

$$A_M = \int_{x_{R_A}}^{x_{F_A}} \frac{l_M y_A dn}{P_{M_A} (x_A P_F - y_A P_P)} \quad (14-51)$$

The application of the crossflow model is illustrated in the next example.

EXAMPLE 14.6

For the conditions of Example 14.5, compute exit compositions for a spiral-wound module that approximates crossflow.

SOLUTION

From Example 14.5: $\alpha_{A,B}^* = 2.98$; $r = 0.1$; $x_{F_A} = 0.21$

From (14-43), using $x_A = x_{F_A}$; $\alpha_{A,B} = 2.60$

An overall module material balance for O_2 (A) gives

$$x_{F_A} n_F = x_{R_A} (1 - \theta) n_F + y_{P_A} \theta n_F \quad \text{or} \quad x_{R_A} = \frac{(x_{F_A} - y_{P_A} \theta)}{(1 - \theta)} \quad (1)$$

Solving (1) and (14-49) simultaneously with a program such as Mathcad, Matlab, or Polymath gives the following results:

θ	x_{R_A}	x_{P_A}	Stage α_S
0.01	0.208	0.407	2.61
0.2	0.168	0.378	3.01
0.4	0.122	0.342	3.74
0.6	0.0733	0.301	5.44
0.8	0.0274	0.256	12.2
0.99	0.000241	0.212	1,120.

Comparing these results to those of Example 14.5, we see that for crossflow, the permeate is richer in oxygen and the retentate is richer in nitrogen. Thus, for a given cut, θ , crossflow is more efficient than perfect mixing, as might be expected.

Also included in the preceding table is the calculated degree of separation for the stage, α_S , defined on the basis of the mole

fractions in the permeate and retentate exiting the stage by

$$(\alpha_{A,B})_S = \alpha_S = \frac{(y_{PA}/x_{RA})}{(1 - y_{PA})/(1 - x_{RA})} \quad (2)$$

Recall that the ideal separation factor, $\alpha_{A,B}^*$, for this example is 2.98. Also, if (2) is applied to the perfect mixing case of Example 14.5, it is found that α_S is 2.603 for $\theta = 0.01$ and decreases slowly with increasing θ until at $\theta = 0.99$, $\alpha_S = 2.548$. Thus, for perfect mixing, $\alpha_S < \alpha^*$ for all θ . Such is not the case for crossflow. In the above table, $\alpha_S < \alpha^*$ for $\theta > 0.2$, and α_S increases with increasing θ . For $\theta = 0.6$, α_S is almost twice α^* .

Calculations of the degree of separation of a binary mixture in a membrane module utilizing cocurrent or countercurrent flow patterns involve the numerical solution of ordinary differential equations. Derivation of these equations and FORTRAN computer codes for their solution are given by Walawender and Stern [27]. A representative solution is shown in Figure 14.10 for the separation of air (20.9 mol% O₂) for conditions of $\alpha^* = 5$ and $r = 0.2$. For a given cut, θ , it is seen that the best separation is achieved with countercurrent flow. The curve for cocurrent flow lies between those for crossflow and perfect mixing. The readily computed crossflow case is considered to be a good, conservative estimate of membrane module performance. The perfect mixing case for binary mixtures is extended to multicomponent mixtures by Stern et al. [29], who present an iterative procedure. As with crossflow, countercurrent flow also offers the possibility of a separation factor for the stage, α_S , defined by (2) earlier, that can be considerably greater than α^* .

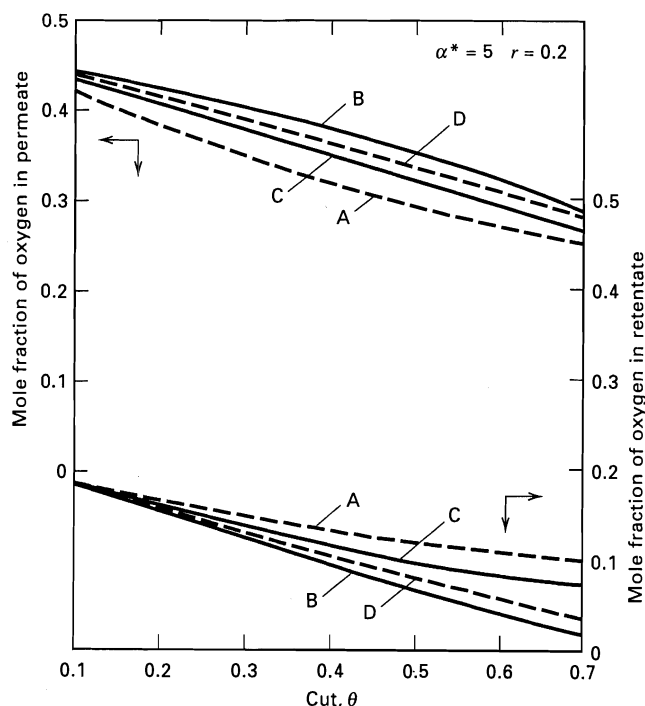


Figure 14.10 Effect of membrane module flow pattern on degree of separation of air. A, perfect mixing; B, countercurrent flow; C, cocurrent flow; D, crossflow.

Cascades

A single membrane module or a number of such modules arranged in parallel or in series without recycle constitutes a single-stage membrane separation process. The extent to which a feed mixture can be separated in a single stage is limited and, as shown in the previous subsection, is determined by the separation factor, α . This factor depends, in turn, on the module flow pattern, the permeability ratio (ideal separation factor), the cut, θ , and the driving force for mass transfer through the membrane. To achieve a higher degree of separation than possible with a single stage, a countercurrent cascade of membrane stages, such as are used in distillation, absorption, stripping, and liquid-liquid extraction, or a hybrid process that couples a membrane separator with another separation operation, such as distillation or adsorption, can be applied. Membrane cascades were presented briefly in Section 5.5. They are now discussed in more detail and illustrated with an example.

A countercurrent recycle cascade of membrane separators, similar to a distillation column, is shown in Figure 14.11a. The feed enters at stage F, somewhere near the middle of the column. Permeate is enriched in components of high permeability in an enriching section, while the retentate is enriched in components of low permeability in a stripping section. The final permeate is withdrawn from stage 1, while the final retentate is withdrawn from stage N . For a cascade, additional factors that affect the degree of separation of the feed are the number of stages and the recycle ratio (permeate recycle rate/permeate product rate). As discussed by Hwang and Kammermeyer [30], it is best to manipulate the cut and reflux rate at each stage so as to force the compositions of the two streams entering each stage to be identical. For example, the composition of retentate leaving Stage 1 and entering Stage 2 would be identical to the composition of permeate flowing from Stage 3 to Stage 2. This corresponds to the least amount of entropy production for the cascade and, thus, the highest second-law efficiency. Such a cascade is referred to as ideal. Calculation methods for cascades are discussed by Hwang and Kammermeyer [30] and utilize the single-stage methods that depend upon the module flow pattern, as discussed in the previous section. The calculations are best carried out with a computer program, but results for a binary mixture can be conveniently displayed on a McCabe-Thiele type diagram in terms of the mole fraction in the permeate leaving each stage, y_i , versus the mole fraction in the retentate leaving each stage, x_i . For a membrane cascade, the equilibrium curve becomes the selectivity curve in terms of the separation factor for the stage, α_S .

In Figure 14.11, it is assumed the pressure drop on the feed or upstream side of the membrane is negligible. Thus, only the permeate must be pumped, if a liquid, or compressed, if a gas, to be sent to the next stage. In the case of gas permeation, compression costs can be high. Thus, membrane cascades for gas permeation are often limited to just two or three stages, with the most common configurations shown in

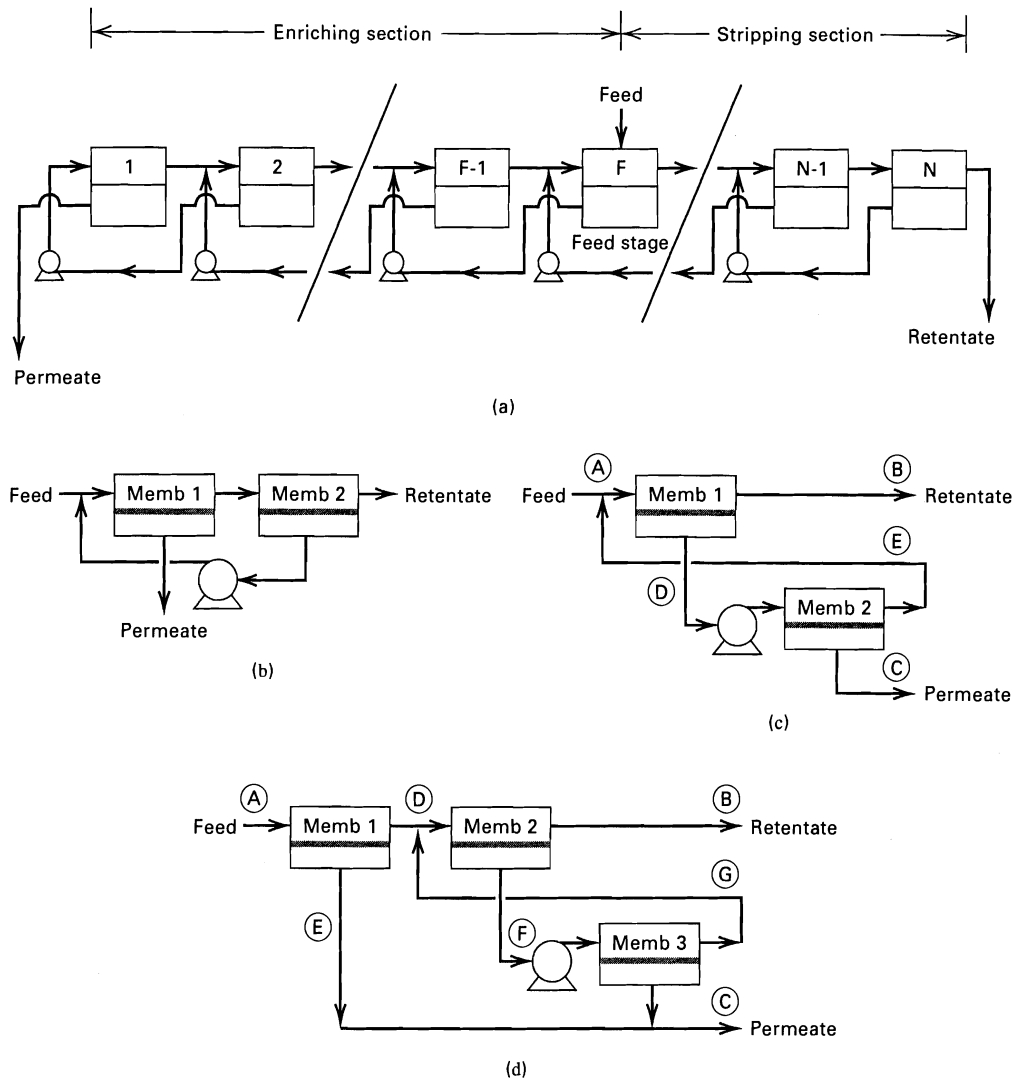


Figure 14.11 Countercurrent recycle cascades of membrane separators. (a) Multiple-stage unit. (b) Two-stage stripping cascade. (c) Two-stage enriching cascade. (d) Two-stage enriching cascade with additional premembrane stage.

Figures 14.11b, c, and d. Compared to a single stage, the two-stage stripping cascade is designed to obtain a more-pure retentate, whereas a more-pure permeate is the goal of the two-stage enriching cascade. The addition of the premembrane stage, shown in Figure 14.11d, may be attractive when the feed concentration is low in the component to be passed preferentially through the membrane, the desired permeate purity is high, the separation factor is low, and/or a high recovery of the more permeable component is desired. An example of the application of the enrichment cascades is given by Spillman [31] for the removal of carbon dioxide from natural gas (assumed to be methane) using cellulose-acetate membranes in spiral-wound modules that approximate crossflow. The ideal separation factor, $\alpha_{\text{CO}_2, \text{CH}_4}^*$ is 21. Results of computer calculations are given in Table 14.8 for a single stage (not shown in Figure 14.11), a two-stage enriching cascade (Figure 14.11c), and a two-stage enriching cascade with an additional premembrane stage (Figure 14.11d). Carbon dioxide flows through the membrane faster than methane. In all three cases, the feed is 20 million (MM) scfd of 7 mol% CO_2 in methane at 850 psig (about 865 psia) and the retentate is 98 mol% in methane. For each stage, the

downstream (permeate side) membrane pressure is 10 psig (about 25 psia). In Table 14.8, for all three cases, stream A is the feed, stream B is the final retentate, and stream C is the final permeate. Case 1 achieves a 90.2% recovery of methane. Case 2 increases that recovery to 98.7%. Case 3 achieves an intermediate recovery of 94.6%. The following degrees of separation are computed from data given in Table 14.8:

Case	α_S for Membrane Stage		
	1	2	3
1	28	—	—
2	28	57	—
3	20	19	44

We can also compute overall degrees of separation for the cascades, α_C , for cases 2 and 3, giving values of 210 and 51, respectively.

External Mass-Transfer Resistances

Thus far, all of the resistance to mass transfer has been assumed to be associated with the membrane. Thus, concentrations in the fluid at the upstream and downstream faces of the

Table 14.8 Separation of N₂ and CH₄ with Membrane Cascades

Case 1: Single Membrane Stage:

	Stream		
	A Feed	B Retentate	C Permeate
Composition (mole%)			
CH ₄	93.0	98.0	63.4
CO ₂	7.0	2.0	36.6
Flow rate (MM SCFD)	20.0	17.11	2.89
Pressure (psig)	850	835	10

Case 2: Two-Stage Enriching Cascade (Figure 14.11c):

	Stream				
	A	B	C	D	E
Composition (mole%)					
CH ₄	93.0	98.0	18.9	63.4	93.0
CO ₂	7.0	2.0	81.1	36.6	7.0
Flow rate (MM SCFD)	20.00	18.74	1.26	3.16	1.90
Pressure (psig)	850	835	10	10	850

Case 3: Two-Stage Enriching Cascade with Premembrane Stage (Figure 14.11d):

	Stream						
	A	B	C	D	E	F	G
Composition (mole%)							
CH ₄	93.0	98.0	49.2	96.1	56.1	72.1	93.0
CO ₂	7.0	2.0	50.8	3.9	43.9	27.9	7.0
Flow rate (MM SCFD)	20.00	17.95	2.05	19.39	1.62	1.44	1.01
Pressure (psig)	850	835	10	840	10	10	850

Note: MM, million.

membrane have been assumed equal to the respective bulk-fluid concentrations on either side of the membrane. When mass-transfer resistances external to the membrane are not negligible, concentration or partial-pressure gradients exist in the boundary layers or films adjacent to the membrane surfaces, as is illustrated for four cases in Figure 14.7. For given bulk-fluid concentrations, the presence of these resistances reduces the driving force for mass transfer across the membrane and, therefore, the flux of penetrant.

In general, the solution-diffusion mechanism for gas permeation is quite slow compared to diffusion in the gas boundary layers or film adjacent to the membrane, so external mass-transfer resistances are negligible and $P_{iF} = P_{i0}$ and $P_{iP} = P_{iL}$ in Figure 14.7d. Because diffusion in liquid boundary layers and films can be slow, concentration polarization cannot be neglected in membrane processes that involve liquids, such as dialysis, reverse osmosis, and pervaporation. The need to consider the effect of concentration polarization is of particular importance in reverse osmosis,

where the effect can reduce the water flux and increase the salt flux, making it more difficult to obtain potable water.

Consider a membrane process of the type in Figure 14.7a, involving liquids with a porous membrane. At steady state, rates of mass transfer of a penetrating species, i , through the three resistances are as follows, assuming no change in the area for mass transfer across the thickness of the membrane,

$$N_i = k_{iF}(c_{iF} - c_{i0}) = \frac{D_{ei}}{l_M}(c_{i0} - c_{iL}) = k_{iP}(c_{iL} - c_{iP})$$

where D_{ei} is given by (14-14).

If these three equations are combined to eliminate the intermediate concentrations, c_{i0} and c_{iL} , we obtain

$$N_i = \frac{c_{iF} - c_{iP}}{\frac{1}{k_{iF}} + \frac{l_M}{D_{ei}} + \frac{1}{k_{iP}}} \quad (14-52)$$

Now consider the membrane process in Figure 14.7b, involving liquids with a dense (nonporous) membrane, for which the solution-diffusion mechanism is used for mass

transfer through the membrane, as given by (14-26). At steady state, for constant mass-transfer area, rates of species mass transfer through the three resistances are:

$$N_i = k_{iF}(c_{iF} - c'_{i0}) = \frac{K_i D_i}{l_M}(c'_{i0} - c'_{iL}) = k_{iP}(c'_{iL} - c_{iP})$$

If these three equations are combined to eliminate the intermediate concentrations, c'_{i0} and c'_{iL} , we obtain

$$N_i = \frac{c_{iF} - c_{iP}}{\frac{1}{k_{iF}} + \frac{l_M}{K_i D_i} + \frac{1}{k_{iP}}} \quad (14-53)$$

in (14-52) and (14-53), where k_{iF} and k_{iP} are mass-transfer coefficients for the feed-side and permeate-side boundary layers or films. The three terms in the denominator of the right-hand side are the resistances to the mass flux. In general, the mass-transfer coefficients depend on fluid properties, flow-channel geometry, and flow regime (turbulent or laminar). In the laminar-flow regime, a long entry region may exist where the mass-transfer coefficient changes with the distance, L , from the entry of the membrane channel. Estimation of the coefficients is complicated by fluid velocities that change because of mass exchange between the two fluids. In (14-52) and (14-53), the membrane resistances, l_M/D_{ei} and $l_M/K_i D_i$, respectively, can be replaced by l_M/P_{Mi} or $\bar{P}_{Mi}$.

Typical mass-transfer coefficients for channel flow can be obtained from the general empirical film-model correlation [32]:

$$N_{Sh} = k_i d_H / D_i = a N_{Re}^b N_{Sc}^{0.33} (d_H / L)^d \quad (14-54)$$

where

$$N_{Re} = d_H v \rho / \mu$$

$$N_{Sc} = \mu / \rho D_i$$

$$d_H = \text{hydraulic diameter}$$

$$v = \text{velocity}$$

The constants a , b , and d are as follows:

Flow Regime	Flow Channel Geometry	d_H	a	b	d
Turbulent, ($N_{Re} > 10,000$)	Circular tube	D	0.023	0.8	0
	Rectangular channel	$2hw/(h+w)$	0.023	0.8	0
Laminar, ($N_{Re} < 2,100$)	Circular tube	D	1.86	0.33	0.33
	Rectangular channel	$2hw/(h+w)$	1.62	0.33	0.33

where

$$w = \text{width of channel}$$

$$h = \text{height of channel}$$

$$L = \text{length of channel}$$

EXAMPLE 14.7

A dilute solution of solute A in solvent B is passed through a tubular-membrane separator, where the feed flows through the tubes. At a certain location, the solute concentrations are

$5.0 \times 10^{-2} \text{ kmol/m}^3$ and $1.5 \times 10^{-2} \text{ kmol/m}^3$, respectively, on the feed and permeate sides. The permeance of the membrane for solute A is given by the membrane vendor as $7.3 \times 10^{-5} \text{ m/s}$. If the tube-side Reynolds number is 15,000, the feed-side solute Schmidt number is 500, the diffusivity of the feed-side solute is $6.5 \times 10^{-5} \text{ cm}^2/\text{s}$, and the inside diameter of the tube is 0.5 cm, estimate the flux of the solute through the membrane if the mass-transfer resistance on the permeate side of the membrane is negligible.

SOLUTION

The flux of the solute is given by the permeance form of (14-52) or (14-53):

$$N_A = \frac{c_{AF} - c_{AP}}{\frac{1}{k_{AF}} + \frac{1}{\bar{P}_{MA}} + 0}$$

$$c_{AF} - c_{AP} = 5 \times 10^{-2} - 1.5 \times 10^{-2} = 3.5 \times 10^{-2} \text{ kmol/m}^3 \quad (1)$$

$$\bar{P}_{MA} = 7.3 \times 10^{-5} \text{ m/s}$$

From (14-54), for turbulent flow in a tube, since $N_{Re} > 10,000$:

$$\begin{aligned} k_{AF} &= 0.023 \frac{D_A}{D} N_{Re}^{0.8} N_{Sc}^{0.33} \\ &= 0.023 \left(\frac{6.5 \times 10^{-5}}{0.5} \right) (15,000)^{0.8} (500)^{0.33} \\ &= 0.051 \text{ cm/s or } 5.1 \times 10^{-4} \text{ m/s} \end{aligned}$$

From (1),

$$N_A = \frac{3.5 \times 10^{-2}}{\frac{1}{5.1 \times 10^{-4}} + \frac{1}{7.3 \times 10^{-5}}} = 2.24 \times 10^{-6} \text{ kmol/s} \cdot \text{m}^2$$

The fraction of the total resistance due to the membrane is

$$\frac{1}{\frac{1}{\frac{7.3 \times 10^{-5}}{5.1 \times 10^{-4}} + \frac{1}{7.3 \times 10^{-5}}}} = 0.875 \text{ or } 87.5\%$$

Concentration Polarization and Fouling

When gases are produced during electrolysis, they accumulate on and around the electrodes of the electrolytic cell, reducing the flow of electric current. This phenomenon is referred to as polarization. A similar phenomenon, *concentration polarization*, occurs in membrane separators when the membrane is permeable to molecules of A, but relatively impermeable to molecules of B. Thus, molecules of B are carried by bulk flow to the upstream surface of the membrane where they accumulate, causing their concentration at the surface of the membrane to increase in a "polarization layer." The equilibrium concentration of B in this layer is reached when its back-diffusion to the bulk fluid on the feed-retentate side equals its bulk flow toward the membrane.

Concentration polarization is most common in pressure-driven membrane separations, such as reverse osmosis and ultrafiltration, where it can reduce the flux of molecules of A through the membrane. The polarization effect can be

particularly serious if the concentration of B attains its solubility limit next to the membrane surface. A precipitate of gel may then form, the result being fouling on the membrane surface or within membrane pores, with a further reduction in the flux of A. In general, concentration polarization and fouling are most severe at high values of the flux of A. Examples of the effects of concentration polarization and fouling and simplified theoretical treatments are given in the sections on reverse osmosis and ultrafiltration.

14.4 DIALYSIS AND ELECTRODIALYSIS

In a dialysis membrane-separation process, shown in Figure 14.12, the feed is a liquid, at pressure P_1 , containing solvent, solutes of type A, and solutes of type B and/or insoluble, but dispersed, colloidal matter. A sweep liquid or wash of the same solvent is fed at pressure P_2 to the other side of the membrane. The membrane is thin with micropores of a size such that solutes of type A can pass through by a concentration driving force. Solute of type B are larger in molecular size than those of type A and pass through the membrane only with difficulty or not at all. This transport of solutes A and B through the membrane is called dialysis. Colloids do not pass through the membrane. With pressure $P_1 = P_2$, the solvent may also pass through the membrane, but by a concentration driving force acting in the opposite direction. The transport of the solvent is called osmosis. By elevating P_1 above P_2 , solvent osmosis can be reduced or eliminated. The products of a dialysis unit (dialyzer) are a liquid *diffusate* (permeate) containing solvent, solutes of type A, and smaller amounts of solutes of type B; and a *dialysate* (retentate) of the solvent and remaining solutes of types A and B, and colloidal matter. Ideally, the dialysis unit would enable a perfect separation between solutes of type A and solutes of type B and any colloidal matter. However, at best only a fraction of the solutes of type A are recovered in

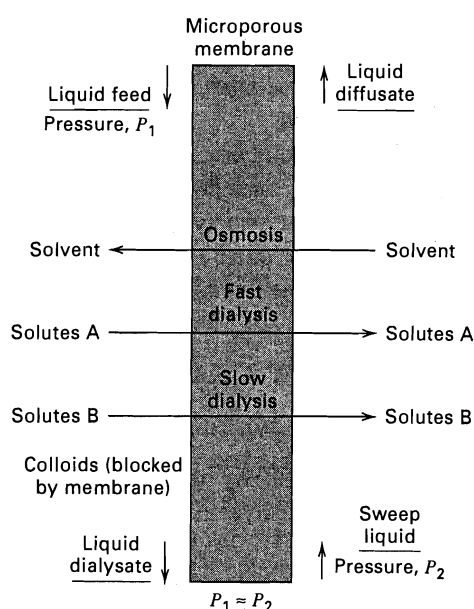


Figure 14.12 Dialysis.

the diffusate, even when solutes of type B do not pass through the membrane.

For example, when dialysis is used to recover sulfuric acid from an aqueous stream containing sulfate salts, the following results are obtained, as reported by Chamberlin and Vromen [33]:

	Streams in		Streams out	
	Feed	Wash	Dialysate	Diffusate
Flow rate, gph	400	400	420	380
H ₂ SO ₄ , g/L	350	0	125	235
CuSO ₄ , g/L as Cu	30	0	26	2
NiSO ₄ , g/L as Ni	45	0	43	0

Thus, about 64% of the H₂SO₄ is recovered in the diffusate, accompanied by only about 6% of the CuSO₄, and essentially no NiSO₄.

Dialysis is closely related to other membrane processes that use other driving forces for separating liquid mixtures, including (1) reverse osmosis, which depends upon a transmembrane pressure difference for solute and/or solvent transport; (2) electrodialysis and electro-osmosis, which depend upon a transmembrane electrical-potential difference for solute and solvent transport, respectively; and (3) thermal osmosis, which depends on a transmembrane temperature difference for solute and solvent transport.

Dialysis is attractive when the concentration differences for the main diffusing solutes are large and the permeability differences between those solutes and the other solute(s) and/or colloids is large. Although dialysis has been known since the work of Graham in 1861 [34], commercial applications of dialysis do not rival reverse osmosis and gas permeation. Nevertheless, dialysis has been applied to a number of separations, including (1) recovery of sodium hydroxide from a 17–20 wt% caustic viscose liquor contaminated with hemicellulose to produce a diffusate of 9–10 wt% caustic; (2) recovery of chromic, hydrochloric, and hydrofluoric acids from contaminating metal ions; (3) recovery of sulfuric acid from aqueous solutions containing nickel sulfate; (4) removal of alcohol from beer to produce a reduced-alcohol beer; (5) recovery of nitric and hydrofluoric acids from spent stainless-steel pickle liquor; (6) removal of mineral acids from organic compounds; (7) removal of low-molecular-weight contaminants from polymers; and (8) purification of pharmaceuticals. Also of great importance is hemodialysis, in which urea, creatine, uric acid, phosphates, and chlorides are removed from blood without removing essential higher-molecular-weight compounds and blood cells. This dialysis device is called an artificial kidney.

Typical microporous-membrane materials used in dialysis are hydrophilic, including cellulose, cellulose acetate, various acid-resistant polyvinyl copolymers, polysulfones, and polymethylmethacrylate, typically less than 50 μm thick and with pore diameters of 15 to 100 $\text{\AA}$. The most common membrane modules are plate-and-frame and hollow-fiber. Compact hollow-fiber hemodialyzers, such as the one shown

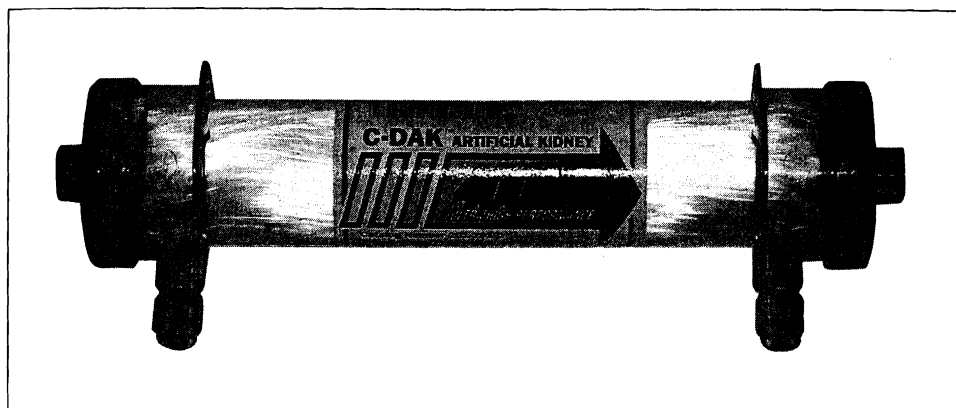


Figure 14.13 Artificial kidney.

in Figure 14.13, which are widely used, typically contain several thousand 200- μm -diameter fibers with a wall thickness of 20–30 μm and a length of 10–30 cm. Dialysis membranes can be thin because pressures on either side of the membrane are essentially equal.

At a differential location in a dialyzer, the rate of mass transfer of solute across the dialysis membrane is given by

$$dn_i = K_i(c_{iF} - c_{iP})dA_M \quad (14-55)$$

where K_i is the overall mass-transfer coefficient, which is given in terms of the individual coefficients from the permeability form of (14-52):

$$\frac{1}{K_i} = \frac{1}{k_{iF}} + \frac{l_M}{P_{M_i}} + \frac{1}{k_{iP}} \quad (14-56)$$

The determination of the membrane area is made by integrating (14-55) taking into account the module flow patterns, the bulk-concentration gradients, and the individual mass-transfer coefficients in (14-56).

One of the oldest membrane materials for use with aqueous solutions is porous cellophane, for which solute permeability is given by (14-14) with $P_{M_i} = D_{e_i}$ and $\bar{P}_{M_i} \cdot l_M$. In the presence of a solution, cellophane will swell to about twice its dry thickness. The wet thickness should be used for l_M . Typical values of parameters given in (14-13) to (14-15) for commercial cellophane are as follows:

Wet thickness = $l_M = 0.004$ to 0.008 cm;

porosity = $\epsilon = 0.45$ to 0.60

Tortuosity = $\tau = 3$ to 5 ;

pore diameter = $D = 30$ to 50 Å

If solute does not interact with the membrane material, the diffusivity, D_{e_i} , in (14-14) is the ordinary molecular diffusion coefficient, which depends only on solute and solvent properties. However, the membrane may have a profound effect on the solute diffusivity if any of a number of membrane–solute interactions occur, including covalent, ionic, and hydrogen bonding; physical adsorption and chemisorption; and membrane polymer flexibility. Thus, it is preferred to measure $\bar{P}_{M_i}$ experimentally using the actual process fluids.

Although the transport of solvents, such as water, which usually occurs in a direction opposite to the solute, could be formulated in terms of Fick's law, it is more common to measure the solvent flux and report the so-called *water-transport number*, which is the ratio of the water flux to the solute flux, with a negative value indicating transport of solvent in the same direction as the solute. The membrane can also interact with the solvent and even curtail solvent transport. Ideally, the water transport number should be a small value less than +1.0. The ideal experimental dialyzer is a batch cell with a variable-speed stirring mechanism on both sides of the membrane so that external mass transfer resistances, $1/k_{iF}$ and $1/k_{iP}$ in (14-56), are made negligible. Stirrer speeds greater than 2,000 rpm may be required.

A common dialyzer is the plate-and-frame type of Figure 14.5a. However, for dialysis applications, the frames are arranged vertically. A typical unit might contain 100 square frames, each 0.75×0.75 m on 0.6-cm spacing, equivalent to 56 m^2 of membrane surface. The dialysis rate for sulfuric acid might be 5 lb/day-ft². More recently developed dialysis units utilize hollow fibers of 200- μm inside diameter, 16- μm wall thickness, and 28-cm length packed into a heat-exchanger-type module to give 22.5 m^2 of membrane area in a volume that might be one-tenth of the volume of an equivalent plate-and-frame unit.

In a plate-and-frame dialyzer, the flow pattern is nearly countercurrent. Because total flow rates change little and solute concentrations are typically small, it is common to estimate the solute transport rate by assuming a constant overall mass-transfer coefficient with a log-mean concentration driving force. Thus, from (14-55):

$$n_i = K_i A_M (\Delta c_i)_{LM} \quad (14-57)$$

where K_i is given by (14-56). This method is applied in the following example.

EXAMPLE 14.8

A countercurrent-flow, plate-and-frame dialyzer is to be sized to process $0.78 \text{ m}^3/\text{h}$ of an aqueous solution of 300 kg/m^3 of H_2SO_4 and smaller amounts of copper and nickel sulfates. A wash water rate of $1.0 \text{ m}^3/\text{h}$ is to be used, and it is desired to recover 30% of the acid at

25°C. From batch laboratory experiments with an acid-resistant vinyl membrane, in the absence of external mass-transfer resistances, a permeance of 0.025 cm/min for the acid and a water transport number of +1.5 are measured. Membrane transport of copper and nickel sulfates is negligible. For these flow rates, experience with plate-and-frame dialyzers indicates that flow will be laminar and the combined external liquid-film mass-transfer coefficients will be 0.020 cm/min. Determine the membrane area required in m².

SOLUTION

$$m_{\text{H}_2\text{SO}_4} \text{ in feed} = 0.78(300) = 234 \text{ kg/h}$$

$$m_{\text{H}_2\text{SO}_4} \text{ transferred} = 0.3(234) = 70 \text{ kg/h}$$

$$m_{\text{H}_2\text{O}} \text{ transferred to dialysate} = 1.5(70) = 105 \text{ kg/h}$$

$$m_{\text{H}_2\text{O}} \text{ in entering wash} = 1.0(1,000) = 1,000 \text{ kg/h}$$

$$m_P \text{ leaving} = 1,000 - 105 + 70 = 965 \text{ kg/h}$$

For mixture densities, assume aqueous sulfuric acid solutions and use the appropriate table in *Perry's Chemical Engineers' Handbook*:

$$\rho_F = 1,175 \text{ kg/m}^3 \quad \rho_R = 1,114 \text{ kg/m}^3 \quad \rho_P = 1,045 \text{ kg/m}^3$$

$$m_F = 0.78(1,175) = 917 \text{ kg/h}$$

$$m_R \text{ leaving} = 917 + 105 - 70 = 952 \text{ kg/h}$$

Sulfuric acid concentrations:

$$c_F = 300 \text{ kg/m}^3 \quad c_{\text{wash}} = 0 \text{ kg/m}^3$$

$$c_R = \frac{(234 - 70)}{952}(1,114) = 192 \text{ kg/m}^3$$

$$c_P = \frac{70}{965}(1,045) = 76 \text{ kg/m}^3$$

The log-mean driving force for H₂SO₄ with countercurrent flow of feed and wash:

$$(\Delta c)_{\text{LM}} = \frac{(c_F - c_P) - (c_R - c_{\text{wash}})}{\ln \left(\frac{c_F - c_P}{c_R - c_{\text{wash}}} \right)} = \frac{(300 - 76) - (192 - 0)}{\ln \left(\frac{300 - 76}{192 - 0} \right)} = 208 \text{ kg/m}^3$$

The driving force is almost constant in the membrane module, varying only from 224 to 192 kg/m³.

From (14-56),

$$K_{\text{H}_2\text{SO}_4} = \frac{1}{\frac{1}{\bar{P}_M} + \left(\frac{1}{k} \right)_{\text{combined}}} = \frac{1}{\frac{1}{0.025} + \frac{1}{0.020}} = 0.0111 \text{ cm/min} \quad \text{or} \quad 0.0067 \text{ m/h}$$

From (14-57), using mass units instead of molar units:

$$A_M = \frac{m_{\text{H}_2\text{SO}_4}}{K_{\text{H}_2\text{SO}_4}(\Delta c_{\text{H}_2\text{SO}_4})_{\text{LM}}} = \frac{70}{0.0067(208)} = 50 \text{ m}^2$$

Electrodialysis

Electrodialysis began in the early 1900s as a modification to dialysis by the addition of electrodes and direct current to increase the rate of dialysis in electrolyte solutions. However, since the 1940s, electrodialysis has developed into a membrane-separation process that differs from dialysis in many ways. Today, electrodialysis refers to an electrolytic process for separating an aqueous, electrolyte feed solution into a concentrate or brine and a dilute or desalted water

(diluate) by means of an electric field and ion-selective membranes. A typical electrodialysis process is shown in Figure 14.14, where the four ion-selective membranes shown are of two types arranged in an alternating-series pattern. The cation-selective membranes (C) carry a negative charge, and thus attract and pass positively charged ions (cations), while retarding negative ions. The anion-selective membranes (A) carry a positive charge that attracts and

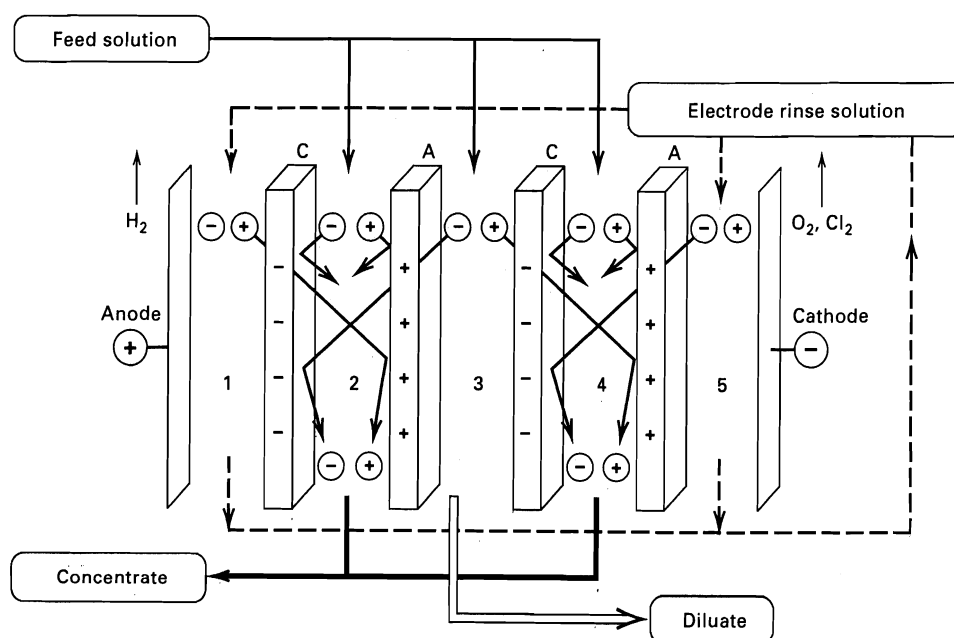
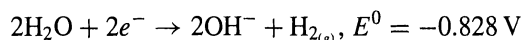


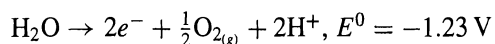
Figure 14.14 Schematic diagram of the electrodialysis process. C, cation transfer membrane; A, anion transfer membrane. [Adapted from W. S. W. Ho and K. K. Sirkar, editors, "Membrane Handbook," Van Nostrand Reinhold, New York (1992).]

permits passage of negative ions (anions). Both types of membranes are impervious to water. The net result is that both anions and cations are concentrated in compartments 2 and 4, from which concentrate is withdrawn, and ions are depleted in compartment 3, from which the diluate is withdrawn. Compartment pressures are essentially equal. Compartments 1 and 5 are bounded on the far sides by the anode and cathode, respectively. A direct-current voltage is applied (e.g., with a battery or direct-current generator) across the anode and cathode, causing current to flow by metallic conduction of electrons through wiring from the anode to the cathode and then through the cell by ionic conduction from the cathode back to the anode. Both electrodes are chemically neutral metals, with the anode being typically stainless steel and the cathode typically platinum-coated tantalum, niobium, or titanium. Thus, the electrodes are neither oxidized nor reduced.

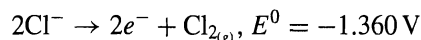
But half reactions must occur at the two electrodes. Typically, the most easily oxidized species is oxidized at the anode and the most easily reduced species is reduced at the cathode. With inert electrodes, the result at the cathode is the reduction of water by the half reaction



The oxidation half reaction at the anode is

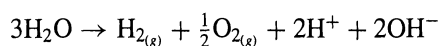


or, if chloride ions are present:

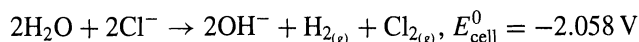


where the electrode potentials are the standard values at 25°C for one molar solution of ions and partial pressures of one atmosphere for the gaseous products. Values of E^0 can be corrected for nonstandard conditions by the Nernst equation.

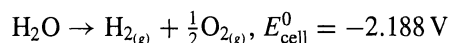
The corresponding overall cell reactions are:



or



The net reaction for the first case is



The electrode rinse solution that circulates through compartments 1 and 5 is typically acidic to neutralize the OH ions formed in compartment 1 and prevent precipitation of compounds such as CaCO_3 and $\text{Mg}(\text{OH})_2$.

The most widely used ion-exchange membranes for electrodialysis, first reported by Juda and McRae [35] in 1950, are (1) cation-selective membranes containing negatively charged groups fixed to a polymer matrix, and (2) anion-selective membranes containing positively charged groups fixed to a polymer matrix. The former, shown schematically in Figure 14.15, includes fixed anions, mobile cations (called counterions), and mobile anions (called co-ions). The latter are almost completely excluded from the polymer

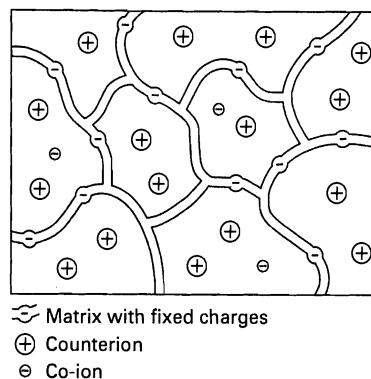


Figure 14.15 Cation-exchange membrane.

[From H. Strathmann, *Sep. and Purif. Methods*, 14(1), 41–66 (1985) with permission.]

matrix by electrical repulsion, called the Donnan effect. For perfect exclusion, only cations are transferred through the membrane. In practice, the exclusion is better than 90%.

A typical cation-selective membrane is made of polystyrene cross-linked with divinylbenzene and sulfonated to produce fixed sulfonate, $-\text{SO}_3^-$, anion groups. A typical anion-selective membrane of the same polymer contains quaternary ammonium groups such as $-\text{NH}_3^+$. Membranes are 0.2–0.5 mm in thickness and reinforced with a screen to provide mechanical stability. The membranes, which are made in flat sheets, contain 30 to 50% water and have a network of pores too small to permit water transport.

A cell pair or unit cell consists of one cation-selective membrane and one anion-selective membrane. Although Figure 14.14 shows an electrodialysis system with two cell pairs, a commercial electrodialysis system is a large stack of membranes patterned after a plate-and-frame configuration that, according to Applegate [2] and the *Membrane Handbook* [6], may contain 100 to 600 cell pairs. In a stack, membranes of from 0.4 to 1.5 m² surface area each are separated by from 0.5 to 2 mm with spacer gaskets. The total voltage or electrical potential applied across the cell includes (1) the electrode potentials discussed earlier, (2) overvoltages due to gas formation at the two electrodes, (3) the voltage required to overcome the ohmic resistance of the electrolyte in each compartment, (4) the voltage required to overcome the resistance in each membrane, and (5) the voltage required to overcome concentration-polarization effects caused by mass-transfer resistances in the electrolyte solutions adjacent to the membrane surface. For large stacks, the latter three voltage increments predominate and depend upon the current density (amps flowing through the stack per unit surface area of membranes). A typical voltage drop across a cell pair is 0.5–1.5 V. Current densities are in the range of 5–50 mA/cm². Thus, a stack of 400 membranes (200 unit cells) of 1 m² surface area each might require 200 V at 100 A. Typically 50 to 90% of brackish water is converted to potable water, depending on concentrate recycle.

As the current density is increased for a given membrane surface area, the concentration-polarization effect increases.

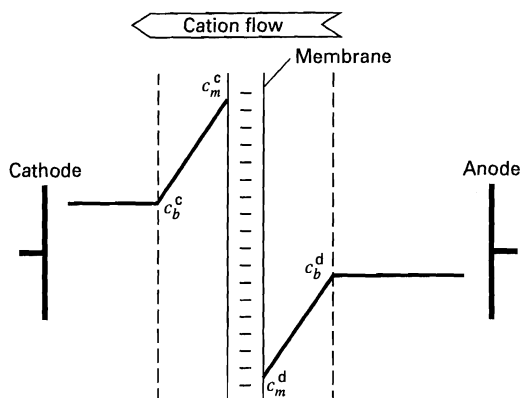


Figure 14.16 Concentration-polarization effects for a cation-exchange membrane.

[From H. Strathmann, *Sep. and Purif. Methods*, 14(1), 41–66 (1985) with permission.]

A schematic diagram of this effect for a single cation-selective membrane is shown in Figure 14.16, where c_m refers to cation concentrations in the membrane, c_b refers to bulk electrolyte cation concentrations, and superscripts c and d refer to concentrate side and dilute side, respectively. The maximum or limiting current density is reached when c_m^d reaches zero. Typically, an electrodialysis cell is operated at 80% of the limiting current density, which is determined by experiment. The corresponding cell voltage or resistance is also determined experimentally.

The amounts of gases formed at the electrodes at the two ends of the stack are governed by *Faraday's law of electrolysis*. During electrolysis, one Faraday (96,520 coulombs) of electricity reduces at the cathode and oxidizes at the anode an equivalent of oxidizing and reducing agent corresponding to the transfer of 6.023×10^{23} (Avogadro's number) electrons through wiring from the anode to the cathode. In general, it takes a very large quantity of electricity to form appreciable quantities of gases in an electrodialysis process.

Of more importance in the design or operation of an electrodialysis process are the membrane area and electrical-energy requirements as discussed by Applegate [2] and Strathmann [36]. The membrane area is estimated from the current density, rather than from a permeability and mass-transfer resistances, by applying Faraday's law:

$$A_M = \frac{FQ\Delta c}{i\xi} \quad (14-58)$$

where

A_M = total area of all cell pairs, m^2

F = Faraday's constant (96,520 amp-s/equivalent)

Q = volumetric flow rate of the diluate (potable water), m^3/s

Δc = difference between feed and diluate ion concentration in equivalents/ m^3

i = current density, amps/ m^2 of a cell pair, usually about 80% of i_{max}

ξ = current efficiency < 1.00

The last variable accounts for the fact that not all of the current is effective in transporting the selected ions through the membranes. Inefficiencies are caused by a Donnan exclusion of less than 100%, some transfer of water through the membranes, current leakage through manifolds, etc.

Power consumption is given by

$$P = IE \quad (14-59)$$

where

P = power, W, I = electric current flow through the stack, and E = voltage across the stack.

The electrical-current flow is given by a rearrangement of (14-58):

$$I = \frac{FQ\Delta c}{n\xi} \quad (14-60)$$

where n is the number of cell pairs.

The main application of electrodialysis is to the desalinization of brackish water in the salt concentration range of 500 to 5,000 ppm (mg/L). Below this range, ion exchange is more economical, whereas above this range, to 50,000 ppm, reverse osmosis is preferred. However, electrodialysis cannot produce water with a very low dissolved-solids content because of the high electrical resistance of dilute solutions. Other applications include recovery of nickel and copper from electroplating rinse water; deionization of cheese whey, fruit juices, wine, milk, and sugar molasses; separation of salts, acids, and bases from organic compounds; and recovery of organic compounds from their salts. Bipolar membranes, prepared by laminating a cation-selective membrane and an anion-selective membrane back-to-back, can be used to produce sulfuric acid and sodium hydroxide from a sodium sulfate solution.

EXAMPLE 14.9

Estimate membrane area and electrical-energy requirements for an electrodialysis process to reduce the salt (NaCl) content of 24,000 m^3 /day of brackish water from 1,500 mg/L to 300 mg/L with a 50% conversion. Assume each membrane has a surface area of 0.5 m^2 and each stack contains 300 cell pairs. A reasonable current density is 5 mA/ cm^2 and the current efficiency is 0.8 (80%).

SOLUTION

Use (14-58) to estimate membrane area

$$F = 96,520 \text{ A/equiv,}$$

$$Q = (24,000)(0.5)/(24)(3,600) = 0.139 \text{ m}^3/\text{s}$$

$$MW_{NaCl} = 58.5, i = 5 \text{ mA/cm}^2 = 50 \text{ A/m}^2$$

$$\Delta c = (1,500 - 300)/58.5 = 20.5 \text{ mmol/L or } 20.5 \text{ mol/m}^3 = 20.5 \text{ equiv/m}^3$$

$$A_M = \frac{(1)(96,520)(0.139)(20.5)}{(50)(0.8)} = 6,876 \text{ m}^2$$

Each stack contains 300 cell pairs with a total area of $0.5(300) = 150 \text{ m}^2$. Therefore, number of stacks = $6,876/150 = 46$ in parallel.

From (14-60), electrical current flow is given by

$$I = \frac{(96,500)(0.139)(20.5)}{(300)(0.8)} \\ = 1,146 \text{ A or } I/\text{stack} = 1,146/46 = 25 \text{ A/stack}$$

To obtain the electrical power, we need to know the average voltage drop across each cell pair. Assume a value of 1 V. From (14-59) for 300 cell pairs.

$$P = (1,146)(1)(300) = 344,000 \text{ W} = 344 \text{ kW}$$

Additional energy is required to pump feed, recycle concentrate, and electrode rinse.

It is also instructive to estimate the amount of feed that would be electrolyzed (say, as water to hydrogen and oxygen gases) at the electrodes. From the half-cell reactions presented earlier, half a molecule of H_2O is electrolyzed for each electron or, 0.5 mol H_2O is electrolyzed for each faraday of electricity.

1,146 amps = 1,146 coulombs/s or $(1,146)(3,600)(24) = 99,010,000$ coulombs/day or $99,010,000/96,520 = 1,026$ faradays/day. This electrolyzes $(0.5)(1,026) = 513$ mol/day of water. The feed rate is $12,000 \text{ m}^3/\text{day}$, or

$$\frac{(12,000)(10^6)}{18} = 6.7 \times 10^8 \text{ mol/day}$$

Therefore, the amount of water electrolyzed is negligible.

14.5 REVERSE OSMOSIS

Osmosis, from the Greek word for “push,” refers to the passage of a solvent, such as water, through a membrane that is much more permeable to the solvent (A) than to the solute(s) (B) (e.g., inorganic ions). The first recorded account of osmosis was given in 1748 by Nollet, whose experiments were conducted with water, an alcohol, and an animal-bladder membrane. The important aspects of osmosis are illustrated by example in Figure 14.17, where all solutions are at 25°C . In the initial condition (a), seawater of approximately 3.5 wt% dissolved salts and at 101.3 kPa is on the left side of the membrane, while pure water at the same pressure is on the right side. The dense membrane is permeable to water, but not to the dissolved salts. By osmosis, water passes from the right side to the seawater on the left side, causing dilution with respect to dissolved salts. At

equilibrium, the condition of Figure 14.17b is reached, wherein some pure water still resides on the right side and seawater, less concentrated in salt, resides on the left side. The pressure, P_1 , on the left side is now greater than the pressure, P_2 , on the right side, with the difference, π , referred to as the *osmotic pressure*.

The process of osmosis is not useful as a separation process because the solvent is transferred in the wrong direction, resulting in mixing rather than separation. However, the direction of transfer of solvent through the membrane can be reversed, as shown in Figure 14.17c by applying a pressure, P_1 , on the left side of the membrane, that is higher than the sum of the osmotic pressure and the pressure, P_2 , on the right side: that is, $P_1 - P_2 > \pi$. Now water in the seawater is transferred to the pure water, and the seawater becomes more concentrated in dissolved salts. This phenomenon, called *reverse osmosis*, can be used to partially remove a solvent from a solute-solvent mixture. As discussed later, an important factor in developing a reverse osmosis separation process is the osmotic pressure, π , of the feed mixture. In general, as discussed in more detail later, π is proportional to the solute concentration. For pure water, $\pi = 0$.

In a reverse-osmosis (RO) membrane-separation process, as shown in Figure 14.18, the feed is a liquid at high pressure, P_1 , containing solvent (e.g., water) and solubles (e.g., inorganic salts and, perhaps, colloidal matter). No sweep liquid is used, but the other side of the membrane is maintained at a much lower pressure, P_2 . A dense membrane, such as an acetate or aromatic polyamide, is used that is permselective for the solvent. To withstand the large pressure differential, the membrane must be thick. Accordingly, asymmetric or thin-wall composite membranes, having a thin, dense skin or layer on a thick, porous support, are used. The products of reverse osmosis are a permeate of almost pure solvent and a retentate of solvent-depleted feed. However, a perfect separation between the solvent and solute is not achieved, since only a fraction of the solvent in the feed is transferred to the permeate.

Reverse osmosis is applied to the desalinization and purification of seawater, brackish water, and wastewater. Prior to 1980, multistage, flash distillation was the main process for the desalinization of water. By 1990, this situation was dramatically reversed, making RO the dominant process for new construction. The dramatic shift from a thermally

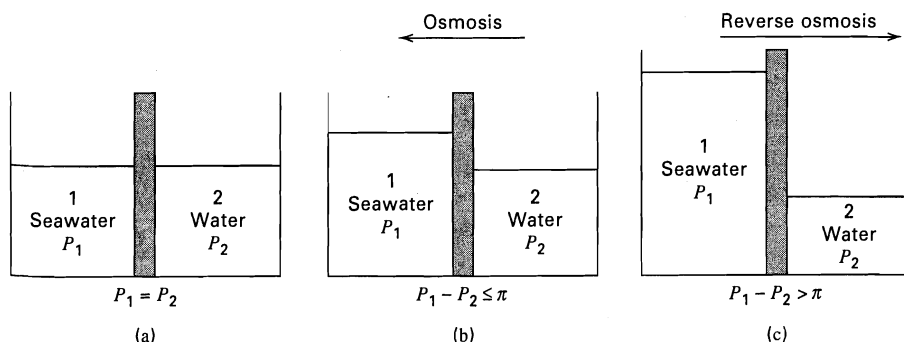


Figure 14.17 Osmosis and reverse-osmosis phenomena. (a) Initial condition. (b) At equilibrium after osmosis. (c) Reverse osmosis.

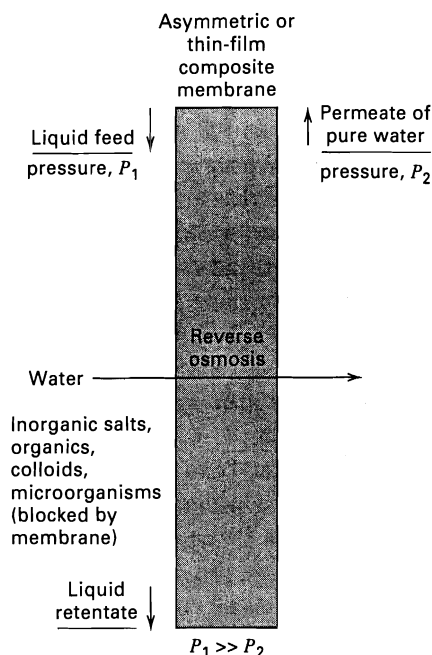


Figure 14.18 Reverse osmosis.

driven process to a more economical pressure-driven process was made possible through the development by Loeb and Sourirajan [7] of an asymmetric membrane that allows pressurized water to pass through at a high rate, while almost preventing transmembrane flows of dissolved salts, organic compounds, colloids, and microorganisms. Today more than 1,000 RO desalting plants are producing more than 750,000,000 gallons per day of potable water worldwide.

According to Baker et al. [5], the use of RO to desalinate water is accomplished mainly with spiral-wound and hollow-fiber membrane modules utilizing cellulose triacetate, cellulose diacetate, and aromatic polyamide membrane materials. Cellulose acetates are susceptible to biological attack, and acidic or basic hydrolysis back to cellulose, making it necessary to chlorinate the feed water and control the pH within the range of 4.5 to 7.5. Polyamides are not susceptible to biological attack and resist hydrolysis in the pH range of 4 to 11. However, polyamides are attacked by chlorine.

The preferred membrane for the desalination of seawater, which contains about 3.5 wt% dissolved salts and has an osmotic pressure of 350 psia, is a spiral-wound, multileaf module of polyamide, thin-film composite operating at a feed pressure of 800 to 1,000 psia. With a transmembrane water flux of 9 gal/ft²-day (0.365 m³/m²-day), this module can recover 45% of the water at a purity of about 99.95 wt%. A typical cylindrical module is 8 inches in diameter by 40 inches long, containing 365 ft² (33.9 m²) of membrane surface. Such modules resist fouling by colloidal and particulate matter, but the seawater must be treated with sodium bisulfate to remove oxygen and/or chlorine.

For the desalination of brackish water containing less than 0.5 wt% dissolved salts, hollow-fiber modules of high packing density, and containing fibers of cellulose acetates or aromatic polyamides, are used if fouling is not serious.

Because the osmotic pressure is much lower (<50 psi), feed pressures can be less than 250 psia. Transmembrane fluxes may be as high as 20 gal/ft²-day.

Other uses of reverse osmosis, usually on a smaller scale than the desalination of water to produce potable water, include (1) the treatment of industrial wastewater to remove heavy-metal ions, nonbiodegradable substances, and other components of commercial value; (2) the treatment of rinse water from electroplating processes to obtain a metal-ion concentrate and a permeate that can be reused as a rinse; (3) the separation of sulfites and bisulfites from effluents in pulp and paper processes; (4) the treatment of wastewater in dyeing processes; (5) the recovery of constituents having food value from wastewaters in food-processing plants (e.g., lactose, lactic acid, sugars, and starches); (6) the treatment of municipal water to remove inorganic salts, low-molecular-weight organic compounds, viruses, and bacteria; (7) the dewatering of certain food products such as coffee, soups, tea, milk, orange juice, and tomato juice; and (8) the concentration of amino acids and alkaloids. In such applications, membranes must have chemical, mechanical, and thermal stability to be competitive with other processes.

As with all membrane processes where the feed being separated is a liquid, three resistances to mass transfer must be considered: the membrane resistance and the two fluid-film or boundary-layer resistances on either side of the membrane. If the permeate is pure solvent, then there is no film resistance on that side of the membrane.

Although the driving force for the transport of water through the dense membrane is the concentration or activity difference in and across the membrane, common practice is to use a driving force based on osmotic pressure. Consider the reverse-osmosis process of Figure 14.17c. At thermodynamic equilibrium, solvent chemical potentials of fugacities on the two sides of the membrane must be equal. Thus,

$$f_A^{(1)} = f_A^{(2)} \quad (14-61)$$

From definitions in Table 2.2, rewrite (14-61) in terms of activities:

$$a_A^{(1)} f_A^0\{T, P_1\} = a_A^{(2)} f_A^0\{T, P_2\} \quad (14-62)$$

For pure solvent, A, $a_A^{(2)} = 1$. For seawater, $a_A^{(1)} = x_A^{(1)} \gamma_A^{(1)}$. Substitution into (14-62) gives

$$f_A^0\{T, P_2\} = x_A^{(1)} \gamma_A^{(1)} f_A^0\{T, P_1\} \quad (14-63)$$

Standard-state, pure-component fugacities f_A^0 increase with increasing pressure. Thus, if $x_A^{(1)} \gamma_A^{(1)} < 1$, then from (14-63), $P_1 > P_2$. The pressure difference $P_1 - P_2$ is shown as a hydrostatic-head difference in Figure 14.17b. This difference, which can be observed experimentally, is defined as the osmotic pressure, π .

To relate π to solvent or solute concentration, we apply the Poynting correction of (2-28), which for an incompressible liquid of specific volume, v_A , gives

$$f_A^0\{T, P_2\} = f_A^0\{T, P_1\} \exp \left[\frac{v_{A,L}(P_2 - P_1)}{RT} \right] \quad (14-64)$$

Substitution of (14-63) into (14-64) gives

$$\pi = P_1 - P_2 = -\frac{RT}{v_{A_L}} \ln(x_A^{(1)} \gamma_A^{(1)}) \quad (14-65)$$

Thus, osmotic pressure is a thermodynamic quantity that replaces activity.

For a mixture, on the feed or retentate side of the membrane, that is dilute in the solute, $\gamma_A^{(1)} = 1$. Also, $x_A^{(1)} = 1 - x_B^{(1)}$ and $\ln(1 - x_B^{(1)}) \approx -x_B^{(1)}$. Substitution into (14-65) gives

$$\pi = P_1 - P_2 = RT x_B^{(1)} / v_{A_L} \quad (14-66)$$

Finally, since $x_B^{(1)} \approx n_B / n_A$, $n_A v_{A_L} = V$, and $n_B / V = c_B$, (14-66) becomes

$$\pi \approx RT c_B \quad (14-67)$$

which was cited in Exercise 1.8. For applications to the reverse osmosis of seawater, Applegate [2] suggests the approximate expression

$$\pi = 1.12T \sum \bar{m}_i \quad (14-68)$$

where π is in psia, T is in K, and $\sum \bar{m}_i$ is the summation of molarities of all dissolved ions and nonionic species in the solution in mol/L. More exact expressions for estimating π are those of Stoughton and Lietzke [38].

In the general case, when reverse osmosis takes place with solute on each side of the membrane, then at equilibrium, $(P_1 - \pi_1) = (P_2 - \pi_2)$. Accordingly, as discussed by Merten [37], the driving force for solvent transport through the membrane is $\Delta P - \Delta\pi$, and the rate of mass transport is

$$N_{H_2O} = \frac{P_{M_{H_2O}}}{l_M} (\Delta P - \Delta\pi) \quad (14-69)$$

where

ΔP = hydraulic pressure difference across the membrane
 $= P_{\text{feed}} - P_{\text{permeate}}$

$\Delta\pi$ = osmotic pressure difference across the membrane
 $= \pi_{\text{feed}} - \pi_{\text{permeate}}$

Often, $\pi_{\text{permeate}} \approx 0$ because the permeate is almost pure solvent.

The flux of solute (e.g., salt) is given by (14-26) in terms of membrane concentrations, and thus is independent of the ΔP across the membrane. Accordingly, the higher the ΔP , the purer the permeate water. Alternatively, the flux of salt may be conveniently expressed in terms of *salt passage*, SP , defined by

$$SP = (c_{\text{salt}})_{\text{permeate}} / (c_{\text{salt}})_{\text{feed}} \quad (14-70)$$

Values of SP decrease with increasing ΔP . *Salt rejection* is given by $SR = 1 - SP$.

For brackish water of 1,500 mg/L as NaCl, at 25°C, (14-68) predicts $\pi = 17.1$ psia. For seawater of 35,000 mg/L as NaCl, at 25°C, (14-68) predicts $\pi = 385$ psia, while Stoughton and Lietzke [38] give 368 psia. From (14-69),

ΔP must be greater than $\Delta\pi$ for reverse osmosis to occur. For the desalinization of brackish water by RO, ΔP is typically 400–600 psi, while for seawater, it is 800–1,000 psi.

The feed water to an RO unit may contain a variety of potential foulants, which are removed prior to passage of the feed through the membrane unit. Otherwise the foulants can adversely affect the performance and reduce the useful life of the membrane. Suspended solids and particulate matter can be removed by screening and filtration. Colloids can be flocculated and filtered. Scale-forming salts require acidification or water softening. Biological materials require chlorination or ozonation. Other organic foulants are removed by adsorption or oxidation.

Concentration polarization is particularly important on the feed side of reverse-osmosis membranes. This effect is illustrated in Figure 14.19, where typical concentrations are shown for water, c_w , and salt, c_s . Because of the high pressure, the activity of water on the feed side is somewhat higher than that of near-pure water on the permeate side, thus providing the necessary driving force for water transport through the membrane. The flux of water to the membrane carries with it salt by bulk flow. However, because the salt cannot readily penetrate the membrane, the concentration of the salt in the liquid adjacent to the surface of the membrane, c_{s_i} , is greater than that in bulk of the feed, c_{s_F} . This difference causes mass transfer of salt by diffusion from the membrane surface back to the bulk feed. The back rate of salt diffusion depends on the mass-transfer coefficient for the film or boundary layer on the feed side. The lower the mass-transfer coefficient, the higher the value of c_{s_i} . The value of c_{s_i} is important because it fixes the osmotic pressure, and influences the driving force for water transport according to (14-69).

Consider steady-state transport of water with back-diffusion of salt. A salt balance at the upstream membrane surface gives

$$N_{H_2O} c_{s_F} (SR) = k_s (c_{s_i} - c_{s_F}) \quad (14-71)$$

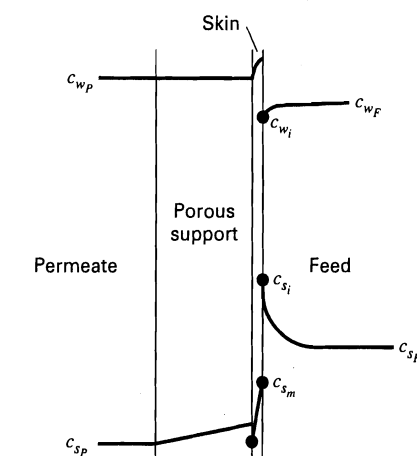


Figure 14.19 Concentration-polarization effects in reverse osmosis.

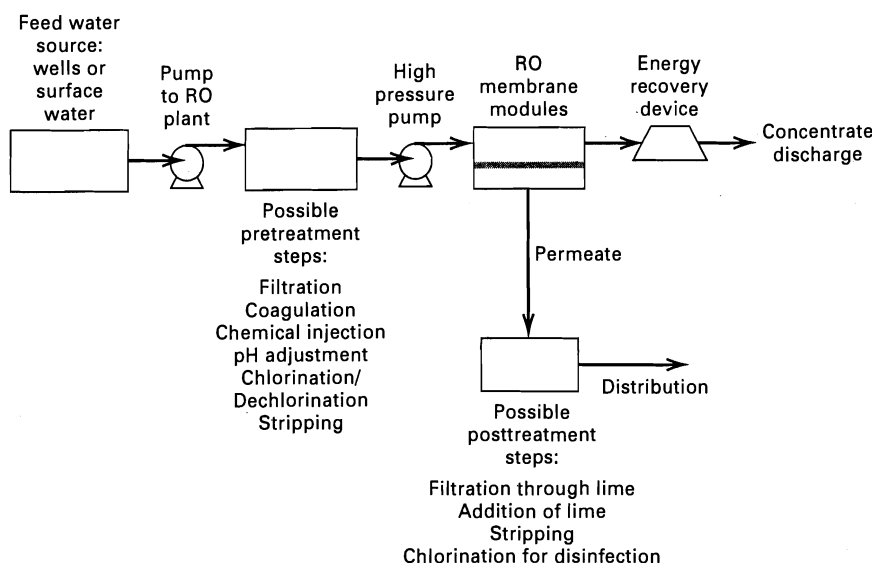


Figure 14.20 Reverse-osmosis process.

Solving for c_{s_i} gives

$$c_{s_i} = c_{s_F} \left(1 + \frac{N_{H_2O}(SR)}{k_s} \right)$$

Values of k_s are estimated from (14-54). As would be expected, the concentration-polarization effect is seen to be most significant for high water fluxes and low mass-transfer coefficients.

A quantitative estimate of the importance of concentration polarization can be derived by defining the following concentration polarization factor, Γ , in terms of (14-7):

$$\Gamma \equiv \frac{c_{s_i} - c_{s_F}}{c_{s_F}} = \frac{N_{H_2O}(SR)}{k_s} \quad (14-72)$$

Values of SR are typically in the range of 0.97 to 0.995. If Γ is greater than, say, 0.2, the effect of concentration polarization may be significant, indicating that design changes to reduce the value of Γ should be considered.

Feed-side pressure drop is also important because, by (14-69), it causes a reduction in the driving force for water transport. Because of the complex geometries used for both spiral-wound and hollow-fiber modules, it is best to estimate pressure drops from experimental data. Feedside pressure drops for spiral-wound modules and hollow-fiber modules range from 43 to 85 and 1.4 to 4.3 psi, respectively [6].

A schematic diagram of a typical reverse-osmosis process for the desalinization of water is shown in Figure 14.20. The source of feed water may be a well or surface water, which is pumped through a series of pretreatment steps to ensure a long membrane life. Of particular importance is pH adjustment. The pretreated water is then fed by a high-pressure-discharge pump to an appropriate parallel-and-series network of reverse-osmosis modules of the spiral-wound or hollow-fiber type. The concentrate, which leaves the membrane system at a high pressure that is 10–15% lower than the inlet pressure, is then routed through a power-recovery turbine, which reduces the net power consumption of the process by 25 to 40% while reducing the pressure of the concentrate to

an appropriate low level. The permeate, which may be 99.95 wt% pure water and about 50% of the feed water, is sent to a series of posttreatment steps before it is ready to drink.

EXAMPLE 14.10

At a certain location in a spiral-wound membrane, the bulk conditions on the feed side are 1.8 wt% NaCl, 25°C, and 1,000 psia, while bulk conditions on the permeate side are 0.05 wt% NaCl, 25°C, and 50 psia. For the particular membrane being used, the permeance values are 1.1×10^{-5} g/cm²-s-atm for H₂O and 16×10^{-6} cm/s for the salt. If mass-transfer resistances are negligible on each side of the membrane, calculate the flux of water in gal/ft²-day and the flux of salt in g/ft²-day. If $k_s = 0.005$ cm/s, estimate the polarization factor.

SOLUTION

Bulk salt concentrations are approximately

$$c_{s_F} = \frac{1.8(1,000)}{58.5(98.2)} = 0.313 \text{ mol/L on feed side}$$

$$c_{s_P} = \frac{0.05(1,000)}{58.5(99.95)} = 0.00855 \text{ mol/L on permeate side}$$

For water transport, using (14-68) for osmotic pressure, noting that dissolved NaCl gives 2 ions per molecule:

$$\Delta P = (1,000 - 50)/14.7 = 64.6 \text{ atm}$$

$$\pi_{\text{feedside}} = 1.12(298)(2)(0.313) = 209 \text{ psia} = 14.2 \text{ atm}$$

$$\pi_{\text{permeate side}} = 1.12(298)(2)(0.00855) = 5.7 \text{ psia} = 0.4 \text{ atm}$$

$$\Delta P - \Delta \pi = 64.6 - (14.2 - 0.4) = 50.8 \text{ atm}$$

$$P_{M_{H_2O}}/l_M = 1.1 \times 10^{-5} \text{ g/cm}^2\text{-s-atm}$$

From (14-69),

$$N_{H_2O} = (1.1 \times 10^{-5})(50.8) = 0.000559 \text{ g/cm}^2\text{-s} \quad \text{or}$$

$$\frac{(0.000559)(3,600)(24)}{(454)(8.33)(1.076 \times 10^{-3})} = 11.9 \text{ gal/ft}^2\text{-day}$$

For salt transport:

$$\Delta c = 0.313 - 0.00855 = 0.304 \text{ mol/L} \quad \text{or} \quad 0.000304 \text{ mol/cm}^3$$

$$P_{M_{NaCl}}/l_M = 16 \times 10^{-6} \text{ cm/s}$$

From (14-26):

$$N_{\text{NaCl}} = 16 \times 10^{-6} (0.000304) = 4.86 \times 10^{-9} \text{ mol/cm}^2\text{-s}$$

or
$$\frac{(4.86 \times 10^{-9})(3,600)(24)(58.5)}{1.076 \times 10^{-3}} = 22.8 \text{ g/ft}^2\text{-day}$$

We see that the flux of salt is very much smaller than the flux of water.

To estimate the concentration-polarization factor, first convert the water flux through the membrane into the same units as the salt mass-transfer coefficient, k_s , i.e., cm/s:

$$N_{\text{H}_2\text{O}} = \frac{0.000559}{1.00} = 0.000559 \text{ cm/s}$$

From (14-70), the salt passage is

$$SP = 0.00855/0.313 = 0.027$$

Therefore, the salt rejection = $SR = 1 - 0.027 = 0.973$

From (14-72), the concentration-polarization factor is

$$\Gamma = \frac{0.000559(0.972)}{0.005} = 0.11$$

Thus, here polarization is not particularly significant.

14.6 GAS PERMEATION

In gas permeation (GP), shown in Figure 14.21, the feed gas, at high pressure P_1 , contains some low-molecular-weight species (MW < 50) to be separated from small amounts of higher-molecular-weight species. Usually a sweep gas is not used, but the other side of the membrane is maintained at a much lower pressure, P_2 , often near-ambient pressure. The membrane, often dense but sometimes microporous, is permselective for certain of the low-molecular-weight species in the feed gas, shown in Figure 14.21 as the A species. If the membrane is dense, these species are

absorbed at the surface and then transported through the membrane by one or more mechanisms. Thus, permselectivity depends on both membrane absorption and the membrane transport rate. Usually all mechanisms are formulated in terms of a partial-pressure or fugacity driving force using the solution-diffusion model of (14-32). The products are a permeate that is enriched in the A species and a retentate that is enriched in B. A near-perfect separation is generally not achievable. If the membrane is microporous, as for example in high-temperature applications, pore size is extremely important because it is usually necessary to block the passage of species B. Otherwise, unless molecular weights of A and B differ appreciably, only a very modest separation is achievable, as was discussed in connection with Knudsen diffusion, (14-22).

Since the early 1980s, applications of GP with dense, polymeric membranes have increased dramatically. Applications include (1) separation of hydrogen from methane; (2) adjustment of H_2 -to-CO ratio in synthesis gas; (3) O_2 enrichment of air; (4) N_2 enrichment of air; (5) removal of CO_2 ; (6) drying of natural gas and air; (7) removal of helium; and (8) removal of organic solvents from air.

Gas permeation must compete with distillation at cryogenic conditions, absorption, and pressure-swing adsorption. Some of the advantages of gas permeation, as cited by Spillman and Sherwin [39], are low capital investment, ease of installation, ease of operation, absence of rotating parts, high process flexibility, low weight and space requirements, and low environmental impact. In addition, if the feed gas is already at so high a pressure that a gas compressor is not needed, then no utilities are required.

Since 1986, the most rapidly developing application for GP has been air separation, for which available membranes have separation factors for O_2 with respect to N_2 of 3 to 7. However, product purities are economically limited to a retentate of 95–99.9% N_2 and a permeate of 30–45% O_2 . Thus, the largest application of GP for air separation is the production of nitrogen rather than oxygen.

Gas permeation also competes very favorably with other separation processes for hydrogen recovery because of the high separation factors achieved. For example, the rate of permeation of hydrogen through a typical dense polymer membrane is more than 30 times that for nitrogen. A typical GP process might achieve a 95% recovery of 90% pure hydrogen from a feed gas containing 60% hydrogen.

Early applications of GP used dense (nonporous) membranes of cellulose acetates and polysulfones, which are still predominant, although polyimides, polyamides, polycarbonates, polyetherimides, sulfonated polysulfones, Teflon, polystyrene, and silicone rubber are also finding applications for temperatures to at least 70°C. Although plate-and-frame and tubular modules can be used for gas permeation, almost all large-scale applications use spiral-wound or hollow-fiber modules because of their higher packing density. Commercial membrane modules for gas permeation are available from more than 20 suppliers. Feed-side pressure is typically 300 to 500 psia, but is as high as 1,650 psia. Typical refinery

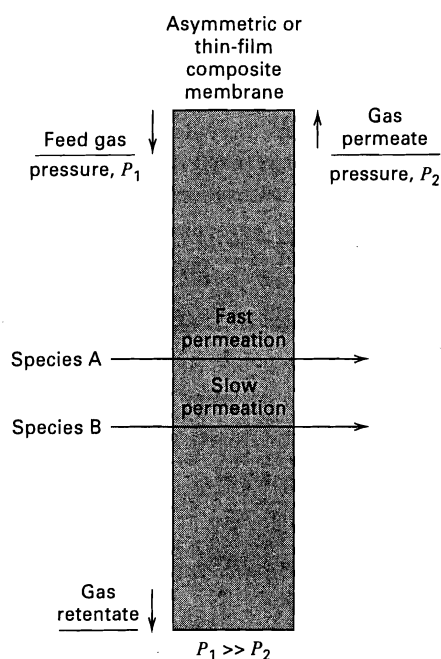


Figure 14.21 Gas permeation.

Table 14.9 Predicted Values of Diffusivity and Solubility of Light Gases in a Glassy and a Rubbery Polymer

Permeant	$D \times 10^{11}$, m ² /s	$S \times 10^4$, gmol/m ³ -Pa	P_M , barrer
Polyvinyltrimethylsilane (Glassy Polymer)			
He	470	0.18	250
Ne	87	0.26	66
Ar	5.1	1.95	30
Kr	1.5	6.22	29
Xe	0.29	20.6	18
Rn	0.07	69.6	15
H ₂	160	0.54	250
O ₂	7.6	1.58	37
N ₂	3.8	0.84	9
CO ₂	4.0	13.6	160
CO	3.7	1.28	14
CH ₄	1.9	3.93	22
C ₂ H ₆	0.12	30.2	10
C ₃ H ₈	0.01	98.1	2.8
C ₄ H ₁₀	0.001	347	1.2
C ₂ H ₄	0.23	17.8	12
C ₃ H ₆	0.038	77.6	9
C ₄ H ₈ (1)	0.0052	293	4.5
C ₂ H ₂	0.58	16.8	32
C ₃ H ₄ (m)	0.17	138.1	70
C ₄ H ₆ (e)	0.053	318.5	50
C ₃ H ₄ (a)	0.15	186.5	83
C ₄ H ₆ (b)	0.03	226.1	20
Polyisoprene (Rubber-like Polymer)			
He	213	0.06	35
Ne	77.4	0.08	18
Ar	14.6	0.58	25
Kr	7.2	1.78	25
Xe	2.7	5.68	45
Rn	1.2	18.7	64
H ₂	109	0.17	54
O ₂	18.4	0.47	26
N ₂	12.2	0.26	10
CO ₂	12.6	3.80	140
CO	12.1	0.38	14
CH ₄	8.0	1.14	27
C ₂ H ₆	3.3	8.13	79
C ₃ H ₈	1.6	25.4	123
C ₄ H ₁₀	1.5	86.4	390
C ₂ H ₄	4.3	4.84	62
C ₃ H ₆	2.7	20.3	163
C ₄ H ₈ (1)	1.5	73.3	333
C ₂ H ₂	5.7	4.64	80
C ₃ H ₄ (m)	4.1	35.3	433
C ₄ H ₆ (e)	2.9	79.6	690
C ₃ H ₄ (a)	4.5	47.4	640
C ₄ H ₆ (b)	3.4	40.0	410

Note: m, methylacetylene; e, ethylacetylene; a, allene; b, butadiene.

applications involve feed-gas flow rates of 20 million scfd, but flow rates as large as 300 million scfd have been reported [40]. When the feed gas contains condensables, it may be necessary to preheat the gas prior to entry into the membrane system to prevent condensation on the membrane as the retentate becomes richer in the high-molecular-weight species. For high-temperature applications where polymers cannot be used, membranes of glass, carbon, and inorganic oxides are available, but are limited in their selectivity.

For dense membranes, external mass-transfer resistances or concentration-polarization effects are generally negligible, and (14-32) with a partial-pressure driving force can be used to compute the rate of species transport through the membrane. As discussed earlier in the subsection on module flow patterns, the appropriate partial-pressure driving force depends on the flow pattern. Cascades of the type discussed earlier are used to increase the degree of separation.

Progress is being made in the development of a method for the prediction of permeability of gases in glassy and rubbery homopolymers, random copolymers, and block copolymers. Teplyakov and Meares [41] present correlations at 25°C for the diffusion coefficient, D , and solubility, S , applied to 23 different gases for 30 different polymers. Predicted values for glassy polyvinyltrimethylsilane (PVTMS) and rubbery polyisoprene are listed in Table 14.9. Typically, D and S agree with experimental data to within $\pm 20\%$ and $\pm 30\%$, respectively.

Gas permeation separators are claimed to be relatively insensitive to changes in feed flow rate, feed composition, and loss of membrane surface area [42]. This claim is tested in the following example.

EXAMPLE 14.11

The feed to a membrane separator consists of 500 lbmol/h of a mixture of 90% H₂ (H) and 10% CH₄ (M) at 500 psia. Permeance values based on a partial-pressure driving force are

$$\bar{P}_{MH} = 3.43 \times 10^{-4} \text{ lbmol/h-ft}^2\text{-psi}$$

and
$$\bar{P}_{MM} = 5.55 \times 10^{-5} \text{ lbmol/h-ft}^2\text{-psi}$$

The flow patterns in the separator are such that the permeate side is well mixed and the feed side is in plug flow. The pressure on the permeate side is constant at 20 psia and there is no pressure drop on the feed side.

(a) Compute the membrane area and permeate purity if 90% of the hydrogen is transferred to the permeate.

(b) For the membrane area determined in part (a), calculate the permeate purity and hydrogen recovery if

(1) the feed rate is increased by 10%.

(2) the feed composition is reduced to 85% H₂.

(3) 25% of the membrane area becomes inoperative.

SOLUTION

The following independent equations apply to all parts of this example. Component material balances:

$$n_{iF} = n_{iR} + n_{iP}, \quad i = \text{H, M} \quad (1,2)$$

Dalton's law of partial pressures:

$$P_k = p_{H_k} + p_{M_k}, \quad k = F, R, P \quad (3,4,5)$$

Partial pressure-mole relations:

$$p_{H_k} = P_k n_{H_k} / (n_{H_k} + n_{M_k}), \quad k = F, R, P \quad (6,7,8)$$

Solution-diffusion transport rates are obtained using (14-32), assuming a log-mean partial-pressure driving force based on the exiting permeate partial pressures on the downstream side of the membrane because of the assumption of perfect mixing on that side:

$$n_{i_P} = \bar{P}_{M_i} A_M \left[\frac{p_{i_F} - p_{i_R}}{\ln \left(\frac{p_{i_F} - p_{i_P}}{p_{i_R} - p_{i_P}} \right)} \right], \quad i = H, M \quad (9,10)$$

Thus, we have a system of 10 equations in the following 18 variables:

$$\begin{array}{cccccc} A_M & n_{H_F} & n_{M_F} & P_F & P_R & P_P \\ \bar{P}_{M_H} & n_{H_R} & n_{M_R} & p_{H_F} & p_{H_R} & p_{H_P} \\ \bar{P}_{M_M} & n_{H_P} & n_{M_P} & p_{M_F} & p_{M_R} & p_{M_P} \end{array}$$

Thus, eight variables must be fixed. For all parts of this example, the following five variables are fixed:

$$\bar{P}_{M_H} \text{ and } \bar{P}_{M_M} \text{ given above}$$

$$P_F = 500 \text{ psia} \quad P_R = 500 \text{ psia} \quad P_P = 20 \text{ psia}$$

For each part, three additional variables must be fixed.

(a)

$$n_{H_F} = 0.9(500) = 450 \text{ lbmol/h}$$

$$n_{M_F} = 0.1(500) = 50 \text{ lbmol/h}$$

$$n_{H_P} = 0.9(450) = 405 \text{ lbmol/h}$$

Solving Equations (1)–(10) above, using a PC program such as MathCad, Matlab, or Polymath, we obtain

$$A_M = 3,370 \text{ ft}^2$$

$$n_{M_P} = 20.0 \text{ lbmol/h} \quad n_{H_R} = 45.0 \text{ lbmol/h} \quad n_{M_R} = 30.0 \text{ lbmol/h}$$

$$p_{H_F} = 450 \text{ psia} \quad p_{M_F} = 50 \text{ psia} \quad p_{H_R} = 300 \text{ psia}$$

$$p_{M_R} = 200 \text{ psia} \quad p_{H_P} = 19.06 \text{ psia} \quad p_{M_P} = 0.94 \text{ psia}$$

(b) Calculations are made in a similar manner using Equations (1)–(10). Results for Parts (1), (2), and (3) are:

	Part		
	(1)	(2)	(3)
Fixed:			
n_{H_F} , lbmol/h	495	425	450
n_{M_F} , lbmol/h	55	75	50
A_M , ft ²	3,370	3,370	2,528
Calculated, in lbmol/h:			
n_{H_P}	424.2	369.6	338.4
n_{M_P}	18.2	25.9	11.5
n_{H_R}	70.8	55.4	111.6
n_{M_R}	36.8	49.1	38.5
Calculated, in psia:			
p_{H_F}	450	425	450
p_{M_F}	50	75	50
p_{H_R}	329	265	372
p_{M_R}	171	235	128
p_{H_P}	19.18	18.69	19.34
p_{M_P}	0.82	1.31	0.66

From the above results, the following are computed:

	Part			
	(a)	(b1)	(b2)	(b3)
Mol% H ₂ in permeate	95.3	95.9	93.5	96.7
% H ₂ recovery in permeate	90	85.7	87.0	75.2

From these results, we see that when the feed rate is increased by 10% (Part b1), the hydrogen recovery drops about 5%, but the permeate purity is maintained. When the feed composition is reduced from 90% to 85% hydrogen (Part b2), the hydrogen recovery decreases by about 3% and the permeate purity decreases by about 2%. With 25% of the membrane area inoperative (Part b3), the hydrogen recovery decreases by about 15%, but the permeate purity is about 1% higher. Overall, percentage changes in hydrogen recovery and purity are less than the percentage changes in feed flow rate, feed composition, and membrane area, thus tending to confirm the insensitivity of gas permeation separators to changes in operating conditions.

14.7 PERVAPORATION

As shown in Figure 14.22, pervaporation (PV) differs from dialysis, reverse osmosis, and gas permeation in that the phase state on one side of the membrane is different from that on the other side. The feed to the membrane module is a liquid mixture (e.g., an alcohol–water azeotrope) at a pressure, P_1 , that is usually ambient or elevated high enough to maintain a liquid phase as the feed is depleted of species A and B to produce the product retentate. A composite membrane is used that is selective for species A, but species B usually has some finite permeability. The dense, thin membrane film is in contact with the liquid side. The retentate is enriched in species B. Generally, a sweep fluid is not used

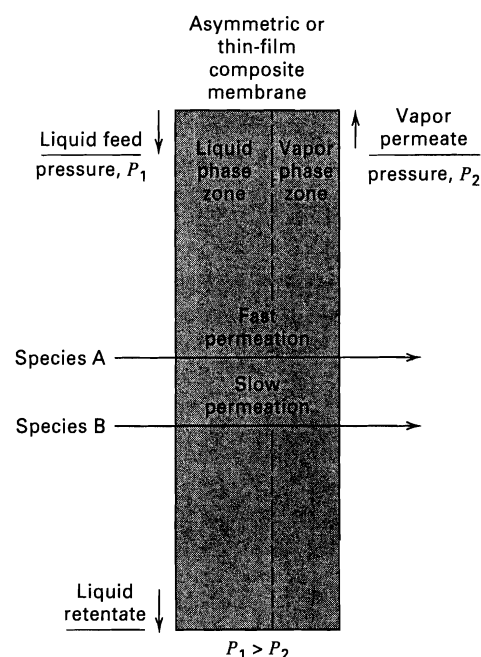


Figure 14.22 Pervaporation.

on the other side of the membrane, but a pressure, P_2 , is maintained at or below the dew point of the permeate, making it vapor. Often, P_2 is a vacuum. Vaporization may occur near the downstream face of the membrane, such that the membrane can be considered to operate with two zones, a liquid-phase zone and a vapor-phase zone, as shown in Figure 14.22. Alternatively, the vapor phase may only exist on the permeate side of the membrane. The vapor permeate is enriched in species A. Overall permeabilities of species A and B depend upon their solubilities in and diffusion rates through the membrane. Generally, the solubilities cause the membrane to swell.

The term pervaporation is a combination of the two words, *permselective* and *evaporation*. It was first reported in 1917 by Kober [43], who studied several experimental techniques for removing water from albumin/toluene solutions. Although the economic potential of PV was shown by Binning et al. [44] in 1961, commercial applications were delayed until the mid-1970s, when adequate membrane materials first became available. Major commercial applications now

include (1) dehydration of ethanol; (2) dehydration of other organic alcohols, ketones, and esters; and (3) removal of organics from water. The separation of organic mixtures, e.g., benzene–cyclohexane, is receiving much attention.

Pervaporation is best applied when the feed solution is dilute in the main permeant because sensible heat of the feed mixture provides the enthalpy of vaporization of the permeant. If the feed is rich in the main permeant, a number of membrane stages may be needed, with a small amount of permeant produced per stage and reheating of the retentate between stages. Even when only one membrane stage is sufficient, the feed may be heated before entering the membrane module.

Many pervaporation separation schemes have been proposed [6], with three of the more important ones shown in Figure 14.23. A hybrid process for integrating distillation with pervaporation to produce 99.5 wt% ethanol from a feed of 60 wt% ethanol is shown in Figure 14.23a. The feed is sent to a distillation column operating at near-ambient pressure, where a bottoms product of nearly pure water and an

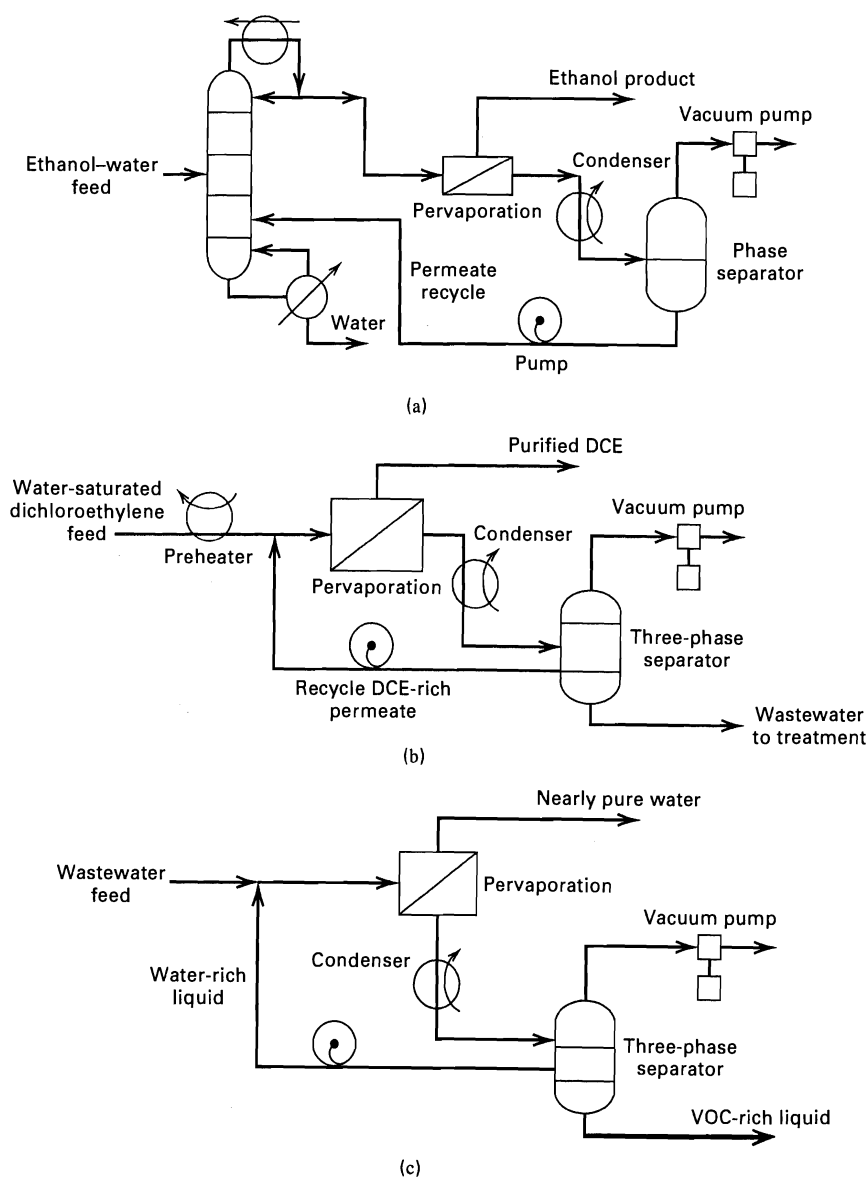


Figure 14.23 Pervaporation processes. (a) Hybrid process for removal of water from ethanol. (b) Dehydration of dichloroethylene. (c) Removal of volatile organic compounds (VOCs) from wastewater.

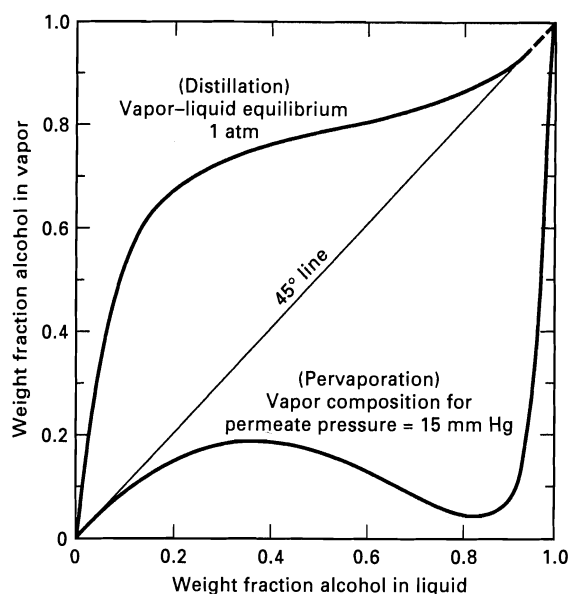


Figure 14.24 Comparison of ethanol–water separabilities.
[From M. Wesslein et al., *J. Membrane Sci.*, **51**, 169 (1990).]

ethanol-rich distillate of 95 wt% is produced. The distillate purity is limited because of the 95.6 wt% ethanol in water azeotrope. The distillate is sent to a pervaporation step where a permeate of 25 wt% alcohol and a retentate of 99.5 wt% ethanol is produced. The permeate vapor is condensed under vacuum and recycled to the distillation column. The vacuum is sustained with a vacuum pump. The dramatic difference in separability of the pervaporation membrane as compared to vapor–liquid equilibrium for distillation is shown with a 45° reference line in Figure 14.24 taken from Wesslein et al. [45]. For pervaporation, the compositions refer to a liquid feed (abscissa) and a vapor permeate (ordinate) at 60°C for a polyvinylalcohol (PVA) membrane and a vacuum of 15 torr. For this membrane, there is no limitation on ethanol purity and the separation index is very high for feeds containing more than 90 wt% ethanol.

A pervaporation process for dehydrating dichloroethylene (DCE) is shown in Figure 14.23b. The liquid feed, which is DCE saturated with water (0.2 wt%), is preheated to 90°C at 0.7 atm and sent to a PVA membrane system, which produces a retentate of almost pure DCE (<10 ppm H₂O) and a permeate vapor of 50 wt% DCE under vacuum. Following condensation, the two resulting liquid phases are separated, with the DCE-rich phase recycled back to the membrane system and the water-rich phase sent to an air stripper, steam stripper, adsorption unit, or hydrophobic, pervaporation, membrane system for residual DCE removal.

Pervaporation can be used for the removal of VOCs (e.g., toluene and trichloroethylene) from wastewater by pervaporation with hollow-fiber modules of silicone rubber, as shown in Figure 14.23c. The retentate is almost pure water (< 5 ppb of VOCs) and the permeate, after condensation, is

(1) a water-rich phase that is recycled to the membrane system and (2) a nearly pure VOC phase.

A pervaporation module may operate with heat transfer or adiabatically with the enthalpy of vaporization supplied by sensible enthalpy of the feed. Consider the adiabatic pervaporation of a binary liquid mixture of components A and B. Assume constant pure-component liquid specific heats, and ignore heat of mixing. For an enthalpy datum temperature of T_0 , an enthalpy balance, in terms of mass flow rates, m , liquid sensible heats, C_P , and heats of vaporization, ΔH^{vap} , gives

$$\begin{aligned} (m_{A_F} C_{P_A} + m_{B_F} C_{P_B})(T_F - T_0) \\ = [(m_{A_F} - m_{A_P}) C_{P_A} + (m_{B_F} - m_{B_P}) C_{P_B}](T_R - T_0) \\ + (m_{A_P} C_{P_A} + m_{B_P} C_{P_B})(T_P - T_0) + m_{A_P} \Delta H_A^{\text{vap}} \\ + m_{B_P} \Delta H_B^{\text{vap}} \end{aligned} \quad (14-73)$$

where enthalpies of vaporization are evaluated at T_P . After collection of terms, (14-73) reduces to

$$\begin{aligned} (m_{A_F} C_{P_A} + m_{B_F} C_{P_B})(T_F - T_R) = (m_{A_P} C_{P_A} + m_{B_P} C_{P_B}) \\ \times (T_P - T_R) + (m_{A_P} \Delta H_A^{\text{vap}} + m_{B_P} \Delta H_B^{\text{vap}}) \end{aligned} \quad (14-74)$$

The temperature of the permeate, T_P , is the permeate dew point at the permeate vacuum upstream of the condenser. The retentate temperature is computed from (14-74).

Membrane selection is critical in the commercial application of PV, when used in the presence of organic compounds. For water permeation, hydrophilic membrane materials are preferred. For example, a three-layer membrane is often used for the dehydration of ethanol, with water being the main permeating species. The support layer is porous polyester, which is cast on a microporous polyacrylonitrile or polysulfone membrane. The final layer, which provides the separation, is dense PVA of 0.1 μm in thickness. This composite combines chemical and thermal stability with adequate permeability. Hydrophobic membranes, such as silicone rubber and Teflon, are preferred when organics are the permeating species.

Commercial membrane modules for PV are almost exclusively of the plate-and-frame type because of the ease of using gasketing materials that are resistant to organic solvents and the ease of providing heat exchange for evaporation and high-temperature operation. However, considerable interest is evident in the use of hollow-fiber modules for the removal of VOCs from wastewater. Because feeds are generally clean and operation is at low pressure, membrane fouling and damage can be minimal, resulting in a useful membrane life of 2–4 years.

Various models for the transport of a permeant through a membrane by pervaporation have been proposed, based on the solution-diffusion model. They all assume equilibrium between the upstream liquid and the upstream membrane surface, and between the downstream vapor and the other side of the membrane. Transport through the membrane follows Fick's law with a concentration gradient of the permeant in

the membrane as the driving force. However, because of the phase change and nonideal-solution effects in the liquid feed, simple equations like (14-55) for dialysis and (14-32) for gas permeation do not apply to pervaporation.

A particularly convenient PV model is that of Wijmans and Baker [46]. They express the driving force for permeation in terms of a partial-vapor-pressure difference. Because pressures on the both sides of the membrane are low, the gas phase follows the ideal-gas law. Therefore, at the upstream membrane surface (1), permeant activity for component i is expressed as

$$a_i^{(1)} = f_i^{(1)} / f_i^{(0)} = P_i^{(1)} / P_i^{s(1)} \quad (14-75)$$

where P_i^s is the vapor pressure at the feed temperature. The liquid on the upstream side of the membrane is generally nonideal. Thus, from Table 2.2:

$$a_i^{(1)} = \gamma_i^{(1)} x_i^{(1)} \quad (14-76)$$

Combining (14-75) and (14-76):

$$P_i^{(1)} = \gamma_i^{(1)} x_i^{(1)} P_i^{s(1)} \quad (14-77)$$

On the downstream vapor side of the membrane (2), the partial pressure is

$$P_i^{(2)} = y_i^{(2)} P_P^{(2)} \quad (14-78)$$

Thus, the driving force can be expressed as $(\gamma_i^{(1)} x_i^{(1)} P_i^{s(1)} - y_i^{(2)} P_P^{(2)})$.

The corresponding permeant flux, after dropping unnecessary superscripts, is

$$N_i = \frac{P_{M_i}}{l_M} (\gamma_i x_i P_i^s - y_i P_P) \quad (14-79)$$

or

$$N_i = \bar{P}_{M_i} (\gamma_i x_i P_i^s - y_i P_P) \quad (14-80)$$

where γ_i and x_i refer to the feed-side liquid, P_i^s is the vapor pressure at the feed-side temperature, y_i is the mole fraction in the permeant vapor, and P_P is the total permeant pressure.

Unlike gas permeation where P_{M_i} depends mainly on the permeant, the polymer, and temperature; the permeability for pervaporation depends additionally on the concentrations of permeants in the polymer, which can be large enough to cause polymer swelling and cross-diffusion effects. For a binary system it is best to back-calculate and correlate the permeant flux with feed composition at a given feed temperature and permeate pressure. Because of these nonideal effects, the selectivity can be a strong function of feed concentration and permeate pressure, causing inversion of selectivity in some cases, as illustrated in the following example.

EXAMPLE 14.12

Wesslein et al. [45] present the following experimental data for the pervaporation of liquid mixtures of ethanol (1) and water (2) at a feed temperature of 60°C for a permeate pressure of 76 mmHg, using a commercial polyvinylalcohol membrane:

wt% ethanol		Total Permeation Flux kg/m ² -h
Feed	Permeate	
8.8	10.0	2.48
17.0	16.5	2.43
26.8	21.5	2.18
36.4	23.0	1.73
49.0	22.5	1.46
60.2	17.5	0.92
68.8	13.0	0.58
75.8	9.0	0.40

At 60°C, vapor pressures are 352 and 149 mmHg for ethanol and water, respectively.

Liquid-phase activity coefficients at 60°C for the ethanol (1)–water (2) system are given by the van Laar equations:

$$\ln \gamma_1 = 1.6276 \left[\frac{0.9232x_2}{1.6276x_1 + 0.9232x_2} \right]^2$$

$$\ln \gamma_2 = 0.9232 \left[\frac{1.6276x_1}{1.6276x_1 + 0.9232x_2} \right]^2$$

Calculate values of permeance for water and ethanol from (14-80).

SOLUTION

For the first row of data, the mole fractions in the feed mixture (x_i) and the permeate (y_i), using molecular weights of 46.07 and 18.02 for ethanol and water, respectively, are

$$x_1 = \frac{0.088/46.07}{\frac{0.088}{46.07} + \frac{(1.0 - 0.088)}{18.02}} = 0.0364$$

$$x_2 = 1.0 - 0.0364 = 0.9636$$

$$y_1 = \frac{0.10/46.07}{\frac{0.10}{46.07} + \frac{0.90}{18.02}} = 0.0416$$

$$y_2 = 1.0 - 0.0416 = 0.9584$$

The activity coefficients for the feed mixture are

$$\gamma_1 = \exp \left\{ 1.6276 \left[\frac{0.9232(0.9636)}{1.6276(0.0364) + 0.9232(0.9636)} \right]^2 \right\} = 4.182$$

$$\gamma_2 = \exp \left\{ 0.9232 \left[\frac{1.6276(0.0364)}{1.6276(0.0364) + 0.9232(0.9636)} \right]^2 \right\} = 1.004$$

From the total mass flux, the component molar fluxes are

$$N_1 = \frac{(2.48)(0.10)}{46.07} = 0.00538 \frac{\text{kmol}}{\text{h} - \text{m}^2}$$

$$N_2 = \frac{(2.48)(0.90)}{18.02} = 0.1239 \frac{\text{kmol}}{\text{h} - \text{m}^2}$$

From (14-80), permeance values are

$$\bar{P}_{M_1} = \frac{0.00538}{(4.182)(0.0364)(352) - (0.0416)(76)}$$

$$= 0.000107 \frac{\text{kmol}}{\text{h} - \text{m}^2 - \text{mmHg}}$$

$$\bar{P}_{M_2} = \frac{0.1239}{(2.004)(1.0 - 0.0364)(149) - (1.0 - 0.0416)(76)}$$

$$= 0.001739 \frac{\text{kmol}}{\text{h} - \text{m}^2 - \text{mmHg}}$$

Results for other feed conditions are computed in a similar manner:

wt% Ethanol		Activity Coefficient in Feed		Permeance, kmol/h-m ² -mmHg	
Feed	Permeate	Ethanol	Water	Ethanol	Water
8.8	10.0	4.182	1.004	1.07×10^{-4}	1.74×10^{-3}
17.0	16.5	3.489	1.014	1.02×10^{-4}	1.62×10^{-3}
26.8	21.5	2.823	1.038	8.69×10^{-5}	1.43×10^{-3}
36.4	23.0	2.309	1.077	6.14×10^{-5}	1.17×10^{-3}
49.0	22.5	1.802	1.158	4.31×10^{-5}	1.10×10^{-3}
60.2	17.5	1.477	1.272	1.87×10^{-5}	8.61×10^{-4}
68.8	13.0	1.292	1.399	7.93×10^{-6}	6.98×10^{-4}
75.8	9.0	1.177	1.539	3.47×10^{-6}	6.75×10^{-4}

The PVA membrane is hydrophilic. Thus, as the concentration of ethanol in the feed liquid increases, the sorption of feed liquid by the membrane decreases, resulting in a reduction of polymer swelling. The preceding results show that as swelling is reduced, the permeance of ethanol decreases more rapidly than that of water, thus increasing selectivity for water. For example, selectivity for water can be defined as

$$\alpha_{2,1} = \frac{(100 - w_1)_P / (w_1)_P}{(100 - w_1)_F / (w_1)_F}$$

where w_1 = weight fraction of ethanol. For cases of 8.8 and 75.8 wt% ethanol in the feed, the selectivities for water are, respectively, 0.868 (more selective for ethanol) and 31.7 (more selective for water).

14.8 ULTRAFILTRATION

Consider an aqueous stream that contains a number of possible dissolved and suspended solutes, including suspended particles like bacteria and blood cells, large, dissolved molecules (e.g., proteins and carbohydrates), and small, dissolved molecules (e.g., salts). It is desired to separate these solutes from the carrier solution. As discussed in an earlier subsection, reverse osmosis is widely used for purifying saltwater. Part of the water passes through the membrane, leaving a retentate (concentrate) that contains most of the salts and some of the water. For recovering the solutes listed above, ultrafiltration and microfiltration (covered in the next subsection) are more commonly used in the manner indicated in Table 14.10.

Table 14.10 Comparison of Reverse Osmosis to Ultrafiltration and Microfiltration

	Microfiltration	Ultrafiltration	Reverse Osmosis
Typical membrane materials	cellulose acetate, polysulfone, ceramics	cellulose acetate, polyamides, polysulfone	cellulose acetate, aromatic polyamides
Pore size (angstroms)	100,000–200	200–10	10–1
Pore size (microns)	10–0.02	0.02–0.001	0.001–0.0001
Membrane pressure drop (psi)	1–10	10–100	100–1,000
Permeate	water + dissolved molecules	water + small molecules	water
Retentate (concentrate)	water + large suspended particles	water + large molecules	water + solutes

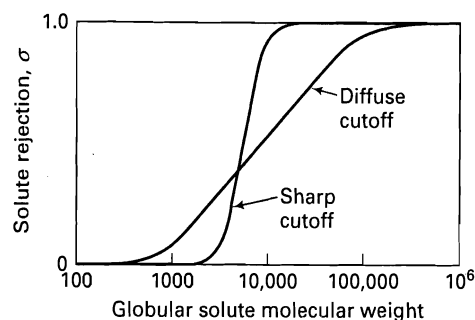


Figure 14.25 Molecular weight cutoff curves.

As with reverse osmosis, microfiltration and ultrafiltration are pressure driven, with the membrane permselective for the solvent (usually water). Microfiltration and ultrafiltration separate mainly by size exclusion of solute(s). Microfiltration retains particles of micron size. Ultrafiltration retains particles of submicron size by ultramicroporous membranes. Typically, ultrafiltration retains solutes in the 300–500,000 molecular weight range, including biomolecules, polymers, sugars, and colloidal particles. Ultrafiltration membranes are characterized by a nominal molecular weight cutoff (MWCO), defined as the smallest globular-solute molecular weight for which the membrane has at least a 90% rejection ($\sigma \geq 0.9$), with most commercial materials operating in the $1,000 \leq \text{MWCO} \leq 100,000$ range. The rejection, σ , is defined in terms of solute concentrations, c_{iP} and c_{iR} on either side of the membrane (i.e., retentate, R , and permeate, P):

$$\sigma_i = 1 - \frac{c_{iP}}{c_{iR}} \quad (14-81)$$

For an ultrafiltration membrane, a retention-cutoff curve is established experimentally. Two generic curves are shown in Figure 14.25. A sharp cutoff is desirable, but more typical is the diffuse cutoff curve, because of the difficulty in producing a membrane with a narrow pore-size distribution.

The flux of solvent through an ultrafiltration membrane depends on the pressure drop across the membrane, flow rates on either side of the membrane, and temperature. It is also a strong function of solute concentration, as shown in Figure 14.26 for experimental data with protein solutes. Often, over a wide range of solute concentration, flux is a function of the logarithm of the solute concentration. Alternatively, flux may be correlated for a given feed with

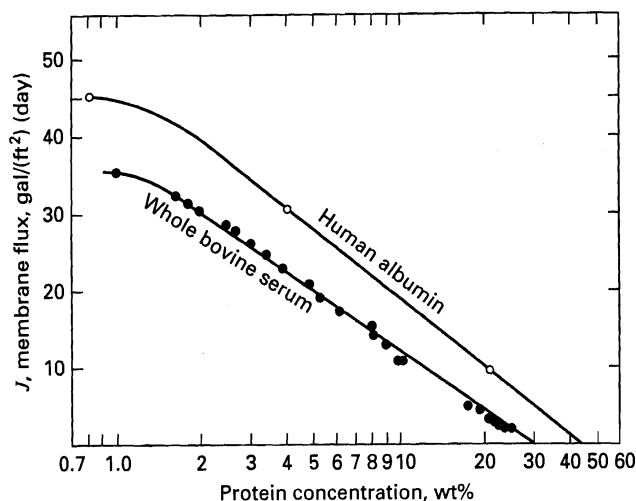


Figure 14.26 Effect of solute concentration on membrane flux.

concentration factor, CF , which is defined in terms of the volumetric flow rates of feed and retentate:

$$CF = \frac{Q_F}{Q_R} \quad (14-82)$$

Concentration polarization, described in Subsection 14.3, can cause a decrease in the permeate flux for ultrafiltration by two mechanisms. In the first, increased rejected solute concentration on the upstream membrane surface increases the osmotic pressure on the retentate side. Because the driving force across membranes for ultrafiltration is the same as for reverse osmosis, i.e., (14-69), the driving force decreases with increasing osmotic pressure on the retentate side. In general, the effect of osmotic pressure in ultrafiltration is only important when concentration polarization occurs because, without polarization, large molecular sizes in the retentate contribute little in moles of solute and, therefore, little to the osmotic pressure as given, approximately, by (14-68). In the second, more generally accepted mechanism, the permeate flux becomes dependent, at least partially, on the mass-transfer resistances of the boundary layer or film and any gel layer on the retentate side of the membrane.

Concentration polarization effects in ultrafiltration are small at low pressures, low-solute concentrations, and high velocities on the retentate side of the membrane. Then, the permeate flux is controlled by transmembrane pressure, in response to which the flux increases linearly. However, as the transmembrane pressure or solute concentration increases or the velocity decreases, a point is reached where this linear relationship no longer holds and permeate flux reaches an asymptotic limiting value, controlled by mass transfer on the retentate side of the membrane. In the extreme case of a thick gel layer, it alone may control the permeate flux. Cheryan [50] and Zeman and Zydney [51] develop theories for predicting limiting permeate flux.

Fouling in ultrafiltration occurs by gel-layer consolidation, adsorption of solutes on the upstream membrane surface, and/or deposition of solutes within the membrane pores. Fouling, unlike concentration polarization, is a function of

time. While fouling can be reversed by periodic cleaning (e.g., back-flushing or chemical treatment), some of the fouling may be irreversible, resulting ultimately in the need to replace the membrane. Zeman and Zydney [51] give a treatment of membrane cleaning.

Process Configurations

An ultrafiltration process is commonly conducted in one of four configurations or combinations thereof: (1) batch ultrafiltration, (2) continuous bleed-and-feed ultrafiltration, (3) batch diafiltration, and (4) continuous bleed-and-feed diafiltration.

Batch Ultrafiltration

A batch configuration is shown in Figure 14.27, where the membrane consists of a number of membrane cartridges in parallel. Initially, the feed tank is filled with a batch, V_F , of feed solution of solute concentration, c_F . The solution is pumped through membrane cartridges, where permeate is continuously removed, but retentate (concentrate) is recycled, usually at a high volumetric flow rate, Q , to minimize fouling, back to the feed tank. Solute concentration in the retentate increases with time, as solvent (usually water) selectively passes through the membrane. As time passes, the retained volume of solution in the system, $V\{t\}$, decreases and its retentate solute concentration, $c_R\{t\}$, increases. Operation is terminated when the desired solute retentate concentration, c_R , is reached. At that point, the feed tank and associated equipment contains the final retentate, V_R , which can be drained to another tank. After cleaning, another batch is processed. The required time for batch processing depends on the membrane area, A , and the flux, J , of permeate through the membrane. The flux decreases with time because it depends strongly on the increasing solute concentration on the upstream side of the membrane as shown in Figure 14.26.

Assume that the batch feed contains completely rejected solutes and only partially rejected solutes. Assume that the flux is a linear function of the logarithm of the concentration

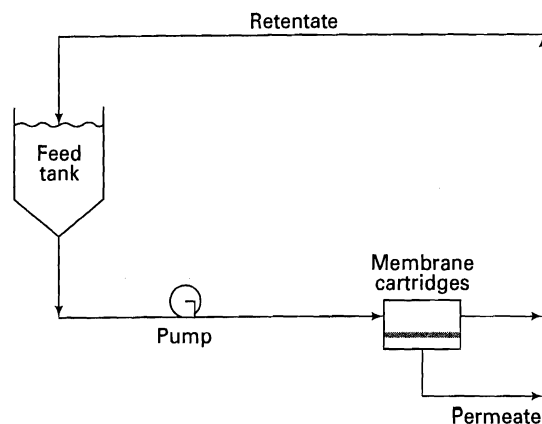


Figure 14.27 Batch ultrafiltration.

factor, CF , as discussed above, where for the batch process of Figure 14.27, the concentration factor is a function of time as defined by

$$CF = \frac{V_F}{\text{retained } V\{t\}} \quad (14-83)$$

Then it can be shown that the average flux, J_{avg} , for the batch process is approximately

$$J_{\text{avg}} = J\{c_F\} - 0.33[J\{c_F\} - J\{c_R\}] \quad (14-84)$$

where values of $J\{c_F\}$ and $J\{c_R\}$ are obtained from experimental data like that in Figure 14.26. The required membrane area as a function of batch processing time, t , is given accordingly by

$$A = \frac{V_P}{t J_{\text{avg}}} = \frac{V_F - V_R}{t J_{\text{avg}}} \quad (14-85)$$

To obtain a material balance for a solute, we note that solute concentration in the retained volume is a function of both the reduction in retained volume and the amount of solute that passes through the membrane. A solute, i , material balance for a differential volume passing through the membrane is, by analogy to (14-45) for gas permeation or to (13-2) for batch distillation,

$$\frac{dV}{V} = \frac{dc_{iR}}{c_{iP} - c_{iR}} \quad (14-86)$$

Combining with (14-81) for the definition of rejection, σ_i ,

$$\frac{dV}{V} = -\frac{dc_{iR}}{\sigma_i c_{iR}} \quad (14-87)$$

Integrating this equation from initial feed to final retentate gives an equation for solute concentration in the retentate as a function of retained volume, where if the retained volume is the final retentate volume, then solute concentration is the concentration in the final retentate volume:

$$c_{iR} = c_{iF} \left(\frac{V_F}{V_R} \right)^{\sigma_i} = c_{iF} (CF)^{\sigma_i} \quad (14-88)$$

The yield, Y_i , of solute, i , defined as the amount of feed solute that is retained in the retentate, is obtained from (14-88):

$$Y_i = \frac{c_{iR} V_R}{c_{iF} V_F} = \left(\frac{V_F}{V_R} \right)^{\sigma_i} \left(\frac{V_R}{V_F} \right) = \left(\frac{V_F}{V_R} \right)^{\sigma_i - 1} = CF^{\sigma_i - 1} \quad (14-89)$$

Application of (14-81) to (14-89) is illustrated in the following example of batch ultrafiltration.

EXAMPLE 14.13

An aqueous feed of 1,000 L is to undergo batch ultrafiltration with a polysulfone membrane. The solute concentrations and their measured rejection values are as follows:

Solute	Type molecule	MW	Concentration, Rejection,	
			c , g/L	σ
Albumin	Globular	67,000	10	1.00
Cytochrome C	Globular	13,000	10	0.70
Polydextran	Linear	100,000	10	0.05

Note that polydextran has the highest molecular weight, but the lowest rejection because it is a linear molecule rather than a globular one. The volume of the final retentate is to be 200 L, which is achieved in a four-hour batch-processing time. Thus, from (14-82), $CF = 1,000/200 = 5$. From experimental measurements, the flux values are 30 L/m²-h at $CF = 1$ and 10 L/m²-h at $CF = 5$. Calculate the solute concentration in the final retentate, yield of each solute, and membrane area. Neglect changes in solution density.

SOLUTION

From (14-84), the average flux = $30 - 0.33(30 - 10) = 23.4$ L/m²-h. The total permeate volume = $1,000 - 200 = 800$ L. From (14-85), for $t = 4$ h,

$$A = \frac{800}{4(23.4)} = 8.55 \text{ m}^2$$

Using (14-89) and (14-88), the yield and concentration of each solute in the final retentate are

Solute	Concentration in final retentate, g/L	% Yield
Albumin	50.0	100.0
Cytochrome C	30.9	61.7
Polydextran	10.8	21.7

Note that although polydextran has a very low rejection, neither the final concentration in the retentate nor the % yield approach zero.

When processing biological materials, the batch configuration may be disadvantageous. Recycle of the retentate can cause damage to proteins and cells. Also, if the batch residence time is too long, unacceptable bacterial growth may occur. Continuous ultrafiltration, which is widely used for large-scale processes, is then preferred.

Continuous Feed-and-Bleed Ultrafiltration

Although, as shown in Figure 14.20, continuous reverse osmosis usually operates in a single-pass mode, continuous ultrafiltration rarely does; instead, it operates in a multipass mode, as shown in Figure 14.28, called single-stage *feed-and-bleed*. This is achieved by recycling, at steady state, a large fraction of the retentate. In effect, feed to the membrane is the sum of fresh feed and recycle retentate. The

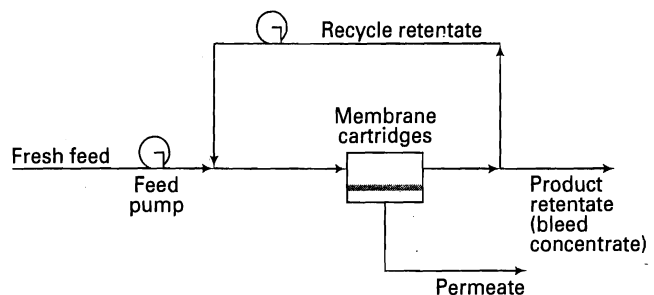


Figure 14.28 Single-stage continuous feed-and-bleed ultrafiltration.

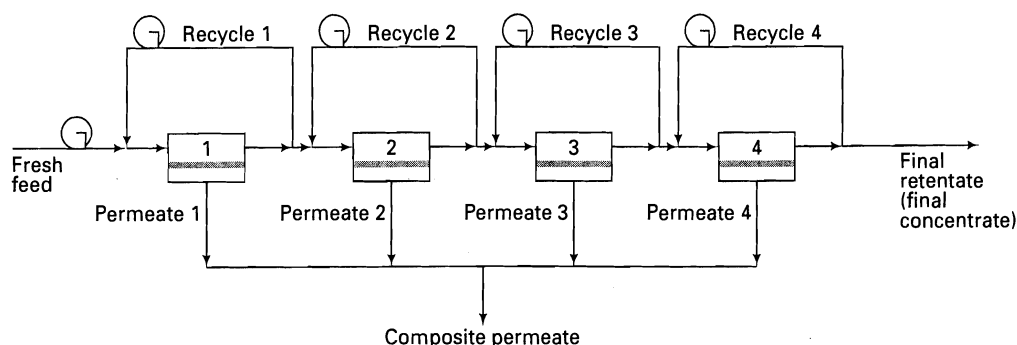


Figure 14.29 Multistage, continuous feed-and-bleed ultrafiltration.

bleed is that portion of the retentate that is not recycled, but is withdrawn as product retentate. At startup the entire retentate is recycled until the desired retentate concentration is achieved, at which time bleed is initiated. The advantages and disadvantages of feed-and-bleed operation, as compared to a single-pass mode (no recycle of retentate), are considered by Cheryan [50] and in detail by Zeman and Zydney [51]. The single-pass mode is usually unsuitable for ultrafiltration because the main product is the concentrate rather than the permeate (as in reverse osmosis), and high yields of permeate are required in order to adequately concentrate solutes in the retentate. Typically, the concentration factor, CF , defined by (14-82), has a value of 10. As a result, a single-pass ultrafiltration requires a very long membrane path or a very large membrane area. A disadvantage of the feed-and-bleed mode, however, is that with the high recycle ratio, the concentration of solutes on the retentate side is at its highest value, resulting, as shown in Figure 14.26, in the lowest flux, with a resulting membrane area larger than that required for the batch mode. To counter this, the feed-and-bleed mode is most often staged as shown for four stages in Figure 14.29, where the retentate (bleed) from each stage is sent to the next stage, while the permeates from the stages are collected into a final composite permeate. Solute concentrations increase incrementally as the retentates pass through the system. The final and highest concentration is only present in the final stage. As a result, retentate concentrations are lower and higher fluxes are achieved, compared to a single-stage, bleed-and-feed system, for all but the final stage. This leads to a smaller total membrane area. In practice, three to four feed-and-bleed stages are usually optimal.

For a single-stage, continuous, bleed-and-feed ultrafiltration, the following material-balance equations apply in terms of volumetric flow rates and concentrations:

Total balance:

$$Q_F = Q_R + Q_P \quad (14-90)$$

Solute total balance:

$$c_{iF} Q_F = c_{iR} Q_R + c_{iP} Q_P \quad (14-91)$$

If the recycle rate is sufficiently high, concentration of the stream flowing on the upstream side of the membrane will be the retentate. Then, if (14-90) and (14-91) are combined with (14-81) and (14-82), rejection in the stage and CF are

constant and based on retentate, such that the following equation applies for computing the solute concentration in the retentate:

$$c_{iR} = c_{iF} \left[\frac{CF}{CF(1 - \sigma_i) + \sigma_i} \right] \quad (14-92)$$

Membrane area is given by (14-85) in continuous-process form,

$$A = \frac{Q_P}{J \{ \text{at } CF \}} \quad (14-93)$$

Yield of solute is given in continuous-process form by combining the definition of yield in (14-89) with (14-82) and (14-92):

$$Y_i = \frac{c_{iR} Q_R}{c_{iF} Q_F} = \frac{1}{CF(1 - \sigma_i) + \sigma_i} \quad (14-94)$$

For the four-stage, continuous, feed-and-bleed ultrafiltration system, shown in Figure 14.29, equations (14-90) to (14-94) are applied to each stage. It is usually assumed that the most desirable multistage system is one in which all stages have the same membrane area, which reduces cost of maintenance. The calculations, as described and illustrated in Example 14.14, are iterative in nature, using an outer loop in which membrane area per stage is assumed, and an inner loop in which an overall concentration parameter is assumed.

Diafiltration

As seen in Figure 14.26, when a high degree of solute separation is desired, the flux may drop to a low value. To overcome low flux levels, when it is necessary to continue removing permeable solutes from solutes of little or no permeability, it is common to employ diafiltration, which involves the addition of solvent (usually water) to the retentate, followed by filtration. Additional solvent dilutes the retentate so as to increase the flux. Thus ultrafiltration is employed to a certain limiting concentration of solutes, followed by diafiltration to further enhance solute separation. The final retentate may not be very concentrated in retained solutes, but it contains a smaller fraction of permeable solutes.

Diafiltration is conducted in the same modes as ultrafiltration, i.e., batch or continuous feed-and-bleed, including multistage systems. The added amount or flow rate of solvent is a variable, whose value, for preliminary calculations,

may be set equal to the amount of permeate obtained during diafiltration.

Consider a batch diafiltration in which retentate from the previous step is added to the feed tank for diafiltration and recycled, without permeate withdrawal from the membrane unit during startup. Dilution solvent is then added at a continuous rate to the feed tank, under perfect-mixing conditions, with permeate withdrawal at a rate equal to the solvent-addition rate. This operation is sometimes referred to as fed-batch or semicontinuous. If the recycle rate is very high, the concentrations of solutes in the membrane unit will be uniform on each side of the membrane, so rejection in the membrane at any instant is given by (14-81), where both concentrations will change with time. Let c_i = the instantaneous concentration of solute in the recycle retentate. Initially, before solvent is added, its value is that of the feed to the diafiltration system, c_{iF} . Let c_{iP} = the instantaneous solute concentration in the permeate leaving the membrane unit. Then (14-81) becomes

$$\sigma_i = 1 - \frac{c_{iP}}{c_i} \quad (14-95)$$

With a constant volume, V_F , in the feed tank, before solvent is added, an instantaneous solute material balance equates the decrease in amount of solute in the feed tank to the amount of solute appearing in the permeate. But permeate flow rate, Q_P , is equal to the solvent addition rate, $Q_S = dV_S/dt$, giving for a solute material balance

$$-V_F \frac{dc_i}{dt} = c_{iP} Q_P = c_{iP} \frac{dV_S}{dt} \quad (14-96)$$

Combining (14-95) and (14-96) to eliminate c_{iP} gives, in integral form over time for diafiltration to the final retentate concentration,

$$\int_{c_{iF}}^{c_{iR}} \frac{dc_i}{c_i} = -\frac{(1 - \sigma_i)}{V_F} \int_0^{V_{S\text{total}}} dV_S \quad (14-97)$$

Integration gives an equation for computing the final retentate concentration:

$$c_{iR} = c_{iF} \exp \left[-\frac{V_{S\text{total}}}{V_F} (1 - \sigma_i) \right] \quad (14-98)$$

Calculations for continuous diafiltration are similar to those for continuous ultrafiltration, as illustrated in Example 14.14.

A major industrial application of ultrafiltration is in processes for manufacturing protein concentrates from skim milk. Skim milk is coagulated to render two products, (1) a thick precipitate called curd, rich in a phosphoprotein called casein, used to make cheese, plastics, paints, and adhesives, and (2) whey (or cheese whey), a watery, residual liquid. One hundred pounds of skim milk yields approximately 10 pounds of curd and 90 pounds of whey. Typically, whey consists, on a mass basis, of 93.35% water; 0.6% true protein (TP) of molecular weight ranging from about 10,000 to 200,000; 0.3% nonprotein nitrogen compounds (NPN); 4.9% lactose (a sugar of empirical formula $C_{12}H_{22}O_{11}$, and

molecular weight of 342, which has an ambient solubility in water of about 10 wt%); 0.2% lactic acid ($C_3H_6O_3$) of molecular weight 90, which is very soluble in water; 0.6% ash (inorganic salts of calcium, sodium, phosphorus, and potassium) of molecular weight ranging from about 20 to 100; and 0.05% butter fat.

Proteins are macromolecules consisting of sequences of amino acids, which contain both amino and carboxylic-acid functional groups. When digested, proteins become sources of amino acids, which are classified as nutritionally essential or nonessential. The *nonessential amino acids* are synthesized by a healthy body from metabolized food. *Essential amino acids* cannot be synthesized by the body, but must be ingested. Amino acids in proteins are the essential building blocks for health because they repair body cells, build and repair muscles and bones, regulate metabolic processes, and provide energy.

The proteins in whey, on a mass basis of total true protein, are betalactoglobulin (50–55%), alpha-lactalbumin (20–25%), immunoglobulins (10–15%), bovine serum albumin (5–10%), and smaller amounts of glycomacropeptide, lactoferrin, lactoperoxidase, and lysozyme. The first five of these eight proteins provide an excellent source of all eight essential amino acids: isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. Approximately 35 wt% of proteins in whey provide amino acids. The nonprotein nitrogen compounds include ammonia, creatine, creatinine, urea, and uric acid, with molecular weights ranging from 17 to 168.

To obtain dry protein concentrate from whey, a number of industrial processes are used. Most involve ultrafiltration, which separates by size exclusion that depends on molecular weight and molecular shape. For separation purposes, the compounds in whey form five groups: (1) true protein and butter fat, (2) nonprotein nitrogen, (3) lactose, (4) lactic acid and ash, and (5) water. A typical process is shown in Figure 14.30. Whey is pumped to ultrafiltration section I, where the exiting retentate (concentrate) contains all of the protein. The other whey-feed components leave mainly in the exiting permeate. The retentate is further concentrated in an evaporator and then spray dried to produce a *whey protein concentrate*. Permeate is pumped to ultrafiltration section II, where all remaining lactose is retained and sent to a second spray dryer to produce *lactose-rich concentrate*, while the permeate is sent to wastewater treatment. Whey-protein concentrate produced by this process contains too high a lactose content for the millions of individuals who are diagnosed as “lactose intolerant” because of susceptibility to digestive disorders, including gassy symptoms, bloating, and diarrhea. To produce so-called *whey protein isolate* of 90–97 wt% protein and almost no lactose or fat, the process of Figure 14.30 is modified by adding additional ultrafiltration. The following example, based on information provided in the 2001 AIChE National Student Design Competition, involves the process design of an ultrafiltration section for producing a protein concentrate.

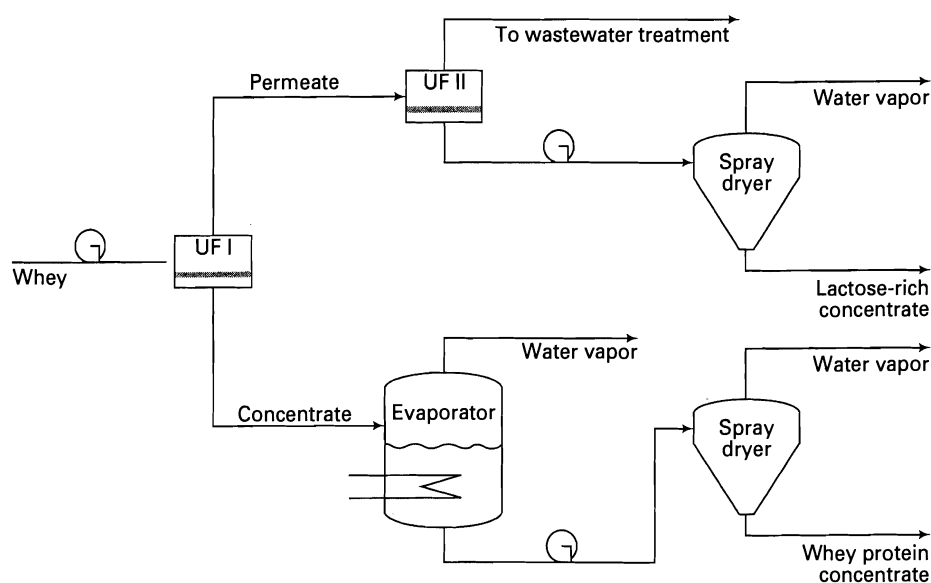


Figure 14.30 Whey process to produce protein and lactose concentrates.

EXAMPLE 14.14

A cheese plant produces a byproduct stream of 1,000,000 lb/day of whey. The whey is to be further processed to obtain a dry powder containing 85 wt% combined TP (true protein) and NPN (nonprotein nitrogen compounds). The process includes three sections: (1) four stages of continuous bleed-and-feed ultrafiltration to reach 55 wt% (dry basis), followed by (2) four stages of continuous diafiltration to reach 75 wt% (dry basis), followed by (3) one stage of batch diafiltration to reach the final 85 wt% (dry basis), with a batch-time limit of 4 hours. Diafiltration must be used above 55 wt% because the retentate from ultrafiltration becomes too viscous above that value. Each section will use PM 10 ultrafiltration hollow-fiber membrane cartridges from Koch Membrane Systems, which are 3 in. in diameter by 40 in. long, with 26.5 ft² of membrane area, at a purchase cost of \$200.00 each. For each cartridge, the recirculation (recycle) rate is 23 gal/min. The number of cartridges is to be the same in each stage for Sections 1 and 2. The inlet pressure to each cartridge is 30 psig, with a crossflow pressure drop of 15 psi and a permeate pressure of 5 psig. For these conditions, the flux through the membrane has been experimentally measured for the whey and correlated as a function of the concentration factor, CF , by the following equation:

$$\text{membrane flux, gal/ft}^2 \cdot \text{day} = 27.9 - 5.3 \ln(CF) \quad (1)$$

where CF , for any stage n , is defined by reference to the fresh feed to Section 1, as $CF_n = F_{\text{Section 1}}/R_n$.

The composition of the whey and measured membrane solute rejections, σ , are as shown in this table:

Component	Wt% in Whey	Flow rate in Whey, lb/day	Solute rejection, σ
Water	93.35	933,500	—
True protein, TP	0.6	6,000	0.970
Nonprotein nitrogen, NPN	0.3	3,000	0.320
Lactose	4.9	49,000	0.085
Ash	0.8	8,000	0.115
Butter fat	0.05	500	1.000

Based on the σ values, the membrane increases the concentration of TP while selectively removing low-molecular-weight solutes of lactose and ash. The density of the whey and all retentate and permeate streams can be assumed to be 8.5 lb/gal. The continuous sections of the process will operate 20 hr per day, leaving 4 hr per day for automatic cleaning to remove accumulated membrane foulants and kill any microorganisms that have attached to equipment surfaces during the 20-hr run period. For each section, calculate:

- (1) Component material balances in lb/day of operation, including dilution water for diafiltration operations.
- (2) Percent recovery from whey of TP and NPN in the intermediate and final 85 wt% concentrate.
- (3) Number of membrane cartridges.

Also, for Section 1, make calculations for a single continuous stage, compare results to those for four stages, and discuss advantages and disadvantages of four stages versus one stage.

SOLUTION

The flow diagram for the specified ultrafiltration–diafiltration process is shown in Figure 14.31. Stream flow rates are those labeled in this diagram.

Single, Continuous, Ultrafiltration Stage for Section 1 to Reach 55 wt% TP + NPN

First compute results for Section 1 using a single stage of continuous, bleed-and-feed ultrafiltration, shown in Figure 14.28. To do this, assume 10 as the value for the concentration factor, CF , and compute by material balance from the whey feed rate, $F_1 = 1,000,000$ lb/day, flow rates of retentate (concentrate), R_1 , and permeate, P_1 . By definition of CF for this type of ultrafiltration, $R_1 = F_1/CF = 1,000,000/10 = 100,000$. Therefore, $P_1 = F_1 - R_1 = 1,000,000 - 100,000 = 900,000$ lb/day. Next, use a mass flow rate form of the yield equation, (14-94), to compute the flow rate of each solute in the concentrate. For example, for TP,

$$\begin{aligned} (m_{\text{TP}})_{R_1} &= \frac{(m_{\text{TP}})_{F_1}}{[CF(1 - \sigma) + \sigma]} \\ &= \frac{6000}{[10(1 - 0.97) + 0.97]} = 4724 \text{ lb/day} \end{aligned}$$

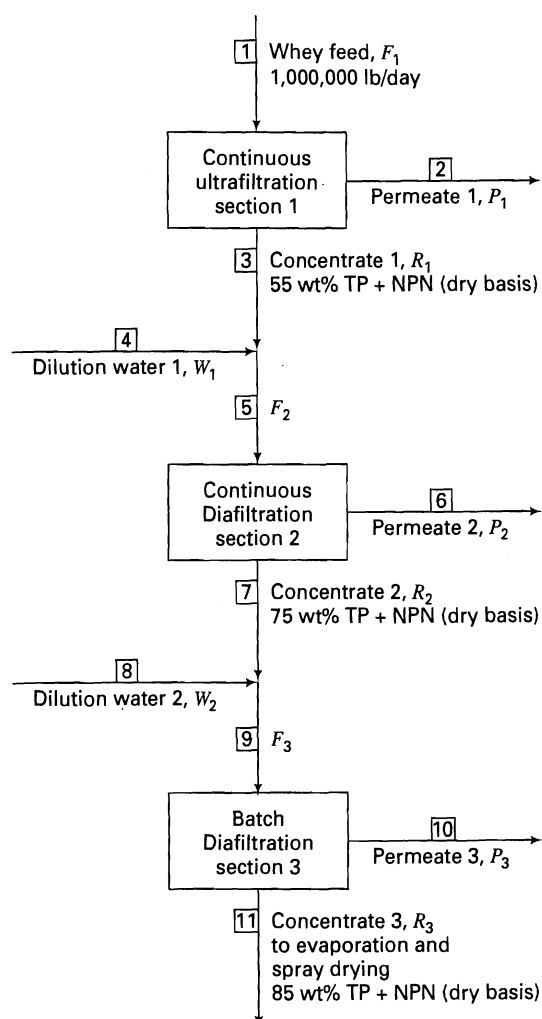


Figure 14.31 Process for Example 14.14.

In a similar manner, flow rates of other solutes in the concentrate, R_1 , are computed with the following results, where the water rate is obtained by difference.

Concentrate for a Single Stage of Continuous Ultrafiltration in Section 1, for an Assumed $CF = 10$

Component	Wt% in Concentrate, C_1 , lb/day	Flow Rate in Concentrate, C_1 , lb/day
Water	88.157	88,157
True protein, TP	4.724	4,724
Nonprotein nitrogen, NPN	0.421	421
Lactose	5.306	5,306
Ash	0.892	892
Butter fat	0.500	500
Total	100.000	100,000

From this table, the wt% TP + NPN in the concentrate on a dry basis is $(4,724 + 421)/(100,000 - 88,157) = 0.4344$ or 43.44 wt% (dry basis). This is less than the 55-wt% target. Therefore, the assumed value of $CF = 10$ is too low. If calculations are made with a spreadsheet, the Solver function can be applied to find the value of CF that meets the 55-wt% target. The result is

$CF = 24.955$, giving the following concentrate:

Concentrate for a Single Stage of Continuous Ultrafiltration in Section 1, for the Correct $CF = 24.955$

Component	Wt% in Concentrate, C_1 , lb/day	Flow Rate in Concentrate, C_1 , lb/day
Water	83.372	33,409
True protein, TP	8.713	3,491
Nonprotein nitrogen, NPN	0.433	174
Lactose	5.335	2,138
Ash	0.899	360
Butter fat	1.248	500
Total	100.000	40,072

The wt% TP + NPN in the concentrate is now $(3,491 + 174)/(40,072 - 33,409) = 0.5500$ or 55.00%, the specified value.

It is also of interest to compute the % yield of TP + NPN, which is

$$(3491 + 174)/(6000 + 3000) = 0.4072 \text{ or } 40.72\%$$

The membrane area for this single stage is computed from (14-93). The permeate rate is $P_1 = F_1 - R_1 = 1,000,000 - 40,072 = 959,928$ lb/day or $959,928/8.5 = 112,933$ gal/day. For 20-h/day operation, the volumetric permeate rate $= 112,933/20 = 5,647$ gal/h. From (1), for the computed CF ,

$$\begin{aligned} \text{membrane flux} &= 27.9 - 5.3 \ln(CF) = 27.9 - 5.3 \ln(24.955) \\ &= 10.85 \text{ gal/ft}^2\text{-day or } 10.85/24 = 0.452 \text{ gal/ft}^2\text{-h} \end{aligned}$$

Therefore, from (14-93), the membrane area $= 5,647/0.452 = 12,490$ ft². Each cartridge has an area of 26.5 ft². Therefore, $12,490/26.5 = 471$ parallel cartridges are needed. Total fresh feed rate based on 20 hr of operation is

$$\frac{1,000,000}{8.5(20)} = 5,882 \text{ gal/h}$$

Fresh feed rate to each cartridge is

$$\frac{5,882}{60(471)} = 0.208 \text{ gal/min/cartridge}$$

The combined flow rate (fresh plus recycle) to each cartridge is $0.208 + 23 = 23.208$ gal/min, which is a desirable high recycle ratio.

To increase the % yield and decrease the number of cartridges, a multistage section is needed. In the problem statement, four stages in series are specified. These are computed next.

Four, Continuous, Ultrafiltration Stages for Section 1 to Reach 55 wt% TP + NPN

The following procedure is based on an equal membrane area for each of the four stages in Figure 14.29. The calculations are accomplished by a double "trial-and-error" (nested iteration) procedure, best carried out using a spreadsheet with a Solver function.

Assume a membrane area per stage. Because the single-stage calculation resulted in an area of approximately 12,500 ft², and we are using four stages, the total area for the four stages will be less. First, assume a total area of 8,000 ft² or 2,000 ft² per stage $= A$. Next, find, by iteration, the overall concentration factor, $\overline{CF}$, that gives the fresh feed rate to the first Stage, as calculated above, of

5,882 gal/h. This is done with a spreadsheet starting from Stage 4 and working backward to Stage 1, using the following equations, where J_n = hourly membrane flux = (1)/24 of Eq. (1), based on a CF using F_1 and R_n . For Stage 4, CF_4 = the assumed $\overline{CF}$ and $R_4 = F_1/CF_4$. Then, for the calculations back to stage 1:

$$P_n = AJ_n, \quad R_{n-1} = P_n + R_n, \quad CF_{n-1} = F_1/R_{n-1}$$

When Stage 1 is reached by calculation of P_1 , the fresh feed rate is computed from $F_1 = P_1 + R_1$. If F_1 is not 5,882 gal/h, new values of $\overline{CF}$ are assumed until the correct value of F_1 is obtained. This iteration can be done with the spreadsheet Solver function. Suppose $\overline{CF} = 20$ is assumed with the assumed $A = 2,000 \text{ ft}^2$. The following results are obtained, where all flows are in gal/h:

$\overline{CF}$	R_4	P_4	R_3	P_3	R_2	P_2	R_1	P_1	F_1
20	294	1002	1296	1657	2953	2021	4974	2250	7224
61.2	96.2	508	604	1320	1924	1831	3755	2127	5882

The tabulated results give $F_1 = 7,224 \text{ gal/h}$, which is too high. Using the Solver function of the spreadsheet, a value of $\overline{CF} = 61.2$ gives the correct F_1 , with the following corresponding computed values of CF_n and J_n :

$\overline{CF}$	CF_4	J_4	CF_3	J_3	CF_2	J_2	CF_1	J_1
61.2	61.2	0.254	9.735	0.660	3.057	0.916	1.566	1.063

However, the assumed value of membrane area per stage may not be correct. To check this, calculations, with a spreadsheet, similar to those above for a single stage are carried out, starting with Stage 1 and proceeding stage-by-stage to Stage 4. Pertinent results for $A = 2,000 \text{ ft}^2$ per stage and $\overline{CF} = 61.2$ are:

Stage	1	2	3	4
Wt% TP + NPN in retentate from stage (dry basis)	17.47	25.46	44.96	74.16

Because the wt% TP + NPN (dry basis) in the retentate from Stage 4 is 74.16%, which is higher than the specified 55 wt%, the calculations must be repeated for other values of membrane area per stage. For each assumed membrane area, a new value of $\overline{CF}$ that gives the correct fresh feed rate must be found. The following spreadsheet results are obtained when iterating on A and $\overline{CF}$:

A, Membrane Area per Stage, ft^2	$\overline{CF}$ for Correct Fresh Feed Rate	Wt% TP + NPN in Final Retentate (dry basis)
2,000	61.2	74.16
1,750	26.9	63.22
1,700	22.5	60.26
1,650	18.8	57.14
1,600	15.7	53.95
1,617	16.65	55.00

These results show that for a continuous, four-stage ultrafiltration system, with equal membrane area per stage, the desired value of 55 wt% (dry basis) for TP + NPN in the retentate (concentrate) from Stage 4 corresponds to $A = 1,617 \text{ ft}^2$ per stage and an overall concentration factor, $\overline{CF}$, of 16.65. From these results, the material balance, which includes the combined permeate from four ultrafiltration stages and the computed retentate (concentrate) of 55 wt% that leaves Stage 4 and becomes the feed to the continuous

diafiltration section to increase the wt% TP + NPN to 75 (dry basis), is as presented in this table:

Concentrate and Combined Permeate from a Four-stage, Continuous Ultrafiltration in Section 1, for a $\overline{CF}$ of 16.65 and $A = 1,617 \text{ ft}^2/\text{Stage}$, which Meets the 55 wt% Specification

Component	Flow Rate of Whey, lb/day	Flow Rate of Concentrate, lb/day	Flow Rate of Combined Permeate, lb/day
Water	933,500	49,897	883,603
True protein, TP	6,000	5,245	755
Nonprotein nitrogen, NPN	3,000	353	2,647
Lactose	49,000	3,476	45,524
Ash	8,000	603	7,397
Butter fat	500	500	0
Total	1,000,000	60,074	939,926

For Section 1, the number of ultrafiltration cartridges required is $1,617/26.5 = 61$ cartridges per stage or a total of 244 cartridges for four stages. The % yield of TP + NPN in the concentrate is $(5,245 + 353)/(6,000 + 3,000) \times 100\% = 62.20\%$. These results compare to 471 cartridges and a % yield of 40.72% for a single stage in Section 1. On both counts, the four-stage system is preferred. A definitive economic analysis needs to include the additional piping and instrumentation costs for a four-module system.

Four, Continuous, Diafiltration Stages for Section 2 to Reach 75 wt% TP + NPN

The following procedure is based on an equal membrane area for each of the four stages, based on a flow diagram similar to Figure 14.29, differing only in the addition of water to the feed to each stage. The calculations for a continuous, multistage, diafiltration system require iteration on a single variable, the added water rate, to achieve the specified 75 wt% TP + NPN. This is conveniently done with a spreadsheet and its Solver function.

For each diafiltration stage, the added water rate, W_n , is the same and equal to W for each stage. Furthermore, for each stage, the permeate rate, P_n is set equal to the added water rate, W . Therefore, the feed rate to stage, F_1 for the first stage and R_{n-1} for the succeeding three stages (i.e., before the added water and the recycle), are all equal to the retentate (concentrate) rate, R_n , sent to the next stage. Thus, all retentate rates are the same, i.e., $F_1 = R_n$. These simplifications result in the same concentration factor, $\overline{CF}$ for every stage:

$$\overline{CF} = \frac{W + F_1}{F_1} \quad (2)$$

For a continuous diafiltration system of n stages, solute component, i , flow rates in the concentrate from the final stage are given by an equation, obtained by applying (14-94) successively to each stage, setting the solute flow rate in the feed to Stages 2, 3, and 4 equal to its flow rate in the retentate from the preceding stage:

$$(m_i)_{R_n} = (m_i)_{F_1} \left[\frac{1}{\overline{CF}(1 - \sigma_i) + \sigma_i} \right]^n \quad (3)$$

Using a spreadsheet, (2) and (3) are used where values of $(m_i)_{F_1}$ are the solute component flow rates in the concentrate leaving Section 1, as given in the above table. Values of the solute rejections, σ_i , are given in the problem statement. A value is assumed for the added water rate to each stage, W , and $\overline{CF}$ is computed from (2). From (3), the values of $(m_i)_{R_n}$ are computed for each solute. The wt% TP + NPN (dry basis) is then calculated. If it is not the specified 75 wt%, a new value of W is assumed. For example, assume a value for W equal to half of the feed rate, F_1 , or $60,074/2 = 30,037$ lb/day. Then, from (1), $\overline{CF} = (30,037 + 60,074)/60,074 = 1.50$. For TP, from (2),

$$(m_{TP})_{R_4} = 5,245 \left[\frac{1}{1.50(1 - 0.97) + 0.97} \right]^4 = 4,942 \text{ lb/day}$$

The calculations are repeated for the other components, and the water rate in the concentrate from stage 4 of Section 2 is determined so that the total concentrate flow rate equals that of the feed, 60,074 lb/day. The wt% TP + NPN in the concentrate is then calculated with a result of 78.2 wt%, which is higher than the specified 75 wt%. Using the Solver function, the correct water rate for each stage is found to be 23,332 lb/day or a total of 93,328 lb/day for the four stages, with a corresponding $\overline{CF} = 1.388$. The resulting material balance, which includes the combined permeate from the four diafiltration stages and the computed retentate (concentrate) of 75 wt% that leaves stage 4 and becomes the feed to the batch diafiltration section to increase the wt% TP + NPN to 85 (dry basis) is as presented in the next table.

Concentrate and Combined Permeate from a Four-stage, Continuous Diafiltration in Section 2, and an Added Water Rate of 23,332 lb/day per Stage, which Meets the 75 wt% Specification

Component	Flow Rate in Feed to Section 2, lb/day	Flow Rate in Concentrate, lb/day	Flow Rate in Combined Permeate, lb/day
Water	49,897	53,214	90,011
True protein, TP	5,245	5,007	238
Nonprotein nitrogen, NPN	353	138	215
Lactose	3,476	1,030	2,446
Ash	603	185	418
Butter fat	500	500	0
Total	60,074	60,074	93,328

From these results, the yield of TP + NPN from diafiltration is $(5,007 + 138)/(5,245 + 353) \times 100\% = 91.91\%$ for an overall yield to this point of $(0.9191)(0.6220) \times 100\% = 57.17\%$. The membrane flux for each stage is obtained from (1). However, the CF used in that equation is the ratio of the whey feed for the process to the retentate rate from the stage, which for the four stages of diafiltration is the same as that for the last stage of the ultrafiltration in Section 1. A value of $CF = 1,000,000/60,074 = 16.65$ applies, which results in a membrane flux of 0.5415 gal/h-ft². The volumetric permeate flow rate per stage = $93,328/[(20)(4)(8.5)] = 137$ gal/h. The membrane area required per stage = $137/0.5415 = 253$ ft². The number of cartridges

per stage = $253/26.5 = 9.5 \approx 10$ cartridges per diafiltration stage for a total of 40 diafiltration cartridges.

These results for four stages of continuous diafiltration may be compared to results obtained in a similar manner with just a single, continuous, diafiltration stage, which gives an added water rate of 167,200 gal/day (compared to 93,328 for four stages) and an overall TP + NPN yield of 55% (compared to 57% for four stages).

A Single, Batch, Diafiltration Stage for Section 3 to Reach 85 wt% TP + NPN

For the final membrane section, a single batch diafiltration is employed with maximum batch time of 4 hr. The common method for carrying out batch diafiltration is to initially fill the feed tank with the feed (concentrate from Section 2) and add water continuously over the 4-hr period to maintain the liquid level in the tank. With a high recycle ratio, the concentration of solutes in the retentate is maintained constant, with the flow rates of the solute component, i , in the daily feed of concentrate given by (14-98) in mass-flow form:

$$(m_i)_R = (m_i)_F \exp \left[-\frac{W}{F}(1 - \sigma_i) \right] \quad (4)$$

where W and F are total amounts of additional water and feed processed during the 4-hr period. To reach the target of 85 wt%, it is particularly important to remove the lactose from the feed. For example, suppose we choose a daily amount of added water equal to the daily amount of feed from Section 2. Then, $W/F = 1$. From the preceding table, that feed contains 1,030 lb/day of lactose, which has a rejection, $\sigma = 0.085$. Substitution into (4) gives

$$(m_{\text{lactose}})_R = 1,030 \exp[-1(1 - 0.085)] = 413 \text{ lb/day}$$

Flow rates of the other solutes in the concentrate from Section 3 are computed in a similar manner, from which the wt% TP + NPN (dry basis) is computed. If this is done with a spreadsheet, the Solver function can be used to determine the added water to achieve 85 wt%. The added water found in this manner is 80,520 lb/day.

The following material balance includes the permeate from the batch diafiltration stage and the computed retentate (concentrate) of 85 wt% (dry basis) from Section 3.

Concentrate and Permeate from a Single-stage Batch Diafiltration in Section 3, for an Added Water Rate of 80,520 lb/day, to Meet the 85 wt% Specification

Component	Flow Rate of Feed to Section 3, lb/day	Flow Rate of Concentrate, lb/day	Flow Rate of Permeate, lb/day
Water	53,214	54,351	79,383
True protein, TP	5,007	4,810	197
Nonprotein nitrogen, NPN	138	55	83
Lactose	1,030	302	728
Ash	185	56	129
Butter fat	500	500	0
Total	60,074	60,074	80,520

From this table the % yield of TP + NPN in Section 3 is $(4,810 + 55)/(5,007 + 138) \times 100\% = 94.56\%$. The overall yield

of TP + NPN from the whey feed is $(4,810 + 55)/(9,000) \times 100\% = 54.06\%$.

The membrane flux is $0.5415 \text{ gal/h-ft}^2$, as in Section 2. The volumetric permeate flow rate over a 4-hour batch operation = $80,520/[(4)(8.5)] = 2,368 \text{ gal/h}$. Therefore, the membrane area required = $2,368/0.5415 = 4,373 \text{ ft}^2$. The number of cartridges needed in Section 3 is $4,373/26.5 = 165$.

14.9 MICROFILTRATION

As shown in Table 14.10, microfiltration is a pressure-driven, microporous membrane process used to retain matter as low as 0.02 micron in size, but more commonly of 0.1–10 microns. The matter may include large colloids, small and solid particles, blood cells, yeast, bacteria and other microbial cells, and very large and soluble macromolecules.

Membrane structures for microfiltration include *screen filters* that collect retained matter on the surface and *depth filters* that trap particles at constrictions within the membrane. As Porter discusses in Schweitzer [52], depth filters include: (1) relatively thick, high-porosity (80–85%) cast-cellulose-ester membranes having an open, tortuous, sponge-like structure; and (2) thin, low-porosity (nominal 10%) polyester or polycarbonate track-etch membranes of a sieve-like structure with narrow distribution of straight-through, cylindrical pores. The latter have a much sharper cutoff, resulting in enhanced separation factors. For example, a Nuclepore™ membrane of type 2 can separate a male-determining sperm from a female-determining sperm. As shown in Table 14.11, Nuclepore™ membranes come in pore sizes from 0.03 to 8.0 microns with water permeate flux

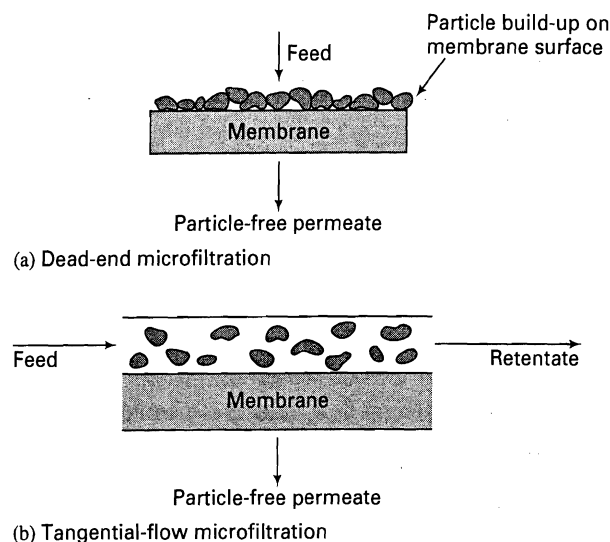


Figure 14.32 Common modes of microfiltration.

rates, at 70°F and a transmembrane pressure difference of 10 psi, ranging from 0.006 to 144 gal/min-ft².

Two different modes of microfiltration are employed, as shown in Figure 14.32. In *dead-end* (in-line) filtration (DEF), which is suitable for batch or semicontinuous filtration of dilute solutions, the entire solvent is forced under pressure through the membrane, in a manner similar to conventional filtration of solid-liquid slurries. With time, as retained material accumulates on or within the membrane, the pressure required to maintain a desired flow rate must increase or permeate flux will decrease. Thus, DEF may operate at constant pressure, constant flux, or a combination thereof. A

Table 14.11 Typical Specifications for Nuclepore™ Track-etch Microfiltration Membranes

Specified Pore Size, μm	Pore-size Range, μm	Nominal Pore Density, Pores/cm ²	Nominal Membrane Thickness, μm	Typical Flow Rates at 10 lb/in ² ΔP , 70°F
				Water, gal/(min)(ft ²)
8.0	6.9–8.0	1×10^5	8.0	144.0
5.0	4.3–5.0	4×10^5	8.6	148.0
3.0	2.5–3.0	2×10^6	11.0	121.0
1.0	0.8–1.0	2×10^7	11.5	67.5
0.8	0.64–0.80	3×10^7	11.6	48.3
0.6	0.48–0.60	3×10^7	11.6	16.3
0.4	0.32–0.40	1×10^8	11.6	17.0
0.2	0.16–0.20	3×10^8	12.0	3.1
0.1	0.08–0.10	3×10^8	5.3	1.9
0.08	0.064–0.080	3×10^8	5.4	0.37
0.05	0.040–0.050	6×10^8	5.4	1.12
0.03	0.024–0.030	6×10^8	5.4	0.006

combination, in which constant-flux operation is employed up to a limiting pressure, followed by constant-pressure operation until a minimum flux is reached, is a superior strategy to improve yield.

In *tangential-flow* (crossflow) *filtration* (TFF), which is more suitable for large-scale, continuous filtration, feed flows along the surface with only a fraction of the solvent passing through the membrane. Ideally, retained matter is carried out of the microfilter with retentate fluid. The tangential-flow mode is usually accompanied by a large retentate recycle in a manner similar to feed-and-bleed operations for ultrafiltration. Typical tangential flow velocities parallel to the membrane surface on the feed-retentate side are in the 3–25-ft/s range. The tangential-flow mode is also used, almost exclusively, for reverse osmosis and ultrafiltration, as discussed above in subsections 14.5 and 14.8. Significant attempts to improve tangential-flow filtration (TFF) have evolved into *high-performance tangential flow filtration* (HPTFF) for separation of macromolecular species. In a study by van Reis et al. [53], significant improvements in yield and purification factors were achieved in the separation of protein mixtures of close molecular weights with HPTFF, wherein the process is optimized with respect to membrane pore-size distribution, membrane chemistry (electrostatic interactions between the membrane material and the solutes), accompanying use of a solution buffer to control pH to obtain the proper electrostatic level, fluid dynamics (including crossflow velocity and transmembrane pressure), and number of stages.

Membrane modules for microfiltration are mainly DEF plate-and-frame and TFF pleated cartridges. In the latter, the membrane is pleated and then folded around a permeate core. Many module types are inexpensive and disposable. A typical disposable cartridge is 2.5 in. in diameter and 10 in. long, with 3 ft² of membrane area. The cartridge may include a prefilter to extend filter life by removing larger particles, leaving the microporous membrane to make the required separation. Compared to TFF, DEF has a lower capital cost, a higher operating cost, and a simpler operation. DEF is most suitable for dilute solutions, while TFF is preferred for concentrated solutions.

Equations for computing TFF microfiltration are those developed for ultrafiltration. This includes batch, continuous feed-and-bleed, and diafiltration operation modes. Equations for DEF microfiltration are those for conventional, batch, solid-liquid, slurry filtration, frequently referred to as cake filtration. These equations are developed as follows, where the only resistances considered are those of the membrane and buildup of particles in a cake on the membrane surface. Any deposition of matter within the membrane is ignored, as is osmotic pressure, which is almost always negligible for microfiltration because of the high solute molecular weight.

At any instant of time, the volumetric permeate flux, J , is

$$J = \frac{1}{A_M} \frac{dV}{dt} = \frac{\Delta P}{\mu(R_m + R_c)} \quad (14-99)$$

where

ΔP = pressure drop across the cake and membrane

μ = permeate viscosity

R_m and R_c = respectively, resistances to permeate flow of membrane and cake

V = permeate volume

A_M = membrane area

Assume that membrane resistance is constant and that the cake can be treated as a growing packed bed of particles, for which the Ergun correlation of Figure 6.37 can be applied. In the low-Reynolds-number region, which is typical of flow through a cake, pressure drop is governed by the Kozeny–Carman equation, which gives for cake resistance.

$$R_c = \frac{150l_c(1 - \epsilon_c)^2}{D_p^2 \epsilon_c^3} = K_1 \frac{l_c(1 - \epsilon_c)^2}{\epsilon_c^3} \quad (14-100)$$

where

l_c = cake thickness, which increases with time

ϵ_c = cake porosity

D_p = effective diameter of the matter in the cake

K_1 = experimental constant for a particular filtration system

The solid matter in the feed, m_c , that will be retained in the cake is

$$m_c = c_F V = \rho_c(1 - \epsilon_c)A_M l_c \quad (14-101)$$

where c_F = the solid matter per unit volume of liquid in the feed.

Substitution of (14-101) into (14-100) to eliminate l_c and substitution of the resulting equation for R_c into (14-99) gives

$$J = \frac{1}{A_M} \frac{dV}{dt} = \frac{\Delta P}{\mu \left[R_m + K_1 \frac{c_F V(1 - \epsilon_c)}{\rho_c A_M \epsilon_c^3} \right]} \quad (14-102)$$

$$= \frac{\Delta P}{\mu \left[R_m + K_2 \frac{c_F V}{A_M} \right]}$$

where K_2 = experimental constant for a particular filtration system.

Constant-Flux Operation

For operation with a constant permeate flux, J , and corresponding volumetric permeate flow rate, pressure drop will increase with time as cake thickness increases. From (14-102), $dV/dt = V/t$, with $V = 0$ at $t = 0$, and variation of pressure drop with time is given by

$$\Delta P = J\mu[R_m + K_2 c_F Jt] \quad (14-103)$$

Initially, pressure depends only on membrane resistance, with $\Delta P = J\mu R_m$. Subsequently, pressure drop increases linearly with time as cake thickness increases.

Constant-Pressure Operation

For operation with a constant pressure drop, permeate flow rate and corresponding permeate flux will decrease with time as cake thickness increases. For this type of operation, the integral of (14-102) applies:

$$\int_0^V \left[R_m + K_2 \frac{c_F V}{A_M} \right] dV = \frac{A_M \Delta P}{\mu} \int_0^t dt \quad (14-104)$$

Integration of (14-104) gives

$$R_m V + \frac{K_2 c_F V^2}{2 A_M} = \frac{A_M \Delta P t}{\mu} \quad (14-105)$$

Equation (14-105) is a quadratic equation in V , which when solved gives the following positive root:

$$V = -\frac{R_m A_M}{K_2 c_F} + \frac{A_M}{K_2 c_F} \left[R_m^2 + \frac{2 K_2 c_F \Delta P t}{\mu} \right]^{\frac{1}{2}} \quad (14-106)$$

Differentiating (14-106) to obtain the permeate flux gives

$$\frac{1}{A_M} \frac{dV}{dt} = J = \frac{\Delta P}{\mu} \left[R_m^2 + \frac{2 K_2 c_F \Delta P t}{\mu} \right]^{-\frac{1}{2}} \quad (14-107)$$

Combined Operation

As mentioned above, improvements in yield can be achieved by a combined operation in which: (1) Constant-flux operation is employed in Stage 1 up to a limiting pressure drop, followed by (2) Constant-pressure operation in Stage 2 until a minimum flux is reached. Let V_{CF} = the volume of permeate obtained during Stage 1 for the time period t_{CF} . Because of the constant permeate flux,

$$V_{CF} = J A_M t_{CF} \quad (14-108)$$

Let ΔP_{UL} = the upper limit of the pressure drop across the cake and membrane. Then, for Stage 2 at constant pressure, (14-104) becomes

$$\int_{V_{CF}}^V \left[R_m + K_2 \frac{c_F V}{A_M} \right] dV = \frac{A_M \Delta P_{UL}}{\mu} \int_{t_{CF}}^t dt \quad (14-109)$$

Integrating (14-109) and solving the resulting quadratic equation for the positive root of V ,

$$V = -\frac{R_m A_M}{K_2 c_F} + \left[\frac{A_M^2 R_m^2}{K_2^2 c_F^2} + \frac{2 A_M}{K_2 c_F} \times \left(R_m V_{CF} + \frac{K_2 c_F V_{CF}^2}{2 A_M} + \frac{A_M \Delta P_{UL} (t - t_{CF})}{\mu} \right) \right]^{\frac{1}{2}} \quad (14-110)$$

where $V - V_{CF}$ = the volume of permeate obtained during Stage 2 for the time period $t - t_{CF}$ during Stage 2.

During Stage 2, the permeate flux decreases with time, according to the following equation, which is obtained by applying to (14-110) the definition of the permeate flux of the left-hand side and middle part of (14-102):

$$J = \frac{A_M \Delta P_{UL}}{K_2 c_F \mu} \left[\frac{A_M^2 R_m^2}{K_2^2 c_F^2} + \frac{2 A_M}{K_2 c_F} \times \left(R_m V_{CF} + \frac{K_2 c_F V_{CF}^2}{2 A_M} + \frac{A_M \Delta P_{UL} (t - t_{CF})}{\mu} \right) \right]^{-\frac{1}{2}} \quad (14-111)$$

Application of the above equations for combined operation is illustrated in the following example.

EXAMPLE 14.15

Diluted skim milk with a protein concentration of 4.3 g/L is to undergo DEF microfiltration. Laboratory experiments have been performed using a cellulose-acetate membrane with an average pore diameter of 0.45 micron and a membrane area of 17.3 cm². For a Stage 1 operation at a constant permeate rate of 15 mL/min, pressure across the cake and membrane increases from 0.3 psi to an upper limit of 20 psi in 400 seconds. The permeate viscosity is 1 cP. If the operation is continued in a second stage at a constant pressure drop at the upper limit until the permeate rate drops to 5 mL/min, estimate the additional time of operation. Also, prepare plots of permeate volume in mL and permeate flux in mL/cm²-min as functions of time in seconds.

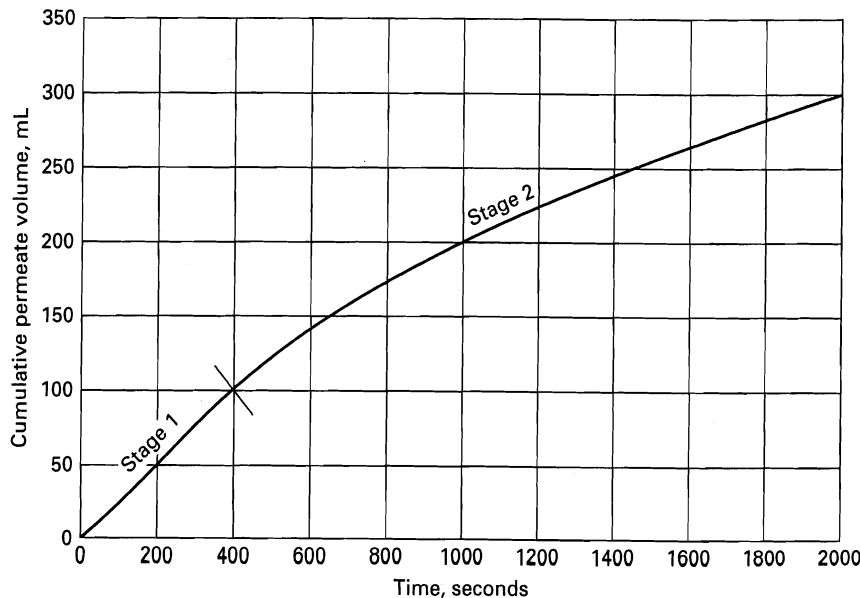


Figure 14.33 Cumulative permeate volume for Example 14.15.

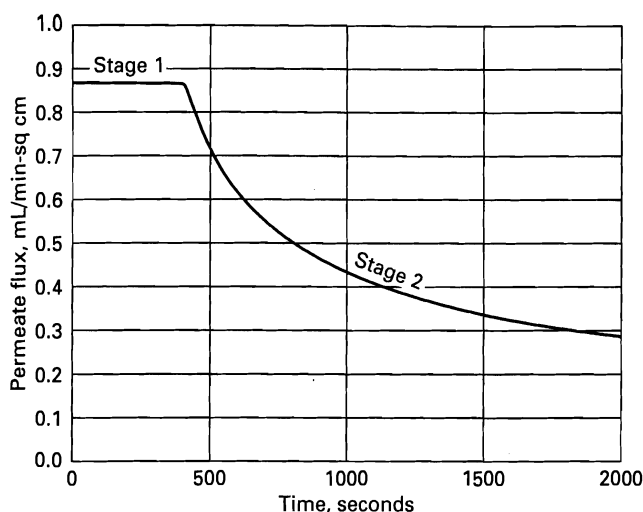


Figure 14.34 Instantaneous permeate flux for Example 14.15.

SOLUTION

In SI units, $c_F = 4.3 \text{ kg/m}^3$, $A_M = 0.00173 \text{ m}^2$, volumetric flow rate in Stage 1 $= 0.25 \times 10^{-6} \text{ m}^3/\text{s}$, P at zero time $= 2068 \text{ Pa}$, P at 400 s $= 137900 \text{ Pa}$, and $\mu = 0.001 \text{ Pa}\cdot\text{s}$. Based on the experimental

results for Stage 1, calculate values of R_m and K_2 , as follows. From (14-103),

$$R_m = \frac{(\Delta P \text{ at } t = 0)}{J\mu} = \frac{2068}{\left(\frac{0.25 \times 10^{-6}}{0.00173}\right)(0.001)} = 1.43 \times 10^{10} \text{ m}^{-1}$$

For $t = 400 \text{ s}$,

$$K_2 = \frac{\Delta P - J\mu R_m}{J^2\mu c_F t} = \frac{\Delta P - (\Delta P \text{ at } t = 0)}{J^2\mu c_F t} = \frac{6895(20 - 0.3)}{\left(\frac{0.25 \times 10^{-6}}{0.00173}\right)^2 (0.001)(4.3)(400)} = 3.78 \times 10^{12} \text{ m/kg}$$

Assume that these values of R_m and K_2 are valid during Stage 2. At the end of Stage 2, the permeate flux $= J = (5 \text{ mL/min})/(17.3 \text{ cm}^2) = 0.289 \text{ mL/min}\cdot\text{cm}^2 = 4.82 \times 10^{-5} \text{ m/s}$.

Solving (14-111) for t gives 2,025 s. Plots of permeate volume and permeate flux versus time, shown in Figures 14.33 and 14.34, are obtained by solving, as a function of time, (14-110) and (14-111) with a spreadsheet during Stage 2.

SUMMARY

1. The separation of liquid and gas mixtures with membranes is an emerging separation operation. Applications began accelerating in the 1980s. The products of separation are retentate and permeate.

2. The key to an efficient and economical membrane separation process is the membrane. It must have good permeability, high selectivity, stability, freedom from fouling, and a long life (2 or more years).

3. Commercialized, membrane-separation processes include dialysis, electrodialysis, reverse osmosis, gas permeation, pervaporation, ultrafiltration, and microfiltration.

4. Most membranes for commercial separation processes are natural or synthetic, glassy or rubbery polymers. However, for high-temperature ($>200^\circ\text{C}$) operations with chemically reactive mixtures, ceramics, metals, and carbon find applications.

5. To achieve high permeability and selectivity, dense, nonporous membranes are preferred. For mechanical integrity, membranes of 0.1 to 1.0 mm in thickness are incorporated as a surface layer or film onto or as part of a much thicker asymmetric or composite membrane.

6. To achieve a high surface area per unit volume, membranes are fabricated into spiral-wound or hollow-fiber modules. Less surface is available in plate-and-frame, tubular, and monolithic modules.

7. Permeation through a membrane can occur by a variety of mechanisms. For a microporous membrane, the mechanisms include bulk flow (with no selectivity), liquid diffusion, gas diffusion, Knudsen diffusion, restrictive diffusion (including sieving), and surface diffusion. For a nonporous membrane, a solution-diffusion mechanism, involving absorption, diffusion, and desorption, is commonly assumed.

8. Flow patterns in membrane modules have a profound effect on overall permeation rates. Idealized flow patterns for which theory

has been developed include perfect mixing, countercurrent flow, cocurrent flow, and crossflow.

9. To overcome the limit of separation in a single membrane-module stage, modules can be arranged in series and/or parallel cascades.

10. In gas permeation, boundary-layer or film mass-transfer resistances on either side of the membrane are usually negligible compared to the membrane resistance. For membrane separation of liquid mixtures, however, external mass-transfer effects and concentration polarization can be significant.

11. For most membrane separators, the component mass-transfer fluxes through the membrane can be formulated as the product of two terms: concentration, partial pressure, fugacity, or activity driving force; and a permeance $\bar{P}_{M_i}$, which is the ratio of the permeability, P_{M_i} , to the membrane thickness, l_M .

12. In the dialysis of a liquid mixture, small solutes of type A are separated from the solvent and larger solutes of type B with a microporous membrane. The driving force is the concentration difference across the membrane. Transport of solvent can be minimized by adjusting pressure differences across the membrane to equal osmotic pressure.

13. In electrodialysis, a series of alternating cation- and anion-selective membranes are used with a direct-current voltage across an outer anode and an outer cathode to concentrate an electrolyte.

14. In reverse osmosis, the solvent of a liquid mixture is selectively transported through a dense membrane. By this means, seawater can be desalinated. The driving force for solvent transport through the membrane is fugacity difference, which is commonly expressed in terms of $\Delta P - \Delta\pi$, where π is the osmotic pressure.

15. In gas permeation, mixtures of gases are separated by differences in permeation rates through dense membranes. The driving force for each component is its partial pressure difference, Δp_i ,

across the membrane. Both permeance and permeability depend on membrane absorptivity for the particular gas species (usually as a Henry's law constant) and species diffusivity through the membrane. Thus, $P_{M_i} = H_i D_i$.

16. In pervaporation, a liquid mixture is separated with a dense membrane by pulling a vacuum on the permeate side of the membrane so as to evaporate the permeate. The driving force may be approximated as a fugacity difference expressed by $(\gamma_i x_i P_i^s - y_i P_P)$.

REFERENCES

1. LONSDALE, H.K., *J. Membrane Sci.*, **10**, 81 (1982).
2. APPLGATE, L.E., *Chem. Eng.*, **91**(12), 64–89 (1984).
3. HAVENS, G.G., and D.B. GUY, *Chem. Eng. Progress Symp. Series*, **64**(90), 299 (1968).
4. BOLLINGER, W.A., D.L. MACLEAN, and R.S. NARAYAN, *Chem. Eng. Progress*, **78**(10), 27–32 (1982).
5. BAKER, R.W., E.L. CUSSLER, W. EYKAMP, W.J. KOROS, R.L. RILEY, and H. STRATHMANN, *Membrane Separation Systems—A Research and Development Needs Assessment*, Report DE 90-011770, Department of Commerce, NTIS, Springfield, VA (1990).
6. HO, W.S.W., and K.K. SIRKAR, Eds., *Membrane Handbook*, Van Nostrand Reinhold, New York (1992).
7. LOEB, S., and S. SOURIRAJAN, *Advances in Chemistry Series*, Vol. 38, *Saline Water Conversion II* (1963).
8. HENIS, J.M.S., and M.K. TRIPODI, U.S. Patent 4,230,463 (1980).
9. WRASIDLO, W.J., U.S. Patent 3,951,815 (1977).
10. BARRER, R.M., *J. Chem. Soc.*, 378–386 (1934).
11. BARRER, R.M., *Diffusion in and through Solids*, Cambridge Press, London (1951).
12. MAHON, H.I., U.S. Patent 3,228,876 (1966).
13. MAHON, H.I., U.S. Patent 3,228,877 (1966).
14. HSIEH, H.P., R.R. BHAVE, and H.L. FLEMING, *J. Membrane Sci.*, **39**, 221–241 (1988).
15. BIRD, R.B., W.E. STEWART, and E.N. LIGHTFOOT, *Transport Phenomena*, John Wiley and Sons, New York, pp. 42–47 (1960).
16. ERGUN, S., *Chem. Eng. Progress*, **48**, 89–94 (1952).
17. BECK, R.E., and J.S. SCHULTZ, *Science*, **170**, 1302–1305 (1970).
18. BECK, R.E., and J.S. SCHULTZ, *Biochim. Biophys. Acta*, **255**, 273 (1972).
19. BRANDRUP, J., and E.H. IMMERGUT, Eds., *Polymer Handbook*, 3rd ed., John Wiley and Sons, New York (1989).
20. LONSDALE, H.K., U. MERTEN, and R.L. RILEY, *J. Applied Polym. Sci.*, **9**, 1341–1362 (1965).
21. MOTAMEDIAN, S., W. PUSCH, G. SENDELBAACH, T.-M. TAK, and T. TANIOKA, *Proceedings of the 1990 International Congress on Membranes and Membrane Processes*, Chicago, Vol. II, pp. 841–843.
22. BARRER, R.M., J.A. BARRIE, and J. SLATER, *J. Polym. Sci.*, **23**, 315–329 (1957).
23. BARRER, R.M., and J.A. BARRIE, *J. Polym. Sci.*, **23**, 331–344 (1957).
24. BARRER, R.M., J.A. BARRIE, and J. SLATER, *J. Polym. Sci.*, **27**, 177–197 (1958).
25. KOROS, W.J., and D.R. PAUL, *J. Polym. Sci., Polym. Physics Edition*, **16**, 1947–1963 (1978).
26. BARRER, R.M., *J. Membrane Sci.*, **18**, 25–35 (1984).
27. WALAWENDER, W.P., and S.A. STERN, *Separation Sci.*, **7**, 553–584 (1972).
28. NAYLOR, R.W., and P.O. BACKER, *AIChE J.*, **1**, 95–99 (1955).
29. STERN, S.A., T.F. SINCLAIR, P.J. GAREIS, N.P. VAHLDECK, and P.H. MOHR, *Ind. Eng. Chem.*, **57**(2), 49–60 (1965).
30. HWANG, S.-T., and K.L. KAMMERMEYER, *Membranes in Separations*, Wiley-Interscience, New York, pp. 324–338 (1975).
31. SPILLMAN, R.W., *Chem. Eng. Progress*, **85**(1), 41–62 (1989).
32. STRATHMANN, H., "Membrane and Membrane Separation Processes," in Vol. A16, *Ullmann's Encyclopedia of Industrial Chemistry*, VCH, FRG, p. 237 (1990).
33. CHAMBERLIN, N.S., and B.H. VROMEN, *Chem. Engr.* **66**(9), 117–122 (1959).
34. GRAHAM, T., *Phil. Trans. Roy. Soc. London*, **151**, 183–224 (1861).
35. JUDA, W., and W.A. MCRAE, *J. Amer. Chem. Soc.*, **72**, 1044 (1950).
36. STRATHMANN, H., *Sep. and Purif. Methods*, **14**(1), 41–66 (1985).
37. MERTEN, U., *Ind. Eng. Chem. Fundamentals*, **2**, 229–232 (1963).
38. STOUGHTON, R.W., and M.H. LIETZKE, *J. Chem. Eng. Data*, **10**, 254–260 (1965).
39. SPILLMAN, R.W., and M.B. SHERWIN, *Chemtech*, 378–384 (June 1990).
40. SCHELL, W.J., and C.D. HOUSTON, *Chem. Eng. Progress*, **78**(10), 33–37 (1982).
41. TEPLYAKOV, V., and P. MEARES, *Gas Sep. and Purif.*, **4**, 66–74 (1990).
42. ROSENZWEIG, M.D., *Chem. Eng.*, **88**(24), 62–66 (1981).
43. KOBER, P.A., *J. Am. Chem. Soc.*, **39**, 944–948 (1917).
44. BINNING, R.C., R.J. LEE, J.F. JENNINGS, and E.C. MARTIN, *Ind. Eng. Chem.*, **53**, 45–50 (1961).
45. WESSLEIN, M., A. HEINTZ, and R.N. LICHTENTHALER, *J. Membrane Sci.*, **51**, 169 (1990).
46. WIJMANS, J.G., and R.W. BAKER, *J. Membrane Sci.*, **79**, 101–113 (1993).
47. RAUTENBACH, R., and R. ALBRECHT, *Membrane Processes*, John Wiley and Sons, New York (1989).
48. RAO, M.B., and S. SIRKAR, *J. Membrane Sci.*, **85**, 253–264 (1993).
49. BAKER, R., *Membrane Technology and Applications*, 2nd ed., John Wiley & Sons, New York (2004).
50. CHERYAN, M., *Ultrafiltration Handbook*, Technomic Publishing Co., Lancaster, PA (1986).
51. ZEMAN, L.J., and A.L. ZYDNEY, *Microfiltration and Ultrafiltration, Principles and Applications*, Marcel Dekker, Inc., New York (1996).
52. SCHWEITZER, P.A., *Handbook of Separation Techniques for Chemical Engineers*, 2nd ed., Section 2.1 by M.C. Porter, McGraw-Hill Book Co., New York (1988).
53. VAN REIS, R. et al., *Biotechnology and Bioengineering*, **56**(1), October 5, 1997.

EXERCISES

Section 14.1

14.1 Explain, as completely as you can, how membrane separations differ from:

- Absorption and stripping
- Distillation
- Liquid-liquid extraction
- Extractive distillation

14.2 For the commercial application of membrane separators discussed at the beginning of this chapter, calculate the permeabilities of hydrogen and methane in barrer.

14.3 A new asymmetric, polyimide, polymer membrane has been developed for the separation of N_2 from CH_4 . At $30^\circ C$, permeance values are 50,000 and 10,000 barrer/cm for N_2 and CH_4 , respectively. If this new membrane is used to perform the separation in Figure 14.35, determine the membrane surface area required in m^2 , and the kmol/h of CH_4 in the permeate. Base the driving force for diffusion through the membrane on the arithmetic average of the partial pressures of the entering feed and the exiting retentate, with the permeate-side partial pressures at the exit condition.

Section 14.2

14.4 A hollow-fiber module has 4,000 ft^2 of membrane surface area based on the inside diameter of the fibers, which are $42 \mu m$ i.d. $\times$ $85 \mu m$ o.d. $\times$ 1.2 m long each. Determine:

- The number of hollow fibers in the module.
- The diameter of the module, assuming the fibers are on a square spacing of $120 \mu m$ center-to-center.
- The membrane surface area per unit volume of module (packing density) m^2/m^3 . Compare your result with that in Table 14.4.

14.5 A typical spiral-wound module made from a flat sheet of membrane material is 0.3 m in diameter and 3 m long. If the packing density (membrane surface area/unit module volume) is $500 m^2/m^3$, determine the center-to-center spacing of the membrane in the spiral, assuming a collection tube 1 cm in diameter.

14.6 A monolithic membrane element, of the type shown in Figure 14.4d, contains 19 flow channels that are 0.5 cm in inside diameter by 0.85 m long. If nine of these elements are placed into a cylindrical module of the type shown in Figure 14.5, determine reasonable values for:

- Module volume in m^3 .
- Packing density in m^2/m^3 . Compare your value with values for other membrane modules given in Table 14.4.

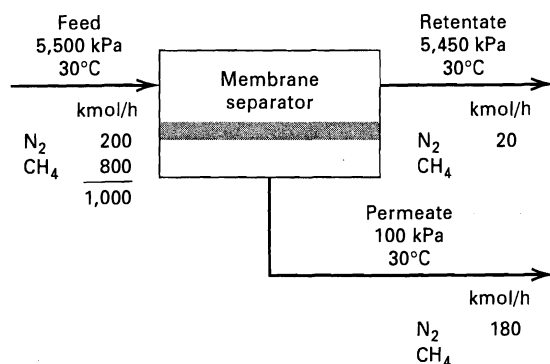


Figure 14.35 Data for Exercise 14.3.

Section 14.3

14.7 Water at $70^\circ C$ is to be passed through a porous polyethylene membrane of 25% porosity with an average pore diameter of 0.3 micron and an average tortuosity of 1.3. The pressures on the downstream and upstream sides of the membrane are 125 and 500 kPa, respectively. Estimate the flow rate of water through the membrane in m^3/m^2 -day.

14.8 A porous-glass membrane, with an average pore diameter of $40 \text{ \AA}$, is to be used to separate light gases at $25^\circ C$ under conditions where Knudsen flow may be dominant. The downstream pressure is 15 psia, while the upstream pressure is not greater than 120 psia. The membrane has been calibrated with pure helium gas, giving a constant permeability of 117,000 barrer over the operating pressure range. Experiments with pure CO_2 over the pressure range give a permeability of 68,000 barrer.

Assuming that helium is in Knudsen flow, predict the permeability of CO_2 . Is the value in agreement with the experimental value? If not, suggest an explanation. Reference: Kammermeyer, K., and L.O. Rutz, *C.E.P. Symp. Ser.*, 55(24), 163-169 (1959).

14.9 Two mechanisms for the transport of gas components through a porous membrane that are not discussed in Section 14.3 or illustrated in Figure 14.6 are (1) partial condensation in the pores by some components of the gas mixture to the exclusion of other components and subsequent transport of the condensed molecules through the pore, and (2) selective adsorption on pore surfaces of certain components of the gas mixture and subsequent surface diffusion across the pores. In particular, Rao and Sircar [48] have found that the latter mechanism provides a potentially attractive means for separating hydrocarbons from hydrogen for low-pressure gas streams. In porous-carbon membranes with continuous pores $4\text{--}15 \text{ \AA}$ in diameter, little pore void space is available for the Knudsen diffusion of hydrogen when the hydrocarbons are selectively adsorbed.

Typically, the membranes are not more than $5 \mu m$ in thickness. Measurements at 295.1 K of permeabilities for five pure components and a mixture of the five components are as follows:

Component	Permeability, barrer		
	As a Pure Gas	In the Mixture	mol% in the Mixture
H_2	130	1.2	41.0
CH_4	660	1.3	20.2
C_2H_6	850	7.7	9.5
C_3H_8	290	25.4	9.4
nC_4H_{10}	155	112.3	19.9
			100.0

A refinery waste gas mixture of the preceding composition is to be processed through such a porous-carbon membrane. If the pressure of the gas is 1.2 atm and an inert sweep gas is used on the permeate side such that partial pressures of feed gas components on that side are close to zero, determine the permeate composition on a sweep-gas-free basis when the composition on the upstream-pressure side of the membrane is that of the feed gas. Explain why the component permeabilities differ so drastically between experiments with the pure gas and the gas mixture.

14.10 A mixture of 60 mol% propylene and 40 mol% propane at a flow rate of 100 lbmol/h and at $25^\circ C$ and 300 psia is to be separated with a polyvinyltrimethylsilane polymer (see Table 14.9 for permeabilities). The membrane skin is $0.1 \mu m$ thick, and spiral-wound modules are used with a pressure of 15 psia on the permeate

side. Calculate the material balance and membrane area in m^2 as a function of the cut (fraction of feed permeated) for:

- (a) Perfect-mixing flow pattern.
(b) Crossflow pattern.

14.11 Repeat part (a) of Exercise 14.10 for a two-stage stripping cascade and a two-stage enriching cascade, as shown in Figure 14.12. However, select just one set of reasonable cuts for the two stages of each case so as to produce 40 lbmol/h of final retentate.

14.12 Repeat Example 14.7 with the following changes:

Tube-side Reynolds number = 25,000

Tube inside diameter = 0.4 cm

Permeate-side mass-transfer coefficient = 0.06 cm/s

How important is concentration polarization?

Section 14.4

14.13 An aqueous process stream of 100 gal/h at 20°C contains 8 wt% Na_2SO_4 and 6 wt% of a high-molecular-weight substance (A). This stream is processed in a continuous, countercurrent-flow dialyzer using a pure water sweep of the same flow rate. The membrane is a microporous cellophane with pore volume = 50%, wet thickness = 0.0051 cm, tortuosity = 4.1, and pore diameter = 31 Å. The molecules to be separated have the following properties:

	Na_2SO_4	A
Molecular weight	142	1,000
Molecular diameter, Å	5.5	15.0
Diffusivity, $cm^2/s \times 10^5$	0.77	0.25

Calculate the membrane area in m^2 for only a 10% transfer of A through the membrane, assuming no transfer of water. What is the percent recovery of the Na_2SO_4 in the diffusate? Use log-mean concentration driving forces and assume that the mass-transfer resistances on each side of the membrane are each 25% of the total mass-transfer resistances for Na_2SO_4 and A.

14.14 A dialyzer is to be used to separate 300 L/h of an aqueous solution containing 0.1 M NaCl and 0.2 M HCl. Laboratory experiments with the microporous membrane to be used give the following values for the overall mass-transfer coefficient K_i in (14-57), for a log-mean concentration driving force:

	K_i , cm/min
Water	0.0025
NaCl	0.021
HCl	0.055

Determine the membrane area in m^2 for 90, 95, and 98% transfer of HCl to the diffusate. For each of the three cases, determine the complete material balance in kmol/h. A sweep of 300 L/h can be assumed.

14.15 A total of 86,000 gal/day of an aqueous solution of 3,000 ppm of NaCl is to be desalinated to 400 ppm by electrodialysis, with a 40% conversion. The process will be conducted in four stages, with three stacks of 150 cell pairs in each stage. The fractional desalination will be the same in each stage and the expected current efficiency is 90%. The applied voltage for the first stage is 220 V. Each cell pair has an area of 1,160 cm^2 . Calculate the current density in mA/cm^2 , the current in A, and the power requirement in kW for the first stage. Reference: Mason, E.A., and T.A. Kirkham, *C.E.P. Symp. Ser.*, **55**(24), 173–189 (1959).

Section 14.5

14.16 A reverse-osmosis plant is used to treat 30,000,000 gal/day of seawater at 20°C containing 3.5 wt% dissolved solids to produce 10,000,000 gal/day of potable water with 500 ppm of dissolved solids, and the balance as brine containing 5.25 wt% dissolved solids. The feed-side pressure is 2,000 psia, while the permeate pressure is 50 psia. A single stage of spiral-wound membranes is used that approximates crossflow. If the total membrane area is 2,000,000 ft^2 , estimate the permeance for water and the salt passage.

14.17 A reverse-osmosis process is to be designed to handle a feed flow rate of 100 gal/min. Three designs have been proposed, differing in the % recovery of potable water from the feed:

Design 1: A single stage consisting of four units in parallel to obtain a 50% recovery

Design 2: Two stages in series with respect to the retentate (four units in parallel followed by two units in parallel)

Design 3: Three stages in series with respect to the retentate (four units in parallel followed by two units in parallel followed by a single unit)

Draw the three designs and determine the percent recovery of potable water for Designs 2 and 3.

14.18 The production of paper involves a pulping step to break down wood chips into cellulose and lignin. In the Kraft process, an aqueous, pulping-feed solution, known as white liquor, is used that consists of dissolved inorganic chemicals such as sodium sulfide and sodium hydroxide. Following removal of the pulp (primarily cellulose), a solution known as weak (Kraft) black liquor (KBL) is left, which is regenerated to recover white liquor for recycle. In the conventional process, a typical 15 wt% (dissolved solids) KBL is concentrated to 45 to 70 wt% by multieffect evaporation. It has been suggested that reverse osmosis might be used to perform an initial concentration to perhaps 25 wt%. Higher concentrations may not be feasible because of the very high osmotic pressure, which at 180°F and 25 wt% solids is estimated to be 1,700 psia. The osmotic pressure for other conditions can be scaled with (14-68) using wt% instead of molality.

A two-stage RO process, shown in Figure 14.36, has been proposed to carry out this initial concentration for a feed rate of 1,000 lb/h at 180°F. A feed pressure of 1,756 psia is used for the first stage to yield a permeate of 0.4 wt% solids. The feed pressure to the second stage is 518 psia to produce water of 300 ppm dissolved solids and a retentate of 2.6 wt% solids. Permeate-side pressure for both stages is 15 psia. Equation (14-69) can be used to estimate membrane area, where the permeance for water can be taken as 0.0134 $lb/ft^2\text{-hr-psi}$ in conjunction with an arithmetic-mean osmotic pressure for plug flow on the feed side. Complete the material balance for the process and estimate the required

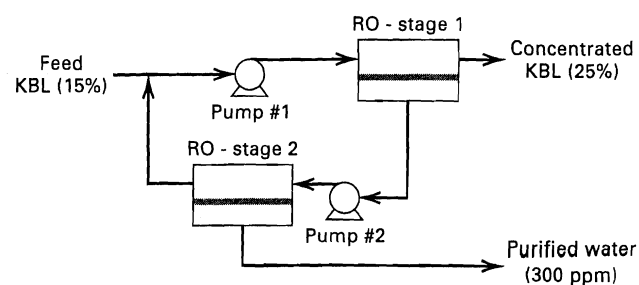


Figure 14.36 Data for Exercise 14.18.

membrane areas for each stage. Reference: Gottschlich, D.E., and D.L. Roberts. Final Report DE91004710, SRI International, Menlo Park, CA, Sept. 28, 1990.

Section 14.6

14.19 Gas permeation can be used to recover VOCs from air at low pressures using a membrane material that is highly selective for the VOCs. In a typical application, 1,500 scfm (0°C, 1 atm) of air containing 0.5 mol% acetone (A) is fed to a spiral-wound membrane module system at 40°C and 1.2 atm. A liquid-ring vacuum pump on the permeate side establishes a pressure of 4 cmHg. A silicone-rubber, thin-composite membrane with a 2- μ m-thick skin gives permeabilities of 4 barrer for air and 20,000 barrer for acetone.

If the retentate is to contain 0.05 mol% acetone and the permeate is to contain 5 mol% acetone, determine the membrane area required in m^2 , assuming crossflow. References: (1) Peinemann, K.-V., J.M. Mohr, and R.W. Baker, *C.E.P. Symp. Series*, **82**(250), 19–26 (1986); (2) Baker, R.W., N. Yoshioka, J.M. Mohr, and A.J. Khan, *J. Membrane Sci.*, **31**, 259–271 (1987).

14.20 The separation of air into nitrogen and oxygen is widely practiced. Cryogenic distillation is most economical for processing 100 to 5,000 tons of air per day, while pressure-swing-adsorption is favorable for 20 to 50 tons/day. For small-volume users requiring less than 10 tons/day, gas permeation finds applications where for a single stage, either an oxygen-enriched air (40 mol% oxygen) or 98 mol% nitrogen can be produced. It is desired to produce 5 tons/day (2,000 lb/ton) of 40 mol% oxygen and nitrogen, ideally of 90 mol% purity, by gas permeation. Assume pressures of 500 psia (feed side) and 20 psia (permeate). Two companies, who can supply the membrane modules, have provided the following data:

	Company A	Company B
Module type	Hollow-fiber	Spiral-wound
$\bar{P}_M$ for O ₂ , barrer/ μ m	15	35
$\bar{P}_{M_{O_2}} / \bar{P}_{M_{N_2}}$	3.5	1.9

Determine the required membrane area in m^2 for each company. Assume that both module types approximate crossflow.

14.21 A joint venture has been underway for several years to develop a membrane process to separate CO₂ and H₂S from high-pressure sour natural gas. Typical feed and product conditions are:

	Feed Gas	Pipeline Gas
Pressure, psia	1,000	980
Composition, mol%:		
CH ₄	70	97.96
H ₂ S	10	0.04
CO ₂	20	2.00

To meet these conditions, the following hollow-fiber membrane material targets have been established:

Selectivity	
CO ₂ –CH ₄	50
H ₂ S–CH ₄	50

where selectivity is the ratio of permeabilities.

$$P_{M_{CO_2}} = 13.3 \text{ barrer,}$$

and membrane skin thickness is expected to be 0.5 μ m.

Make calculations to show whether the targets can realistically meet the pipeline-gas conditions in a single stage with a reasonable

membrane area. Assume a feed-gas flow rate of 10×10^3 scfm (0°C, 1 atm) with crossflow.

Reference: Stam, H., in *Future Industrial Prospects of Membrane Processes*, L. Cecille and J.-C. Toussaint, Eds., Elsevier Applied Science, London, pp. 135–152 (1989).

Section 14.7

14.22 Pervaporation is to be used to separate ethyl acetate (EA) from water. The feed rate is 100,000 gal/day of water containing 2.0 wt% EA at 30°C and 20 psia. The membrane is dense polydimethylsiloxane with a 1- μ m-thick skin in a spiral-wound module that approximates crossflow. The permeate pressure is 3 cmHg. The total measured membrane flux at these conditions is 1.0 L/ m^2 -h with a separation factor given by (14-36) of 100 for EA with respect to water. A retentate of 0.2 wt% EA is desired for a permeate of 45.7 wt% EA. Determine the required membrane area in m^2 and estimate the temperature drop of the feed. Reference: Blume, I., J.G. Wijans, and R.W. Baker, *J. Membrane Sci.*, **49**, 253–286 (1990).

14.23 For a temperature of 60°C and a permeate pressure of 15.2 mmHg, Wesslein et al. [45] measured a total permeation flux of 1.6 kg/ m^2 -h for a 17.0 wt% ethanol in water feed, giving a permeate of 12 wt% ethanol. Otherwise, conditions were those of Example 14.12. Calculate the permeances of ethyl alcohol and water for these conditions. Also, calculate the selectivity for water.

14.24 The separation of benzene (B) from cyclohexane (C) by distillation at 1 atm is impossible because of a minimum-boiling-point azeotrope at 54.5 mol% benzene. However, extractive distillation with furfural is feasible. For an equimolar feed, cyclohexane and benzene products of 98 and 99 mol%, respectively, can be produced. Alternatively, the use of a three-stage pervaporation process, with selectivity for benzene using a polyethylene membrane, has received attention, as discussed by Rautenbach and Albrecht [47]. Consider the second stage of this process where the feed is 9,905 kg/h of 57.5 wt% B at 75°C. The retentate is 16.4 wt% benzene at 67.5°C and the permeate is 88.2 wt% benzene at 27.5°C. The total permeate mass flux is 1.43 kg/ m^2 -h and the selectivity for benzene is 8. Calculate the flow rates of retentate and permeate in kg/h and the required membrane surface area in m^2 .

Section 14.8

14.25 Based on the problem statement of Example 14.14, calculate for just Section 1 the component material balance in pounds per day of operation, the percent recovery (yield) from the whey of the TP and NPN in the final concentrate, and the number of cartridges required if only two stages are used instead of four.

14.26 Based on the problem statement of Example 14.14, design a four-stage diafiltration section to take the 55 wt% concentrate from Section 1 and achieve the desired 85 wt% concentrate, thus eliminating Section 3.

Section 14.9

14.27 Using the membrane and feed conditions of and values for R_m and K_2 determined in Example 14.15 for DEF microfiltration, compute and plot the permeate flux and cumulative permeate volume as a function of time. Assume a combined operation with Stage 1 at a constant permeate rate of 10 mL/min to an upper-limit pressure drop of 25 psi, followed by Stage 2 at this pressure drop until the permeate rate drops to a lower limit of 5 mL/min.

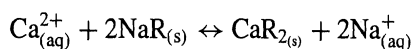
Chapter 15

Adsorption, Ion Exchange, and Chromatography

Adsorption, ion exchange, and chromatography are *sorption* operations, in which certain components of a fluid phase, called solutes, are selectively transferred to insoluble, rigid particles suspended in a vessel or packed in a column. Sorption, which is a general term introduced by J.W. McBain [*Phil. Mag.*, **18**, 916–935 (1909)], includes selective transfer to the surface and/or into the bulk of a solid or liquid. Thus, absorption of gas species into a liquid and penetration of fluid species into a nonporous membrane are also sorption operations. In a general sorption process, the sorbed solutes are referred to as *sorbate*, and the sorbing agent is the *sorbent*.

In an *adsorption* process, molecules, as shown in Figure 15.1a, or atoms or ions, in a gas or liquid diffuse to the surface of a solid, where they bond with the solid surface or are held there by weak intermolecular forces. The adsorbed solutes are referred to as *adsorbate*, whereas the solid material is the *adsorbent*. To achieve a very large surface area for adsorption per unit volume, highly porous solid particles with small-diameter interconnected pores are used, with the bulk of the adsorption occurring within the pores.

In an *ion-exchange* process, as shown in Figure 15.1b, ions of positive charge (*cations*) or negative charge (*anions*) in a liquid solution, usually aqueous, replace dissimilar and displaceable ions of the same charge contained in a solid *ion exchanger*, which also contains immobile, insoluble, and permanently bound co-ions of the opposite charge. Thus, ion exchange can be cation or anion exchange. Water softening by ion exchange involves a cation exchanger, in which the following reaction occurs to replace calcium ions with sodium ions.



where R is the residual material of the ion exchanger. The exchange of ions is reversible and does not cause any permanent change to the structure of the solid ion exchanger. Thus, it can be used and reused unless fouled by organic compounds in the liquid feeds that attach to exchange sites on and within the ion exchanger. The ion-exchange concept can be extended to the removal of essentially all inorganic salts from water by a two-step process called *demineralization* or

deionization. In the first step, a cation resin exchanges hydrogen ions for cations such as calcium, magnesium, and sodium. In the second step, an anion resin exchanges hydroxyl ions for strongly and weakly ionized anions such as sulfate, nitrate, chloride, and bicarbonate. The hydrogen and hydroxyl ions that enter the water combine to form water. Regeneration of the cation and anion resins is usually accomplished with sulfuric acid and sodium hydroxide, respectively.

In *chromatography*, the sorbent may be a solid adsorbent, an insoluble, nonvolatile, liquid absorbent contained in the pores of a granular solid support, or an ion exchanger. In either case, the solutes to be separated move through the chromatographic separator, with an inert, eluting fluid, at different rates because of repeated sorption, desorption cycles.

During adsorption and ion exchange, the solid separating agent becomes saturated or nearly saturated with the molecules, atoms, or ions transferred from the fluid phase. To recover the sorbed substances and allow the adsorbent to be reused, it is regenerated by desorbing the sorbed substances. Accordingly, these two separation operations are carried out in a cyclic manner. In chromatography, regeneration occurs continuously, but at changing locations in the separator.

Adsorption processes may be classified as *purification* or *bulk separation*, depending on the concentration in the feed fluid of the components to be adsorbed. Although there is no sharp dividing concentration, Keller [1] has suggested 10 wt%. Early applications of adsorption involved only purification. For example, adsorption with charred wood to improve the taste of water has been known for centuries. The decolorization of liquid solutions by adsorption with bone char and other materials has been practiced for at least five centuries. Adsorption of gases by a solid (charcoal) was first described by C.W. Scheele in 1773. Commercial applications of bulk separation by gas adsorption began in the early 1920s, but did not escalate until the 1960s, following the inventions by Milton [2] of synthetic molecular-sieve zeolites, which provide high adsorptive selectivity, and by Skarstrom [3] of the *pressure-swing cycle*, which made possible the efficient operation of a fixed-bed, cyclic, gas adsorption process. The commercial-scale bulk separation

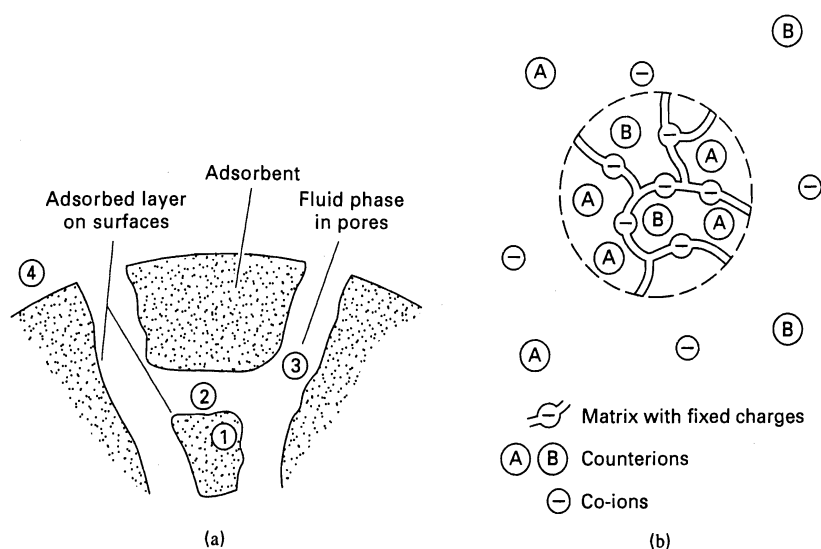


Figure 15.1 Sorption operations with solid-particle sorbents. (a) Adsorption. (b) Ion exchange.

of liquid mixtures also began in the 1960s, following the invention by Broughton and Gerhold [4] of the simulated moving bed for adsorptive separation.

The use of ion exchange dates back to at least the time of Moses, who, while leading his followers out of Egypt into the wilderness, sweetened the bitter waters of Marah with a tree [Exodus 15:23–26]. In ancient Greece, Aristotle observed that the salt content of water is reduced when it percolates through certain sands. Systematic studies of ion exchange were published in 1850 by both Thompson and Way, who experimented with cation exchange in soils before the discovery of the existence of ions.

The first major application of ion exchange, which occurred about 100 years ago, was for water treatment to remove the ions responsible for water hardness, such as calcium. Initially, the ion exchanger was a porous, natural, mineral zeolite containing silica. In 1935, synthetic, insoluble,

polymeric-resin ion exchangers were introduced. Today they are dominant for water-softening and deionizing applications, but natural and synthetic zeolites still find some use.

Since the invention of chromatography by M.S. Tswett [5], a Russian botanist, in 1903, it has found widespread use as an analytical and preparative laboratory technique. Tswett separated a mixture of structurally similar yellow and green chloroplast pigments in leaf extracts by dissolving the extracts in carbon disulfide and passing the solution through a column packed with chalk particles. The pigments were separated by color. Hence, the name chromatography was coined by Tswett in 1906 from the Greek words *chroma*, meaning color, and *graphe*, meaning writing. Chromatography has revolutionized the laboratory chemical analysis of liquid and, particularly, gas mixtures. The large-scale, commercial applications described by Bonmati et al. [6] and Bernard et al. [7], however, did not begin until the 1980s.

15.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- List the major types of porous adsorbents and their most significant properties.
- Explain why a few grams of porous adsorbent can have an adsorption area as large as a football field.
- Differentiate between chemisorption and physical adsorption.
- Explain how ion-exchange resins work.
- List types of sorbents used in chromatography.
- Compare three major expressions (so-called isotherms) used for correlating single-component adsorption equilibria.
- List steps involved in adsorption of a solute, and which steps may control rate of adsorption.
- Estimate external and internal rates of adsorption.
- Describe major modes for contacting the adsorbent with a fluid containing solute(s) to be adsorbed.
- Describe major methods for regenerating adsorbent.
- Calculate vessel size or residence time for any of the major modes of slurry adsorption.
- List and explain assumptions for ideal fixed-bed adsorption and explain the concept of width of mass-transfer zone.

- Explain the concept of breakthrough in fixed-bed adsorption.
- Calculate bed height, bed diameter, and cycle time for fixed-bed adsorption.
- Compute separations for a simulated moving bed operation.
- Make design calculations for ion-exchange cycles.
- Calculate rectangular and Gaussian-distribution pulses in chromatography.

Industrial Example

The pressure-swing gas adsorption process is primarily used for the dehydration of air and for the separation of air into nitrogen and oxygen. A small unit for the dehydration of compressed air is described by White and Barkley [8] and shown in Figure 15.2. The unit consists of two fixed-bed adsorbers, each 12.06 cm in diameter and packed with 11.15 kg of 3.3-mm-diameter Alcoa F-200 activated-alumina beads to a height of 1.27 m. The external porosity (void fraction) of the bed is 0.442 and the alumina-bead bulk density is 769 kg/m³.

The unit operates on a 10-min cycle, with 5 min for adsorption of water vapor from the air and 5 min for regeneration, which consists of depressurization, purging of the water vapor, and a 30-s repressurization. While one bed is adsorbing, the other bed is being regenerated. The adsorption (drying) step takes place with air entering at 21°C and 653.3 kPa (6.45 atm) with a flow rate of 1.327 kg/min, passing up through the bed with a pressure drop of 2.386 kPa. The dew-point temperature of the air at system pressure is reduced from 11.2 to -61°C by the adsorption process. During the 270-second period of purging, about a third of the dry air leaving one bed is directed to the other bed as a downward-flowing purge to regenerate the adsorbent. The purge is exhausted at a pressure of 141.3 kPa. By conducting the

purge flow countercurrent to the entering air flow, the highest degree of water-vapor desorption is achieved.

Other equipment, shown in Figure 15.2, includes an air compressor; an aftercooler; piping and valving to switch the beds from one step in the cycle to the other; a coalescing filter

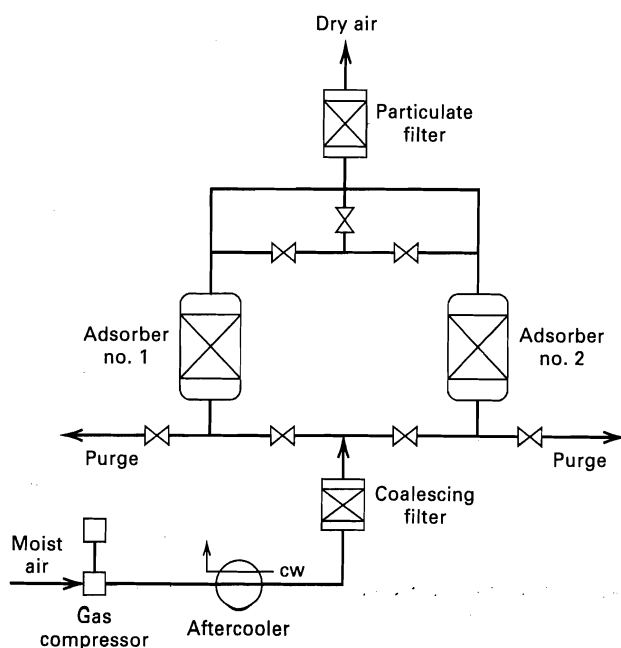


Figure 15.2 Pressure-swing adsorption for the dehydration of air.

Table 15.1 Industrial Applications of Sorption Operations

1. Adsorption

Gas purifications:

- Removal of organics from vent streams
- Removal of SO₂ from vent streams
- Removal of sulfur compounds from gas streams
- Removal of water vapor from air and other gas streams
- Removal of solvents and odors from air
- Removal of NO_x from N₂
- Removal of CO₂ from natural gas

Gas bulk separations:

- N₂/O₂
- H₂O/ethanol
- Acetone/vent streams
- C₂H₄/vent streams
- Normal paraffins/isoparaffins, aromatics
- CO, CH₄, CO₂, N₂, A, NH₃/H₂

Liquid purifications:

- Removal of H₂O from organic solutions
- Removal of organics from H₂O
- Removal of sulfur compounds from organic solutions
- Decolorization of solutions

Liquid bulk separations:

- Normal paraffins/isoparaffins
- Normal paraffins/olefins
- p*-xylene/other C₈ aromatics
- p*- or *m*-cymene/other cymene isomers
- p*- or *m*-cresol/other cresol isomers
- Fructose/dextrose, polysaccharides

2. Ion Exchange

- Water softening
- Water demineralization
- Water dealkalization
- Decolorization of sugar solutions
- Recovery of uranium from acid leach solutions
- Recovery of antibiotics from fermentation broths
- Recovery of vitamins from fermentation broths

3. Chromatography

- Separation of sugars
- Separation of perfume ingredients
- Separation of C₄–C₁₀ normal and isoparaffins

to remove aerosols from the entering air; and a particulate filter for the exiting dry air to remove adsorbent fines. If the dry air is needed only at low-to-moderate pressures, an air turbine can be installed to recover energy while reducing the air pressure.

During the 5-min adsorption period of the cycle, the capacity of the adsorbent for water must not be exceeded. In this example, the water content of the air is reduced from 1.27×10^{-3} kg H₂O/kg air to the very low value of 9.95×10^{-7} kg H₂O/kg air. To achieve this exiting water vapor content, only a small fraction of the adsorbent capacity is utilized during the adsorption step, with most of the adsorption occurring in the first 0.2 m of the 1.27-m bed height.

The bulk separation of gas and liquid mixtures by adsorption is an emerging separation operation. Important progress is being made in the development of new and more-selective

adsorbents and in more-efficient operation cycles. In addition, attention is being directed to hybrid systems that include membrane and other types of separation steps. Already, the three sorption operations addressed in this chapter have found numerous applications, some of which are listed in Table 15.1, compiled from listings in Rousseau [9]. The applications cover a very wide range of species molecular weight.

This chapter discusses sorbents, including their equilibrium, sieving, transport, and kinetic properties with respect to solutes being removed from solutions; techniques for conducting cyclic operations; and equipment configuration and design. Both equilibrium-stage and rate-based models are developed. Although emphasis is on adsorption, the basic principles of ion exchange and chromatography are also presented. Further descriptions of the three sorption operations are given by Rousseau [9] and Ruthven [10].

15.1 SORBENTS

To be suitable for commercial applications, a sorbent should have (1) high selectivity to enable sharp separations, (2) high capacity to minimize the amount of sorbent needed, (3) favorable kinetic and transport properties for rapid sorption, (4) chemical and thermal stability, including extremely low solubility in the contacting fluid, to preserve the amount of sorbent and its properties, (5) hardness and mechanical strength to prevent crushing and erosion, (6) a free-flowing tendency for ease of filling or emptying vessels, (7) high resistance to fouling for long life, (8) no tendency to promote undesirable chemical reactions, (9) the capability of being regenerated when used with commercial feedstocks that contain trace quantities of high-molecular-weight species that are strongly sorbed and difficult to desorb, and (10) relatively low cost.

Adsorbents

Most solids are able to adsorb species from gases and liquids. However, only a few have a sufficient selectivity and capacity to make them serious candidates for commercial adsorbents. Of considerable importance is a large specific surface area (area per unit volume), which is achieved by adsorbent manufacturing techniques that result in solids with a microporous structure. By the definition of the International Union of Pure and Applied Chemistry (IUPAC), a micropore is <20 Å, a mesopore is 20–500 Å, and a macropore is >500 Å (50 nm). Typical commercial adsorbents, which may be granules, spheres, cylindrical pellets, flakes, and/or powders of size ranging from 50 µm to 1.2 cm, have specific surface areas from 300 to 1,200 m²/g. Thus, just a few grams of adsorbent can have a surface area equal to that of a football field (120×53.3 yards or 5,350 m²)! Such a large area is made possible by a particle porosity from 30 to 85 vol% with average pore diameters from 10 to 200 Å. To quantify

this, consider a cylindrical pore of diameter d_p and length L . The surface area-to-volume ratio is

$$S/V = \pi d_p L / (\pi d_p^2 L / 4) = 4/d_p \quad (15-1)$$

If the fractional particle porosity is ϵ_p and the particle density is ρ_p , the specific surface area, S_g , in area per unit mass of adsorbent is

$$S_g = 4\epsilon_p / \rho_p d_p \quad (15-2)$$

Thus, if ϵ_p is 0.5, ρ_p is 1 g/cm³ = 1×10^6 g/m³, and d_p is 20 Å (20×10^{-10} m), substitution into (15-2) gives $S_g = 1,000$ m²/g, a desirable value.

Depending upon the type of forces between the fluid molecules and the molecules of the solid, adsorption may be classified as *physical adsorption* (van der Waals adsorption) or *chemisorption* (activated adsorption). Physical adsorption from a gas occurs when the intermolecular attractive forces between molecules of a solid and the gas are greater than those between molecules of the gas itself. In effect, the resulting adsorption is like condensation, which is exothermic and thus is accompanied by a release of heat. The magnitude of the heat of adsorption can be less than or greater than the heat of vaporization, and changes with the extent of adsorption. Physical adsorption, which may be a monomolecular (unimolecular) layer, or may be two, three or more layers thick (multimolecular), occurs rapidly. If unimolecular, it is reversible; if multimolecular, such that capillary pores are filled, hysteresis may occur. The density of the adsorbate is of the order of magnitude of the liquid rather than the vapor state. As physical adsorption takes place, it begins as a monolayer, becomes multilayered, and then, if the pores are close to the size of the molecules, capillary condensation occurs, and the pores fill with adsorbate. Accordingly, the maximum capacity of a porous adsorbent can be more related to the pore volume than to the surface area. However, for gases at temperatures above their critical temperature, adsorption is confined to a monolayer.

In contrast, chemisorption involves the formation of chemical bonds between the adsorbent and adsorbate in a monolayer, often with a release of heat much larger than the heat of vaporization. Chemisorption from a gas generally takes place only at temperatures greater than 200°C and may be slow and irreversible. Commercial adsorbents rely on physical adsorption; catalysis relies on chemisorption.

Adsorption from a liquid is a more difficult phenomenon to measure experimentally or describe. When the fluid is a gas, experiments are conducted with pure gases or with mixtures. The amount of gas adsorbed in a confined space is determined from the measured decrease in total pressure. When the fluid is a liquid, no simple procedure for determining the extent of adsorption from a pure liquid exists; consequently, experiments are only conducted using liquid mixtures, including dilute solutions. When porous particles of adsorbent are immersed in a liquid mixture, the pores, if sufficiently larger in diameter than the molecules in the liquid, fill with liquid. At equilibrium, because of differences in the extent of physical adsorption among the different molecules of the liquid mixture, the composition of the liquid in the pores differs from that of the bulk liquid surrounding the adsorbent particles. The observed exothermic heat effect is referred to as the *heat of wetting*, which is much smaller than the heat of adsorption from the gas phase. As with gases, the extent of equilibrium adsorption of a given solute increases with concentration and decreases with temperature. Chemisorption can also occur with liquids.

Listed in Table 15.2 are six major types of solid adsorbents in use. Included are the nature of the adsorbent and representative values of the mean pore diameter, d_p ; particle porosity (internal void fraction), ϵ_p ; particle density, ρ_p ; and specific surface area, S_g . In addition, for some adsorbents, the capacity for adsorbing water vapor at a partial pressure of 4.6 mmHg in air at 25°C is listed, as taken from Rousseau [9]. Not included is the specific pore volume, V_p , which can

be computed from the other properties by

$$V_p = \epsilon_p / \rho_p \quad (15-3)$$

Also not included in Table 15.2, but of interest when the adsorbent is used in fixed beds, are the bulk density, ρ_b , and the bed porosity (external porosity), ϵ_b , which are related:

$$\epsilon_b = 1 - \frac{\rho_b}{\rho_p} \quad (15-4)$$

In addition, the true solid density (also called the crystalline density), ρ_s , can be computed from a similar expression:

$$\epsilon_p = 1 - \frac{\rho_p}{\rho_s} \quad (15-5)$$

The specific surface area of an adsorbent, S_g , is measured by adsorbing gaseous nitrogen, using the well-accepted BET method (Brunauer, Emmett, and Teller [11]). Typically, the BET apparatus operates at the normal boiling point of N_2 (−195.8°C) by measuring the equilibrium volume of pure N_2 physically adsorbed on several grams of the adsorbent at a number of different values of the total pressure in the vacuum range of 5 to at least 250 mmHg. Brunauer, Emmett, and Teller derived a theoretical equation to model the adsorption by allowing for the formation of multimolecular layers. Furthermore, they assumed that the heat of adsorption during monolayer formation (ΔH_{ads}) is constant and that the heat effect associated with subsequent layers is equal to the heat of condensation (ΔH_{cond}). The BET equation is

$$\frac{P}{v(P_0 - P)} = \frac{1}{v_m c} + \frac{(c - 1)}{v_m c} \left(\frac{P}{P_0} \right) \quad (15-6)$$

where

P = total pressure

P_0 = vapor pressure of adsorbate at test temperature

v = volume of gas adsorbed at STP (0°C, 760 mmHg)

Table 15.2 Representative Properties of Commercial Porous Adsorbents

Adsorbent	Nature	Pore Diameter d_p , Å	Particle Porosity, ϵ_p	Particle Density ρ_p , g/cm ³	Surface Area S_g , m ² /g	Capacity for H ₂ O Vapor at 25°C and 4.6 mmHg, wt% (Dry Basis)
Activated alumina	Hydrophilic, amorphous	10–75	0.50	1.25	320	7
Silica gel:	Hydrophilic/					
Small pore	hydrophobic,	22–26	0.47	1.09	750–850	11
Large pore	amorphous	100–150	0.71	0.62	300–350	—
Activated carbon:	Hydrophobic,					
Small pore	amorphous	10–25	0.4–0.6	0.5–0.9	400–1200	1
Large pore		>30	—	0.6–0.8	200–600	—
Molecular-sieve carbon	Hydrophobic	2–10	—	0.98	400	—
Molecular-sieve zeolites	Polar-hydrophilic, crystalline	3–10	0.2–0.5	1.4	600–700	20–25
Polymeric adsorbents	—	40–25	0.4–0.55	—	80–700	—

v_m = volume of monomolecular layer of gas adsorbed at STP

c = constant related to the heat of adsorption
 $\approx \exp[(\Delta H_{\text{cond}} - \Delta H_{\text{ads}})/RT]$

Experimental data for v as a function of P are plotted, according to (15-6), as $P/[v(P_0 - P)]$ versus P/P_0 , from which v_m and c are determined from the slope and intercept of the best straight-line fit of the data. The value of S_g is then computed from

$$S_g = \frac{\alpha v_m N_A}{V} \quad (15-7)$$

where

N_A = Avogadro's number = 6.023×10^{23} molecules/mol

V = Volume of gas per mole at STP conditions (0°C , 1 atm) = $22,400 \text{ cm}^3/\text{mol}$

The quantity α is the surface area covered per adsorbed molecule. If we assume spherical molecules arranged in close two-dimensional packing, the projected surface area is:

$$\alpha = 1.091 \left(\frac{M}{N_A \rho_L} \right)^{2/3} \quad (15-8)$$

where

M = molecular weight of the adsorbate

ρ_L = density of the adsorbate in g/cm^3 , taken as the liquid at the test temperature

Although the BET surface area may not always represent the surface area available for adsorption of a particular molecule, the BET test is reproducible and widely used in the characterization of adsorbents.

The specific pore volume, typically cm^3 of pore volume/g of adsorbent, is determined for a small mass of adsorbent, m_p , by measuring the volumes of helium, V_{He} , and mercury, V_{Hg} , displaced by the adsorbent. The helium is not adsorbed, but fills the pores. At ambient pressure, the mercury cannot enter the pores because of unfavorable interfacial tension and contact angle. The specific pore volume, V_p , is then determined from

$$V_p = (V_{\text{Hg}} - V_{\text{He}})/m_p \quad (15-9)$$

The particle density is obtained from

$$\rho_p = \frac{m_p}{V_{\text{Hg}}} \quad (15-10)$$

The true solid density is obtained from

$$\rho_s = \frac{m_p}{V_{\text{He}}} \quad (15-11)$$

The particle porosity is then obtained from (15-3) or (15-5).

The distribution of pore volume over the range of pore size, which is of great importance in adsorption, is measured by mercury porosimetry for large-diameter pores ($>100 \text{ \AA}$); by gaseous-nitrogen desorption for pores of $15\text{--}250 \text{ \AA}$ in diameter; and by molecular sieving, using molecules of different diameter, for pores $<15 \text{ \AA}$ in diameter. In mercury porosimetry, the extent of mercury penetration into the pores

is measured as a function of applied hydrostatic pressure. A force balance along the axis of a straight pore of circular cross-section for the pressure and the interfacial tension between the mercury and the adsorbent surface gives

$$d_p = -\frac{4\sigma_I \cos \theta}{P} \quad (15-12)$$

where for mercury: σ_I = interfacial tension = 0.48 N/m and θ = contact angle = 140° . With these values, (15-12) becomes

$$d_p(\text{\AA}) = \frac{21.6 \times 10^5}{P(\text{psia})} \quad (15-13)$$

Thus, forcing mercury into a $100\text{-\AA}$ -diameter pore requires a very high pressure of $21,600 \text{ psia}$.

The nitrogen desorption method for determining pore-size distribution in the more important $15\text{--}250\text{-\AA}$ -diameter range is an extension of the BET method described earlier for measuring specific surface area. By increasing the nitrogen pressure above 600 mmHg , the multilayer adsorbed films reach the point where they bridge the pore, resulting in capillary condensation. At $P/P_0 = 1$, the entire pore volume is filled with nitrogen. Then, by reducing the pressure in steps, nitrogen is desorbed selectively, starting with the larger pores. This selectivity occurs because of the effect of pore diameter on the vapor pressure of the condensed phase in the pore, as given by the Kelvin equation:

$$P_p^s = P^s \exp \left(-\frac{4\sigma v_L \cos \theta}{RT d_p} \right) \quad (15-14)$$

where

P_p^s = vapor pressure of liquid in pore

P^s = the normal vapor pressure of liquid on a flat surface

σ = surface tension of liquid in pore

v_L = molar volume of liquid in pore

Thus, the vapor pressure of the condensed phase in the pore is less than its normal vapor pressure for a flat surface. The effect of d_p on P_p^s can be significant. For example, for liquid nitrogen at -195.8°C , $P^s = 760 \text{ torr}$, $\sigma = 0.00827 \text{ N/m}$, $\theta = 0$, and $v_L = 34.7 \text{ cm}^3/\text{mol}$. Equation (15-14) then becomes

$$d_p(\text{\AA}) = 17.9 / \ln(P^s/P_p^s) \quad (15-15)$$

From (15-15), for $d_p = 30 \text{ \AA}$, $P_p^s = 418 \text{ torr}$, a reduction in vapor pressure of almost 50%. At $200 \text{ \AA}$, the reduction is only about 10%. At 418 torr pressure, only pores less than $30 \text{ \AA}$ in diameter remain filled with liquid nitrogen. For greater accuracy in applying the Kelvin equation, a correction is needed for the thickness of the adsorbed layer. The use of this correction is discussed in detail by Satterfield [12]. For a monolayer, this thickness for nitrogen is about 0.354 nm , corresponding to a P/P_0 in (15-6) of between 0.05 and 0.10 . At $P/P_0 = 0.60$ and 0.90 , the adsorbed thicknesses are 0.75 and 1.22 nm , respectively. The correction is applied by subtracting twice the adsorbed thickness from d_p in (15-14) and (15-15).

EXAMPLE 15.1

Using data from Table 15.2, determine the volume fraction of pores in silica gel (small-pore type) filled with adsorbed water vapor when its partial pressure is 4.6 mmHg and the temperature is 25°C. At these conditions, the partial pressure is considerably below the vapor pressure of 23.75 mmHg. In addition, determine whether the amount of water adsorbed is equivalent to more than a monolayer, if the area of an adsorbed water molecule is given by (15-8) and the specific surface area of the silica gel is 830 m²/g.

SOLUTION

Take 1 g of silica gel particles as a basis. From (15-3) and data in Table 15.2, $V_p = 0.47/1.09 = 0.431$ cm³/g. Thus, for 1 g, pore volume is 0.431 cm³. From the capacity value in Table 15.2, amount of adsorbed water = $0.11/(1 + 0.11) = 0.0991$ g. Assume density of adsorbed water is 1 g/cm³, volume of adsorbed water = 0.0991 cm³, fraction of pores filled with water = $0.0991/0.431 = 0.230$, and surface area of 1 g = 830 m². From (15-8):

$$\alpha = 1.091 \left[\frac{18.02}{(6.023 \times 10^{23})(1.0)} \right]^{2/3} = 10.51 \times 10^{-16} \text{ cm}^2/\text{molecule}$$

$$\text{Number of water molecules adsorbed} = \frac{(0.0991)(6.023 \times 10^{23})}{18.02} = 3.31 \times 10^{21} \text{ molecules}$$

$$\text{Number of water molecules in a monolayer for } 830 \text{ m}^2 = \frac{830(100)^2}{10.51 \times 10^{-16}} = 7.90 \times 10^{21}$$

Therefore, only $3.31/7.90$ or 42% of one monolayer is adsorbed.

The four most widely used adsorbents in decreasing order of commercial usage are carbon (activated and molecular-sieve), molecular-sieve zeolites, silica gel, and activated alumina. In Table 15.2, activated alumina, Al₂O₃, which includes activated bauxite, is made by removing water from hydrated colloidal alumina. Activated alumina has a moderately high specific surface area, with a capacity for adsorption of water sufficient to dry gases to less than 1 ppm moisture content. Because of its great affinity for water, activated alumina is widely used for the removal of water from gases and liquids.

Silica gel, SiO₂, which is made from colloidal silica, has a high surface area and high affinity for water and other polar compounds. Related silicate adsorbents include magnesium silicate, calcium silicate, various clays, Fuller's earth, and diatomaceous earth. Silica gel is also highly desirable for water removal. Both small-pore and large-pore types are available.

Activated carbon is made by processes that involve the partial oxidation of a number of materials, including coconut shells, fruit nuts, wood, coal, lignite, peat, petroleum residues, and bones. Because activated carbon is hydrophobic and has a high specific surface area, it is particularly useful for processes involving nonpolar and weakly polar organic molecules. Macropores within the carbon particles help transfer molecules to the micropores. Two commercial grades are produced, one with large pores for liquid applications and one with small pores for gas adsorption. As shown

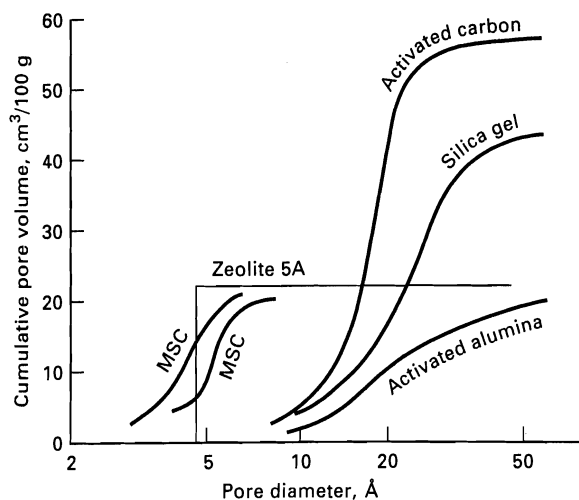
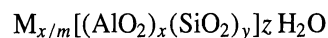


Figure 15.3 Representative cumulative pore-size distributions of adsorbents.

in Table 15.2, activated carbon is relatively hydrophobic and has a large surface area. Accordingly, it has found wide application for the purification and separation of gas and liquid mixtures containing nonpolar and weakly polar organic compounds, which adsorb much more strongly than water. In addition, the bonding strength of adsorption on activated carbon is low, resulting in a low heat of adsorption and ease of regeneration of the adsorbent.

Unlike activated carbon, which typically has pore diameters starting from 10 Å, molecular-sieve carbon (MSC) has much smaller pores ranging from 2 to 10 Å, making it possible to separate N₂ from air. The small pores, in one process, are made by depositing coke in the pore mouths of activated carbon.

Most commercial adsorbents have a range of pore sizes, as shown in Figure 15.3, where the cumulative pore volume is plotted against pore diameter. Exceptions are the molecular-sieve zeolites, which are crystalline, inorganic polymers of aluminosilicates and alkali or alkali-earth cation elements, such as Na, K, Mg, and Ca, with the general stoichiometric, unit-cell formula



where M is the cation with valence m , z is the number of water molecules in each unit cell, and x and y are integers such that $y/x \geq 1$. The cations balance the charge of the AlO₂ groups, each having a net charge of -1 . To activate the zeolite, the water molecules are removed by raising the temperature or pulling a vacuum. This leaves the remaining atoms spatially intact in interconnected, cage-like structures with six identical window apertures each, the size of which ranges from 3.8 to about 10 Å, depending on the cation and the crystal structure. These apertures act as sieves, which permit small molecules to enter the crystal cage, but exclude large molecules. Thus, compared to the other types of adsorbents, molecular-sieve zeolites are highly selective because all apertures have the same size. The properties and applications of five of the most commonly used molecular-sieve

Table 15.3 Properties and Applications of Some Molecular-Sieve Zeolites

Designation	Cation	Unit Cell Formula	Aperture Size, Å	Typical Applications
3A	K ⁺	K ₁₂ [(AlO ₂) ₁₂ (SiO ₂) ₁₂]	2.9	Drying of reactive gases
4A	Na ⁺	Na ₁₂ [(AlO ₂) ₁₂ (SiO ₂) ₁₂]	3.8	H ₂ O, CO ₂ removal; air separation
5A	Ca ²⁺	Ca ₅ Na ₂ [(AlO ₂) ₁₂ (SiO ₂) ₁₂]	4.4	Separation of air; separation of linear paraffins
10X	Ca ²⁺	Ca ₄₃ [(AlO ₂) ₈₆ (SiO ₂) ₁₀₆]	8.0	Separation of air; removal of mercaptans
13X	Na ⁺	Na ₈₆ [(AlO ₂) ₈₆ (SiO ₂) ₈₆]	8.4	

zeolites are given in Table 15.3, taken from Ruthven [13]. The zeolites separate not only by molecular size and shape, but also by polarity. Thus, they can also separate molecules of similar size. Some zeolites have circular apertures, whereas others have elliptical apertures. Adsorption in zeolites is actually a selective and reversible filling of crystal cages, so surface area is not a pertinent factor. Although naturally occurring zeolite minerals have been known for more than 200 years, molecular-sieve zeolites were first synthesized by Milton [2], who used very reactive materials at temperatures of 25–100°C.

The structure of the unit cell of a type A zeolite is shown in Figure 15.4a as a three-dimensional structure of silica and alumina tetrahedra, each formed by four oxygen atoms surrounding a silicon or aluminum atom. Oxygen and silicon atoms have two negative and four positive charges, respectively, causing the tetrahedra to build uniformly in four directions. Aluminum, with a valence of 3, causes the alumina tetrahedron to be negatively charged. The added cation provides the balance. In Figure 15.4a, an octahedron of tetrahedra is evident with six faces, with one near-circular window aperture at each face. A type X zeolite is shown in Figure 15.4b. This unit-cell structure results in a larger window aperture. Monographs by Barrer [14] and Breck [15] cover many important aspects of zeolites.

Of lesser commercial importance are polymeric adsorbents. Typically, they are spherical beads, 0.5 mm in diameter, made from microspheres about 10⁻⁴ mm in diameter. They are produced by polymerizing styrene and divinylbenzene for adsorbing nonpolar organics from aqueous solutions, and by polymerizing acrylic esters for adsorbing polar solutes. They are regenerated by leaching with organic solvents.

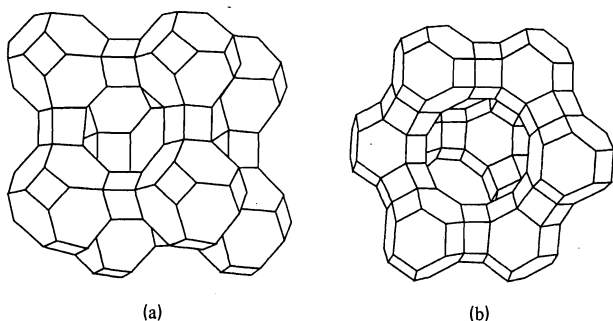


Figure 15.4 Line structures of molecular-sieve zeolites: (a) Type A unit cell. (b) Type X unit cell.

Ion Exchangers

The first ion exchangers were naturally occurring inorganic aluminosilicates (zeolites), used in experiments in the 1850s to exchange between ammonium ions in fertilizers and calcium ions in soils. Industrial water softeners using zeolites were introduced about 1910. However, the zeolites were unstable in the presence of mineral acids. The instability problem was solved by Adams and Holmes [16] in 1935, when they synthesized the first organic-polymer, ion-exchange resins by the polycondensation of phenol and aldehydes. Depending upon the nature of the phenolic group, the resin contains either sulfonic ($-\text{SO}_3^-$) or amine ($-\text{NH}_3^+$) groups, used for the reversible exchange of cations or anions. Today, the most widely used ion exchangers are synthetic, organic-polymer resins based on styrene- or acrylic-acid-type monomers, as described by D'Alelio in U.S. Patent 2,366,007 (Dec. 26, 1944).

Ion-exchange resins are generally solid gels in spherical or granular form, which consist of (1) a three-dimensional polymeric network, (2) ionic functional groups attached to the network, (3) counterions, and (4) a solvent. Strong-acid, cation-exchange resins and strong-base, anion-exchange resins that are fully ionized over the entire pH range are based on the copolymerization of styrene and a cross-linking agent, divinylbenzene, to produce the three-dimensional, cross-linked structure shown in Figure 15.5a. The degree of cross-linking is governed by the ratio of divinylbenzene to styrene. Weakly acid, cation exchangers are sometimes based on the copolymerization of acrylic acid and methacrylic acid, as shown in Figure 15.5b. These two cross-linked copolymers swell in the presence of organic solvents and have no ion-exchange properties.

To convert the copolymers to water-swelling gels with ion-exchange properties, ionic functional groups are added to the polymeric network by reacting the copolymers with various chemicals. For example, if the styrene-divinylbenzene copolymer is sulfonated, as shown in Figure 15.6a, a cation-exchange resin, as shown in Figure 15.6b, is obtained with ($-\text{SO}_3^-$) groups permanently attached or fixed to the polymeric network to give a negatively charged matrix and exchangeable, mobile, positive hydrogen ions (cations). The hydrogen ion can be exchanged on an equivalent basis with other cations, such as Na⁺, Ca²⁺, K⁺, or Mg²⁺, to maintain neutrality of the polymer. For example, two H⁺ ions are

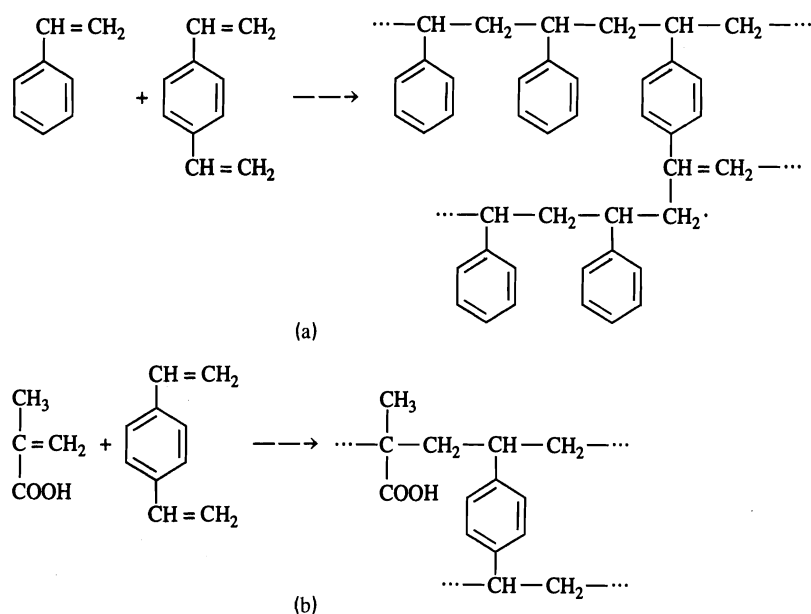


Figure 15.5 Ion-exchange resins: (a) Resin from styrene and divinylbenzene; (b) Resin from acrylic and methacrylic acid.

exchanged for one Ca^{2+} ion. The exchangeable ions are called *counterions*. The liquid whose ions are being exchanged also contains other ions of unlike charge, such as Cl^- for a solution of NaCl , where Na^+ is exchanged. These other ions are called *co-ions*. Often the liquid treated is water, which dissolves to some extent in the resin and causes it to swell. Other solvents, such as methanol, are also soluble in the resin. If the styrene-divinylbenzene copolymer is

chloromethylated and then aminated, a strong-base, anion-exchange resin is formed, as shown in Figure 15.6c, which can exchange Cl^- ions for other anions, such as OH^- , HCO_3^- , SO_4^{2-} , and NO_3^- .

Commercial ion exchangers in the hydrogen, sodium, and chloride form are available under the trade names of Amberlite, Duolite, Dowex, Ionac, and Purolite. Typically, they are in the form of spherical beads from about 40 μm

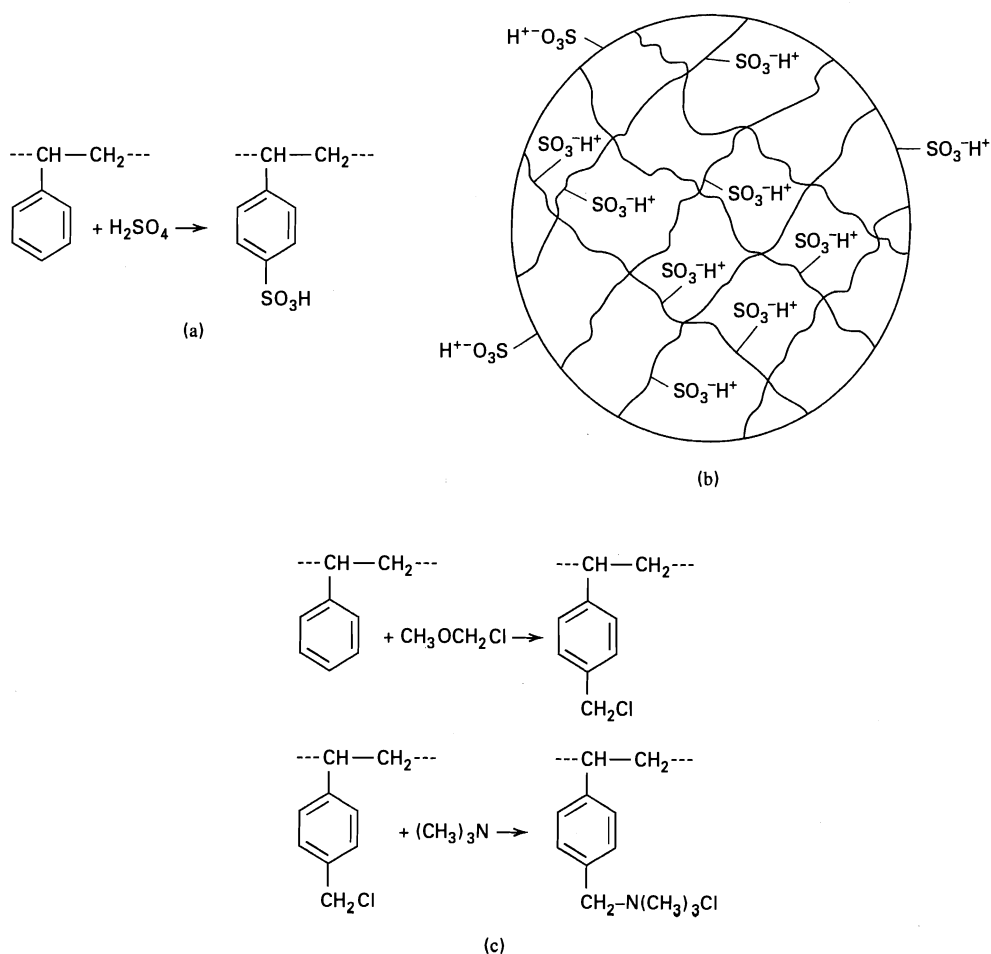


Figure 15.6 Introducing ionic functional groups into resins. (a) Sulfonation to a cation exchanger. (b) Fixed and mobile ions in a cation exchanger. (c) Chloromethylation and amination to an anion exchanger.

to 1.2 mm in diameter. When saturated with water, the beads have typical moisture contents from 40 to 65 wt%. When water-swollen, they have a particle density of 1.1–1.5 g/cm³. When packed into a bed, they have bulk densities from 0.56 to 0.96 g/cm³ with fractional bed porosities of 0.35–0.40.

When water is demineralized by ion exchange, potential organic foulants must be removed from the water feed. As discussed by McWilliams [17], this can be accomplished by coagulation, clarification, prechlorination, and the use of ion-exchanger traps.

The maximum ion-exchange capacity of a strong-acid cation or strong-base anion exchanger is stoichiometric, based on the number of equivalents of mobile charge in the resin. Thus, 1 mol H⁺ is one equivalent, whereas 1 mol Ca²⁺ is two equivalents. The exchanger capacity is usually quoted as eq/kg of dry resin or eq/L of wet resin. The wet capacity depends on the water content and degree of swelling of a given resin, whereas the dry capacity is fixed. For the copolymer of styrene and divinylbenzene, the maximum capacity is determined on the assumption that each benzene ring in the resin contains one sulfonic-acid group.

EXAMPLE 15.2

A commercial, ion-exchange resin is made from 88 wt% styrene and 12 wt% divinylbenzene. Estimate the maximum ion-exchange capacity in eq/kg resin (same as meq/g resin).

SOLUTION

Basis: 100 g of resin before sulfonation.

	<i>M</i>	g	gmol
Styrene	104.14	88	0.845
Divinylbenzene	130.18	12	0.092
		100	0.937

Therefore, sulfonation at one location on each benzene ring requires 0.937 mol of H₂SO₄ to attach the sulfonic acid group (*M* = 81.07) and split out one water molecule. This is 0.937 equivalents with the addition in weight of 0.937(81.07) = 76 g. Total dry weight of sulfonated resin = 100 + 76 = 176 g maximum ion-exchange capacity, or

$$\frac{0.937}{(176/1,000)} = 5.3 \text{ eq/kg(dry)}$$

Depending on the extent of cross-linking, resins from copolymers of styrene and divinylbenzene are listed as having actual capacities of from 3.9 (high degree of cross-linking) to 5.5 (low degree of cross-linking). Although a low degree of cross-linking favors dry capacity, almost every other ion-exchanger property, including wet capacity and selectivity, is improved by cross-linking, as discussed by Dorfner [18].

Sorbents for Chromatography

Sorbents (called *stationary phases*) for chromatographic separations come in a wide variety of forms and chemical

compositions because of the many ways in which chromatography is applied. Figure 15.7 shows a classification of analytical chromatographic systems, taken from Sewell and Clarke [19]. The mixture to be separated, after injection into the carrier fluid to form the *mobile phase*, may be a liquid (*liquid chromatography*) or a gas (*gas chromatography*). Often, the mixture is initially a liquid, but is vaporized without decomposition by the carrier gas, giving a gas mixture for the mobile phase. Gas carriers are inert and do not interact with the sorbent or components of the feed. Liquid carriers (solvents) can interact and must be selected carefully.

The stationary sorbent phase is a solid, a liquid supported on or bonded to a solid, or a gel. With a porous-solid adsorbent, the mechanism or mode of separation is adsorption. If an ion-exchange mechanism is desired, a synthetic, polymer-resin ion exchanger is used. With a polymer gel or a microporous solid, a separation based on sieving, called *exclusion*, can be applied. Unique to chromatography are the liquid-supported or liquid-bonded solids, where the mechanism is absorption into the liquid, also referred to as a *partition mode* of separation or *partition chromatography*. With mobile liquid phases, there is a tendency for the stationary liquid phase to be stripped or dissolved. Therefore, methods of chemically bonding the stationary liquid phase to the solid bonding support have been developed.

All sorbents can be used in columns. In packed columns >1 mm inside diameter, the sorbents are in the form of particles. In capillary columns <0.5 mm inside diameter, the sorbent is the inside wall or is a coating on the wall. If the inside wall of the capillary is liquid coated, the capillary column is referred to as a wall-coated, open-tubular (WCOT) column. If the coating is a layer of fine particulate support material to which a liquid adsorbent is added, the column is called a support-coated, open-tubular (SCOT) column. If the wall is coated with a porous adsorbent only, the column is referred to as a porous-layer, open-tubular (PLOT) column.

Each type of sorbent can be applied to sheets of glass, plastic, or aluminum for use in *thin-layer* (or planar) *chromatography* or to a sheet of cellulose material for use in *paper chromatography*. If a pump, rather than gravity, is used to pass a liquid mobile phase through a packed column, the name *high-performance liquid chromatography* (HPLC) is used.

The two most common adsorbents used in chromatography are porous alumina and porous silica gel. Of lesser importance are carbon, magnesium oxide, and various carbonates. Alumina is a polar adsorbent and is preferred for the separation of components that are weakly or moderately polar, with the more polar compounds retained more selectively by the adsorbent and, therefore, eluted from the column last. In addition, alumina is a basic adsorbent, preferentially retaining acidic compounds. Silica gel is less polar than alumina and is an acidic adsorbent, preferentially retaining basic compounds, such as amines. Carbon is a nonpolar (apolar) stationary phase with the highest attraction for larger nonpolar molecules.

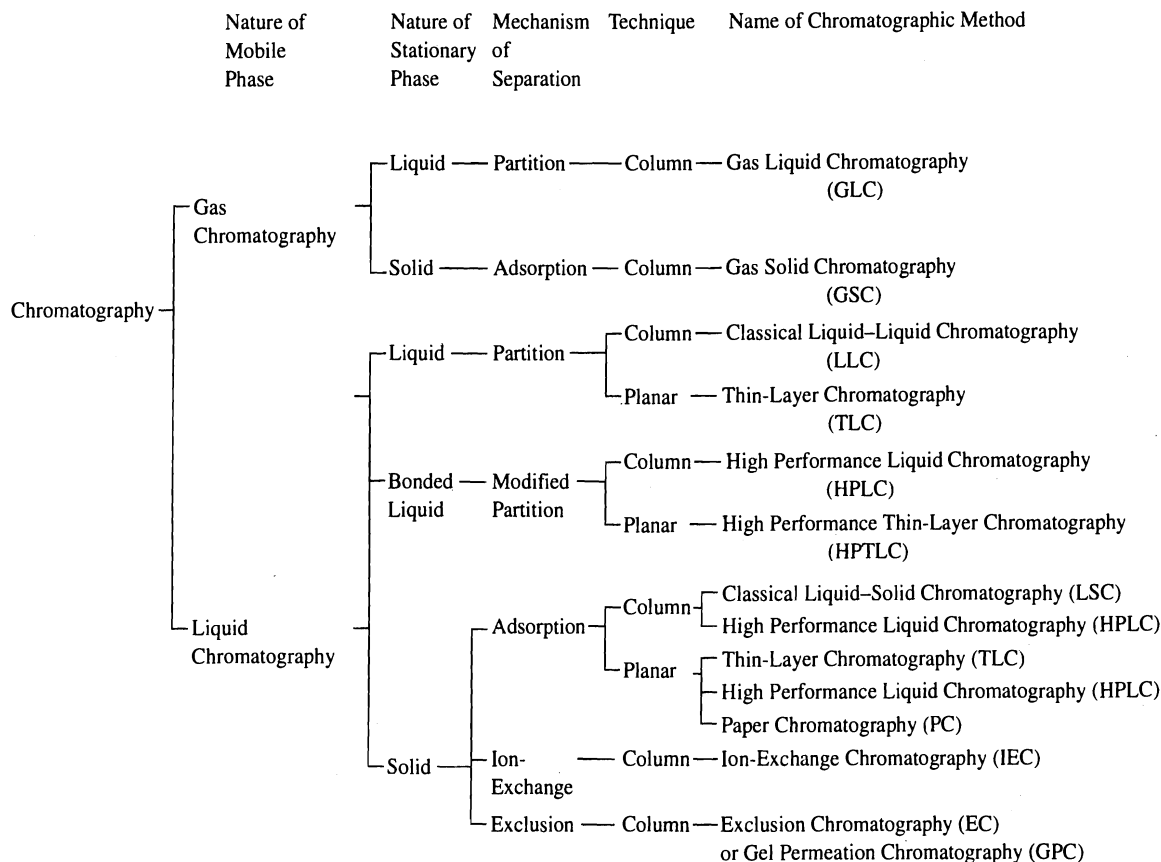


Figure 15.7 Classification of analytical chromatographic systems.

[From P.A. Sewell and B. Clarke, *Chromatographic Separations*, John Wiley and Sons, New York (1987) with permission.]

Adsorbent-type sorbents are better suited for the separation of a mixture on the basis of chemical type (e.g., olefins, esters, acids, aldehydes, alcohols) than for separation of individual members of a homologous series. For the latter, partition chromatography is preferred, wherein an inert-solid support, often silica gel, is coated with a liquid phase. For application to gas chromatography, the liquid must be non-volatile. For liquid chromatography, the stationary liquid phase must be insoluble in the mobile phase. Since this is difficult to achieve, the stationary liquid phase is usually bonded to the solid support. An example of a bonded phase is the result of reacting silica with a chlorosilane. Both monofunctional and bifunctional silanes are used, as shown in Figure 15.8, where R is a methyl (CH_3) group and R' is a hydrocarbon chain (C_6 , C_8 , or C_{18}) where the terminal CH_3 group is replaced with a polar group, such as $-\text{CN}$ or $-\text{NH}_2$.

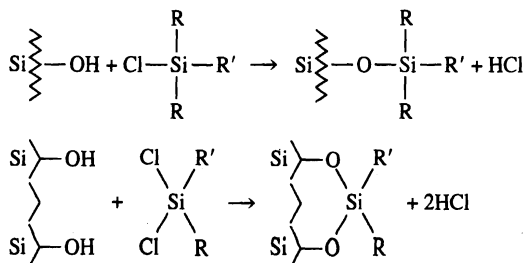


Figure 15.8 Bonded phases from the reaction of surface silanol groups with (a) Monofunctional and (b) Bifunctional chlorosilanes.

If the resulting stationary phase is more polar than the mobile phase, the technique is referred to as *normal-phase chromatography*. Otherwise, the name *reverse-phase chromatography* is used.

In liquid chromatography, the order of elution from the column of the solutes in the mobile phase can also be influenced by the solvent carrier of the mobile phase by matching the solvent polarity with the solutes and using more-polar adsorbents for less-polar solutes and less-polar adsorbents for more-polar solutes.

EXAMPLE 15.3

For the separation of each of the following mixtures, select an appropriate mode of chromatography from Figure 15.7: (a) gas mixture of O_2 , CO , CO_2 , and SO_2 , (b) vaporized mixture of anthracene, phenanthrene, pyrene, and chrysene, and (c) aqueous solution containing Ca^{2+} and Ba^{2+} .

SOLUTION

- Use gas-solid chromatography, that is, with a gas mobile phase and a solid-adsorbent stationary phase.
- Use partition or gas-liquid chromatography, that is, with a gas mobile phase and a bonded liquid coating on a solid for the stationary phase.
- Use ion-exchange chromatography, that is, with a liquid as the mobile phase and polymer resin beads as the stationary phase.

15.2 EQUILIBRIUM CONSIDERATIONS

In adsorption, a dynamic phase equilibrium is established for the distribution of the solute between the fluid and the solid surface. This equilibrium is usually expressed in terms of (1) concentration (if the fluid is a liquid) or partial pressure (if the fluid is a gas) of the adsorbate in the fluid and (2) solute *loading* on the adsorbent, expressed as mass, moles, or volume of adsorbate per unit mass or per unit BET surface area of the adsorbent. Unlike vapor-liquid and liquid-liquid equilibria, where theory is often applied to estimate phase distributions, particularly in the form of K -values for the former type of equilibrium, no acceptable theory has been developed to estimate fluid-solid adsorption equilibria. Thus, it is necessary to obtain experimental equilibrium data for a particular solute, or mixture of solutes and/or solvent, and a sample of the actual solid-adsorbent material of interest. If the data are taken over a range of fluid concentrations at a constant temperature, a plot of solute loading on the adsorbent versus concentration or partial pressure in the fluid, called an *adsorption isotherm*, is made. This equilibrium isotherm places a limit on the extent to which a solute is adsorbed from a given fluid mixture on an adsorbent of given chemical composition and geometry for a given set of conditions. The rate at which the solute is adsorbed is also an important consideration and is discussed in Section 15.3.

Pure Gas Adsorption

For pure gases, experimental physical-adsorption isotherms have shapes, that are classified into five types by Brunauer et al. [20], as shown in Figure 15.9 and discussed in considerable detail by Brunauer [21]. The simplest isotherm is Type I, which corresponds to unimolecular adsorption, as characterized by a maximum limit in the amount adsorbed. This type applies often to gases at temperatures above their critical temperature. The more complex Type II isotherm is associated with multimolecular adsorption of the BET type and is observed for gases at temperatures below their critical temperature and for pressures below, but approaching, the saturation pressure (vapor pressure). The heat of adsorption for the first adsorbed layer is greater than that for the succeeding layers, each of which is assumed to have a heat of adsorption equal to the heat of condensation (vaporization). Both Types I and II are desirable isotherms, exhibiting strong adsorption.

The Type III isotherm in Figure 15.9, with its convex nature, is undesirable because the extent of adsorption is low except at high pressures. According to the BET theory, it corresponds to multimolecular adsorption where the heat of adsorption of the first layer is less than that of succeeding layers. Fortunately, this type of isotherm is rarely observed, an example being the adsorption of iodine vapor on silica gel. In the limit, as the heat of adsorption of the first layer approaches zero, adsorption is delayed until the saturation pressure is approached.

The derivation of the BET equation (15-6) assumes that an infinite number of molecular layers can be adsorbed.

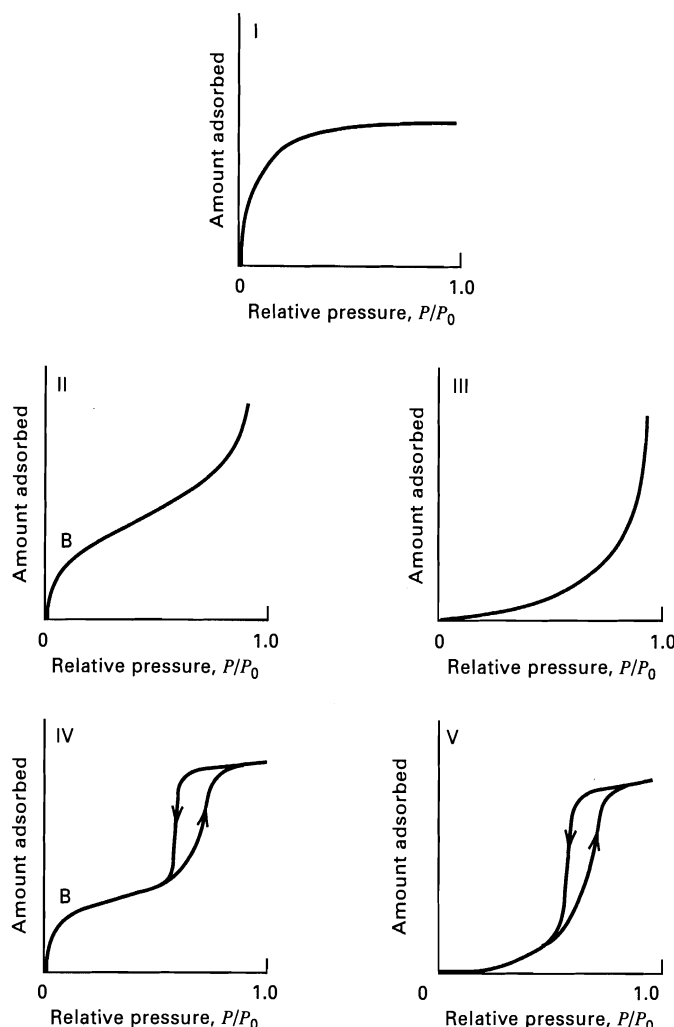


Figure 15.9 Brunauer's five types of adsorption isotherms. (P/P_0 = total pressure/vapor pressure.)

Thus, the equation precludes the possibility of capillary condensation. In a development by Brunauer et al. [20], subsequent to the BET equation, the number of layers is restricted by pore size, and capillary condensation is assumed to occur at a reduced vapor pressure in accordance with the Kelvin equation (15-14). The resulting equation is quite complex, but predicts adsorption isotherms of Types IV and V in Figure 15.9, where we see that the maximum extent of adsorption occurs before the saturation pressure is reached. Type IV is the capillary-condensation version of Type II; Type V is the capillary-condensation version of Type III.

As shown in Figure 15.9, a hysteresis phenomenon can occur in multimolecular adsorption regions for isotherms of types IV and V. The upward adsorption branch of the hysteresis loop is due to simultaneous, multimolecular adsorption and capillary condensation. Only capillary condensation occurs during the downward desorption branch of the loop. Hysteresis can also occur throughout any isotherm when strongly adsorbed impurities are present. Thus, measurements of pure-gas adsorption require adsorbents with clean pore surfaces, normally achieved by preevacuation.

Physical adsorption data of Titoff [22] for ammonia gas on charcoal, as discussed by Brunauer [21], are shown in Figure 15.10. The five adsorption isotherms of Figure 15.10a

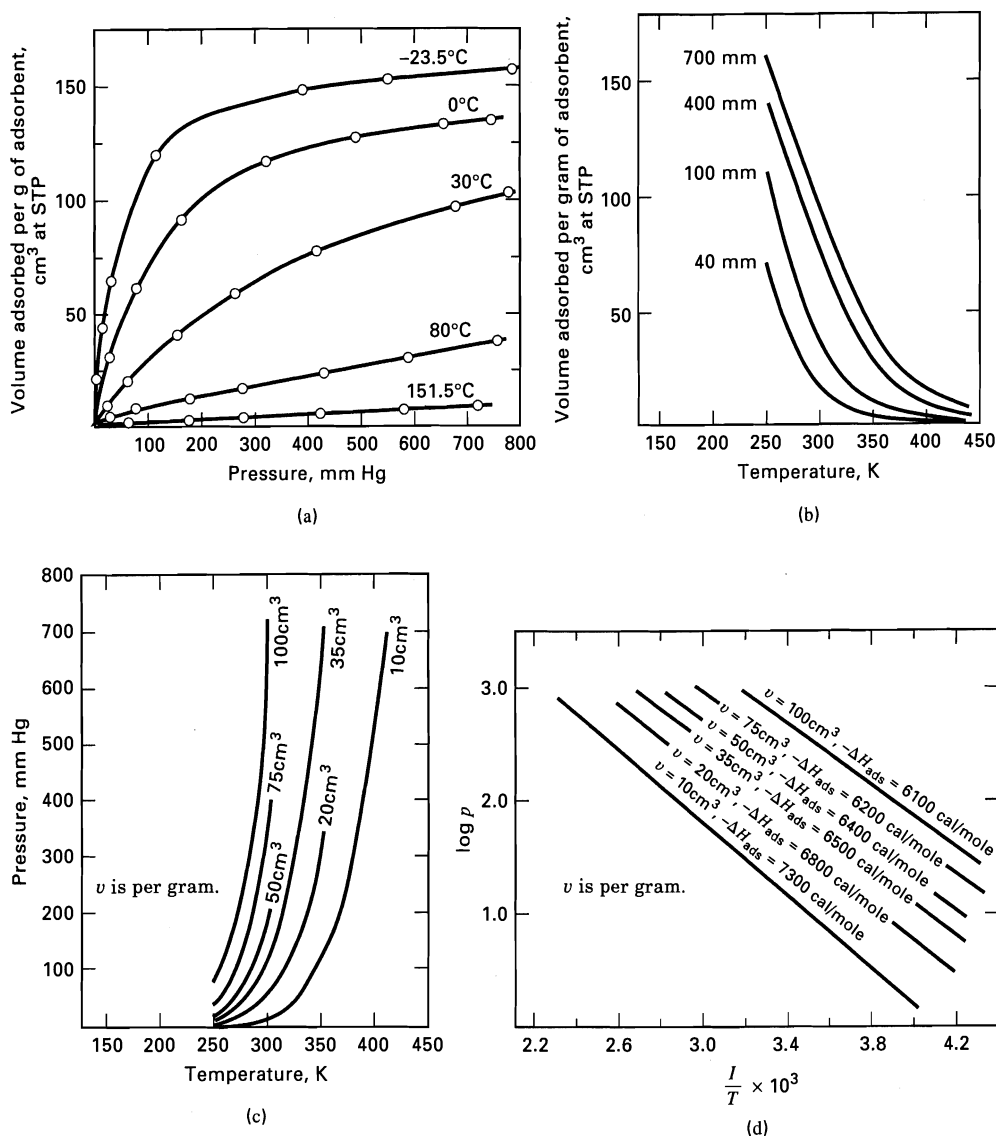


Figure 15.10 Different displays of adsorption equilibrium data for NH_3 on charcoal. (a) Adsorption isotherms. (b) Adsorption isobars. (c) Adsorption isosteres. (d) Isosteric heats of adsorption.

[From S. Brunauer, *The Adsorption of Gases and Vapors*, Vol. I, Princeton University Press (1943) with permission.]

cover pressures from vacuum to almost 800 mmHg and temperatures from -23.5 to 151.5°C . For ammonia, the normal boiling point is -33.3°C and the critical temperature is 132.4°C . For the lowest-temperature isotherm, up to 160 cm^3 (STP) of ammonia per gram of charcoal is adsorbed, which is equivalent to $0.12\text{ g NH}_3/\text{g charcoal}$. All five isotherms are of Type I. When the amount adsorbed is low ($<25\text{ cm}^3/\text{g}$), the isotherms are almost linear and the following form of Henry's law, called the *linear isotherm*, is obeyed:

$$q = kp \quad (15-16)$$

where q is equilibrium loading or amount adsorbed/unit mass of adsorbent (specific amount adsorbed), k is an empirical, temperature-dependent constant, and p is the partial pressure of the component in the gas. As the temperature increases, the amount adsorbed decreases because of Le Chatelier's principle for an exothermic process. This is shown more clearly for the same data in the crossplot of *adsorption isobars* in Figure 15.10b, where absolute temperature is employed. A third method of displaying the

experimental data is in the form of *adsorption isosteres*, also obtained by crossplotting, as shown in Figure 15.10c. These curves, representing constant amounts adsorbed, resemble vapor-pressure plots, for which the adsorption form of the Clausius–Clapeyron equation,

$$\frac{d \ln p}{dT} = \frac{-\Delta H_{\text{ads}}}{RT^2} \quad (15-17)$$

or

$$\frac{d \log p}{d(1/T)} = \frac{-\Delta H_{\text{ads}}}{2.303RT} \quad (15-18)$$

is applied to determine the heat of adsorption; which is negative because the effect is exothermic. The result is shown in Figure 15.10d, where it is seen that $-\Delta H_{\text{ads}}$ is initially $7,300\text{ cal/mol}$, but decreases as the amount adsorbed increases, reaching $6,100\text{ cal/mol}$ at $100\text{ cm}^3/\text{g}$. These values can be compared to the heat of vaporization of NH_3 , which at 30°C is $4,600\text{ cal/mol}$.

Experimental, adsorption-isotherm data for 18 different pure gases and a variety of solid adsorbents are summarized and analyzed by Valenzuela and Myers [23]. The data show

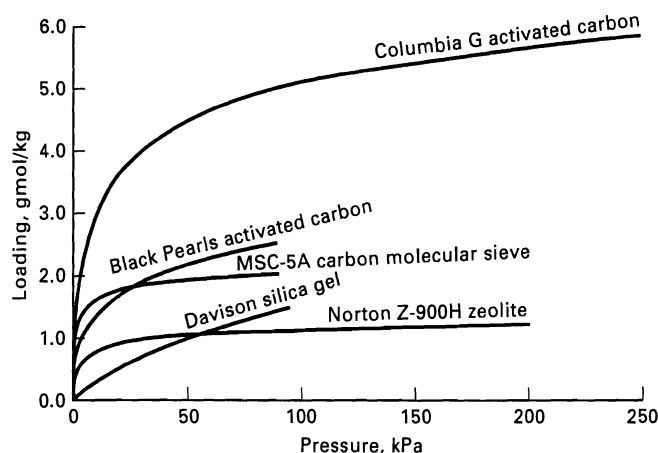


Figure 15.11 Adsorption isotherms for pure propane vapor at 298–303 K.

that adsorption isotherms for a given pure gas at a fixed temperature vary considerably with the adsorbent. This is shown for pure propane vapor in the narrow temperature range of 25–30°C in Figure 15.11 for pressures up to about 101.3 kPa. The highest specific adsorption is with Columbia G-grade activated carbon, while the lowest is with Norton Z-900H, a zeolite molecular sieve. Columbia G-grade activated carbon has about twice the adsorbate capacity of Cabot Black Pearls activated carbon.

The literature data compiled by Valenzuela and Myers [23] also show that for a given adsorbent, the loading depends strongly on the gas. This is illustrated in Table 15.4 for a temperature of 38°C and a narrow pressure range of 97.9 to 100 kPa from the data of Ray and Box [24] for Columbia L activated carbon. Included in the table are normal boiling points and critical temperatures. As might be expected, the species are adsorbed in approximately the inverse order of volatility.

The correlation of experimental adsorption isotherms for pure gases is the subject of a number of published articles and books. As summarized by Yang [25], approaches have ranged from empirical to theoretical. For practical applications, the classical equations of Freundlich and Langmuir

Table 15.4 Comparison of Equilibrium Adsorption of Pure Gases on 20–40 mesh Columbia L Activated Carbon Particles ($S_g = 1,152 \text{ m}^2/\text{g}$) at 38°C and ~1 atm

Pure gas	q , mol/kg	T_b , °F	T_c , °F
H ₂	0.0241	−423.0	−399.8
N ₂	0.292	−320.4	−232.4
CO	0.374	−313.6	−220.0
CH ₄	0.870	−258.7	−116.6
CO ₂	1.64	−109.3	87.9
C ₂ H ₂	2.67	−119	95.3
C ₂ H ₄	2.88	−154.6	48.6
C ₂ H ₆	3.41	−127.5	90.1
C ₃ H ₆	4.54	−53.9	196.9
C ₃ H ₈	4.34	−43.7	216.0

are still dominant because of their simplicity and ability to correlate isotherms of Type I in Figure 15.9.

Freundlich Isotherm

The equation attributed to Freundlich [26], but which was actually devised earlier by Boedecker and van Bemmelen, according to Mantell [27], is empirical and nonlinear in pressure:

$$q = kp^{1/n} \quad (15-19)$$

where k and n are temperature-dependent constants. Generally, n lies in the range of 1 to 5. With $n = 1$, (15-19) reduces to the Henry's law equation (15-16). Experimental q - p isothermal data can be fitted to (15-19) by a nonlinear, curve-fitting computer program or by converting (15-19) to a linear form as follows, and using a graphical method or a linear-regression program:

$$\log q = \log k + (1/n) \log p \quad (15-20)$$

If the graphical method is employed, the data are plotted as $\log q$ versus $\log p$. The best straight line through the data has a slope of $(1/n)$ and an intercept of $\log k$. In general, k decreases with increasing temperature, while n increases with increasing temperature and approaches a value of 1 at high temperatures. Equation (15-19) is derived by assuming a heterogeneous surface with a nonuniform distribution of the heat of adsorption over the surface, as discussed by Brunauer [21].

Langmuir Isotherm

The Langmuir equation [28], which is restricted to Type I isotherms, is derived from simple mass-action kinetics, assuming chemisorption. Assume that the surface of the pores of the adsorbent is homogeneous ($\Delta H_{\text{ads}} = \text{constant}$) and that the forces of interaction between adsorbed molecules are negligible. Let θ be the fraction of the surface covered by adsorbed molecules. Therefore, $(1 - \theta)$ is the fraction of the bare surface. Then, the net rate of adsorption is the difference between the rates of adsorption on the bare surface and desorption on the covered surface:

$$dq/dt = k_a p(1 - \theta) - k_d \theta \quad (15-21)$$

At equilibrium, $dq/dt = 0$ and (15-21) reduces to

$$\theta = \frac{Kp}{1 + Kp} \quad (15-22)$$

where K is the adsorption-equilibrium constant ($= k_a/k_d$). Here,

$$\theta = q/q_m \quad (15-23)$$

where q_m is the maximum loading corresponding to complete coverage of the surface by the gas. Thus, the Langmuir adsorption isotherm is restricted to a monomolecular layer. Combining (15-23) with (15-22), we obtain the *Langmuir isotherm*:

$$q = \frac{Kq_m p}{1 + Kp} \quad (15-24)$$

At low pressures, if $Kp \ll 1$, (15-24) reduces to the linear Henry's law form (15-16), while at high pressures where $Kp \gg 1$, $q = q_m$. At intermediate pressures, (15-24) is nonlinear in pressure. Although originally devised by Langmuir for chemisorption, (15-24) has been widely applied to physical adsorption data.

The quantities K and q_m in (15-24) are treated as empirical constants, obtained by fitting the nonlinear equation directly to experimental data or by employing the following linearized form, numerically or graphically:

$$\frac{p}{q} = \frac{1}{q_m K} + \frac{p}{q_m} \quad (15-25)$$

Using (15-25), the best straight line is drawn through a plot of points p/q versus p , giving a slope of $(1/q_m)$ and an intercept of $1/(q_m K)$. If the theory is reasonable, K should change rapidly with temperature, but q_m should not because it is related through v_m by (15-7) to the specific surface area of the adsorbent, S_g . It should be noted that the Langmuir isotherm predicts an asymptotic limit for q at high pressure, whereas the Freundlich isotherm does not.

Other Adsorption Isotherms

Valenzuela and Myers [23] fit pure gas adsorption-isotherm data to the more complex three-parameter isotherms of (1) Toth:

$$q = \frac{mp}{(b + p^t)^{1/t}} \quad (15-26)$$

where m , b , and t are constants for a given adsorbate-adsorbent system and temperature, and (2) Honig and Reyerson (called the UNILAN equation):

$$q = \frac{n}{2s} \ln \left[\frac{c + pe^s}{c + pe^{-s}} \right] \quad (15-27)$$

where n , s , and c are constants for a given adsorbate-adsorbent system and temperature. The Toth and UNILAN isotherms reduce to the Langmuir isotherm for $t = 1$ and $s = 0$, respectively.

EXAMPLE 15.4

The following experimental data for the equilibrium adsorption of pure methane gas on activated carbon (PCB from Calgon Corp.) at 296 K were obtained by Ritter and Yang [*Ind. Eng. Chem. Res.*, **26**, 1679–1686 (1987)]:

q , cm ³ (STP) of CH ₄ /g carbon	45.5	91.5	113	121	125	126	126
$P = p$, psia	40	165	350	545	760	910	970

Fit the data to: (a) the Freundlich isotherm, and (b) the Langmuir isotherm. Which isotherm provides a better fit to the data?

SOLUTION

By using the linearized forms of the isotherm equations, a spreadsheet or other computer program can be used to do a linear regression to obtain the constants.

(a) Using (15-20), we obtain $\log k = 1.213$, $k = 16.34$, $1/n = 0.3101$, and $n = 3.225$.

Thus, the Freundlich equation is $q = 16.34 p^{0.3101}$.

(b) Using (15-25), we obtain $1/q_m = 0.007301$, $q_m = 137.0$, $1/(q_m K) = 0.5682$, and $K = 0.01285$.

Thus, the Langmuir equation is $q = \frac{1.760p}{1 + 0.01285p}$

The predicted values of q from the two isotherms are as follows:

p , psia	q , cm ³ (STP) of CH ₄ /g carbon		
	Experimental	Freundlich	Langmuir
40	45.5	51.3	46.5
165	91.5	79.6	93.1
350	113	101	112
545	121	115	120
760	125	128	124
910	126	135	126
970	126	138	127

For this example, the Langmuir isotherm fits the data significantly better than the Freundlich isotherm. Average percent deviations, in q , are computed to be 1.01% and 8.64%, respectively. One reason for the better fit of the Langmuir isotherm is the trend of the data to an asymptotic value for q at the highest pressures.

Gas Mixtures and Extended Isotherms

Commercial applications of physical adsorption involve mixtures rather than pure gases. If the adsorption of all components in the gas except one (A) is negligible, then the adsorption of A is estimated from its pure gas-adsorption isotherm using the partial pressure of A. If the adsorption of two or more components in the mixture is significant, the situation is quite complicated. Experimental data show that one component can increase, decrease, or have no influence on the adsorption of the other, depending on interactions of adsorbed molecules. A simple theoretical treatment is the extension of the Langmuir equation by Markham and Benton [29], who neglect interactions and assume that the only effect is the reduction of the vacant surface area for the adsorption of A because of the adsorption of other components. Consider a binary gas mixture of A and B. Let θ_A = fraction of the surface covered by A and θ_B = fraction of the surface covered by B. Then, $(1 - \theta_A - \theta_B)$ = fraction of vacant surface. At equilibrium:

$$(k_A)_a p_A (1 - \theta_A - \theta_B) = (k_A)_d \theta_A \quad (15-28)$$

$$(k_B)_a p_B (1 - \theta_A - \theta_B) = (k_B)_d \theta_B \quad (15-29)$$

Solving these equations simultaneously, and combining the results with (15-23) for each component, gives

$$q_A = \frac{(q_A)_m K_A p_A}{1 + K_A p_A + K_B p_B} \quad (15-30)$$

$$q_B = \frac{(q_B)_m K_B p_B}{1 + K_A p_A + K_B p_B} \quad (15-31)$$

where $(q_i)_m$ is the maximum amount of adsorption of species i for coverage of the entire surface. Equations (15-30) and (15-31) are readily extended to a multicomponent mixture of j components:

$$q_i = \frac{(q_i)_m K_i p_i}{1 + \sum_j K_j p_j} \quad (15-32)$$

In a similar fashion, as shown by Yon and Turnock [30], the Freundlich equation can be combined with the Langmuir equation to give the following extended relation for gas mixtures:

$$q_i = \frac{(q_i)_0 k_i p_i^{1/n_i}}{1 + \sum_j k_j p_j^{1/n_j}} \quad (15-33)$$

where $(q_i)_0$ is the maximum loading, which may differ from $(q_i)_m$ for a monolayer. Equation (15-33) represents data for nonpolar, multicomponent mixtures in molecular sieves reasonably well. Unfortunately, Broughton [31] has shown that the extended-Langmuir equation lacks thermodynamic consistency; such is also the case for the extended-Langmuir-Freundlich equation. Accordingly, both (15-32) and (15-33) are frequently referred to as nonstoichiometric isotherms. Nevertheless, for practical application, their simplicity often makes them the isotherms of choice. In particular, both the extended Langmuir and Freundlich adsorption isotherms of (15-32) and (15-33) are frequently referred to as *constant-selectivity-equilibrium* equations because they predict a separation factor (selectivity), $\alpha_{i,j}$, for each pair of components, i, j , in a multicomponent mixture that is constant for a given temperature and independent of mixture composition. For example, (15-32) gives

$$\alpha_{i,j} = \frac{q_i/q_j}{p_i/p_j} = \frac{(q_i)_m K_i}{(q_j)_m K_j}$$

As with multicomponent (three or more components) vapor-liquid and liquid-liquid phase equilibria, experimental data for binary and multicomponent gas-solid adsorbent equilibria are scarce and less accurate than corresponding pure gas data. Valenzuela and Myers [23] include experimental data on adsorption of gas mixtures from nine published studies on 29 binary systems, for which pure gas-adsorption isotherms were also obtained. They also describe procedures for applying the Toth and UNILAN equations to multicomponent mixtures based on the ideal-adsorbed-solution (IAS) theory of Myers and Prausnitz [32]. Unlike the extended-Langmuir equation (15-32), which is explicit in the amount adsorbed, the IAS theory, though more accurate, is not explicit and requires an iterative solution procedure. Additional experimental data for higher-order (ternary and/or higher) gas mixtures are given by Miller, Knaebel, and Ikels [33] for 5A molecular sieves and by Ritter and Yang [34] for activated carbon. Yang [25] presents a discussion of existing theories on adsorption of gas mixtures, together with comparisons of these theories with mixture data for activated carbon and zeolites. The data on zeolites are the

most difficult to correlate, with the simplified statistical thermodynamic model (SSTM) of Ruthven and Wong [35] giving best results.

EXAMPLE 15.5

The experimental work of Ritter and Yang, cited in Example 15.4, also includes adsorption isotherms for pure CO and CH₄, and a binary mixture of CH₄(A) and CO(B). For the pure gases, Ritter and Yang give relations over a temperature range of 296–480 K, for the two Langmuir constants. At 294 K, these constants are as follows:

	$q_m, \text{cm}^3(\text{STP})/\text{g}$	K, psi^{-1}
CH ₄	133.4	0.01370
CO	126.1	0.00624

With these constants, use the extended-Langmuir equation to predict the specific adsorption volumes (STP) of CH₄ and CO for a vapor mixture of 69.6 mol% CH₄ and 30.4 mol% CO at 294 K and a total pressure of 364.3 psia. Compare the results with the following experimental data of Ritter and Yang:

Total volume adsorbed, $\text{cm}^3(\text{STP})/\text{g}$	114.1
Mole fractions in adsorbate:	
CH ₄	0.867
CO	0.133

SOLUTION

$$p_A = y_A P = 0.696(364.3) = 253.5 \text{ psia}$$

$$p_B = y_B P = 0.304(364.3) = 110.8 \text{ psia}$$

From (15-30):

$$q_A = \frac{133.4(0.0137)(253.5)}{1 + (0.0137)(253.5) + (0.00624)(110.8)} = 89.7 \text{ cm}^3(\text{STP})/\text{g}$$

$$q_B = \frac{126.1(0.00624)(110.8)}{1 + (0.0137)(253.5) + (0.00624)(110.8)} = 16.9 \text{ cm}^3(\text{STP})/\text{g}$$

The total amount adsorbed = $q = q_A + q_B = 89.7 + 16.9 = 106.6 \text{ cm}^3(\text{STP})/\text{g}$, which is 6.6% lower than the experimental value. Estimated mole fractions in the adsorbate are $x_A = q_A/q = 89.7/106.6 = 0.841$ and $x_B = 1 - 0.841 = 0.159$. These adsorbate mole fractions deviate from the experimental values by a mole fraction of 0.026. For this example, the extended-Langmuir isotherm gives reasonable results.

Liquid Adsorption

When porous adsorbent particles are immersed in a pure gas, the pores fill with the gas, and the amount of adsorbed gas is determined by the decrease in total pressure. With a liquid, the pressure does not change, and no simple experimental procedure has been devised for determining the extent of adsorption of a pure liquid. If the liquid is a homogeneous binary mixture, it is customary to designate one component the solute (1) and the other the solvent (2). The assumption is then made that the change in composition of the bulk liquid in contact with the porous solid is due entirely to adsorption

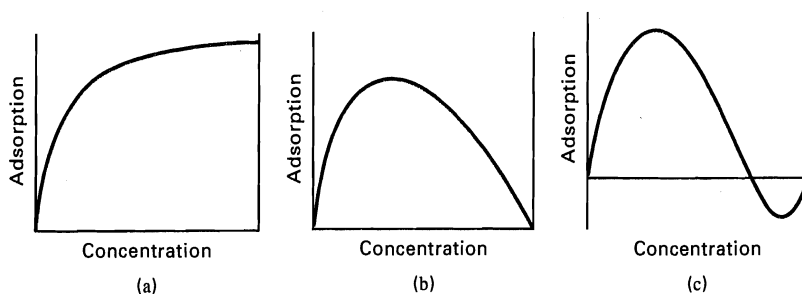


Figure 15.12 Representative isotherms of concentration change for liquid adsorption.

of the solute. That is, adsorption of the solvent is tacitly assumed not to occur. If the liquid mixture is dilute in the solute, the consequences are not serious. If, however, experimental data are obtained over the entire concentration range, the distinction between solute and solvent is arbitrary and the resulting adsorption isotherms, as discussed by Kipling [36], can exhibit curious shapes that are unlike those obtained for pure gases or gas mixtures. To demonstrate this, let

n^0 = total moles of binary liquid brought into contact with adsorbent

m = mass of adsorbent

x_1^0 = mole fraction of solute in the mixture before contact with adsorbent

x_1 = mole fraction of solute in the bulk solution after adsorption equilibrium is achieved

q_1^e = apparent moles of solute adsorbed per unit mass of adsorbent

A solute material balance, assuming no adsorption of solvent and a negligible change in the total moles of liquid mixture, gives

$$q_1^e = \frac{n^0(x_1^0 - x_1)}{m} \quad (15-34)$$

If data are obtained at constant temperature over the entire concentration range and then processed with (15-34) and plotted as adsorption isotherms, the resulting curves are not of the expected type shown in Figure 15.12a. Instead, curves of the type shown in Figures 15.12b and c are obtained, where negative adsorption appears to occur in Figure 15.12c. Such isotherms are probably best referred to as *composite isotherms* or *isotherms of concentration change*, as suggested by Kipling [36]. Likewise, the adsorption loading, q_1^e , of (15-34) is more correctly referred to as the *surface excess*.

Under what conditions are composite isotherms of the form shown in Figures 15.12b and c obtained? This is shown by several examples in Figure 15.13, where various combinations of hypothetical adsorption isotherms for solute (A) and solvent (B) are shown together with the resulting composite isotherms. Thus, when the solvent is not adsorbed, as seen in Figure 15.13a, a composite curve without negative adsorption is obtained. In all other cases of Figure 15.13, negative values of the surface excess are obtained.

Valenzuela and Myers [23] tabulate literature values for the equilibrium adsorption of 25 different binary-liquid

mixtures. With one exception, all 25 mixtures give composite isotherms of the forms shown in Figures 15.12b and c. The one exception is a mixture of cyclohexane and *n*-heptane with silica gel, for which the surface excess is almost negligible (0 ± 0.05 mmol/g) over the composition range of $x_1 = 0.041$ to 0.911. They also include literature references to 354 sets of binary-mixture data, 25 sets of ternary-mixture data, and 3 sets of data for higher-order mixtures.

When data for the binary mixture are only available in the dilute region, the amount of adsorption, if any, of the solvent may be constant and all changes in the total amount adsorbed are due to just the solute. In that case, the adsorption isotherms are of the form of Figure 15.12a, which resembles the form obtained with pure gases. It is then common to fit the data with concentration forms of the Freundlich equation

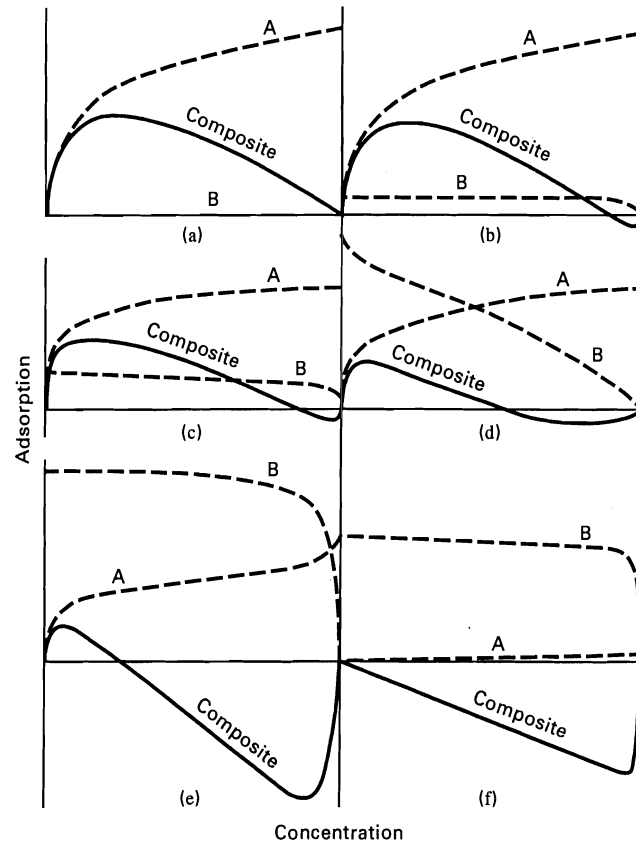


Figure 15.13 Origin of various types of composite isotherms for binary liquid adsorption.

[From J.J. Kipling, *Adsorption from Solutions of Non-electrolytes*, Academic Press, London (1965) with permission.]

(15-19) or the Langmuir equation (15-24):

$$q = kc^{1/n} \quad (15-35)$$

$$q = \frac{Kq_m c}{1 + Kc} \quad (15-36)$$

Candidate systems for this case are small amounts of organic compounds dissolved in water and small amounts of water dissolved in hydrocarbons. For liquid mixtures that are dilute in two or more solutes, the multicomponent adsorption may be estimated from a concentration form of the extended Langmuir equation (15-32) based on the constants, q_m and K , obtained from experiments on the single solutes. However, when solute-solute interactions are suspected, it may be necessary to determine the constants from multicomponent data. As with gas mixtures, the concentration form of (15-32) also predicts a constant selectivity for each pair of components in a multicomponent mixture.

EXAMPLE 15.6

Small amounts of VOCs in water can be removed by adsorption. Generally, two or more VOCs are present. An aqueous stream containing small amounts of acetone (1) and propionitrile (2) is to be treated with activated carbon. Single-solute equilibrium data available from Radke and Prausnitz [37] have been fitted to the Freundlich and Langmuir isotherms, (15-35) and (15-36), with the average deviations indicated, for solute concentrations up to 50 mmol/L:

Acetone in Water (25°C):		Absolute Average Deviation of q , %
$q_1 = 0.141c_1^{0.597}$	(1)	14.2
$q_1 = \frac{0.190c_1}{1 + 0.146c_1}$	(2)	27.3
Propionitrile in water (25°C):		
$q_2 = 0.138c_2^{0.658}$	(3)	10.2
$q_2 = \frac{0.173c_2}{1 + 0.0961c_2}$	(4)	26.2

where

q_i = amount of solute adsorbed, mmol/g

c_i = solute concentration in aqueous solution, mmol/L

Use these single-solute results with an extended Langmuir-type isotherm to predict the equilibrium adsorption in a binary-solute aqueous system containing 40 and 34.4 mmol/L, respectively, of acetone and propionitrile at 25°C with the same adsorbent. Compare the results with the following experimental values from Radke and Prausnitz [37]:

$$q_1 = 0.715 \text{ mmol/g}, \quad q_2 = 0.822 \text{ mmol/g}, \quad \text{and} \quad q_{\text{total}} = 1.537 \text{ mmol/g}$$

SOLUTION

From (15-32), the extended Langmuir isotherm for the liquid phase is

$$q_i = \frac{(q_i)_m K_i c_i}{1 + \sum_j K_j c_j} \quad (5)$$

From (2), $(q_1)_m = 0.190/0.146 = 1.301 \text{ mmol/g}$.

From (4), $(q_2)_m = 0.173/0.0961 = 1.800 \text{ mmol/g}$.

From (5):

$$q_1 = \frac{1.301(0.146)(40)}{1 + (0.146)(40) + (0.0961)(34.4)} = 0.749 \text{ mmol/g}$$

$$q_2 = \frac{1.800(0.0961)(34.4)}{1 + (0.146)(40) + (0.0961)(34.4)} = 0.587 \text{ mmol/g}$$

$$q_{\text{total}} = 1.336 \text{ mmol/g}$$

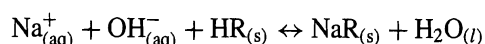
Compared to experimental data, the percent deviations for q_1 , q_2 , and q_{total} , respectively, are 4.8%, -28.6%, and -13.1%. Better agreement is obtained by Radke and Prausnitz using an IAS theory. It is expected that a concentration form of (15-33) would also give better agreement, but that requires that the single-solute data be refitted for each solute to a Langmuir-Freundlich isotherm of the form

$$q = \frac{q_0 kc^{1/n}}{1 + kc^{1/n}} \quad (6)$$

Ion Exchange Equilibria

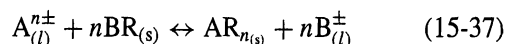
Ion exchange differs from adsorption in that one sorbate (a counterion) is exchanged for a solute ion, and the exchange is governed by a reversible, stoichiometric, chemical-reaction equation. Thus, the selectivity of the ion exchanger for one counterion over another may be just as important as the capacity of the ion exchanger. Accordingly, for ion exchange, we apply the law of mass action to obtain an equilibrium ratio rather than fit data to a sorption isotherm such as the Langmuir or Freundlich equation.

As discussed by Anderson [38], two cases are important. In the first case, the counterion initially in the ion exchanger is exchanged with a counterion from an acid or base. For example:



Note that the hydrogen ion leaving the ion exchanger immediately reacts with the hydroxyl ion in the aqueous solution to form water, leaving no counterion on the right-hand side of the reaction. Accordingly, the ion exchange will continue until the aqueous solution is depleted of sodium ions or the ion exchanger is depleted of hydrogen ions.

In the second case, which is more common than the first, the counterion being transferred from the ion exchanger to the fluid phase remains as an ion. For example, the exchange of counterions A and B is expressed by the reaction



where A and B must both be either cations (positive charge) or anions (negative charge). For this case, at equilibrium, we can define a conventional chemical-equilibrium constant according to the law of mass action:

$$K_{\text{A,B}} = \frac{q_{\text{AR}}^n c_{\text{B}}^{n\pm}}{q_{\text{BR}}^n c_{\text{A}}^{n\pm}} \quad (15-38)$$

where molar concentrations c_i and q_i refer to the liquid and ion-exchanger phases, respectively. The constant, $K_{\text{A,B}}$ is

not a rigorous thermodynamic equilibrium constant because (15-38) is written in terms of concentrations instead of activities. Although (15-38) could be corrected by including activity coefficients, it is usually applied in the form shown, with $K_{A,B}$ referred to as a *molar selectivity coefficient* for A entering the ion exchange resin and displacing B. For the resin phase, concentrations are in equivalents per unit mass or unit bed volume of ion exchanger. For the liquid solution, concentrations are in equivalents per unit volume of solution. For dilute liquid solutions, $K_{A,B}$ is reasonably constant for a given pair of counterions and a particular resin of a given degree of cross-linking.

When exchange is between two counterions of equal charge, (15-38) can be reduced to a simple equation in terms of just the equilibrium concentrations of A in the liquid solution and in the ion exchange resin. Because of (15-37), the total concentrations C and Q in equivalents of counterions in the liquid solution and the resin, respectively, remain constant during the exchange process. Accordingly:

$$c_i = Cx_i/z_i \quad (15-39)$$

$$q_i = Qy_i/z_i \quad (15-40)$$

where x_i and y_i are equivalent fractions, rather than mole fractions, of A and B, such that

$$x_A + x_B = 1 \quad (15-41)$$

$$y_A + y_B = 1 \quad (15-42)$$

and z_i = valence of counterion i . Combining (15-38) to (15-42) results in

$$K_{A,B} = \frac{y_A(1 - x_A)}{x_A(1 - y_A)} \quad (15-43)$$

Thus, at equilibrium, x_A and y_A are independent of the total equivalent concentrations C and Q . Such is not the case when the two counterions are of unequal charge, as in the exchange of Ca^{2+} and Na^+ . A derivation for this general case gives

$$K_{A,B} = \left(\frac{C}{Q}\right)^{n-1} \frac{y_A(1 - x_A)^n}{x_A(1 - y_A)^n} \quad (15-44)$$

For unequal counterion charges, we see that $K_{A,B}$ depends on the ratio C/Q and on the ratio of charges, n .

When experimental data for $K_{A,B}$ for a particular binary system of counterions with a particular ion exchanger are not available, the method of Bonner and Smith [39], as modified by Anderson [38], is used for screening purposes or preliminary calculations. In this method, the molar selectivity coefficient is

$$K_{ij} = K_i/K_j \quad (15-45)$$

where values for relative molar selectivities K_i and K_j are given in Table 15.5 for cations with an 8% cross-linked strong-acid resin and in Table 15.6 for anions with strong-base resins. For values of K in these tables, the units of C and Q are, respectively, eq/L of solution and eq/L of bulk bed volume of water-swelled resin.

A typical cation-exchange resin of the sulfonated styrene-divinylbenzene type, such as Dowex 50, as described by

Table 15.5 Relative Molar Selectivities for Cations with 8% Cross-linked Strong-Acid Resin

Li^+	1.0	Zn^{2+}	3.5
H^+	1.3	Co^{2+}	3.7
Na^+	2.0	Cu^{2+}	3.8
NH_4^+	2.6	Cd^{2+}	3.9
K^+	2.9	Be^{2+}	4.0
Rb^+	3.2	Mn^{2+}	4.1
Cs^+	3.3	Ni^+	3.9
Ag^+	8.5	Ca^{2+}	5.2
UO_2^{2+}	2.5	Sr^{2+}	6.5
Mg^{2+}	3.3	Pb^{2+}	9.9
		Ba^{2+}	11.5

Table 15.6 Approximate Relative Molar Selectivities for Anions with Strong-Base Resins

I^-	8	OH^- (Type II)	0.65
NO_3^-	4	HCO_3^-	0.4
Br^-	3	CH_3COO^-	0.2
HSO_4^-	1.6	F^-	0.1
NO_2^-	1.3	OH^- (Type I)	0.05–0.07
CN^-	1.3	SO_4^{2-}	0.15
Cl^-	1.0	CO_3^{2-}	0.03
BrO_3^-	1.0	HPO_4^{2-}	0.01

Bauman and Eichhorn [40] and Bauman, Skidmore, and Osmun [41], has an exchangeable ion capacity of 5 ± 0.1 meq/g of dry resin. As shipped, the water-wet resin might contain 41.4 wt% water. Thus, the wet capacity is $5(58.6/100) = 2.9$ meq/g of wet resin. If the bulk density of a drained bed of wet resin is 0.83 g/cm^3 , the bed capacity is 2.4 eq/L of resin bed.

As with other separation processes, a separation factor, $S_{A,B}$, which ignores the valence of the exchanging ions, can be defined for an equilibrium stage. For the binary case, in terms of equivalent ionic fractions:

$$S_{A,B} = \frac{y_A(1 - x_A)}{x_A(1 - y_A)} \quad (15-46)$$

which is identical to (15-43). Experimental data for an exchange between Cu^{2+} (A) and Na^+ (B) (counterions of unequal charge) with Dowex 50 cation resin over a wide range of total-solution normality at ambient temperature are shown in terms of y_A and x_A in Figure 15.14, from Subba Rao and David [42]. At low, total-solution concentration, the resin is highly selective for copper ion, whereas at high, total-solution concentration, the selectivity is reversed to favor sodium ion slightly. A similar trend was observed by Selke and Bliss [43, 44] for exchange between Ca^{2+} and H^+ using a similar resin, Amberlite IR-120. This sensitivity of the selectivity is shown dramatically in Figure 15.15, from Myers and Byington [45], where the natural logarithm of the separation factor, $S_{\text{Cu}^{2+}, \text{Na}^+}$, as computed from the data of Figure 15.14 with (15-46), is plotted as a function of

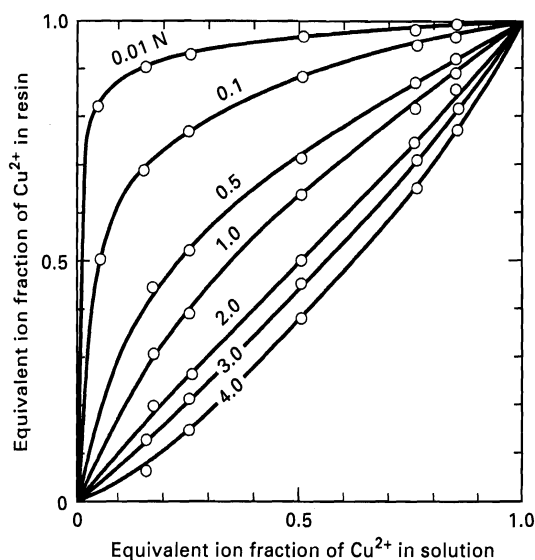


Figure 15.14 Isotherms for ion exchange of Cu^{2+} and Na^+ on Dowex 50-X8 as a function of total normality in the bulk solution. [From A.L. Myers and S. Byington, *Ion Exchange Science and Technology*, M. Nijhoff, Boston (1986) with permission.]

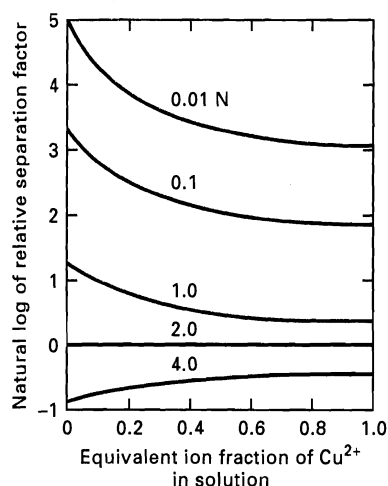


Figure 15.15 Relative separation factor of Cu^{2+} and Na^+ for ion exchange on Dowex 50-X8 as a function of total normality in the bulk solution.

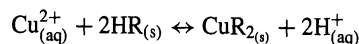
[From A.L. Myers and S. Byington, *Ion Exchange Science and Technology*, M. Nijhoff, Boston (1986) with permission.]

equivalent ionic fraction, $x_{\text{Cu}^{2+}}$. For dilute solutions of Cu^{2+} , $S_{\text{Cu}^{2+},\text{Na}^+}$ ranges from about 0.5 at a total concentration of 4 N to 60 at 0.01 N. In terms of $K_{\text{Cu}^{2+},\text{Na}^+}$ of (15-44), with $n = 2$, the corresponding variation is computed to be only from about 0.6 to 2.2.

EXAMPLE 15.7

An Amberlite IR-120 ion-exchange resin similar to that of Example 15.2, but with a maximum ion-exchange capacity of 4.90 meq/g of dry resin, is used to remove cupric ion from a waste stream containing 0.00975-M CuSO_4 (19.5 meq Cu^{2+} /L solution). The spherical resin particles range in diameter from 0.2 to just over 1.2 mm. The equilibrium ion-exchange reaction is of the divalent-monovalent

type:



As ion exchange takes place, the milliequivalents of cations in the aqueous solution and in the resin remain constant.

Experimental measurements by Selke and Bliss [43, 44] show an equilibrium curve of the type of Figure 15.14 at ambient temperature that is markedly dependent on the total equivalent concentration of the aqueous solution, with the following equilibrium data for the cupric ion with a 19.5 meq/liter solution:

c , meq Cu^{2+} /L Solution	0.022	0.786	4.49	10.3
q , meq Cu^{2+} /g Resin	0.66	3.26	4.55	4.65

These data follow a highly nonlinear isotherm.

- From the data, compute the molar selectivity coefficient, K , at each value of c for Cu^{2+} and compare it to the value estimated from (15-45) using Table 15.5.
- Predict the milliequivalents of Cu^{2+} exchanged at equilibrium from 10 L of 20 meq Cu^{2+} /L using 50 g of dry resin with 4.9 meq of H^+ /g.

SOLUTION

- Selke and Bliss do not give a value for the resin capacity, Q , in eq/L of bed volume. Assume a value of 2.3. From (15-44):

$$K_{\text{Cu}^{2+},\text{H}^+} = \left(\frac{C}{Q} \right) \frac{y_{\text{Cu}^{2+}}(1 - x_{\text{Cu}^{2+}})^2}{x_{\text{Cu}^{2+}}(1 - y_{\text{Cu}^{2+}})^2}$$

where $(C/Q) = 0.0195/2.3 = 0.0085$

$$x_{\text{Cu}^{2+}} = c_{\text{Cu}^{2+}}/19.5 \quad \text{and} \quad y_{\text{Cu}^{2+}} = q_{\text{Cu}^{2+}}/4.9$$

Using the above values of c and q from Selke and Bliss:

q , meq Cu^{2+} /g	$x_{\text{Cu}^{2+}}$	$y_{\text{Cu}^{2+}}$	$K_{\text{Cu}^{2+},\text{H}^+}$
0.66	0.00113	0.135	1.35
3.26	0.0403	0.665	1.15
4.55	0.230	0.929	4.04
4.65	0.528	0.949	1.30

The average value of K is 2.0. The values in Table 15.5 when substituted into (15-45) predict

$$K_{\text{Cu}^{2+},\text{H}^+} = 3.8/1.3 = 2.9$$

which is somewhat higher.

- Assume a value of 2.0 for $K_{\text{Cu}^{2+},\text{H}^+}$ with $Q = 2.3$ eq/L. The total solution concentration, C , is 0.02 eq/L. Equation (15-44) becomes

$$2.0 = \left(\frac{0.02}{2.3} \right) \frac{y_{\text{Cu}^{2+}}(1 - x_{\text{Cu}^{2+}})^2}{x_{\text{Cu}^{2+}}(1 - y_{\text{Cu}^{2+}})^2} \quad (1)$$

Initially, the solution contains $(0.02)(10) = 0.2$ equivalents of cupric ion with $x_{\text{Cu}^{2+}} = 1.0$. Let a = equivalents of Cu exchanged. Then, at equilibrium, by material balance:

$$x_{\text{Cu}^{2+}} = \frac{0.02 - (a/10)}{0.02} \quad (2)$$

$$y_{\text{Cu}^{2+}} = \frac{(a/50)}{0.0049} \quad (3)$$

Substitution of (2) and (3) into (1) gives

$$2.0 = 0.0087 \frac{\left[\frac{(a/50)}{0.0049} \right] \left[1 - \frac{0.02 - (a/10)}{0.02} \right]^2}{\left[\frac{0.02 - (a/10)}{0.02} \right] \left[1 - \frac{(a/50)}{0.0049} \right]^2} \quad (4)$$

Solving (4), a nonlinear equation, for a gives 0.1887 equivalents of Cu exchanged. Thus, $0.1887 / [(0.020)(10)] = 0.944$ or 94.4% of the cupric ion is exchanged.

Equilibria in Chromatography

As discussed in Section 15.1, separation by chromatography involves sorption mechanisms of many types, including adsorption on porous solids, absorption or extraction (partitioning) in liquid-supported or bonded solids, and ion exchange in synthetic resins. Thus, at equilibrium, depending upon the sorption mechanism, equations such as (15-19), (15-24), (15-32), and (15-33) for gas adsorption; (15-35) and (15-36) for liquid adsorption; (6-17) to (6-20) for gas absorption; (8-1) for liquid extraction; and (15-38), (15-43), and (15-44) for ion exchange apply.

When the equilibrium (distribution or partition) constant is defined as

$$K_i = q_i / c_i \quad (15-47)$$

where q is concentration in the stationary phase and c is concentration in the mobile phase, solutes with the highest equilibrium constants will elute from the chromatographic column at a slower rate than solutes with the smallest equilibrium constants.

15.3 KINETIC AND TRANSPORT CONSIDERATIONS

For the adsorption of a solute onto the porous surface of an adsorbent, the following steps are required:

1. External (interphase) mass transfer of the solute from the bulk fluid by convection, through a thin film or boundary layer, to the outer, solid surface of the adsorbent
2. Internal (intrapphase) mass transfer of the solute by pore diffusion from the outer surface of the adsorbent to the inner surface of the internal porous structure
3. Surface diffusion along the porous surface
4. Adsorption of the solute onto the porous surface

For chemisorption, which involves bond formation, the rate of the fourth kinetic step may be slow and even controlling; for physical adsorption, however, step 4 is almost instantaneous because it depends only on the collision frequency and orientation of the molecules with the porous surface. Thus, only the first three steps need be considered here.

During regeneration of the adsorbent, the reverse of the four steps occurs, where the rate of physical desorption is instantaneous. Adsorption and desorption are accompanied

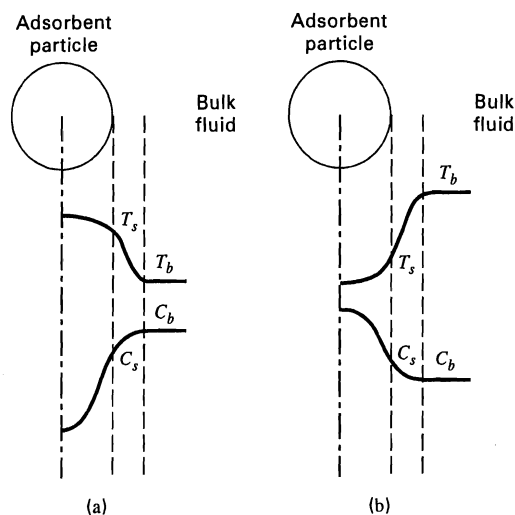


Figure 15.16 Solute concentration and temperature profiles for a porous adsorbent particle surrounded by a fluid: (a) Adsorption. (b) Desorption.

by heat transfer because of the exothermic heat of adsorption and the endothermic heat of desorption. However, although external mass transfer is limited to a convective mechanism, external heat transfer from the particle outer surface occurs not only by convection through the film or boundary layer surrounding each solid particle in the bed, but also by thermal radiation between particles when the fluid is a gas, and by conduction at points of contact by adjacent particles. Conduction and radiation mechanisms for heat transfer can also exist within the particle, in addition to convective heat transfer by the fluid within the pores.

In a fixed bed of adsorbent particles, solute concentration and temperature change continuously with time and location. For a given particle at a particular time, profiles of temperature and solute concentration in the fluid are as shown in Figures 15.16a and b for adsorption and desorption, respectively, where subscripts b and s refer to bulk fluid and particle outer surface, respectively. The fluid concentration gradient is usually steepest within the particle, whereas the temperature gradient is usually steepest in the fluid film or boundary layer surrounding the particle. Thus, although the major resistance to heat transfer is usually external to the adsorbent particle, the major resistance to mass transfer usually resides in the adsorbent particle. All four gradients in Figure 15.16 approach asymptotic values at the end points.

External Transport

Rates of convective mass and heat transfer between the outer surface of a particle and the surrounding bulk fluid during an adsorption process are given, respectively, by

$$n_i = \frac{dN_i}{dt} = k_c A (c_{b_i} - c_{s_i}) \quad (15-48)$$

$$q = \frac{dQ}{dt} = h A (T_s - T_b) \quad (15-49)$$